CROSSLIGHT Software Inc.

Crosslight Device Simulation Software

- General Description

Updated: 2008.7

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Part I

OVERVIEW

Chapter 1

INTRODUCTION

1.1 How to use this manual

If you have not installed the software, you may read the manual up to Chapter 2, and then install the software. After installation of the software, you may read Chapter 3 to get started.

If you already have experience with the software and want to know more about the models implemented in the packages, please go to the relevant chapters in the manual. Part II is to give a description of physical and numerical models common to all of the different device simulators and is recommended for all users.

1.2 What is Crosslight Software

Crosslight Software includes a number of computer aided design (CAD) simulation packages (PICS3D, LASTIP, APSYS). These packages are designed to simulate semiconductor electronic or optoelectronic devices. We shall also refer to these simulation software packages as **simulators**.

Crosslight simulators are based on finite element analysis in 2/3 dimensions. They involve a large number of sophisticated physical and numerical models. Our purpose here is to give a complete description of these models.

1.3 What is LASTIP

LASTIP (LASer Technology Integrated Program) is a powerful device simulation program designed to simulate the operation of a semiconductor laser in two dimensions (2D). Given the structural and material properties, it produces a large amount of simulation data to describe the lasing characteristics. Based on well established physical models, it provides the user with a quantitative insight into various aspects of a semiconductor laser.

It can be used as a computer aided design (CAD) tool to optimize existing lasers or to assess new designs. With the physical models and advanced capabilities of LASTIP, the user can concentrate on device optimization and design while leaving all the numerical modeling work to the computers.

1.3.1 Applications

LASTIP can be used to model the electrical and optical behaviors on a 2D cross section of all types of semiconductor lasers emitting at any wavelength and from any semiconductor material.

1.3.2 Capabilities

LASTIP solves the appropriate differential equations for both quantum well and bulk semiconductor lasers. LASTIP can be used to analyze a large number of physical variables. These include (but are not limited to) the following:

- 1) Light versus current (L-I) characteristics.
- 2) Current versus voltage (I-V) characteristics.
- 3) 2D potential, electric field and current distributions.
- 4) 2D distributions of electron and hole concentrations.
- 5) Band diagrams under various bias conditions.
- 6) Quantum well subband structure with valence mixing model.
- 7) 2D distributions of occupancy and concentration of deep level traps in a semiconductor.
- 8) 2D optical field distribution.
- 9) 2D local optical gain distribution.
- 10) Modal gain spectrum as a function of current.
- 11) Spontaneous emission spectrum as a function of current.
- 12) Far-field distribution.

- 13) All of the above as a function of time (transient model).
- 14) All of the above at different temperatures.

1.4 What is APSYS

APSYS is a general purpose two-dimensional (2D) finite element analysis and modeling software program for semiconductor devices. It includes many advanced physical models and offers a very flexible modeling and simulation environment for modern semiconductor devices. Advanced features include hot carrier transport, heterojunction models and nonisothermal analysis.

1.4.1 Applications

APSYS can be applied to the modeling and analysis of almost all devices except semiconductor lasers (which are simulated by our other products LASTIP and PICS3D). These include the following devices based on silicon and compound materials.

- 1. Silicon MOSFET's, bipolar transistors and CCD's.
- 2. SiGe HBT's and HBT's of AlGaAs and InGaAsP.
- 3. GaAs MESFET's and Photodectors.
- 4. Devices involving multiple quantum wells.
- 5. Light Emitting Diodes (LED's).

1.4.2 Capabilities

APSYS solves self-consistently the hydrodynamic equations, the heat transfer equations as well as the conventional drift-diffusion equations. Data generated by APSYS include the following:

- 1)Current versus voltage (I-V) characteristics.
- 2)2D potential, electric field and current distributions.
- 3)2D distributions of electron and hole concentrations.
- 4)2D distributions of hot carrier temperatures in the hydrodynamic model.
- 5)2D distributions of lattice temperature for the heat transfer model.

- 6)Band diagrams under various bias conditions.
- 7)Results of AC small signal analysis for any frequency range. Extraction of 2-port AC parameters such S- and Y- parameters.
- 8)Quantum well subband structure with valence mixing model for quantum devices.
- 9)2D distributions of occupancy and concentration of deep level traps in a semiconductor.
- 10)2D optical field distribution for photonic devices such as photodetectors.
- 11)Spontaneous emission spectrum as a function of current for LED's.
- 12)All of the above as a function of time (transient model).
- 13)All of the above at different environment temperatures.

1.5 What is PICS3D

PICS3D (Photonic Integrated Circuit Simulator in 3D) is a three dimensional (3D) simulator for laser diodes and related waveguiding photonic devices/circuits. Based on 2/3D finite element analysis, it solves the semiconductor and optical wave equations to provide an accurate description of the device characteristics. When calibrated with specific material/process, it can be used as a computer-aided design tool to optimize existing devices or to assess new designs.

1.5.1 Applications

PICS3D is designed to simulate a variety of semiconductor optoelectronic devices including the following.

- 1. Fabry-Perot (FP) lasers.
- 2. Distributed Feedback (DFB) lasers.
- 3. Distributed Bragg Reflector (DBR) lasers.
- 4. Semiconductor Optical Amplifiers.
- 5. Waveguide photodetectors.
- 6. Vertical Cavity Surface Emitting Lasers (VCSELs).

- 7. External cavity lasers.
- 8. Fiber grating lasers.
- 9. Electrode absorption modulators.
- 10. Multisection/Multielectrode DFB or DBR lasers.
- 11. Multisection photonic integrated circuit combining more than one of the above devices.

1.5.2 Capabilities

PICS3D usually combines solutions of the semiconductor equations in multiple 2D cross sections of a waveguiding device with the longitudinal mode solution to construct a quasi-3D model of a photonic device. Alternatively, full 3D solution of the semiconductor equations may be used at a higher computing cost. Both quantum well and bulk semiconductor lasers can be modeled. Advanced quantum well models (such as k.p theory) are implemented, which allow the user to model strained quantum wells. The semiconductor materials implemented include GaAs/AlGaAs, InGaAsP grown on InP and GaAs, InGaAs/GaAs, InGaP and many others. The capabilities of simulation include (but are not limited to) the following.

- 1)Light versus current (L-I) characteristics.
- 2)3D potential, electric field and current distributions.
- 3)3D distributions of electron and hole concentrations.
- 4)Band diagrams under various bias conditions.
- 5)Quantum well subband structure with valence mixing model.
- 6)3D distributions of occupancy and concentration of deep level traps in a semiconductor.
- 7)3D optical field distribution.
- 8)3D local optical gain distribution.
- 9) Full multiple mode emission spectra at different power levels.
- 10) Lasing wavelength, output power and longitudinal photon density distribution as a function of bias current.
- 11) Characteristics of DFB lasers with spatial and spectral hole burning effects.

- 12) Full multi-mode simulation of DFB lasers.
- 13) Relative Intensity Noise (RIN), Frequency Noise (FM) and spectral linewidth under different bias conditions.
- 14) Static tuning and dynamic modulation characteristics of single- and multi-electrode DFB or DBR lasers.
- 15) Second harmonic distortion in a laser system under direct current modulation.

Chapter 2

INSTALLATION

2.1 System Requirements

The basic system requirements are IBM-compatible PC with 256 MB of memory, at least 200 MB of hard disk space on c: drive and 250 MB of swap space. It is recommended that memory of over 250 MB be available since virtual memory/swap space is much slower.

It is also recommended that the more stable, later versions of Windows 2000 or XP be used. For higher perfromance on large scale computations, we recommend use multi-cpu options or 64-bit computers.

2.2 Installation

The installshield will guide you through the installation process. Version-specific instructions may come with installation CD or with downloads. The program may be launched from the Start ->Programs menu.

Chapter 3

BASIC FILES AND PROCEDURES

3.1 For the Impatient

We assume you know how to use a personal computer installed with the Microsoft Windows operating system. Using the Word program on Windows means opening a file with extension .doc on the computer.

To start the Crosslight software, (suppose you are using LASTIP package, APSYS and PICS3D follow the same routine), click on

```
Start ->All Program -> Lastip_200x.xx
```

The Simucenter window will pop up, which is the main Graphics User Interface of Crosslight Software. Any license problems will stop the program at this moment. We have produced several introductory movies to help users get started. Please click on HELP from the main menu and select "Movies: Getting Started", and have fun with the movies!

After watching the movie, you may want to explore the software package by yourself. First run the example input files under the "examples" directory that comes with the installation. In many cases, adjusting device parameters in an example results in structures similar to your target device.

Start the SimuCenter (standing for SimuLastip, SimuApsys and SimuPics3d) program. Click on File->Open and browse and select a existing project in the lastip_examples (apsys_examples, pics3d_examples), load the project by double clicking on .sol, .gain or .spj file. Use the mouse to highlight the input files and apply the simulation program on them in the following order, by right clicking on each files, respectively:

1. Run the "*.layer" files to generate the .geo file.

2. Run the "*.geo" files to generate the mesh.

3. Run the "*.mplt" files to inspect the mesh generated if you wish to know if the mesh is reasonable. Optionally you can run "*.gain" files to preview the gain spectrum or other critical physical variables.

4. Run the main equation solver with input files in "*.sol" .

5. Analyze and plot the results with the input files of " *.plt". Alternatively, you may open the .std output files with the CrosslightVew program as you will be prompted when clicking on the .std files.

6. Alternatively, you may start CrosslightView by clicking on "Action -> View Results – > CrosslightView"

3.2 How Crosslight Simulation Program Works

A simulation is controlled by one or more input files. Applying the executable of the program on these input files activates the numerical simulation. An input file contains a collection of statements recognized by the program. The statements have been invented by designers of Crosslight to allow the user to interact with the program. There are three basic input file types with different file extensions: .layer, .sol and .plt . These are used to input the device geometry and generate the mesh, solve the equations and plot the resutls, repsectively. A number of user interface programs have been created to facilitate the creation of these input files. These are described in the following sections. Depending on purposes and detailed structure, other related input file types are also created.

3.2.1 Input / Output Files

The overall file structure is show in Fig. 3.1 As we discussed previously, the main input files are .layer, .sol and .plt. There are also other auxiliary input / output files that the user need to know in order to use the simulation software effectively. We give a detailed description of the most important input / output files as follows.

1. .geo. The main input file that describe the full details of the device geometry and the initial mesh allocation.

2. .sol. The main input file that defines the material properties and controls the bias and other conditions of main equation solver.

3. .layer. An important auxiliary input file that uses the layer structure description to generate the .geo, .doping, .mater and .mplt files,

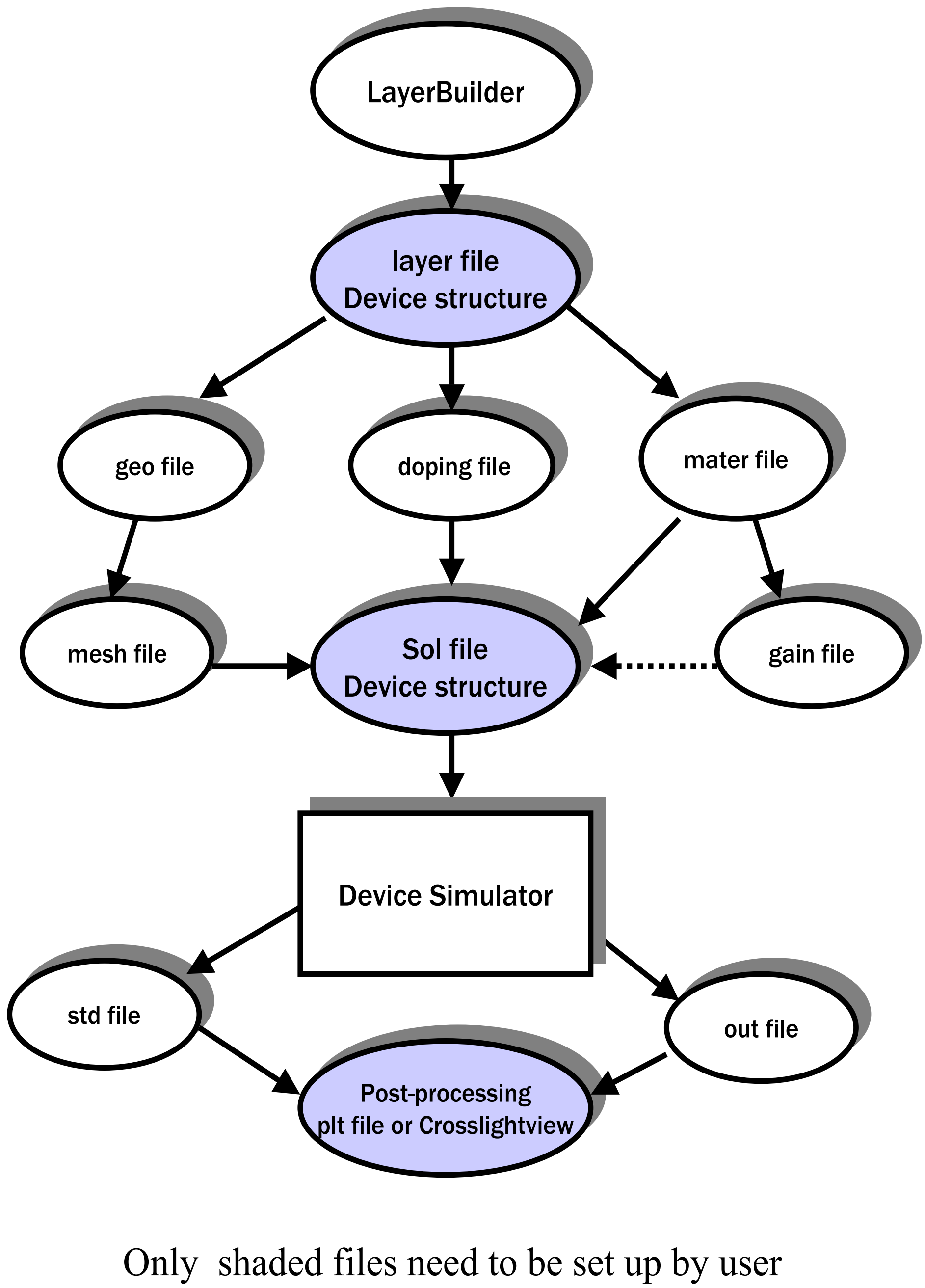


Figure 3.1: Overall file structure and flow chart of Crosslight device simulators

4. .doping which contains doping information that is to be included in the .sol file.

5. .mater which contains material information that is to be included in the .sol file.

6. .mplt can be used to plot the mesh generated from .geo file.

7. .gain file is another important auxiliary input file that can be used to preview the optical gain spectrum, spontaneous emission rate spectrum, quantum well subbands, and other critical physical properties. This may be used by the user to do some preliminary estimate before the full simulation is performed.

8. .out files may appear as .out_0001, .out_0002, etc.. These are numerical output data from the main equation solver. They can be used by the .plt program to be plotted. These output files are not meant to be understood by the user.

9. .std files may appear as .std_0001, .std_0002, etc.. These are another form of numerical output data from the main equation solver. They can be used by the CrosslightView program to be displayed in 3D color graphics. These output files can be understood by the user should there were such a need.

10. .plt file is used to plot the data in .out output files. The program calls the public domain software GNUPLOT to display the graphics in various computer platforms and printers.

3.2.2 The direct approach

Since the core simulation software interacts directly with the input files (.layer, .sol and .plt), the direct approach is to directly edit/control input files line by line. The input files may be copied from an existing example or be created by one of the set up utility programs. The direct approach is recommended for users who wish to use the software continuously for a relatively long period of time and wish to use the full power of the software, because

1. It allows the user to use the full capabilities of the software.

2. Most examples are given in input files and can be copied and modified for new simulations.

3. It is easier to get support for simulation. If an error is encountered, the user can simply e-mail the input file for assistance.

4. The input files are the same for all computer platforms. This makes it easy to upgrade to a new computer in the future.

3.2.3 The Setup Utility Programs

The direct use of main input files .layer, .sol and .plt may turned out to be difficult for new users. The main difficulty lies in the large number of adjustable parameters the user has to deal with for these input files. However, only a small number of parameters needs to be adjusted for commonly used structures and simulation requirements. To reduce the number of parameters the user needs to define, we can use the Setup Utility Programs (SUP's) to generate simplified version of the input files.

The SUP's are text based interactive programs that can be run from any computer OS. These include the following:

1. SetupLayer program that prompts the user to enter material layer thickness, doping and material macro library. It will generate the .layer file which may then be used to generate .geo, .doping, .mater and .mplt.
2. SetupApsys, SetupLastip, or SetupPics3d program that can be used to produce the .sol, .gain and .plt input files.

3.2.4 The graphic user interface

The text based input files .geo, .sol and .plt may be difficult for new users to create from scratch. For those users who prefer to run a program using the "point and click", a number of Graphic User Interface (GUI) programs are available (depending on the computer platforms).

At the time of printing of this manual, the following GUI's are available:

1. SimuCenter. A GUI that activates the main executable apsys.exe and also integrates various device setup utilities and graphic input / output programs.
2. LayerBuilder which generates layers of material used to build a device. Compatible with the .layer input file format which is to be explained shortly in this chapter.
3. GeoEditor which allows the user to draw a complex device in irregular shapes. Compatible with the .geo input file format.
4. CrosslightView. A 3D color graphic display tool using the OpenGL technology. Compatible with the .std file format.

Please refer to the manuals of them respectively for details, which can be accessed from the HELP of the Simucenter.

3.3 Creating a Good Mesh

The first step in running a device simulation is to load or create a simulation mesh based on the geometry and material properties of the cross section of your device. Usually, the mesh data file is given a .msh file extension and it contains the finite element mesh coordinates and node numbers. The mesh data file is used by the main simulation input file (.sol file) with the statement **load_mesh**.

The .msh data file is generated by an input file with extension ".geo" which contains all the details about the geometry and material of the device. The ".geo" input file provides the most versatile and flexible geometric data specification for finite element analysis. Virtually any device geometry or material variation may be handled by the ".geo" data format.

Since the ".geo" file requires the specification of absolute coordinates of the device geometry, it may appear tedious for some simple structures, especially devices with parallel layers. Therefore we have created a separate program call LAYER to help generate the ".geo" file easily. The LAYER program is controlled by a more simple input file with file extension ".layer".

The following subsections describe the ".geo" and and ".layer" input file formats. Serious users of the simulator should study the ".geo" input file format. For beginners or casual users, ".layer" input file may be sufficient.

3.3.1 SetupLayer

The Setup Utility Program (SUP) SetupLayer may easily be used to generate the .layer file. To run this SUP to generate a .layer file (say test1.layer), On DOS prompt, type

```
setuplayer.exe -layer test1 [enter]
```

From Simucenter, right click on the .layer file and choose "set up layer by command"

Then all you need is to answer the questions prompted by the SUP. We illustrate its use by setting up a very simple 1D laser diodes of the following structure. For other devices with an active layer, the procedure is similar.

```
--------------------------------------
1 um  Al(0.5)Ga(0.5)As  p=1.e24
--------------------------------------
0.2 um  GaAs  undoped;  active region
--------------------------------------
```

```
1 um  Al(0.5)Ga(0.5)As  n=1.e24
--------------------------------------
```

Please note that when we enter the layer structure, we start from bottom as layer number 1 and go all the way up to the top.

The SUP runs as follows:

```
DOS>Apsys -setlayer -layer test1


 Setting up file:test1.layer

 We assume a 1D layer structure on the xy-plane.

 Please enter the device half-width (um)
 (We assume you have a symmetric device
    and you only need to simulate half of it)
 [typical:1-5 um]
1.5

   Please enter the layer structure starting from
   bottom as layer 1

 For layer number   1

 Please enter layer thickness (um)
1

 Please select doping (1/m^3):
 n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
 p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
3

 Is this layer an active layer (y/n)?
n
```

```
 Please enter layer bulk material macro to be used
  (1) gaas -- bulk GaAs
  (2) inp -- bulk InP
  (3) ingaas -- bulk InGaAs
  (4) algaas -- bulk AlGaAs
  (5) ingaasp -- bulk InGaAsP (matched to InP)
  (6) ingaasp_xy -- bulk InGaAsP (strained)
  (7) Enter another macro name other than the above
4
 Do you wish to grade Al [y/n] ?
n
 Please enter the Al composition [0-1]
0.5

 Is there grating structure in this layer? [y/n]
n
 Next?
   (1) next layer
   (2) go back to a previous layer
       (so that I can re-enter parameters)
   (3) repeat the previous few layers
       (as in a MQW device)
   (4) generate input files and  quit
   (5) quit
1

 For layer number   2

 Please enter layer thickness (um)
0.2

 Please select doping (1/m^3):
 n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
 p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
6

 Is this layer an active layer (y/n)?
y

 Please enter active layer material macro
```

```
   quantum subbands, gain, etc.

 Please enter the active region material system
 (1) AlGaAs/AlGaAs --- quantum well
 (2) InGaAs/AlGaAs --- strained quantum well
 (3) InGaAsP/InP --- InGaAsP/InGaAsP strained
     quantum well/barrier; substrate=InP.
 (4) AlGaAs --- bulk AlGaAs
 (5) InGaAsP --- bulk InGaAsP lattice matched to InP
 (6) InGaAsP/GaAs --- InGaAsP/InGaAsP strained
     quantum well/barrier; substrate=GaAs.
 (7) InGaAlAs/InP  --- InGaAlAs/InGaAlAs strained
     quantum well/barrier; substrate=InP.
 (8) GaN --- bulk GaN (9) AlGaN --- bulk AlGaN
 (10) InGaN --- bulk InGaN
 (11) InGaN/AlGaN -- strained well/barrier;
      substrate = GaN.
 (12) InGaAlP/GaAs --- In(1-xw-yw)Ga(xw)Al(yw)P
      /In(1-xb-yb)Ga(xb)Al(yb)P; substrate=GaAs
 (13) InGaN/InGaN --- Strained well grown on material
      lattice matched to GaN.
4
 Enter Al composition (0 to 1)
0
 Enter Temperature (Kelvin)
300

 Please enter layer bulk material macro to be used
  (1) gaas -- bulk GaAs
  (2) inp -- bulk InP
  (3) ingaas -- bulk InGaAs
  (4) algaas -- bulk AlGaAs
  (5) ingaasp -- bulk InGaAsP (matched to InP)
  (6) ingaasp_xy -- bulk InGaAsP (strained)
  (7) Enter another macro name other than the above
4
 Please enter the Al composition [0-1]
0.

 Is there grating structure in this layer? [y/n]
n
 Next?
   (1) next layer
```

```
   (2) go back to a previous layer
       (so that I can re-enter parameters)
   (3) repeat the previous few layers
       (as in a MQW device)
   (4) generate input files and  quit
   (5) quit
1

 For layer number   3

 Please enter layer thickness (um)
1

 Please select doping (1/m^3):
 n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
 p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
9

 Is this layer an active layer (y/n)?
n

 Please enter layer bulk material macro to be used
  (1) gaas -- bulk GaAs
  (2) inp -- bulk InP
  (3) ingaas -- bulk InGaAs
  (4) algaas -- bulk AlGaAs
  (5) ingaasp -- bulk InGaAsP (matched to InP)
  (6) ingaasp_xy -- bulk InGaAsP (strained)
  (7) Enter another macro name other than the above
4
 Do you wish to grade Al [y/n] ?
n
 Please enter the Al composition [0-1]
0.5

 Is there grating structure in this layer? [y/n]
n
 Next?
   (1) next layer
   (2) go back to a previous layer
```

```
       (so that I can re-enter parameters)
   (3) repeat the previous few layers
       (as in a MQW device)
   (4) generate input files and  quit
   (5) quit
4
 Please enter the total number of mesh points
   for all the layers;  typical [20-40]
40
 Enter the maximum average mesh spacing:
   typical [0.1 - 0.5 um]
0.1
 Generating .layer file
```

The result of the test1.layer looks like the following

```
$file:test1.layer
begin_layer
column column_num=1 w=  0.150000E+01 mesh_num=2  r=1.
bottom_contact column_num=1 from=0 to=  0.150000E+01 &&
  contact_num=1 contact_type=ohmic
$
layer_mater  macro_name=algaas               &&
  var1=  0.500000E+00 &&
  column_num=1
layer d=  0.100000E+01 n= 16 &&
  n_doping1=  0.1000E+25 &&
  r=  0.8000E+00 xp1=0 xp2=1
$
layer_mater  macro_name=algaas               &&
  var1=  0.000000E+00 &&
  active_macro=AlGaAs               &&
  avar1=  0.000000E+00 &&
  avar2=  0.300000E+03 &&
  column_num=1
layer d=  0.200000E+00 n=  6 &&
  r= -0.1200E+01 xp1=1 xp2=1
$
layer_mater  macro_name=algaas               &&
  var1=  0.500000E+00 &&
  column_num=1
```

```
layer d=  0.100000E+01 n= 16 &&
  p_doping1=  0.1000E+25 &&
  r=  0.1200E+01 xp1=1 xp2=0
$
top_contact column_num=1 from=0 to=  0.150000E+01 &&
  contact_num=2 contact_type=ohmic
end_layer
```

We will explain the .layer file in the following subsection.

3.3.2 Growing the layers: .layer files

A natural way to define the device geometry is to use the ".layer" file which may be used to generate the more complicated ".geo" file. For the .geo file the user must calculate the coordinates of each point. This may become tedious for structures with many layers. Some simplification comes in when a device consists of many layers parallel to each other. For this type of layered structure, the device is divided into layers and columns. All we need to specify is the material and thickness of each layer and there is no need to calculate the coordinates of each point of a polygon.

After the user has created this .layer file, a separate program which comes with the installation package called LAYER (which may also be activate by Apsys -layer) performs the conversion to the larger and more tedious ".geo" format.

The first statement in the ".layer" file is **begin_layer** and the last **end_layer**. In this method, the device geometry is divided into columns and layers, which results in the division of the device into polygons (mostly rectangles), the origin of each at its lower left-hand corner. Modifications may then be made to the shape of each polygon with the constraint that the upper and lower edges of polygon be parallel (see the Reference Manual for a description of the commands used to go about this, *i.e.* upper_w#, lower_w#, angle).

We give brief description of how the ".layer" file is constructed here. An easy way of learning to use the .layer file is to follow one or two examples that come with the .layer files.

1. We use the statement "column" to define the width and mesh lines in the y-direction for each column. If we are treating a 1-d simulation, we only need to put 2 mesh lines in the column.

2. We use "bottom_contact" to define the contact location at the bottom of a column.

3. We list all the layers using the "layer" statement. We start from the bottom layer and go all the way up. The "layer" defines the layer thickness, mesh number used, the mesh ratio, and extra mesh points at the end of each layer (if applicable). Optionally the doping concentration can be specified within the **layer**. The material for the layer can be defined in one of the following two ways.

4. A material number to be consistent with material macro in .sol file may be assigned using "m#=". For example, column number 1 of this layer is assigned material number 3: "m1=3".

5. Or we strongly recommend the use of the statement **layer_mater** to directly define the material macro for this layer. This is especially convenient for graded material because the user does not need to figure out the absolution coordinates of the grading.

6. We use "top_contact" to define the contact location on the top of a column.

If doping concentration is defined for any of the layers, a file with extension .doping will be produced when the LAYER program is applied. The .doping contains all the detailed doping profile statements. This .doping file can be included in the .sol file using **include** statement.

Similarly, if **layer_mater** is used to define the material macros in the .layer file, a file with extension .mater will be produced. Again the **include** statement may be used to include it in the .sol input file.

Here is a very simple .layer file that is used to define the geometry of a device with three polygons and two different materials.

```
begin_layer
$
$ w=width
$
column column_num=1 w=0.5 mesh_num=3 r=1.0
$
bottom_contact column_num=1 from=0 to=0.5 contact_num=1
$
$ starting from the bottom and go up
$ AlGaAs (m=1);  GaAs (m=2)
$
$begin listing
$ d=layer thickness, n=mesh points, r=mesh ratio
$ m1=material number for column 1, xp1=1 or xp2=1 puts extra mesh points
$ at the end of the layer.
```

```
$
layer d=1.5   n=10  r=0.8     m1=1 xp2=1
layer d=0.8   n=10  r=1.0     m1=2 xp1=1 xp2=1
layer d=1.5   n=10  r=1.2     m1=1 xp1=1
$end listing
$
top_contact column_num=1 from=0 to=1 contact_num=2
$
end_layer
```

Please note that we have neither defined the material nor the doping in the above file. We will have to set the material in .sol to be consistent with the material number assignment in the above. We also need to define the doping in .sol.

A much better approach is to specify both the material and doping all at once in the .layer as in the following statement. The following input file fragment shows how graded material with n-type doping of $1 \times 10^{24} m^{-3}$ is defined. Here the first variable of the Bulk Material Macro (BMM) algaas is graded from 0.71 to 0.33 within a layer thickness of 0.1462 um.

```
$
layer_mater macro_name=algaas grade_var=1 &&
  grade_from=0.71 grade_to=0.33 column_num=1
layer d=0.1462 n=6  r=0.8 xp2=1 n_doping1=1.e24
$
```

For material layer which has optical gain or when quantum well subbands are involved, the layer should be defined as an active layer. The **active_macro** is used to defined the Active Layer Macro (ALM). The corresponding variables for the ALM must be specified with parameters "avar#". An example of active layer is given as follows.

```
$
layer_mater macro_name=gaas column_num=1 &&
  active_macro=AlGaAs/AlGaAs avar1=0. avar2=0.33 avar3=300.
layer d=0.0076 n=4  r=1.0 xp1=1 xp2=1
$
```

Please note that an active layer (or a non-active quantum well layer) must be specified with both BMM and ALM.

3.3.3 Dealing with polygons: .geo files

This section is for experienced user, you may skip it at first.

In a .geo input file, a mesh is created with two sections of statements.

The first section begins with **begin_geometry** and ends with **end_geometry**. The statements in this section are **point** and **polygon**. The basic building blocks of a geometry description are the points and polygons. Be sure to draw a schematic of your device with labels for each point and polygon before you start, since you will likely need it in setting up the input file. The statement **edge_curve** may be used to bend the edge of a polygon to fit a rounded curve.

The second section begins with **begin_meshgen** and ends with **end_meshgen**. The statements in this section include **put_mesh**, **double_mesh** [optional], **multi_double** [optional].

It is important to divide a complicated device into simple polygons (either tetrahedrons or triangles). Tetrahedrons are preferred because they are easier to refine. You definitely need a separate polygon if the material is different; for example, GaAs and AlGaAs can **NOT** be described by a single polygon.

In some cases you may wish to divide the same material into different polygons because the shape is not a simple polygon. Try to avoid simple polygons having two adjacent internal angles both greater than 90 degrees, since the finite elements generated within such a polygon will contain a large amount of undesirable obtuse triangles.

The user must assign each polygon with a material number. Different materials must be given different numbers. The same type of material (say GaAs) with two different regions of doping must not be considered two different materials. The reason is that the SIMULATOR treats the boundary between two different materials with thermionic emission theory and assumes it has a bandgap discontinuity. An insulator must be given a larger material number than that for a semiconductor.

The basic operation is the **put_mesh** statement. It directly allocates the mesh along the edge of a polygon. Note that the mesh lines will automatically propagate to adjacent polygons and there is no need to put mesh on a polygon with mesh extended from other polygons already.

Optionally you can use **double_mesh** to double the mesh density in a specified region of a polygon. Note that the parameters **extra_point_1** and **extra_point_2** can be used to put additional mesh lines near the abrupt heterojunction to define a sharp heterojunction. In APSYS the abrupt junction is as sharp as the mesh spacing straddled across the material boundary.

It is recommended that the user practice setting up a simple mesh to get used to the mesh generator before attempting a more complicated design.

An unsatisfactory grid is the major cause of non-convergence. Of course convergence can be improved by making a fine grid everywhere throughout the device, but there are two constraints. First APSYS allows a limited number of the grid points (4400 in the current version and may be changed upon request). Second, CPU time increases non-linearly with the number of grid nodes. Consequently, solutions with a large number of grid nodes may be prohibitively long – only put a fine mesh where necessary. A fine grid is usually necessary at junctions and other areas where electrical quantities change rapidly with distance. If convergence fails, look for an area of the device that may be active and yet have a coarse grid, modify the grid with the statement **refine_mesh** and re-run the simulation.

APSYS provides powerful techniques to refine the mesh adaptively based on spatial distribution of physical variables or at different bias points. These are described in other parts of the documentation.

The following is a list of the basic statements one must use in the geometry (.geo) input file:

point

polygon

put_mesh

mesh_output

3.4 Setting up a simulation

To be specific, we will explain SetupApsys only. The procedures for other packages are similar.

Setting up the .sol, .gain and .plt file can be painful due to the large number of control parameters involved. A simple way to set up these is to run the interactive program SetupApsys. All you need to do is to answer a number of questions and the input files will be generated for you.

As we mentioned earlier that the SUP uses reduced number of input parameters to help set up the input files. However, you may still need to modify the generated .sol or .plt file to ensure optimal results.

The use of SetupApsys is as follows. Suppose that you already generated the geometry information using SetupLayer or one of the GUI's (LayerBuilder or GeoEditor). Then you need to run the .layer file (if you have not already done so) to produce the .geo file. Run the .geo file to generate the mesh data in .msh file (say test1.msh). The SetupApsys program will need the geometry information contained in test1.msh and also the doping and material information in test1.doping and test1.mater, respectively.

3.4.1 Setting up the .sol file

To generate the .sol file that is used to control the main simulator, you need to have the .msh data file generated from the .geo file. On DOS type

```
setupapsys.exe -sol test1
```

On Simucenter, right click on .sol file and choose "setup by command". Then, simply answer the questions prompted by the SUP and set the bias to be applied, and the contact to be biased, etc.

3.5 Input Statement Syntax

In case you wish to modify the input files, you will need to have some basic understanding of the input statement syntax.

A statement starts with a keyword followed by parameters specified with " =". For statements used to describe the material properties (such as the bandgap), a material statement overrides a previous material statement with the same keyword.

Comments start with " $". A line to be continued terminates with " &&". Several symbols within a statement are invisible to the simulation program and therefore can be used freely to group input data. These symbols are Space, " ,", " (", ")"," [", "]", " {" and " }". One can use [ ] to group two or more numbers together, for example. The length of a line in a statement must **NOT** exceed 80 characters. For a longer statement, the continuation symbol && must be used at the end of the line to be continued to the next line.

The user is referred to the Reference Manual for detailed information on the syntax of statements, parameters and mathematical functions.

3.6 Previewing Physical Properties

Before the main equation solver is used (which may take a long time), you may wish to preview the physical properties such as the optical gain/absorption spectrum. This is accomplished by running Apsys on the file "*.gain". The file starts with the statement **begin_gain** and ends with **end_gain**.

The following is a list of the basic statements you may use in the .gain input file:

```
plot_data
active_reg
```

```
gain_wavel
gain_density
gain_spon
sp.rate_wavel
```

3.7 Solving the Basic Equations

The main input file for the equation solver, *.sol, should start with **begin** and end with **end**. The statements needed to run a simulation can be classified into three categories which should be placed in the following order as detailed in the subsections. A full list of a .sol file is given below:

```
$file:camel.sol
$***********
begin


$input and output directions
load_mesh mesh_inf=camel.msh
output sol_outf=camel.out

$ Material and doping files
include file=camel.doping
include file=camel.mater

more_output ac_data=yes

$newton_par is used to control the Newton solver
newton_par damping_step=5. var_tol=1.e-9 res_tol=1.e-9 &&
  max_iter=100 opt_iter=15 stop_iter=50 print_flag=3 &&
  mf_solver=2

$ Simulation actions start here
$First get equilibrium state
$ equilibrium solution is counted as scanline=1
equilibrium

newton_par damping_step = 2 var_tol = 1.000000e+000  &&
    res_tol = 1.000000e-004 max_iter = 30 opt_iter = 15  &&
    stop_iter = 15 print_flag = 3 change_variable = no  &&
    mf_solver = 0
```

```
$ramping up valtage
$ The first scan statement is counted as scanline=2

scan var=voltage_1 value_to=1.6 print_step=0.2 &&
  init_step=0.01 min_step=1.e-5 max_step=0.05

end
```

3.7.1 Input / output direction

This category includes statements **load_mesh** and **output**. These define the simulation mesh and direct the output data. We recommend that the mesh data file have the same base file name as the .sol file. For example, if the .sol is test1.sol, the mesh file should be named test1.msh. Similarly, the output should be named test1.out.

Using the same base file name ensures that the GUI programs (if they are to be used) will detect and handle the files correctly.

3.7.2 Material and doping statements

The second category includes statements used to define the material, doping profiles (**doping profile**) and to describe the contacts (**contact**). Typically, the statements are **load_macro**, **getactive_layer** and **contact**)

When we define a material, we really just attribute one or more material macro libraries to it. For an active region, we must attribute two macros to it: the bluk material macro (BMM) and the active layer macro (ALM).

The load_macro and get_active_layer actually represent a group of statements used to define the semiconductor material. Each statement is used to describe one property of the material (*e.g.* band gap). Since there are a large number of these statements, they are collected together as macros in the file " C:
crosslight
crosslight.mac", which has also been added to the end of this manual, Appendix B.

If you have better knowledge of the material parameters than those listed in the macro library, you are encouraged to revise the library or override the parameters in the macro by re-issuing an input material statement of proper keyword after the call of the material macro. Please also see the comments in Appendix B.

If you wish to know exactly what material parameters are being used within a macro, you can look at the macro file at "c:\crosslig\crosslight.mac".

3.7.3 Simulation Action

Simulation action means solving the discretized non-linear partial differential equation (PDE's) at a specific bias condition using the Newton's method. The statements that cause such action are **equilibrium** and **scan**. The former means solving the equation at zero bias. The latter can be used to apply a voltage or current bias, let the clock running for transient simulation, and set other control variables.

The **newton_par** is used to supply parameters for the **equilibrium** and **scan** statements.

The Newton's method means reducing the errors of the non-linear equations and the variables using non-linear iterative method.

During run-time, the maximum equation error (which indicates how well the equations are satisfied) and the maximum variable error (which indicates the error on the variables to be solved) are printed. The headings of the table which appears on the screen and their meanings are:

it# : number of iterations.

Eqn-errori : error on the discretized differential equations. If the error is zero, it means that the discretized differential equations have been solved exactly. To be more specific, let the Poisson equation at node i be written as $F_i(V_1, V_2, ..., V_k, ..) = 0$ where V_k is the potential at node k. The value of the equation should be zero for the exact solution. The deviation of this equation from zero is the Eqn-error.

Var-errori : variable error. In the above example,the potential V_i is solved iteratively. Suppose we have gone through m steps to get $V_i^{(m)}$. The variable error is, at this point, $\Delta V_i = V_i^{(m)} - V_i^{(m-1)}$. Var-error should be zero when the solution has converged to a stable solution.

Eqn : gives the equation number and position of maximum error (see node#, eq#).

Variable : gives the equation number and position of maximum error on the variables of the differential equation being solved. This may be the potential (in units of thermal voltage V_t, about 1/40 of a volt), the electron quasi-Fermi level (in V_t), or the hole quasi-Fermi level (in V_t).

: represents one of the four discretized differential

eq : equations, Poisson's equation (eq#1), the electron continuity equation (eq#2), the hole continuity equation (eq#3), or the discretized photon rate equation (eq#5). Note that eq#4 is the electron energy equation, which is not used in APSYS but in other Crosslight products.

node # : used to indicate the location of the node which gives rise to maximum error in the solution. The location of the node number can be found in the .msh file, where the data is listed as

x y node number

At each iteration, APSYS examines the equation errors and determines which has the largest error, *i.e.*,the equation which is having the most difficulty converging. It prints the value of the error, the equation number, and the associated node number. Through this information, one may determine which equation is causing a convergence problem. If the Poisson equation has maximum error, some part of the device may need a finer mesh.

The run-time message indicating whether a bias is applied ("solving equations at equilibrium" or "solving equations with bias") has a value corresponding to those defined by the **scan** statement in the ".sol" file. Note that APSYS has implemented an automated procedure in the **scan** statement for choosing the optimal bias point based on convergence information during run-time. More detail on this procedure is discussed in the Reference Manual. If convergence is achieved, the bias for which it occurred is reported. If it fails to converge, no bias is reported.

The following pages show a typical solver run-time print-out in the standard output computer device.

```
-------------------------------------------------------------

             #    ######   #####  #     #  #####
            # #   #     # #     #  #   #  #     #
           #   #  #     # #         # #   #
          #     # ######   #####     #     #####
          ####### #             #    #          #
          #     # #       #     #    #    #     #
          #     # #        #####     #     #####

                     Version: 3.7.1
                 Release Date: Nov 2000
      Copyright (c) 1995-2000, Crosslight Software Inc.
```

```
                   Simulation for Device:
                         camel.sol
-------------------------------------------------------------
Date(d/m/y):          11          12         2000
Time(s/m/h):           8          19            1

======Statement: load_mesh=====

Loading mesh file: camel.msh
Mesh points =          210

======Statements of load_macro (band_gap,etc)=====
======Statements of doping=====
======Statements of contact=====

Initializing  parameters.
(Please wait)

======Statement: output=====


======Statement: equilibrium=====


Solving equations at equilibrium

Error report for equations and variables:
 it#   eqns       potential
    1  0.858E+02  0.271E+02
    2  0.707E+02  0.238E+02
    3  0.558E+02  0.192E+02
    4  0.413E+02  0.244E+02
    5  0.328E+02  0.199E+02
    6  0.246E+02  0.149E+02
    7  0.163E+02  0.994E+01
    8  0.808E+01  0.498E+01
    9  0.937E-01  0.502E-01
   10  0.511E-03  0.223E-03
   11  0.151E-07  0.630E-08
   12  0.130E-09  0.603E-10
   13  0.122E-10  0.565E-11
  Save bias data and continue.
```

```
  Solver converged at
Voltage: (Volt)  0.0000E+00  0.0000E+00
Current: (A/m)   0.0000E+00  0.0000E+00

 Printing 2D data to output file
     Data file:camel.out_01

 Data set #           1 printed at
Voltage: (Volt)  0.0000E+00  0.0000E+00
Current: (A/m)   0.0000E+00  0.0000E+00


 ======Statement: scan=====


 Solving equations with bias.

Changing: voltage_1            with step:   0.2000E+00
 Error report for equations and variables:
  it#   eqns       potential  elec       hole       other
     1  0.412E+02  0.867E+01  0.395E+01  0.629E+01  0.000E+00
     2  0.250E+02  0.936E+00  0.856E+00  0.801E+00  0.000E+00
     3  0.410E-01  0.818E-01  0.224E+00  0.664E+00  0.000E+00
     4  0.112E-02  0.478E-02  0.424E-02  0.392E-01  0.000E+00
     5  0.397E-05  0.194E-03  0.115E-03  0.936E-03  0.000E+00
     6  0.624E-08  0.535E-05  0.333E-05  0.275E-04  0.000E+00
     7  0.477E-11  0.496E-06  0.309E-06  0.254E-05  0.000E+00
   Save bias data and continue.

Changing: voltage_1            with step:   0.2000E+00
  Solver converged at
Voltage: (Volt)  0.2000E+00  0.0000E+00
Current: (A/m)  -0.1230E-07  0.1230E-07

... etc. ...
... etc. ...
... etc. ...


Changing: voltage_1            with step:   0.5000E-01
  Solver converged at
Voltage: (Volt)  0.1500E+01  0.0000E+00
```

```
Current: (A/m)  -0.4140E-04  0.4140E-04

 Printing 2D data to output file
     Data file:camel.out_02

 Data set #           2 printed at
Voltage: (Volt)  0.1500E+01  0.0000E+00
Current: (A/m)  -0.4140E-04  0.4140E-04

 Generating .std files
 Completed .std files
 Date(d/m/y):          11          12        2000
 Time(s/m/h):          23          19           1
```

Usually doing a simulation means arranging the statements in the input file such that the voltage or current bias of the device is ramped from zero to a desirable value. The following statements should appear in the main solver input file in the following order:

load_mesh

output

contact

load_macro

get_active_layer

active_reg

doping

(sor_par)

(init_wave)

newton_par

equilibrium

newton_par

scan

3.8 Output Data Organization

To be specific, we shall take APSYS as an example to illustrate the output data organization here. The output data for LASTIP is completely the same. The data for PICS3D is somewhat different in that one bias point corresponds to two .out files after the **scan3d** statement is activated.

All of the simulation results of APSYS are written and stored in files defined by the statement **output** in "*.sol". There are two types of output data: One type is the older .out file and the other is the .std file.

The output data are divided into bias data (or scan_data) and structural data (or position dependent data or xy_data, which describe the distribution of physical variables over space or frequency domain). Bias data includes quantities such as contact current, contact voltage and laser power. Structural data includes spatial dependent variables such as potential, electron concentration, current distributions and optical gain spectrum.

APSYS prints and stores all bias data but only prints selected structural data because of disk storage limits. Consider a situation where the user ramps up the bias from 0 volt to 1 volt. APSYS may have to solve the equations in 4 steps (at 0, 0.3, 0.6 and 1 volt) before the bias reaches 1 volt. In such a case all 4 bias data points (*e.g.*, I-V data) are stored and printed in a single output data file. The user may only want to store the structural data at 0 and 1 volt exclusively since these files are large.

The APSYS convention is to store the bias data under a file defined in the **output** statement. The .std file will be automatically generated once the .out file is defined. All structural data (or xy_data) are stored under the same file name with extension " _####", where #### can range from 0001 to 9999. As an example, if the output file is given a name "device1.out", then all bias and structural data will be stored in " device1.out_0001", " device1.out_0002", " device1.out_0003", *etc.*

The data generated by APSYS is organized such that every time APSYS prints a set of structural data it also prints the bias data cumulated since the printing of the previous set of structural data. APSYS groups the printed structural data and the cumulated bias data together and assigns a "data set number" to them. The user refers to the output data with such a data set number.

Consider the above example. The file device1.out_0001 is obtained at 0 volt, device1.out_0002 at 1 volt and device1.out_03 at 1.5 volt. The file extension number is the same as the data set number. The data in the three data files (for .out or .std) are as follows:

```
Data set number 1:  device1.out_0001 or device1.std_0001
[Bias data (scan data):]
0 volt, 0 Amp.
structural data obtained at 0 volt.
```

```
Data set number 2: device1.out_0002 or device1.std_0002
[Bias data (scan data):]
0.3 volt, 0.6 Amp.
0.6 volt, 1.2 Amp.
1.0 volt, 2.2 Amp.
structural data obtained at 1 volt.


Data set number 3:  device1.out_0003 or device1.std_0003
[Bias data (scan data):]
1.2 volt, 3.1 Amp.
1.4 volt, 3.2 Amp.
1.5 volt, 3.5 Amp.
structural data obtained at 1.5 volt.
```

Therefore all data in a APSYS simulation are referred to by a " data set number". The bias condition of a particular printed data set can be controlled by the user and this information is stored in the " message file" with extension " .msg". Upon completion of a simulation with input file " gaas1.sol", the user will find the bias condition for each printed data set in the file " gaas1.sol.msg". The data set numbers are needed when the user wishes to plot the data at a particular bias condition.

The control of data set printing and its relation to bias is given by the **scan** statement which determines the bias to be applied and at what interval shall the results be printed. Please refer to the Reference Manual for more details of **scan** statement.

3.9 Analyzing the Results

If the equation solver has converged up to a certain data set and has printed the data to its output files, you can start looking at the results of the converged data sets without having to wait until the simulation is complete.

You can create an input file labeled "*.plt" for the post processor starting with **begin_pstprc** and ending with **end_pstprc**. Use **plot_1d**, **plot_2d**, **plot_3d**, and **plot_scan** to plot the physical quantities of interest to you.

3.9.1 Setting up the .plt file

There are currently two graphic tools for plotting the simulation data. The first is the GNUPLOT public domain software that can be run on all type of computers. The second is the CrosslightView program that uses the .std output data.

The .plt file instructs APSYS program on what program to be plotted. It can be generated by (on DOS)

```
setuplatip.exe -plt test1
```

On Simucenter, right click on .plt file and choose "setup .plt". The following statements may appear in the post processor input file:

```
plot_data
get_data
plot_1d
plot_2d
plot_3d
plot_scan
plot_ac_curr
plot_ac_parameters
```

Here is the full list of a typical .plt file used for postprocessing

```
$file:camel.plt
$ **************
begin_pstprc
plot_data plot_device=postscript

$ ---- plot structure data at first data set (equilibrium)

get_data  main_input=camel.sol sol_inf=camel.out &&
  xy_data=(1 1)
plot_1d variable=band from=(0.5 0.0) to=(0.5, 2.3)

$ ---- plot structure data at last data set (max. voltage bias)

get_data  main_input=camel.sol sol_inf=camel.out &&
  xy_data=(9 9)
plot_1d variable=band from=(0.5 0.0) to=(0.5, 2.3)

plot_1d variable=elec_curr_y from=(0.5 0.0) to=(0.5, 2.3)

$ Plot bias dep. quantities using scanline specification.
$ Please make sure scan_data range is large enough to cover
$ the intended scanline.
```

```
get_data  main_input=camel.sol sol_inf=camel.out &&
  scan_data=(1 9)
plot_scan scan_var=voltage_1 variable=total_curr_1 scanline=2

$ ====> AC analysis as a post-processing step
$ AC analysis at max. voltage

get_data  main_input=camel.sol sol_inf=camel.out &&
  xy_data=(9 9)

ac_voltage log_freq1=1  log_freq2=10 contact_num=1
plot_ac_curr variable=capacitance_1
plot_ac_curr variable=conductance_1

ac_voltage current_distr=yes log_freq1=1 log_freq2=1

plot_1d_ac_curr variable=elec_curr_y imag_part=no from=(0.5 0.) to=(0.5  2.3)

$ We plot capacitance-voltage and conductance-voltage here at 1 Mhz
$ To generate reasonable AC vs. bias plots, please save sufficient
$ number of data sets.  AC analysis plots must start from data set
$ number 2.

get_data  main_input=camel.sol sol_inf=camel.out &&
  scan_data=(2 9)

ac_voltage log_freq1=6 log_freq2=7 contact_num=1 &&
  freq_point=2 versus_bias=yes scanline=2

set_plot scan_var=voltage_1
plot_ac_curr variable=capacitance_1
plot_ac_curr variable=conductance_1

$ For purpose of demonstrating the plotting,
$ let us plot y parameters assuming input=output=eletrode No. 1

ac_parameters versus_bias=yes log_freq1=6.  log_freq2=7. &&
  input_contact=1 output_contact=1 freq_point=2 scanline=2
plot_ac_parameters parameter_type=y smith_chart=no

end_pstprc
```

Chapter 4

CONVERGENCE ISSUES

We are mainly concerned with the convergence of the drift-diffusion equations in this chapter. When the equation error and the variable error refuse to reduce no matter how small the biasing step is, the simulation cannot be continued. The simulator will give an error message indicating the simulation is stalled. There are several possible causes of convergence difficulties, discussed in the following sections.

Please report all cases of non-convergence in its simplest structure to Crosslight for technical assistance. The simpler the structure, the easier it is to debug.

4.1 Choice of voltage or current bias

The choice of voltage or current bias affects the convergence of the 2D solver. The rule is to use voltage bias for devices with high resistance and use current bias for those with low resistance. For example, a diode under forward bias has low resistance. Current bias should be used in this case. On the other hand, a reverse biased diode has high resistance and voltage bias should be used.

The following discussion is mainly for a forward biased p-n junction for which we wish to injection a certain amount of current. However, the basic principle is also applicable to other type of devices.

The p-n junction of a diode has very high junction resistance at low bias. The current or the conductivity of the junction increases exponentially as the applied forward bias voltage approaches the bandgap of the junction material.

From the basic properties of the forward biased diode discussed above, one can arrange the simulation procedure to achieve optimal performance of a simulation by starting with a voltage bias because the resistance of the diode is high at low bias. In this way the current may easily be pumped up to a level comparable to, but still below, the lasing threshold. This corresponds to ramping up the voltage to about

80-90% of the bandgap of the active region material. Then one should turn on the current bias all the way to its maximum value.

The simulator may be driven into a non-convergent situation if the above procedures are not followed. Here are the most common errors and causes for laser diodes with small band gap and low contact resistance.

1) The voltage bias is too low, or not applied at all, before the current bias is turned on. Under zero or low bias condition, the p-n junction has a high resistance. Such a condition means that even a small control current applied afterward can cause a large change or fluctuation in the internal voltage of the junction, and may result in instability in the main Newton solver. Therefore one should apply an adequate voltage (80-90% of bandgap) before turning on the current bias.

2) The applied voltage bias is too high. When the voltage bias is greater than the turn-on threshold of the diode, the injection current starts to increase at an exponential rate. A small increment in the voltage at such a high bias can cause a huge increase in the current and can drive the simulator into divergence.The same situation occurs in breakdown of a MOSFET at high drain voltage.

For wide bandgap material with high material/contact resistance, the story is somewhat different. This is especially true for vertical cavity surface emitting lasers (VCSEL's) where the resistance due to DBR mirrors are quite high. In these type of devices, the resistance is high even under forward bias condition. Therefore, use of voltage bias all the way from zero bias to above threshold may have more advantage in achieving convergence.

4.2 Physical boundary conditions

When the boundary condition is not reasonable, it may cause unconvergence situation. One such example ocurr when an Ohmic contact is too close to an active region or an area with large splitting of IMREF's. The Ohmic contact is an ideal boundary that forces the IMREF's to the same equilibrium level. Such an abrupt change may be hard for the continuity equation to follow and therefore, lead to unconvergence.

Another common case of unconvergence is in thermal simulation where an external heat resistor is attached to a contact which is too close to the self-heating region.

4.3 Coarse mesh

A coarse mesh is a common cause of non-convergence. It is wise to stop the simulation and plot the distribution of potential and carrier concentrations obtained from the previous data set or the equilibrium solution. Inspection of the plots may enable you to locate rough mesh points in regions where solutions vary rapidly with distance. Refinement of the mesh using the statement **refine_mesh** may improve the chance of getting a converged solution.

If a refined mesh does not help, try to reduce the complexity of the device (*e.g.* try a 1D device with similar structure). Testing a simple device may help you to find out the cause of the problem in the more complex device. Sometimes you may find that the solution has difficulty converging for a particular type of junction. This can be caused by approximations made to boundary conditions, or by unrealistic conditions unintentionally imposed.

4.4 Use of basic variables

The default basic variables in our simulator are the potential and quasi-Fermi levels. In some structures with many current blocking or high resistance regions, the solution at low bias is difficult to obtain with high accuracy. The reason is that under high resistance, the current is so small that to satisfy the current continuity equation within a small but finite tolerance, there are actually many possible solutions for the quasi-Fermi levels. As a consequence, the variables tend to vary between these many solutions and refuse to converge to a unique solution.

The trouble is in the use of quasi-Fermi levels in regions where both the current and minority carrier densities are small. In these regions the quasi-Fermi level is not small numerically and it may fluctuate and cause convergence problem in the solution. Devices of this type includes the laser diodes (LD) with highly doped current blocking layers, junction field effect transistors (JFET) and charge coupled devices (CCD).

The choice of electron and hole concentrations as the basic solution variable avoids the fluctuation problem mentioned above. The minority carrier concentration density is too low to show up in the error vector and we do not care about the accuracy of the minority carriers in those regions with low current densities.

The **scan** statement provides a parameter **change_variable** to allow the user to switch from Fermi levels to carrier concentrations. However, use carrier concentration also has drawbacks. The accuracy for minority carriers may not be accurate. The variable may vary 20 orders of magnitudes. As a result, the inversion of the Jacobian matrix may cause numerical overflow problem. Therefore, some of the off-diagonal matrix elements must be reduced artificially. More details about change variable may be found in the Reference Manual.

4.5 Slow transient tecnique

For structures with high resistivity or with complicated high doping, convergence may be difficult at low bias before current is turned on. For example InGaN MQW structure with strong polarization interface charges, are known to be difficult to converge.

One useful technique is the slow transient trick which turns out to be effective. It works as follows:

For example, we may have an original difficult step:

```
scan var=voltage_1 value=-3.5
```

We may improved it as:

```
scan var=voltage_1 value=-3.5 &&
  var2=time value2_to=1
```

Such a technique has been used to tackle some conventionally difficult cases such as floating poly gate in MOSFET, and highly insulating organic/insulator structures.

There are physical and numerical basis for such a method: highly insulating regions are inherently unstable against current continuity equation simply because the system can easily flips to a different state without causing any steady state current. In reality, a slow transient causes charging - discharging current which drives the system into the correct physical steady state.

4.6 Initial guess

As is well known for Newton's method, a poor initial guess solution may cause the solution to diverge. Such a situation is normally not a concern because the simulator has automated the procedure of finding the best guess solution. If a solution is not found with an initial guess solution from a previous bias, then the simulator automatically reduces the bias step and tries a new Newton iteration.

As mentioned previously, simulations of some structures with highly doped current blocking layers have difficulty converging at low bias because of high resistance. A poor initial guess may cause problems in this situation. One may wish to avoid the solution at low bias by applying a large initial bias step. If the device structure is relatively simple this trick often works. In a case where the device structure is complicated and the doping of the current blocking layer is high, this trick may fail. One of the reasons for such a failure is that the guess solution (equilibrium solution) is too far away from the solution at high bias. Such a condition is worsened by high doping at the current blocking layer.

A trick to overcome the convergence difficulty is to solve for an " easy device" at high bias first. Then the solution from the " easy device" is used to initialize the solution for the " target device" at high bias. Here the " easy device" has the same mesh and geometric structure as the " target device" except the current blocking layer is lightly doped. For the simulator, a structure with a lightly doped blocking layer is an easy structure.

4.7 Convergence for current-blocking structures

Convergence may be difficult for structures which use n-p-n or p-n-p junction to block the current flow for a certain region. This kind of design is commonly used in buried heterostructure (BH) lasers.

This type of structure requires special attention since the use of quasi-Fermi levels as the default variables often causes convergence difficulties. As mentioned in previous section, the change of variable from Fermi levels to carrier concentrations may avoid such convergence problems. However, if the Fermi levels solution is desired for any reason, or if the use of new variables fails, the following techniques may be used to force the convergence of the solver using quasi-Fermi levels.

current-blocking structures have difficulty converging because more material regions are involved. The heavy doping in the current blocking layer also tends to cause trouble, because the current flow condition there is very different from that in the other regions. To guarantee good convergence, we must supply the simulator with a good guess solution at a reasonably high bias voltage. Slowly ramping up the bias voltage does not always help because low bias itself is a problem, as was discussed in previous sections.

We recommend the following trick whenever a heavily doped current-blocking structure shows non-convergence. First, create an identical structure with lightly doped current blocking layer. Declare this lightly doped current blocking layer as having a "new_doping" instead of the usual "doping". Run the simulator for this structure and ramp up the voltage to 90 % of the bandgap of the active region (this should not be a problem since we have artificially reduced the doping of the blocking layer). At this point, ramp up the doping density in the blocking layer (marked as "new_doping") to a realistic doping level. Now the current device structure/doping is what we intend to simulate. We continue the simulation with current bias and other standard simulation tasks.

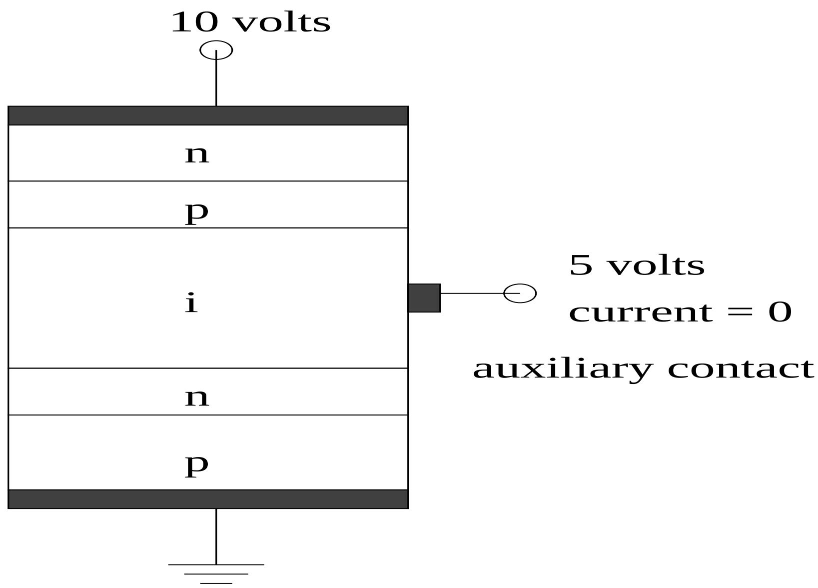


Figure 4.1: Schematic for the use of auxiliary contact for a p-n-i-p-n reverse biased structure.

4.8 Use of Auxiliary Ohmic Contacts

A powerful technique to force the simulation software to converge is the use of auxiliary ohmic contacts. For some complex devices, certain high resistance areas are far away from any boundaries. For example, a p-n-i-p-n structure under reverse bias condition has its i-region well isolated from the ohmic contacts. As a result, the variables may fluctuate in the i-region and we may have a convergence problem no matter how fine the mesh spacing is.

In such a case, we may add an auxiliary ohmic contact to be attached to a single mesh point in the isolated high resistant region. When we use voltage control on the auxiliary contact, the SIMULATOR should be easy to converge because the new contact prevents the variables from fluctuating in the high resistant region. However, the voltage control on the contact may cause current to flow in or out of it while in reality, no current should exist in such a contact. To eliminate the effects of the auxiliary contact, we can use current control to reduce its current to zero after the desired bias has been obtained.

For example, in the above p-n-i-p-n structure (see Fig. 4.1), we wish to reverse bias to breakdown at about 10 volts. We can simultaneously ramp up the voltages at the auxiliary contact to about 5 volts to ensure convergence. After the device breaks down and current starts to flow at around 10 volts of total bias, we can use current control to force the current at the auxiliary contact to be zero, thus eliminate its effects on the device. Without the the auxiliary contact, the above device would be difficult to bias to breakdown.

4.9 Bandgap reduction technique

Wide bandgap materials are hard to converge since their intrinsic carrier densities are low. The reverse junction current is orders of magnitude lower than their narrow bandgap counter parts. The problem is more serious if impact ionization is involved because the surge of ionization current in reverse junction may never be detected by the numerical solver since the reverse current is below the numerical floating point accuracy.

One technique is to reduce the bandgap first, then achieve the bias current you need, and finally increase the semiconductor bandgap to the realistic value.

For example, you may set reduce the bandgap by 20% at equilibrium as follow:

```
equilibrium bandgap_reduction=0.2
```

Then, you increase the voltage and current to the operating point as usual.

Finally, increase the bandgap back while holding the bias current using:

```
scan var_num=2 2_variables=(bandgap_reduction current_1) ...
```

Since the I-V curve you obtain this way is under reduced bandgap and thus not a realistic result, you need to ramp down the current again (with the correct bandgap) to obtain the desired I-V curve.

4.10 Negavtive differential mobility issues

For material with negative differential mobility due to transition of electrons from the Γ band to L and/or X bands, convergence may become a problem when the applied voltage is high. Examples of such problems are a HEMT under high voltage bias and a laser diode under high current injection condition.

The steady state solution can be driven into non-convergence because of the negative differential resistance due to the special field dependence mobility as shown in Fig. 4.2 (as defined by the "n.gaas" velocity model in gaas macro).

In reality, the device may be driven into unstable or oscillation states (Gunn effect) at some bias point. We rarely see such kind of negative I-V characteristics in steady state measurements. A possible explanation is that the field/current adjusts itself in the region of negative resistance in the form of transient redistribution and the steady state terminal I-V curve bypasses the peak of negative resistance.

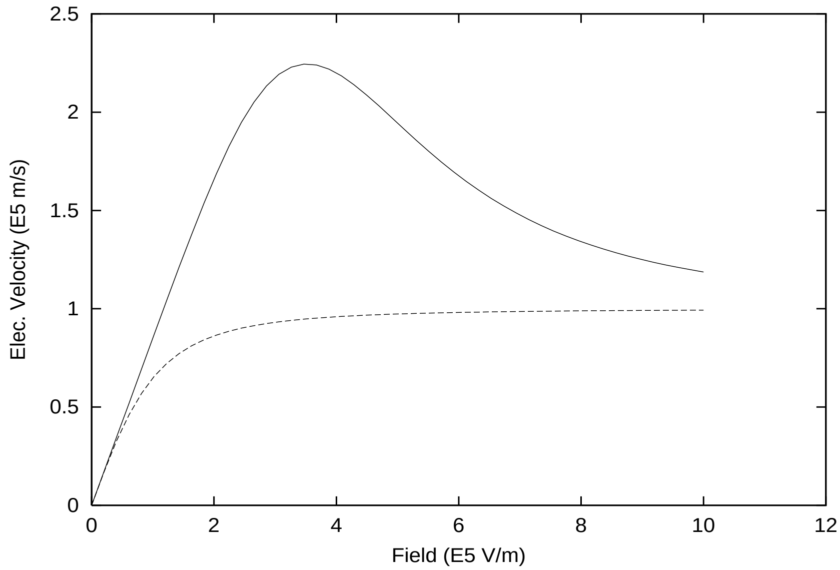


Figure 4.2: Field dependence electron velocity of undoped GaAs. The solid curve is the "n.gaas" mobility model while the dash curve is the "beta" model.

Thus, to overcome the non-convergence in steady state simulation in such as situation, we suggest using a different form of mobility model (such as the "beta" model, as illustrated by dash line in Fig. 4.2).

An alternative method to overcome the non-convergence is to use transient simulation while biasing the FET. This method means the simulator and the computer hardware must work harder.

Part II

PHYSICAL AND NUMERICAL MODELS

Chapter 5

DRIFT-DIFFUSION MODEL

This chapter provides a description of the basic theories used in the all of Crosslight's simulation software packages. The governing model of semiconductor is the the drift-diffusion (DD) model. The variety of physical phenomena in a semiconductor require many different physical models. However, the drift-diffusion model is the most basic because many of the well known characteristics of a semiconductor device such as the capability to amplify electrical signal may be explained using the DD theory. This chapter will also explain the boundary conditions and basic properties such as mobility and pure resitor behavior.

5.1 Basic equations

The basic equations[1] used to describe the semiconductor device behavior are Poisson's equation,

$$-\nabla \cdot \left(\frac{\epsilon_0 \epsilon_{dc}}{q} \nabla V \right) = -n + p + N_D(1 - f_D) - N_A f_A + \sum_j N_{tj}(\delta_j - f_{tj}), \tag{5.1}$$

and the current continuity equations for electrons and holes,

$$\nabla \cdot J_n - \sum_j R_n^{tj} - R_{sp} - R_{st} - R_{au} + G_{opt}(t) = \frac{\partial n}{\partial t} + N_D \frac{\partial f_D}{\partial t}, \tag{5.2}$$

$$\nabla \cdot J_p + \sum_j R_p^{tj} + R_{sp} + R_{st} + R_{au} - G_{opt}(t) = -\frac{\partial p}{\partial t} + N_A \frac{\partial f_A}{\partial t}. \tag{5.3}$$

These equations govern the electrical behavior (*e.g.*, I-V characteristics) of a semiconductor device.

Optionally, two more equations are used to described the carrier energy w or the carrier temperature distribution. The carrier temperature is not to be confused with the lattice temperature. The former is a description of how the carrier distribution deviates from Fermi-Dirac distribution. Such a model is also referred to as hydrodynamic model. Our theory here is based on the formulation derived by Azoff [5][7]. For simplicity, we only present the equations for hot electrons. The corresponding equations for hot holes are completely analogous.

$$\nabla \cdot S + R_n w - \nabla E_c \cdot J_n + \frac{n(w - w_0)}{\tau_w} + \frac{\partial (nw)}{\partial t} = 0, \tag{5.4}$$

$$S = -\frac{5}{3} J_n w - \frac{10}{9} \mu_n n w \nabla w. \tag{5.5}$$

where w is the total energy of an electron and $w_0 = 3kT/2$ is the electron energy at equilibrium. S is the electron energy flux intensity and τ_w is the energy relaxation time.

Definitions of symbols are listed in the nomenclature at the end of this chapter. More details about the derivation and discretization of the hydrodynamic model is given in the Apendix.

The primary function of our simulator is to solve these equations self-consistently for the electrostatic potential, V, the electron and hole concentrations, n and p, respectively, the electron and hole energy, W_n and W_p, respectively.

From theories of semiconductor device physics [1] the carrier flux densities J_n and J_p in Equations (5.2) and (5.3) can be written as functions of the carrier concentration and the quasi-Fermi levels:

$$J_n = n\mu_n \nabla E_{fn}, \tag{5.6}$$

$$J_p = p\mu_p \nabla E_{fp}, \tag{5.7}$$

where μ_n and μ_p are mobilities of electrons and holes, respectively. For convenience, we use carrier flux density and current density interchangeably even though they differ by a factor of electron charge.

For hydrodynamic model, the expression for the electron (or hole) current is modified:

$$J_n = \mu_n \left\{ -n\nabla[\psi + \chi + \gamma_n] + \frac{2}{3}\nabla(nw) - nw\nabla ln(m_n) \right\}, \tag{5.8}$$

More details about the current flow expression in the hydrodynamic model is given in the Appendix.

5.1.1 SRH and Auger recombination

The software treats carrier recombination due to deep level traps [Shockley-Read-Hall (SRH) recombination] using:

$$R_n^{tj} = c_{nj} n N_{tj}(1 - f_{tj}) - c_{nj} n_{1j} N_{tj} f_{tj}, \tag{5.9}$$

$$R_p^{tj} = c_{pj} p N_{tj} f_{tj} - c_{pj} p_{1j} N_{tj} (1 - f_{tj}). \tag{5.10}$$

where n_{1j} is the electron concentration when the electron quasi-Fermi level coincides with the energy level E_{tj} of the jth trap. A similar definition applies to p_{1j}. Under transient conditions, the following trap dynamic equation is valid [6]:

$$N_{tj} \frac{\partial f_{tj}}{\partial t} = R_n^{tj} - R_p^{tj}. \tag{5.11}$$

The capture coefficients c_{nj} and c_{pj} for electrons and holes relates to the lifetime of the carrier due the jth recombination center by the following relation:

$$\frac{1}{\tau_{nj}} = c_{nj} N_{tj}, \tag{5.12}$$

$$\frac{1}{\tau_{pj}} = c_{pj} N_{tj}. \tag{5.13}$$

One can show that Equations (5.9) to (5.13) are equivalent to the standard expression of the SRH model [1] under steady state conditions.

The capture coefficients can be further expressed as

$$c_{nj} = \sigma_{nj} \overline{v}_n, \tag{5.14}$$

$$\overline{v}_n = \sqrt{\frac{8kT}{\pi m_n}}, \tag{5.15}$$

$$c_{pj} = \sigma_{pj} \overline{v}_p, \tag{5.16}$$

$$\overline{v}_p = \sqrt{\frac{8kT}{\pi m_p}}. \tag{5.17}$$

A trap (or recombination center) is completely specified by its density N_{tj}, capture cross sections σ_{nj} and σ_{pj}, and energy level E_{tj}.

The Auger recombination rate is given by

$$R_{au} = (C_n n + C_p p)(np - n_i^2), \tag{5.18}$$

where the Auger coefficients C_n and C_p depends on the type of material simulated.

5.1.2 Carrier statistics

The electron and hole concentrations in semiconductors are defined by Fermi-Dirac distributions and a parabolic density of states which, when integrated, yield [1]

$$n = N_c F_{1/2} \left(\frac{E_{fn} - E_c}{kT} \right), \tag{5.19}$$

$$p = N_v F_{1/2} \left(\frac{E_v - E_{fp}}{kT} \right), \tag{5.20}$$

where $F_{1/2}$ is the Fermi integral of order one-half. For the convenience of numerical evaluation, the approximation proposed by Bednarczyk and Bednarczyk is used [2]:

$$F_{1/2}(x) \approx \left(e^{-x} + \xi(x)\right)^{-1}, \tag{5.21}$$

$$\xi(x) = \frac{3}{4}\sqrt{\pi}\,[\nu(x)]^{3/8}, \tag{5.22}$$

$$\nu(x) = x^4 + 50 + 33.6x\left\{1 - 0.68exp[-0.17(x+1)^2]\right\}. \tag{5.23}$$

This expression is accurate to within 0.4% of error in all ranges.

In the limit of low carrier concentration Equations (5.19) and (5.20) reduce to the familiar Boltzmann statistics:

$$n = N_c exp\left(\frac{E_{fn} - E_c}{kT}\right), \tag{5.24}$$

$$p = N_v exp\left(\frac{E_v - E_{fp}}{kT}\right), \tag{5.25}$$

The program uses the more general Fermi-Dirac statistics of Equations (5.19), (5.20) and (5.21) by default.

5.1.3 Incomplete ionization of impurities

The simulation program can accurately account for the incomplete ionization of shallow impurities in semiconductors. The occupancies f_D and f_A are used to describe the degree of ionization. It is assumed that the shallow impurities are in equilibrium with the local carriers and therefore the occupancy of the shallow impurities can be described by

$$f_D = \frac{1}{1 + g_d^{-1} exp[(E_D - E_{fn})/kT]}, \tag{5.26}$$

$$f_A = \frac{1}{1 + g_a exp[(E_A - E_{fp})/kT]}, \tag{5.27}$$

where the subscripts D and A are used to denote shallow donors and acceptors, respectively.

From the discussions of Section 5.1.1, the occupancy of a deep level trap can be determined through the trap dynamic equation, Equation (5.11). In general the deep trap is not in equilibrium with the carriers (*i.e.*, the trap does not share the same quasi-Fermi level as the carriers). From Eqs (5.9), (5.10) and (5.11), one obtains the following expression for the trap occupancy under steady state conditions:

$$f_{tj} = \frac{c_{nj}n + c_{pj}p_{1j}}{c_{nj}(n + n_{1j}) + c_{pj}(p + p_{1j})}. \tag{5.28}$$

In the case of surface states or surface recombination centers, the software allows for the distribution of dense traps near the surface region. This provides a mechanism for the surface charge states as well as for surface recombination. Fermi level pinning effects on a semiconductor surface can be modeled using this approach.

In a transient simulation the trap occupancy is a function of time, depending on the trap capture rates as well as on the local carrier concentrations. The program uses Equation (5.11) to determine the trap states at each transient time step.

By default the simulation software assumes incomplete ionization of dopants with energy level defined by the **level** parameter in the **doping** statement. The dopants of a specific material may be forced to be fully ionized with the command **full_ionization**.

5.1.4 Poole-Frenkel model of incomplete ionization

The Poole-Frenkel effect (also Frenkel-Poole effect or field induced emission) was originally used to describe field dependent thermionic emission from traps in the bulk of an insulator [1]. This mechanism is, however, equally applicable to incomplete ionization of impurities in semiconductors.

Under an electric field F, the electrostatic potential near an impurity center is modified and the effective work function or the ionization energy is reduced by:

$$\Delta E_{PF} = \sqrt{\frac{qF}{\pi \epsilon_0 \epsilon}}. \tag{5.29}$$

The Poole-Frenkel model is implemented by shifting the ionization energy by ΔE_{PF}. The field dependence of this shift introduces some complication into the discretized Poisson's equation. The reason is that at each node, additional work must be done to search for the maximum field component surrounding the node.

Another complication associated with the Poole-Frenkel model occurs at the ohmic contacts, because the simulator solver is set up such that the built-in potential at an ohmic contact is always fixed, or in other words, the dopant energy level or the Imref at the contact is fixed relative to the band edges. Such an assumption contradicts with the Poole-Frenkel model which changes the ionization energy and the Imrefs as the applied field is varied. To overcome this difficulty, the ionization energy at the contact is allowed to be shifted to a constant value before the simulation. Without such a shift in the ionization energy at the contact, the ohmic contact may behave like a Schottky contact because of a large fixed built-in potential.

The Poole-Frenkel model is important when the ionization energy is large and the temperature is low (*e.g.*, 100 meV at 100K). Without such a model, the semiconductor may have an unrealistic high resistance.

By default, this model is turned off and can be enabled using the parameter **pf_model** of the **doping** statement.

5.1.5 Carrier mobility

The carrier mobilities μ_n and μ_p account for the scattering mechanism in electrical transport. In general the mobility is a function of the electrical field [1]. Our simulation software provides several analytical formulas of field dependent mobility.

- 1. The simplest mobility model uses constant mobilities μ_{0n} and μ_{0p} for electrons and holes, respectively, throughout each material region in the device.

- 2. Another simplified field dependent mobility model is the two-piece mobility model:

$$\begin{aligned} \mu_n &= \mu_{0n}, \quad for\ F < F_{0n}; && (5.30) \\ \mu_n &= v_{sn}/F, \quad for\ F \geq F_{0n}; && (5.31) \\ v_{sn} &= \mu_{0n} F_{0n}, && (5.32) \end{aligned}$$

  for the electron mobility. F_{0n} is a threshold field beyond which the electron velocity saturates to a constant. Similar expressions can be defined for holes:

$$\begin{aligned} \mu_p &= \mu_{0p}, \quad for\ F < F_{0p}; && (5.33) \\ \mu_p &= \mu_{0p} F_{0p}/F, \quad for\ F \geq F_{0p}; && (5.34) \\ v_{sp} &= \mu_{0p} F_{0p}. && (5.35) \end{aligned}$$

- 3. A commonly used mobility model has the following form for electrons and holes, respectively:

$$\mu_n = \frac{\mu_{0n}}{(1 + (\mu_{0n} F/v_{sn})^{\beta_n})^{1/\beta_n}}, \tag{5.36}$$

$$\mu_p = \frac{\mu_{0p}}{(1 + (\mu_{0p} F/v_{sp})^{\beta_p})^{1/\beta_p}}. \tag{5.37}$$

- 4. Many III-V compound semiconductors [(*e.g.*, GaAs) exhibit negative differential resistance due to the transition of carriers into band valleys with lower mobility [1]. The software has implemented the following field dependent model for this case:

$$\mu_n = \frac{\mu_{0n} + (v_{sn}/F_{0n})(F/F_{0n})^3}{1 + (F/F_{0n})^4}. \tag{5.38}$$

  Besides the field dependence of the mobility, another important effect is the impurity dependence of the low field mobility [1]. The program uses the following formulas

$$\mu_{0n} = \mu_{1n} + \frac{(\mu_{2n} - \mu_{1n})}{1 + \left(\frac{N_D + N_A + \sum_j N_{tj}}{N_{rn}}\right)^{\alpha_n}}, \tag{5.39}$$

$$\mu_{0p} = \mu_{1p} + \frac{(\mu_{2p} - \mu_{1p})}{1 + \left(\frac{N_D + N_A + \sum_j N_{tj}}{N_{rp}}\right)^{\alpha_p}}. \quad (5.40)$$

where the parameters μ_{1n} and μ_{2n} are fitting parameters from experimental data.

- 5.Poole-Frenkel type of field enhanced mobility model. Higly localized carriers with small mobility can be excited by external field to make hopping movements. Commonly used for organic semiconductors, the following mobility model is implemented.$\mu = \mu_0 exp[(F/F_{cr})^{px}]$

In many applications, the mobility is anisotropic and also depends on the vertical field. A number of vertical field dependent model (such as the Lombardi model) are implemented within the statement of **mobility_xy**.

5.2 Boundary Conditions

The boundary conditions for the electrical part [*i.e.*, the first three equations Equations (5.1) to (5.3)] include ohmic contacts, Schottky contacts, Neumann (reflective) boundaries, lumped elements and current controlled contacts. For the hot carrier equations, the boundary condition here is the contact carrier temperature.

5.2.1 Ohmic contact

Ohmic contacts are implemented as simple Dirichlet boundary conditions, where the surface potential and electron and hole quasi-Fermi levels (V_s, E^s_{fn},E^s_{fp}) are fixed. The minority and majority carrier quasi-Fermi potentials are equal and set to the applied bias of the electrode, *i.e.*,

$$\phi^s_n = \phi^s_p = -E^s_{fn} = -E^s_{fp} = V_{applied}. \quad (5.41)$$

The potential V_s at the boundary is fixed at a value consistent with zero space charge, *i.e.*,

$$-n + p + N_D(1 - f_D) - N_A f_A + \sum_j N_{tj}(\delta_j - f_{tj}) = 0. \quad (5.42)$$

With the solution of V_s and Equation (5.41) one can use Equations (5.19) and (5.20) to calculate n_s and p_s.

It should be pointed out that the purpose of creating an Ohmic contact in a simulation is to have a boundary condition which does not disturb the area of simulation but provides a path for current flow. This is in contrast to the Schottky contact which can be either injecting or depleting carriers, depending on the polarity of the bias.

In most device simulations, the ideal Ohmic contact with a charge neutral assumption is sufficient as a good carrier injector if the contact area is highly doped or if the carrier injection requirement is not too high. If the vicinity of the contact has low doping or if a high level of carrier injection is required, the above ideal Ohmic contact ceases to be a valid model for an realistic Ohmic contact made from metal. In such a case, the simulator refuses to inject carriers into the active region no matter how large the bias current is. A technique to get around this problem is to highly dope the vicinity of the ideal Ohmic contact region.

5.2.2 Schottky contacts

Schottky contacts to the semiconductor are defined by a Schottky barrier height of the electrode metal and a surface recombination velocity. The surface potential at a Schottky contact is defined by

$$V_s = \chi - \chi_{ref} - \phi_b + V_{applied}. \tag{5.43}$$

where χ and χ_{ref} are the electron affinities of the semiconductor and a reference material, respectively, and ϕ_b is the Schottky barrier height.

In general, the quasi-Fermi levels E^s_{fn} and E^s_{fp} are no longer equal to $V_{applied}$ and are defined by a current boundary condition at the surface instead:

$$J_{sn} = \gamma_n \overline{v}_n^{therm}(n_s - n_{eq}), \tag{5.44}$$
$$J_{sp} = \gamma_p \overline{v}_p^{therm}(p_s - p_{eq}), \tag{5.45}$$

where J_{sn} and J_{sp} are the electron and hole currents at the contact, n_s and p_s are the actual surface electron and hole concentrations, and n_{eq} and p_{eq} are the equilibrium electron and hole concentrations if infinite surface recombination velocities are assumed (*i.e.* $\phi_n = \phi_p = V_{applied}$). The thermal recombination velocities are given by

$$\overline{v}_n^{therm} = \sqrt{\frac{kT}{2m_n\pi}}, \tag{5.46}$$
$$\overline{v}_p^{therm} = \sqrt{\frac{kT}{2m_p\pi}}, \tag{5.47}$$

The constants γ_n and γ_p provide a mechanism to include any correction (*e.g.* that due to tunneling) to the standard theory.

5.2.3 Abrupt heterojunctions

An abrupt junction is also considered a boundary condition in the program because most physical quantities are discontinuous across an abrupt heterojunction. Similar to a Schottky contact, an abrupt heterojunction exhibits non-ohmic behavior, such as the effect of rectification. The physical model used by the software for current transport across the junction is the thermionic emission model [1]. A similar expression for the current flux across the junction can be defined as

$$J_{hn} = \gamma_{hn}\overline{v}_{bn}^{therm}(n_b - n_{b0}), \tag{5.48}$$
$$J_{hp} = \gamma_{hp}\overline{v}_{bp}^{therm}(p_b - p_{b0}), \tag{5.49}$$

where n_b and p_b denote the electron and hole concentrations, respectively, on the barrier side of the junction. n_{b0} and p_{b0} are the corresponding concentrations when the quasi-Fermi levels are the same as those on the opposite side. These equations ensure that, when the quasi-Fermi levels on both sides of the barriers are the same, the net current is zero. The derivation of the boundary conditions of the above equations can be found in Refs. [3] and [4].

The thermal recombination velocities are given by

$$\overline{v}_{bn}^{therm} = \sqrt{\frac{kT}{2m_{bn}\pi}}, \tag{5.50}$$
$$\overline{v}_{bp}^{therm} = \sqrt{\frac{kT}{2m_{bp}\pi}}. \tag{5.51}$$

Other symbols have similar meanings as those in Section 5.2.2.

The abrupt heterojunction model described here is important to accurately model the thermionic emission behavior (*e.g.* temperature dependence) of a device using an abrupt heterojunction as a means of carrier confinement. The abrupt junction model has its limitations, however, especially when both sides of the junction are heavily doped with the same type of dopant (*e.g.* donors). In such a case the Fermi level is strongly pinned at the band edges on both sides, and the potential barrier is forced to form a sharp peak above the Fermi level. Since the peak region in the barrier is strongly depleted of carriers, the resistance is very high according to Equation (5.6). If this junction happens to be reverse biased (*i.e.* the carriers on the barrier side tend to be more depleted when the bias is applied), the high resistance can cause an unrealistically large voltage drop there.

Fortunately, the difficulty associated with a highly doped abrupt heterojunction does not occur very often near the active regions, because junctions there are rarely heavily doped with the same dopant on both sides. If the heavily doped contact region has an abrupt heterojunction, problems may arise. An example of this is GaAs/AlGaAs, which may have a heavily doped GaAs substrate under the heavily doped cladding region.

In most cases the abrupt junction model implemented in the program is adequate. In setting up a new device structure one may ignore the problem mentioned above unless the simulation result shows an unrealistically large voltage drop. The user may be able to spot such a problem just from simulator's run-time printout; if the printout shows that passing a few mA through a diode causes a voltage drop of 10 volts on the device, for example, then one should locate which abrupt junction is causing trouble by looking at the band diagram, then replace the barrier of the abrupt junction by a graded barrier without affecting any active regions. Usually a graded barrier with grading length of 100 Å or so is adequate to replace the heavily doped junction that is causing the trouble.

This limitation of the program arises from the assumption of drift-diffusion as the key transport mechanism. In reality the main mechanism of carrier transport across the thin and sharp barrier is quantum tunneling.

5.2.4 Neumann boundaries

Along the outer non-contacted edges of simulated devices, homogeneous reflecting Neumann boundary conditions are imposed so that current only flows out of the device through the contacts. Additionally, in the absence of surface charge along such edges, the normal electric field component goes to zero. In a similar fashion, current is not permitted to flow from the semiconductor into an insulating region; further, at the interface between two different materials, the difference between the normal components of the respective electric displacements must be equal to any surface charge present along the interface.

5.2.5 Lumped elements

Lumped elements have been created to cut down the number of grid points to discretize some device structures, thereby saving CPU time. Lumped resistance might be useful in a simulation of a semiconductor device structure with a substrate contact located far away from the active region. If the whole structure were to be simulated, a tremendous number of grid points, probably more than half, would be wasted to account for a purely resistive region of the device. Because CPU time is generally a superlinear function of the number of grid points, including such regions can be quite expensive. Applying this boundary condition creates an extra unknown (V_s), the voltage on the semiconductor contact, which if defined by Kirchoff's equation:

$$\frac{V_{applied} - V_s}{R_s} - \sum_{i=1}^{N_b} (J_n + J_p) = 0, \tag{5.52}$$

where N_b is the number of boundary grid points associated with the electrode of interest. Note that this auxiliary equation, due to the currents inside the summation, has dependencies on the values of potential and carrier concentrations at the nodes on the electrode as well as all nodes directly adjacent to the electrode. It is important to remember that a lumped element contact will have a single voltage (or potential-adjusted for possible doping non-uniformity) associated with the entire electrode.

The simulator should be used as much as possible to calculate any resistance components that might be included as lumped elements. For instance, in the case of substrate resistance of a semiconductor device, one could just simulate the n-substrate with ohmic contacts at both ends. From the plot of the contact current versus voltage, the resistance can be directly extracted from the slope.

5.2.6 Current boundary condition

The current boundary condition is a very important option in the simulation of device. For example, when a laser diode is biased beyond threshold, the gain and carrier concentration essentially saturate. This also causes the bias voltage to saturate, while the current continues to rise because of stimulated emission. Under such conditions, operating the simulator at a voltage-controlled mode would cause the current to jump to huge values for a small increase in voltage.

In some devices (not necessarily lasers), the terminal current is a multi-valued function of the applied voltage. This condition implies that for some voltage boundary conditions, a numerical procedure may end up, depending on the initial guess solution, with a solution in either of two distinct and stable states. Furthermore, a condition of primary interest is at the trigger point, where $dI/dV = 0$, which is difficult to obtain with a simple voltage boundary condition. Additionally, it is nearly impossible to compute any solutions in the negative resistance regime with voltage inputs.

A possible alternative to the difficult situation mentioned above would be to define a current controlled boundary condition, since voltage can be thought of as a single-valued function of the terminal current. Such a boundary condition has been implemented in the software, as an auxiliary equation with an additional unknown boundary potential. Like the lumped element case, a Kirchoff equation is written at the contact points:

$$J_{source} - \sum_{i=1}^{N_b}(J_n + J_p) = 0. \tag{5.53}$$

Note that unlike the lumped element case, J_{source} is a constant specified by the user and has no dependence on the boundary potential V_s (the V_s dependence is buried in the summation). Since the full Newton's method is used for the solution of all equations in the program, no degradation of convergence has been observed for current controlled simulations.

For the case of semiconductor lasers, it is strongly recommended that the simulator be switched to current controlled mode before the threshold (which is possible to estimate by experience).

For regions where dI/dV is small, the voltage boundary condition is definitely preferable. In the case of operating regimes where dI/dV is large, the current boundary condition may be preferable. It is not uncommon for the negative resistance regime of a certain device to have a slope dI/dV very close to zero. Such behavior should be considered when using a current source to trace out an entire I-V curve. It might be preferable to switch back to a voltage source after passing the trigger point.

5.3 Pure Resistor Regions and Metal Macros

Starting from versions later than 2007.3, Crosslights device simulation program approximates a pure resistor region as a special kind of semiconductor. The following properties are assigned to a metal macro or pure resistor:

- 1 The default bandgap is a small 0.1 eV.
- 2 The default electron and hole relative effective masses are 10.
- 3 The metal macro is undoped by default.

Due to the heavy mass and small bandgap, the special semiconductor (or metal macro) above always pins the Fermi levels at the mid-gap. As a result, equal and large amount of electrons and holes are available for conduction purpose. The mobility is computed from the conductivity specified in the metal macros. We find that such a kind of semiconductor behaves exactly like a metal. Depending on the band alignment with a real semiconductor, the metal macro is capable of injecting electrons and/or holes.

With such a definition, the work function of a real metal is approximately equal to the affinity of the small bandgap metal macro. The difference is only 0.05 eV, negligible for many practical purposes. The work function of a real metal is related to the affinity of of metal macro by the following formula:

```
real_metal_work_function=metal_macro_affinity + 0.05(eV)
```

Such an approach of treating a real metal as a fake semiconductor has the advantage of sharing all the sophisticated semiconductor models, such as AC analysis and quantum tunneling, with real semiconductors.

One should exercise caution when setting up quantum tunneling between a metal macro and a semiconductor barrier. When using the propagation matrix approach, the program uses the default density of state (DOS) mass of 10 which is larger than most real metal masses. The calculated tunneling current for carriers from the metal to semiconductor (or reverse bias of a Schottky junction) may be underestimated bue to such a large mass. One may correct the default metal mass using the command **cond_dos_mass_ratio_n** or **cond_dos_mass_ratio_p**.

This command means to define the ratio of conduction mass over DOS mass. It may be used to define conduction mass of the metal for tunneling. For example, if the real metal has a relative effective mass of 0.5 while the default DOS mass for a metal macro is 10, a value of 0.05 may be used for in above command.

5.4 Bandgap Narrowing Effect

It is observed experimentally that the shrinkage of bandgap occurs when impurity concentration is particulary high, eg. $n_{imp} > 10^{23} m^{-3}$. This effect is so called bandgap narrowing effect which is ascribed to the emerging of the impurity band formed by the overlaped impurity states. The bandgap narrowing model proposed by Slotboom is given by

$$\Delta E_g = E_{ref} \left\{ \ln \frac{n_{imp}}{n_{ref}} + \sqrt{\ln^2 \frac{n_{imp}}{n_{ref}} + 0.5} \right\} \tag{5.54}$$

where E_{ref}, n_{ref}andn_{imp} represent energy parameter, density parameter and impurity concentration, respectively.

In case of silicon, E_{ref} and n_{ref} were obtained by fitting experiment, which yeilds

$$E_{ref} = 0.009 \quad [V] \tag{5.55}$$

$$n_{ref} = 10^{23} \quad [m^{23}] \tag{5.56}$$

It should be noted that the bandgap narrowing effect is doping dependent effect, whereas the many body effect, which reduces band gap as well, is carrier dependent effect.

It should also be pointed out that the macro system in Crosslight software provides an easy way for user to define any kind of bandgap narrowing model since dopant density is an internal variable automatically passed to any macro functions, including the bandgap.

5.5 Trap models

A deep level trap located with the semiconductor bandgap is described by the charge property (deep donor or deep acceptor) as appears in the Poissons equation we described previously. It is also described by a rate equation to detail the carrier trapping and detrapping:

$$R_n^{tj} = c_{nj} n N_{tj}(1 - f_{tj}) - c_{nj} n_{1j} N_{tj} f_{tj}, \tag{5.57}$$
$$R_p^{tj} = c_{pj} p N_{tj} f_{tj} - c_{pj} p_{1j} N_{tj}(1 - f_{tj}). \tag{5.58}$$

The simulation software can simulate the behavior of any number of trap levels within the bandgap. In the extreme of densely populated traps such as those appearing in amorphous materials, it is more convenient to use a continuous function to describe the energy distribution of the trap levels. The software currently supports the Gaussian and exponential trap distribution. The continuous trap models are defined by parameters within the **doping** statement.

When the photon energy of light is less than the semiconductor bandgap, it is still possible to generate photo-carriers if deep level traps are sensitive to light. Electrons and holes being trapped in the deep level trap centers require less than bandgap energy to be excited to the conduction or valence bands. We shall also refer to such a process as trap photo-excitation.

In such a case, the Schokley-Read-Hall recombination terms of the drift-diffusion must be revised to include photo-carrier generation terms due to emission from traps (G_{tn} and G_{tp}):

$$R_n^{tj} = c_{nj} n N_{tj}(1 - f_{tj}) - c_{nj} n_{1j} N_{tj} f_{tj} - G_{tn}, \tag{5.59}$$
$$R_p^{tj} = c_{pj} p N_{tj} f_{tj} - c_{pj} p_{1j} N_{tj}(1 - f_{tj}) - G_{tp}. \tag{5.60}$$

We will derive the generation term G_{tn} from the consideration that it must be proportional to trap density and electron occupancy. It must also be proportional to light intensity.

Let us recall the photon generation term of direct electron-hole pairs:

$$G_{eh} = v_g S \alpha \tag{5.61}$$

where v_g is the group velocity of light, S the photon density and α the absorption coefficient.

In analogy, we can write down the trap excitation term as

$$G_{tn} = v_g S N_{tj} f_{tj} \sigma_{xn} \tag{5.62}$$

where σ_{xn} has the unit of area and we shall refer to it as trap photo-excitation cross section. Please note that this cross section times the amount of trapped electrons (i.e., $N_{tj} f_{tj}$) may be compared with usual bulk material absorption coefficient in units of 1/m.

Similarly, trap excitation for holes is given by

$$G_{tp} = v_g S N_{tj}(1 - f_{tj})\sigma_{xp} \tag{5.63}$$

Chapter 6

NUMERICAL TECHNIQUES AND 3D SIMULATION

This chapter discusses numerical issues relating to solving the drift-diffusion equations on a discretized 2/3 dimensional finite element mesh. The 3D mesh may be regarded as a collection of 2D planes with carefully constructed plane-to-plane finite element connectivity. The reduction of 3D into 2D in case of cylindrical symmetry is also discussed.

6.1 Numerical Techniques

6.1.1 Introduction

All Crosslight software packages involve solving the drift-diffusion equations. Usually, additional equations related to optics are solved in addition to the DD equations. Since substantial amount of computation time is spent on the DD equations, we shall concentrate mainly on the numerical techiques related to them here.

The first three partial differential equations (PDE's), Equations (5.1) to (5.3), describe the electrical behavior of the bulk or quantum well semiconductor devices. Poisson's equation, Equation (5.1), governs the electrostatic potential, and the continuity equations for the electron and holes, Equations (5.2)-(5.3), govern the carrier concentrations. These differential equations are discretized as described in the next section. The resulting set of equations is coupled and nonlinear. Consequently, there is no method to solve the equations in one direct step. Instead, solutions must be obtained by a nonlinear iteration method, starting from some initial guess. The solution methods are detailed in subsection 6.1.3, and the choice of initial guess is explained in subsection 6.1.4.

6.1.2 Discretization

To solve the device equations on a computer, they must be discretized on a simulation grid. That is, the continuous functions of the PDE's are represented by vectors of function values at the nodes, and the differential operators are replaced by suitable difference operators. Instead of solving for four unknown functions (potential, electron and hole concentrations, and the electron or hole energy), the simulator solves for $3N$ or $4N$ unknown numbers, depending on the model choice, where N is the number of grid points.

The key to discretizing the differential operators on a general triangular grid is the box method [33]. Each equation is integrated over a small polygon enclosing each node, yielding 4N nonlinear algebraic equations for the unknown potential, concentrations and the wave amplitude. The integration equates the flux into the polygon with the sources and sinks inside it, so that conservations of current and electric flux are built into the solution. The integrals involved are performed on a triangle by triangle basis, leading to a simple and elegant way of handling general surfaces and boundary conditions. In this case the integral is simply replaced by a summation of the integrand evaluated at the node multiplied by the area surrounding it.

In the case of the continuity equations, the carrier fluxes must be evaluated with care; the classic finite difference formulas are modified as first demonstrated by Scharfetter and Gummel (SG-formula) in 1969 [34]. From the user's point of view, the discretization is completely automatic and no intervention is required.

For the hydrodynamic model, the SG-formula for discretization must be modified. The details are worked out in the Appendix.

6.1.3 Newton's method

We discuss the numerical solution of drift-diffusion equations. In Newton's method, all of the variables in the problem are allowed to change during each iteration, and all of the coupling between variables is taken into account. Due to this, the Newton algorithm is very stable, and the solution time is nearly independent of bias conditions. The basic algorithm is a generalization of the Newton-Raphson method for the root of a single equation. Equations (5.1)–(5.3) can be written as

$$F_v^j(V^{j1}, E_{fn}^{j1}, E_{fp}^{j1}) = 0, \tag{6.1}$$

$$F_n^j(V^{j1}, E_{fn}^{j1}, E_{fp}^{j1}) = 0, \tag{6.2}$$

$$F_p^j(V^{j1}, E_{fn}^{j1}, E_{fp}^{j1}) = 0, \tag{6.3}$$

where j runs from 1 to N, and $j1$ includes j itself plus its surrounding mesh points. Equations (6.1) to (6.3) represent a total number of $3N$ equations, which is sufficient to solve for $3N$ variables: (V^1, E_{fn}^1, E_{fp}^1, V^2, E_{fn}^2, E_{fp}^2, ..., V^N, E_{fn}^N, E_{fp}^N).

Once the equations are discretized in the above form, standard Newton techniques can be used to solve for them. These involve the evaluation of the Jacobian matrix to linearize Equations (6.1) to (6.3), followed by linear solution of the linearized equations (involving LU factorization of the matrix), and finally, nonlinear iteration to get the final solution. Since the Jacobian matrix is a sparse matrix, sparse matrix solution techniques are used to improve the computation speed.

The major acceleration of a Newton iteration is the Newton-Richardson method, whereby the Jacobian matrix is only refactored when necessary. When it is not necessary to factorize, the iterative method using the previous factorization is employed. The iterative method is extremely fast provided the previous factorization is reasonable. Frequently the Jacobian need only be factorized only once or twice per bias point using the Newton-Richardson method, as opposed to the twenty to thirty times required in the conventional Newton method. The decision to refactor is made on the basis of the decrease per step of the maximum error of both the equation residuals and the variable differences. This mechanism has been automated within the program and no user intervention is needed.

6.1.4 Initial guess and adaptive bias stepping

Several types of initial guess solutions are used in the simulator. The first is the simple charge neutral assumption used to obtain the first (equilibrium) bias point. For the wave equation, the guess solution is a delta function at the center of the active region. This is the starting point of any device simulation. Any later solution with applied bias needs an initial guess of some type, obtained by modifying one or two previous solution(s). When only one previous solution is available, the solution currently loaded is used as the initial guess, modified by setting the applied bias at the contact points. The same is true for the wave equation. When two previous solutions with two different biases are available, it is possible to obtain a better initial guess by interpolating the two previous solutions to the present one according to the two previous bias points and the present bias. All of these steps are automated within the program and no user intervention is needed. A schematic of the biasing strategy is given in Figure 6.1.

As in any Newton method, convergence strongly depends on the wise choice of the initial guess solution. In principle, the Newton nonlinear iteration should always converge as long as the initial guess is close enough to the solution. The closer the initial guess is to the solution, the fewer nonlinear iterations are required to reach convergence. The program takes this fact into account and implements an adaptive method to control the bias step after the successful convergence of the previous solution.

Assuming that the previous solution takes K_1 number of iterations for a bias step of ΔV_1 to reach convergence and the user believes (by experience) that the optimal

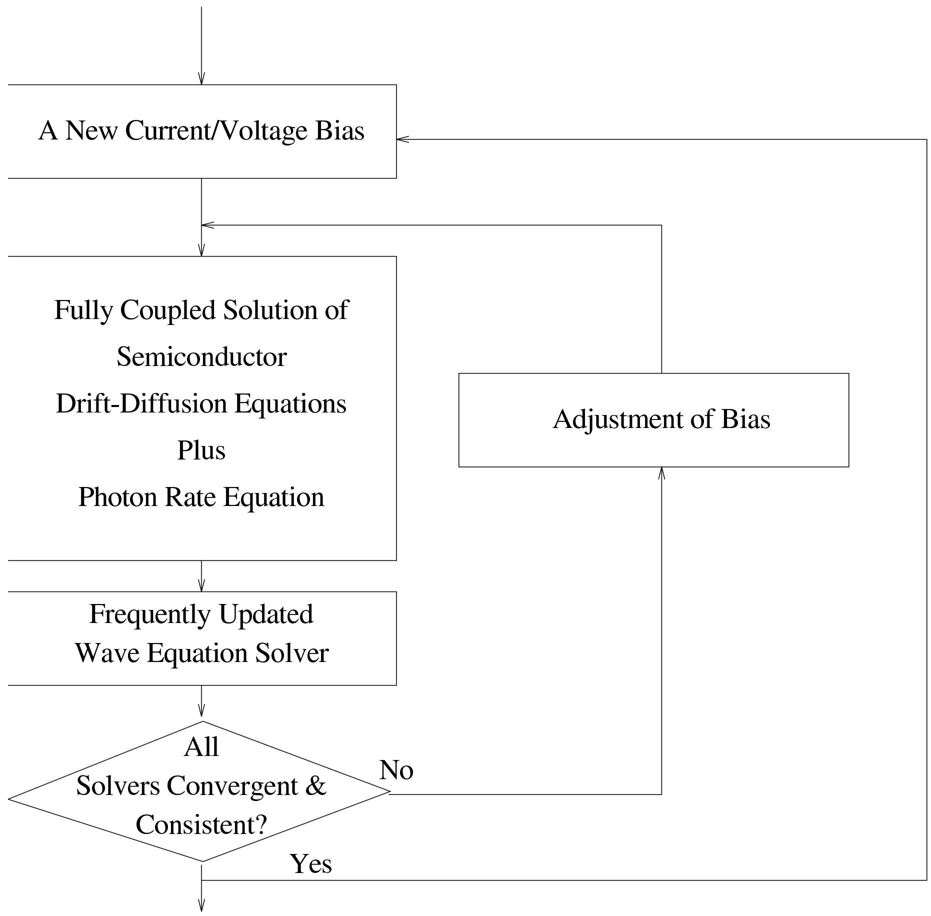


Figure 6.1: Flow diagram of how the simulator performs a typical simulation.

number of iterations should be K_0, the current bias step ΔV_2 is adjusted to $\Delta V_2 = \Delta V_1 K_0 / K_1$. In this way an optimal step can be determined such that the nonlinear Newton iteration is controlled to converge within K_0 iterations. Again this procedure is completely automated for the user.

6.1.5 Transient simulation

The method used to discretize the differential equations in time is the backward Euler method. Its basic approximation is the following equation:

$$\frac{\partial S}{\partial t} = \frac{S(t) - S(t - \Delta t)}{\Delta t}. \tag{6.4}$$

This method has the advantage of being highly stable. Strictly speaking, the simulation system has more than 5 equations when the trap dynamic equation, Equation (5.11), is also included in a transient simulation. Equation (5.11) is only dependent on local variables and can be solved before the major Newton step involving all mesh points.

Once all the equations are discretized in terms of solution of the previous time step, one can treat the solution of the present time step using the same method as for the steady state solution, *i.e.* one can linearize the discretized equations and use Newton's method to solve them.

6.1.6 Mesh issues

Correct allocation of the grid is a crucial issue in device simulation. The number of nodes in the grid (N) has a direct influence on the simulation time, where the number of arithmetic operations necessary to achieve a solution is proportional to N^p where p usually varies between 1.5 and 2. Because the different parts of a device have very different electrical and optical behavior, it is usually necessary to allocate a fine grid to some regions and a coarse grid to others. As far as possible, it is desirable not to allow the fine grid in some regions to spill into regions where it is unnecessary, in order to maintain a reasonable simulation time. We have developed a simple algorithm to generate and refine the grid as described below.

The first step in mesh generation is to specify the device boundaries and the region boundaries for each material. The program uses triangles and tetrahedrons as the basic building blocks to describe an arbitrary device. Furthermore, the edge of each polygon can be bent to the desired shape. The basic operation in the mesh generator after the boundaries have been defined is to draw lines parallel to the edges of the polygons (see Figure 6.2). This way one obtains small tetrahedrons which can be bisected into two triangles (the basic elements in the finite element method).

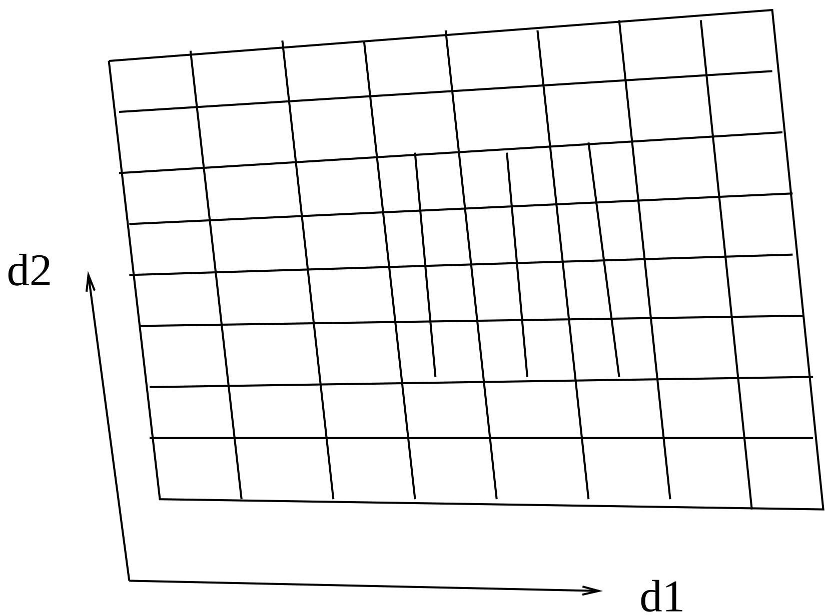


Figure 6.2: Schematics for the basic operations in the mesh generator. Two mesh lines in the d_2 direction, are used to double the mesh near the center of the polygon.

Another basic operation is doubling the mesh density in a specified region as shown in the example of Figure 6.2. This is accomplished by drawing a new line between the old mesh lines. Repeating this operation, one can manually generate a mesh with any variation of mesh densities.

The simulator has provided another practical approach to automatic mesh generation and refinement. The mesh is refined according to the variation of specified physical quantities. In a case where the electron concentrations between two adjacent nodes are greater than a value specified by the user, the program will automatically allocate additional mesh points between the two nodes. More detail can be found in the description of the statement **refine_mesh** in the Reference Manual.

We have also developed a means of reducing computation time by dividing large grids into smaller sections for the initial Newton iterations. This approach is similar to that used in non-linear multigrid method. However, the stability and effectiveness of this method is still the subject of research.

6.2 The Cylindrical Coordinate System

6.2.1 Introduction

There is good reason to establish a cylindrical coordinate system for device simulation. Many devices are fabricated with cylindrical symmetry. Many types of photo-detectors and most of the vertical cavity surface emitting lasers have such symmetry. The original three dimensional coordinate system can be reduced to a two dimensional one. We first consider our differential equation systems under a general coordinate system. Then we apply the theory to the cylindrical system.

6.2.2 Some general considerations for an arbitrary coordinate system

Our purpose is to re-write the drift-diffusion equation, usually written in cartesian coordinates, (x,y,z), in the new coordinate system (q_1, q_2, q_3) which are orthogonal to each other. We start by considering a small distance along each of the three coordinates:

$$ds_j = H_j dq_j, H_j = \sqrt{\left(\frac{\partial x}{\partial q_1}\right)^2 + \left(\frac{\partial y}{\partial q_2}\right)^2 + \left(\frac{\partial z}{\partial q_3}\right)^2}. \tag{6.5}$$

Then the gradient of a scalar variable u is given by

$$\nabla u = \frac{1}{H_1}\frac{\partial u}{\partial q_1}\mathbf{e_1} + \frac{1}{H_2}\frac{\partial u}{\partial q_2}\mathbf{e_2} + \frac{1}{H_3}\frac{\partial u}{\partial q_3}\mathbf{e_3} \tag{6.6}$$

where $\mathbf{e_j}$ is used to denote the unit vector for each of the new coordinates.

Let us derive the expression for the divergence, $\nabla \cdot \mathbf{A}$ of an arbitrary vector $\mathbf{A}$.

We consider a small volume defined within from q_1 to $q_1 + dq_1$, q_2 to $q_2 + dq_2$, and q_3 to $q_3 + dq_3$. We are going the evaluate the out-going flux though the surfaces of this small volume. The outgoing flux of $\mathbf{A}$ through the surfaces at q_1 and $q_1 + dq_1$ is given by

$$(A_1 H_2 H_3)|_{q_1+dq_1} dq_2 dp_3 - (A_1 H_2 H_3)|_{q_1} dq_2 dp_3 = \frac{\partial}{\partial q_1}(A_1 H_2 H_3) dq_1 dq_2 dp_3 \qquad (6.7)$$

Similarly, the flux through surfaces at q_2 and $q_2 + dq_2$ is given by

$$\frac{\partial}{\partial q_2}(A_2 H_3 H_1) dq_1 dq_2 dp_3 \qquad (6.8)$$

and the flux through surfaces at q_2 and $q_2 + dq_2$ is given by

$$\frac{\partial}{\partial q_3}(A_3 H_1 H_2) dq_1 dq_2 dp_3 \qquad (6.9)$$

The divergence is defined as the the total out-going flux divided by the volume under consideration:

$$\nabla \cdot \mathbf{A} = \frac{1}{H_1 H_2 H_3}\left[\frac{\partial}{\partial q_1}(H_2 H_3 A_1) + \frac{\partial}{\partial q_2}(H_3 H_1 A_2) + \frac{\partial}{\partial q_3}(H_1 H_2 A_3)\right] \qquad (6.10)$$

6.2.3 Cylindrical coordinate system

The relation between the old (x_1, y_1, z_1) and the new coordinate system (r, θ, z_r) is given by

$$\begin{aligned} x_1 &= r cos\theta \\ y_1 &= r sin\theta \\ z_1 &= z_r \end{aligned} \qquad (6.11)$$

To avoid confusion with the (x,y,z) system used in our simulation software, and have used a new set of symbols here. In our simulator, y corresponds to z_1 or z_r, and x corresponds to r.

It is easy to find

$$H_r = 1, H_\theta = r, H_z = 1. \qquad (6.12)$$

Therefore, the gradient is given by

$$\nabla u = \frac{\partial u}{\partial r}\mathbf{e_r} + \frac{1}{r}\frac{\partial u}{\partial \theta}\mathbf{e}_\theta + \frac{\partial u}{\partial z_r}\mathbf{e_z} \qquad (6.13)$$

and the divergence is given by

$$\nabla \cdot \mathbf{A} = \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_r) + \frac{\partial}{\partial \theta}(A_\theta) + \frac{\partial}{\partial z_r}(rA_z)\right] \tag{6.14}$$

or

$$\nabla \cdot \mathbf{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial}{\partial \theta}A_\theta + \frac{\partial}{\partial z_r}A_z \tag{6.15}$$

6.2.4 Application to devices with cylindrical symmetry

For device with cylindrical symmetry, $\partial over \partial\theta$ yields zero. The operators gradient and divergence becomes:

$$\nabla u = \frac{\partial u}{\partial r}\mathbf{e_r} + \frac{\partial u}{\partial z_r}\mathbf{e_z} \tag{6.16}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_r) + \frac{\partial}{\partial z_r}(rA_z)\right] \tag{6.17}$$

Therefore it is easy to see that in the cylindrical system, all we need to do is to replace $\mathbf{A}$ by r$\mathbf{A}$ and multiply $\frac{1}{r}$ in front of the differential operators.

Our next task is to convert the new differential equation into a suitable discretization scheme for the cylindrical system. Let us recall how we discretize the drift-diffusion equation in the old coordinate system. We evaluate the gradient (the electric field or the current density vector) for each boundary of the finite box which we create for the neighbouring elements. We then sum up the total flux and divide it by the area of the finite box. In the new system, we need to multiply the flux by r which is evaluated at the boundary point. We also divide the total flux by the area of the finite box and by r evaluated at the center of the box (or the center point).

It is obvious that we should keep in mind that the missing dimension is no longer infinite but has rotational symmetry. For example, we need to integrate current density over the radiate to get the total current.

6.3 Modeling 3D Current Flow in Semiconductors

6.3.1 Introduction

With all the simulation power and sophistication, Crosslight software packages are mostly written for a two-dimension cross section of a semiconductor device. We usually take a cross section where physical quantities vary the most. We created sophisticated techniques to vary the mesh density and grid boundaries to fit all types of device geometries.

In previous versions of PICS3D, the optical wave is allowed to interact in all three dimensions. However, current flow in z-direction is forbidden. The isolation of current flow allows us to decouple the drift-diffusion equation and save a large amount of computation time.

With increasing computing power and decreasing cost of modern computers (especially high end PC's), our next natural step is to lift the limitation for current flow and to create a full 3D model for drift-diffusion and energy transport equations.

Some of the outstanding issues must be resolved to extend from 2D to 3D. These include the following.

- The user input/interface should be a natural extension of 2D simulation. All material macros in 2D should be usable in 3D. All physical models and boundary conditions must be extendable to 3D.
- A useful 3D simulation should be able to handle geometry and material variation in the z-dimension easily.
- Smart memory management scheme must be developed so that the 3D package can be run on anything from a laptop to a super computer.

We have successfully overcome most of the difficulties and created an option for modeling current flow in all three dimensions. We will call this new capability "3D-Current Flow" or "3D-Flow" option to differentiate it from our previous 3D model with limitation on current flow in the z-direction. In the following subsections, we described the basic design ideas behind the 3D-Flow option.

6.3.2 3D structure and material input

It is most convenient to input the geometry and material information based on a description of many 2D segments since we already know how to input the 2d structure. We also know how to create and refine the 2D mesh.

As usual, we find the parallel cross sections of a device that have the most variation in physical variables and define them as x-y planes. The z-direction has a relatively slow variation. For convenience of structural data input, we only need to describe the segment that has uniform material properties along the z-direction.

If the material properties such as bandgap, doping, effective masses are uniform from z1 to z2, we call it a "segment" (or a "z-segment"). Within each segment, there can be different "x-y planes" (or a "plane"). The physical variables such as potential and carrier concentrations may vary from plane to plane although the material properties may be the same within the segment. Sometime it is also necessary to set z1=z2 with only one plane in a segment.

The designation of segments and planes are completely described by the statement **3dflow_segment**.

For different segments, we can assign different mesh, material, and doping properties, assuming they are the same within each segment. The statements of material and doping for each segment bracketed using the **begin_zmater** and the **end_zmater** statements.

Within each segment, it is also possible to define different electrodes provided the electrodes are carefully numbered so as not to confuse with electrodes in other segments.

6.3.3 3D mesh generation

The basic idea for 3D mesh generation is to let the user generate and refine mesh for each 2D segment while the 3D mesh connecting different 2D meshes are generated automatically. We have created a sophisticated algorithm to automatically generate prisms and tetrohedrons connecting the 2D planes.

Ideally, complete automation for 3D meshes should be achievable regardless of heterostructures and material types. However, at this stage of development, the user still needs to provide some guidance to the simulator as to how the 3D meshes should be formed.

We need to introduce the concept of "taper lines" as illustrated in Fig. 6.3. Consider a structure with a taper extending from points A1-B1 on segment 1 to A2-B2 on segment 2. We defined a "taper line" A1-C1 on segment 1 and a corresponding taper line A2-C2 (point C2 is hidden behind the figure). These taper lines force the the simulator to divide the mesh in the two segments into two different regions on the left and right sides. The 3D mesh points on the taper lines are forced to connect to each other through prisms and/or tetrohedrons. The taper lines are critical especially if we are dealing with semiconductor / insulator interface. Within the regions divided by taper lines, the simulator automatically creates connecting prisms and tetrohedrons.

6.3.4 3D simulator control

Once we have input the 3D material data and created the 3D meshes, it is rather straight forward to control the simulator. We need to apply a voltage or a current to one or more electrodes. The output simulation data may be viewed using the conventional .plt files (with statements such as **lplot_xy**, **vplot_xy**, etc.). we can also use the CrosslightView program to plot and analyze the data.

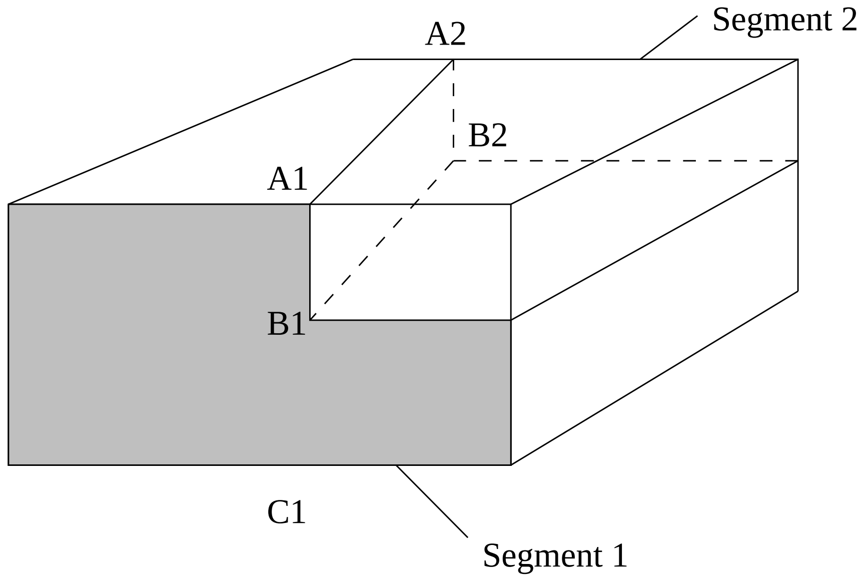


Figure 6.3: Schematics of taper lines in two different segments.

Chapter 7

AC ANALYSIS AND HIGH FREQUANCY PARAMETERS

For many users of Crosslight simulation software, assessment of high frequency performance of their devices is a key reason of the modeling. Analysis of high frequency behavior of a semiconductor device is more than solving the wave propagation in a passive medium. The response of a semiconductor to high frequency modulation is rather complex and requires special transformation of the drift-diffusion equations. This chapter gives an outline of the AC analysis theory for semiconductor devices.

7.1 Theory

7.1.1 Introduction

AC small signal analysis for semiconductor equations was first developed by Laux [56] in 1985. A modified version of this technique including the deep level trap model was later developed by Li and Dutton [57]. We shall follow their formulation in this section. We derive the formulas in subsection 7.1.2. The numerical techniques are given in section 7.2.

7.1.2 Basic formulas

We re-write the basic semiconductor equations including the trap dynamic model as follows.

$$-\nabla\cdot\left(\frac{\epsilon_0\epsilon_{dc}}{q}\nabla V\right)=-n+p+N_D(1-f_D)-N_Af_A+\sum_j N_{tj}(\delta_j-f_{tj}), \qquad (7.1)$$

$$\nabla \cdot J_n - \sum_j R_n^{tj} - R_{sp} - R_{st} - R_{au} + G_{opt}(t) = \frac{\partial n}{\partial t} + N_D \frac{\partial f_D}{\partial t}, \tag{7.2}$$

$$\nabla \cdot J_p + \sum_j R_p^{tj} + R_{sp} + R_{st} + R_{au} - G_{opt}(t) = -\frac{\partial p}{\partial t} + N_A \frac{\partial f_A}{\partial t}. \tag{7.3}$$

$$R_n^{tj} - R_p^{tj}. = N_{tj} \frac{\partial f_{tj}}{\partial t} \tag{7.4}$$

where

$$R_n^{tj} = c_{nj} n N_{tj} (1 - f_{tj}) - c_{nj} n_{1j} N_{tj} f_{tj}, \tag{7.5}$$
$$R_p^{tj} = c_{pj} p N_{tj} f_{tj} - c_{pj} p_{1j} N_{tj} (1 - f_{tj}). \tag{7.6}$$

The basic variables involved in the AC analysis are V, the potential; n and p, the electron and hole concentrations, respectively; and f_{tj}, the trap occupancy of the jth trap. To simplify our derivation, we use the following notation to represent Equations (7.1) to (7.4).

$$F_1(V, U_n, U_p, f_{tj}) = 0 \tag{7.7}$$

$$F_2(V, U_n, U_p, f_{tj}) = \left(\frac{\partial n}{\partial U_n} + N_D \frac{\partial f_D}{\partial U_n} \right) \frac{\partial U_n}{\partial t} \tag{7.8}$$

$$F_3(V, U_n, U_p, f_{tj}) = \left(-\frac{\partial p}{\partial U_p} + N_A \frac{\partial f_A}{\partial U_p} \right) \frac{\partial U_p}{\partial t} \tag{7.9}$$

$$H_j(U_n, U_p, f_{tj}) = N_{tj} \frac{\partial f_{tj}}{\partial t} \tag{7.10}$$

where we have changed the variables n and p into U_n and U_p, the normalized quasi-Fermi levels, respectively. For the electron or hole transport equations (F_2 or F_3) the first term represents on the right hand side, $\frac{\partial n}{\partial t}$ or $\frac{\partial p}{\partial t}$ giving the time dependence of electron or hole concentration, while the second term gives the time dependent increase of carrier concentration because of trap emptying.

In general, the AC input signal (voltage) and solution variables are expressed as:

$$\xi = \xi_0 + \Delta \xi e^{i \omega t} \tag{7.11}$$

where $\xi_0$is the DC solution and $\Delta\xi$ is the complex AC solution with ω being 2πf. ξ can represent V, U_n, U_p, or f_{tj} as complex solutions. By substituting ξ back into the trap equation and doing a Taylor series expansion to first order at the DC solution, Equation (7.4) becomes:

$$\frac{\partial H_j}{\partial U_n} \Delta U_n + \frac{\partial H_j}{\partial U_p} \Delta U_p + \frac{\partial H_j}{\partial f_{tj}} \Delta f_{tj} = N_{tj} i \omega \Delta f t j \tag{7.12}$$

solving for Δf_{tj} gives:

$$\Delta f_{tj} = \left(\frac{\partial H_j}{\partial U_n} \Delta U_n + \frac{\partial H_j}{\partial U_p} \Delta U_p \right) \left(N_{tj} i\omega - \frac{\partial H_j}{\partial f_{tj}} \right)^{-1} \tag{7.13}$$

$$\Delta f_{tj} = \frac{-\frac{\partial H_j}{\partial U_n} \left(\frac{\partial H_j}{\partial f_{tj}} + i\omega N_{tj} \right) \Delta U_n - \frac{\partial H_j}{\partial U_p} \left(\frac{\partial H_j}{\partial f_{tj}} + i\omega N_{tj} \right) \Delta U_p}{(N_{tj}\omega)^2 + \left(\frac{\partial H_j}{\partial f_{tj}} \right)^2} \tag{7.14}$$

or for simplicity, rewriting as:

$$\Delta f_{tj} = F_n(\omega)\Delta U_n + F_p(\omega)\Delta U_p \tag{7.15}$$

Similarly expanding the Poison equation, electron and hole continuity equations as a first order Taylor Series and substituting the above expression for Δf_{tj} we obtain the following equations,

$$\frac{\partial F_1}{\partial V}\Delta V + \frac{\partial F_1}{\partial U_n}\Delta U_n + \frac{\partial F_1}{\partial U_p}\Delta U_p + \frac{\partial F_1}{\partial f_{tj}}(F_n \Delta U_n + F_p \Delta U_p) = 0 \tag{7.16}$$

$$\frac{\partial F_2}{\partial V}\Delta V + \frac{\partial F_2}{\partial U_n}\Delta U_n + \frac{\partial F_2}{\partial U_p}\Delta U_p + \frac{\partial F_2}{\partial f_{tj}}(F_n \Delta U_n + F_p \Delta U_p) = \left(\frac{\partial n}{\partial U_n} + N_D \frac{\partial f_D}{\partial U_n} \right) i\omega \Delta U_n \tag{7.17}$$

$$\frac{\partial F_3}{\partial V}\Delta V + \frac{\partial F_3}{\partial U_n}\Delta U_n + \frac{\partial F_3}{\partial U_p}\Delta U_p + \frac{\partial F_3}{\partial f_{tj}}(F_n \Delta U_n + F_p \Delta U_p) = \left(-\frac{\partial p}{\partial U_p} + N_A \frac{\partial f_A}{\partial U_p} \right) i\omega \Delta U_p \tag{7.18}$$

We rewrite these 3 equations in 3×3 matrix form as:

$$(J + D)\xi = 0 \tag{7.19}$$

for each mesh point in the semiconductor. The J matrix or Jacobian contains the first 3 terms in each equation while D contains the remaining terms concerning the traps and RHS of the equation. Using this notation with $\xi = (\Delta V, \Delta U_n, \Delta U_p)$ the D matrix at each point is of the form:

$$D = \begin{pmatrix} 0 & \frac{\partial F_1}{\partial f_{tj}} F_n(\omega) & \frac{\partial F_1}{\partial f_{tj}} F_p(\omega) \\ 0 & -(\frac{\partial n}{\partial U_n} + N_D \frac{\partial f_D}{\partial U_n}) i\omega + \frac{\partial F_2}{\partial f_{tj}} F_n(\omega) & \frac{\partial F_2}{\partial f_{tj}} F_p(\omega) \\ 0 & \frac{\partial F_3}{\partial f_{tj}} F_n(\omega) & -(-\frac{\partial p}{\partial U_p} + N_A \frac{\partial f_A}{\partial U_p}) i\omega + \frac{\partial F_3}{\partial f_{tj}} F_p(\omega) \end{pmatrix} \tag{7.20}$$

Consider the boundary points and the internal mesh points separately so that:

$$J \rightarrow j + J_B \tag{7.21}$$

and

$$D \rightarrow D + D_B \tag{7.22}$$

The matrix equation can then be rewritten in a separated form for internal and boundary mesh points as follows:

$$(J + D)\xi + (J_B + D_B)\xi_B = 0 \tag{7.23}$$

AC Dirichlet boundary conditions imply that ΔU_n and ΔU_p=0 so that the AC voltage drive at the contact has the boundary solution:

$$\xi_B = \begin{pmatrix} \Delta V_B \\ 0 \\ 0 \end{pmatrix} \tag{7.24}$$

where ΔV_B is the applied AC voltage of one volt. Since the first column in the D matrix must be zero the matrix equation becomes:

$$(J + D)\xi = -J_B \cdot \xi_B \equiv B \tag{7.25}$$

In order to create equations with all real numbers, the D and ξ complex matrices are written with the real and imaginary components separated:

$$D = D_r + iD_i \tag{7.26}$$

$$\xi = \xi_r + i\xi_i \tag{7.27}$$

The matrix is thus rewritten as:

$$(J + D_r + iD_i)(\xi_r + i\xi_i) = B \tag{7.28}$$

Equating real and imaginary components provides the 2 equations

$$(J + D_r)\xi_r - D_i\xi_i = B \tag{7.29}$$

$$D_i\xi_r + (J + D_r)\xi_i = 0 \tag{7.30}$$

which can be rewritten as the expanded matrix:

$$\begin{pmatrix} J + D_r & -D_i \\ D_i & J + D_r \end{pmatrix} \begin{pmatrix} \xi_r \\ \xi_i \end{pmatrix} = \begin{pmatrix} B \\ 0 \end{pmatrix} \tag{7.31}$$

7.2 Numerical techniques in AC analysis

Our task has been reduced to solving the expanded matrix equation detailed in the previous subsection. The AC solution matrix $(\xi_r\xi_i)$ is easily solved using the sparse matrix inverter to invert the expanded Jacobian and multiply by the matrix on the RHS. The inversion can be slow because the expanded Jacobian J+D is twice the order of original J matrix obtained from the DC solution. We use direct solution techniques to perform the matrix inversion so that convergence is guaranteed for all frequencies.

In our simulator, the AC analysis is implemented as a post-processor, i.e., the analysis is done after the DC solution is obtained from the main solver input file (with .sol extension). This is done in two steps:

- The AC analysis matrix is constructed from Jacobian matrix elements used for DC simulation. To enable the data transfer from the usual DC analysis, we use a statement **more_output** to export data needed by the AC analysis to data files.
- The AC analysis is activated by the input file with extension .plt. The input file must specify which data set (thus the DC bias condition) is used for the AC analysis. The statement **ac_voltage** is used to apply a one volt AC signal on one of the contacts (or electrodes). The resulting AC current data are stored in an output file defined by the user.
- Once the AC current is calculated, a statement **plot_ac_curr** is used to plot results of the AC analysis such as conductance and capacitance.

The basic outputs are the complex AC currents at all the electrodes and the AC current distribution, in response to the applied AC voltage (one volt) at one of the electrodes. The following important quantities can be extracted from the AC current:

- The conductance matrix elements are computed from the real part of the electrode AC current with one volt applied AC bias at one of the electrodes.
- The capacitance matrix elements are computed from the imaginary part of the electrode AC current, divided by ω, with one volt applied AC bias at one of the electrodes.
- The current gain can be computed from ratio of the AC current magnitudes between two electrodes, with one volt applied AC bias at one of the electrodes.
- The voltage gain is slightly more complicated. It must be done in two steps. For example, we wish to obtain the voltage gain $\Delta V_2/\Delta V_1$. This may be written as $\Delta V_2/\Delta V_1 = \frac{(\Delta I_2/\Delta V_1)}{(\Delta I_2/\Delta V_2)}$ which means we must apply the **ac_voltage** twice to two different electrodes. The voltage gain is obtained by taking the ratio of the two values of AC current at electrode 2 from the two AC voltage biases.

7.3 Two-Port Network Parameters

Linear networks, or nonlinear networks operating with signals sufficiently small to cause the networks to respond in a linear manner, can be completely characterized

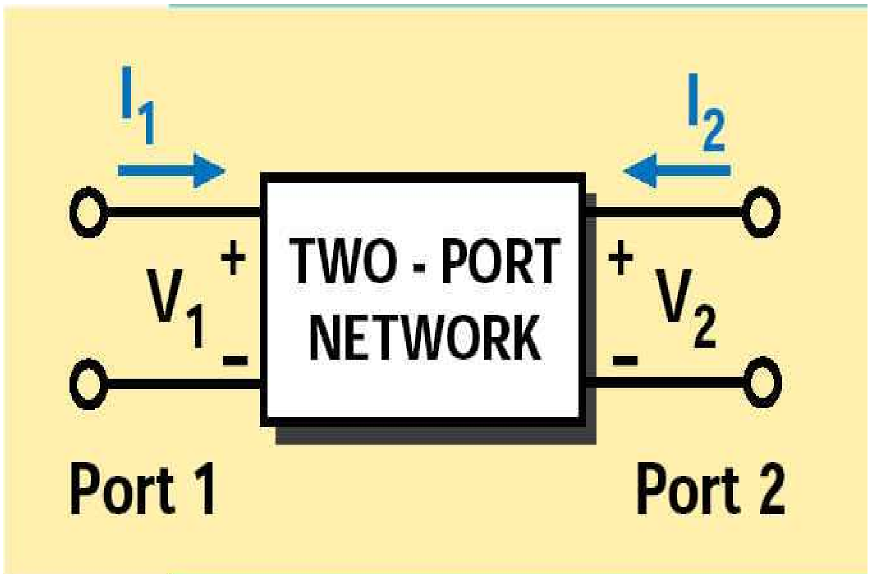


Figure 7.1: Schematics of a general 2-port network. Please note the direction of current flow.

by parameters measured at the network terminals (ports) without regard to the contents of the networks. Once the parameters of a network have been determined, its behavior in any external environment can be predicted without regard to the contents of the network. We shall introduce some useful 2-port parameters in this section and explain how they are extracted from DC simulations.

A general 2-port network is indicated in Fig. 7.1. Please note that the positive direction of the current flow is different than in the simulator and all 2-port parameters related to current flow direction will be different by a sign from current flow calculated from the simulator which assigns a positive sign to all current flowing out of a device.

In an AC analysis of semiconductor devices, it is common to share one contact for the input and output. For example, it is often convenient to use the common emitter configuration in a bipolar transistor. By default, all contacts of a device have zero AC voltage bias from a DC simulation. Thus, it is sufficient to specify a single contact as input or output since all other contacts are grounded in AC voltages. Such a situation is ideal for the computation of the Y-parameters which are defined as follows (referring to Fig. 7.1).

$$\begin{aligned} I_1 &= Y_{11}V_1 + Y_{12}V_2 \\ I_2 &= Y_{21}V_1 + Y_{22}V_2 \end{aligned} \tag{7.32}$$

It is often convenient to express the Y-parameters relative to the system characteristic admittance $Y_c = 1/Z_c$ where Z_c is the system characteristic impedance (often set to be 50 Ohm). We denote the relative admittance using lower case letters here, for example, $y_{11} = Y_{11}/Y_c$.

Based on our discussions in previous sections, the complete set of Y-parameters can be obtained if we excite the device twice, once at the input and once at the output contact. Once the Y-parameters are obtained, the important S-parameters can be computed from the following equations (please refer to any text books on microwave theories):

$$\begin{aligned}
s_{11} &= \frac{(1-y_{11})(1+y_{22})+y_{12}y_{21})}{(1+y_{11})(1+y_{22})-y_{12}y_{21}} \\
s_{12} &= \frac{-2y_{12}}{(1+y_{11})(1+y_{22})-y_{12}y_{21}} \\
s_{21} &= \frac{-2y_{21}}{(1+y_{11})(1+y_{22})-y_{12}y_{21}} \\
s_{22} &= \frac{(1+y_{11})(1-y_{22})+y_{12}y_{21}}{(1+y_{11})(1+y_{22})-y_{12}y_{21}}
\end{aligned} \tag{7.33}$$

The theories discussed in this section strongly suggest a convenient way of studying the 2-port parameters: if you wish to understand the S-parameters, always go back to the Y-parameters which may be easily understood from a simple AC-voltage excitation of the input or output. Also, a convenient way to check the correctness of 2-port parameters should be to plot the Y-parameters which may be compared with AC-current responses from single-terminal excitations, keeping in mind the sign difference mentioned earlier in this section.

7.4 AC Photo-response

In standard AC analysis, the AC source is voltage applied to one of the electrodes. The source of perturbation appears as the **B** vector on the right hand side of Equation 7.28. The component of **B** is non-zero only at boundary nodes where the AC voltage is applied.

When a device is illuminated by a modulated optical signal, similar vector can be set up. The contribution of the optical signal comes from the generation- recombination term which is distributed throughout the whole area sensitive to the light signal. Therefore, the vector **B** is non-zero in all mesh points where there is light illumination.

To activate the AC photo-response model, the approach is similar to that of AC voltage. Instead of using **ac_voltage**, we use **ac_light**. The results are displayed as AC current from a certain electrode, in the same manner as in case of AC voltage bias.

7.5 Laser Diode Modulation Response

Laser diode modulation response is a special case of AC analysis with additional variable of photon density. Similar to the AC photo-response described in the previous section, the same photon-carrier interaction is through the same generation/stimulated recombination term: $R_{st} = v_g g_m S(x, y, z)$ where v_g is the group velocity of light, g_m is the modal optical gain and S is the bulk photon density. In the case of photo-sensitive devices, the optical gain becomes negative and carriers are generated instead of annihilated. An additional photon rate equation must be solved together with all other equations of the drift-diffusion system.

To activate the modulation model, AC small voltage excitation (through **ac_voltage**) is used. The corresponding AC current is computed and used in the plotting the laser diode modulation response, which is measured in per unit modulation current.

Chapter 8

QUANTUM WELLS AND INTERBAND TRANSITIONS

This chapter presents the quantum well models and related interband optical transitions in Crosslight simulation software. Different levels of approximations for quantum well models are described along with the corresponding optical gain/absorption calculations. The theory of many-body gain enhancement implemented in the software is also explained near the end of the chapter.

8.1 Quantum Well Models

8.1.1 Introduction

Quantum well models are among the most sophisticated in Crosslight Software. There are different levels of approximations. The simple (also the default) quantum model assumes a a single, symmetric, flat-band and step-wise potential profile. Parabolic subbands are assumed. The valence mixing model takes into account coupling of different hole subbands, resulting in non-parabolic subbands. The complex MQW model removes the restriction of symmetry and decoupling. This allows us to treat coupling between wells. The self-consistent MQW model removes the restriction of flat-band and couples the potential with charge density in a self-consistent manner.

8.1.2 Simple quantum wells

When a confining potential well gets narrow, it is neccessary to treat it as a quantum well. We consider the case of a simple quantum well without external applied field. All the quantum levels for the sub-bands of Γ, L, light holes and heavy holes (and crystal field holes for wurtzite) are computed from well known formulas in quantum mechanics for a square quantum well [10].

As external field is applied, we allow the quasi-Fermi level to vary as a function of distance. The density of states and quantum levels are assume to be the same as if there is no applied field.

The quantum well levels are calculated at every bias point during an actual simulation because the bandgap of the active region is a function of carrier density. As the bias increases, the higher carrier density in the quantum well reduces the bandgap and changes the quantum well depth. Therefore the quantum levels must be re-computed at every bias point in a self-consistent simulation.

Note that by default, the heavy and light holes are assumed to be parabolic and decoupled. Simplification has been made so that, by default, all the in-plane effective masses of heavy hole or light hole subbands are assumed to be same. If the user wishes to alter the effective mass of a particular subband, the software allows him/her to do so with the "modify_qw" statement.

8.1.3 Carrier concentration for quantum wells

We use the following expression for the density of electrons and holes in a quantum well:

$$n = \sum_j \rho_j^0 kTln\left[1 + e^{(E_{fn}-E_j)/kT}\right] + unconfined\ electrons \tag{8.1}$$

$$p = \sum_i \rho_i^0 kTln\left[1 + e^{(E_i-E_{fp})/kT}\right] + unconfined\ holes \tag{8.2}$$

where the subscript i denotes all confined states for the different hole bands, and j designates those for the Γ and L bands. The number of unconfined carriers are calculated using Fermi statistics as described in Section 5.1.2.

For the simple quantum model, we assume that the electron states outside the quantum wells can be treated as unbound states so that three dimensional Fermi statistics can be applied to these regions for the carrier density.

8.1.4 Anisotropic parabolic approximation

Inclusion of biaxial strain in a quantum-well has become a common practice for many heterojunction devices. Strain offeres an additional degree of freedom and produces some desirable effects, such as a lower threshold current for a laser diode.

Strain is known to cause the valence band of a III-V semiconductor to split into separate light hole (LH) and heavy hole (HH) bands (or CH for wurtsite structure) which are strongly non-parabolic. A rigorous treatment of strain effects on the band structure is rather complicated [13] as the gain function involves an integration over k-space.

We may approximate the non-parabolic band structure by an anisotropic parabolic one. One can show that with proper choices of parameters, a good approximation to the non-parabolic band structure can be obtained. This approximation greatly simplifies the calculation of gain, spontaneous emission and carrier concentration, and it allows one to incorporate strain effects into the program.

For zinc-blende structure, a simplified analytical band structure of a strained quantum well has recently been developed [13] using an efficient decoupling method to transform the original 4×4 valence band Hamiltonian into two blocks of 2×2 upper and lower Hamiltonians. As a result of the decoupling, an analytical expression can be derived [13]. To be consistent with Crosslight convention in 2D simulations of semiconductor devices, the y-axis (equivalent to the z-axis in [13]) is defined to be perpendicular to the quantum-well plane and the z-axis to be along the direction of light propagation.

We use the following expression for the bulk valence band energy [13]:

$$-E = \frac{\hbar^2 \gamma_1}{2m_0}(k_x^2 + k_y^2) \tag{8.3}$$

$$\pm \left[\left(\frac{\hbar^2 \gamma_2}{2m_0}(k_x^2 - 2k_y^2) + \zeta \right)^2 + 3 \left(\frac{\hbar^2 \gamma_2}{2m_0} k_x^2 \right)^2 + 12 \left(\frac{\hbar^2 \gamma_3}{2m_0} \right)^2 k_x^2 k_y^2 \right]^{1/2}, \tag{8.4}$$

$$\tag{8.5}$$

$$\begin{aligned} \varepsilon &= \frac{a_0 - a(x)}{a_0}, && (8.6)\\ \delta E_{sh} &= b(1 + 2c_{12}/c_{11})\varepsilon, && (8.7)\\ \zeta &= \frac{1}{2}\delta E_{sh}, && (8.8) \end{aligned}$$

where the valence band is given in the $k_x - k_y$ plane. The information on strain is contained in ζ; compressive strain is represented by a negative ζ.

The valence band mixing effect for bulk material has been taken into account since the coupling between heavy and light holes has been included in the above equations.

Note that the in-plane direction for Equation (8.4) is the [100] crystal direction, and is different from the [110] direction. For many applications, it is desirable to average the energy dispersion in both the [100] and [110] directions. One can show that the average can be approximated by replacing γ_2 in the second term under the square root of Equation (8.4) by $(\gamma_2 + \gamma_3)/2$. This is called the axial approximation. The simulator provides an option to turn on this approximation.

To convert the non-parabolic bulk band structure into a suitable form, the software fits the valence band to the following anisotropic parabolic band expression, over a range large enough to cover all the optical transitions in k-space:

$$E = -\frac{\hbar^2}{2m_{vx}m_0}k_x^2 - \frac{\hbar^2}{2m_{vy}m_0}k_y^2 \; . \tag{8.9}$$

The quality of the fit is found to be reasonable for practical cases.

Such an approximation is appropriate not only because of the reasonable quality of the fit, but also because the approximation has not lost the basic effects of strain, $i.e.$, it causes the symmetry in the band structure to be broken. The splitting of the heavy and light hole bands is described as [13]:

$$\begin{aligned}
\delta E_{hy} &= 2a(1 - c_{12}/c_{11})\varepsilon, && (8.10)\\
E_g^{hh} &= E_g^u + \delta E_{hy} - \delta E_{sh}, && (8.11)\\
E_g^{lh} &= E_g^u + \delta E_{hy} \\
&\quad + \frac{1}{2}\left[\delta E_{sh} + \Delta - \sqrt{\Delta^2 + 9\delta E_{sh}^2 - 2\delta E_{sh}\Delta}\right]. && (8.12)
\end{aligned}$$

Once the band structure is simplified to an anisotropic parabolic form, the program uses a one-band model to solve for the subbands of the quantum well from the effective masses perpendicular to the quantum well plane. By default, the mixing between different subbands is ignored and all the subbands of heavy holes (or light holes) are assumed to have the same effective in-plane mass. The resulting subbands from such a theory agree qualitatively with k.p theories with multiband full mixing effects, especially for subbands near the band edges and near the Γ center. Since the most important subbands are those near the band edges, the subband model here is expected to be reasonable.

This approach of a one-band model with parabolic subbands has been used extensively for many years and is included in many frequently cited papers (see for example, [14]-[32]).

Once the parabolic subbands are found, one can apply conventional approaches to treat the carrier concentration and the optical transition probabilities. Specifically, the effective mass perpendicular to the plane (m_{vy}) determines the quantum subband levels (or quantum confinement effects) and the optical transition energies. The (2D) density of states and joint density of states of each subband depends on the effective mass in the plane (m_{vx}).

For those users who wish to adjust in-plane effective masses for different subbands, so that the theory here agrees with other k.p models, the program provides an option to do so. That is, the user can shift a subband level or change a subband in-plane mass to match any other theory.

8.1.5 Carrier density in anisotropic parabolic approximation

We use the effective masses parallel to the plane of the quantum well for the evaluation of carrier concentrations.

$$n = \sum_j \rho_j^{x0} kTln\left[1+e^{(E_{fn}-E_j)/kT}\right] + unconfined\ electrons, \tag{8.13}$$

$$p = \sum_i \rho_i^{x0} kTln\left[1+e^{(E_i-E_{fp})/kT}\right] + unconfined\ holes, \tag{8.14}$$

where the subscript i denotes all confined states for the heavy and light holes, and j designates those for the Γ and L bands. The unconfined carriers are calculated using Fermi statistics as described in Section 5.1.2. Note that the effective mass perpendicular to the plane, (m_{vy}), affects the quantum levels and therefore the distribution of carriers in the quantum well.

8.1.6 Valence mixing

Early theories of bulk and quantum well semiconductor lasers assumed parabolic subbands for both the conduction and valence bands. Many early classic papers on quantum well semiconductor lasers used the concept of effective mass, which is equivalent to the parabolic band approximation. A partial list of publications includes Refs. [14]-[32]. The reason for the use of parabolic band, regardless of its inaccuracy, is very simple: the computation of the energy subbands is not the end of the story and one has to use the subbands to calculate other more interesting physical quantities, such as the optical gain. Calculation of the optical gain for a non-parabolic subband is not easy, and one is much better off staying with a parabolic band approximation.

More recent theories of quantum well subbands are mostly based on k.p theory with valence band mixing effects. These usually involve solving a 4×4 [13], 6×6 [17] or 8×8 [19, 20] Hamiltonian of the Luttinger-Kohn type, and imposing an envelope function approximation in solving the quantum well subband structures (see, for example, [21]). The valence subband structures obtained from this approach are complicated and heavily mixed in many cases. For some cases, the effective masses of some of the hole subbands are negative at the Γ point in k-space, making it impossible to use analytical approaches.

Generation of the subbands using a valence mixing model is not difficult and has been done previously by many authors [13, 17, 18, 19, 20]. In contrast, the computation of quasi-Fermi level and optical transition probabilities is not easy because the subbands are non-parabolic and the wave functions can be heavily mixed.

The detailed formulation of k.p theory for quantum well subband structure calculations has been described in many publications; We ahve closely followed the formulas of S.L Chuang [13]. Details will not be repeated here; rather, an outline of physical models is presented to give an overview of the theory.

In k.p theory for a bulk crystal, the basic problem is the following: given the energy level and wave function at a symmetry point of the k-space (for example, the Γ point), how does one calculate the band structure near such a symmetry point?

Based on the symmetry properties of the crystal, a 4×4 or 6×6 Luttinger-Kohn Hamiltonian for the Γ valence band can be constructed, depending on how many valence bands are involved. As a solution of the bulk Luttinger-Kohn Hamiltonian, the energy dispersion is expressed in terms of Luttinger numbers and quadratic equations in k [15] [16]. Note that the energy is accurate to the k^2 term and no further. The wave function at a general k-point is expressed as a linear combination of the wave function at the symmetry point.

If the solid crystal is under perturbation, one may assume that the perturbed wave function can still be expressed as a linear combination of the wave functions at the symmetry point. Within the envelope function approximation, the wave function of the perturbed crystal (due to the potential of a quantum well or of an impurity) can be written as (see Eq. (IV.12) in [15]):

$$\Psi = \sum_{j=1} F(\mathbf{r})\phi_j(\mathbf{r}), \tag{8.15}$$

where ϕ is the periodic part of the Bloch function obtained at the Γ point in the absence of any perturbation. In the work of Luttinger and Kohn [15], the perturbing impurity potential was assumed to be "gentle" and involving only tens of meV. Today the application of the envelope function approximation has been very much broadened.

For a quantum well, the envelope function $F(\mathbf{r})$ in the linear combination is determined by two boundary conditions: the continuity of the wave function and the continuity of the current density. As a result of the solution of the envelope function, the subband energy dispersion and the wave function can be obtained. Due to the non-vanishing off-diagonal terms in the Luttinger-Kohn Hamiltonian, the wave function is a mixture of heavy hole and light hole functions (hence the name valence mixing). In general, the valence subband structure obtained from such a model has the following features.

1) The energy and wave function at the Γ point are the same as those derived from the decoupled methods discussed in previous sections. This makes it possible to label

each subband as heavy hole and light hole based on the base function there. For this reason, many concepts developed in the effective mass approximation can still be used. For example, the selection rule and matrix element for optical transitions at or near the Γ point can still be used.

2) The key difference of this model is the anti-crossing behavior of the subbands in the valence mixing model, *i.e.*, the subband of the light hole can not cross that of the heavy hole. This causes some major discrepancies for structures where light holes and heavy holes are close. The software has captured this major feature of anti-crossing by adjusting the in-plane effective masses of the subbands so that the subbands do not cross each other.

3) Some of the subbands have negative effective masses at the Γ point. In the effective mass theory for optical gain calculation, negative masses are not allowed. Therefore, at first glance, there is nothing one can do about this in the effective mass theory. If one takes another look at the subband structure over a larger k-range, the effective mass eventually returns to a positive value. If a large enough k fitting range is considered, one can still approximate the funny-looking subband by a parabolic band with positive effective mass. The approximation is not so bad considering the density of states is to be broadened by a scattering lifetime, and the exact location of the energy level is meaningless in a practical calculation anyway.

In summary, Crosslight simulator has two levels of models regarding strains and valence mixing effects. By default, the simulator fits the non-parabolic bulk valence band dispersion with two parabolic curves in each direction to obtain anisotrpic effective masses. Efficient computation of carrier density and interband optical transition may be achieved this way. By setting **valence_mixing=yes** in statement if **active_reg** or **set_active_reg**, a full computation of subbands using k.p theory is performed. Carrier densities and interband optical transitions are obtained using numerical integral over the non-parabolic subbands, resulting in longer computation time.

8.1.7 Complex MQW active regions

In previous subsections, it has been implied that quantum wells in an MQW system do not decouple with each other. This is a good approximation if the wells are far apart. There are circumstances in which we need to consider the effect of coupling between quantum wells. Also, it is may also be desireble to have quantum well design with non-symmetric barriers.

Initially, it appeared that extension of a non-decoupled model to a coupled one should be straightforward: just use the potential for two wells instead of one. However in an actually simulation, we must solve problem of localizing the carriers in a complex coupled MQW structure.

Consider the simple case of two wells. As the two wells are brought closer together, the degenerate subbands start to split. From the view point of wave mechanics, the wave function of each energy level belong to both wells. However, drift-diffusion theory (classical theory) require that we must know where the carriers are located.

Therefore, we face the task of deciding for each coupled confined state, which well does the carrier belong. Similarly, for optical interband transition, the process takes place in the whole coupled complex structure. But again, we must decide for each transition, which well does it take place.

The situation is like quantum tunneling where we have a contradiction: the wave nature in a quantum theory versus the particle nature in drift-diffusion theory. We must create a reasonable method to bridge the gap.

For carrier density calculation, we localize the coupled confined states as follows. Suppose the probability of finding an electron of a confined state in well 1 is p_1 and that of well 2 is p_2. In general, $p_1 + p_2 < 1$ (the remaining probability is spent in the barriers). We regard the electron is localized in well 1 with a probability of $p_1/(p_1 + p_2)$. Similar consideration is given for holes.

For optical transition from the valence band to the conduction band, we regard the optical process take place in well 1 with a chance of $p_1q_1/(p_1q_1 + p_2q_2)$, where q_1 and q_2 are probabilities of finding the hole in well 1 and 2, respectively.

The complex active region is declared by the statements **begin_complex** and **end_complex**. This can be used to define coupled wells and MQW with uneven barriers.

8.1.8 Self-consistent carrier density model

8.1.9 localized confined states

For problems of multiple quantum wells (MQW), the default approach in our simulator is to solve the quantum confined states in flat-band condition (assuming no electric field). When there is local variation of potential, we assume it is small enough that flat-band solution is still valid but only introduces a local correction to the confined energy level (see Fig. 8.1). For simplicity of numerical computation, we also assume that carrier density is only confined within the well and is computed according to the local Fermi level and local confined energy level. The following formula for confined 2D carrier electron density is used:

$$\begin{aligned} n_{2D}(x,y) &= \frac{1}{d_w}\sum_j \rho_j^0 kT ln\left[1 + exp[(E_{fn}(x,y) - E_j(x,y))/kT]\right], \quad \textit{inside well} \\ &= 0, \quad \textit{outside well} \end{aligned} \tag{8.16}$$

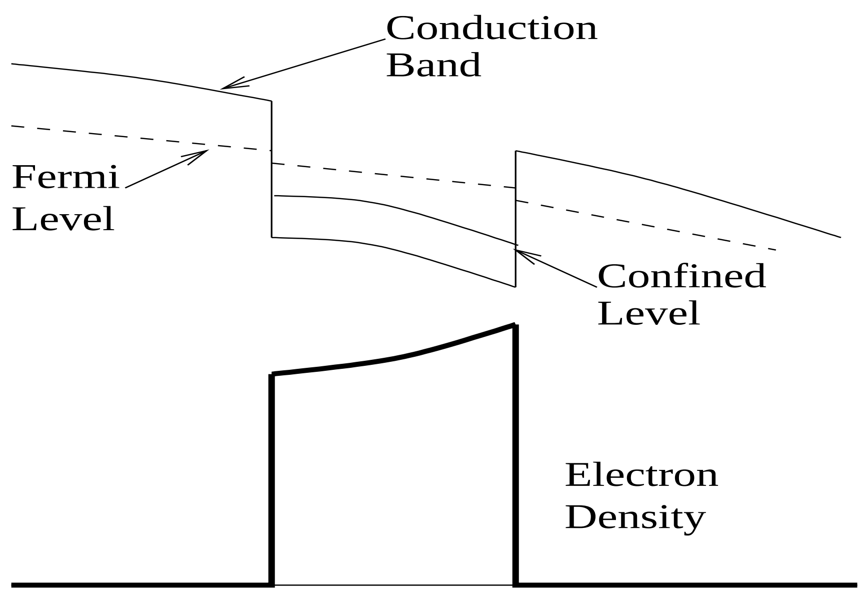


Figure 8.1: Default model of quantum well carrier density calculation.

where d_w is the well thickness. The subscript j denotes all confined states. Please note that both the Fermi level and confined level are assumed to be spatial dependent. ρ_j^0 is the 2D density of states. The variation of the confined level $E_j(x, y)$ follows that of potential, much in the same way the conduction band edge follows the potential change (see Fig. 8.1).

Please note that the spatial variation in this model is uniform except for variations introduced by variation in Fermi level and confined level. We shall refer to this model as "uniform profile" density model.

The above model is simple to implement and efficient to compute. It gives the right MQW behavior if the realistic band structure is close to flat-band condition and we do not care about the details of the carrier density within the scale of the quantum well. Such is the case of most MQW lasers under lasing condition when the band structure is close to flat-band under forward bias and high injection condition.

However, we may run into trouble if the details of carrier density indeed matter when the well is heavily tilted to one-side or another due to strong electrical field such as piezo-electric field. We will consider this case in the next subsection.

The above discussion is limited to electron density but discussion for hole density is completely similar.

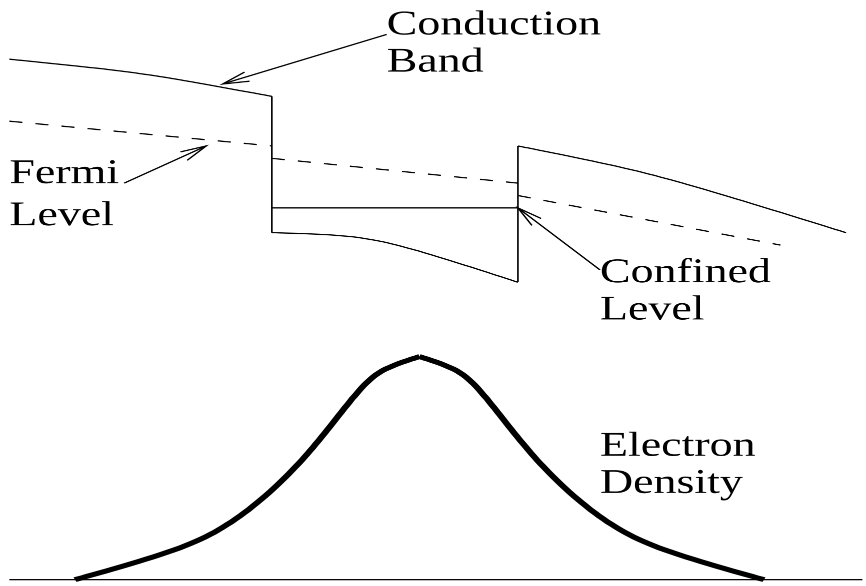


Figure 8.2: Schematics of the self-consistent carrier density model for a single quantum well.

8.1.10 Selfconsistent carrier density distribution

When the potential well is under strong electric field such as in the case of piezo-electric field, the well will be tilted to one side. The confined wave function will also be tilted to one side (see Fig. 8.2). Furthermore, the optical transition between conduction and valence band will have a different dipole moment than under flat-band condition. The tilting of the well will have strong consequences to current overflow which may affect the device performance.

As an option in our software, the self-consistent electron density is given as

$$n_{2D}(x,y) = \sum_j g_n^j(y)\rho_j^0 kTln\left[1 + exp[(E_{fn}(x,y) - E_j)/kT]\right] \tag{8.17}$$

where $g_n^j(y)$ is the electron wave function assuming the well is parallel to x-axis. The confined level E_j is no longer a function of position. Computation of the above density distribution requires more work because the density at one point depends on a global function $g_n^j(y)$. We also have to update the wave function and confined energy levels at different biases.

Similar expressions can be obtained for the hole subbands.

To achieve complete self-consistency, the following procedures are used:

- 1) At equilibrium, solve the potential using flat-band approximation. The default uniform profile model is used for the density. This gives us the initial potential distribution.

- 2) At equilibrium, solve the potential again with potential obtained in 1) above. The self-consistent density model is used. A new density profile is obtained in this step.
- 3) We iterate 2) above until numerical self-consistency is achieved for density profile and potential distribution.
- 4) Once we achieve self-consistency in equilibrium after a number of numerical iterations, we increase the bias and repeat 2) and 3) above.

8.1.11 Cases of valence mixing and complex MQW

In the case of valence mixing model, the valence subbands are no longer parabolic and we must abandon the formulas given in the previous subsections. The confined hole density can be written as an integral over the non-parabolic subbands $E_j(k)$.

$$p = \sum_j g_p^j(y) \frac{1}{2\pi} \int_{-\infty}^{E_k(0)} [1 - f_v(E_j(k), E_{fp})] dk^2 \tag{8.18}$$

where f_v is the Fermi function for the valence band.

For complex MQW with coupled wells, the same expression can be used except that we need to use a proper density partition scheme to make sure that the correct portion of carrier density from a quantum level be allcoated to each well.

8.1.12 Quantum wells in quantum-MOS and HEMT

When the gate of an MOS structure is under bias, the interface of silicon and SiO2 forms a triangular-well for electrons or holes, depending on the polarity of the bias. Similar situation arises in high mobility electron field effect transistors (HEMT) where the internal electric field causes the heterojunction to form a trianglular well to confine the 2D electron gas system.

A triangular well formed in such a way may be regarded as a special case of a quantum well discussed above. To derive such a structure from a simple symmetric flat-band quantum well, we take the following steps: 1) The left barrier is lowered to the point of zero barrier height; 2) External and internal electric field is allowed to tilt the potential in such a way that a triangular well is formed between the bottom of the well and right barrier.

Thus, a realistic quantum well model for quantum-MOS and HEMT may be achieved if we apply the complex-MQW and self-consistent models to a non-symmetric quantum well. Since our models are well established for quantum wells with two barriers, we prefer to treat triangular wells as if they have a left-barrier of zero height.

In the case of quantum-MOS, some special treatment is required since the conduction band should be treated with anisotropic electron masses which may affect the conduction band quantum states, depending on the crystal orientation with respect to the SiO2.

8.2 Interband Optical Transition

8.2.1 Introduction

Many direct-bandgap devices handled by Crosslight simulation software interact with light in one way or another. For indirect bandgap devices such as silicon MOS, this section may be skipped.

For active devices such as laser diodes and LED's, the interband optical gain and spontaneous emission spectrum are important. For photo-sensitive devices such as photo-detectors, the interband absorption (opposite of gain) is important.

In this section, we start with the basic theory of optical gain based on a parabolic band structure. The parabolic band model is the default model and is also our starting point for more complex optical gain models. The parabolic band restriction may be removed if a numerical integral is used to replace the effective masses.

Note that the formulas for quantum wells we consider here are rather general and are also applicable to bulk material. In the case of bulk material, only one subband need to be considered with three-dimensional reduced density of states.

8.2.2 Interband transition model

Similar to the derivation of the absorption coefficients for a solid, the local gain due to a transition from a conduction band labeled j to a valence band labeled i can be written as [9],

$$\begin{aligned} g_{ij}(E_{ij}^0) &= \int P_{ij}\left(\frac{\epsilon_1}{\overline{n}c}\right)\rho_{ij}dE_{ij} \\ &= \int \left(\frac{2\pi}{\hbar}\right) \mid H_{ij}\mid^2 (f_j - f_i)\delta(E_{ij}-\hbar\omega)\left(\frac{\epsilon_1}{\overline{n}c}\right)\rho_{ij}dE_{ij} \\ &= \left(\frac{2\pi}{\hbar}\right) \mid H_{ij}\mid^2 (f_j' - f_i')\left(\frac{\epsilon_1}{\overline{n}c}\right)\rho_{ij} \end{aligned} \tag{8.19}$$

$$\rho_{ij} = \rho_{ij}^0 h(\hbar\omega - E_{ij}^0), \tag{8.20}$$

$$\mid H_{ij}\mid^2 = \left(\frac{q}{m_0}\right)^2 \left(\frac{2\hbar\omega}{4\epsilon_1\epsilon_0\omega^2}\right) M_{ij}^2. \tag{8.21}$$

where $\overline{n}$ is the real part of the refractive index. f_i and f_j are the Fermi functions for the ith and the jth levels, respectively, and f_i' and f_j' are given by

$$f_i' = \left\{1 + exp\left[(E_i^0 - \frac{m_{ij}}{m_i}(E - E_{ij}^0) - E_{fp})/kT\right]\right\}^{-1}, \tag{8.22}$$

$$f_j' = \left\{1 + exp\left[(E_j^0 + \frac{m_{ij}}{m_j}(E - E_{ij}^0) - E_{fn})/kT\right]\right\}^{-1}. \tag{8.23}$$

Note that the above formulas for the gain are rather general and are applicable to quantum wells as well as to bulk material. In the case of bulk material, only one pair of energy bands need be considered. Three-dimensional reduced density of states and bulk momentum matrix elements are to be used. In the case of quantum wells, many subbands in both the valence band and the conduction band need to be considered. 2D density of states and quantum well momentum matrix elements are to be used. Therefore one can consider the expression for the bulk material as a special case of that for quantum wells. The following discussion assumes the formulas for quantum wells.

The gain function is the sum of g_{ij} over all the subbands for the allowed transitions. The software uses the following expression for the density of electrons and holes in the quantum well:

$$n = \sum_j \rho_j^0 kTln\left[1 + e^{(E_{fn}-E_j)/kT}\right] + unconfined\ electrons \tag{8.24}$$

$$p = \sum_i \rho_i^0 kTln\left[1 + e^{(E_i-E_{fp})/kT}\right] + unconfined\ holes \tag{8.25}$$

where the subscript i denotes all confined states for the heavy and light holes, and j designates those for the Γ and L bands. The number of unconfined carriers are calculated using Fermi statistics as described in Section 5.1.2.

At the current stage of model development the software assumes that the electron states outside the quantum wells can be treated as unbound states so that three dimensional Fermi statistics can be applied to these regions for the carrier density. This assumption is reasonable in most applications, and greatly simplifies the simulation since the potential profile is generally a strong function of bias voltages.

All the quantum levels for the sub-bands of Γ, L, light holes and heavy holes are computed from well known formulas in quantum mechanics for a square quantum well [10]. The quantum well levels are calculated at every bias point during an actual simulation because the bandgap of the active region is a function of carrier density. As the bias increases, the higher carrier density in the quantum well reduces the bandgap and changes the quantum well depth. Therefore the quantum levels must be re-computed at every bias point in a self-consistent simulation.

Note that by default, the heavy and light holes are assumed to be decoupled and can be treated separately. Simplification has been made so that, by default, all the in-plane effective masses of heavy hole or light hole subbands are assumed to be same. If the user wishes to alter the effective mass of a particular subband, the simulator allows him/her to do so with the "modify_qw" statement. Further discussion on the effect of valence band mixing is given in Sections 8.1.4 and 8.1.6.

The simulator uses TE (transverse electromagnetic) mode as the default. The user should set the polarization mode to TM (transverse magnetic) mode if a large tensile strain is present in the material. We use the following anisotropic dipole moment for the heavy and light hole transitions in a quantum well [11]:

$$A_{hh} = \frac{3+3cos^2(\theta_e)}{4} \quad (TE), \tag{8.26}$$

$$A_{lh} = \frac{5-3cos^2(\theta_e)}{4} \quad (TE), \tag{8.27}$$

$$A_{hh} = \frac{3-3cos^2(\theta_e)}{2} \quad (TM), \tag{8.28}$$

$$A_{lh} = \frac{1+3cos^2(\theta_e)}{2} \quad (TM), \tag{8.29}$$

$$M_{hh} = A_{hh}O_{ij}M_0 \ , \tag{8.30}$$

$$M_{lh} = A_{lh}O_{ij}M_0 \ , \tag{8.31}$$

where A_{hh} and A_{lh} are the quantum well dipole moment enhancement factors. M_0 is the dipole moment of the bulk material given by the following expression:

$$M_0 = \frac{1}{6}\frac{m_0}{m_e}\frac{E_{g0}(E_{g0}+\Delta)}{E_{g0}+2\Delta/3}. \tag{8.32}$$

$cos(\theta_e)$ is defined as

$$cos(\theta_e) = E_{ej}/[E_{ej} + m_r/m_e(E - E_{ij})] \quad forE > E^0_{ij}, \tag{8.33}$$

$$cos(\theta_e) = 1 \quad forE \leq E^0_{ij}. \tag{8.34}$$

where E_{ej} is the electron confined subband energy of level j and m_r is the reduced mass.

A few interesting points about dipole enhancement are discussed here. The above equations show that the heavy hole transition is favored under TE polarization while the light hole is favored for TM polarization. With a particular polarization (TE or TM) the bulk moment is split between the heavy and light holes and they average out to the bulk moment. It is interesting to note that the light hole under TM has a larger dipole moment than the heavy hole under TE. Therefore, from the point of view of the larger dipole moment, the light hole transition is very attractive. A major drawback for light hole transition is that, due to light mass, the quantum mechanic wave function tends to spread out more than for the heavy hole. As a result, the overlap integral (O_{ij} above) is reduced.

Broadening of intra-band scattering significantly reduces the local gain function and must be considered. To describe the broadening of the quantum levels, the software uses a line shape function $L()$ in a convolution integral with the optical gain. The final expression for the gain function in the quantum well then becomes

$$g_{qw} = \sum_{i=j} \int g_{ij}(E_x)\tau/\hbar L\left[(E_x - E_{ij}^0)/(\hbar/\tau)\right] dE_x. \tag{8.35}$$

The simplest line shape function is the Lorentzian shape function:

$$L(E_x - E_{ij}^0) = \frac{1}{\pi}\frac{\Gamma_0}{(E_x - E_{ij}^0)^2 + \Gamma_0^2}, \tag{8.36}$$

where Γ_0 is the constant half width of the shape function.

The expression in Equation (8.35) cannot be integrated analytically and an efficient numerical approach for the evaluation of the integral is necessary to perform the 2-D simulation. The program has approximated the Fermi functions f_i and f_j with linear combinations of four exponential terms, which gives an error of less than 0.04% . The Lorentzian shape function is approximated with a linear combination of three exponential terms to give an error of less than 1.2% for a range of -8 to +8 for a standard Lorentzian function.

Equation (8.35) is used to evaluate the stimulated emission and dielectric constants in the basic equations at all the points of the 2D mesh.

The corresponding local loss can be written as

$$\alpha_{qw} = -\alpha_{fn}n - \alpha_{fp}p - \alpha_{act}, \tag{8.37}$$

where the first two terms are due to free carrier absorption in the quantum well. α_{act} is an adjustable background loss term to account for losses in the active quantum well for mechanisms other than interband transitions and free carrier absorption.

When a laser is biased near threshold, the current is mainly determined by the spontaneous emission rate and/or Auger recombination. Therefore the evaluation of the spontaneous emission rate is important. The following expression for the spontaneous emission rate is used [12]:

$$r_{sp}^{qw}(E) = \sum_{i=j}\left(\frac{2\pi}{\hbar}\right) \mid H_{ij} \mid^2 f_j'(1 - f_i')D(E)\rho_{ij}, \tag{8.38}$$

where D(E) is the optical mode density in the material which has a refractive index of $\overline{n}$, given by [12],

$$D(E) = \frac{\overline{n}^3 E^2}{\pi^2\hbar^3 c^3}. \tag{8.39}$$

The software has used the same broadening line shape functions for the spontaneous emission spectrum in Equation (8.38) as for the gain function. This allows the user to compare his/her experimental data with the broadened spontaneous spectrum. The broadened spontaneous emission spectrum is, however, not used in the main simulation. The reason is that since carrier recombination involves emission at all frequencies, it is only necessary to integrate the unbroadened spectrum in Equation (8.38) over all possible frequencies:

$$R_{sp}^{qw} = \int_0^{\infty} r_{sp}^{qw}(E)dE. \tag{8.40}$$

Note that Equations (8.35) and (8.40) are given in terms of the quasi-Fermi levels which can be treated directly as variables for the Newton's method used by the software.

The choice of τ is important since it reduces the peak gain and directly determines the threshold current. Also note that the lasing frequency is determined by the maximum of the modal gain g_m at the present injection level rather than the *local* gain g, since it is the modal gain that appears in the rate equation. The former is an average gain over the optical field. For simplicity the software uses the wavelength at the peak mode gain as the single wavelength appearing in the wave equation.

The effective bandgap for optical gain calculations is reduced because of exchange effects [14]. Such a bandgap shrinkage is given by: $\Delta E_g = A_x \left(\frac{n+p}{2}\right)^{1/3}$. Since this term increases with carrier concentration, it causes a "red shift" tendency in the optical gain spectrum as the carrier concentration is increased. The peak optical gain also has a "blue shift" tendency due to the band filling effect (*i.e.*, the Fermi level separation becomes larger as more subbands are filled). The blue shift effect is usually stronger than the red shift effect due to bandgap shrinkage, and one often observes a blue shift in experiment [37] [38].

In the case of anisotropic parabolic approximation for strained quantum wells, the optical gain formulas can simply be extended by using the in-plane quantities (labeled "x") as follows. The local gain due to a transition from the j^{th} in the conduction band to i^{th} in the valence band is

$$\begin{aligned} g_{ij}(E_{ij}^0) &= \left(\frac{2\pi}{\hbar}\right) \mid H_{ij} \mid^2 (f_j' - f_i') \left(\frac{\epsilon_1}{\overline{n}c}\right) \rho_{ij}^x, & (8.41)\\ \rho_{ij}^x &= \rho_{ij}^{x0} h(\hbar\omega - E_{ij}^0), & (8.42)\\ \mid H_{ij} \mid^2 &= \left(\frac{q}{m_0}\right)^2 \left(\frac{2\hbar\omega}{4\epsilon_1\epsilon_0\omega^2}\right) M_{ij}^2, & (8.43)\end{aligned}$$

where f_i and f_j are the Fermi functions for the i^{th} and the j^{th} levels, respectively. The gain function is the sum of the g_{ij} over all allowed transitions.

Similar to the treatment of the gain function, the spontaneous emission rate can be written as

$$r_{sp}^{qw}(E) = \sum_{i,j} \left(\frac{2\pi}{\hbar}\right) \mid H_{ij} \mid^2 f_j'(1 - f_i')D(E)\rho_{ij} \ , \tag{8.44}$$

where $D(E)$ is the optical mode density.

The program assumes that the dipole moment of a strained quantum well system is the same as that for its unstrained counterpart. This assumption should be accurate near the Γ point. As in the case of the unstrained quantum well, the overlap integral in the dipole moment is assumed to be k-independent in this model.

8.2.3 Landsberg model of gain broadening

The simplest model for gain broadening is to use the following Lorentzian function in a convolution integral:

$$g(E) = \int g^0(E_1)F_l(E_1 - E)dE_1, \tag{8.45}$$

where g^0 is the gain function without broadening. The Lorentzian function is given by the expression:

$$F_l(E_1 - E) = \frac{1}{\pi}\frac{\Gamma_0}{\Gamma_0^2 + (E_1 - E)^2}, \tag{8.46}$$

where Γ_0 is the halfwidth of the broadened energy level and is assumed constant. The assumption of constant Γ_0 means that the gain spectrum is broadened to the same degree across the whole spectrum. The advantage of this model is that it is relatively simple to implement and results in a most stable solution in the 2D simulations.

According to Landsberg's model, the broadening across the spectrum should not be uniform.

The basic mechanism for Landsberg's model is carrier-carrier scattering. The original formulas were derived for bulk material [39] [40] and were extended successfully by Zielinski *et.al.* [37] [38] for quantum well laser modeling. This model is phenomenological and is the most simple form of non-Lorentzian shape function. The success of such a model in comparison with experiment is very encouraging [37] [38].

According to Landsberg's model [39] [40] [41], the halfwidth Γ is a complicated function of the transition energy which has its maximum value at the bottom of the band (or band edge) and decreases to zero as the energy approaches the quasi-Fermi level. Therefore the halfwidth is a function of both the transition energy and the carrier density. The software has used the following approximation according to Martin and Stormer [41]:

$$F_l(E_1 - E) = \frac{1}{\pi}\frac{\Gamma(E_1)}{\Gamma(E_1)^2 + (E_1 - E)^2}, \tag{8.47}$$

where

$$\Gamma(E_1) = \Gamma_0 \left[1 - 2.229 \frac{E_1}{E_{fn} - E_{fp}} + 1.458 \left(\frac{E_1}{E_{fn} - E_{fp}} \right)^2 - 0.229 \left(\frac{E_1}{E_{fn} - E_{fp}} \right)^3 \right]$$

$$for\ E_{fn} - E_{fp} \geq 0\ \&\ 0 \leq E_1 \leq E_{fn} - E_{fp} \tag{8.48}$$

For bulk material a Γ_0 value of 1.2 meV was found. For quantum wells the broadening is much larger (up to 30 meV). The implementation of this model is important for accurate modeling of the gain spectrum, especially for quantum wells. The maximum broadening Γ_0 in the Landsberg model is treated as a fitting parameter for the case of quantum wells.

Note that Equation 8.48 is valid only when the peak gain is positive, *i.e.*, only when the difference between the quasi-Fermi levels is greater than the bandgap of the quantum well subbands. The transition energy must also be between the band gap and this difference. To extend the spectrum to other conditions, the formulas proposed by Zielinski *et.al.* [37] are used:

$$g(E) = I_{conv}(g^0, E) \qquad for E < E_{fn} - E_{fp}, \tag{8.49}$$

$$g(E) = I_{conv}(g^0, E) + g^0(E) \qquad for E \geq E_{fn} - E_{fp}, \tag{8.50}$$

with

$$I_{conv}(g^0, E) = \sum_{i,j} \int_{E_{ij}}^{E_{fn}-E_{fp}} \frac{1}{\pi} \frac{\Gamma(E_1) dE_1}{\Gamma(E_1)^2 + (E_1 - E)^2}$$

$$for\ E_{fn} - E_{fp} \geq E_{ij}, \tag{8.51}$$

and

$$I_{conv}(g^0, E) = 0$$

$$for\ E_{fn} - E_{fp} < E_{ij}. \tag{8.52}$$

The two-piece function is continuous across the spectrum because because $g^0(E) = 0$ when $E = E_{fn} - E_{fp}$.

In the current version of the software, no broadening is imposed on the gain of the bulk material for two reasons. First, the level broadening coefficient is relatively small (1.2meV), and second, the unbroadened gain spectrum does not have a sharp edge near the bandgap as in the quantum well case. Therefore the error caused by not using broadening in bulk material gain is relatively small.

8.2.4 Landsberg broadening and experimental data

This Section gives an example of how the user can calibrate the gain model with experimental data. For a description of the program's commands and syntax, please refer to the Reference Manual. The gain function here uses the Landsberg broadening formula, discussed in Section 8.2.3, which is based on well established physical models for emission spectra from recombination of electron-hole pairs. It uses an energy and carrier density dependent half width in the Lorentzian broadening function. The broadening results in a more symmetric quantum well gain spectrum than the Lorentzian broadening function with constant Γ. The Landsberg model is usually in close agreement with experiment and is strongly recommended when an experimental gain spectrum is used for the purpose of calibration.

The experimental gain data are taken from Figure 1 of E. Zielinski, *et.al.*, *J. Quantum Electron.*, **25**, No. 6, 1407 – 1416 (1989) (referred to as the PAPER). Since the details of the multiple quantum well system were not given in the PAPER, except for the composition and well width, a rough guess (or fitting) of the optical confinement factor is used according to the information accompanying the published data.

Normally when the detailed structure of the laser is known one can perform a full simulation and plot the optical field distribution. The optical confinement factor is automatically printed on the graphical output.

Here, two sets of experimental data for quantum well and bulk InGaAs/InP are used to compare with the results from the gain module of the software. Detailed input files can be found in the standard examples that come with installation of the software. The input files are named "lands.gain", which models the multiple quantum well device, and "lands2.gain", which models the bulk device.

Implemention of the Landsberg broadening model is very simple. The parameter **broadening=landsberg** is set in the statement **active_reg** in "*.sol". First we consider an InGaAs/InP quantum well structure of well thickness 0.0105 μm. A carrier scattering lifetime of 0.2×10^{-13} seconds is used, and a background loss of 45 cm^{-1} is assumed for this structure. The result of the program's gain spectrum together with the experimental data from the PAPER is given in Figure 8.3.

In another example, the bulk InGaAs/Inp structure is considered. The background loss is again assumed to be 45 cm^{-1}. The optical gain spectrum for the bulk material along with experimental data from the PAPER is given in Figure 8.4.

To plot experimental data on the same plot as the simulated gain curve, the parameter **include_data** is used in the file "lands.gain". The quantum well model of the software agrees with the published data very well using the same carrier density as that suggested in the PAPER. The carrier density for the bulk material is slightly different from the value suggested in the PAPER in order to obtain a good fit of the data. This slight difference may be due to the lack of broadening for bulk material gain in the program. Note that the current version of the software assumes

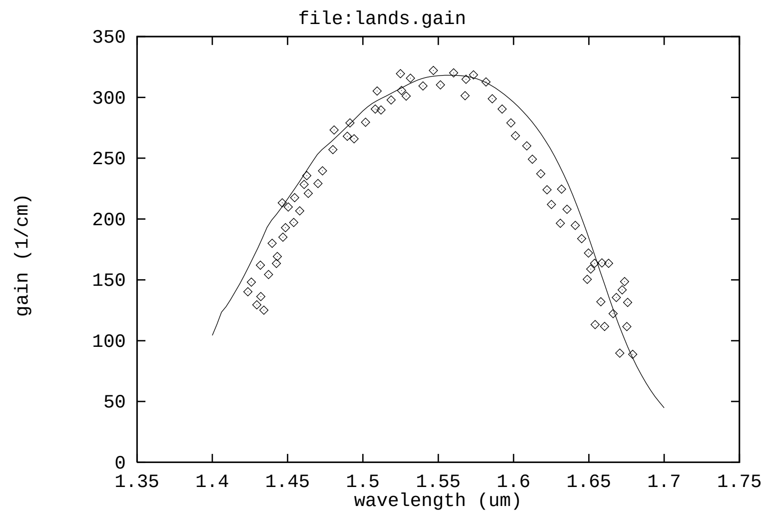


Figure 8.3: Gain spectrum of InGaAs/InP quantum well structure as compared with experimental data. Landsberg type gain broadening is used in the gain model.

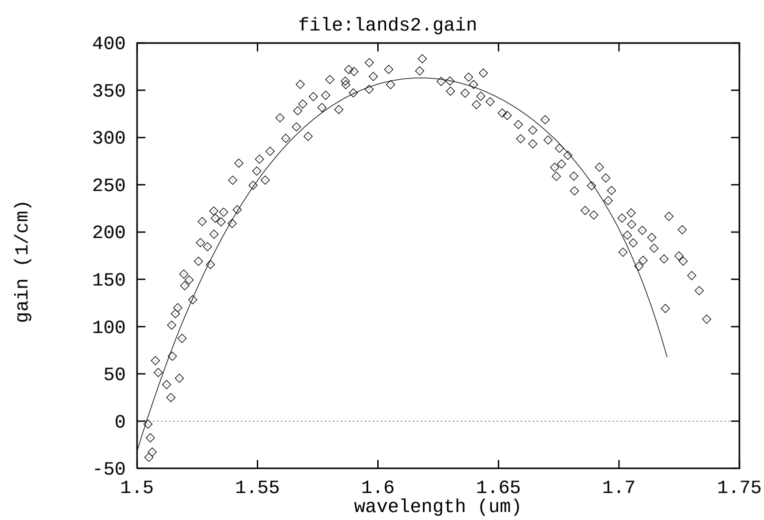


Figure 8.4: Gain spectrum for bulk InGaAs/InP as compared with experimental data.

the broadening effect is negligible for bulk material (which is a good approximation compared with quantum well gain). In both cases the quality of the fit and the values of the parameters are reasonable according to the information given in the PAPER.

8.2.5 Gain integral with valence mixing

In the case of valence mixing, the valence bands are not parabolic and we must use a different formula for the optical gain spectrum. Based on theories developed in Ref. [80], the gain spectrum can be expressed in numerical integration over k_t.

$$g(E) = \frac{g_0}{2\pi t E} \sum_{i,j} \int_0^{\infty} \frac{(\pi/\Gamma) f_{dip}(k_t) M_b (f_j - f_i) dk_t^2}{1 + (E_{cj}(k_t) - E_{kpi}(k_t) - E)^2/\Gamma^2} \tag{8.53}$$

where t is the quantum well thickness; $\Gamma = \hbar/\tau_{scat}$ is broadening due to intraband scattering relaxation time τ_{scat}; E_{cj} is the jth conduction subband; E_{kpi} is the ith valence subband from the k.p calculation; the sum is over all possible valence and conduction subbands; g_0 is a constant defined as

$$g_0 = \frac{\pi q^2 \hbar}{\varepsilon_0 c m_0^2 \overline{n}} \tag{8.54}$$

where q is free electron charge; $\overline{n}$ is the real part of the refractive index; all other symbols have their usual meanings.

M_b is the bulk dipole momentum given by:

$$M_b = \frac{1}{6} \frac{m_0 q}{m_c} \frac{E_{g0}(E_{g0} + \Delta_{so})}{E_{g0} + (2/3)\Delta_{so}} \tag{8.55}$$

where E_{g0} is the unstrained bandgap; m_c is the effective mass of the conduction band; Δ_{so} is spin-orbit coupling energy.

The dipole factor due to non-parabolic subbands are given by the following overlap integral for the case of symmetric well:

$$\begin{aligned} f_{dip} &= (3/2)[< c|g_1(k_t,z) >^2 + (1/3) < c|g_2(k_t,z) >^2] \quad (TE) \\ f_{dip} &= 2 < c|g_2(k_t,z) >^2 \quad (TM) \end{aligned} \tag{8.56}$$

where $|c>$ is the conduction band wave function; $|g_1>$ and $|g_2>$ are the valence band envelop function.

f_j and f_i are the Fermi functions expressed as follows:

$$\begin{aligned} f_j^{-1} &= 1 + exp\left[\frac{1}{kT}\left(E_{cj0} + \frac{\hbar^2 k_t^2}{2 m_0 m_c} - E_{fn}\right)\right] \\ f_i^{-1} &= 1 + exp\left[\frac{E_{kpi}(k_t)}{kT}\right] \end{aligned} \tag{8.57}$$

where E_{cj0} is the bottom of jth conduction subband.

8.2.6 Non-linear gain model

Conventional optical gain is a concept based on linear response theory, *i.e.*, only first order perturbation is used to treat a system under an external optical field. For semiconductor lasers, it is often necessary to include the higher order effects (or non-linear effects). The optical gain tends to decrease as a function of photon density when non-linear effects are included. Such an effect is also called gain suppression.

The non-linear gain effect is responsible for the spectral hole burning effect in semiconductor lasers [8].

The non-linear gain effect is commonly expressed as [8].

$$g = g_0(1 - \epsilon S). \tag{8.58}$$

where ϵ is the non-linear gain suppression coefficient which is not to be confused with the dielectric constant, and S is the photon density. Since the correction term is typically of the order of a few percent, one can use the following alternative form:

$$g = \frac{g_0}{1 + \epsilon S} \tag{8.59}$$

Such a formula has the numerical advantage that the correction term can never make the optical gain negative. Equation (14.19) has the same format as the gain saturation for a simple, two-level system [45].

In the case of multi-lateral modes, the photon density S is replaced by the sum of the densities of all the modes.

8.3 Refractive Index Model

8.3.1 Index change due to interband transitions

In many applications, especially DFB lasers in fiber optic communications, the change of refractive index as a function of injection carriers is an important parameter. It affects the laser linewidth, FM noise and modulation characteristics. Therefore it is desirable for laser designers to evaluate the index change for a particular device structure. The program implements such a change in refractive index due to interband optical transitions based on the standard Kramers-Kronig formulas.

A modified version of the Kramers-Kronig formula relating the real and imaginary parts of the refractive index of an arbitrary material can be written as [46]:

$$\Delta\overline{n}(E) = -\frac{c\hbar}{\pi e}\int_0^\infty \frac{\Delta g(E') - \Delta g(E)}{(E' - E)(E' + E)} dE' \tag{8.60}$$

The integral above is difficult to evaluate numerically because of the singularity at $E = E'$. To obtain an accurate integral, we separate it into the following three parts:

$$\begin{aligned}
\Delta\overline{n}(E) &= -\frac{c\hbar}{\pi e}\int_0^{\infty}\frac{\Delta g(E') - \Delta g(E)}{(E'-E)(E'+E)}dE' \\
&= -\frac{c\hbar}{\pi e}\int_0^{E-\varepsilon}\frac{\Delta g(E') - \Delta g(E)}{(E'-E)(E'+E)}dE' \\
&\quad -\frac{c\hbar}{\pi e}\int_{E+\varepsilon}^{\infty}\frac{\Delta g(E') - \Delta g(E)}{(E'-E)(E'+E)}dE' \\
&\quad -\frac{c\hbar}{\pi e}\int_{E-\varepsilon}^{E+\varepsilon}\frac{\Delta g(E') - \Delta g(E)}{(E'-E)(E'+E)}dE'
\end{aligned} \tag{8.61}$$

where ε is a small number. The third term may be written as

$$-\frac{c\hbar}{\pi e}\frac{dg}{dE}\int_{E-\varepsilon}^{E+\varepsilon}\frac{1}{E'+E}dE' \tag{8.62}$$

and this can be evaluated analytically.

To ensure the correct implementation of the standard model, we have compared our calculations with experimental data obtained by C.H. Henry *et.al.* [46] for a bulk GaAs material system. The calculation is done at transparency conditions. Note that in Henry's paper, the transparent density was estimated to be about $0.8 \times 10^{24} m^{-3}$ while our calculation gives a value of $1.5 \times 10^{24} m^{-3}$. Since our value of transparent density agrees with values from a more recent theory by Vahala *et.al.*[47], we have decided to use this theoretical value here. Our result (also included in LASTIP's examples under examples/index_change/henry.gain) agrees well with that from Henry's paper (see Figure 8.5).

Once the index change is calculated, it is trivial to compute the alpha factor (or the linewidth enhancement factor), which is very important for DFB lasers and related devices used in telecommunications. Again we have calculated the index change for bulk GaAs (see Figure 8.6) and find that the results are similar to those from theoretical calculations by Vahala *et.al.* [47].

It is clear from Figures 8.5 and 8.6 that the index change is very sensitive to the wavelength being used. Therefore any effort to improve the alpha factor or the index change should take into account the spectral dependence. A small shift in the operating wavelength may mean a large improvement or degradation.

8.3.2 Other models of index change

Crosslight software alos implemented for the following forms of index change model.

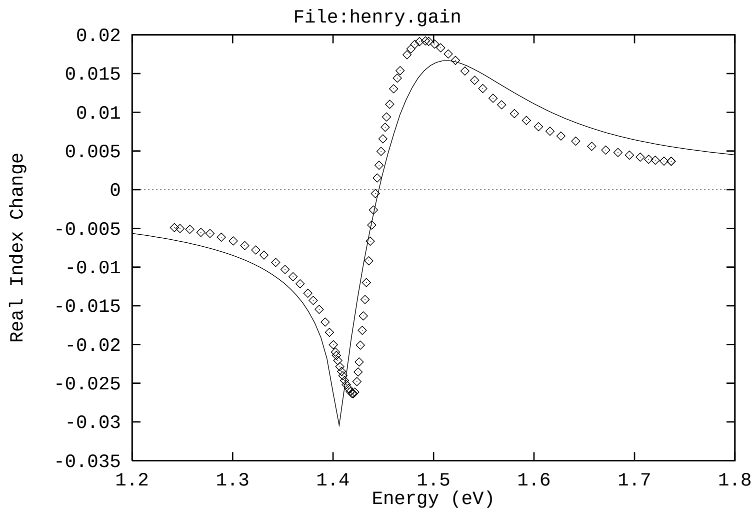


Figure 8.5: Index change with respect to equilibrium. The index change is evaluated at transparent conditions for bulk GaAs material. The points are experimental data from the work of C.H. Henry *et.al*.

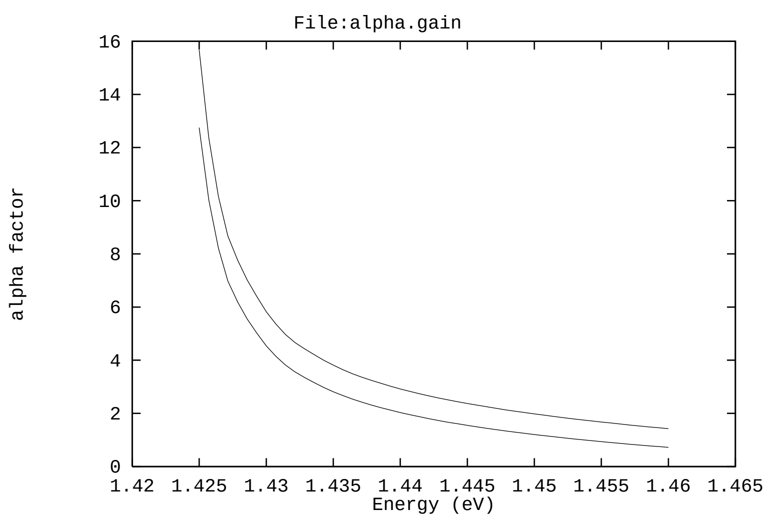


Figure 8.6: The linewidth enhancement factor (alpha factor) as a function of wavelength at two different carrier densities ($1.8 \times 10^{24} m^{-3}$ and $2.5 \times 10^{24} m^{-3}$).

- 1) Use a line with enhancement factor α_{lw} to compute the index change based on local gain change as follows.

$$\Delta\overline{n}(E) = -\frac{\alpha_{lw}}{2k_0}(g - g_0) \tag{8.63}$$

- 2) Free-carrier/plasma part:

$$\Delta\overline{n}(E) = -\frac{q^2 n}{2m_e\omega^2\varepsilon_0\varepsilon} \tag{8.64}$$

- 3) Linear model:

$$\Delta\overline{n}(E) = [a_n^{(a)} + b_n^{(a)}(T - 300)](n - n_0) + [a_p^{(a)} + b_p^{(a)}(T - 300)](p - p_0) \tag{8.65}$$

8.4 Optical Gain with Many Body Coulomb Interaction

Coulomb interaction between electrons and holes confined in an active region of laser diodes can significantly enhance it's optical gain value and influence the emission spectrum. In particular, it has been reported that, Coulomb enhancement is stronger in short wavelength blue-green ZnCdSe and InGaN quantum well lasers compare to the long wavelength III-V lasers, due to the lower value of dielectric constant in wide-bandgap materials[83]. We discover that another major reason for wurtzite material system to have higher manybody (MB) effect is that the due to imbalance of density of states between the conduction and the valence bands, the occupany is much different in the two bands, which causes the enhancement factor to increase.

The role of Coulomb interaction between electrons and holes confined into an active region of laser diode can be explained within a simple physical picture; while Coulomb force attracts electrons and holes at a closer distance, e-h radiative recombination rate increases, which manifests in enhancement in intensity of spontaneous emission as well as in gain magnitude. In addition Coulomb interaction modifies spontaneous emission and gain spectra of laser diodes by shifting the gain peak towards lower energies.

One of the inherent characteristics of electron-hole plasma is screening of the Coulomb potential. The modified, screened Coulomb potential $V_s(q,\omega)$, as described in the momentum and frequency domain, relates to an unscreened Coulomb potential, V_q, as:

$$V_s(q,\omega) = \frac{V_q}{\varepsilon(q,\omega)}, \tag{8.66}$$

where $\varepsilon(q,\omega)$ is the longitudinal dielectric function. If one neglects the dynamic aspect of screening contained in a frequency dependence of the dielectric function,$\varepsilon(q,\omega)$, one can obtain the following dielectric function using the plasmon-pole approximation[84]:

$$1/\varepsilon(q) = 1 - \omega_{pl}^2/\omega_q^2, \tag{8.67}$$

where ω_{pl} is the plasma frequency, while ω_q is an effective plasmon mode frequency.

In particular, in case of electrons and holes confined in a quantum well, one should use a two-dimensional (2D) Coulomb interaction, given by:

$$V_q = e^2/2\varepsilon_0 q, \tag{8.68}$$

and 2D plasma frequency:

$$\omega_{pl}^2 = \frac{e^2 N_{2D} q}{2\varepsilon_b \varepsilon_0 m_r} \tag{8.69}$$

where $N_{2D} = n \times w_{qw}$ is the 2D carrier density in a quantum well width of w_{qw}, ε_b is the static background dielectric constant in the area of quantum well; $1/m_r = 1/m_e + 1/m_h$, is the reduced effective mass of conduction band electrons and valence band holes in directions parallel to the quantum well plane. The effective plasmon frequency ω_q is given in 2D case by:

$$\omega_q^2 = \omega_{pl}^2(1 + q/\kappa^2) + \frac{C_{pl}}{4}\left(\frac{\hbar q^2}{2mr}\right)^2, \tag{8.70}$$

where C_{pl} is a unitless constant typically in a range 1-4, while κ^2 is the square of inverse screening length for electrons and holes confined in the quantum well, which can be derived as[85]:

$$\kappa^2 = e^2/(\pi\hbar^2\varepsilon_b\varepsilon_0)\sum_{i,j}[m_{cj}f_e(E_{cj})/d_{cj} + m_{vi}f_h(E_{vi})/d_{vi}], \tag{8.71}$$

Here, summation $\sum_{i,j}$ is over all sub-bands characterized by energies, parallel masses, and a wave-function average width, E_{cj}, m_{ej}, d_{cj}, and E_{vi}, m_{vi}, d_{vi} for jth conduction band and ith valence band.

$$d_{nk} = \pi\hbar/\sqrt{2m_n E_{nk}}, \tag{8.72}$$

where index n= c or v for conduction and valence bands, respective.

Consider, first, the effect of carrier Coulomb interaction on the spectrum of e-h interband recombination, often ascribed as bandgap-renormalization effect. It's physical meaning is "band-gap narrowing" effect, which can partially be explained by the screening of conduction band electron-electron Coulomb repulsion by positively charged valence band holes available near electrons at close proximity. This Coulomb Hole (CH) contribution to the band-gap reduction is approximately given in 2D by:

$$\Delta E_{CH} = -2E_0 a_0 \kappa \times ln\left[1 + \sqrt{32\pi N_{2D}/C_{pl}\kappa^3 a_0}\right] \tag{8.73}$$

where a_0 is the Bohr radius of quantum well electron-hole exciton, given by:

$$a_0 = 4\pi\hbar^2\varepsilon_b\varepsilon_0/e^2 m_{rj} \tag{8.74}$$

and E_0 is corresponding effective Rydberg energy:

$$E_0 = \hbar^2/(2m_{rj}a_0^2). \tag{8.75}$$

Additional contribution to Coulomb hole band-gap renormalization is a self-energy correction to e-h plasma due to exchange interaction, which also results in reduction of band-gap energy as:

$$\Delta E_{gSX} = -\frac{2E_0 a_0}{\kappa}\int_0^\infty dkk\frac{1 + C_{pl}\kappa a_0 k^2/32\pi N_{2D}}{1 + q/\kappa + C_{pl}a_0 k^3/32\pi N_{2D}}[f_e(E_{cj\mathbf{k}}) + f_h(E_{vj\mathbf{k}})] \tag{8.76}$$

Therefore the net band-gap narrowing due to Coulomb-hole and exchange interaction is:

$$\Delta E_g = \Delta E_{gCH} + \Delta E_{gSX} \tag{8.77}$$

and the renormalized band-gap:

$$E_{g\ ren} = E_g + \Delta E_g \tag{8.78}$$

This effective band-gap energy reduction can be observed in a red shift of all recombination energy values $E_c v(\mathbf{k})$ specified at a given parallel momentum $\mathbf{k}$ of jth-conduction and ith-valence quantum well subbands, as:

$$E_{cv}(\mathbf{k}) = E_g + \Delta E_g + E_{cj\mathbf{k}} + E_{vi\mathbf{k}} \tag{8.79}$$

where $E_{cj\mathbf{k}}$ and $E_{vi\mathbf{k}}$ are energies of electrons and holes from jth-subband of conduction and ith-subband of valence band quantum well in the active region.

The spectral wave function, $g(E_{cv})$ consists of a sum of $g_{ji}(E_{cv})$ contributions from transitions between jth-subband electrons and ith-subband holes, given by [85]:

$$\begin{aligned} g_0(E_{cv}) &= \sum_{j,i} g_{ji}(E_{cv}) = \pi e^2\hbar/(\varepsilon_0 c^3 m_0^2 n_{av})\sum_{j,i}(1/E_{cv})|M_{ji}(E_{cv})|^2\rho_{r,ji} \\ &\quad \times[f_e(E_{cj\mathbf{k}}) + f_h(E_{vj\mathbf{k}}) - 1], \end{aligned} \tag{8.80}$$

where $|M_{ji}(E_{cv})|^2$ is the transition matrix element at transition energy E_{cv} , $\rho_{r,ji}$ is the reduced density of states between jth-conduction subband, and ith-valence subband; n_{av} is the background refractive index. In practice laser diode spectra are broadened, primarily due to the strong carrier-carrier scattering and other intraband scattering mechanisms, o as well as lattice disorder. Consequently, gain becomes a convolution of an un-broaden gain function of Eq. (8.80) with a broadening function L(x), as:

$$g_B(\hbar\omega) = \int_{E_{g0}}^\infty g_0(E_{cv})L(\hbar\omega - E_{cv})dE_{cv} \tag{8.81}$$

where E_{g0} is an energy of the lowest e-h recombination transition between ground levels of quantum well in conduction and valence band of active region. Typically, a Lorenzian broadening function shape is assumed in L(x), given by:

$$L(\hbar\omega - E_{cv}) = (1/\pi)\Gamma_{cv}(\hbar\omega, E_{cv})/[\Gamma_{cv}^2(\hbar\omega, E_{cv}) + (\hbar\omega - E_{cv})^2], \tag{8.82}$$

and parameterized by Lorenzian width, Γ_{cv}, which is related to scattering time by $\Gamma_{cv} = \hbar/\tau_{cv}$ (note that some sources define Γ_{cv} as equal $\hbar/2\tau_{cv}$). For simplicity, we choose Γ_{cv} to be a constant parameter in our calculations (i.e., we neglect its dependence upon $\hbar\omega$ and E_{cv} energies).

Finally, if Coulomb interaction is included in gain spectral function derivation, $g(\hbar\omega)$ of equation (8.81) becomes:

$$g(\hbar\omega) = Real\left\{\int_{E_{g0}}^{\infty} \frac{g_0(E_{cv})}{1 - q_1(E_{cv},\hbar\omega)}\left[1 - i\frac{(E_{cv}-\hbar\omega)}{\Gamma_{cv}}\right] L(\hbar\omega - E_{cv})dE_{cv}\right\} \quad (8.83)$$

where

$$q(E_cv,\hbar\omega) = \frac{-ia_0E_0E_{cv}}{\pi\kappa|M_{ji}(E_{cv})|} \int_0^{\infty} dk'k'\frac{|M_{ji}(E_{cv'}|}{E_{cv'}} \times \frac{f_e(E_{cj\mathbf{k'}}) + f_h(E_{vi\mathbf{k'}}) - 1}{(\Gamma_{cv} + i(E_{cv} - \hbar\omega)} \times \Theta(\mathbf{k},\mathbf{k'}) \quad (8.84)$$

and

$$\Theta(\mathbf{k},\mathbf{k'}) = \int_0^{2\pi} d\theta \frac{1 + C_{pl}\kappa a_0 q^2/32\pi N_{2D}}{1 + q/\kappa + C_{pl}\kappa a_0 q k^3/32\pi N_{2D}}, \quad (8.85)$$

and

$$q^2 = k^2 + k'^2 - 2kk'cos\theta, \quad (8.86)$$

and θ is the angle between in-plane vectors **k** and **k'**.

The Eq. (8.83) gives the final integral formula for Coulomb enhanced gain spectral function, numerically calculated in Crosslight simulation software. User has an option to include, band-gap renormalization effect as defined in Eq. (8.77) and independently Coulomb contribution as formulated in Eq.(8.83), along with a custom definition of ε_b, C_{pl}, Γ_{cv}, and n_{av} values as input parameters.

8.5 Exciton Electro-absorption in Quantum Wells

8.5.1 Introduction

It is well known, from semiconductor physics, that an inter-band light absorption is not a simple derivative of a free electron-hole (e-h) pair generation. Both e-h energy, as well as their motion are affected by mutual Coulomb attraction, and some of those effects were already considered in a context of many-body Coulomb correction to QW optical gain (Section 8.4 of this manual). In extreme case, strongly correlated, hydrogen atom-like, bound electron-hole state, called exciton, can be formed in semiconductors. It is characterized by a ground state binding energy, or effective Rydberg :

$$R_y^* = \frac{m_r e^4}{2(4\pi\epsilon_s\hbar)^2}, \quad (8.87)$$

and a spacial average separation distance, or Bohr radius:

$$a_0^* = \frac{4\pi\epsilon_s(\hbar)^2}{m_r e^2}, \tag{8.88}$$

where $m_r = m_e m_h/(m_e + m_h)$ is the electron-hole reduced mass, ϵ_s is a substrate static dielectric permittivity, ($\epsilon_s = \kappa\epsilon_0$, with κ dielectric constant). Excitons can be experimentally observed in a near band-gap absorption edge spectral region, and their spectrum can be effectively varied in a controlled fashion (modulated) by application of an external electrostatic field. This can be directly utilized in electro-absorption modulators for fiber-optics application. In particular, an effective modulator can be based on Quantum-Confined Stark Effect (QCSE)effect observed in thin QW and MQW heterostructures. Due to dimensional confinement realized in these structures, such an effect is quite pronounced, implying strong overlap of e-h wave function and enhancement of the oscillator strength of interband transitions, along with the rise in exciton binding energy. For example in case of GaAs/AlGaAs QW structures excitonic absorbtion becomes an important distinct component of a near band-edge absorption spectra at room temperature, while GaAs bulk excitons are typically observed only at low temperatures.

At higher densities (strong excitation limit) attractive Coulomb force between electron-hole pairs becomes screened by free carriers, and excitons ultimately decompose into free electron-hole plasma and absorption saturates. Therefore, proper model of screening in quasi 2D electron-hole plasmas, confined into semiconductor QW, and MQW region, becomes important in electro-absorption modulator modeling especially at high excitation limit.

8.5.2 Screened Coulomb Potential in Quantum Wells

Many-body physics and their theoretical treatment of Coulomb interaction in quasi 2D QW electrons and hole systems is quite complex, and is usually treated within reasonable approximations. A well known 3D screened Coulomb interaction potential, given by Yukawa potential with screening constant parameter, κ is as follow:

$$V_s^{3d}(r) = \frac{q^2}{4\pi\epsilon_s}\frac{\exp(-\kappa r)}{r} \tag{8.89}$$

often becomes inadequate when applied to calculation of exciton binding energy in 2D electron-hole heterostructure systems. In particular, 3D-Coulomb potential overestimates long range screening effect, while a 2D-Screened Coulomb potential

$$V_s^{2d}(r) = \frac{q^2}{4\pi\epsilon_s}\left[\frac{1}{r} - \kappa\int_0^\infty \frac{dq J_0(qr)}{(q+\kappa)}\right], \tag{8.90}$$

underestimates it. Here, zero-order Bessel function, $J_0(qr)$, is used. It turns out that for thin electron and hole layers as implemented in semiconductor inversion layers, accumulation layers or MQW heterostructures, we need an intermediate potential $V_s(r)$ between $V_s^{2d}(r)$ and $V_s^{3d}(r)$, which we shall refer to as a *quasi-2D screened Coulomb potential*. The electron and hole spatial density distribution across such layers (in z-direction) is defined by density probability function, given by envelope wave function square, as $|\phi(z_e|^2$, and $|\phi(z_h|^2$. First we shall use a single pole approximation at low frequency limit to derive a screened Coulomb potential in momentum space, which can be expressed as:

$$V_s(q) = \frac{V_q^{2d}}{\varepsilon(q)} F(q) \tag{8.91}$$

where V_q^{2d} is an unscreened 2D Coulomb potential

$$V_q^{2d} = \frac{e^2}{2\varepsilon_b \varepsilon_0 q}, \tag{8.92}$$

$F(q)$ is a form-factor, which represents the deviation of the Coulomb potential from a perfect 2D case (of zero thickness layer limit). The form factor is usually expressed in some forms of overlap integral of the quantum well wave functions. The mathematical details of its evaluation are out of the scope of this document.

The dielectric function:

$$\frac{1}{\epsilon(q)} = 1 - \frac{\omega_{pl}^2}{\omega_q^2}, \tag{8.93}$$

and plasma frequency

$$\omega_{pl}^2 = \frac{e^2}{2\varepsilon_b \varepsilon_0} \times qF(q)[\sum_i \frac{n_{c,i}}{m_{c,i}} + \sum_j \frac{n_{v,j}}{m_{v,j}}]. \tag{8.94}$$

Again, similarly to Eq. (8.70) of Section 8.4, the effective plasmon frequency is given by:

$$\omega_q^2 = \omega_{pl}^2 \left(1 + \frac{1}{\kappa F(q)}\right) + \frac{C_{pl}}{4} \left(\frac{\hbar q^2}{2m_r}\right)^2, \tag{8.95}$$

Subsequently, From $V_s(q)$ of Eq. (8.116) we can find quasi-2D Coulomb potential in a real space, using Fourier-Bessel transform of $V_s(q)$, which gives:

$$V_s(r) = \int_0^\infty q V_s(q) J_0(qr) dq \tag{8.96}$$

The quasi-2D Coulomb potential, $V_s(r)$ of Eq. (8.117), can be then employed to obtain binding energy of exciton in QW and MQW, and seems the most appropriate in use for this applications. Nevertheless we left to the user an option, to use in his simulation exciton models using 3D and 2D Coulomb potentials as defined in Eqs. (8.114), for comparative studies.

8.5.3 Exciton Binding Energy in Quantum Well

In our treatment, the confining potential along z-direction, along with wave-functions and corresponding energy values QW sub-bands are based on selfconsistent solution of a Schroedinger and Poisson Equations. The solution is later applied to calculate binding energy of exciton via the form-factor, and finally absorption spectrum of excitons in MQW or QW can be calculated. At this time we limited our derivation to the ground, 1s, excitonic states, but this can be easily extended into excited exciton state as well as to excitons at higher QW levels, in the future.

Consider a wave function for a ground state exciton in a QW as:

$$\Phi(z_e, z_h, r) = \phi(z_e)\phi(z_h)\phi_{exc}(r), \tag{8.97}$$

where $\phi_{exc}(r)$ is a chosen trial wave function for a 1s exciton in QW to be:

$$\phi_{exc}(r) = (\frac{2}{\pi})^{1/2}\frac{1}{\lambda_{exc}}\exp(-r/\lambda_{exc}). \tag{8.98}$$

Here λ_{exc} represents an in-plane exciton Bohr radius, while $\phi(z_e)$,$\phi(z_h)$ are ground state electron and heavy or light hole envelope wave functions. Coulomb attraction of e-h pairs is assumed to be week, compare to the effect of QW confinement in z-direction. As a result, electron and hole wave functions are assumed to be unaffected by long range Coulomb interaction, which only modifies their energy values(true in the limit of 2D exciton Bohr radius, larger than the size o the z-confinement).

We choose to use Ritz variational method for binding energy estimate. The method relies on minimization of the exciton energy upon the trial function parameter, λ_{exc}:

$$E_{exc}(\lambda_{exc}) = E_g + E_{e0} + E_{ho} - E_{b,exc}, \tag{8.99}$$

which is equivalent to finding a maximum in exciton binding energy,E_{bexc}:

$$E_{b,exc}(\lambda_{exc}) = \frac{<\Phi|\hat{\mathbf{H}}|\Phi>}{<\Phi|\Phi>}, \tag{8.100}$$

where

$$\hat{H} = -\frac{\hbar^2}{2m_r}(\nabla_r^2 + \nabla_z^2) + V_{QW}(z) + V_s(r), \tag{8.101}$$

and after partial integration:

$$E_{b,exc}(\lambda_{exc}) = \frac{\hbar^2}{2m_r\lambda_{exc}^2} + <\phi_{exc}(r)|V_s(r)|\phi_{exc}(r)> \tag{8.102}$$

with respect to the 2D Bohr radius λ_{exc} parameter of a trial function $\phi_{exc}(r)$ given by Eq.(8.98).

8.5.4 Quantum Well Exciton Absorption

A general absorption coefficient formula can be applied to the exciton transition as well, as given by:

$$\alpha(\hbar\omega) = A_0 \sum_n |\varphi_n(0)|^2 \delta(E_n + E_g - \hbar\omega)n, \tag{8.103}$$

where

$$A_0 = \frac{\pi e^2 |\hat{\mathbf{e}} \cdot \mathbf{p}_{cv}|^2}{n_r c \epsilon_0 \omega m_0^2} \tag{8.104}$$

$\mathbf{p}_{cv}$ is an interband transition matrix element in QW, n_r is a refractive index, c-speed of light,$\phi_n(r)$ is a normalized wave function of exciton. Here, summation over n extends to both bound and continuum state excitons. For QW bound exciton states we have the oscillator strength given by :

$$\phi_n(0) = \frac{1}{\pi a_0^2 (n - \frac{1}{2})^3}, \tag{8.105}$$

and

$$E_n = \frac{R_y}{(n - \frac{1}{2})^2} \tag{8.106}$$

Initially, it is sufficient to include only ground state heavy and light hole excitons (n=1), although in general case the bound exciton absorption coefficient becomes:

$$\alpha_B(\hbar\omega) = A_0 \sum_{n=1}^{\infty} \frac{2}{R_y a_0^2 \pi (n - \frac{1}{2})^3} \times \delta[\epsilon + \frac{1}{(n - \frac{1}{2})^2}], \tag{8.107}$$

where $\epsilon = (E - E_g)/R_y$.

The continuum-state exciton contribution to absorption is:

$$\alpha_C(\hbar\omega) = A_0 \int_0^{\infty} dE \frac{S_{2D}(E)}{2\pi R_y a_0^2} \delta(E + E_G - \hbar\omega), \tag{8.108}$$

where

$$S_{2D}(\epsilon) = \frac{2}{1 + \exp(-2\pi/\epsilon)} \tag{8.109}$$

is known as Sommerfeld enhancement factor.

The total absorption is equal $\alpha_B(\hbar\omega) + \alpha_C(\hbar\omega)$:

$$\alpha_{exc}(\hbar\omega) = \frac{A_0}{2\pi R_y a_0^2} [4 \times \sum_{n=1}^{\infty} \{ \frac{1}{(n - \frac{1}{2})^3} \delta(\epsilon + \frac{1}{(n - \frac{1}{2})^2})] + S_{2D}(\epsilon)\} \tag{8.110}$$

Finally, we include the finite line-width effect by introducing the Lorentzian broadening function to obtain

$$\alpha_{exc}(\hbar\omega) = \frac{A_0}{2\pi R_y a_0^2} \left[4 \times \sum_{n=1}^{\infty} \frac{1}{\pi (n - \frac{1}{2})^3} \frac{\gamma}{\left(\epsilon + (n - \frac{1}{2})^{-2}\right)^2 + \gamma^2} + \int_0^{\infty} \frac{d\epsilon'}{\pi} \frac{\gamma S_{2D}(\epsilon')}{(\epsilon' - \epsilon)^2 + \gamma^2} \right] \tag{8.111}$$

8.6 Franz - Keldysh Effect

Franz-Keldysh Effect (*FKE*) is a widely known physics phenomena observed in inter-band absorption spectra of bulk semiconductors at the presence of external, dc-electro-static field, **F**. It is a also often named the photo-assisted Zener-tunnelling and relates to *DC-Stark Effect* in bulk semiconductors. It is similar to the *Quantum-Confined Stark Effect*,(*QCSE*) observed in semiconductor Quantum Wells, (*QW*) . Both, *Franz-Keldysh effect* adn *QCSE*, found application in modern electro-absorbtion-modulators.

8.6.1 Solution of the Schrödinger Equation

Consider the case of a single non-interacting particle charge, q in a uniform external dc-electrostatic field acting along z-direction, $\mathbf{F} = [0, 0, F]$. It can be described by the *Schrödinger Equation* :

$$\left(\frac{-\hbar^2}{2m_r}\nabla^2 + q\mathbf{F}\cdot\mathbf{r}\right)\Phi(\mathbf{r}) = \mathrm{E}\Phi(\mathbf{r}), \tag{8.112}$$

where $m_r = m_e * m_h/(m_e + m_h)$. is the reduced effective mass. One can search the solution of Eq.(8.112) in the form of:

$$\Phi(\mathbf{r}) = \frac{e^{ik_x + ik_y}}{\sqrt{A}}\Phi_{E_z}(z), \tag{8.113}$$

where the z-dependent part of the wave function $\Phi(z)$ obeys:

$$\left(\frac{-\hbar^2}{2m_r}\frac{d^2}{dz^2} + qFz\right)\Phi(z) = \mathrm{E}_z\Phi(z), \tag{8.114}$$

with a total energy relative to the bottom of the band given by:

$$E = \frac{-\hbar^2}{2m_r}(k_x^2 + k_y^2) + E_z. \tag{8.115}$$

The solution to the Eq. (8.114) can be expressed in terms of Airy Functions $Ai(Z)$, satisfying the proper normalization and boundary condition of exponential decay in the limit $z \to +\infty$, where variable:

$$Z = \sqrt[3]{\frac{2m_r qF}{\hbar^2}} \times \left(z - \frac{E_z}{qF}\right) \tag{8.116}$$

Electron and hole states in an external electrostatic field $\mathbf{F} = \hat{z}F$ can be characterized by the the wave function for a given set of quantum numbers (kx, ky, Ez) satisfying Eq. (8.113).

8.6.2 Absorption Spectrum

The inter-band absorption calculation can be obtained from a sum over all electron-hole states:

$$\alpha(\hbar\omega) = A_0 \sum_n |\varphi_n(0)|^2 \delta(E_n + E_g - \hbar\omega), \tag{8.117}$$

where

$$A_0 = \frac{\pi q^2 |\hat{\mathbf{e}} \cdot \mathbf{p}_{cv}|^2}{n_r c \epsilon_0 \omega m_0^2} \tag{8.118}$$

Substituting the sum over all quantum number states by the integration over E_z continuous spectrum in Eq. (8.113), one can obtain :

$$\alpha(\hbar\omega) = A_0 \times 2 \sum_{k_x,k_y} \int_0^\infty dE_z |\varphi(0)|^2 \delta\left[\frac{\hbar^2}{2m_r}(k_x^2 + k_y^2) + E_z + E_g - \hbar\omega\right], \tag{8.119}$$

For typical III-V compound semiconductor zincblend structure we can substitute into Eq. (8.119) $E_t = \frac{\hbar^2}{2m_r}[k_x^2 + k_y^2]$ and use quasi-continuous integration over dE_t instead of sum

$$\frac{2}{A} \sum_{k_x,k_y} = 2 \int \frac{d^2\mathbf{k}_t}{(2\pi)^2} = \frac{m_r}{\pi\hbar^2} \int dE_t \tag{8.120}$$

One can obtain absorption coefficient from Eq. (8.119) as follows:

$$\alpha(\hbar\omega) = \frac{A_0}{2\pi} \frac{m_r}{\pi\hbar^2} \sqrt{\hbar\theta_F} \int_\eta^\infty d\eta Ai^2 = \frac{A_0}{2\pi} \frac{m_r}{\pi\hbar^2} \sqrt{\hbar\theta_F}[-\eta Ai^2(\eta) + Ai'^2(\eta)], \tag{8.121}$$

where $Ai'(\eta)$ is the derivative of $Ai(\eta)$ over η, $\hbar\theta_F = \sqrt[3]{\frac{\hbar^2 q^2 F^2}{2m_r}}$, and $\eta = \frac{E_g - \hbar\omega}{\hbar\theta_F}$

Fig. 8.7 illustrates the net result of applying an electrostatic field on inter-band absorption spectra. The apparent oscillations above the band-gap energy and enhanced exponential decay in below-bandgap energy region are signature of the Franz-Keldysh effect. Fig. 8.8 compares the absorption spectrum with and without the external field as a result of the Franz-Keldysh effect.

8.7 Interband Optical Transition Model for Quantum Dots

This section formulates the optical gain and spontaneous emission spectra for a quantum dot device. We consider an ensemble of quantum dots (QDOT) within a multiple quantum well system. For a self-assembled quantum dots device [128], the dots are closely stacked up on top of each other and there may be strong quantum mechanical coupling between the them. In the following formulation, we assume that

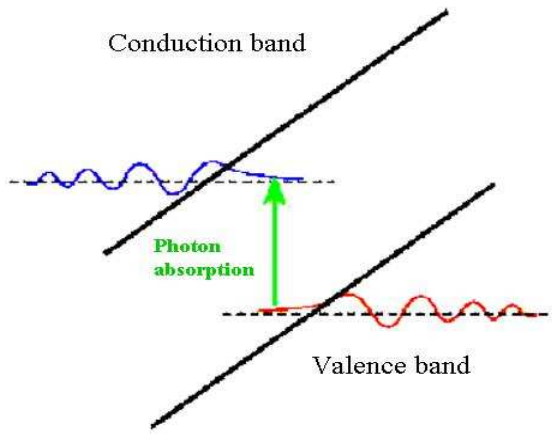


Figure 8.7: Concept of Franz-Keldysh effect of photo-assisted absorption in bulk semiconductors in external electrostatic field. The electron and hole wave-functions become Airy functions, which in external electrostatic field "tunnel" into the band-gap region allowing for overlap of electron and hole wave-functions and absorption of photons with energies below the band-gap energy.

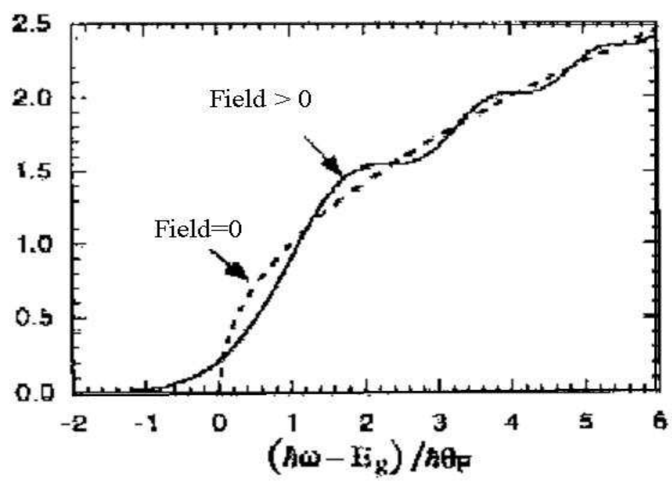


Figure 8.8: The associated absorption spectrum for finite field $F \neq 0$ (solid curve). The dashed line is the free electron and hole absorption spectrum without applied electrostatic field.

the optical spectrum theory for quantum well has been well established [80] and we shall focus on quantum dots.

The formulas here are similar to those in [129]. In a quantum complex containing both wells and dots, the total 2D area; carrier concentrations can be expressed as

$$n_{2d} = n_{qw} + n_{qd} \tag{8.122}$$

The QDOT electron concentration can be written as

$$n_{qd} = \sum_{i,s} 2N_{qd}(s) f_c(E_{i,s}) \tag{8.123}$$

where the sum i is over all energy levels and s is over quantum dots of difference sizes. Please note the factor of two taking into account spin degeneracy. $N_{qd}(s)$ is the areal density of quantum dots with size s. The Fermi function is given by

$$f_c(E_{i,s}) = \frac{1}{1 + exp[(E_{i,s} - E_{fn})/kT]} \tag{8.124}$$

In general, analytical expression for energy level $E_{i,s}$ is not available and one has to solve a 3D numerical eigen value problem to find the confined energies and associated wave functions ψ_{is} for a 3D dot embeded within a well [129]. The APSYS software includes numerical solutions of quantum states for a general 3D quantum dot structure.

Having solved the quantum states of the 3D dot/well complex, we proceed to formulate the optical gain spectrum as follows:

$$g_{qd}(E) = \sum_s \sum_{i,j} \frac{\pi q^2 \hbar M_b^2 N_{dq}}{E \varepsilon_0 c_0 m_0^2 n_r t_{cmplx}} | < \psi_{is} | \psi_{js} > |^2 g_{cp} G_s(E - E_{ijs})(f_c - f_v) \tag{8.125}$$

where g_{cp} is a multiple dot coupling factor which is used avoid double counting. It is inversely proportional to the number of coupling dots. For example, if we have two dots coupling to each other, the number of states will be doubled and we need to divide by two to avoid double counting. M_b is the bulk matrix element and t_{cmplx} is the thickness of the well/dot complex. The sum is over all possible confined electron and hole states and over all dot sizes. The last factor is the populaton inversion factor determined by quasi-Fermi levels of electrons and holes. Other symbols have their usual meanings. The function G_s is a Gaussian broadening function representing intraband and inhomogeneous broadening defined as follows:

$$\begin{aligned} \Delta &= \hbar/\tau \\ G_s(E) &= \frac{1}{\Delta\sqrt{2\pi}} exp\left[-\frac{(E - E_{ijs})^2}{2\Delta^2}\right] \end{aligned} \tag{8.126}$$

The material gain of the active well/dot complex contains contribution from the quantum well, i.e., from states above the quantum dot. Depending on the detailed structure and film quality, the states of the quantum wells may not be perfectly extended through the while well. One may regard some of the states from the quantum wells or the wetting layers as being truncated by the quantum dots. Thus it is convenient to introduce a factor to control or describe the contribution from the wells as follows

$$g_{2d} = r_{qw} g_{qw} + g_{qd} \tag{8.127}$$

where g_{qw} and g_{qd} are optical gain contribution from quantum wells and dots, respectively. r_{qw} is a factor between zero and unity related to the areal ratio of the well area over that of the dots. The above optical gain formulas are important for application involving laser diodes (LD) and vertical cavity surface emitting lasers (VCSEL).

Having found the formula for the optical gain, we can express the spontaneous emission rate using the relation between optical gain and spontaneous emission [2]. The spontaneous emission is given as follows:

$$r_{qd}^{sp}(E) = \frac{q^2 n_r E M_b^2}{\pi \varepsilon_0 m_0^2 \hbar^2 c_0^3} | < \psi_{is} | \psi_{js} > |^2 g_{cp} G_s(E - E_{ijs}) N_{qd}(s)/t_{cmplx} f_c (1 - f_v) \tag{8.128}$$

The total spontaneous emission rate should also include contribtion from the quantum wells in analogy with the optical gain:

$$r_{2d}^{sp}(E) = r_{qw} r_{qw}^{sp}(E) + r_{qd}^{sp}(E) \tag{8.129}$$

Integration of the above over all photon energies results in the total carrier radiative recombination rate as a dominant recombination mechanism for many devices based on forward biased p-n junctions:

$$R_{sp} = \int r_{2d}^{sp}(E) dE \tag{8.130}$$

The spontaneous emission model is important for applications involving light emitting diode (LED), LD and VCSEL. For LED, the EL spectrum comes from the spontaneous emission rate in Eq. 8.129 directly while radiative recombination rate in Eq. 8.130 determines the internal quantum efficiency (IQE). For LD and VCSEL, the radiative recombination rate is often the dominant leakage mechanism upon which the threshold behavior depends.

The software also offers a model for a quantum well to include a quantum dot-like density of states (DOS). This may be used to account for high brightness LED from InGaN-based MQW where a pure quantum well model can not explain the high brightness which is thought to come from localized, dot-like states associated with indium segregation.

The QD model in the software usually requires a separate calculation of 3D quantum states with given shapes such as disk or sphere. The QD-like model skips such details and makes it easier to manage: the model uses a DOS reduction factor to represent 3D quantum confinements which result in lower DOS than pure quantum well.

The DOS reduction factor is defined as the ratio of total number of quantum states in a dot over that in the well. Such a factor many be estimated if shapes of the dots are known. For a simple case, one may use states in a quantum box against those in a quantum well.

To make analogy with reduced DOS in quantum well against those in the bulk, the actual reduction ratio can be estimated if the potential profile of the well is known. Polarization charge and self-consistent calculation may be enabled in such QD like states.

Chapter 9

IMPACT IONIZATION AND TUNNELING

This chapter describes our model of impact ionization and quantum tunneling. We will show that there is similarity between interband tunneling and impact ionization. Both result in generation of electron and hole pairs from a reverse biased p-n junction. Intraband tunneling involves only one type of carrier and the main issue is how to reconcile the quantum nature of wave mechanics with the classical behavior of drift-diffusion.

9.1 Impact Ionization Model

9.1.1 Instroduction

Impact ionization is defined through the following expression for generation rate (see Ref. [1]):

$$G = \alpha_n n v_n + \alpha_p p v_p. \tag{9.1}$$

where α_n is the electron ionization rate defined as the number of electron-hole pairs generated by an electron per unit distance traveled. α_p is similarly defined for holes. Both α_n and α_p are strongly dependent on the electric field. Equation (9.1) is somewhat difficult to implement in a drift-diffusion model because the it is not directly concerned with the velocity. Since impact ionization occurs only in high field region where the drift terms dominates, Selberherr [58] suggested the use of J/q in place of nv for ease of implementation:

$$G = \alpha_n J_n/q + \alpha_p J_p/q \tag{9.2}$$

The issue remains how to choose values of α_n and α_p. A commonly used expression was proposed by Chynoweth [59] as follows:

$$\alpha = \alpha_n^{\infty} exp[-(F_{cn}/F)] \tag{9.3}$$

This formula was later generalized to the following [58]:

$$\alpha = \alpha_n^{\infty} exp[-(F_{cn}/F)^{\kappa_n}] \tag{9.4}$$

Similar expression can be written down for the holes. Usually, the above formulas is valid for a certain field range and we need more than one set of parameters for the whole field range.

Another theory of impact ionization is called the Baraff's three-parameter theory. It takes into account the temperature dependence. A convenient formula for the Baraff's theory was given by Crowell and Sze [60] as follows,

$$\alpha\lambda = exp[g(r,x)] \tag{9.5}$$

$$\begin{aligned} g(r,x) \quad = \quad & (11.5r^2 - 1.17r + 3.9 \times 10^{-4})x^2 \\ & (46r^2 - 11.9r + 1.75 \times 10^{-2})x \\ & -757r^2 + 75.5r - 1.92 \qquad (9.6) \\ & \qquad (9.7) \end{aligned}$$

where

$$r =< E_p > /E_I \tag{9.8}$$

and

$$x = E_I/qF\lambda \tag{9.9}$$

$< E_p >$ is the optical phonon energy and λ is the optical phonon scattering mean free path. E_I is the ionization energy for which the following formula was suggested [60]:

$$E_I = 3E_g/2 \tag{9.10}$$

The temperature dependence of $< E_p >$ and λ are given by

$$< E_p >= E_p tanh\left(\frac{E_p}{2kT}\right) \tag{9.11}$$

$$\lambda = \lambda_0 tanh\left(\frac{E_p}{2kT}\right) \tag{9.12}$$

To summarize, for the Chynoweth model, we need to specify α_n^{∞} (ionization rate at infinite field), F_{cn} (critical field), and κ_n (field exponent) for electrons and similar quantities for holes.

For the Baraff's model, we need to specify the zero temperature mean free path λ_0, and the optical phonon energyE_g.
The software has implemented both the Baraff's model and the Chynoweth's model. All the parameters are expected to be supplied by the simulation user, although the default for silicon is include. This approach is used to reflect the reality that impact ionization model parameters can vary widely, even for well known material such as silicon, as we will show in the next section.

9.1.2 Some Useful Parameters for Impact Ionization Models

The following parameters for the Baraff's model are taken from Sze [1]:

Ge, (both electrons and holes)

Optical-phonon energy E_p (eV)	0.037
Phonon mean free path λ_0 ($\mathring{A}$)	105

Si

Optical-phonon energy E_p(eV)	0.063
Phonon mean free path λ_0 ($\mathring{A}$)	76 (electron)
Phonon mean free path λ_0 ($\mathring{A}$)	55 (hole)

GaAs

Optical-phonon energy E_p(eV)	0.035
Phonon mean free path λ_0 ($\mathring{A}$)	58

The following parameters of the Chynoweth's model are taken from Grant for silicon [61]:

$$\alpha_n = 2.6 \times 10^6 exp(-1.43 \times 10^6/F) cm^{-1} \tag{9.13}$$

for $F < 2.4 \times 10^5 V/cm$.

$$\alpha_n = 6.2 \times 10^5 exp(-1.08 \times 10^6/F) cm^{-1} \tag{9.14}$$

for $2.4 \times 10^5 V/cm < F < 5.3 \times 10^5 V/cm$

$$\alpha_n = 5.0 \times 10^5 exp(-0.99 \times 10^6/F) cm^{-1} \tag{9.15}$$

for $F > 5.3 \times 10^5 V/cm$.

For holes:

$$\alpha_p = 2.0 \times 10^6 exp(-1.97 \times 10^6/F)cm^{-1} \tag{9.16}$$

for $2.0 \times 10^5 V/cm < F < 5.3 \times 10^5 V/cm$.

$$\alpha_p = 5.6 \times 10^5 exp(-1.32 \times 10^6/F)cm^{-1} \tag{9.17}$$

for $F > 5.3 \times 10^5 V/cm$.

The following parameters of the Chynoweth's model are taken from Overstraeten and de Man for silicon [62]:

For electrons:

$$\alpha_n^\infty = 7.03E5\ 1/cm \tag{9.18}$$
$$F_{cn} = 1.231E6\ V/cm \tag{9.19}$$
$$for\ 1.75E5 < F < 6.0E5\ V/cm \tag{9.20}$$

For holes

$$\alpha_p^\infty = 1.582E6\ 1/cm \tag{9.21}$$
$$F_{cn} = 2.036E6\ V/cm \tag{9.22}$$
$$for\ 1.75E5 < F < 4.0E5\ V/cm \tag{9.23}$$

$$\alpha_p^\infty = 6.71E5\ 1/cm \tag{9.24}$$
$$F_{cn} = 1.692E6\ V/cm \tag{9.25}$$
$$for\ 4.E5 < F < 4.0E5\ V/cm \tag{9.26}$$

The above model parameters for silicon are plotted in Fig. 9.1 as impact ionization coefficient for electrons and holes. It is clear that the impact ionization coefficient vary substantially depending on the source of parameters.

9.1.3 Impact Ionization of Compound Semiconductors

The impact ionization coefficients of of compound semiconductors are less well known than those of silicon. A convenient model is the formulas by Chynoweth:

$$\alpha = \alpha_n^\infty exp[-(F_{cn}/F)] \tag{9.27}$$

for electrons. The formula for holes is similar. We fit the above formula to the experimental data published in Ref. [1] and obtain the results in Table 9.1.

For InGaAsP material lattice matched to InP, we can use linear interpolation of InP and In(0.53)Ga(0.47)As.

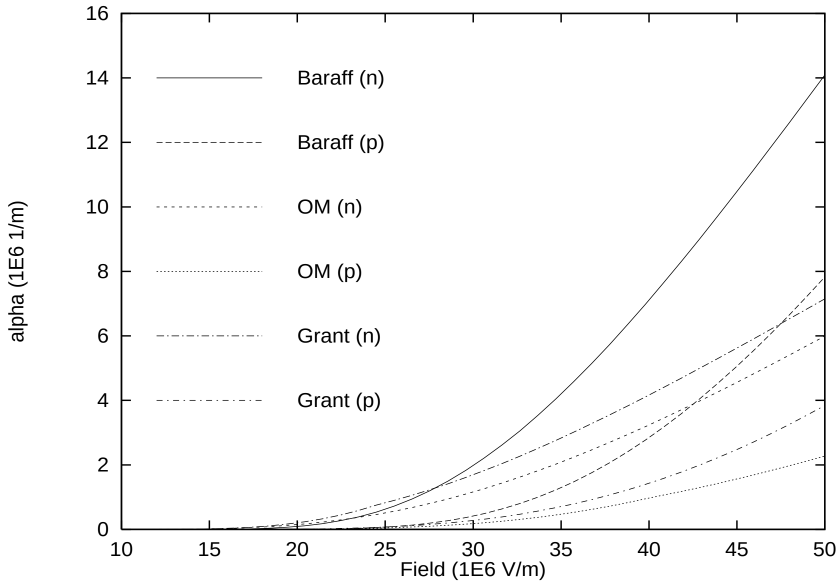


Figure 9.1: Electron and hole impact ionization coefficients of silicon for different models. Baraff is for Baraff's three-parameter theory; OM is for the parameters of Chynoweth's formula from the work of Overstraeten and de Man; Grant is for parameters of Chynoweth's formula from the work of Grant. n and p are used to denote parameters for electrons and holes, respectively.

Material	α_n^∞	F_{cn}	α_p^∞	F_{cp}
GaP	.738E+09	.202E+09	.738E+09	.202E+09
InP	.836E+09	.155E+09	.157E+09	.994E+08
In(0.53)Ga(0.47)As	.344E+10	.833E+08	.125E+11	.102E+09
GaAs(0.88)Sb(0.12)	.997E+07	.250E+08	.920E+07	.316E+08

Table 9.1: Impact ionization rates for some compound semiconductors

9.2 Intraband Quantum Tunneling

9.2.1 Introduction

In highly doped heterojunction or Schottky contacts, an important current transport mechanism is the quantum tunneling effect. In highly doped Schottky contacts of heterojunctions, the quantum barriers are so thin that significant amount of tunneling current goes through the barriers. It is also the key transport mechanism in realistic Ohmic contact which is essentially a highly doped Schottky contact.

We derived the formulas for tunneling current at the top of the barrier in subsection 9.2.2. The formulas are generalized to an arbitrary point in subsection 9.2.3. The tunneling transparencies for a few important cases are given in subsection 9.2.4. Finally, we discuss the numerical treatment of quantum tunneling in drift-diffusion model in subsection 9.2.5.

We use the standard WKB theory to give formulas for the transparency in subsection 9.2.3. Discussion of implementation of the tunneling current is given in subsection 9.2.4.

9.2.2 Tunneling current at top of barrier

Analytical formulas incorporating tunneling effects at the top of the potential barrier were derived by Grinberg et. al. [52] whose approach we shall follow here. Similar to [52] we assume that Boltzmann statistics can be used because the quasi-Fermi level is usually much lower than the top of the barrier. We use the Schematic in Fig. 9.3 and consider the drift current on top of the barrier X_m

$$J = qvn_m + qvn_m\alpha_T \tag{9.28}$$

where α_T is the tunneling coefficient to be derived. The velocity v is dependent on mobility or the thermionic emission properties at the top of the barrier.

Here the basic understanding is that the quasi-Fermi level is relatively flat around the barrier such that the energy distribution of the carriers can be expressed by a simple Boltzmann function (or exponential function). We assume that the carriers with energy between U_0 and U_m are capable of tunneling through the barrier and appears at the other side of the it, giving rise to the additional term in Eq. (9.28). For carriers with energy greater than U_m, they are accounted for by the standard drift-diffusion model.

We define an energy dependent carrier distribution n_E such that

$$n = \int n_E dE \tag{9.29}$$

Using Boltzmann distribution

$$n_E = g(E)exp\left(\frac{U_m - E}{kT}\right) \tag{9.30}$$

where $g(E)$ is the density of states which is a much slower function of energy than the exponential function. For numerical convenience, we ignore the slower energy dependence and replace it by an average constant n_{Em} so that Eqn. 9.29 still holds. As we shall see later, our final results do not depend on the choice of constant n_{Em}.

As we are able to express the energy distribution as follows

$$n_E = n_{Em}exp\left(\frac{U_m - E}{kT}\right) \tag{9.31}$$

the tunneling current can be expressed as

$$J_{tun} = \int_{U_0}^{U_m} qvn_E D_T(E)dE \tag{9.32}$$

$$= qvn_{Em}\int_{U_0}^{U_m} exp\left(\frac{U_m - E}{kT}\right) D_T(E)dE \tag{9.33}$$

where $D_T(E)$ is the energy dependent tunneling transparency. We note that the carrier density at the top of the potential barrier can be expressed in term of n_{Em}:

$$n_m = \int_{U_m}^{\infty} n_{Em}exp\left(\frac{U_m - E}{kT}\right) dE \tag{9.34}$$

$$= n_{Em}kT \tag{9.35}$$

Therefore the tunneling current can be written as

$$J_{tun} = qvn_m(kT)^{-1}\int_{U_0}^{U_m} exp\left(\frac{U_m - E}{kT}\right) D_T(E)dE \tag{9.36}$$

The tunneling coefficient α_{Tm} at the top of the barrier is given by

$$\alpha_{Tm} = (kT)^{-1}\int_{E_0}^{U_m} exp\left(\frac{U_m - E}{kT}\right) D_T(E)dE \tag{9.37}$$

9.2.3 Tunneling current at an arbitrary point

In a 2D simulation, all physical quantities vary in space. We have to assume that the tunneling current also varies in space to make the model compatible with the 2D drift-diffusion model. In another word we have to decide what the tunneling effect is on the current at an arbitrary point away from the top of the barrier. It would be wrong to assume that the same tunneling current appears everywhere in the whole device, although this is the simple case for many textbook tunneling examples.

We must point out that the treatment of tunneling mechanism we give here is of semi-classical nature. The reason is that quantum tunneling assumes the carrier is described by a wave function and can not be localized accurately, while the drift-diffusion model assumes that the carriers are localized and have a definite spatial distribution. Therefore, we should try to satisfy both requirements by choosing a proper area for the tunneling model. If the area of consideration is too small, it conflicts with the wave nature of the problem. If the area is too large, the model contradicts with particle nature of drift-diffusion model. It is clear that the transport mechanism at a point far from the quantum barrier is unrelated to the quantum tunneling.

Again we assume that only those carriers with energy between U_0 and U_m contribute to tunneling. We wish to relate the tunneling current at an arbitrary point to the local current density.

We give the distribution function at an arbitrary point x,

$$n_E(x) = n_{Ex} exp\left(\frac{U(x) - E}{kT}\right) \tag{9.38}$$

The tunneling current can be expressed by

$$\begin{aligned} J_{tun} &= \int_{U_0}^{U_m} qvn_E(x) D_T(E) dE && (9.39)\\ &= qvn_{Ex} \int_{U_0}^{U_m} exp\left(\frac{U(x) - E}{kT}\right) D_T(E) dE && (9.40)\end{aligned}$$

Using the same derivation as in the previous subsection, we re-write the current as

$$\begin{aligned} J_{tun} &= qvn_{Ex} \int_{U_0}^{U_m} exp\left(\frac{U(x) - E}{kT}\right) D_T(E) dE && (9.41)\\ &= qvn(x)(kT)^{-1} \int_{U_0}^{U_m} exp\left(\frac{U(x) - E}{kT}\right) D_T(E) dE && (9.42)\\ &= qvn(x)(kT)^{-1} exp\left(\frac{U(x) - U_m}{kT}\right) \int_{U_0}^{U_m} exp\left(\frac{U_m - E}{kT}\right) D_T(E) dE && (9.43)\\ &= qvn(x) exp\left(\frac{U(x) - U_m}{kT}\right) \alpha_{Tm} && (9.44)\end{aligned}$$

The above formula suggest that all we need to do is to multiply the tunneling coefficient at the top of the barrier by a Boltzmann exponential factor $exp\left(\frac{U_m - E}{kT}\right)$. This is consistent with our common sense that in regions farther away from the barrier, the tunneling effect is less pronounced.

9.2.4 Tunneling transparency

In the current version of the our simulation program, we have included the following models for the tunneling transparency.

- 1. Transparency for a rectangular barrier. Following Ref. [53], for a rectangular barrier of barrier height U_m and thickness d_0, we have the following exact solution for the tunneling transparency:

$$D(E) = \left(1 + \frac{sinh^2\left(\sqrt{\frac{2m^*(U_m - E)}{\hbar^2}} a\right)}{4(E/U_m)(1 - E/U_m)}\right)^{-1} \tag{9.45}$$

- 2. Abrupt heterojunction barrier. One side of the potential barrier has a vertical wall and the other side has a profile described by a quadratic formula [54]. The formula was worked out and is given as follows [52] [54] :

$$D(E) = exp\left(-\frac{U_m}{E_{00}}\left[\sqrt{1-R_x}+\frac{1}{2}R_x ln(R_x) - R_x ln(1+\sqrt{1-R_x})\right]\right) \tag{9.46}$$

$$E_{00} = \frac{\hbar q}{2}\sqrt{\frac{\mid N_d - N_a \mid}{m^*\varepsilon}} \tag{9.47}$$

$$R_x = E/U_m \tag{9.48}$$

  where N_a and N_d are the doping concentrations for acceptors and donors, respectively. U_m and E are measured from the lowest point of the potential profile on the side with a smooth profile.

- 3. Arbitrary barrier with smooth potential profile on both sides. The formula derived from WKB theory can be found in Ref. [55].

$$D(E) = exp\left(-\frac{2}{\hbar}\int_a^b \sqrt{2m^*[U(x)-E]}dx\right) \tag{9.49}$$

  Please note that the requirement for the above formula is that the potential profile must be smooth. Significant error may arise if this requirement is not satisfied. For example, we compare the exact solution with the WKB theory in Fig. 9.2 for a rectangular barrier with abrupt vertical wall on both sides. We find that substantial error can result for such a case.

- 4. Propagation matrix method may be used to treat arbitrary barrier. This method cuts up the potential barrier into piece-wise constant and find the analytical solution in the form of propagation matrices. This method is rather accurate but the computation time required is relatively long.

9.2.5 Numerical evaluation of tunneling current

In a 2D simulation of semiconductor, the evaluation of the tunneling current is not simple since the formulas we developed are all one-dimensional. We suggest the following steps for treating the tunneling current:

- 1)Identify a significant potential barrier. This usually occurs in Schottky contacts, heterojunctions and between quantum wells. The user should mark down a rectangular area with which the tunneling current distribution is to be calculated.

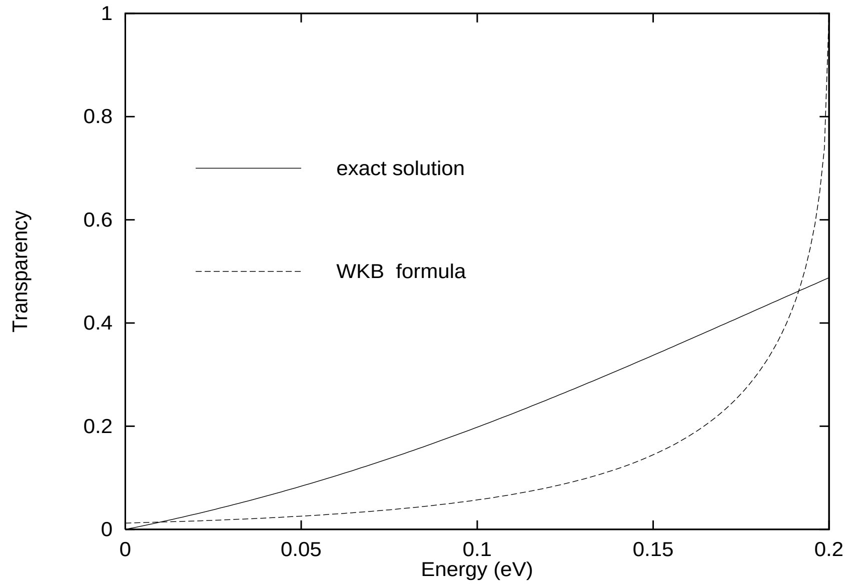


Figure 9.2: Comparison of the transparency from the WKB with the exact solution for a rectangular potential barrier. The barrier thickness is 20 Å and the barrier height is 0.2 eV. The relative effective mass of the carrier is assumed to be 0.2.

- 2)Select a one dimensional segment for the calculation. Since the potential distribution is 2D, selection of the 1D segment reduces the amount of calculation involved.
- 3)A current enhancement factor $1 + \alpha_T$ is calculated along the selected 1D segment.
- 4)Multiply current flow component along the 1D segment with $1 + \alpha_T$ distribution within the rectangular area. For a current flux between two nodes in an arbitrary direction with an angle θ_T from the 1D segment, we multiply $1 + \alpha_T cos(\theta_T)$.

9.2.6 Coupled MQW and intraband tunneling

We wish to discuss the relation between coupled MQW and intraband tunneling. The two models are closely related to each other. They both come from the solution of the same quantum mechanical wave equation.

The quantum levels of a coupled MQW are solutions of confined states of the wave equation whereas intraband tunneling deals with solution of unconfined states. The carriers tunnel through the barriers of an MQW system via standing waves of confined states while traveling waves are used to describe the intraband tunneling effects in the above subsections.

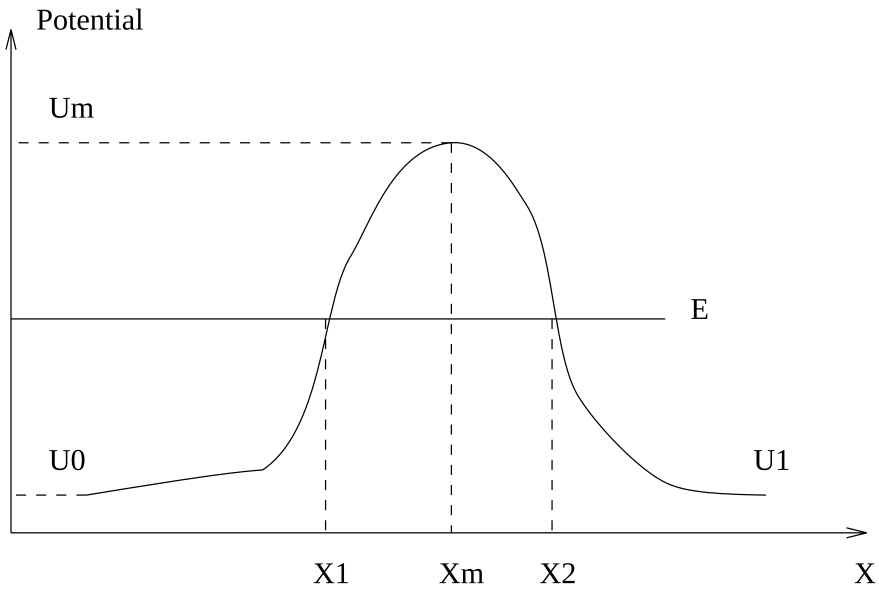


Figure 9.3: Schematic for the potential barrier

In the tunneling model, the carrier states are assumed to be unconfined on both sides of the barrier. One must not use both models to describe the same quantum state. For example, if confined states are already used to described a coupled MQW system with standing waves distribution between the wells, one must not apply the above tunneling model to treat carrier transport between the wells within the same region.

Similarly in the case of quantum-MOS, the quantum states in the triangular well under the gate have already been treated as confined (or partly confined if quantum wave leakage to gate is considered) states. Since the penetration of the wave function from the confined states into the oxide/gate has already been considered, there is no need to apply tunneling model to these confined states. Alternatively, if direct tunneling method is used to treat the current flow through the barrier between confined states in the triangular well and the unconfined states at the gate of an MOS, the the carriers penetrating the oxide must be removed from the transport equations to avoid double counting the tunneling current.

9.2.7 High resistant heterojunction and quantum tunneling

We consider a situation of current injection in the classical drift-diffusion model when high doping is used around an abrupt heterojunction. The peak of the potential barrier depletes the mobile carriers while ionized dopants generates a high internal electric field layer. The high internal field creates a steep potential profile which pushes the barrier higher. Since all mobile carriers are depleted within the barrier, the resistivity of this barrier layer is extremely high. A small current will create a

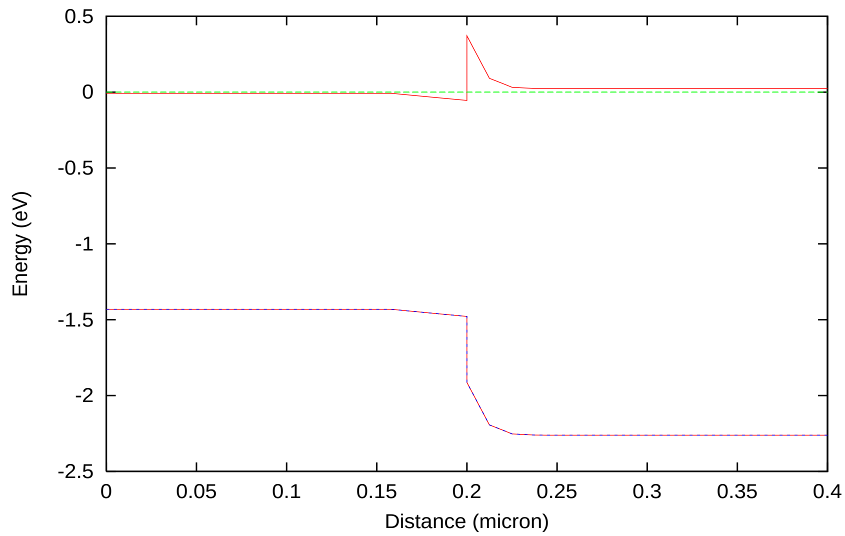


Figure 9.4: Band diagram of a highly doped GaAs/InGaAlP junction in equilibrium.

large voltage drop. Worse still, if the carriers flow from the abrupt side of the barrier, the abrupt potential profile prevents any carriers to fill up depletion layer from the abrupt side while the applied bias adds to the internal field and further depletes the other side of the barrier and hence widens the depletion layer. As a result, it becomes extremely difficult to inject any amount of current into the device with even tens of volts.

To illustrate the above barrier-depletion/voltage-drop effect, we simulate (using drift-diffusion model only) a highly n-doped junction of GaAs/InGaAlP with electrons flowing from the side of GaAs (smaller bandgap). The ideal abrupt junction forms a sharp barrier as shown in Fig. 9.4 in equilibrium. Fig. 9.5 illustrates the narrow depletion layer in the plot of electron concentration. As voltage biased is used to inject barriers from the side of GaAs, the carriers are unable to flow into the depletion layer while the applied field adds to the internal field to create a wider depletion layer as indicated in Fig. 9.6. The corresponding band diagram shows a large voltage drop in the widened depletion layer (see Fig. 9.7).

In reality, a huge voltage drop is not observed in a highly doped heterojunction due to the following reasons. First, the heterojunction is never ideally abrupt and we can conclude that the high barrier in Fig. 9.4 is exaggerated. It can easily be shown in a simulation that the barrier height is extremely sensitive to the abruptness of the junction. Secondly, the quantum tunneling effect makes it easy for mobile carriers to go over the barrier.

The analysis above lead us to propose the following approaches of modeling highly doped abrupt junctions. If we really wish to have a realistic model of transport

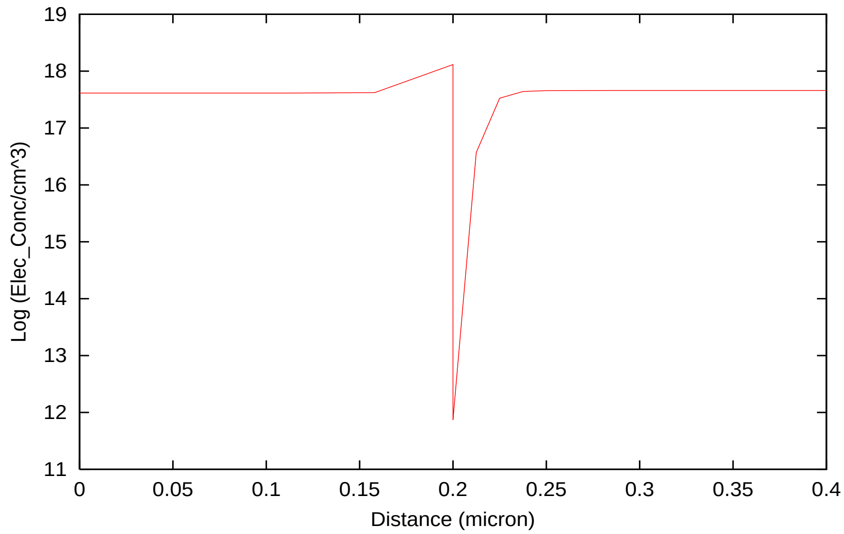


Figure 9.5: Electron concentration distribution of the heterojunction in equilibrium.

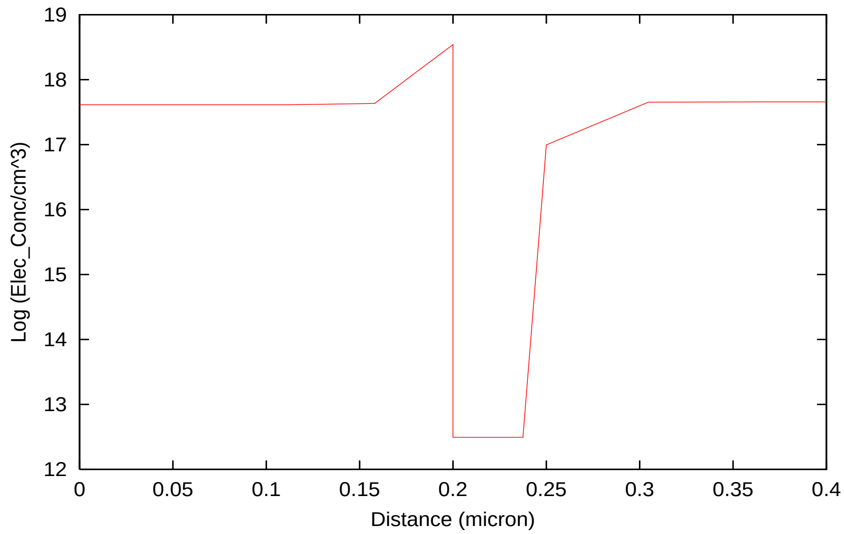


Figure 9.6: Electron concentration distribution of the heterojunction at a bias of five volts.

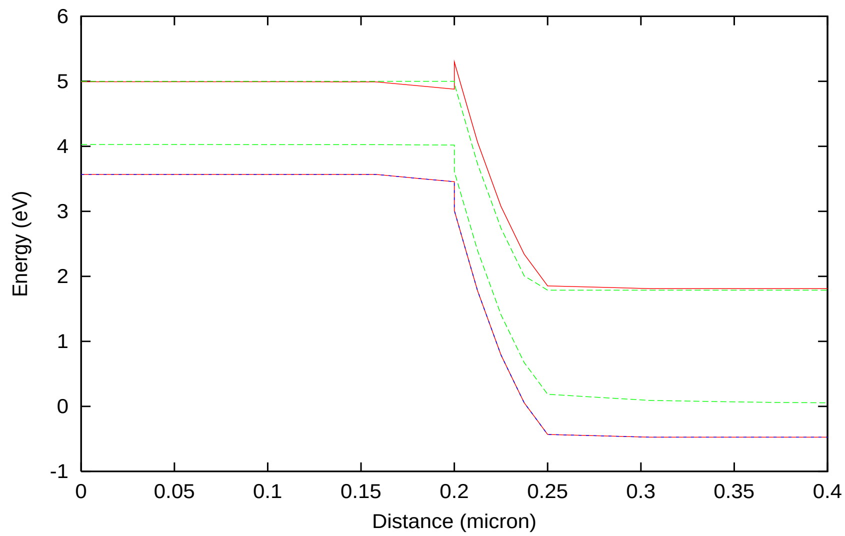


Figure 9.7: Band diagram of the heterojunction at a bias of five volts.

through a highly doped junction, we should specify abruptness of the junction accurately based on material interface quality analysis/experiment. Then we are ready to apply the quantum tunneling model described in this section. We find that using tunneling on ideally abrupt junction usually underestimates the amount of current and still results in a voltage drop larger than that observed in experiment. We thus conclude that ideally abrupt junction model is inherently flawed under high doping condition.

On the other hand, if we wish to avoid dealing with the issues of deciding on the abruptness and using tunneling altogether, we can artificially grade the junction over a distance large enough to avoid forming a barrier but small enough not to disturb other part of the device. For example, if the junction in the above example is far away from the active region of a laser diode, it is safe to grade the junction over a distance of 100 Åwithout upsetting the active region while avoiding the formation of the barrier totally.

9.3 Interband Tunneling in Semiconductors

9.3.1 Theory

Consider the case of a direct tunneling between valence and conduction bands of semiconductor in external applied electric field. A p-n junction designed for such a purpose is commonly referred to as a tunnel junction (TJ) or an Esaki junction. Such type of tunneling is also referred to as Zener tunneling as in a Zener tunneling

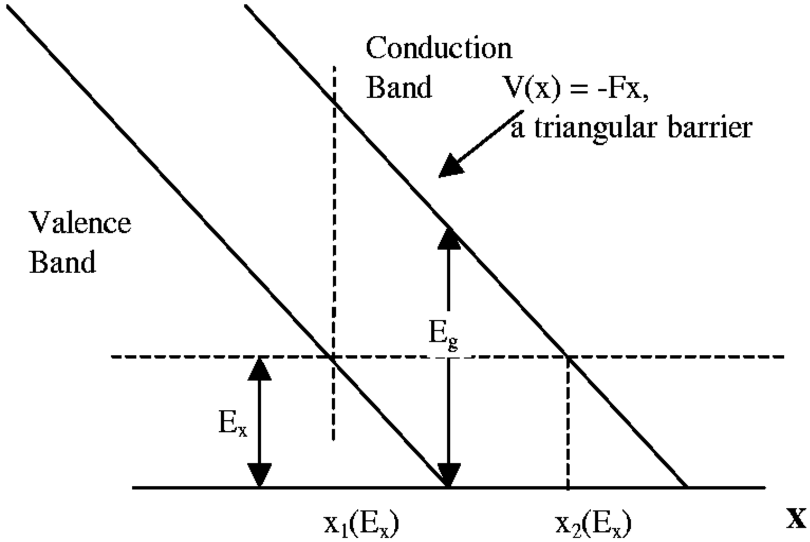


Figure 9.8: Schematic illustrating the Zener tunneling process.

diode under reverse bias. We assume a direct gap semiconductor with a parabolic band with a minimum or a maximum at a central Γ-point at ($\mathbf{k}$=0).

When strong external electric field is applied to a semiconductor, tunneling of electrons between conduction and valence band can take place generating electron-hole pairs. Newly created carriers will drift in a strong electric field and can result in impact ionization. Thus both impact ionization and Zener tunneling electron-hole generation rate should be included in drift-diffusion current solver together.

Similar to impact ionization, generation rate originating from Zener tunneling can be treated in a local electric field approximation, which is a reasonable approximation considering the tunneling distance is typically less than 100 Angstrom for a semiconductor of 1eV bandgap energy. Zener tunneling is usually observed for semiconductors at a field greater than $F=10^6$ V/cm.

Consider the electron state of energy E, tunneling from the valence to the conduction band as presented in Fig 9.8. Within the forbidden bandgap where E obeys $E_g > E > 0$, we have decayed wave function with imaginary wave vector $k_x = i\kappa$.

Neither the exact dependence of k_x, nor the potential barrier that the tunneling electron experiences, upon distance, x, is known through the forbidden gap. Similarly the effective mass remains an unknown and questionable parameter, while electron tunnels through the forbidden gap. Nevertheless, we can usually assume the tunneling mass takes a value somewhere between the conduction band mass and valence band mass. Following the derivation of band-to-band tunneling current in Refs. [69], [70] and [71], we will calculate the probability of tunneling within WKB approximation as follows:

$$D = exp[-2J(E_{||})] \tag{9.50}$$

where:

$$J(E_{||}) = \int_{x_1}^{x_2} \kappa(x)dx. \tag{9.51}$$

$\kappa(x)$ is the decay constant of the tunneling electron wave function, given by:

$$\hbar^2\kappa^2/2m^* = |E_{||} - U|, \tag{9.52}$$

where $E_{||}$, and $E_{\perp}$ are electron kinetic energies in directions along and perpendicular to the tunneling direction, respectively.

After some derivation that are explained in Ref. [71], we can obtain the tunneling probability as follows:

$$D = exp(-2J) = P_0 exp(-E_{\perp}/\underline{E}) \tag{9.53}$$

where:

$$J(E_{||}) = \int_{x_1}^{x_2} |(2m^*/\hbar^2)[(E_g/2)^2 - (\varepsilon_c)^2]/E_g + E_{\perp}|^{\frac{1}{2}} dx \tag{9.54}$$

and

$$P_0 = exp\left[\frac{\pi m^{*\frac{1}{2}}(E_g)^{3/2}}{2(2)^{\frac{1}{2}} qF\hbar}\right] = exp\left(-\frac{E_g}{4\underline{E}}\right) \tag{9.55}$$

$$\underline{E} = \frac{(2)^{\frac{1}{2}} qF\hbar}{2\pi m^{*\frac{1}{2}}(E_g)^{\frac{1}{2}}} \tag{9.56}$$

P_0 has a meaning of the tunneling probability with a zero perpendicular (to x-direction) momentum. $\underline{E}$ is a measure of significance of perpendicular momentum range in Eq. (9.54). In other words, if $\underline{E}$ is small, the only electrons that can tunnel through the bandgap barrier are the electrons with perpendicular momentum near zero. Typically $\underline{E}$ is in a range of 5-100 meV. The effective tunneling mass is defined here, as in Ref. [71]:

$$m^* = 2m_c m_v/(m_c + m_v) \tag{9.57}$$

9.3.2 Application to reverse biased p-n junction diode

Next formula for band-to-band tunneling will be applied to calculations of a tunneling current density for a reverse biased n^+ p^+ junction. Following derrivations in [71], consider the situation of a reverse biase p-n junction as in Fig. 9.9.

Let dE be an energy increment corresponding to incremental dx, then: $dE = qFdx$. If electron tunnels across the junction at energy E , the perpendicular motion kinetic

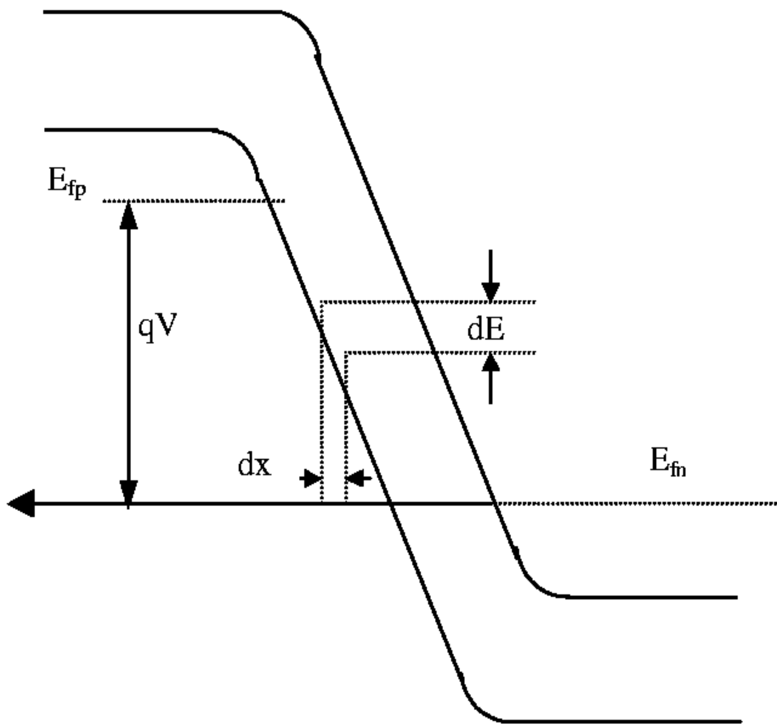


Figure 9.9: Schematic of a reverse biased Zener diode.

energy must obey $E_\perp < E$. The incident flux per unit volume with perpendicular momentum ranging from $k_\perp$ to $(k_\perp + dk_\perp)$ is:

$$\Phi = (q^2F/\hbar)\cdot(1/4\pi)^3\cdot(2\pi k_\perp dk_\perp)\cdot f_E(E) \tag{9.58}$$

where $f_E(E)$ is a Fermi function. Using the relation $E_\perp = h^2\mathbf{k}_\perp{}^2/2m^*$ we can obtain:

$$\Phi = q^2Fm^*/(2\pi^2\hbar^3)f_E(E)dE_\perp \tag{9.59}$$

The tunneling flux must include also the necessity of final electron state being unoccupied, thus has to be multiplied by $(1 - f'_E(E))$ factor. As a result, the differential current density of a tunneling flow becomes:

$$dI/A = q^2Fm^*/(2\pi^2\hbar^3)P_0exp(-E_\perp/\underline{E})[f_E(E) - f'_E(E)]dE_\perp dx, \tag{9.60}$$

where dI/A = (incident flux) × (probability of tunneling) × (volume).

We can also modify the Eq. (9.60) using electrostatic force qF = dE/dx thus :

$$dI/A = qm^*/(2\pi^2\hbar^3)P_0exp(-E_\perp/\underline{E})[f_E(E) - f'_E(E)]dE_\perp dE \tag{9.61}$$

This current needs to be integrated with respect to $E_\perp$ in the limits of $0 < E_\perp < E_{max}$:

$$E_{max} = min(E_{vp} - E), (E_{cn} - E) \tag{9.62}$$

However, we simplify the integration of equation (9.61) by approximation of E_{max} to become an infinity, which is justified as long as $E \gg \underline{E}$.

As typically $\underline{E} \approx$ 5-50 meV, the contribution to the tunneling process of those electrons with energy $E \approx \underline{E}$ can be neglected due to the small magnitude of an electric field near the edges of depletion layer of reverse biased p-n junction (See Fig. 9.9). Thus integration of equation (9.61) gives:

$$dI/A = qm^*/(2\pi^2\hbar^3)P_0\underline{E}[f_E(E) - f'_E(E)]dE \tag{9.63}$$

or using dE = qF dx,

$$dI/A = q^2Fm^*/(2\pi^2\hbar^3)P_0\underline{E}[f_E(E) - f'_E(E)]dx \tag{9.64}$$

or approximately:

$$dI/A = q^2Fm^*/(2\pi^2\hbar^3)P_0\underline{E}dx \tag{9.65}$$

where we used $[f_E(E) - f'_E(E)] = 1$ for the range of reverse p-n junction.

Equation (9.65) can be rewritten in terms of e-h pair generation rate (in units: $m^{-3}s^{-1}$):

$$G_{Zen}(F) = dI/A/(qdx) = qFm^*/(2\pi^2\hbar^3)P_0\underline{E} \tag{9.66}$$

where

$$P_0 = exp\left[\frac{\pi m^{*\frac{1}{2}}(E_g)^{3/2}}{2(2)^{\frac{1}{2}}qF\hbar}\right] = exp\left(-\frac{E_g}{4\underline{E}}\right) \tag{9.67}$$

and

$$\underline{E} = \frac{(2)^{\frac{1}{2}}qF\hbar}{2\pi m^{*\frac{1}{2}}(E_g)^{\frac{1}{2}}} \tag{9.68}$$

This expression for a local field dependent generation rate can be subsequently substituted into the drift diffusion equation solver as a carrier geneartion term which can be activated using the command **zener**.

9.3.3 Non-local model and forward biased tunnel junction

The model we developed in previous subsections were converted into volume integral over the whole junction area for each mesh point, making it a local carrier generation term in the drift-diffusion equation. We find that such an implementation may suffer from numerical non-convergence near zero bias since the generation term does not have a zero-current offset mechanism. Basically, the high field within the junction causes a high generation rate even at zero bias. Without current flow to balance out the carrier generation, the program may run into convergence difficult.

We have developed an alternative and more stable implementation of the TJ model based on a form of direct flow between two reference points on both sides of the TJ. The reference points are of the order of 10 nm over which quantum effects are most pronounced. These points are close enough so that the TJ band profile is not distorted by the external potential profile. They are far enough apart so that the conduction and valence bands can be regarded as being flat. The choice of such a range is similar to our intraband tunneling model where an effective tunneling range must be chosen properly. Such a non-local model has been included in the command **equiv_tunnel_junc** and can be activated by setting the parameter **use_physical_model**.

The formulas for forward biased tunnel junction take the same form as those for the reverse one except that the approximation we made in Eq. (9.61) for infinite integration range for $dE_\perp$ is no longer in use. Furthermore, the Fermi occupancy factor $[f_E(E) - f'_E(E)]$ is much smaller than unity and must be carefully evaluated. The forward biased TJ has also been included in the **equiv_tunnel_junc**.

A note on application of forward TJ is due here. We find that the forward current behavior is strongly affected by potential and current injection conditions surrounding the TJ area. For many cases of non-linear injection condition, the negative resistance (N-shape) expected for forward TJ does not appear. The reason is that non-linear injection (such as through an n-i junction or a Schottky barrier) may overshadow a narrow and weak N-shape peak. To produce the negative resistance behavior from the software, we should first start with a pure TJ without perturbation of external doping profile and contact effects. For example, we should start with a TJ where the n-side and p-side are reasonably uniform in doping and with ohmic contacts on both sides. Such idea conditions surely produce an N-shape in forward I-V curve (with careful small biasing steps) given sufficient carrier population inversion across the junction. Then, we gradually introduce more realistic doping profiles, one step at a time, while maintaining the N-shape at each step.

Chapter 10

FINER POINTS ABOUT QUANTUM WELLS

10.1 Band Offset In Strained Quantum Well

10.1.1 Source of confusion

In a quantum well without strain in the well and barrier, the band offset definition is just the fraction of bandgap discontinuity in the conduction band:

$$O_{ff} = \Delta E_c/(\Delta E_c + \Delta E_v) \tag{10.1}$$

Confusion comes in as soon as HH and LH bands are split by strained in the well and/or barrier in the valence band. There are many ways of defining the band offset in strained MQW system, among them three are explained as follows.

1) Use of unstrained bandgap discontinuity:

$$O_{ff} = \Delta E_c^u/(\Delta E_c^u + \Delta E_v^u) \tag{10.2}$$

where superscript ^{u} is used to denote unstrained quantities (see Fig. 10.1). This convention is used by [13] and is most convenient for theoretical derivations, since the valence band potential can be written as one minus band offset times unstrained bandgap difference, plus terms induced by strain. This band offset can also be related to the work function of the unstrained material.

2) Use of HH valence band:

$$O_{ff} = \Delta E_c/(\Delta E_c + \Delta E_v^{hh}) \tag{10.3}$$

3) The program uses the highest valence band to define the band offset (see Fig.

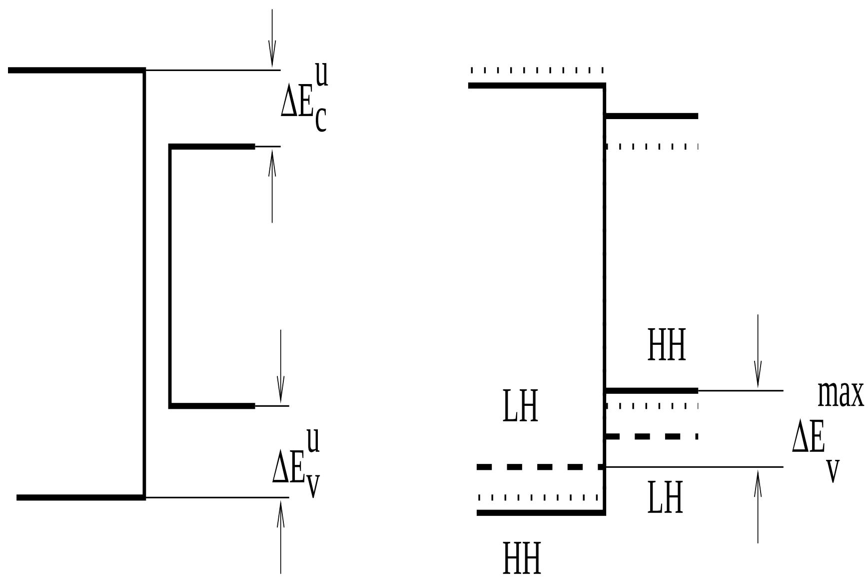


Figure 10.1: Schematics illustrating the definition of band offset.

10.1) as follows:

$$O_{ff} = \Delta E_c / (\Delta E_c + \Delta E_v^{max}) \tag{10.4}$$

where

$$\Delta E_v^{max} = Max(E_v^{hh}, E_v^{lh}) - Max(E_{vb}^{hh}, E_{vb}^{lh}) \tag{10.5}$$

where the subscript $_b$ has been used to denote quantities in the barrier. The advantage of using the highest valence band is that it reflects the hole transport and density of states status since the holes are mostly likely to stay in the highest valence band whenever there is a split of HH and LH bands. Therefore this band offset will most likely reflect experimental band offset values.

10.1.2 Theoretical considerations

To compare band offset values of 1) and 3) above, we consider the bandgap variation when there is strain. We shall call the band offset in 1) above as internal band offset O_{ffi} (a value internally known to the program) while the band offset in 3) is called external band offset O_{ffx} (or simply band offset). For simplicity, we ignore the effects of split-off band and write the bandgap variation as a function of strain as follows:

$$E_g^{hh} \quad = \quad E_g^u + \delta E_{hy} - \delta E_{sh}, \tag{10.6}$$

$$E_g^{lh} \;=\; E_g^u + \delta E_{hy} + \delta E_{sh} \tag{10.7}$$

Please note the both δE_{hy} and δE_{sh} are positive for compressive strain. We calculate the conduction band offset:

$$\Delta E_c = O_{ffi}(E_{gb}^u - E_g^u) - (1-v)\delta E_{hy} + (1-v)\delta E_{hyb} \tag{10.8}$$

$$\Delta E_v = (1 - O_{ffi})(E_{gb}^u - E_g^u) - v\delta E_{hy} + v\delta E_{hyb} + |\delta E_{sh}| - |\delta E_{shb}| \tag{10.9}$$

where v is the valence band partition of the hydrostatic strain term. Please note that by using the absolute value above, we have taken the highest valence band to calculate the valence band discontinuity (convention 3). The offset (external) is related to the internal offset by

$$O_{ffx} = \Delta E_c/(\Delta E_c + \Delta E_v) \tag{10.10}$$

Equations 10.8, 10.9 and 10.10 can be used to convert external and internal band offsets. Let us consider the simple case of unstrained barrier:

$$O_{ffx} = \frac{O_{ffi}(E_{gb}^u - E_g^u) - (1-v)\delta E_{hy}}{(E_{gb}^u - E_g^u) - \delta E_{hy} + |\delta E_{sh}|} \tag{10.11}$$

This can be further arranged as

$$O_{ffx} = O_{ffi} + \frac{O_{ffi}(\delta E_{hy} - |\delta E_{sh}|) - (1-v)\delta E_{hy}}{\Delta E_{gmax}} \tag{10.12}$$

where

$$\Delta E_{gmax} = (E_{gb}^u - E_g^u) - \delta E_{hy} + |\delta E_{sh}| \tag{10.13}$$

which is the maximum bandgap discontinuity taking the highest valence band in the well. Further arrangement:

$$O_{ffx} = O_{ffi} + \frac{-O_{ffi}|\delta E_{sh}| - [(1-v) - O_{ffi}]\delta E_{hy}}{\Delta E_{gmax}} \tag{10.14}$$

According to Ref. [13], $(1-v)$ is taken as $2/3$ and the O_{ffi} is around 0.4. Thus the factor $[(1-v) - O_{ffi}]$ is positive. For InGaAsP system, the second term in the numerator dominates and we expect that for the case of compressive strain (positive δE_{hy}), the band offset (external) is smaller than the internal band offset, while it is larger than the internal band offset for the case of tensile strain.

10.2 Intraband Relaxation Times and Gain Broadening

10.2.1 Introduction

The influence of line shape broadening on optical gain can be described by

$$G(\hbar\omega) = \int_{E'_g}^{\infty} g(E_{cv}) L(\hbar\omega - E_{cv}) dE_{cv} \tag{10.15}$$

here $G(\hbar\omega)$ is the gain coefficient as a function of photon energy $\hbar\omega$. $L(\hbar\omega - E_{cv})$ is the lineshape broadening function characterized by the intraband relaxation. This lineshape function is often represented as

$$L(\hbar\omega - E_{cv}) = \frac{1}{\pi} \frac{\hbar/\tau_{in}}{(\hbar\omega - E_{cv})^2 + (\hbar/\tau_{in})^2}, \tag{10.16}$$

where τ_{in} is called the intraband relaxation time, which is regarded as the reciprocal intraband scattering probability, and $\hbar/\tau_{in}$ is the broadening factor we will study in this section. Previously, we just assume τ_{in} to be a constant of the order of 10^{-13} seconds. The lack of accurate model for this parameter causes significant uncertainty in the optical gain.

Our theory here is limited to the case of intraband relaxation due to carrier-carrier and carrier-LO-phonon scattering in quantum wells. For details, please consult references [64] and [65].

We consider optical transition between subband energy $E_{vik_\parallel}$ in the valence band and $E_{cjk_\parallel}$ in conduction band. The energy is expressed as

$$E_{cik_\parallel} = E_{ci} + \hbar^2 k_\parallel^2 / 2m_c \tag{10.17}$$

$$E_{vjk_\parallel} = E_{vj} + \hbar^2 k_\parallel^2 / 2m_v \tag{10.18}$$

where the (E_{ci} and E_{vj}) are the quantized energy levels, m_c and m_v are the effective masses. $k_\parallel$ is the wave vector parallel to the well interface.

When discussing scattering mechanisms occurring in conduction band and valence band, it is convenient to introduce broadening factors Γ_c and Γ_v, respectively. According to theory of Green's function [64],[65], these are related to the decay rate of amplitude of the wave function. For example, the expectation value of the amplitude of the electron wave is approximated as $\psi_c \propto exp(-\Gamma_c t/\hbar)$ in the conduction band, including thermal and quantum- mechanical statics. Thus, the number of electrons at this energy level decays as $exp(-2\Gamma_c t/\hbar)$. The decay rate in the valence band is obtained similarly. Therefore, the broadening factors are related to the intraband relaxation times (a measure of particle decay probability) in the conduction and valence bands (τ_c and τ_v) by

$$\hbar/\tau_c = 2\Gamma_c(E_{cik_{\|}}) \tag{10.19}$$

$$\hbar/\tau_v = 2\Gamma_v(E_{vjk_{\|}}) \tag{10.20}$$

In optical transition, the dipole moment between electron and hole is proportional to $\psi_c\psi_v^* \propto exp[-(\Gamma_c + \Gamma_v)t/\hbar]$. Thus, the intraband relaxation time determining the spectral broadening, which is the relaxation time of the dipole moment is obtained as

$$\begin{aligned} \hbar/\tau_{in} &= \Gamma_v + \Gamma_c \\ &= (\hbar/\tau_c + \hbar/\tau_v)/2 \end{aligned} \tag{10.21}$$

In the following subsections, broadening factors $\hbar/\tau_c$ and $\hbar/\tau_v$ determined by carrier-carrier and carrier-LO phonon scattering are calculated, which are dominant in lightly-doped compound semiconductors.

10.2.2 Carrier-Carrier Scattering

The intraband relaxation time in the conduction band (τ_c) due to carrier-carrier scattering is calculated from the imaginary part of self energy using the Green's function method[64]. In the present analysis, the calculation is based on second-order self energy. Only the product between two matrix elements with the same the momentum change is taken into account in various terms of the self energy.

In this treatment, $\hbar/\tau_c$ is calculated by Fermi's Golden Rule and the spectral broadening factor for scattering in conduction band is written as

$$\hbar/\tau_c = (2\pi) \sum_{k'_{\|}p'_{\|}} \sum_{ij} \mid V_{cn}(k_{\|}k'_{\|}, ii'jj') \mid^2 \cdot A_c(B_c + C_c) \tag{10.22}$$

with

$$A_c = \delta(E_{cik_{\|}} + E_{njp_{\|}} - E_{ci'k'_{\|}} - E_{nj'p'_{\|}}) \tag{10.23}$$

$$B_c = f_c(E_{ci'k'_{\|}})f_n(E_{nj'p'_{\|}})\left[1 - f_n(E_{njp_{\|}}\right] \tag{10.24}$$

$$C_c = \left[1 - f_c(E_{ci'k'_{\|}})\right]\left[1 - f_n(E_{nj'p'_{\|}})\right] f_n(E_{njp_{\|}}) \tag{10.25}$$

where the suffix n refers to the conduction ($n = c$) or valence ($n = v$) band, i, i', j, and j' are subband numbers, the δ-function represents the energy conservation, and V_{cn} is the matrix element of the interaction given below.

A simple illustration is possible for the above formula. The scattering processes are schematically shown in Figs. 10.2.2(a) and 10.2.2(b), for $n = c$.

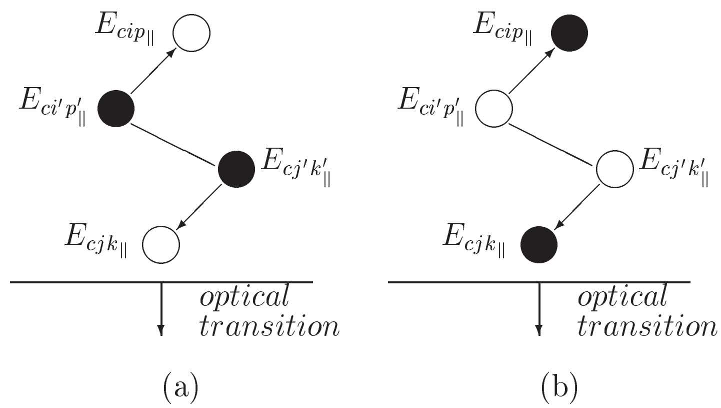


Figure 10.2: Schematic illustration of electron-electron scattering in the conduction band. (a) Formation of a hole at energy $E_{cjk_\parallel}$ under the optical transition is intercepted due to collision of two electrons at $E_{cj'k'_\parallel}$ and $E_{ci'p'_\parallel}$, resulting in spectral broadening. (b) Formation of an electron at energy $E_{cjk_\parallel}$ under the optical transition is scattered due to collision of two holes.

In Fig. 10.2.2(a), two electrons at $E_{ci'p'_\parallel}$ and $E_{cj'k'_\parallel}$ collide with each other, and are scattered into two holes at $E_{cip_\parallel}$ and $E_{cjk_\parallel}$. Formation of a hole at $E_{cjk_\parallel}$ during optical transition is intercepted by this process, resulting in spectral broadening.

In Fig. 10.2.2(b), optical transition of an electron at $E_{cjk_\parallel}$ is intercepted by the collision of two holes at $E_{cj'k'_\parallel}$ and $E_{ci'p'_\parallel}$.

A process for $n = v$ in Eq. (10.22) is similarly given. Electron-electron scattering and electron-hole scattering correspond to $n = c$ and $n = v$, respectively. However, the formula for the spectral broadening factor caused by another scattering electron-LO phonon ($n = LO$) is radically different, which we will discuss in the next subsection.

For the interaction matrix element V_{cn} in (2.16), we use the screened Coulomb potential, which is the result of static and long wavelength limit of the many-body interaction. Deviation from the exact calculation is discussed below

$$V_{cn}(k_\parallel k'_\parallel, ii'jj') = \int\int \psi^*_{ci'k'_\parallel}(r_1)\psi^*_{nj'p'_\parallel}(r_2)\frac{e^2\exp(-\lambda_s r)}{4\pi\epsilon r} \cdot\psi_{cik_\parallel}(r_1)\psi_{njp_\parallel}(r_2)dr_1 dr_2 \tag{10.26}$$

where e is the electron charge, ϵ is the static dielectric constant, $r = | r_1 - r_2 |$, and λ_s is the inverse screening length given below; ψ's are the electron wave functions, which are approximated as follows in the well potential:

$$\psi_{njk_\parallel}(r) \simeq u_{nk}(r)\Phi_{nj}(z)\exp(ik_\parallel \cdot r_\parallel)/\sqrt{S} \tag{10.27}$$

with

$$\Phi_{nj}(z) = \sqrt{2/L_{nj}}\sin(j\pi z/L_{nj}) \tag{10.28}$$

where the z axis is perpendicular to the well interface, S is the interface area of the sample, u is the periodic part of the bulk Bloch function, $k_{\|}$ and $r_{\|}$ are the components of the wave vector and position vector parallel to the interface, respectively, and L_{nj} is the effective well width given by

$$\begin{aligned} L_{nj} &= \pi / k_{nj\perp} \\ &= \pi\hbar/\sqrt{2m_n E_{nj}} \end{aligned} \tag{10.29}$$

where $k_{nj\perp}$ is the z component of the wave vector.

The inverse screening length λ_s is given as follows for the carrier injection case, where electrons and holes simultaneously exist:

$$\begin{aligned} \lambda_s^2 &= -\frac{e^2}{\epsilon}\left\{\int_{E_{c1}}^{\infty}\frac{\partial f_c}{\partial E_c}g_c dE_c + \int_{E_{v1}}^{\infty}\frac{\partial f_v}{\partial E_v}g_v dE_v\right\} \\ &= \frac{e^2}{\pi\hbar^2\epsilon}\sum_j\{m_c f_c(E_{cj})/L_{cj} + m_v f_v(E_{vj})/L_{vj}\} \end{aligned} \tag{10.30}$$

where g_c and g_v are the step-like density-of-states of the conduction and valence bands, respectively. We sum over all the subbands and both heavy hole and light hole.

Substituting Eq. (10.27) into Eq. (10.26) the matrix element is calculated as

$$\begin{aligned} V_{cn}(k_{\|}k'_{\|}, ii'jj') &= \frac{e^2}{2\epsilon S}\frac{\delta(k_{\|}-k'_{\|}, p'_{\|}-p_{\|})}{\sqrt{g_{\|}^2+\lambda_s^2}} \\ &\cdot \int\int \Phi^*_{ci'}(z_1)\Phi_{ci}(z_1)\Phi^*_{nj'}(z_2)\Phi_{nj}(z_2) \\ &\cdot \exp(- \mid z_1 - z_2 \mid \sqrt{g_{\|}^2+\lambda_s^2}) dz_1 dz_2 \end{aligned} \tag{10.31}$$

where the δ-notation represents the momentum conservation within the plane parallel to the interface, and $g_{\|} = k_{\|} - k'_{\|}$.

Generally, the possibility of scatterings in the same subband is much higher than in the different subbands, so we only consider one subband in either the conduction or valence bands ($i' = i, j = j'$). The absolute square of the matrix element is calculated as

$$\begin{aligned} \mid V_{cn}(k_{\|}k'_{\|}, iij'j') \mid^2 &= (e^2/2\epsilon S L_e)^2 \\ &\cdot \delta(k_{\|}-k'_{\|}, p'_{\|}-p_{\|}) T(g_{\|}, k_{ci\perp}, k_{nj'\perp}) \end{aligned} \tag{10.32}$$

where L_e is the minimum of the effective well widths of the four wave functions in (10.31), and

$$T(g_{\|}, k_{ci\perp}, k_{nj'\perp}) = \left[\frac{2}{\alpha} + \frac{\delta(k_{ci\perp}, k_{nj'\perp})}{\alpha + 4k_{ci\perp}^2} - \frac{2}{L_e\sqrt{\alpha}}A_t(B_t + C_t)\right]^2 \tag{10.33}$$

with

$$A_t = 1 - \exp(-L_e\sqrt{\alpha}) \tag{10.34}$$

$$B_t = \frac{1}{\alpha} + \frac{\alpha + 4k_{ci\perp} + 4k^2_{nj'\perp}}{(\alpha + 4k^2_{ci\perp})(\alpha + 4k^2_{nj'\perp}})) \tag{10.35}$$

$$C_t = \frac{L_e k_{ci\perp}(L_e k_{ci\perp} - \pi)(\alpha - 4k^2_{nj'\perp})}{4(\alpha + 4k^2_{ci\perp})(\alpha + 4k^2_{nj'\perp})\left[\exp(L_e\sqrt{\alpha}) - 1\right]} \tag{10.36}$$

$$\alpha = g^2_{\|} + \lambda^2_s \tag{10.37}$$

The first and second terms within the square bracket of Eq. (10.33) are essentially the same as those in bulk semiconductors. These terms arise from the forward and backward waves in the standing wave along the z axis given by Eq. (10.28), the third term expressed by Eqs. (10.34)- (10.36), which approaches zero at the limit $L \to \infty$, is peculiar to the quantum-well structures, and results from the localization of the wave function. Equation (10.33) coincides with that for the perfect two-dimensional case at the limit $L \to 0$ due to the existence of the third term.

Replacing the summation for $k'_{\|}$ and $p'_{\|}$ in with the integral for the wave vector parallel to the well interface, the broadening factor $\hbar/\tau_{cn}$ is given as follows:

$$\hbar/\tau_{cn} = \frac{m_c e^4}{8\pi^5\hbar^2\epsilon^2}(\frac{m_n}{m_c})^2 \int_0^{2\pi} d\phi \int_0^{\infty} dk'_{\|} \int_0^{\infty} du A_\tau(B_\tau + C_\tau) \tag{10.38}$$

with

$$A_\tau = T(g_{\|}, k_{ci\perp}, k_{nj'\perp})(k^2_{ci\perp}k'_{\|} \mid \beta \mid / g^2_{\|}) \tag{10.39}$$

$$B_\tau = f_c(E_{cik'_{\|}}) f_n(E_{nj'p'_{\|}}) \left[1 - f_n(E_{nj'p_{\|}})\right] \tag{10.40}$$

$$C_\tau = \left[1 - f_c(E_{cik'_{\|}})\right]\left[(1 - f_n(E_{nj'p'_{\|}})\right] f_n(E_{nj'p_{\|}}) \tag{10.41}$$

and

$$g^2_{\|} = k^2_{\|} + k'^2_{\|} - 2k_{\|}k'_{\|}\cos\phi \tag{10.42}$$

$$\beta = k'^2_{\|} - k'_{\|}k_{\|}\cos\phi + (m_c/m_n - 1)g^2_{\|}/2 \tag{10.43}$$

$$E_{cik'_{\|}} = (\hbar^2/2m_c)(k'^2_{\|} + k^2_{ci\perp}) \tag{10.44}$$

$$\begin{aligned} E_{nj'p'_{\|}} &= (\hbar^2/2m_n)[(m_n/m^2_c(u^2+1) \\ &\quad + g^2_{\|} - 2(m_n/m_c)\beta + k^2_{ci\perp}] \end{aligned} \tag{10.45}$$

$$E_{nj'p_{\|}} = E_{cik'_{\|}} + E_{nj'p'_{\|}} - E_{cik_{\|}} \tag{10.46}$$

where the parabolic band structure is assumed, and the summation with respect to n corresponds to electron-electron scattering and electron-hole scattering for $n = c$ and $n = v$, respectively.

In the same manner, carrier-carrier scattering also occurs for holes in the valence band, corresponding to the processes in the conduction band shown in Figure 10.2.2. The spectral broadening factor for the valence band $\hbar/\tau_v$ is calculated by the same procedure as above, and obtained by replacing the suffix c with v.

10.2.3 Carrier-Phonon Scattering

The spectral broadening factor in the conduction band due to carrier-LO phonon scattering $\hbar/\tau_{cLO}$ is calculated in a manner similar to carrier-carrier scattering as

$$\hbar/\tau_{cLO} = (2\pi)\sum_{k'_\parallel}\sum_{j'} \mid V_{cLO}(k_\parallel k'_\parallel, jj') \mid^2 (A_q + B_q) \tag{10.47}$$

$$A_q = \delta(E_{cjk_\parallel} - E_{cj'k'_\parallel} \pm \hbar\omega_{LO}) f_c(E_{cj'k'_\parallel}) \tag{10.48}$$

$$B_q = \delta(E_{cj'k'_\parallel} - E_{cjk_\parallel} \pm \hbar\omega_{LO}) \left[1 - f_c(E_{cj'k'_\parallel})\right] \tag{10.49}$$

where $\hbar\omega_{LO}$ is the phonon energy, $+\hbar\omega_{LO}$ and $-\hbar\omega_{LO}$ correspond to emission and absorption of phonons, respectively, and V_{cLO} is the matrix element of the carrier-LO phonon scattering, and is given below.

The scattering processes given by (10.48) and (10.49) are schematically shown in Figures 10.2.3(a) and 10.2.3(b), respectively.

Assuming the electron wave function of (10.27), the square of the matrix element $\mid V_{cLO}(k_\parallel k'_\parallel, jj') \mid^2$ is given as follows, taking into account the screening effect:

$$\mid V_{cLO}(k_\parallel k'_\parallel, jj') \mid^2 = \sum_q (e^2\hbar\omega_{LO}/2V)(1/\epsilon_\infty - 1/\epsilon) C_q \cdot D_q \tag{10.50}$$

with

$$C_q = \left[q/(q^2+\lambda_s^2)\right]^2 \left\{ \begin{array}{c} n_q+1 \\ n_q \end{array} \right\} \delta(k_\parallel - k'_\parallel, q_\parallel) \tag{10.51}$$

$$D_q = \mid \int \Phi^*_{cj}(z)\Phi_{cj'}(z) e^{-iq_\perp z} dz \mid^2 \tag{10.52}$$

where q, $q_\parallel$, and $q_\perp$ are the phonon wave vector and its components parallel and perpendicular to the well interface, respectively. ϵ and ϵ_∞ are the static and optical dielectric constants, respectively, V is the volume of the system, and the factors $n_q + 1$ and n_q correspond to the emission and absorption of phonons, respectively, where n_q is the phonon number per mode q given by

$$n_q = 1/(\exp(\hbar\omega_{LO}/kT) - 1. \tag{10.53}$$

The intensity of the phonon mode confined in the well is assumed to be much smaller than that of the bulk mode, and is neglected in (10.50)-(10.52).

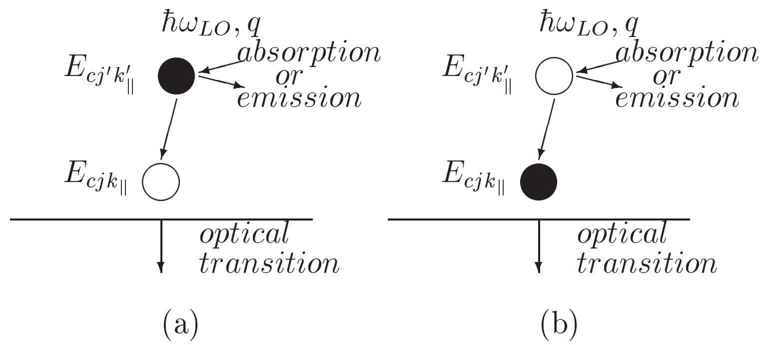


Figure 10.3: Schematic illustration of electron-LO phonon scattering in the conduction band. (a) Formation of a hole at energy $E_{cjk_{\parallel}}$ under the optical transition is intercepted by an electron at $E_{cj'k'_{\parallel}}$ which emits or absorbs a phonon. (b) Formation of an electron at energy $E_{cjk_{\parallel}}$ under the optical transition is scattered by a hole $E_{cj'k'_{\parallel}}$ which emits or absorbs a phonon.

Replacing the summation for $k'_{\parallel}$ and q in (10.47)-(10.52) with the integral for the wave vector, $\hbar/\tau_{cLO}$ is calculated for one subband case ($j = j' = i$) as

$$\hbar/\tau_{cLO} = (m_c e^2/2\pi^2\hbar^2)(1/\epsilon_\infty - 1/\epsilon)\hbar\omega_{LO}(A_o + B_o) \tag{10.54}$$

with

$$A_o = \left[n_q + 1 - f_c(E_{cik_{\parallel}} - \hbar\omega_{LO})\right] \cdot \int_0^{\phi_{max}} \frac{(I(a_1) + I(a_2))d\phi}{\sqrt{k_{\parallel}^2 \cos^2\phi - 2m_c\omega_{LO}/\hbar}} \tag{10.55}$$

$$B_o = \left[n_q + f_c(E_{cik_{\parallel}} + \hbar\omega_{LO})\right] \cdot \int_0^{\pi} \frac{I(a_3)d\phi}{\sqrt{k_{\parallel}^2 \cos^2\phi + 2m_c\omega_{LO}/\hbar}} \tag{10.56}$$

and

$$\phi_{max} = \cos^{-1}\left(\sqrt{2mc\omega_{LO}/\hbar k_{\parallel}^2}\right) \tag{10.57}$$

$$I(a) = \int_0^\infty \frac{ab(b^2x^2 + a^2)\sin^2 x dx}{(b^2x^2 + a^2 + 1)^2 x^2 (x^2/\pi^2 - 1)^2} \tag{10.58}$$

$$a_1 = \gamma - \sqrt{\gamma^2 - \delta^2} \tag{10.59}$$

$$a_2 = \gamma + \sqrt{\gamma^2 - \delta^2} \tag{10.60}$$

$$a_3 = \gamma + \sqrt{\gamma^2 + \delta^2} \tag{10.61}$$

$$b = (k_{\|}/\lambda_s)\cos\phi \tag{10.62}$$

$$\delta = \sqrt{2m_c\omega_{LO}/\hbar\lambda_s^2} \tag{10.63}$$

The process of numerical calculation is similar to that for carrier-carrier scattering. In the same manner, carrier-phonon scattering also occur for holes in the valence band corresponding to the process in the conduction band shown in Figure 10.2.3. The spectral broadening factor in the valence band caused by the hole-LO phonon scattering is obtained by replacing the suffix c with v in the above equations. At the same time, the subband suffix together with the suffix v should be j, other than i.

Equations (10.38) and (10.54) are two basic formulas for calculating the broadening factors with respect to the carrier-carrier and carrier-phonon scatterings, respectively.

Please note that two new parameters comes into the carrier-phonon scattering. One is the optical dielectric constant ϵ_∞ and the other is the longitudinal optical phonon energy $\hbar\omega - LO$. As an indication, we use $\hbar\omega_{LO} = 34.E - 3$ eV and $\epsilon_\infty = 11.4$ for InGaAsP/InP system, and $\hbar\omega_{LO} = 46.E - 3$ eV and $\epsilon_\infty = 10.4$ for GaAs/AlGaAs system. For accurate setting of these two parameters, please use the **tau_model** statement.

10.2.4 Nomenclature

We list the symbol definitions separately here due to the large number of new symbols in this section.

$D_c(E), D_v(E)$	Energy spectra of electron and hole, which have energy E.
E_{cj}, E_{vj}	Quantized energy levels in conduction and valence bands.
$E_{cjk_\parallel}, E_{vjk_\parallel}$	Energies of electron and hole under the optical transition at the subband j.
E, E_{cv}	Photon energy and energy difference between conduction and valence bands under the optical transition.
E_{fc}, E_{fv}	Quasi-Fermi levels of conduction and valence bands.
E_g	Bulk bandgap energy of semiconductor.
e	Electronic charge.
$\epsilon, \epsilon_\infty, \epsilon_o$	Static dielectric constant and optical dielectric constant and dielectric constant of conductor in vacuum.
$f_c(), f_v()$	Fermi distribution functions in conduction and valence bands.
g_c, g_v, g_{cv}	Step-like-densities-of-states of electron (c) and hole (v) and the step-like-density-of-states electron-hole pair.
$\Gamma_c, \Gamma_v, \Gamma_{in}$	Half broadening factors in conduction (c) and valence (v) bands, and average broadening factor, their relationship is $\Gamma_{in} = \Gamma_c + \Gamma_v$.
$\hbar$	Plank constant.
k	Boltzmann constant.
$k_\parallel, k_\perp$	Wave vectors parallel and perpendicular to the well interface.
$k_{f_\parallel}$	Component of the Fermi wave vector parallel to the well interface.
$L()$	Lorentzian line shape function for optical gain broadening.
L_w	Well width.

L_{cj}, L_{vj}, L_e	Effective well widths at the jth levels in conduction (c) and valence (v) bands, and the minimum of the four effective well widths.
λ_s	Inverse screening length.
m_0	Free electron mass.
m_c, m_v, m_{cv}	Effective masses of electron (c) and hole (v) and reduced effective mass between electron and hole, and their relationship is $1/m_{cv} = 1/m_c + 1/m_v$.
μ_o	Permeability of semiconductor of the vacuum.
n_r	Refractive index of semiconductor.
N	Injected electron density.
P	Injected hole density.
ω, ω_{LO}	Angular frequencies of light wave and longitudinal optical (LO) phonon wave.
$\psi' s$	Electron wave functions.
$\Phi' s$	Complex amplitudes of the electron wave functions.
$r, r_{\parallel}$	Position vector and the element of position vector parallel to well interface.
$< R_{cv}^2(E_{cv}) >$	Averaged square of electric dipole moment.
S	Interface area of sample.
T	Absolute temperature.
$\tau_c, \tau_v, \tau_{in}$	Intraband relaxation times in the conduction (c) and valence (v) bands and the total intraband relaxation time (in), and their relationship is $2/\tau_{in} = 1/\tau_c + 1/\tau_v$.
$\hbar/\tau_c, \hbar/\tau_v, \hbar/\tau_{in}$	Broadening factors caused by carrier-carrier and carrier-LO phonon scatterings in conduction (c) and valence (v) bands, and the total broadening factor including summation over the conduction and the valence bands.
u	Periodic part of the bulk Bloch function.
V_{cn}	Matrix elements of the interaction in conduction (n=c) and valence (n=v) bands.
V_{in}	Injection potential. Obviously eV_{in} is injected carrier energy.

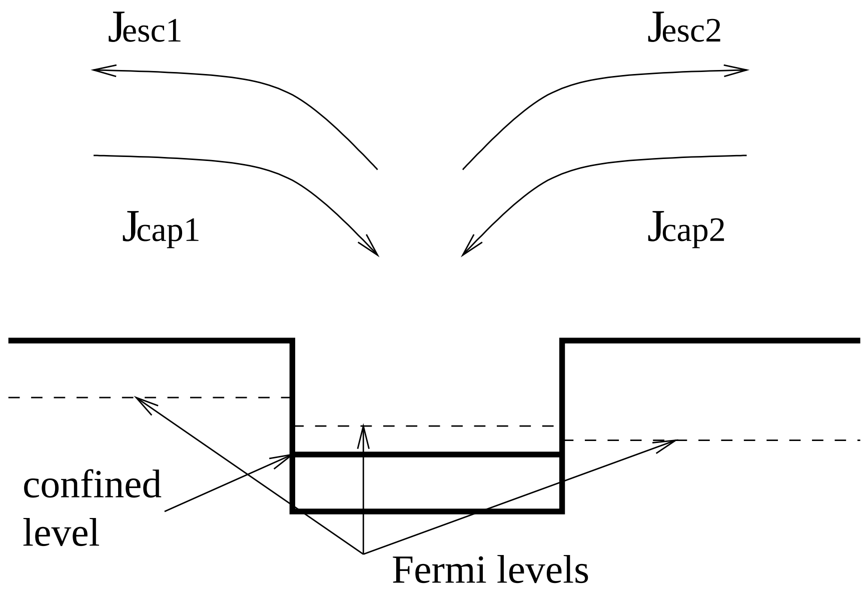


Figure 10.4: Schematics of capture and escape currents in real coordinate space.

10.3 Quantum Well Capture and Escape Processes

10.3.1 Carrier Capture and Escape in Real Space

As described in previous sections, the carrier transport across a heterojunction is via thermionic emission. For a quantum well, the carriers have to go across two heterojunctions (or two interfaces) via the same thermionic emission mechanism.

Some researches prefer to describe carrier transport across a quantum well in terms of "carrier capture" and "carrier escape" [79]. We shall also describe the carrier transport across a quantum well in the same technical terms.

In schematic in Fig. 10.4, we describe the events in real space (or in spatial coordinates). We denote the current components in term of capture and escape processes. In thermionic theory, the capture components I_{cap1} and I_{cap2} do not experience any scattering by the quantum well interfaces (in semi-classical approximation) and can be written as

$$J_{cap1} = \gamma_{12n}\overline{v}_{1n}^{therm} n_1, \tag{10.64}$$

$$J_{cap2} = \gamma_{23n}\overline{v}_{3n}^{therm} n_3, \tag{10.65}$$

where $\overline{v}_{1n}^{therm}$ is the thermal velocity for carriers in region 1 (left barrier) and n_1 is the carrier density there. γ_{12n} is a correction factor due to the semi-classical approximation (for neglecting quantum well potential scattering). Similar definition applies to the right barrier.

The computation of escape components are less straight forward because the carriers with energies less than the barrier are strongly reflected by the barrier. However,

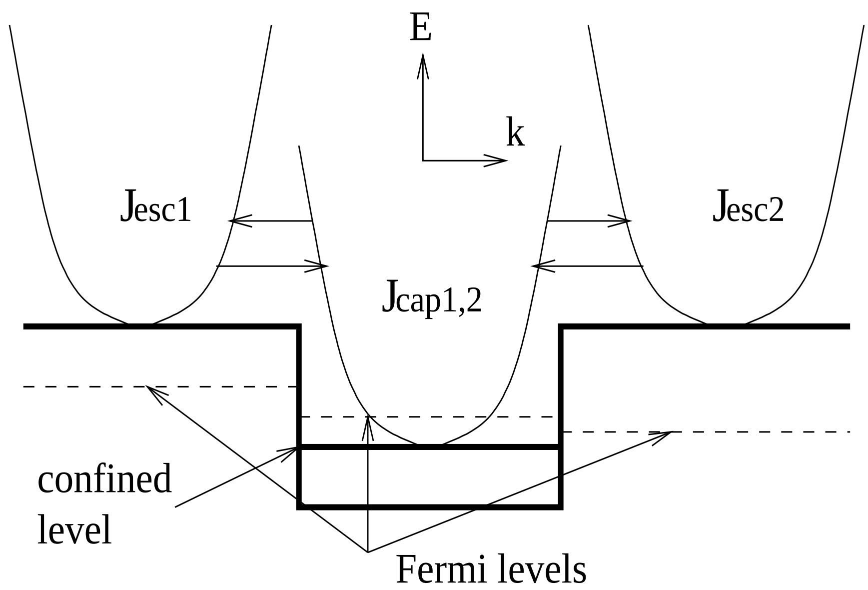


Figure 10.5: Schematics of capture and escape currents in energy-wave vector coordinate space assuming elastic scattering.

we make the observation that the escape component must balance out the capture component if Fermi levels on both sides are equal (such is the case in equilibrium). Thus, the escape components can be written as:

$$J_{esc1} = \gamma_{12n} \overline{v}_{1n}^{therm} n_{10}, \tag{10.66}$$

$$J_{esc2} = \gamma_{23n} \overline{v}_{3n}^{therm} n_{30}, \tag{10.67}$$

where n_{10} is the carrier density on the left barrier with Fermi level set to equal to that in the well (region 2). Similar arguments apply to J_{esc2}.

10.3.2 Carrier Capture and Escape in k-Space

The capture/escape event in the thermionic emission model above can also be described in energy space. The explanation of thermionic emission capture/escape model within the frame work of quasi-equilibrium (i.e., use of quasi-Fermi levels) is as follows. A carrier transfers with the same kinetic energy (elastic process) between confined and unconfined states with thermal velocity. After the transfer, it is instantly thermalized with local temperature and becomes part of the local carrier distribution described by the a quasi-Fermi level.

Similar to Ref. [79] we assume that transition of carriers between unconfined states and confined states have the same initial and final energies (see Fig. 10.5). The capture current can be written as follows in the notation of Ref [79].

$$J_{cap} \propto \int_{E_c^b}^{\infty} \rho_u(E) f(E, E_{fn}^u) \rho_c(E)[1 - f(E, E_{fn}^c)] dE \tag{10.68}$$

where u and c are for unconfined and confined states. E_c^b and E_{fn} are the energy at the barrier and Fermi level, respectively. $\rho(E)$ is the density of states.

Usually, the Fermi level is below barrier and we can safely set

$$[1 - f(E, E_{fn}^c)] \approx 1 \tag{10.69}$$

We also know that 2D density of states $\rho_c(E)$ is a constant in E. Thus the capture current can be written as

$$J_{cap} \propto \int_{E_c^b}^{\infty} \rho_u(E) f(E, E_{fn}^u) dE = n_u \tag{10.70}$$

which is just the unconfined carrier density at the barrier. This is consistent with Equations 10.64 and 10.65.

Once we have the expressions for the capture components, the escape components can be obtained using the argument that the total current flow must be zero if Fermi levels are flat.

In conclusion, the carrier capture/escape model in our simulator is that of elastic carrier scattering process followed by instant local thermalization which has been reported to agree with experiment (see Ref. [79]).

10.4 Dynamic Behavior of Quantum Capture and Escape

It is generally believed that the dynamic behavior (i.e., high speed modulation characteristics) of a laser diode depends on how fast carriers can be captured from unconfined states into confined states of the quantum well [87]. In a microscopic theory of quantum capture, energy distributions in both energy and space can be solved to give a complete description of the capturing process due to carrier-phonon interactions. The capturing process can be described by the following characteristic capture time τ_{cap} [**?**]:

$$\frac{1}{\tau_{cap}} = \int_{E_{final}} S_{3D,2D} \frac{g_{2D}}{L_{norm}} dE_{2D} \tag{10.71}$$

where E_{2D} is the 2D energy, E_{final} the energy of the final states, $S_{3D,2D}$ the scattering probability between 3D and 2D states, g_{2D} the density of state in the well, and L_{norm} the normalization distance of the confined wave function. Detailed carrier-photon interaction must be taken into account to evaluate $S_{3D,2D}$ above. Solving the microscopic energy-space equations also produce the spectral hole burning effect which were thought to further limit the modulation band width.

We consider the dynamic effect associated with the quantum capture within the thermionic capture/escape model. We show that it is possible to compare the phenomenological carrier capturing theory in our simulation software with the more complex microscopic energy-space equation model. We shall derive the characteristic capturing time as follows.

Consider a simplified version of our carrier transport model for electrons in a quantum well. The simplified continuity equation with thermionic emission can be written as

$$\frac{dN_w}{dt} = \gamma \frac{v_{th}^{(b)}}{d_w} N_b - \gamma \frac{v_{th}^{(w)}}{d_w} N_w - \frac{N_w}{\tau_a} \tag{10.72}$$

where N_w and N_b are electron concentrations in the well and at the barrier, respectively. d_w is the quantum well width. $v_{th}^{(w)}$ and $v_{th}^{(b)}$ are the mass-dependent thermionic velocities in the well and at the barrier, respectively. τ_a is the interband recombination life time due to radiative, stimulated, SRH, and Auger recombination effects. γ is the heterojunction capture coefficient used to modify the thermal velocity.

Suppose we apply a pulse to the injection current which causes the barrier concentration N_b to behave like a step function. We wish to see how the quantum well carriers respond. Solving Eqn. (10.72) shows that the carriers in the well undergo the following transient behavior:

$$exp\{-[\gamma v_{th}^{(w)}/d_w + 1/\tau_a]t\} \tag{10.73}$$

Since the $1/\tau_a$ term is about five orders smaller and can be ignored, we find that the characteristic capturing time in our simulator can be expressed simply as

$$\frac{1}{\tau_{cap}} = \frac{\gamma v_{th}^{(w)}}{d_w} \tag{10.74}$$

Comparison of Equations (10.74) and (10.71) leads us to conclude that the the heterojunction capture coefficient γ is proportional to the microscopic scattering probability between 3D and 2D states, and thus can be used as a convenient parameter to measure the deviation of our thermionic model from the more complex microscopic energy-space model. The theoretical analysis of hole capturing is completely similar and shall be omitted here.

We consider a numerical example of idea thermionic emission with $\gamma = 1$. For a GaAs quantum well of 100 $\mathring{A}$, we find that the electron and hole capturing times are 0.1 and 0.3 pico seconds, respectively. These values seem to be reasonable compared to other estimates (see for example, [87]). As we pointed out in the last subsection that thermionic emission model involves the assumption of instant local thermalization, it is no surprise that these values are on the fast side. The actual capturing process should be smaller (with $\gamma < 1$).

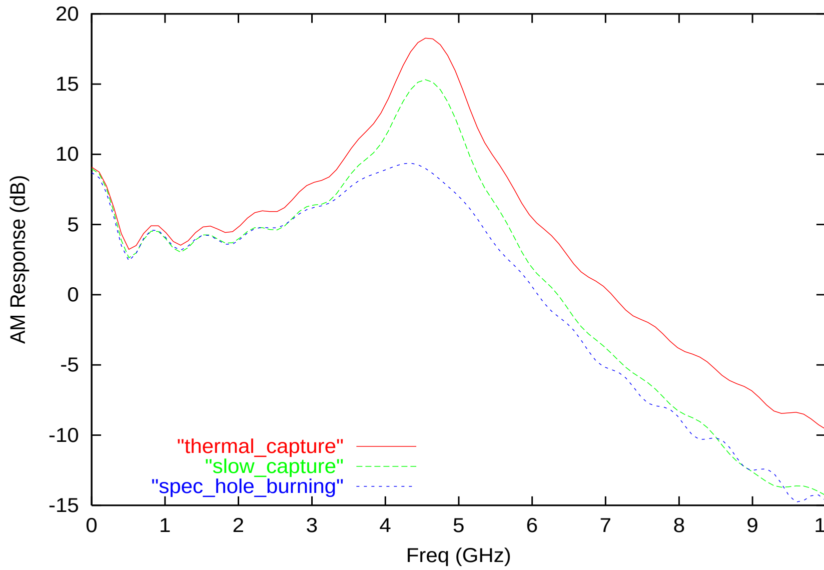


Figure 10.6: Numerical simulation of the modulation response of a 76 ÅSQW GaAs/AlGaAs laser diode. The capturing speed is reduced by a factor of 0.01 and the non-linear gain constant is set to $5 \times 10^{-23}\ m^3$.

Having resolved the carrier-capturing dynamics, we consider the spectral hole burning effect. In our simulator, this is expressed as a factor multiplied to the material optical gain:

$$g(\omega, S) = \frac{g_0(\omega)}{1 + \epsilon_{nl} S} \tag{10.75}$$

where S is the bulk photon density and ϵ_{nl} is an empirical constant also called non-linear gain saturation constant.

To show that both the capturing and spectral hole burning effects are important for the modulation band width, we use the Fourier transform of a small-step transient simulation to produce the modulation response in Figure 10.6. We find that spectral hole burning has similar effect as a slower capturing time.

10.4.1 Direct Flying-over Transport Model

When quantum wells (QW) or quantum dots (QD) are much smaller than the mean free path of the carriers (typically several nm), there is large chance the carriers directly fly over the QW or QD before being thermalized with carriers inside them. Given the mean free path n for electron estimated from the conductivity (Drude model) near the QW/QD, we can write the probability of flying-over the QW/QD as $exp(-D_{qw}/\lambda_n)$. This model may be enabled using the **q_transport** command. Similar models can be enabled for holes.

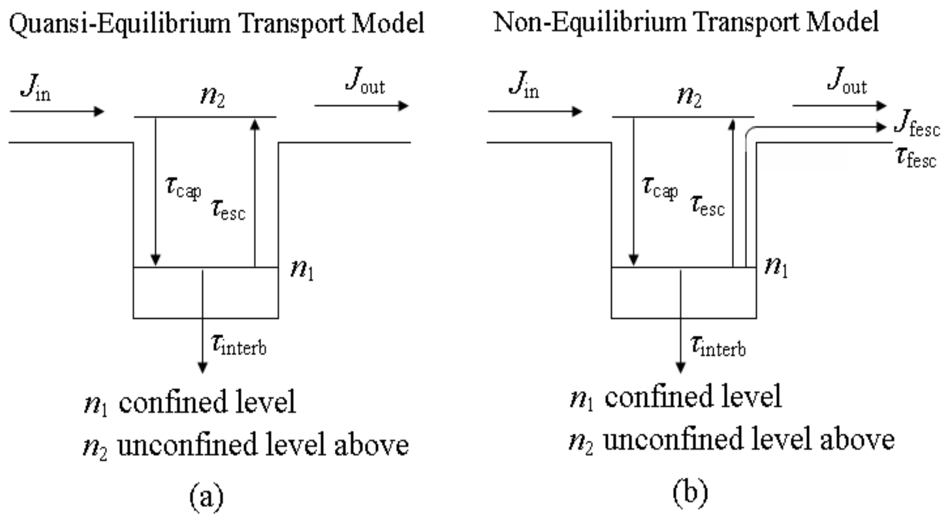


Figure 10.7: Schematic of Quansi- and Non-equilibrium transport model

10.4.2 Fractional Density Quantum Well/Dot Escape Theory

Crosslight also implements a new non-equilibrium carrier escape scheme to describe the phenomenon that the injected carriers escape from the quantum dots (QDs) or the quantum wells (QWs) before being thermalized with the local IMREF. A fraction of total carrier density in QWs or QDs is used to represent the density of hot carriers escaping from QWs or QDs. The theory is based on Ref.[130].

Assuming that the carrier capture/escape processes in QWs or QDs can be described by the following two-level quantum capture/escape model in the traditional quansi-equilibrium transport theory as shown in Fig.10.7(a), where n_1, n_2 are the density of carriers on the two levels. J_{in}, J_{out} are the injection and outflow density of carriers. τ_{cap}, τ_{esc}, τ_{interb} are the carrier lifetime of capture by the confined level, escape and interband recombination from the confined level, respectively. Then, the rate equation of carriers in QWs or QDs in the quasi-equilibrium transport model is

$$\frac{dn_t}{dt} = -\frac{n_t}{\tau_{interb}} + J_{in} + J_{out} \tag{10.76}$$

where $n_t = n_1 + n_2$. In our non-equilibrium carrier escape scheme, a fraction of the confined carriers are free to go directly via average conductivity with average local velocity: V_{fesc}. Then the carrier transport process can be shown as in Fig.10.7(b), the rate equation of carriers in QWs or QDs is

$$\frac{dn_t}{dt} = -\frac{n_t}{\tau_{interb}} + J_{in} + J_{out} - J_{fesc} \tag{10.77}$$

where J_{fesc} is the carrier density escaping from wells or dots before being thermalized, given by

$$J_{fesc} = \frac{F_{esc} n_t V_{fesc}}{d} \tag{10.78}$$

where F_{esc} is the fraction of total carrier density escaping from the wells or dots. d is the thickness of the wells or dots.

According to Ref.[131], the lifetime of carriers escaping before being thermalized can be determined by a similar form:

$$\tau_{fesc} = \frac{d}{V_{fesc} F_{esc}} \tag{10.79}$$

where $V_{fesc} = 2\pi\hbar/dm^*$, m^* is the effective mass. If $F_{esc} = 0.1$, $d = 3nm$, $m^* = 0.1$, then $\tau_{fesc} = 12fs$, which is close to the quantum well tunneling escape time as shown in Ref.[131]. This means that our fractional density escape scheme is equivalent to the traditional time constant escape scheme.

Chapter 11

THERMAL EFFECTS IN SEMICONDUCTORS

This chapter deals with thermal effects in semiconductors. As we will show that a variety of heating sources exist in semiconductors. It is critical that these sources are accurately formulated when modeling devices where self-heating is important. Examples of such devices are high power transistors and high power laser diodes.

11.1 Isothermal temperature variation

Crosslight software allows the user to vary the isothermal temperature. The following material parameters are temperature dependent: bandgap, mobility, nonradiative recombination, carrier distribution, *etc.* the program allows the user to incorporate all possible temperature variations into a device simulation via uder-defined material macros.

11.2 Heat flow and temperature distribution

11.2.1 Expression of heat flux

It is well known that the heating effect is very important for semiconductor lasers in almost all applications. The heat generated by a semiconductor laser often forces the designer to include an additional cooling system and therefore increase the cost of the application. The heating effect is more important for high power semiconductor lasers where the device temperature determines the maximum power.

From the point of view of simulation and modeling, the concern is two-fold. First, we must find out the temperature distribution from all possible heat sources. This involves a much larger simulation area than the small region near the p-n junction. We must consider how the heating power flows though the whole substrate as well as from any wire bonds. Secondly, we must consider how the heating affects the laser performance. This means we must accurately evaluate the degradation of power, efficiency, *etc.*, due to the the non-uniform temperature distribution. This is not a trivial task because virtually all variables and material parameters are temperature dependent. Our goal is to provide a thermal modeling environment so that all possible temperature dependences can be taken into account.

We are concerned with the generation and flow of lattice heating power in a semiconductor. We first consider the heat flux and then the heat source.

We introduce the thermal conductivity such that the heating power flux (in Watt/m^2) is given by

$$J_h = -\kappa \nabla T \tag{11.1}$$

Conservation of energy requires that the temperature distribution satisfy the following basic thermal equation:

$$C_p \rho \frac{\partial T}{\partial t} = -\nabla \cdot J_h + H \tag{11.2}$$

or

$$C_p \rho \frac{\partial T}{\partial t} = \nabla \cdot \kappa \nabla T + H \tag{11.3}$$

where C_p is the specific heat and ρ is the density of the material. H is the heat source.

11.2.2 Thermoelectric power and thermal current

The temperature gradient induces a current of the form [48] [49] [50] :

$$-\, q\mu_n n P_n \nabla T \tag{11.4}$$

to the electron current and

$$-\, q\mu_p p P_p \nabla T \tag{11.5}$$

to the hole current.

The thermoelectric powers, P_n and P_p are given by

$$P_n = \frac{k_B}{q}\left[-\frac{5}{2} - \nu + ln\left(\frac{n}{N_c}\right)\right] \tag{11.6}$$

and

$$P_p = \frac{k_B}{q}\left[-\frac{5}{2} - \nu + ln\left(\frac{p}{N_v}\right)\right] \tag{11.7}$$

where ν is the exponent used in the field-dependent relaxation time. For phonon scattering [51],

$$\nu = -1/2 \tag{11.8}$$

11.3 Heat sources

The heat source can be separated into Joule heat, generation/recombination heat, and Thomson and Peltier heating terms. Following the suggestion of Ref. [48] we discuss these terms in more detail in the following subsections.

11.3.1 Joule heat

There are two sources of Joule heating. One is from the steady state or low frequency part of the electrical field:

$$H_{Joule-dc} = \frac{J_n^2}{q\mu_n n} + \frac{J_p^2}{q\mu_p p} \tag{11.9}$$

The other part comes from the optical frequency. When the optical wave passes a lossy semiconductor, the wave is absorbed by the lossy material. The absorbed energy can either generate electron hole pairs or be dissipated and become Joule heat. We also refer to this type of heat source as the optical part of the Joule heating. The internal loss of a semiconductor laser is the cause of the optical part of the Joule heating while the band to band absorption is the source for electron-hole pair generation in photo-detectors.

A simple derivation of this term follows. The optical power dissipated per unit volume can be expressed as:

$$power\ loss = \sigma_{op} F_{op}^2 = \epsilon_2 \epsilon_0 \omega F_{op}^2. \tag{11.10}$$

Where the optical field F_{op} is the root mean square of the oscillating field. For convenience a constant γ is introduced such that the complex wave amplitude is related to the electric field by

$$\mid F \mid^2 = \gamma \mid W \mid^2 . \tag{11.11}$$

To determine γ we use the following basic relation:

$$S\hbar\omega = \int \epsilon_0 \epsilon_1 \mid F \mid^2 dv = \epsilon_0 \gamma \int \epsilon_1 \mid W \mid^2 dv = \epsilon_0 \gamma < \epsilon_1 > . \tag{11.12}$$

The power loss becomes

$$power\ loss = S\hbar\omega^2 \epsilon_2 \mid W \mid^2 / < \epsilon_1 > . \tag{11.13}$$

or in terms of material internal loss and local index:

$$\epsilon_2 = \overline{n}_1 \alpha_i / k_0 \tag{11.14}$$

$$H_{Joule-op} = S\hbar\omega^2 \overline{n_1} \alpha_i \mid W \mid^2 / (k_0 < \epsilon_1 >). \tag{11.15}$$

11.3.2 Recombination heat

When an electron-hole pair recombines, the energy either converts to a photon (radiative) or turns into heat (non-radiative). In a real device, most of the photons emitted by spontaneous radiative recombination are eventually absorbed by the semiconductor and are converted into heat. Therefore we assume the spontaneous recombination is a heat source.

The stimulated recombination in a semiconductor laser should **not** be considered a heat source because the photons from this term are coherently amplified and are eventually emitted to the outside of the laser cavity.

For each electron-hole pair recombined, the heat released is the difference between the quasi-Fermi levels:

$$H_{rec} = (R_{trap} + R_{Aug} + R_{spon})(E_{fn} - E_{fp}) \tag{11.16}$$

11.3.3 Thomson and Peltier heat

The Thomson heat comes from the change in thermoelectric power when an electron-hole pair recombines:

$$H_T = qR_{total}T(P_p - P_n) \tag{11.17}$$

where P_p and P_n are thermoelectric power for hole and electrons, respectively.

The Peltier heat is related to the spatial variation in the thermoelectric power:

$$H_P = -T(J_n \nabla P_n + J_p \nabla P_p) \tag{11.18}$$

11.4 Temperature dependent parameters

Having established the theoretical basis for thermal modeling, we must also supply accurate temperature dependent parameters. Here we make no assumptions and treat all parameters as temperature dependent. The problem is that we do not know all of the temperature dependences for all of the materials. The user-accessible material parameter macro library allows the user to enter the temperature dependence into any one of the parameters at any time before the simulation.

11.5 Thermal boundary

We need to establish some realistic boundary conditions for the heat transport equation. In our simulator they are separately defined for electrical contacts and other parts of device.

For electrical contacts they are:

- 1) Thermal contact type 1. The contact is a heat sink assumed to have a lattice temperature given by the parameter **lattice_temp**.
- 2) Thermal contact type 2. The contact is assumed to have an out-going heat power flow given by the parameter **heat_flow**. The lattice temperature of the contact is unknown and is to be determined by the thermal equation solver.
- 3) Thermal contact type 3. The contact is assumed to be connected to a thermal conductor with its conductance given by parameter **thermal_cond**. This thermal conductor is connected to a heat sink with a fixed temperature given by parameter **extern_temp**. Please note that if this contact is also desclared as having an external circuit with a resistor, then the heat flow due to self-heating in this resistor will be taken into account with the following formula $\frac{1}{2}RI^2$ where R is the resistor and I is the electric current. Again the lattice temperature of the contact is unknown and is to be determined by the thermal equation solver.

For other parts of simulated device boundary conditions are implemented through the thermal interfaces of type 1-3 defined similarly to thermal contacts.

Chapter 12

Waveguide Optical Modes

12.1 Introduction

Many optoelectronic devices simulated by Crosslight software contain a waveguide. It is thus necessary to solve the lateral optical modes of a waveguide. Since the optical modes couple to the drift-diffusion model through carrier-dependent refractive index, efficient solution of the wave eqaations are critical.

Our basic assumption in dealing with 2D simulation of optical wave is that it propagates along with z-dimension with a factor $exp(j\beta z) = exp(jk_0^2 n_{eff} z)$ where n_{eff} is the complex effective index. The wave equation can then be reduced in dimension by variable separation.

When the waveguide dimension of an edge type of semiconductor laser is comparable to or larger than the wavelength, the lateral optical modes can be described by the scalar wave equation as follows:

$$\nabla^2 W(x,y) + k_0^2(\varepsilon(x,y) - n_{eff}^2)W(x,y) = 0 \tag{12.1}$$

where $\varepsilon(x,y)$ is the optical dielectric constant (not to be confused with the static dielectric constant). Our basic task is find the effective index and the related optical mode $W(x,y)$.

There are several numeric methods we use to solve the optical modes. Here is a listing:

- SOR iterative method. Convenient for simple single confined mode.
- Effective index method (EIM) is a quasi-1D model which is effective for devices with index varying slowly in one direction.
- Enhanced effective index method (EEIM) which is an extension of EIM. It can be used to radiative boundary.

- Analytical approximation by Gaussian function.
- User defined data file.
- Arnoldi restarted method which is a highly sophisticated direction eigen solution method useful for multiple lateral modes.

It is also possible to solve the above wave equation in vectorial form as an option.

12.2 Scalar Wave Equation For Lateral Modes

12.2.1 Effective index of a scalar mode

Our basic assuption in dealing with 2D simulation of optical wave is that it propagates along with z-dimension with a factor $exp(j\beta z) = exp(jk_0^2 n_{eff} z)$ where n_{eff} is the complex effective index. The wave equation can then be reduced in dimension by variable separation.

When the waveguide dimension of an edge type of semiconductor laser is comparable to or larger than the wavelength, the lateral optical modes can be described by the scalar wave equation as follows:

$$\nabla^2 W(x,y) + k_0^2(\varepsilon(x,y) - n_{eff}^2)W(x,y) = 0 \tag{12.2}$$

where $\varepsilon(x,y)$ is the optical dielectric constant (not to be confused with the static dielectric constant). Our basic task is find the effective index and the related optical mode $W(x,y)$. Using the scalar equation implies that all vectorial modes (TE/TM) obeys the same equation with the same boundary condition. For MQW structure edge lasers, this assumption is reasonable since the polarization dependence comes in mostly from the optical gain, not from the wave amplitude distribution.

As we will describe in the section on numerical methods, we use three methods for solving the scalar wave equation. The first is to use the SOR iterative method to solve the discretized eigen value problem. The 2nd is the effective index method (EIM) that decomposes the scalar wave equation into two separate one-dimensional wave equations. The EIM method solves a single mode 1D wave equation in the y-direction for a serious of y-cuts. Then it uses the eigen values from these y-cuts as the index distribution along the x-direction and solves a multi-mode wave equations in the x-direction. The third and also the latest is the direct solution of sparse eigen matrix problem using the advanced technique of Arnoldi method.

The enhanced effective index method (EEIM) option, which will be described in a later section, may be used to treat case of radiative boundary.

For problem of single confined lateral mode, the default solver of SOR iterative method is still the most convenient and efficient. For problem of nearly one dimensional in index distribution, the EIM method is efficient and accurate. For multiple confined mode simulation, the recommended method is the Arnoldi direct sparse eigen solver.

12.2.2 Multi-lateral mode model

A semiconductor laser can be driven into multi-lateral mode operation at high current injection levels, if higher order lateral modes exist in the waveguide structure. The 2D module solves the wave equation for the lateral modes at every current bias and solves, self-consistently, the photon densities of the modes. An additional photon rate equation for each additional lateral mode is included. Note that the lateral modes are coupled directly though the drift-diffusion equation via the stimulated recombination term and through the nonlinear gain effect as described later in this chapter.

For multimode solution of the scalar wave equation, we recommend the Arnoldi method as the most reliable solution. For multimode solution involving radiative modes, EEIM is the only choice in this version of the simulator.

12.3 Vectorial Helmholtz wave equation

12.3.1 Basic Vectorial Wave Equations

In this section, we derive the vectorial wave equation following Ref. [115]. For a harmonic guided wave propagating in the positive z direction, we consider the vector fields in the follow notation:

$$\overline{E}(x,y,z,t) = (E_x, E_y, E_z)exp[j(\omega t - \beta z)] \tag{12.3}$$

$$\overline{H}(x,y,z,t) = (E_x, E_y, E_z)exp[j(\omega t - \beta z)] \tag{12.4}$$

$$\overline{D} = \varepsilon\overline{E} \qquad \overline{B} = \mu\overline{H} \tag{12.5}$$

From Maxwell's equations for source-free regions,

$$\nabla\cdot\overline{D} = 0, \qquad \nabla\cdot\overline{B} = 0 \tag{12.6}$$

$$\nabla\times\overline{E} = -\frac{\partial\overline{B}}{\partial t} = -j\omega\mu\overline{H} \tag{12.7}$$

$$\nabla\times\overline{H} = \frac{\partial\overline{D}}{\partial t} = j\omega\varepsilon\overline{E} \tag{12.8}$$

we obtain

$$\nabla\times\nabla\times\overline{E} = \nabla(\nabla\cdot\overline{E}) - \nabla^2\overline{E} = \omega^2\mu\varepsilon\overline{E} = k^2\overline{E} \tag{12.9}$$

where $k = 2\pi n/\lambda$ is the wave number. Thus,

$$\nabla \cdot \overline{D} = 0, \ or \ \nabla \cdot (k^2 \overline{E}) = 0 \tag{12.10}$$

$$\overline{E} \cdot \nabla k^2 + k^2 \nabla \cdot \overline{E} = 0 \tag{12.11}$$

$$\nabla \cdot \overline{E} = -(1/k^2) \overline{E} \cdot \nabla k^2 \tag{12.12}$$

Using the Maxwell's equations above, we obtain the vectorial Helmholtz wave equation.

$$\nabla^2 \overline{E} + k^2 \overline{E} + \nabla \left(\frac{\overline{E} \cdot \nabla k^2}{k^2} \right) = 0 \tag{12.13}$$

Let us consider the transverse component of the above equation with the notation:

$$\overline{E}_T = (E_x, E_y, 0), \quad \nabla_T = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, 0 \right) \tag{12.14}$$

and we note that in most cases, it is reasonable to assume that $\frac{\partial k^2}{\partial z} = 0$. We obtain,

$$\nabla_T^2 \overline{E}_T + (k^2 - \beta^2) \overline{E}_T + \nabla_T \left(\frac{\overline{E}_T \cdot \nabla_T k^2}{k^2} \right) = 0 \tag{12.15}$$

which is the vectorial wave equation.

The x component of the above is as follows:

$$\frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + (k^2 - \beta^2) E_x + \frac{\partial}{\partial x} \left(\frac{E_x}{k^2} \frac{\partial k^2}{\partial x} + \frac{E_y}{k^2} \frac{\partial k^2}{\partial y} \right) = 0 \tag{12.16}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial E_x}{\partial x} + \frac{E_x}{k^2} \frac{\partial k^2}{\partial x} \right) + \frac{\partial^2 E_x}{\partial y^2} + (k^2 - \beta^2) E_x + \frac{\partial}{\partial x} \left(\frac{E_y}{k^2} \frac{\partial k^2}{\partial y} \right) = 0 \tag{12.17}$$

Our final expression for the x component of the wave equation is as follows:

$$\frac{\partial}{\partial x} \left(\frac{1}{k^2} \frac{\partial}{\partial x} (k^2 E_x) \right) + \frac{\partial^2 E_x}{\partial y^2} + (k^2 - \beta^2) E_x + \frac{\partial}{\partial x} \left(\frac{E_y}{k^2} \frac{\partial k^2}{\partial y} \right) = 0 \tag{12.18}$$

Similarly, the y component can be written as follows:

$$\frac{\partial^2 E_y}{\partial x^2} + \frac{\partial}{\partial y} \left(\frac{1}{k^2} \frac{\partial}{\partial y} (k^2 E_y) \right) + (k^2 - \beta^2) E_y + \frac{\partial}{\partial y} \left(\frac{E_x}{k^2} \frac{\partial k^2}{\partial x} \right) = 0 \tag{12.19}$$

Once the transverse components are solved, we can obtain the z component from the zero-divergence as follows:

$$E_z = \left(\frac{1}{j\beta k^2} \right) \nabla_T \cdot (k^2 \overline{E}_T) \tag{12.20}$$

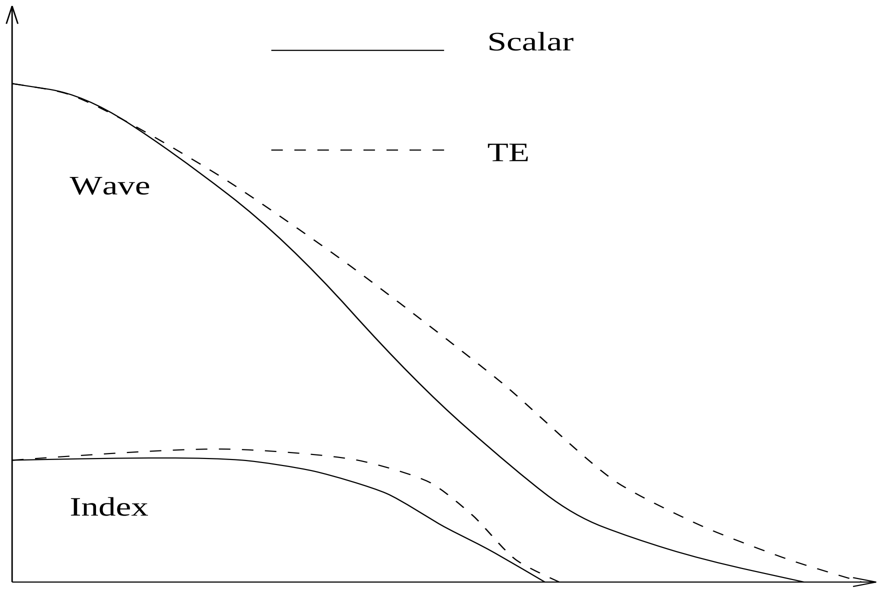


Figure 12.1: Schemtics of the right half of the optical mode along the lateral direction. The solid lines are the refractive index and the scalar mode. The dashed lines are the "vectorial effective index" and the corresponding wave distribution.

12.3.2 Interpretation of Vectorial Solutions

In this subsection, we attempt to provide a simple prediction or interpretation of the vectorial wave solution in comparison with the scalar solution. We only consider the solution of $(E_x, 0, E_z)$ for the TE mode here since the same discusion can be made for the TM mode exchanging the x-axis with the y-axis. We re-write the vectorial wave equation equation as follows:

$$\frac{\partial}{\partial x}\left[\frac{1}{k^2}\frac{\partial(k^2)}{\partial x}E_x + \frac{\partial E_x}{\partial x}\right] + \frac{\partial^2 E_x}{\partial y^2} + (k^2 - \beta^2)E_x + = 0 \tag{12.21}$$

$$\frac{\partial}{\partial x}\left[\frac{1}{k^2}\frac{\partial(k^2)}{\partial x}E_x + \frac{\partial E_x}{\partial x}\right] + \frac{\partial^2 E_x}{\partial y^2} + (k^2 - \beta^2)E_x + = 0 \tag{12.22}$$

$$\frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + (k^2 - \beta^2)E_x + \frac{\partial}{\partial x}\left[\frac{E_x}{k^2}\frac{\partial(k^2)}{\partial x}\right] = 0 \tag{12.23}$$

We find that the vectorial wave equation can be written in the same form as the scalar one if we introduce a "vectorial effective index", $k_{ve} = 2\pi n_{ve}/\lambda$.

$$\frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + (k_{ve}^2 - \beta^2)E_x = 0 \tag{12.24}$$

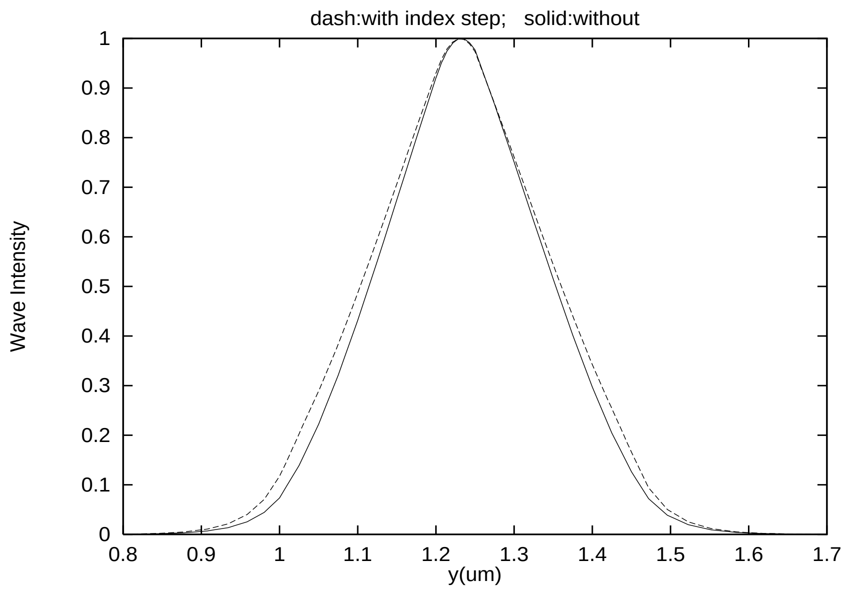


Figure 12.2: A simple laser structure used to understand the increased index caused by the vectorial term. The active layer is 0.05 μm GaAs. The wave guiding layer is 0.2 μm Al(0.2)Ga(0.8)As and the cladding layer is Al(0.8)Ga(0.1)As. The increased index step is placed at a distance of 0.2 μm from the active layer to simulate the effect of vectorial term.

$$k_{ve}^2 = k^2 + \frac{1}{E_x}\frac{\partial}{\partial x}\left[\frac{E_x}{k^2}\frac{\partial(k^2)}{\partial x}\right] \tag{12.25}$$

Therefore, interpretation of the vectorial wave equation can be made using the scalar equation and the "vectorial effective index". Let us consider qualitatively the effect of vectorial wave using the schematic in Fig. 12.1. We consider only the case of continuous index profile because the physical conclusion is the same for abrupt index profile.

Based on Fig. 12.1, we decide that $\frac{\partial(k^2)}{\partial x}$ is negative in positions with index change. Since $E_x(x)$ decays in the positive x-direction, we conclude that Eq. (12.25) yields increased effective index in locations of index change. The increased effective index in positions of index change can be used to interpret or predict the effect of vectorial wave equations as compared with the scalar equation. In a device with index change far from the center of the mode (see Fig. 12.2), the increased index of the vectorial term causes the wave to spread out more than scalar wave. If the index change is close to the mode center (see Fig. 12.4), more optical confinement is provided by the use of the vectorial wave. As a result, the vectorial wave mode is more confined than the scalar mode.

To demonstrate the effect of the increased index, we set up a series of scalar wave simulations as shown in Figs. 12.2 to 12.4. We artificially put an increased index in

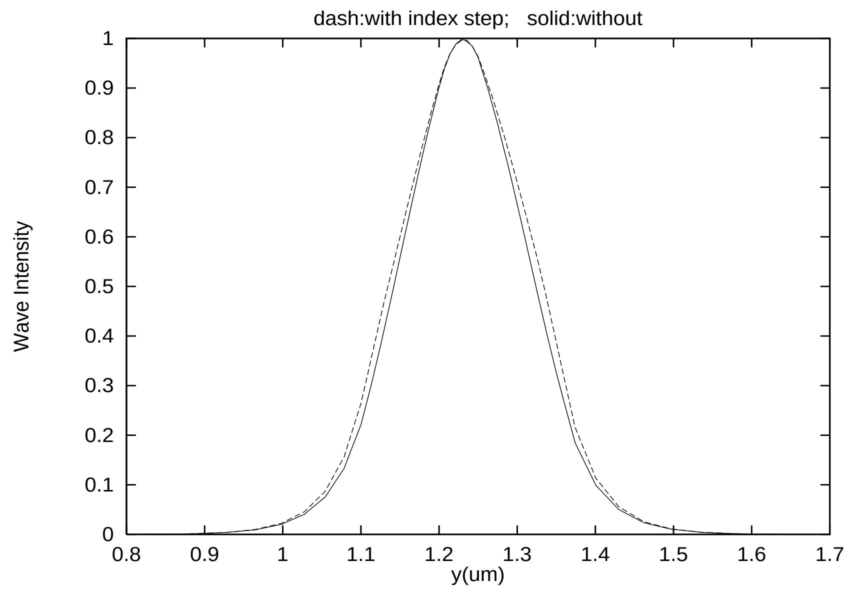


Figure 12.3: A simple laser structure used to understand the increased index caused by the vectorial term. The active layer is 0.05 μm GaAs. The wave guiding layer is 0.1 μm Al(0.2)Ga(0.8)As and the cladding layer is Al(0.8)Ga(0.1)As. The increased index step is placed at a distance of 0.1 μm from the active layer to simulate the effect of vectorial term.

three different positions in the simulations. We find that as the position of the index step moves closer to the center of the mode, more confinement is provided.

This simulation series show that in most laser diode structures, the TE mode produces a mode which spreads out more in the x direction than the scalar mode. The reason is that the index change in the x-direction is usually located far from the mode center. For example, a typical buried heterostructure laser may uses a two micron wide active region width to provide the lateral electrical/optical confinement. The index change is about one micron away from the mode center. On the other hand, the TM mode often produces more confined optical mode because the transverse index change is much closer to the mode center. It is common to have index change about 0.1 microns away from the mode center.

We also discover that for ridge waveguide laser using the TE polarization, the vectorial wave solver yield little difference than the scalar one because the lateral index change is weak and far away from the mode center. This may be regarded as a justification for using the scalar wave equation.

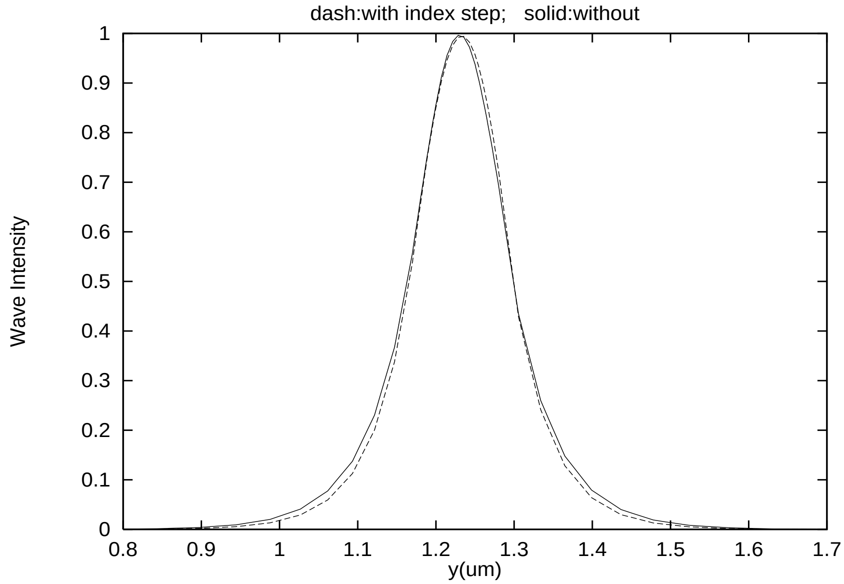


Figure 12.4: A simple laser structure used to understand the increased index caused by the vectorial term. The active layer is 0.05 μm GaAs. The wave guiding layer is 0.03 μm Al(0.2)Ga(0.8)As and the cladding layer is Al(0.8)Ga(0.1)As. The increased index step is placed at a distance of 0.03 μm from the active layer to simulate the effect of the vectorial term.

12.4 Far-field computation

After solving for the waveguided optical modes, it is straight-forward to compute the far-field emission pattern. The problem is basically that of a diffraction: *i.e.*, given the distribution of a propagating beam at a 2D cross section, calculate the distribution at a distance from this cross section. The formulation for this problem has been well established in classical electrodynamics and the user is referred to standard textbooks on electrodynamics.

A semiconductor laser emits light in the form of a narrow spot of elliptical cross section. The spatial-intensity distribution of the emitted light near the laser facet is known as the *near-field*. During its propagation, the spot grows in size due to beam divergence. The dimensions of the elliptical spot and its divergence angles are important beam parameters associated with the laser mode. The *far-field* pattern is the angular distribution of beam intensity far from the laser facet. It is an important factor in the design of laser geometries since it directly affects the coupling of power from the laser source to the object being illuminated.

Figure 14.1 gives a schematic representation of the far-field emission of a semiconductor laser. The full angles at half power are $\theta_\perp$ and $\theta_{||}$ in the directions perpendicular to and along the laser junction plane, respectively. Typically, $\theta_{||}$ is of the order of 10°, whereas $\theta_\perp$ is considerably larger (35 to 65°), depending on the detailed near-field

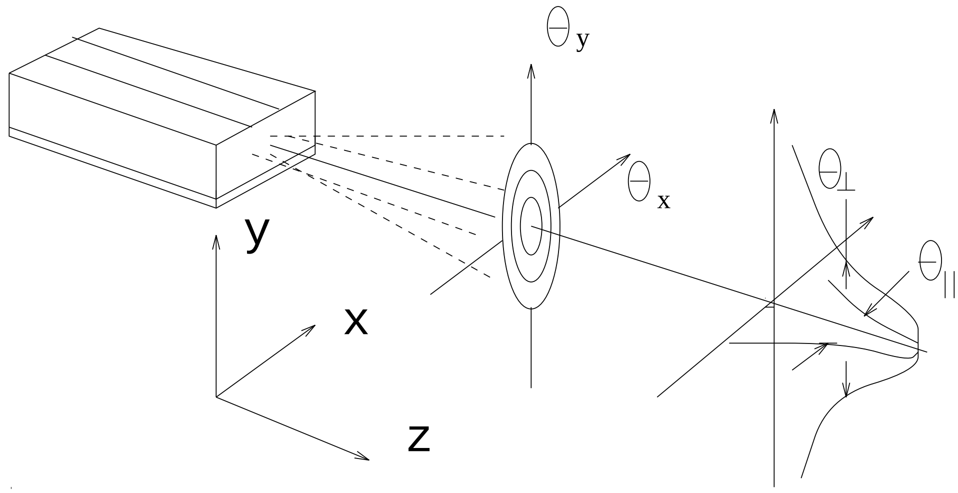


Figure 12.5: Schematic representation of the far-field emission of a semiconductor laser.

distribution.

The program solves the wave equation and the semiconductor equations self-consistently and yields accurate near-field distribution as the routine output data. This example describes the far-field model as an additional option of the software.

Refs. [1] and [8] have given mathematical formulas for the far-field distribution based on the solution to the wave equation in free space. Since these mathematical expressions were derived for broad-area lasers only (*i.e.*, assuming the x-dimension is infinite), the program has extended their formulas to an arbitrary two-dimensional situation using the same solution approach as suggested in [1] and [8]. This feature allows the user to obtain the far-field distribution immediately after the near-field is computed.

As an example, we consider a ridge waveguide GaAs/AlGaAs laser emitting at about 0.78 μm. It has a nonplanar ridge which makes it necessary to use the finite element feature of the software to handle its complex geometry. The near-field pattern was solved using a self-consistent approach for the wave equation, the semiconductor equations, and the rate equation altogether.

The far-field pattern is computed based on the near-field solution and is presented in Figure 14.2 as a contour plot of the far-field power. The contour lines are evenly

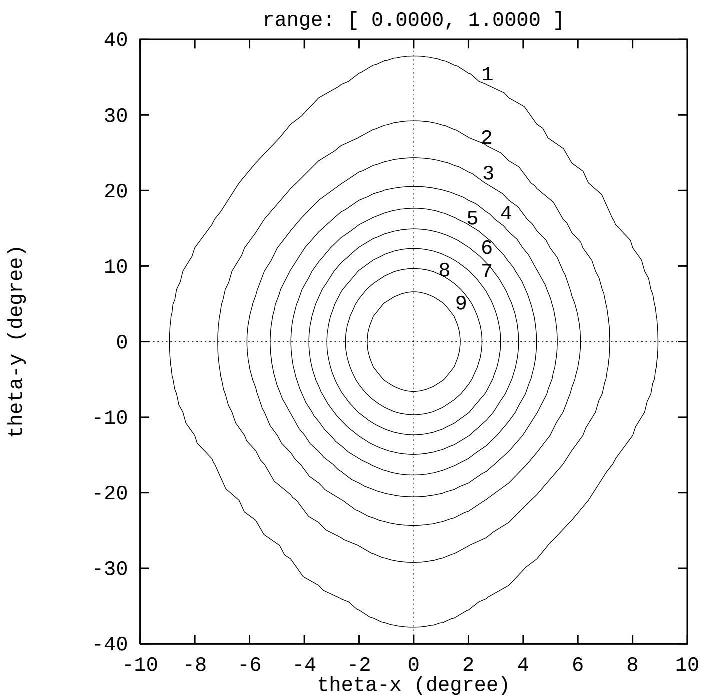


Figure 12.6: An example of the far-field emission pattern from a non-planar ridge waveguide GaAs/AlGaAs laser emitting at 0.78 μm.

spaced in power level (at 10% power intervals). The angles $\theta_\perp$ and $\theta_{||}$ can be directly extracted from the contour plot.

To activate the far-field option as a post-processing step (in .plt file), the solution generated by running "*.sol" is required.

To summarize, a useful option has been made available to software users for the far-field behavior of a semiconductor laser. It is capable of accurately computing the far-field emission patterns from lasers of arbitrary geometries and materials.

12.5 Enhanced Effective Index Method

12.5.1 Introduction

In the standard effective index method (EIM), all boundaries are assumed to be zero or decay exponentially. As a result, radiative modes are excluded. In this section, we will describe an extension of the EIM which allows us to treat radiative modes as well as confined modes.

The standard set up is as illustrated in Fig. 12.7. We follow the solution approach of Ref. [78] and assume the wave is TE-polarized. It is convenient to solve the H field in this case. The mode is assumed to be confined in y-direction, i.e., the top and bottom boundaries have zero field intensities.

The left boundary is assumed to be either even (with derivative zero) or odd (with field zero). This is most suitable for device with symmetry axis place on the left of the simulation area.

The radiative boundary is assumed to be on the right side. This will be most convenient for cases such as ARROW wave guide type of lasers and other antiguiding lasers.

12.5.2 Rectangular model

For an arbitrary device structure, it is always possible to divide into columns and rows. The more the division, the more accuracy can be achieved with cost of computation time. The simulation program is designed such that the user specifies the columns and the program determines the rows automatically. Usually, it is efficient to divide the structure into 2 or 3 columns according to the geometry of the device. For example, a ridge waveguide laser is best divided into two columns with boundary at the ridge wall.

According to Ref. [78], the basic wave equation for the TE mode can be represented by the following magnetic field equation.

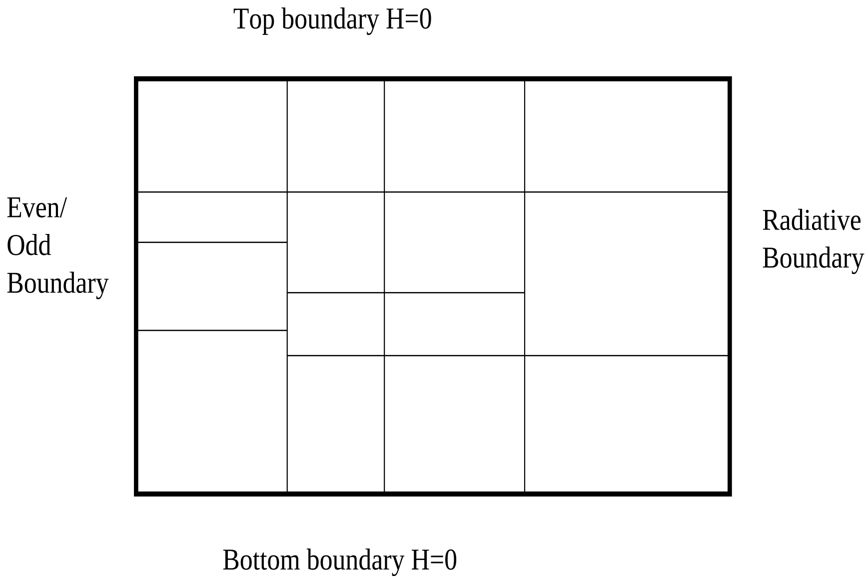


Figure 12.7: Boundary conditions and division of columns and rows in the enhanced effective index method (EEIM).

$$\nabla^2 H(x, y) = k_0^2(\bar{\varepsilon} - \varepsilon)H(x, y) \tag{12.26}$$

The horizontal boundary between row 1 and row 2 is

$$H_1 = H_2; \frac{\partial H}{\partial y} \mid_1 = \frac{\partial H}{\partial y} \mid_2 \tag{12.27}$$

The vertical boundary between column 1 and column 2 is

$$H_1 = H_2; \frac{1}{\varepsilon_1}\frac{\partial H}{\partial y} \mid_1 = \frac{1}{\varepsilon_2}\frac{\partial H}{\partial y} \mid_2 \tag{12.28}$$

In each vertical segment, only horizontal boundaries between dielectrics occur, and ε is now a step function of y. We separate the variable as follows.

$$H(x, y) = \alpha(x)\beta(y) \tag{12.29}$$

The function β is a vertical eigenfunction for this vertical segment with eigenvalue δ, satisfying

$$\frac{d^2\beta}{dy^2} = k_0^2(\delta - \varepsilon(y))\beta \tag{12.30}$$

and α then satisfies a similar equation:

$$\frac{d^2\alpha}{dx^2} = k_0^2(\bar{\varepsilon} - \delta)\alpha \tag{12.31}$$

Although there are, in general, an infinite number of eigenfunctions β for each vertical segment, we include only some finite number N of these to construct a general solution. The general solution in segment j can be written

$$H_j(x,y) = \sum_{i=1}^{N}[a_i exp(k_0\sqrt{\overline{\varepsilon} - \delta_i}x) + b_i exp(-k_0\sqrt{\overline{\varepsilon} - \delta_i}x)]\beta_i(y) \tag{12.32}$$

Our task is to determine eigen value δ and function β in the first step, and $\overline{\varepsilon}$ and the coefficients of a_i and b_i in the second step.

12.5.3 Numerical approach

Our first step is to solve Eq. 12.30 within each column. The analytical expression for β is expressed in term of unknown δ. Assuming there are M layers, we have 2M number of unknown coefficients in this expression. The horizontal boundary condition gives a set of 2M equations that can be solved by setting the determinant to be zero. The zeros of the determinant is used to find the roots of δ. Use of singular decomposition is then used to determine the 2M coefficients. We shall call this procedure as searching for roots in y-direction (y-root).

Zero boundary on top and bottom will produce a complete set of eigen vectors that can be used to construct any wave function. Thus, the more β base functions we choose, the more accurate the wave function to be constructed. However, use of more base function in y-direction slows down the solution substantially. Here we will choose a limited N base functions as β_i, (i=1,2,...,N).

Once we have the β base functions, we can construct the final H-field as in Eq. 12.32. For each column j, the H-field consists of 2N coefficients of a_i and b_i. The vertical boundary conditions can be used to write down two equations for H-fields connecting H-field in jth column with that in the (j+1)th column. The boundary condition on the right boundary requires that traveling wave to the left be zero. This boundary condition can be used to model both radiating wave to the right or for confined wave with decaying wave.

If we multiply the two equations by $\beta_k(y)$ and integrate over y, we will get 2N equations relating j and (j+1)th columns. Using this technique for all columns will produce sufficient number of equations to determine all the a_i and b_i coefficients and the modal eigen solution for $\overline{\varepsilon}$. We shall call this procedure as searching for modes in x-direction (x-root) . This step generates the final results for the optical modes.

The modal optical dielectric constant can be used to determine the modal optical gain as follows:

$$g_{eeim} = -Im(\overline{\varepsilon})k0/n_{eff} \tag{12.33}$$

where $n_{eff} = Re(\sqrt{\overline{\varepsilon}})$ is the real part of modal effective index. We shall call this EEIM modal gain to differentiate it with the usual modal gain which we approximate by averaging the local gain with the weight of wave intensity. We shall refer to the latter as averaging modal gain.

The use of averaging modal gain has great numerical advantage of being stable against the Newton's method used by the drift-diffusion equations. But the averaging gain is inaccurate when there is radiative loss in the case of leaky modes. Direct use of EEIM gain will cause numerical problems near lasing condition. As a compromise, we estimate the radiation loss at transparent condition by setting all imaginary parts of the dielectric constants to zeros and use the EEIM gain obtained this way as the radiation loss. We will use this estimated radiation loss to correct the averaging gain through out the whole simulation.

In summary, we have been able to used the EEIM to estimate the radiation loss and use it to correct the averaging modal gain.

12.5.4 Estimate of radiative loss

Using the EEIM module, we are able to find the radiative modes and the effective index (or eigen value of the wave equation) associated with it. The modal loss/gain is represented by the imaginary part of the eigen value.

In many cases, we need to discuss device performance in terms of radiative loss and material loss separately. The issue here is how we can separate the radiative loss from the material loss since the imaginary part of the eigen value of the wave equation comes from both material and radiative losses.

We recall how we calculate the modal gain/loss in the absence of radiative loss. To a good approximation, we calculate the modal gain/loss by averaging the local material gain/loss using the wave intensity as the weight function. This approach simplifies and stabilizes the numerical procedures. Such method of wave intensity averaging can not be extended to the case when there is radiative loss. The reason is that the wave extends to infinity where there is radiative loss.

In our simulator, we separate the radiative loss as the different between the total gain/loss and the material gain/loss from averaged local material gain/loss.

We use two methods to estimate the radiative loss. The first method of estimate is to set all material to be lossless (by setting imaginary part of indices to be zero). We then apply the EEIM method to compute the total loss with is just the imaginary part of the eigen value of wave solution. Since the material loss is artificially set to zero, the total loss from the EEIM is the radiative loss we needed. We refer to such a method of estimate as the "pure index estimate" (we use purely real index here).

An alternative method of radiative loss estimate is to compute at every bias both the total gain/loss from the EEIM and also the averaged local gain/loss. The difference between the two is the bias dependent radiative loss. Since the local complex index varies as a function of bias, the radiative loss also changes with bias. We shall refer to this method of estimate as the "bias-dependent estimate".

The "pure index estimate" is most convenient since we only get one value for each device. In most cases, it gives a good representation of the radiative losses in the same device when there are material losses.

The "bias-dependent estimate" is more realistic since the radiative loss depends also on local material gain/loss. Since the numerical fluctuation of EEIM as a function of bias can be quit large, the relative error from this method may also be substantial as compared with the pure index estimate.

To maintain numerical stability, we still use the method of averaging the local gain to obtain the modal material gain/loss. Then the radiative loss from either of the above estimate methods is subtracted from the material modal gain/loss to obtain the total modal gain/loss which is needed to compute the lasing characteristics.

We use the pure index method as our default setting the simulator.

The pure index method is the most stable since we only have one number to substrate from the material gain. The numerical fluctuation of the bias-dependent method may be a problem when calculating the light vs. current characteristics. So we freeze the value of the radiative loss when the total gain near threshold. In both methods, we freeze the optical wave function from the EEIM when the device is biased near the threshold. This is to ensure a smooth light intensity as a function of bias.

12.5.5 Substrate radiative loss

A common simulation issue is to study the radiative loss caused by high refractive index substrate. So the direction of radiative wave is downwards in -y direction. Since our EEIM model assumes confined y-modes and radiative x-modes, we must first rotate our device counter-clockwise by 90 degrees. This may be achieved by use of GeoEditor GUI program.

12.5.6 Choice of optical window in y-direction

If the material high index in y-direction is always well centered, it does not matter what size of solution region you choose in y-direction (optical boundary in init_wave statement). However, there are cases where some columns have no optical confinements in y-direction at all.

For example, a ridge waveguide laser rotated by -90 degrees will have the substrate region with no optical confinement. In such as case, the base modes in y-direction depends very much on the choice of optical window. If the window size in y is too small, it obviously will affect the y-modes. If the size is too large, many y-modes will be needed to match the base modes from other columns with optical confinements in y. With proper choice of optical window in y-direction, it is possible get reasonable modal solutions using only 3-5 y-base modes.

12.5.7 Run-time messages

The EEIM model is invoked at every bias step to ensure the variation of modes as a function of bias is taken into account. When using the EEIM model, the simulator prints out messages stating the status of the execution of the EEIM model procedures. Here we list the Run-time messages when EEIM is invoked for the first time as lines with > sign. Our comments are also given in the listing.

```
> -------Start EEIM Model--------
> With initial root search (this may take a while).

Without any previous knowledge of the roots (or modes),  we have
to search the complex delta (for y-modes) and epsilon (for x-modes)
using a point by point approach.   This may take a while.

> ---Estimate of radiation loss (pure index)---

As a first time use of EEIM in the program, we estimate
the radiation loss.  This is done by setting all imaginary parts
of the local index to zero.  The imaginary part of
epsilon (eigenvalue) is used to estimate the radiation modal
loss.

> Save base in "tmpb101.dat"
> Save base in "tmpb102.dat"
> Save base in "tmpb103.dat"
> Save base in "tmpb201.dat"
> Save base in "tmpb202.dat"
> Save base in "tmpb203.dat"
...
> Save base in "tmpb701.dat"
> Save base in "tmpb702.dat"
> Save base in "tmpb703.dat"

The y-base modes are computed and stored in "tmpbijk.dat",  where
i, j, k are integers for ith column and y-base mode number jk.   You may
use the following gnuplot command to view the y-mode:
plot  "tmpbijk.dat" w l.
```

```
Start to search for roots for the final optical field.

> Initial x-search found roots=           5

Found 5 possible roots for x-direction.

> Solve wave in x direction.
>   Find the following possible roots.
>No./Iter/Error=  1   80  0.7779E-02
>   root= (10.2363356873790,1.027566003195660E-003)
>     EEIM gain[1/m]=  -2374.09673169544
>   Save wave in "tmploss01.dat"
>No./Iter/Error=  2   80  0.6136E-03
>   root= (10.2118941158888,1.060170706117816E-003)
>     EEIM gain[1/m]=  -2452.35642688367
>   Save wave in "tmploss02.dat"
>No./Iter/Error=  3   80  0.1955E-04
>   root= (10.1705711295711,1.119772469334270E-003)
>     EEIM gain[1/m]=  -2595.48221708482
>   Save wave in "tmploss03.dat"

We have found the above 3 modes which are saved in "tmplossij.dat".
You may use the following gnuplot commands to plot modes
to see if these are physical:

set parametric
splot "tmplossij.dat" w l.

> Radiation loss for pure index model:
> Mode        Loss(1/m)
>           1   2374.09673169544

We estimate the radiation loss for the requested first mode as above.
This loss will be used to correct the material gain
to get the actual total modal gain.


> ---End of estimate of radiation loss---

Once we finished the "pure index estimate",  we proceed to
use EEIM model for optical modes are each bias with run-time
message similar to the above.
```

```
> Save base in "tmpb101.dat"
> Save base in "tmpb102.dat"
> Save base in "tmpb103.dat"
...

> -------End of EEIM Model--------
```

When the EEIM model is invoked in the next bias step, the roots found in the previous bias are used as initial guess. If the initial guess does not produce stable solution, the program will re-invoke the EEIM model again with initial root search. Initial root search usually take up most of the time.

Please note that our simulator always gives the pure index estimate the first time EEIM is invoked. Then, the modes are solved at every bias steps. If pure index estimate is not used, radiative loss at each bias step is calculated and printed at different bias steps.

12.6 Perfectly Matched Layer Boundary

The finite element (FEM) discretization of the wave equation results in an eigen value matrix problem of the order of the number of mesh points (for scalar equations). The matrix problem can easily be set up if the modes are well confined within a finite region and their intensities decay as a function of distance from the mode centers. We can obtain a well defined eigen matrix by deleting the nodes of the boundary assuming zero wave intensities there. The situation for leaky modes is more complicated for finite element analysis: there is no simple way to truncate the mesh to establish a well defined eigen matrix.

The idea of perfectly match layer (PML) is to create an artificial layer to absorb the radiating wave without reflecting it back. Since the wave decays to zero intensity within the PML, we have converted the problem into one of confined modes if our mesh extends to the outer boundary of the PML.

We have implemented a model of anisotropic PML due to Sacks and others [Z. S. Sacks, D.M. Kingsland, R. Lee, and Jin-Fa Lee, ”A perfectly matched anisotropic absorber for use as an absorbing boundary condition,” IEEE Trans. Antennas Propagate., vol.43, (no.12), IEEE, pp.1460-1463, Dec. 1995.] In the anisotropic PML model, the artificial PML has a complex anisotropic dielectric constant. If the complex anisotropic dielectric constant is chosen properly, the PML method should enable us to apply FEM to radiative mode solutions which are more accurate and powerful than EEIM.

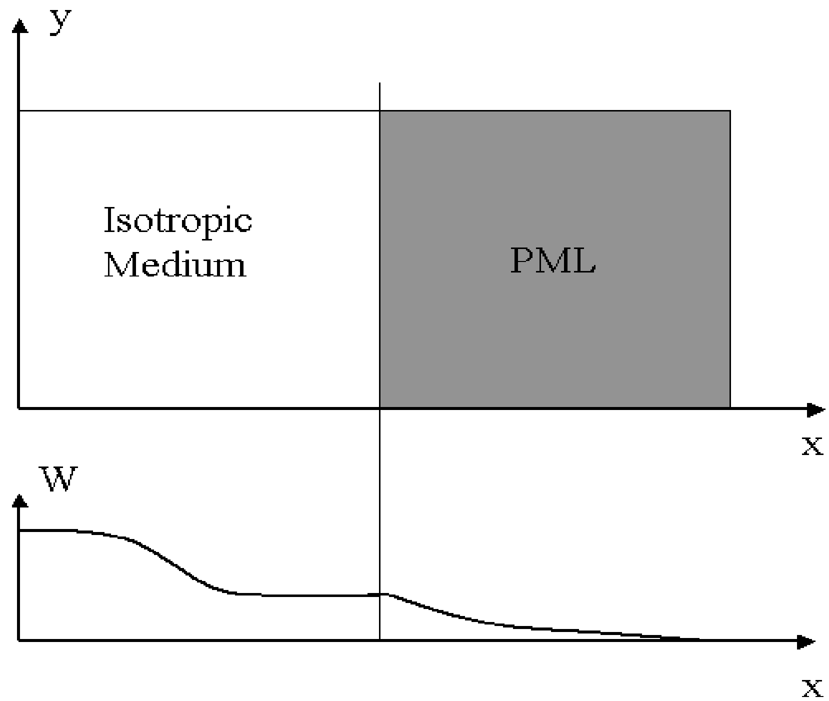


Figure 12.8: Schematics of the perfectly matched layer the decay of the wave within it.

We define an artificial layer with an anisotropic medium tensor as follows

$$\frac{[\varepsilon]}{\varepsilon} = \frac{[\mu]}{\mu} = [\Lambda] = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \tag{12.34}$$

where ε and μ are permitivities of the isotropic medium to be matched. Without loss of generality, we consider the situation as indicated in Fig. 12.8. As proved by Sacks et. al., the reflection from the PML may be eliminated if the permitivity tensor components have the following relationship.

$$c = b = 1/a \tag{12.35}$$

As we show in Appendix A, the wave equation in the PML can be written as follows.

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} + k^2\vec{E} = 0 \tag{12.36}$$

Consider a scalar field being one of the components:

$$E(x, y, z) = W(x, y)exp(j\beta z) \tag{12.37}$$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) W(x, y) + (k^2 - \beta^2)W(x, y) = 0 \tag{12.38}$$

Let us study the asymptotic behavior of the wave in the PML. Usually, the radiative wave occurs when the index near the PML is higher than the effective index, or :$Re(k^2 - \beta^2) > 0$. The wave behaves like

$$W \sim exp(jc\sqrt{k^2 - \beta^2}) \tag{12.39}$$

Thus, the imaginary part of c will cause the wave to decay within the PML so that we can truncate our eigen matrix on the outer boundary of the PML.

The criteria for setting the permitivity parameter c is to attenuate the wave without causing reflection. Theoretically, any value of c can be used without causing reflection. However in numerical simulation, realization of a theory can only be approximate because of discretization using finite number of mesh points. Thus, to minimize reflection, we recommend using unity as the real value of c, i.e., the same real reflective index in the PML as in the isotropic medium.

When choosing the imaginary part of c, please note the two extremes: If Im(c) is too large, the substantial difference between the index of the isotropic medium and the PML may cause significant reflection and you may observe standing wave ripples in the isotropic medium near the PML. On the other hand, if Im(c) is too small, the attenuation of the wave within the PML is so weak that it requires a thick PML layer and many mesh points within the PML. If the thickness of the PML is not large enough, the truncation of the eigen matrix within the PML will cause reflection within the PML which may propagate all the way back to the isotropic medium. Again, we may observe standing wave ripples in the isotropic medium. Therefore the PML should be adjusted in such a way that standing wave ripples do not appear in the isotropic medium.

Based on some of our initial tests, a value of 0.02 for the imaginary part of c goes well with a PML layer thickness of 5 microns and a mesh number of 50 in the direction towards the PML.

12.7 Optical Pumping and Incident Light

In Crosslight's simulator, it is possible to consider a device operating at two different wavelengths: one as an input optical pump and the other as output emitter. More specifically, the shorter wavelength is for the pump and the longer wavelength for emission. For LASTIP and PICS3D, the optical pumping corresponds to optically pumped laser diodes or VCSEL's while in APSYS, it corresponds to optically pumped LED's.

Figure 12.9 shows that for a typical active region under a specific carrier injection condition, it is possible for the material to exhibit interband absorption (or negative gain, thus e-h pair generation) at shorter wavelength while yielding optical gain at the peak gain wavelength.

Our accurate optical gain spectrum model is evaluated at every mesh point of the active regions (or passive regions declared as "active") which respond to different optical wavelengths to treat them either as a pump or emitter. The refraction and absorption of the incident pump light passing through the device is processed as plane wave passing a multiple optical layer system. The simulator figures out the correct amount of carriers generated by the pump which in turn causes population inversion and lasing action.

It is also possible to apply this option to VCSEL's. The simulator accepts input pumping power in the form of single wavelength or of user-defined continuous spectrum.

As in the key emission wavelength, the incident wave can be treated as a guided wave and thus all previous discussion of lateral optical mode may be applied without modification. The important command to control the incident light power is the statement **light_power**.

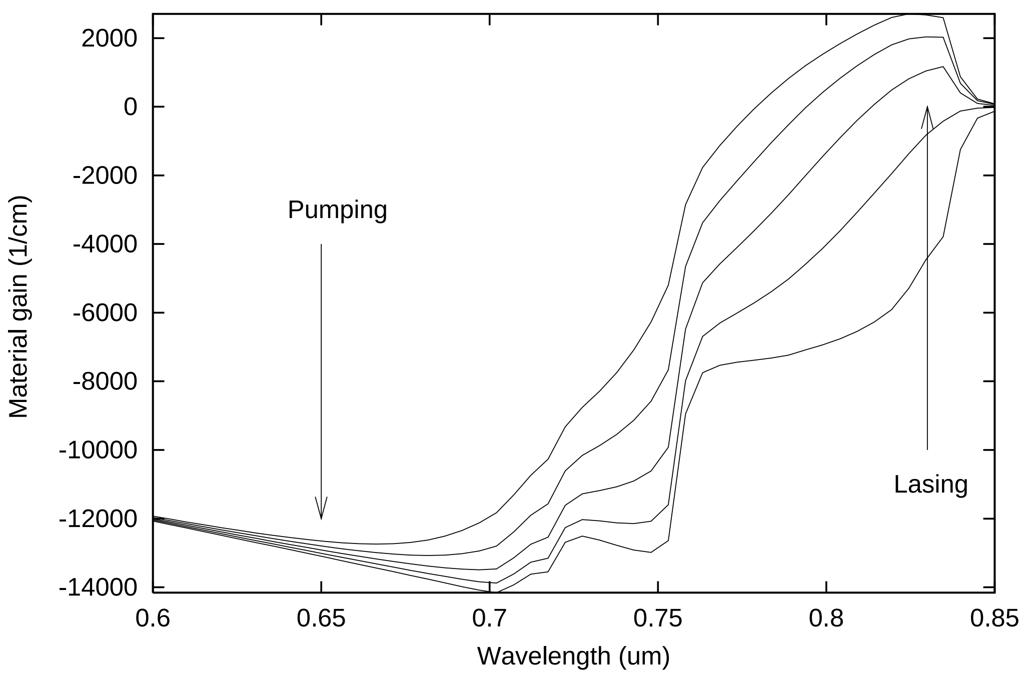


Figure 12.9: Optical gain spectrum and indication of pumping and lasing wavelengths.

Chapter 13

WURTZITE STRAINED MQW

13.0.1 Introduction

Recently, wurtzite strained quantum wells have been studied intensively because of applications in GaN blue green lasers and LED's. Previous band structures models are applicable for zincblende compounds. In this section, we will describe MQW band structure and optical transition models for wurtzite crystals here. We follow the work of S.L. Chuang [66] closely here. In the following subsections, we will describe the bulk wurtzite band structure, the MQW subband model and the optical transitions.

We must point out a couple of major differences between zincblende and wurtzite crystals here. The first issue is the base lattice. For zinc-blende material system, it is usually well known what the basic lattice constant is (for example, GaAs or InP) and the strain is well defined in each layer once we know the substrate lattice. Whereas in wurtzite system, a bulk GaN-based material layer is usually grown on a sapphire substrate. A significant amount of lattice mismatch between the substrate and the GaN-based layer results in a large dislocation or defect density. Then an MQW system is grown on the bulk layer. Depending on the competition of dislocation or defect between the material layers, It is not always clear what the base lattice constant of the MQW should be. Although the default lattice base constant in our software is that of GaN, it may not be true for many systems. For example, if a thick bulk layer of AlGaN is grown on a sapphire substrate before the MQW is placed on top, the base lattice constant should be that of bulk AlGaN instead of GaN. For this reason, our active layer macros for wurtzite MQW provide a statement to describe the base lattice so that the users can adjust it according to individual material systems.

Another important issue is material parameters. As we will see below, there are many times more band structure and strain parameters for wurtzite than zinc-blende material system. Since wurtzite semiconductor band structures and optical properties are less understood, and many bandstructure parameters are not available, our expectation of the accuracy of the simulation should be lower than that for zinc-blende material system.

13.0.2 Bulk Band Structure

Consider a strained wurtzite crystal pseudomorphically grown along the c-axis (z axis) on another thick wurtzite layer. The base lattice constant is a_0 and the original lattice constant of the layer under consideration is a. The strain tensor in the well region has the following elements:

$$\begin{aligned}
\varepsilon_{xx} &= \varepsilon_{yy} = \tfrac{a_0 - a}{a} \\
\varepsilon_{zz} &= -\tfrac{2C_{13}}{C_{33}}\varepsilon_{xx} \\
\varepsilon_{xy} &= \varepsilon_{yz} = \varepsilon_{zx}
\end{aligned} \tag{13.1}$$

The conduction bands can be characterized by a parabolicband model with different electron effective masses m_e^t and m_e^z perpendicular and parallel to the c-growth direction, respectively. The hydrostatic energy shift in the conduction band can be written as

$$P_{ce} = a_{cz}\varepsilon_{zz} + a_{ct}\varepsilon_{xx} + \varepsilon_{yy} \tag{13.2}$$

We take $a_{cz} = a_{ct} = a_c$ for simplicity.

Band structure for the valence band is more complicated. Using the k.p method Chuang and Chang [67] has derived a 6 by 6 Hamiltonian which has been blockdiagonalized in to the following upper and lower Hamiltonians.

$$H^v_{6\times 6}(k) = \begin{bmatrix} H^U_{3\times 3}(k) & 0 \\ 0 & H^L_{3\times 3}(k) \end{bmatrix} \tag{13.3}$$

where

$$H^U = \begin{bmatrix} F & K_t & -iH_t \\ K_t & G & \Delta - iH_t \\ iH_t & \Delta + iH_t & \lambda \end{bmatrix} \tag{13.4}$$

$$H^L = \begin{bmatrix} F & K_t & iH_t \\ K_t & G & \Delta + iH_t \\ -iH_t & \Delta - iH_t & \lambda \end{bmatrix} \tag{13.5}$$

When there is strain, the energy bands of shift in a complicated manner. We need to use a common reference energy level when writing down the Hamiltonian. Following the convention for the wurtzite structure, if E_c^0 is the unstrained conduction band edge, the reference valence band level is as follows:

$$E_v^0 = E_c^0 - (E_g + \Delta_1 + \Delta_2) \tag{13.6}$$

When there is strain, the conduction band is shifted by an amount P_{ce} and the new reference level is:

$$E_v^0 = E_c - (E_g + \Delta_1 + \Delta_2 + P_{ce}) \tag{13.7}$$

Using the above reference energy, the matrix elements are defined as follows:

$$\begin{aligned}
F &= \Delta_1 + \Delta_2 + \lambda + \theta \\
G &= \Delta_1 - \Delta_2 + \lambda + \theta \\
\lambda &= \lambda_k + \lambda_\varepsilon \\
\lambda_k &= \frac{\hbar^2}{2m_0}(A_1 k_z^2 + A_2 k_t^2) \\
\lambda_\varepsilon &= D_1 \varepsilon_{zz} + D_2(\varepsilon_{xx} + \varepsilon_{yy}) \\
\theta &= \theta_k + \theta_\varepsilon \\
\theta_k = &= \frac{\hbar^2}{2m_0}(A_3 k_z^2 + A_4 k_t^2) \\
\theta_\varepsilon &= D_3 \varepsilon_{zz} + D_4(\varepsilon_{xx} + \varepsilon_{yy}) \\
K_t &= \frac{\hbar^2}{2m_0} A_5 k_t^2 \\
H_t &= \frac{\hbar^2}{2m_0} A_6 k_t k_z \\
\Delta &= \sqrt{2}\Delta_3
\end{aligned} \tag{13.8}$$

Please note that the base vectors $|1>$ to $|6>$ of the above can be expressed by spherical harmonics $Y_{lm}(l=1)$ [66]. For the upper Hamiltonian, $|1>$, $|2>$, $|3>$ corresponds to heavy hole (HH), light hole (LH) and crystal field split hole (CH), respectively. At the zone center (k=0), the HH subbands are decoupled from the LH and CH subbands, while there is always a coupling between the LH and CH bands.

In our simulation software, the above Hamiltonian is solved to generate the bulk band structure from which the density of states are evaluated for bulk macros. To achieve maximum efficiency, the bulk valence band structure is fitted to a parabolic band for each of HH, LH and CH bands under any strain condition. Such fitted parabolic bands are used for modeling the non-active bulk regions. For MQW active region, a more rigorous approach is used which we shall describe below.

13.0.3 MQW Model - Effective Mass Approximation

In general, the dispersion of bulk valence bands is non-parabolic and anisotropic with strong mixing or anti-crossing behavior in the direction perpendicular (or transverse) to the c-axis. We need an effective mass model for the computation of density of states and for a simplified MQW model.

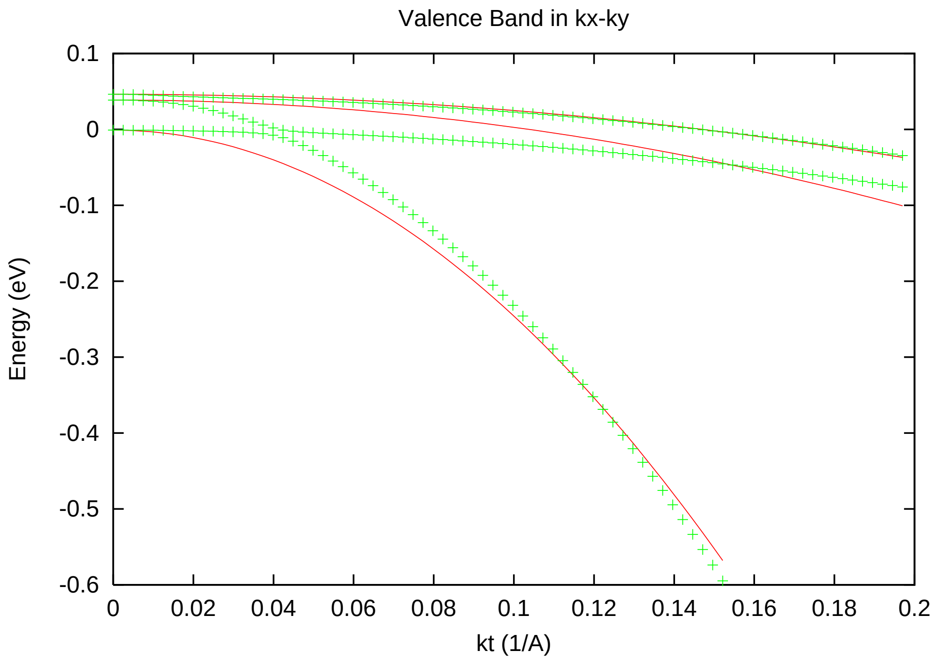


Figure 13.1: Transverse valence bands (points) fitted to effective mass model.

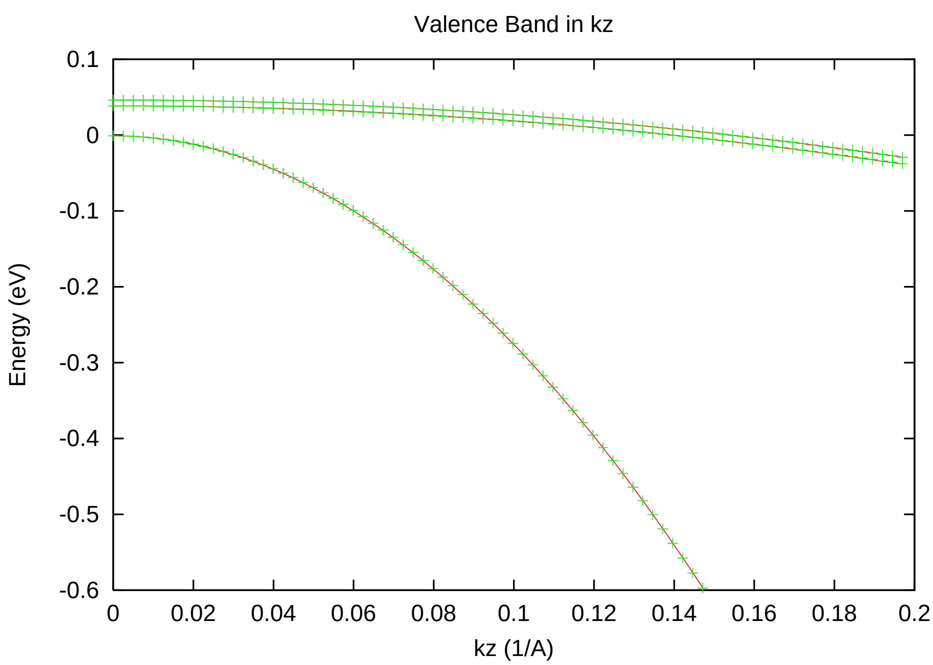


Figure 13.2: Valence bands along c-axis (points) fitted to effective mass model.

An efficient approximation is to take the effective masses fitted from the bulk band structure. This approach takes into account the anti-crossing behavior but the quality of the fit can be poor near some range for the transverse direction (see Fig. 13.1 and Fig. 13.2).

Based on Ref. [67], we have also implemented the following analytical effective mass model for the valence band:

1) Within a range of small k (a situation when the valence band is lightly populated by holes), the following effective masses hold:

$$\begin{aligned} m0/m^z_{hh} &= -(A_1 + A_3) \\ m0/m^z_{lh} &= -\left[A_1 + \left(\frac{E^0_2 - \lambda_e}{E^0_2 - E^0_3}\right) A_3\right] \\ m0/m^z_{ch} &= -\left[A_1 + \left(\frac{E^0_3 - \lambda_e}{E^0_3 - E^0_2}\right) A_3\right] \end{aligned} \tag{13.9}$$

$$\begin{aligned} m0/m^t_{hh} &= -(A_2 + A_4) \\ m0/m^t_{lh} &= -\left[A_2 + \left(\frac{E^0_2 - \lambda_e}{E^0_2 - E^0_3}\right) A_4\right] \\ m0/m^t_{ch} &= -\left[A_2 + \left(\frac{E^0_3 - \lambda_e}{E^0_3 - E^0_2}\right) A_4\right] \end{aligned} \tag{13.10}$$

where $E^0_i (i = 1, 2, 3)$ are the valence band edges at k=0. The parabolic bands of this model are shown in Figs. 13.3 and 13.4.

2) For a large range of k (a situation when the valence band is heavily populated by holes), the following effective masses formulas are valid:

$$\begin{aligned} m0/m^z_{hh} &= -(A_1 + A_3) \\ m0/m^z_{lh} &= -(A_1 + A_3) \\ m0/m^z_{ch} &= -A_1 \end{aligned} \tag{13.11}$$

$$\begin{aligned} m0/m^t_{hh} &= -(A_2 + A_4 - A_5) \\ m0/m^t_{lh} &= -(A_2 + A_4 + A_5) \\ m0/m^t_{ch} &= -A_2 \end{aligned} \tag{13.12}$$

The parabolic bands of this model are shown in Figs. 13.5 and 13.6.

3) A compromise of the above two models is to average them. The parabolic bands with average masses are shown in Figs. 13.7 and 13.8.

The default setting in Crosslight software is 3) above. The user may adjust the setting of the effective mass model through statement **modify_wurtzite** . The choice of model is based on how heavily the valence bands are populated by holes, or the amount of hole density.

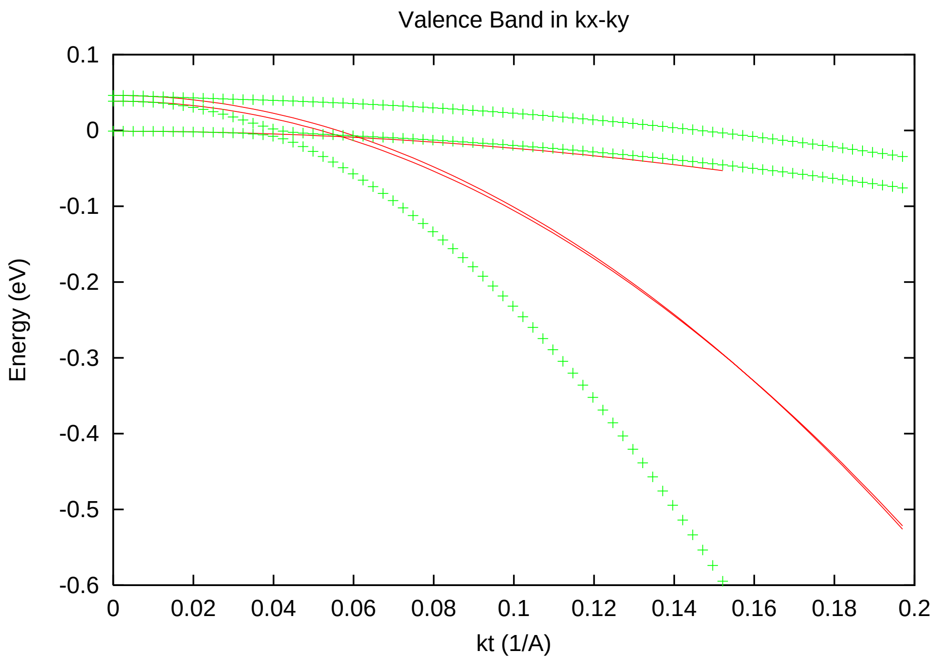


Figure 13.3: Transverse valence bands (points) as compared with analytical effective mass model of small k-range.

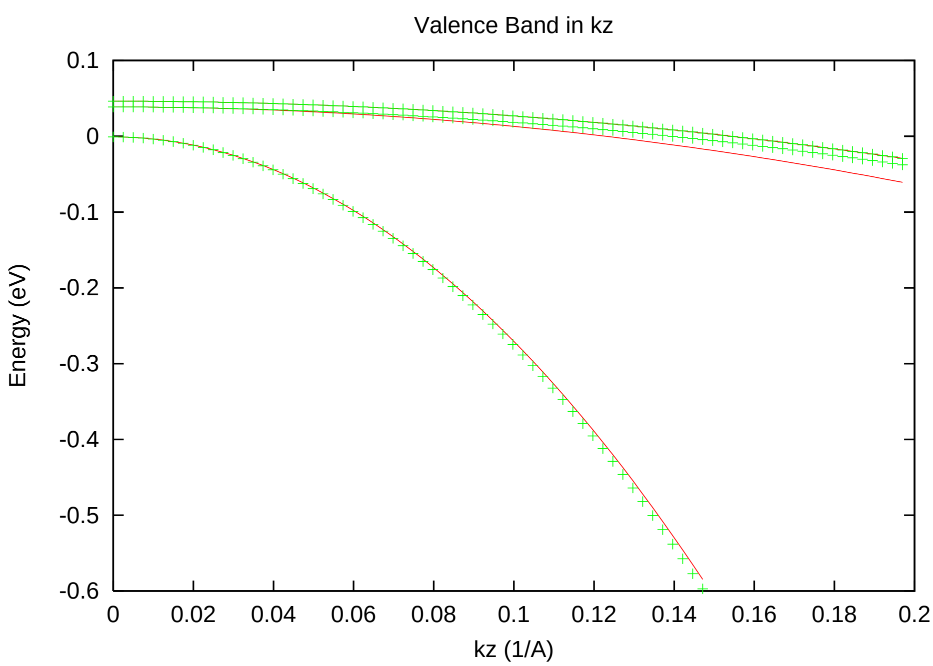


Figure 13.4: c-axis valence bands (points) as compared with analytical effective mass model of small k-range.

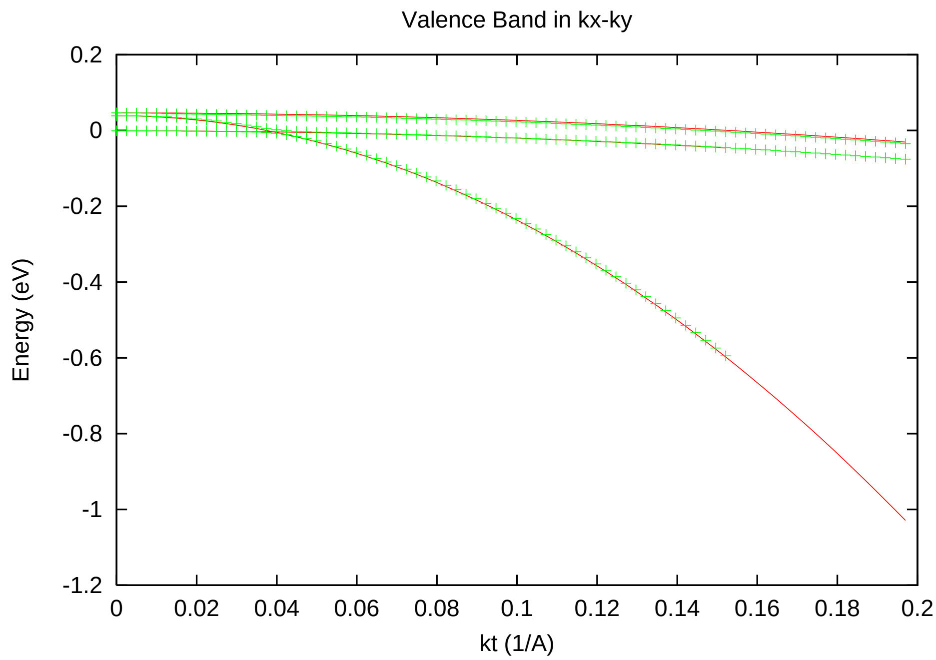


Figure 13.5: Transverse valence bands (points) as compared with analytical effective mass model of large k-range.

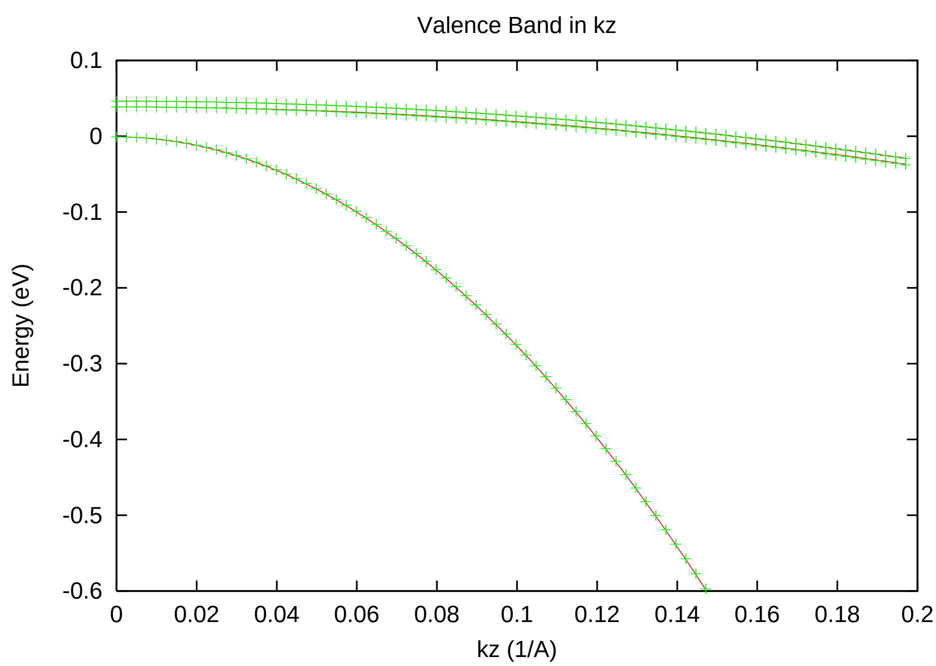


Figure 13.6: c-axis valence bands (points) as compared with analytical effective mass model of large k-range.

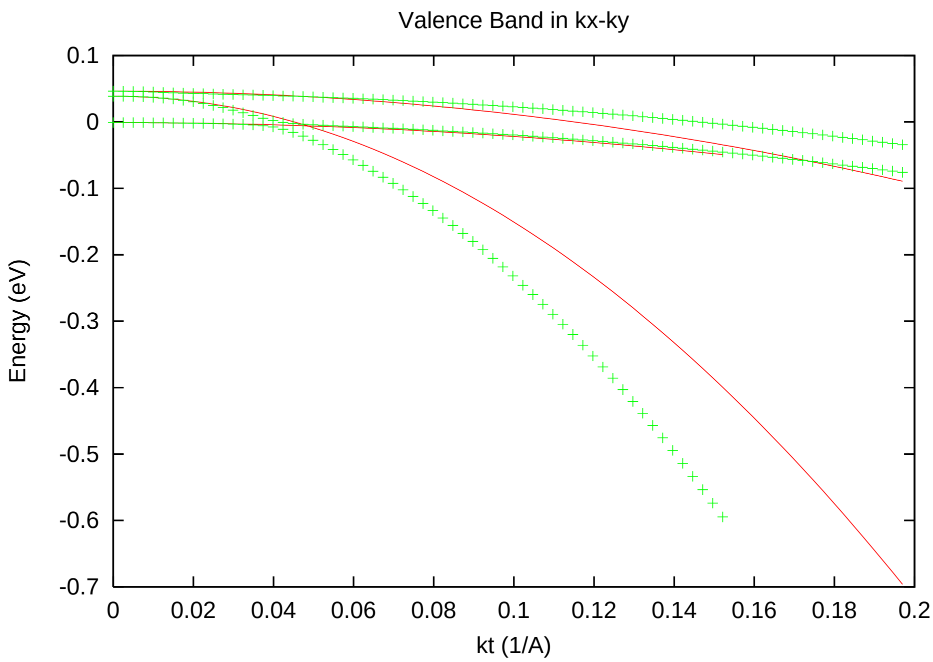


Figure 13.7: Transverse valence bands (points) as compared with analytical effective mass model of averaged masses.

Since the crystal symmetry of wurtzite is different from that of zincblende, the dipole moment in effective mass approximation for zincblende can not be used here and we need to use more accurate models, such as k.p theory, to compute the dipole moment for optical transitions. However, study of results in k-dependent dipole moment from k.p theory for a large number of structures leads us to propose the following simplified dipole moment enhancement factors:

$$\begin{aligned} A_{hh} &= 3/2 \\ A_{lh} &= \frac{3cos^2(\theta_e)}{2} \\ A_{ch} &= 0 \end{aligned} \tag{13.13}$$

for TE and

$$\begin{aligned} A_{hh} &= 0 \\ A_{lh} &= 3 - 3cos^2(\theta_e) \\ A_{ch} &= 3 \end{aligned} \tag{13.14}$$

for TM. $cos^2(\theta_e)$ is defined in the same way as in zincblende. Please note that we formulate the above factors in such a way that the following conservation rule is obeyed:

$$2 \times A_h^{TE} + A_h^{TM} = 3, (h = hh, lh, orch) \tag{13.15}$$

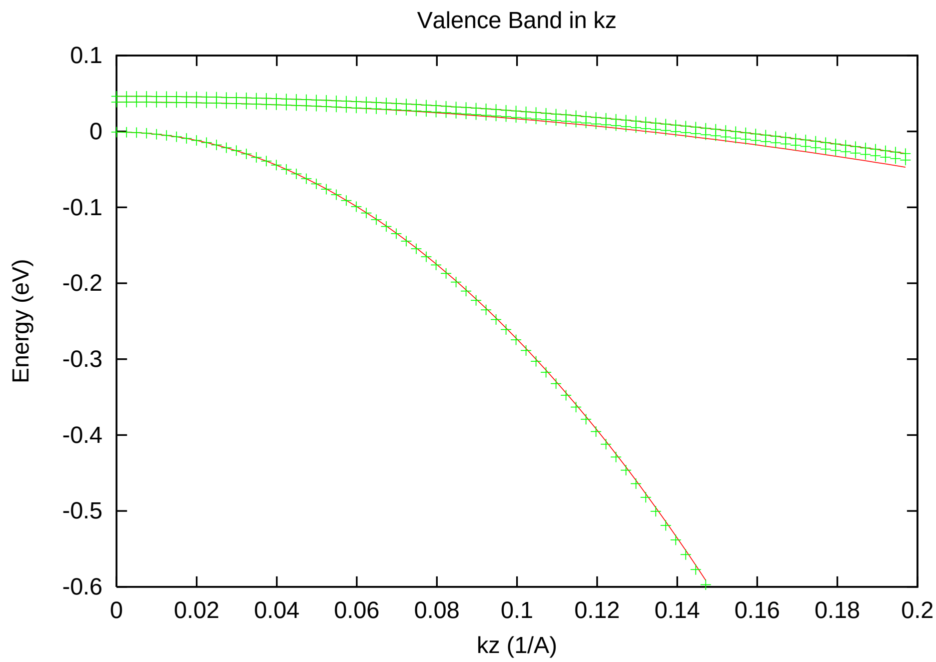


Figure 13.8: c-axis valence bands (points) as compared with analytical effective mass model of averaged masses.

The dipole moment is expressed as

$$\begin{aligned} M_{hh} &= A_{hh} O_{ij} M_b \\ M_{lh} &= A_{lh} O_{ij} M_b \\ M_{ch} &= A_{ch} O_{ij} M_b \end{aligned} \tag{13.16}$$

where O_{ij} is wave function overlap integral and M_b is bulk dipole moment given by [66] as follows:

$$\begin{aligned} M_b^{TM} &= \frac{m_0}{6} E_{pz} \\ E_{pz} &= \left(\frac{1}{m_e^z} - 1 \right) \frac{(E_g + \Delta_1 + \Delta_2)(E_g + 2\Delta_2) - 2\Delta_3^2}{E_g + 2\Delta_2} \end{aligned} \tag{13.17}$$

$$\begin{aligned} M_b^{TE} &= \frac{m_0}{6} E_{px} \\ E_{px} &= \left(\frac{1}{m_e^t} - 1 \right) \frac{E_g[(E_g + \Delta_1 + \Delta_2)(E_g + 2\Delta_2) - 2\Delta_3^2]}{(E_g + \Delta_1 + \Delta_2)(E_g + \Delta_2) - \Delta_3^2} \end{aligned} \tag{13.18}$$

13.0.4 MQW Model – Valence Mixing

The valence mixing treatment is given in this subsection. Within the envelope function approximation, we write the wave function as follows.

$$\Psi_m^U(z;k_t) = \frac{e^{i\mathbf{k_t}\cdot\mathbf{r_t}}}{\sqrt{A}}\left(g_m^{(1)}(z;k_t)|1> + g_m^{(2)}(z;k_t)|2> + g_m^{(3)}(z;k_t)|3>\right) \tag{13.19}$$

$$\Psi_m^L(z;k_t) = \frac{e^{i\mathbf{k_t}\cdot\mathbf{r_t}}}{\sqrt{A}}\left(g_m^(4)(z;k_t)|4> + g_m^{(5)}(z;k_t)|5> + g_m^{(6)}(z;k_t)|6>\right) \tag{13.20}$$

The valence subbands are determined by the following coupled differential equations:

$$\sum_{j=1}^{3}\left(H_{ij}^U\left(k_z = -i\frac{\partial}{\partial z}\right) + \delta_{ij}E_v^0(z)\right)g_m^{(j)}(z;k_t) = E_m^U(k_t)g_m^{(i)}(z;k_t) \tag{13.21}$$

The valence band discontinuity is represented by discontinuity in $E_v^0(z)$ (the reference energy). Similar equations can be written down for the lower Hamiltonian. It can be shown that MQW with reflection symmetry $E_v^0(z) = E_v^0(-z)$, the upper and lower Hamiltonians have the same band structures.

In our simulation software, we use finite difference method to solve the coupled differential equations. This gives us the valence mixing subbands.

The dipole moments, taking into account the upper and lower Hamiltonian degeneracy, are given as follows:

$$M_{nm}^{TE} = M_b\frac{3}{4}\left[<\phi_n|g_m^{(1)}>^2 + <\phi_n|g_m^{(2)}>^2\right] \tag{13.22}$$

$$M_{nm}^{TM} = M_b\frac{3}{2}<\phi_n|g_m^{(3)}>^2 \tag{13.23}$$

13.0.5 Nomenclature

We list some critical symbol definitions here because these are specific to wurtzite band structure.

a, c	Lattice constants of Hexagonal structure.
E_g	Bandgap.
E_g	Bandgap.
$\Delta_1, \Delta_2, \Delta_3$	Energy parameters.
Δ_{so}	Spin-orbit coupling energy.
m_e^z, m_e^t	Conduction band effective masses along c-axis (z) and transverse (x-y) direction, respective.
$A_i, (i = 1, ..., 6)$	Valence band effective mass parameters.
a_h, a_c, a_v	Hydrostatic deformation potentials of total, conduction, and valence parts, respectively.
$D_i, (i = 1, ..., 4)$	Valence band shear deformation potentials.
C_{13}, C_{33}	Elastic stiffness constants.

13.1 Nomenclature

A_x	Coefficient for band-gap narrowing effects.
a_0, a	Lattice constants of GaAs (a_0) and InGaAs (a).
B	Radiative recombination coefficient.
B_1,B_2,B_z	Complex wave amplitudes of the optical wave propagating from right to left in a laser.
B_k	Constant used in mathematical fittings.
b	Shear deformation potential.
C_n, C_p	Auger recombination coefficients, for electrons (n) and holes (p).
c_{nj}, c_{pj}	Electron (n), hole (p) capture coefficients of the j th deep trap.
c	Velocity of light in vacuum.
c_{11}, c_{12}	Elastic constants.
D_c, D_v	Density of states of the conduction and valence band.
E_D, E_A	Shallow donor (D) and acceptor (A) levels.
E_i^0, E_j^0, E_{ij}^0	Energy minimums of the i th and the j th levels, and the difference between them, . The i refers to the valence band sub-band levels and j refers to the conduction band sub-band levels.
E, E_{ij}	Photon energy and energy difference between the i th and j th levels.
E_{fn}, E_{fp}	Quasi-Fermi energies of electrons (n) and holes (p).
E_g	Bandgap.
E_{g0}	Unstrained bandgap.
ΔE_{PF}	Shift in ionization energy of a dopant due to Poole-Frenkel effect.
δE_{hy}	Hydrostatic strain energy.
δE_{sh}	Energy associated with shear strain.
F	Electric field intensity.
F_l	Lorentzian shape function.
F_{0n},F_{0p}	Threshold electric field used in the electron (n) and hole (p) mobility models.
$F_{1/2}$	Fermi integral of order one-half.
f_D, f_A	Occupancy of the donor (D) and acceptor (A) levels.
f_{tj}	Occupancy of the j th deep trap level.
f_i, f_j	Fermi functions for the i th and the j th level.
f_i', f_j'	Integrated Fermi functions for the i th and the j th levels.

g	Interband local gain.
g_{ij}, g	Local gain due to transition between the i th and the j th levels, and the local gain of the material.
g_d, g_a	Degeneracies of the shallow donor (d) and acceptor (a).
H_{ij}	Hamiltonian matrix element between the i th and the j th states.
$h(x)$	step function of variable x.
h,$\hbar$	Plank's constants.
I_1, I_2	The integrals needed to evaluate the quantum well gain and spontaneous emission rate.
J_n, J_p	Electron (n) and hole (p) current flux densities.
J_{sn}, J_{sp}	Electron (n) and hole (p) current flux densities on the surface.
J_{hn}, J_{hp}	Electron (n) and hole (p) current flux densities across the heterojunction.
J_{source}	Current flux source.
k	Boltzmann constant.
L	Laser cavity length.
$L()$	Lines shape function for gain broadening.
m_0	Electron mass
m_i, m_j, m_{ij}	Relative effective masses of the i th and the j th level and the reduced effective mass between the i th and the j th level. Their relation is defined by $1/m_{ij} = 1/m_i + 1/m_j$. Indices i and j refer here to quantum well sub-bands.
m_n,m_p	Bulk effective masses for electrons (n) and holes (p).
m_e,m_h	The same as m_n and m_p, respectively.
m_{bn},m_{bp}	Bulk effective masses for electrons (n) and holes (p) on the barrier side of the heterojunction.
m_{zl}	Relative effective mass of the L-band used in the quantum level calculation.
m_{vx}, m_{vy}, m_{vz}	Effective hole masses in the x, y and z direction.
M_0, M_{ij}, M_{lh}, M_{hh}	Momentum matrix elements for bulk material, between the i th and the j th states, involving light- and heavy-hole transitions, respectively.
N	Total number of grid points in the simulation space. Also used for the total number of longitudinal layers.
N_b	Number of grid points associated with a boundary of interest.
N_D, N_A	Doping density of shallow donors (D) and shallow acceptors (A).
N_{tj}	Density of the j th deep trap.
n	Electron concentration or density.
n_b	Electron concentration or density on the barrier side of the hetero-junction.
n_{b0}	Electron concentration or density on the barrier side of the hetero-junction when its quasi-Fermi level coincides with that on the other side.

n_s	Electron concentration or density on the surface.
$\overline{n}$	Real part of refractive index.
n_i	Intrinsic carrier density.
n_{2D}	The surface concentration of a quantum well, equal to the bulk density times the layer thickness.
N_{rn}, N_{rp}	Reference doping density in electron (n) and hole (p) mobility equations.
P_{ij}	Probability of a transition from the i th to the j th level.
p	Hole concentration or density.
p_b	Hole concentration or density on the barrier side of the heterojunction.
p_{b0}	Hole concentration or density on the barrier side of the heterojunction when its quasi-Fermi level coincides with that on the other side.
p_s	Hole concentration or density on the surface.
Q^j_{em}	Energy density of the jth emitted longitudinal mode.
Q^j_{cav}	Energy density of the jth longitudinal mode within the laser cavity.
q	Electronic charge.
R^{tj}_n, R^{tj}_p	Electron (n) and hole (p) recombination rates per unit volume through the j th deep trap.
R_{sp}, R^b_{sp}, R^{qw}_{sp}	Spontaneous recombination rate per unit volume, also called the radiative recombination rate, and the same quantity in bulk (b) and in the quantum well (qw).
r^{qw}_{sp}	Frequency dependent spontaneous recombination rate for the quantum well.
R_s	Resistance associated with a boundary condition.
R_{st}	Stimulated recombination rate per unit volume.
R_{au}	Auger recombination rate per unit volume.
r_m,r^{eff}_m	Single facet reflectivity and effective reflectivity of a laser.
T	Absolute temperature.
T_{scat}	The scattering kernel for carrier-carrier scattering.
t	Thickness of the quantum well.
V	Electrical potential.
V_s	Electrical potential on the surface.
$V_{applied}$	Applied electrical potential.
v	Volume.
$\overline{v}_n$, $\overline{v}_p$	Average thermal velocity of electrons (n) and holes (p). The average is taken over all three dimensions.
$\overline{v}^{therm}_n$, $\overline{v}^{therm}_p$	Average thermal velocity of electrons (n) and holes (p). The average is taken only over the dimensional of interest.
$\overline{v}^{therm}_{bn}$, $\overline{v}^{therm}_{bp}$	Average thermal velocity of electrons (n) and holes (p) located on the barrier side of the heterojunction. The average is taken only over the dimensional of interest.
v_{sn},v_{sp}	Saturation velocity for electrons (n) and holes (p).
W	Carrier energy.
W_n	Electron energy.

W_n	Hole energy.
W_j^e, W_i^h	Effective width of the wave functions for electron and hole in the jth and ith subbands, respectively.
x, y	Spatial coordinates used in the model. The x-axis is parallel to the quantum well. x is also used elsewhere as the Al composition in AlGaAs or the In composition in InGaAs.
α α_{qw}	Local loss coefficient due to loss other than interband recombination, and local loss in the quantum well.
α_{int}	Internal loss defined as the weighted average of the local loss α.
α_n, α_p	Constants used in doping dependent mobility functions for electrons (n) and holes (p).
α_{fn}, α_{fp}	Coefficients of the free carrier absorption for electrons (n) and holes (p) in the quantum well.
α_0	Absorption coefficient for regions outside the quantum well.
β, β_1, β_2	Complex eigenvalues of the wave equation and its real (1) and imaginary (2) parts. They are also considered effective indices.
α_{em}	Optical loss coefficient due to emission of light in the lasing mode.
α_{emj}	Optical loss coefficient due to emission of light in the jth longitudinal mode.
β_p	Constant used in the hole mobility equation.
Δ	Spin-orbit splitting of the valence band energy.
Δ_{ij}^a	Transition energy shift term in the Asada broadening model.
δ	Constant. Value is 1 for deep donors and 0 for deep acceptors when used to represent electrical charge. Also used as a δ-function.
ϵ_0	Dielectric constant of vacuum.
ϵ_{dc}	Relative DC or low frequency dielectric constant.
ϵ, ϵ_1, ϵ_2	Complex optical dielectric constant and its real (1) and imaginary (2) parts. ϵ is also used, under a completely context, for non-linear gain coefficient.
ϵ_2^g, ϵ_2^α	Imaginary parts of the dielectric constant corresponding to gain due to interband transition and loss due to other mechanisms, respectively.
ε	Strain.
Γ	Energy level broadening parameter, or the half width of the Lorentzian broadening function.
Γ_0	The maximum energy level broadening parameter, or the half width of the Lorentzian broadening function.
Γ_a	The energy level broadening parameter, or the half width of the Lorentzian broadening function, as used in the Asada's broadening model.
$\gamma_1, \gamma_2, \gamma_3$	Luttinger numbers.

γ	Constant relating the optical wave amplitude to electric field.
γ_n,γ_p	Adjustable constants for the electron (n) and hole (p) thermionic emission model, used to account for tunneling and other effects.
γ_{hn},γ_{hp}	Adjustable constants for the electron (n) and hole (p) thermionic emission model for heterojunctions, used to account for tunneling and other effects.
χ	Affinity.
χ	Affinity of a reference material.
λ	Optical wavelength.
λ_j	Optical wavelength of the jth longitudinal mode.
μ_n, μ_p	Mobility of electrons (n) and holes (p).
μ_{0n}, μ_{0p}	Low field electron (n) and hole (p) mobilities.
μ_{1n}, μ_{1p}	Minimum electron (n) and hole (p) mobilities.
μ_{2n}, μ_{2p}	Maximum electron (n) and hole (p) mobilities.
ν	Function used in the Fermi-Direct statistics.
ρ_i, ρ_j, ρ_{ij}	Density of states for the i th and the j th levels, and the reduced density of states for transition between the j th and the i th levels.
ϕ_n,ϕ_p	Quasi-Fermi potential for electrons (n) and holes (p).
ϕ_b	Schottky barrier height.
ρ_i^0, ρ_j^0, ρ_{ij}^0	Constants for the two-dimensional density of states for the i th and the j th levels, and reduced density of states for transition between the i th and the j th levels. For example $\rho_i^0 = m_i/\pi\hbar^2 t$.
σ_{nj}, σ_{pj}	Electron (n) and hole (p) capture cross sections of the j th deep trap.
τ	Intra-band scattering lifetime.
τ_{nj},τ_{pj}	Electron (n) and hole (p) life time due to the jth trap.
τ_h, τ_v	Hole-hole scattering life time for the Asada gain broadening model. $\tau_h = 2\tau_v$.
ω	Angular frequency of the optical wave.
ω_a,ω_β	SOR parameters used for the wave equation.
ξ	Function used in the Fermi-Dirac statistics.
ζ, ζ_n, ζ_p	Variable related to the effective masses. ζ is also used for half the shear strain energy for a different context.
$<\ \ >$	Average with $\mid W \mid^2$ as weight.

Part III

PRODUCT SPECIFIC MODELS

Chapter 14

LASTIP SPECIFIC MODELS

14.1 Introduction

This chapter provides a description of the theory specific to LASTIP program. Wr shall describe the photon rate equation for edge emitting semiconductor lasers. Some fine details such as far field computation and optically pumped laser modeling are also described.

14.2 Stimulated Emission and Rate Equation

This section describes the stimulated emission rate and the rate equation from the definition of group velocity as the speed of photon energy movement along the laser cavity (along the z-direction).

The stimulated emission rate, R_{st} is defined as the rate of photon emission per unit volume as a result of material gain experienced by the traveling light wave. By energy conservation, the emitted photon must cause an increase in the power, P, of the traveling wave by the same amount:

$$dP = dz(\hbar\omega) \int R_{st} dx dy \tag{14.1}$$

Since modal gain g_m relates to the power increase by the following relation:

$$g_m = \frac{1}{P} \frac{dP}{dz} \tag{14.2}$$

we obtain the following expression:

$$g_m P = (\hbar\omega) \int R_{st} dx dy \tag{14.3}$$

Using group velocity as the traveling speed of the optical power, we obtain

$$P = \hbar\omega v_g S \tag{14.4}$$

$$\frac{1}{v_g} = \frac{\partial k}{\partial \omega} \tag{14.5}$$

where S is the linear photon density S (also called photon number in the 2D simulation plane). We introduce the normalized 2D optical wave function W so that

$$\int \mid W \mid^2 dxdy = 1 \tag{14.6}$$

and

$$g_m = \int g(x,y) \mid W \mid^2 dxdy \tag{14.7}$$

Note that the wave function W has been normalized such that the above integral yields the Γ factor if the local gain is uniform in the active region.

From Eqs (14.3) to (14.7) we obtain

$$\int R_{st} dxdy = \int v_g S g(x,y) \mid W \mid^2 dxdy \tag{14.8}$$

which implies our result

$$R_{st} = v_g g(x,y) S \mid W \mid^2 \tag{14.9}$$

or in terms of the bulk photon density s:

$$R_{st} dxdy = v_g g s \tag{14.10}$$

In a 2D simulation, Eq. (14.9) is used more often.

The rate equation for the linear photon density S (also called photon number in the 2D simulation plane) in the laser cavity may be viewed as a statement of conservation of energy, *i.e.*

$$\begin{aligned} stimulated\ emission \quad & - internal\ loss \quad - emitted\ power \\ +spontaneous\ emission \quad = \quad & power\ increase \end{aligned} \tag{14.11}$$

The "power increase" in the above equation is the rate of linear photon density ($\partial S/\partial t$). Equation (14.11) can be detailed as

$$stimulated\ emission = \int R_{st} dxdy = v_g g_m S \tag{14.12}$$

Internal loss is defined as the modal loss experienced by the mode of the traveling wave. This loss term is usually denoted by α_i and it can often be determined experimentally. Similar to modal gain, the internal loss can be written as

$$\alpha_{int} \approx \int \mid W \mid^2 \alpha dxdy \tag{14.13}$$

Defining an effective loss coefficient α_{em} due to emitted power loss from the device, the emitted power in Equation (14.11) can be written in the same form as Equation (14.12):

$$emitted\ power = v_g \alpha_{em} S \tag{14.14}$$

In the special case of a Fabry-Perot cavity, the emitted power loss equals the facet mirror loss, and we obtain the following familiar formula:

$$emitted\ power = v_g S \frac{1}{2L} ln\left(\frac{1}{r_{m1} r_{m2}}\right). \tag{14.15}$$

where r_{m1} and r_{m2} are the power reflectivities of the two facets.

The power due to spontaneous emission is simply the integration of the spontaneous emission rate over the cavity:

$$spontaneous\ emission = c_m \int R_{sp} dx dy, \tag{14.16}$$

where c_m is the fraction of spontaneous emission coupled into the lasing mode.

Summation of the above terms yields the familiar rate equation:

$$v_g(g_m - \alpha_{int} - \alpha_{em})S + c_m \int R_{sp} dx dy = \partial S/\partial t \tag{14.17}$$

The derivation given above makes it clear that LASTIP is only concerned with the total number of photons in the cavity so that it can compute the stimulated recombination rate used in the basic equations. The rate equation averages out most details of longitudinal distribution of the optical fields. Therefore the rate equation is expected to be accurate if the z-dimension is uniform.

In cases where strong variation of physical variable along the z-direction, such as DFB and DBR lasers, the rate equation will be inaccurate for a complete description of the laser cavity. However, the 2D rate equation may still be used as a phenomenological model if proper values of effective mirror reflectivity can be chosen. For example, if the mirror reflectivity of the DBR mirror is known, the rate equation in this section, thus the LASTIP software, can still be used.

14.3 Far-Field Computation

The far-field beam distribution can be computed from the near-field distribution. The problem is basically that of a diffraction: *i.e.*, given the distribution of a propagating beam at a 2D cross section, calculate the distribution at a distance from this cross section. The formulation for this problem has been well established in classical electrodynamics and the user is referred to standard textbooks on electrodynamics.

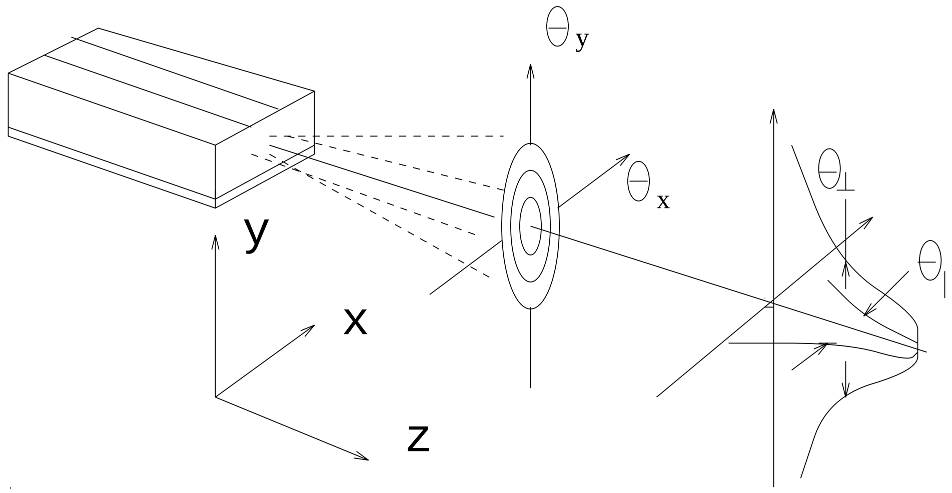


Figure 14.1: Schematic representation of the far-field emission of a semiconductor laser.

14.3.1 An example of far-field computation

A semiconductor laser emits light in the form of a narrow spot of elliptical cross section. The spatial-intensity distribution of the emitted light near the laser facet is known as the *near-field*. During its propagation, the spot grows in size due to beam divergence. The dimensions of the elliptical spot and its divergence angles are important beam parameters associated with the laser mode. The *far-field* pattern is the angular distribution of beam intensity far from the laser facet. It is an important factor in the design of laser geometries since it directly affects the coupling of power from the laser source to the object being illuminated.

Figure 14.1 gives a schematic representation of the far-field emission of a semiconductor laser. The full angles at half power are $\theta_\perp$ and $\theta_{||}$ in the directions perpendicular to and along the laser junction plane, respectively. Typically, $\theta_{||}$ is of the order of 10°, whereas $\theta_\perp$ is considerably larger (35 to 65°), depending on the detailed near-field distribution.

LASTIP solves the wave equation and the semiconductor equations self-consistently and yields accurate near-field distribution as the routine output data. This example describes the far-field model as an additional option of LASTIP.

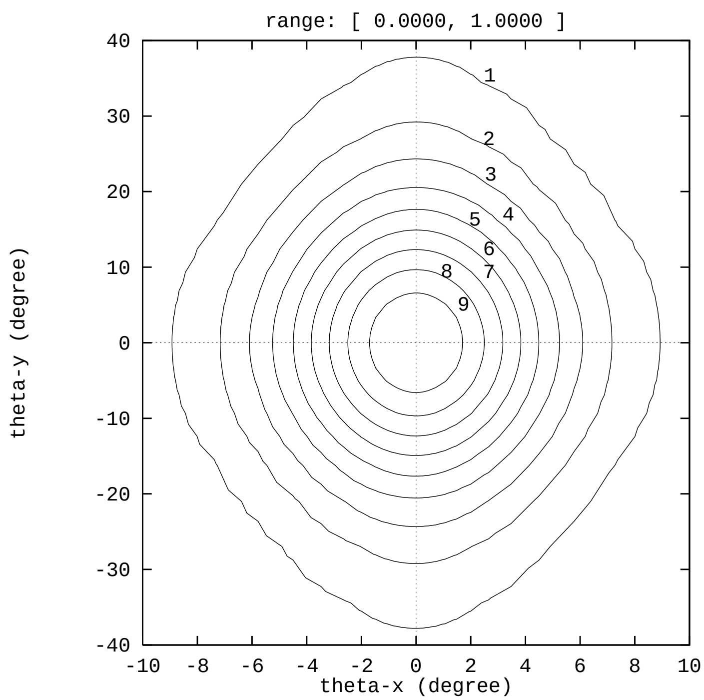


Figure 14.2: An example of the far-field emission pattern from a non-planar ridge waveguide GaAs/AlGaAs laser emitting at 0.78 μm.

Refs. [1] and [8] have given mathematical formulas for the far-field distribution based on the solution to the wave equation in free space. Since these mathematical expressions were derived for broad-area lasers only (*i.e.*, assuming the x-dimension is infinite), LASTIP has extended their formulas to an arbitrary two-dimensional situation using the same solution approach as suggested in [1] and [8]. This feature allows the user to obtain the far-field distribution immediately after the near-field is computed.

As an example, we consider a ridge waveguide GaAs/AlGaAs laser emitting at about 0.78 μm. It has a nonplanar ridge which makes it necessary to use the finite element feature of LASTIP to handle its complex geometry. The near-field pattern was solved using a self-consistent approach for the wave equation, the semiconductor equations, and the rate equation altogether.

The far-field pattern is computed based on the near-field solution and is presented in Figure 14.2 as a contour plot of the far-field power. The contour lines are evenly spaced in power level (at 10% power intervals). The angles $\theta_\perp$ and $\theta_{||}$ can be directly extracted from the contour plot.

To activate the far-field option as a post-processing step (in .plt file), the solution generated by running "*.sol" is required.

To summarize, a useful option has been made available to LASTIP users for the far-field behavior of a semiconductor laser. It is capable of accurately computing the far-field emission patterns from lasers of arbitrary geometries and materials.

14.4 Non-Linear Gain Model

Conventional optical gain is a concept based on linear response theory, *i.e.*, only first order perturbation is used to treat a system under an external optical field. For semiconductor lasers, it is often necessary to include the higher order effects (or non-linear effects). The optical gain tends to decrease as a function of photon density when non-linear effects are included. Such an effect is also called gain suppression.

The non-linear gain effect is responsible for the spectral hole burning effect in semiconductor lasers [8].

The non-linear gain effect is commonly expressed as [8].

$$g = g_0(1 - \epsilon S). \tag{14.18}$$

where ϵ is the non-linear gain suppression coefficient which is not to be confused with the dielectric constant, and S is the photon density. Since the correction term is typically of the order of a few percent, one can use the following alternative form:

$$g = \frac{g_0}{1 + \epsilon S} \tag{14.19}$$

Such a formula has the numerical advantage that the correction term can never make the optical gain negative. Equation (14.19) has the same format as the gain saturation for a simple, two-level system [45].

In the case of multi-lateral modes, the photon density S is replaced by the sum of the densities of all the modes.

14.5 Temperature Variation Model

LASTIP allows the user to vary the isothermal temperature. The following material parameters are temperature dependent: band gap, mobility, nonradiative recombination, optical gain, carrier distribution, *etc*. LASTIP allows the user to incorporate all possible temperature variations into a device simulation.

Since the temperature dependence of many of the compounds used for semiconductor lasers are not well known, LASTIP assumes that all materials grown on GaAs have the same temperature dependence as GaAs, and all materials grown on InP have the same temperature dependence as InP. Obviously these assumptions are inaccurate.

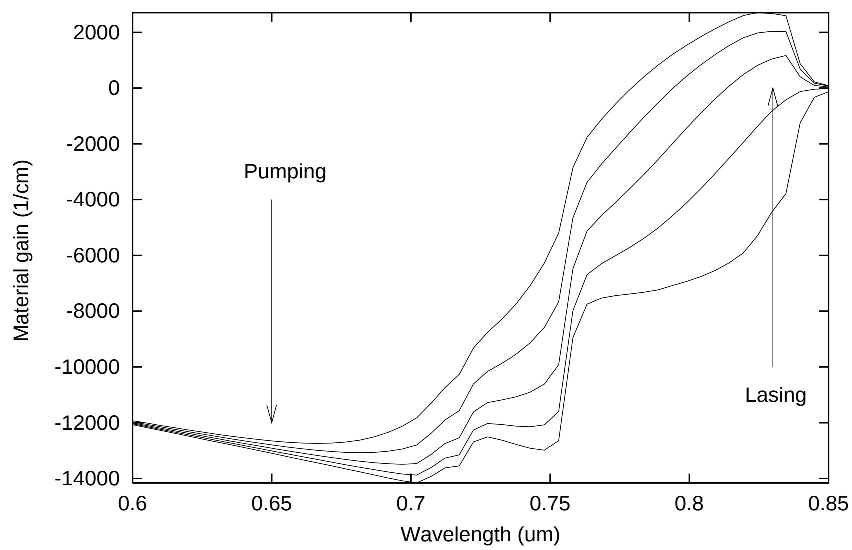


Figure 14.3: Indication of simutaneous pumping and lasing operation on the same gain curve.

LASTIP still implements such a temperature dependence in the hopes that it will serve as a convenient example to allow users to modify the temperature dependence in the material macro library.

14.6 Optically Pumped Laser

It is possible to simulate optically pumped laser if we allow the semiconductor material to respond to two different wavelengths. As illustrated in Fig. 14.3, under a high injection condition, it is always possible to find two different regimes on the same curve with opposite signs of the gain/loss.

For the same active region, if light of shorter wavelength is absorbed, it is possible to have positive gain at the same time. Thus the carrier generated from the absorption may be used to provide the carrier injection needed for population inversion for lasing action at longer wavelength.

Please note that if the wavelength is much shorter than the lasing wavelength, it is possible for passive layers (cladding layers) to absorb light and produce photocarriers. Thus, the user should determine which layers are involved in interband optical transition. These layers should be declared as "active" even if some of them do not generate positive gain. Once a semiconductor layer is declared "active", the simulator will automatically determines the sign and intensity of the interband optical transitions. Photo-carriers are generated automatically if absorption occurs.

Chapter 15

APSYS SPECIFIC MODELS

This chapter provides a description of the theories unique to the APSYS simulation software. Physical and numerical models common to all simulation packages of Crosslight are detailed in a separate **Crosslight Software General Description**.

15.1 Time-Dependent Traveling Wave Equations

15.1.1 The Coupled Mode Theory

In this section we consider a time-dependent (or large signal model) situation where the mode amplitude varies with time as a result of external time dependent bias. We assume that the external bias source has a frequency much smaller than the optical frequency ω. This assumption is reasonable since the typical optical frequency is 3×10^{14} Hz, while a typical external modulation biasing source is no more than 10 GHz (1×10^{10} Hz).

We derive the large signal model along the same lines as for CW. We consider a propagating field of the following form:

$$\begin{aligned} E(y,z,t) &= [R(z,t)exp(-j\beta_0 z) + L(z,t)exp(j\beta_0 z)]\phi(y)exp(j\omega t) \\ &+ \Delta E(y,z,t)exp(j\omega t) \end{aligned} \tag{15.1}$$

The optical oscillation term $exp(j\omega)$ did not appear in the equations for CW because they all cancel out. As we mentioned earlier, the mode amplitude varies slowly. Therefore, we are able to decouple the slow and fast terms and only consider the component for one slow frequency $\Omega << \omega$. That is, we can replace $R(z,t)$ by $R_\Omega(z)exp(j\Omega t)$. Once we have solved $R_\Omega(z)$, the original $R(z,t)$ can be obtained by an inverse Fourier transfer:

$$R(z,t) = \int R_\Omega(z)exp(j\Omega t)d\Omega \tag{15.2}$$

Similar arguments can be used for all other slow varying quantities. The propagating field becomes:

$$\begin{aligned} E(y,z,t) &= [R_\Omega(z)exp(-j\beta_0 z) + L_\Omega(z)exp(j\beta_0 z)]\phi(y)exp[j(\omega+\Omega)t] \\ &+ \Delta E_\Omega(y,z)exp[j(\omega+\Omega)t] \end{aligned} \tag{15.3}$$

Formally, the above expression is identical to that for CW except ω is replace by $(\omega+\Omega)$. Since the slow frequency Ω is much smaller than ω, all the derivation steps for CW applies to our large signal analysis here. All we need to do is to replace ω by $(\omega+\Omega)$ everywhere in the derivation. We end up with the following coupled mode theory for one slow frequency component:

$$\left\{\frac{\partial}{\partial z} + j\frac{\Delta\Omega}{v_g} + j\frac{\Delta\omega}{v_g} - \frac{g_m-\alpha_i}{2} + h_1\right\} R_\Omega(z) + (h_1 + jh_2)L_\Omega(z) = 0 \tag{15.4}$$

$$\left\{\frac{\partial}{\partial z} - j\frac{\Delta\Omega}{v_g} - j\frac{\Delta\omega}{v_g} + \frac{g_m-\alpha_i}{2} - h_1\right\} L_\Omega(z) - (h_1 + jh_2)R_\Omega(z) = 0 \tag{15.5}$$

where g_m is the modal gain. h_1 and h_2 are coupling coefficient from 1st and 2nd (if applicable) oder diffractions. To perform the inverse Fourier transform, we multiply the above equations with $exp(j\Omega t)$ and integrate over Ω. We obtain the following time dependent coupled wave equations:

$$\left\{\frac{\partial}{\partial z} + \frac{1}{v_g}\frac{\partial}{\partial t} + j\frac{\Delta\omega}{v_g} - \frac{g-\alpha_i}{2} + h_1\right\} R(z,t) + (h_1 + jh_2)L(z,t) = 0 \tag{15.6}$$

$$\left\{\frac{\partial}{\partial z} - \frac{1}{v_g}\frac{\partial}{\partial t} - j\frac{\Delta\omega}{v_g} + \frac{g-\alpha_i}{2} - h_1\right\} L(z,t) - (h_1 + jh_2)R(z,t) = 0 \tag{15.7}$$

15.1.2 Optical Amplifier Model

Optical amplifier is a special case where the coupling coefficient is negligible so that we only consider the traveling wave in one direction. For an optical amplifier, we are only interested in a single optical frequency so that $\Delta\omega = 0$ in the above equations. We consider the much simplified equation as follows:

$$\left\{\frac{\partial}{\partial z} + \frac{1}{v_g}\frac{\partial}{\partial t} - \frac{g_m-\alpha_i}{2}\right\} R(z,t) = 0 \tag{15.8}$$

If we assume that the optical gain and the internal loss are constants, we have the following simple solution:

$$R(z,t) = F_R(z - v_g t)exp\left(\frac{g_m-\alpha_i}{2}z\right) \tag{15.9}$$

where $F_R(z - v_g t)$ is the envelop function determined by the initial boundary condition of the wave guide.

In general, the gain and loss are functions of position and time. Therefore, we can use the following numerical technique to solve the above equation.

We divide the longitudinal direction into many uniform subsections with distance of Δz. Suppose we have solved for the wave at t_0 as $R_0(z_k)$ where k is the subsection label. We then choose the next time internal as $\Delta t = \Delta z / v_g$. The distribution of wave at Z_k for the next time interval is determined by $R_0(z_{k-1})$ times the exponential factor which depends on the local gain (thus the importance of self-consistency).

If the time interval is different from $\Delta z / v_g$, then we must take linear interpolations. For example, if the time interval is much smaller than $\Delta z / v_g$, then the propagation of wave at Z_k is from a point between Z_{k-1} and Z_k.

For simplicity, we ignore the phase part of the traveling wave equation and only consider the power or photon density traveling wave:

$$\left\{ \frac{\partial}{\partial z} + \frac{1}{v_g} \frac{\partial}{\partial t} - (g_m - \alpha_i) \right\} S_l(z,t) = 0 \tag{15.10}$$

where S_l is the linear photon density.

15.1.3 Self-consistency Issues of Traveling Wave Model

We consider a simple case of wave propagation in yz-plane and solve the drift-diffusion equation on this 2D plane. The optical power propagates according to the following formula:

$$\begin{aligned} S_l(z_k, t) &= S_l(z_{k-1}, t - \Delta t) exp \left[\int_{z_{k-1}}^{z_k} (g_m(z) - \alpha_i(z)) dz \right], \\ & k = 1, 2, ..., N_z \end{aligned} \tag{15.11}$$

here the integral affects all of the yz-mesh points between z_{k-1} to z_k.

To obtain a self-consistent solution, we must solve the above N_z equations with the drift-diffusion equation. The coupling is mainly through the stimulated recombination term which we must work out in detail:

$$R_{st}(y, z) = v_g g(y, z) s(y, z) \tag{15.12}$$

where g is the local gain and s is the local bulk photon density. At each subsection at z_k, we solve for the transverse mode as $\phi_k(y)$ so that

$$\int |\phi_k(y)|^2 dx dy = 1 \tag{15.13}$$

Then we can use the linear density to express the stimulated recombination as follows:

$$R_{st}(y, z_k) = v_g g(y, z_k)|\phi_k(y)|^2 S_l(z_k) \tag{15.14}$$

If a yz-mesh point is between two z-mesh points, we just use linear interpolation.

For the drift-diffusion solver, we must enhance the matrix by an order of N_z and derive new Jacobian matrix with dependence on $S_l(z_k)$.

15.1.4 Implementation in APSYS

Implementation of the optical amplifier model is rather convenient in APSYS because APSYS has already had model for photo sensitive devices with light as input. The difference is that we use guided traveling wave to describe the optics. Also we use stimulated recombination term to describe both gain and loss of the guided traveling wave. For amplifier model, up to two wavelengths of input light source can be used. These are denoted "light" and "light2". This is convenient if one light source is pulsed and the other is CW background light. In APSYS transient simulation we need to use the multiple variable scan capability to variable time, current, light and light2 simultaneously.

15.1.5 Other Related Documents

The optical amplifier module uses the optical gain function as a function of carrier density and optical frequency. The user is refer to the LASTIP User's Manual for a detailed description of the optical gain function. The optical mode solution along the y-direction using the effective index method can also be found in the above Manual.

The detailed derivation of the coupled mode theory can be found in the PICS1D or PICS3D User's Manuals. All of the above documentation can be obtained free of charge for our customers from Crosslight Software Inc..

15.2 Theories of Light Emitting Diodes

15.2.1 Introduction

Light emitting diode (LED) is different from laser diode (LD) and deserves special treatment for the following reasons: 1) LED operates well below lasing threshold while theories of lasers almost always assumes lasing condition; 2) Absence of simulated recombination in LED; 3) Continuous emission spectrum must be considered for LED while longitudinal modes of laser only requires a limited number of lasing wavelengths; 4) The randomness of spontaneous recombination in LED in contrast to the coherence of laser modes which affects the near and farfield models.

In this section, we attempt to establish the basic theories for LED emission for emission in one direction, ie., emission vertical to the semiconductor surface. This allows us to use a simple 1D model to formulate our theory. Once the basic formulas are established, we will extend it to emisison in other directions.

Many of the theories here were initially formulated by Henry [63] for laser diode and semiconductor optical amplifiers. We find that the same theories are also applicable for LED's. Interested readers are encouraged to consult Ref. [63].

15.2.2 Fabry-Perot Cavity and LED

This section is devoted to power emission for light emitting diodes (LED's). LED may be regarded as a special case of Fabry-Perot laser. Without loss of generality, we consider the following device configuration: The z-direction has a waveguiding structure which supports confined lateral/traverse modes. We later relax this condition to include the case with no optical confinement. Please note that in LED, we must treat as many modes as possible since they all contribute to the emission power because of the non-coherent nature of spontaneous emission, while in the case of LD, only a limited number of highly coherent and highly powered modes are considered.

For simplicity, we assume throughout this section that the LED does not have any optical coatings so that light emitts directly from the semiconductor to the air.

The theoretical bases regarding the wave function, the Wronskian and the "Diffusion" coefficient and its relation to semiconductor spontaneous recombination are covered in Appendix D. We shall only use the results here. Also pardon our detailed, step-by-step derivation here since we wish to ensure the correct algebra as well as to educate our readers.

We consider a waveguide with well confined lateral / transverse modes (we will call it **xy-mode** model). Later, we will relax this constrain and consider a case only confined transverse modes in y direction exist (we call it **y-mode** model). Finally, we consider a case where there are no confined modes (we call it **uniform model**).

For a Fabry-Perot cavity from 0 to L, the two basic wave functions Z_1 and Z_2 are as follows:

$$Z_1(z) = r_1 exp(jkz) + exp(-jkz) \tag{15.15}$$

$$Z_2(z) = exp(jkz) + r_2 exp(2jkL - jkz)] \tag{15.16}$$

We construct the following Wronskian:

$$\begin{aligned} W &= Z_1(z)Z_2'(z) - Z_2(z)Z_1'(z) \\ &= [r_1 exp(jkz) + exp(-jkz)] \end{aligned}$$

$$
\begin{aligned}
& [jkexp(jkz) - jkr_2exp(2jkL - jkz)] \\
& -[exp(jkz) + r_2exp(2jkL - jkz)] \\
& [jkr_1exp(jkz) - jkexp(-jkz)] \\
= & -r_1exp(jkz)jkr_2exp(2jkL - jkz) \\
& +exp(-jkz)]jkr_1exp(jkz) \\
& +r_1exp(jkz)jkexp(-jkz) \\
& -r_2exp(2jkL - jkz)jkr_1exp(jkz) \\
= & 2jk[1 - r_1r_2exp(2jkL)] \qquad (15.17)
\end{aligned}
$$

Consider the solution of the field in terms of the Green's function:

$$
g(z, z_s)W_n = Z_1(z)Z_2(z_s)\theta(z_s - z) + Z_2(z)Z_1(z_s)\theta(z - z_s) \qquad (15.18)
$$

$$
\begin{aligned}
E(z)W_n &= \int dz_sf(z_s)[Z_1(z)Z_2(z_s)\theta(z_s - z) + Z_2(z)Z_1(z_s)\theta(z - z_s)] \\
&= Z_1(z)\int_z^L Z_2(z_s)dz_sf(z_s) + Z_2(z)\int_0^z Z_1(z_s)dz_sf(z_s) \qquad (15.19)
\end{aligned}
$$

where $f(z_s)$ is the noise term due to spontaneous emission the details of which can be found in Appendix D.

Consider the squared ensemble average

$$
\begin{aligned}
< |E(z)W_n|^2 > &= < [Z_1(z)\int_z^L Z_2(z_s)dz_sf(z_s) \\
&+ Z_2(z)\int_0^z Z_1(z_s)dz_sf(z_s)] \\
& [Z_1^*(z)\int_z^L Z_2^*(z_t)dz_tf^*(z_t) \\
&+ Z_2^*(z)\int_0^z Z_1^*(z_t)dz_tf^*(z_t)] > \qquad (15.20)
\end{aligned}
$$

We note that the cross terms of the above four terms are zero since noise sources are uncorrelated.

$$
\begin{aligned}
< |E(z)W_n|^2 > &= |Z_1(z)|^2\int_z^L |Z_2(z_s)|^2dz_sf_0(z_s) \\
&+ |Z_2(z)|^2\int_0^z |Z_1(z_s)|^2dz_sf_0(z_s) \\
&= |Z_1(z)|^2\int_z^L |Z_2(z_s)|^2dz_s2D_{FF}(z_s) \\
&+ |Z_2(z)|^2\int_0^z |Z_1(z_s)|^2dz_s2D_{FF}(z_s) \qquad (15.21)
\end{aligned}
$$

As usual, we ignore the interference terms when we consider energy flow so that

$$
|Z_1(z)|^2 \approx r_1^2exp(-2k''z) + exp(2k''z) \qquad (15.22)
$$

$$|Z_2(z)|^2 \approx exp(-2k''z) + r_2^2 exp(-2k''(2L-z)] \tag{15.23}$$

We note that $|W_n|^2$ varies between $4k^2(1-r_1r_2)^2$ and $4k^2(1+r_1r_2)^2$ rapidly. To a good approximation, we take $|W_n|^2 = 4k^2$ to simplify matters.

The linear photon density (in units of 1/m) is given by

$$s(z)\hbar\omega\Delta\omega = \sum_n 2\varepsilon_0 n n_g < |E(z)|^2 > \tag{15.24}$$

where the summation n is over different lateral modes. The power emission spectrum may be calculated from left and right facets as follows:

$$P_L(\omega)Delta\omega = \frac{1-r_1^2}{1+r_1^2}\sum_n v_g 2\varepsilon_0 n n_g < |E(0)|^2 > \tag{15.25}$$

where v_g and n_g are the group velocity and group index, respectively.

At the left facet

$$< |E(0)|^2 >= |Z_1(0)|^2/|W_n|^2 \int_0^L |Z_2(z_s)|^2 dz_s 2D_{FF}(z_s) \tag{15.26}$$

$$P_L(\omega) = \frac{1-r_1^2}{1+r_1^2} v_g 2\varepsilon_0 n n_g \frac{(1+r_1^2)}{4k^2} \int_0^L |Z_2(z_s)|^2 dz_s 2D_{FF}(z_s) \tag{15.27}$$

As we derived in Appendix D that the diffusion coefficient

$$2D_{FF} = \frac{\pi\hbar}{\varepsilon_0 v_g n^2}\hbar\omega < n|r_{sp}(\omega)|n > . \tag{15.28}$$

where r_{sp} is the spontaneous emission rate at a photon frequency. $|n >$ is used to denote the wave function integral for the nth lateral mode.

$$P_L(\omega) = \sum_n \frac{1-r_1^2}{1+r_1^2} v_g 2\varepsilon_0 n n_g \frac{(1+r_1^2)}{4k^2}\frac{\pi\hbar}{\varepsilon_0 v_g n^2} \int_0^L |Z_2(z_s)|^2 dz_s \hbar\omega < n|r_{sp}(\omega)|n > . \tag{15.29}$$

Our key result for spontaneous emission power for an ω interval is as follows:

$$P_L(\omega)\Delta\omega = \sum_n (1-r_1^2)\frac{2n_g\pi}{n}\frac{1}{4k^2}\int_0^L |Z_2(z_s)|^2 dz_s(\hbar\omega) < n|r_{sp}(E)|n > \Delta E. \tag{15.30}$$

The above formula is applicable for a waveguide with limited number of lateral / transverse modes (the xy-mode model). It can be adapted for different circumstances as follows.

We consider a device cross section with active region large enough to accommodate many different lateral modes in both x and y direction (the plane wave model or uniform model). Let us now count the number of allowable lateral modes at frequency ω. Please note the following relations: If we assume a large rectangular device active cross section,

$$(\omega/c)^2 = k_z^2 + k_x^2 + k_y^2 \tag{15.31}$$

At a fixed emission frequency, if we consider all emissions in $\pm z$ direction, the allowable lateral states are confined within a circle of

$$k_x^2 + k_y^2 < (\omega n/c)^2 \tag{15.32}$$

Thus,

$$\sum_n = A\pi k^2/(4\pi^2) = Ak^2/(4\pi) \tag{15.33}$$

where A is the active region cross section. We assume all the modes overlaps with a uniform active region perfectly (uniform model) so that $< n|n >= 1$.

So far, all of our models based on the Green's function are one-dimensional. Thus the wave number k should be understood as k_z if the actual propagation direction is not in z-direction. We make a further approximation to replace k_z by a certain average $< k_z >$ and remove the summation altogether:

$$P_L(\omega)\Delta\omega = (1 - r_1^2)\frac{n_g}{2n}\frac{k^2}{4 < k_z >^2} A \int_0^L |Z_2(z_s)|^2 dz_s (\hbar\omega) r_{sp}(E)\Delta E. \tag{15.34}$$

The internal emission power from spontaneous emission is

$$P_{spon}(\omega)\Delta\omega = A \int_0^L dz_s (\hbar\omega r_{sp}(E))\Delta E. \tag{15.35}$$

The ratio of $\zeta_{ext} = P_L/P_{spon}$ gives us the LED external efficiency at a frequency in the left direction for such a highly multimode device.

If we assume that r_{sp} is reasonably uniform in z-direction, the external efficiency in one direction can be written as

$$\zeta_{ext} = (1 - r_1^2)\frac{n_g}{2n}\frac{k^2}{4 < k_z >^2} \int_0^L |Z_2(z_s)|^2 dz_s/L \tag{15.36}$$

We choose $k^2/ < k_z >^2 = 4/3$ so that in the limit of zero facet reflectivity and transparent material, the external efficiency for one facet is 1/6, as it should be if the device is in a cubic shape.

Thus

$$P_L(\omega)\Delta\omega = (1 - r_1^2)\frac{n_g}{6n} A \int_0^L |Z_2(z_s)|^2 dz_s (\hbar\omega) r_{sp}(E)\Delta E. \tag{15.37}$$

and the external efficiency for a case of uniform spontaneous emission medium is

$$\zeta_{ext} = (1 - r_1^2)\frac{n_g}{6n} \int_0^L |Z_2(z_s)|^2 dz_s/L \tag{15.38}$$

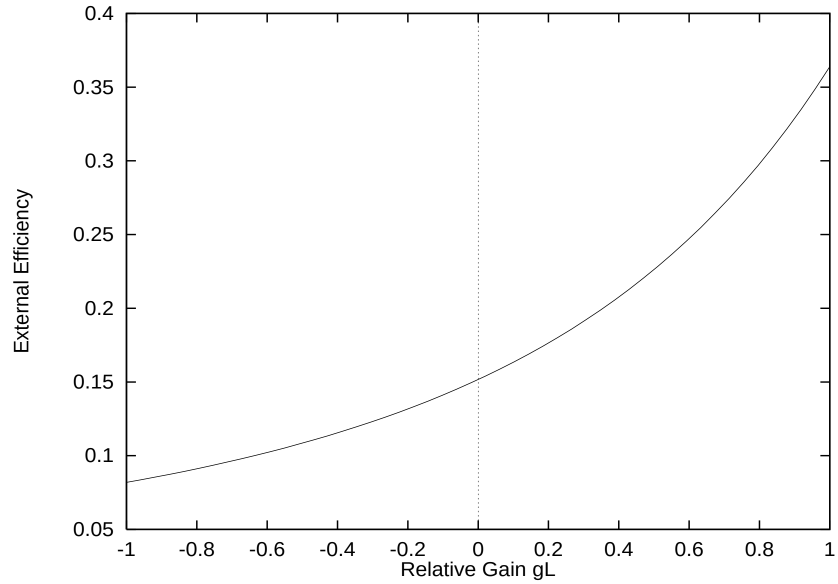


Figure 15.1: External efficiency versus relative gain (gL) of an LED with large active region cross section supporting many lateral modes.

The integral is evaluated as follows:

$$\begin{aligned}\int_0^L |Z_2(z)|^2 dz/L &= \int_0^L dz/L[exp(-2k''z) + r_2^2 exp(-2k''(2L-z)] \\ &= \frac{1}{2k''L}[1 - exp(-2k''L)] \\ &- \frac{1}{2k''L} r_2^2 exp(-4k''L)[1 - exp(2k''L)] \end{aligned} \tag{15.39}$$

To gain some insight into how the external efficiency of an LED depends on the gain/loss of the material, we plot the single facet external efficiency of an LED with large active region cross section supporting many lateral modes (uniform model). Parameters are chosen as follows: $r1^2 = r_2^2 = 0.3$, $n = n_g = 3.4$. The result is shown in Fig. 15.1. As expected, the efficiency goes up as gain is increased. At transparency, the external efficiency amounts to 1/6 for a single facet which accounts for 100 percent of light emission in one direction.

Similar expression can be written for traveling waves in other directions. For example

$$P_R(\omega)\Delta\omega = \sum_n (1-r_1^2)/(1+r_2^2) \frac{2n_g\pi}{n} \frac{1}{4k^2} Z_2(L) \int_0^L |Z_1(z_s)|^2 dz_s(\hbar\omega) < n|r_{sp}(E)|n > \Delta E. \tag{15.40}$$

The integral is evaluated as follows:

$$\int_0^L |Z_1(z)|^2 dz/L = \int_0^L dz/L[r_1^2 exp(-2k''z) + exp(2k''z)]$$

$$
\begin{aligned}
&= \frac{1}{2k''L} r_1^2 [1 - exp(-2k''L)] \\
&- \frac{1}{2k''L} [1 - exp(2k''L)]
\end{aligned}
\tag{15.41}
$$

$$
P_R(\omega)\Delta\omega = \sum_n (1-r_2^2) \frac{2n_g\pi}{n} \frac{1}{4k^2} exp(-2k''L) \int_0^L |Z_1(z_s)|^2 dz_s (\hbar\omega) < n|r_{sp}(E)|n > \Delta E.
\tag{15.42}
$$

$$
\begin{aligned}
exp(-2k''L) \int_0^L |Z_1(z)|^2 dz/L &= \frac{1}{2k''L} r_1^2 exp(-2k''L)[1 - exp(-2k''L)] \\
&+ \frac{1}{2k''L} [1 - exp(-2k''L)] \\
&= \frac{1}{2k''L} [1 - exp(-2k''L)] \\
&- \frac{1}{2k''L} r_1^2 exp(-4k''L)[1 - exp(2k''L)]
\end{aligned}
\tag{15.43}
$$

which is completely consistent with our result for the left facet. Thus we will only consider emission in one direction without loss of generality.

15.2.3 Internal reflection and LED emission power

The result in Eq. 15.37 is obtained under rather ideal condition and assumes that all emission in $\pm z$ can escape from the LED. The result is useful for qualitative analysis regarding the dependence of gain/loss. However, the efficiency from this result is overestimated because in reality, only a small fraction of the light in $\pm z$ can escape from the LED simply because of the internal reflection anlge is much smaller than 90 degrees. For a more acccurate theory, we will re-evaluate Eq. 15.30 taking into account the internal reflection angle.

If the effective refractive index of the material is n_{e1}, the maximum internal reflcetion angle θ_{max} (see also Fig. 15.2):

$$
sin\theta_{max} = 1/n_{e1}
\tag{15.44}
$$

For the uniform model, we should only consider the wave vectors within this angle.

$$
\sum_n = \int_0^{k/n_{e1}} A dk_r^2/(4\pi)
\tag{15.45}
$$

where

$$
k_r^2 = k_x^2 + k_y^2
\tag{15.46}
$$

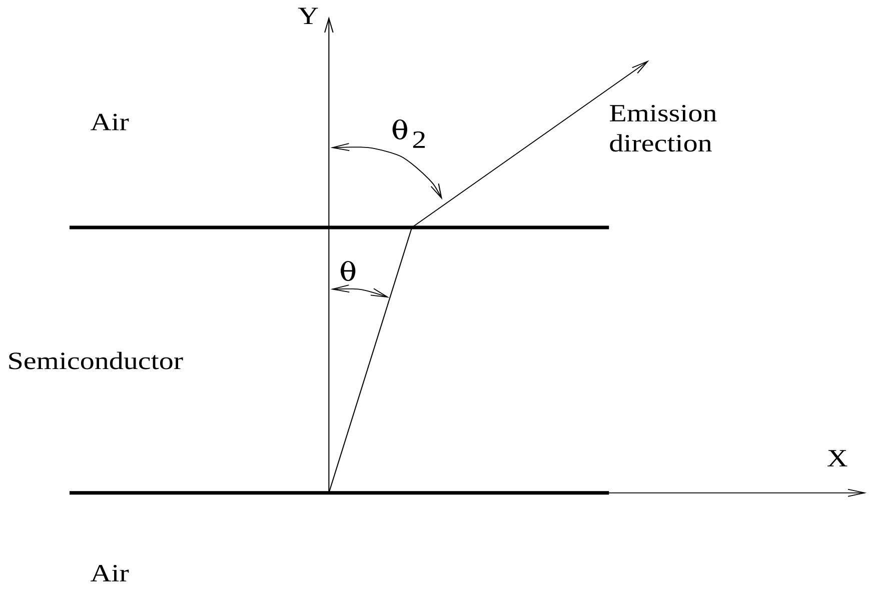


Figure 15.2: Schemtics showing the angle of refraction for an LED.

Please note that the wave number in Eq. 15.30 is to be understood as k_z. We need to evaluate the numerical factor as follows:

$$\begin{aligned} f_{cone} &= \\ &= \frac{1}{8}\int_0^{k/n_{e1}} dk_r^2/(k^2 - k_z^2) \\ &= \frac{1}{8} ln\left(\frac{n_{e1}^2}{n_{e1}^2 - 1}\right) \end{aligned} \tag{15.47}$$

The power emission from facet 1 (left) is

$$P_L(\omega)\Delta\omega = (1 - r_1^2)\frac{An_g}{8n} ln\left(\frac{n_{e1}^2}{n_{e1}^2 - 1}\right)\int_0^L |Z_2(z_s)|^2 dz_s(\hbar\omega) r_{sp}(E)\Delta E. \tag{15.48}$$

Please note that the above emission power is much smaller than the ideal case in Eq. 15.37 because the new numerical factor is much smaller:

$$\frac{1}{8} ln\left(\frac{n_{e1}^2}{n_{e1}^2 - 1}\right) < 1/6 \tag{15.49}$$

It is clear that for an LED with simple configuration, the main power limitation is set by the internal reflection angle. Most of the emission power is lost due to internal facet reflection and absorption. Only a small fraction (around 10 percent) escapes through a small cone vertical to the LED surface.

The reduction factor in Eq. (15.47) has a problem of consistency with the factor of 1/6 because the former does not reduce to the latter if the index of refraction tends to unity. Therefore, we should use the same approximation of $< k_z^2 > /k^2 = 3/4$ and obtain:

$$
\begin{aligned}
f_{cone} &= \\
&= \frac{1}{8 < k_z^2 >} \int_0^{k/n_{e1}} dk_r^2 \\
&= \frac{1}{6n_{e1}^2}
\end{aligned}
\tag{15.50}
$$

The above formula of cone reduction factor is preferred because it reduces to the idea case of no internal reflection if the index tends to unity. For most applications in LED emission to the top surface, the size of the active region facing the top is large enough to use the uniform model and thus Eq. 15.50 is our model of choice. The simulation program has switch to allow the user to include or exclude the effect of total internal reflection. By default, the idea case of no internal reflection is used so that the power emission is calculated by setting index to unity in Eq. 15.50. This represents an overestimate of emission in one direction. On the other hand, use of total reflection to assume that all lights being totally reflected are lost is an under-estimate because modern LED's often have special geometric design to enable the lights being totally reflected to change directions and to escape the device.

In some cases, such as emitting from the sides of a thin device with single mode confining waveguide, it is useful to consider a case where only one direction (say y-direction) has confined modes (y-mode model) while the x-direction is unconfined with a large device dimension L_x. We can simplify mode summation in the x-direction as follows.

Recall our basic result

$$
P_L(\omega)\Delta\omega = \sum_{nx}\sum_{ny}(1-r_1^2)\frac{2n_g\pi}{n}\frac{1}{4k^2}\int_0^L |Z_2(z_s)|^2 dz_s(\hbar\omega) < n|r_{sp}(E)|n > \Delta E.
\tag{15.51}
$$

The number of available modes in the x-direction is simply

$$
\sum_{nx} = \int_0^{k/n_{e1}} L_x dk_r/(2\pi)
\tag{15.52}
$$

Thus we need to evaluate the integral:

$$
\begin{aligned}
g_l = & \quad \frac{1}{4}\int_0^{k/n_{e1}} dk_r/(k^2-k_r^2) \\
= & \quad \frac{1}{8} ln\left(\frac{n_{e1}+1}{n_{e1}-1}\right)
\end{aligned}
\tag{15.53}
$$

Then we obtain our result for y-mode model:

$$P_L(\omega)\Delta\omega = \sum_{ny}(1-r_1^2)\frac{n_g}{n}\frac{L_x g_l}{k}\int_0^L |Z_2(z_s)|^2 dz_s(\hbar\omega) < n|r_{sp}(E)|n > \Delta E. \quad (15.54)$$

15.2.4 Far field distribution

We use a rather simple model to evaluate the far field distribution. We consider three effects:

- 1.) The optical path is different (longer) when the emission angle is not vertical. The longer the optical path, the more power absorption occurs.
- 2.) The power reflection r_1^2 is a function of emission engle.
- 3.) Light direction change due to refraction causes a solid angle within the semicondcutor to be magnified upon refraction. More details are discussed as follows.

It is reasonable to assume that the spontaneous emission is random and have uniform intensity in all directions within the semiconductor. Consider emission within a solid angle $\Delta\theta\Delta\phi$. The θ angle is is determined by the following equation (see also Fig. 15.2):

$$n_{e1}cos\theta\Delta\theta = cos\theta_2\Delta\theta_2 \quad (15.55)$$

Thus the solid angle is amplified and the light intensity is reduced by a factor

$$\left(\frac{cos\theta_2}{n_{e1}cos\theta}\right)\left(\frac{sin\theta}{sin\theta_2}\right) \quad (15.56)$$

where the 2nd factor is due to angular integration in $\Delta\phi$.

15.3 Ray Tracing Simulation

15.3.1 Introduction

Ray tracing (RT) is a modeling technique based on treating light waves as an ensemble of geometrical rays reflected and refracted off material boundaries. This technique is most suitable for optical systems where geometrical nature of the optics is more important than the wave nature. A good example is light emitting diode where the light source is spontaneous emission with randomized phase information while the coherent effect is not significant since the dimension of the cavity is much larger than the wavelength. In this section, we shall describe the theoretical backgrounds of the RT technique in the APSYS software.

A unique feature of our ray tracing technique is the geometrical treatment combined with a wave optics approach, taking into account refraction, reflection, absorption and interference effects. Our model is able to simulate in 2/3 dimensions the angular distribution of the transmitted power of an LED, domed LED, and light energy flux distribution inside a device for further computation by main APSYS module.

An absorbed photon can produce a electron-hole pair. Then the electron-hole pair can produce a photon again through spontaneous emission. This process can continue on and on and is named photon recycling effect. The photon recycling effect is taken into account through interaction of the RT results with the generation-recombination terms of the drift-diffusion module of the main APSYS solver, as we will describe in later subsections.

15.3.2 Basic ideas

The basic assumption in RT is that a ray of light travels in a straight line within a uniform medium. For simplicity consider a simple cubic crystal. Let us put in the center a point source of light which emits rays of light in the form of many straight lines. The following can happen to a ray:

- It returns back into crystal completely due to total internal reflection and no fraction of light power goes outside.
- It splits into reflected and transmitted rays.
- It transmits to the outside completely.

The last outcome is possible only when the external medium has the same refractive index as the crystal. The first two depend on incident light angle and relative refractive indices of the external medium and the crystal.

Also due to the wave nature of light there is a dependence on polarization of the electro- and magnetic- field vectors with respect to the plane of incidence. If the electric field vector is normal to the plane of incidence, the ray is said to have transverse electric (TE) polarization. On the other hand, if the magnetic field vector is normal to the plane of incidence, the ray is said to have transverse magnetic (TM) polarization.

For example, if our crystal has a refractive index of 3.5 (typical for GaAs), the reflection coefficient (amplitude ratio) of the TE- and TM- polarized light as a function of the angle of incidence can be plotted as Figure 15.3. As you can see if light impinges a surface at an angle greater then around 16.5 degrees in our case, it is totally reflected back. Such an angle is called critical or total internal reflection angle.

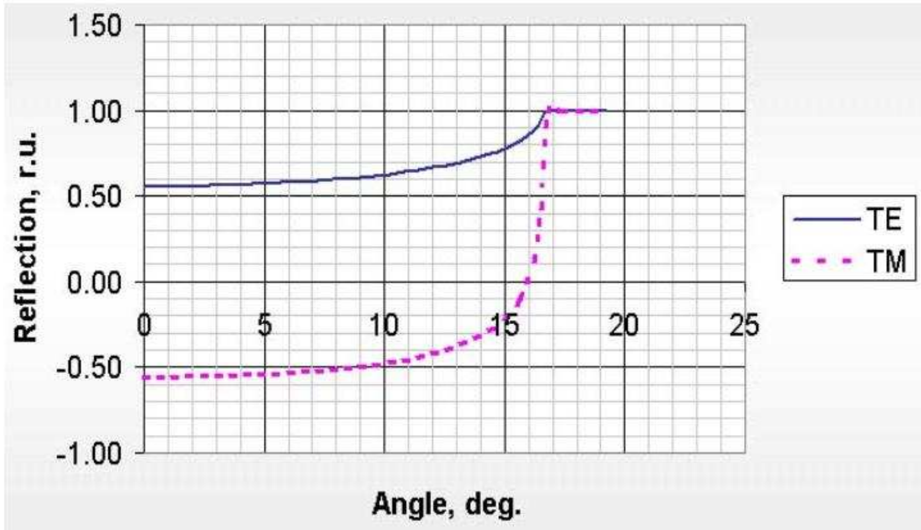


Figure 15.3: The reflection coefficient (amplitude ratio) TE- and TM- polarized light as a function of angle of incidence

For light source from spontaneous emission, as is the case in an LED, the light wave does not have a definite polarization. A ray emitted from such a source should be regarded as having random polarization. To simulate such as source, two approaches can the used: The first is to treat a ray as having random polarization, the second is the treat it as having fifty-fifty TE-TM model (i.e. 50% TE-portion and 50% TM-portion in wave). The latter approach was implemented because both of them produce the same results.

Most real optoelectronic devices consist of a multi-box structure rather than a single crystal box. It is even difficult to imagine how a single ray of light undergoes multiple reflection and refraction in a multi-box structure. The situation is more complicated if multiple rays from multiple source points (see Figure 15.4) must be considered. The RT model tracks down all those rays within the multi-box structure until they are either absorbed by the material or emitted outside of the device.

Presently, the RT program can handle a multi-box structure with some boxes containing distributed Bragg reflectors (DBR). Metal contacts of different kinds can also be supported.

15.3.3 Coherent and incoherent lights

Our RT model needs to treat both coherent and incoherent waves and we shall start with some discussion about the coherence of light here. Based on Fourier transform

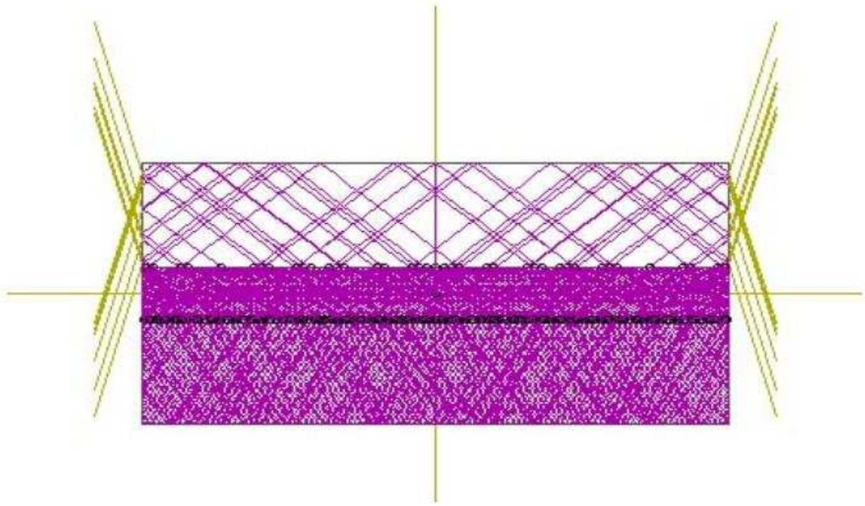


Figure 15.4: The rays undergo multiple reflections/refractions within a device until some of them will are absorbed and others emitted outside.

theories in the analysis related to the Wiener-Khintchine theorem[72], we can express the coherence length of the wave as follows

$$L = \frac{c}{\Delta f} = -\frac{\lambda^2}{\Delta\lambda} \tag{15.57}$$

where Δf (and thus $\Delta\lambda$) is the spectral line-width of the quasi-monochromatic radiation. The coherence length may be used to judge the suitability of geometric or wave approaches of the simulation.

Take the example of a GaAs LED emitting at 0.85 μm. A typical line-width of the spontaneous emission of the LED is 0.05 μm. The coherence length using the above formula works out to be about 14 μm, which is smaller than the dimension of a typical LED. We thus conclude that geometrical optics approach treating rays as independent and non-interfering waves is valid.

The optical power carried by a ray traveling in a semiconductor decays due to the absorption by the material. Its intensity can be written as a function of distance Z as follows:

$$I = I_0 \exp(-4\pi k Z/\lambda) \tag{15.58}$$

where I_0 is the wave intensity of the light source, k is the extinction coefficient defined as the imaginary part of the complex refractive (optical) index. λ is the wavelength of the light in free space. Figure 15.5 shows a ray being traced in an absorbing media.

When dealing with coherent waves, we use a ray model that carries polarization (TE/TM), field strength and phase factor. The resultant amplitudes of transmitted

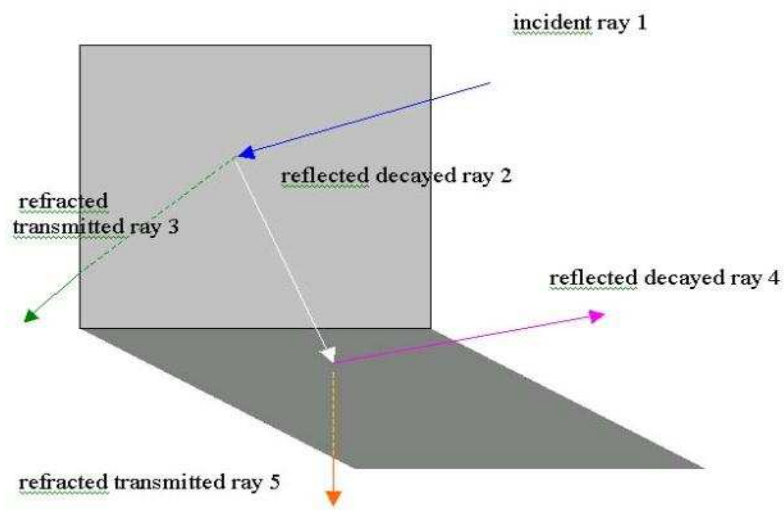


Figure 15.5: Ray tracing in an absorbing media

waves are converted back into units of power [73]. The following equation shows the electric field of a simple plane-polarized plane wave propagating along Z-axis:

$$E = \mathcal{E} \exp(i(\omega t - (2\pi N/\lambda)Z)) \tag{15.59}$$

where $\mathcal{E}$ is the vector amplitude, ω - angular frequency of this wave, N - complex refractive indexes.

To compare two physical approaches in our simulation, let us consider a single ray carrying with it some initial power (Figure 15.6).

To simulate the complicated structures that consist of a main structure, multi-layer structures, and metal contacts, we combine the geometrical treatment of the non-coherent light ray propagation and coherent interactive plane-polarized harmonic waves.

15.3.4 Reflection and refraction

Let us consider simple boundary (Figure 15.7)

The following is Fresnel amplitude reflection and transmission coefficient at oblique incidence for TE-polarized wave [73]:

$$r_e = \frac{N_1 \cos\alpha - N_2 \cos\beta}{N_1 \cos\alpha + N_2 cos\beta} \tag{15.60}$$

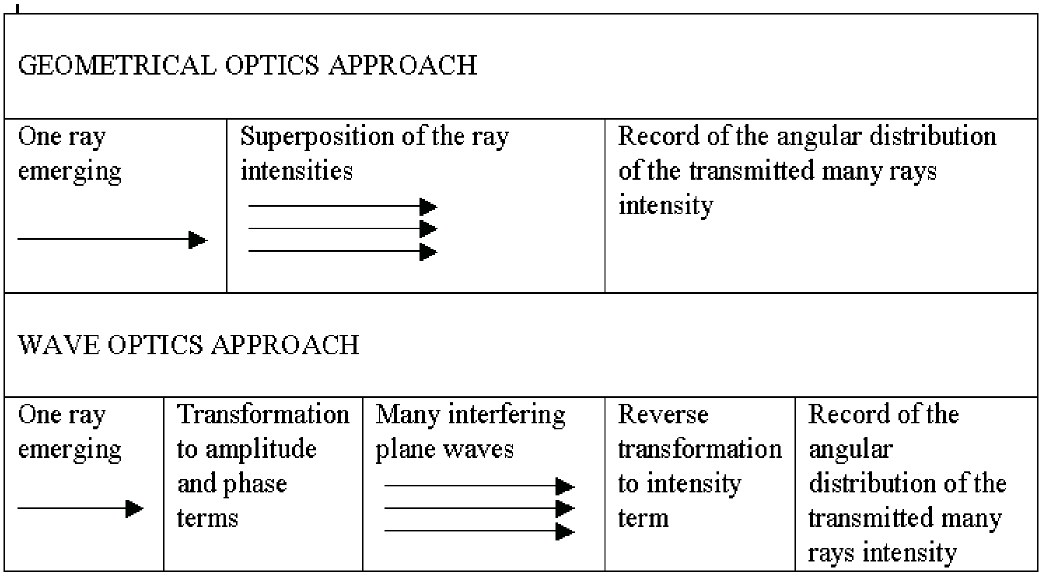


Figure 15.6: The difference in simulation for two physical approaches

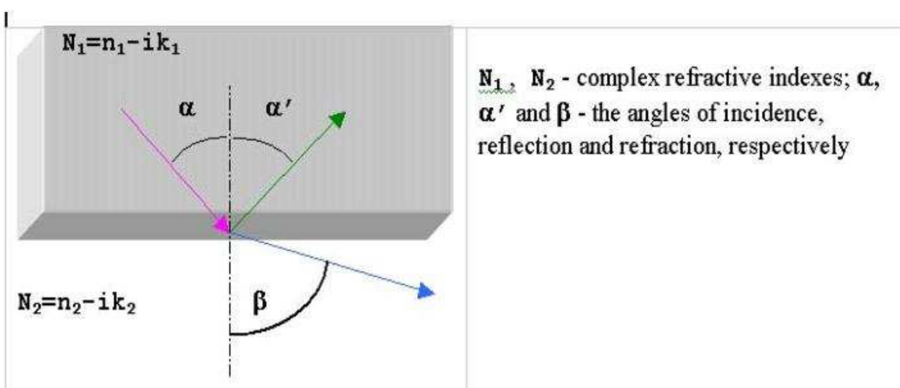


Figure 15.7: Propagation of a light ray from a dense to a rarer medium ($n1 > n2$)

$$t_e = \frac{2N_1 \cos\alpha}{N_1 \cos\alpha + N_2 cos\beta} \tag{15.61}$$

for TM-polarized wave:

$$r_m = \frac{N_1 \cos\beta - N_2 \cos\alpha}{N_1 \cos\beta + N_2 cos\alpha} \tag{15.62}$$

$$t_m = \frac{2N_1 \cos\beta}{N_1 \cos\beta + N_2 cos\alpha} \tag{15.63}$$

The resultant amplitude transmission and reflection coefficient t and r are converted into transmittance and reflectance using the expressions:

$$R_e = R_m = rr^* \tag{15.64}$$

$$T_e = \frac{tt^* Re(N_2) \cos\beta}{N_1^* \cos\alpha} \tag{15.65}$$

$$T_m = \frac{tt^* Re(N_2) \cos\alpha}{N_1^* \cos\beta} \tag{15.66}$$

where * denotes complex conjugate, Re - real part of complex numbers. The above power transmittance and reflectance here are used in both approaches mentioned above.

15.3.5 Ray tracing program design

Since a large number of rays needs to be generated and traced, it is efficient to separately simulate such substructures as DBR's and store the substructure response to incident rays to a special data base. This way, the substructure may be used by the main RT program as a black-box with well defined input/output. Figures 15.8, 15.9 show in general how the program works.

Different models are implemented to simulate the emission source of a LED in the 2D ray tracing module. The light source can be a point source, a line source or continuous source. Also there is an option to emit the power from every active mesh-point generated by the APSYS main simulator. This can enhance the simulation accuracy.

The line source model in the 2D ray tracing module becomes a model of plane source in the 3D ray tracing module. According to this model light emits from a few equal-spaced points with uniform spatial angular distribution of rays directions.

Two different models are implemented to simulate metal contacts. One is a model of an opaque mirror with user defined power reflectance. Another model is to consider the metal electrode as semi-transparent with a certain complex refractive index and to treat it as a thin layer. When an electromagnetic radiation is incident from semiconductor onto the surface of a significant absorbing medium such as metal

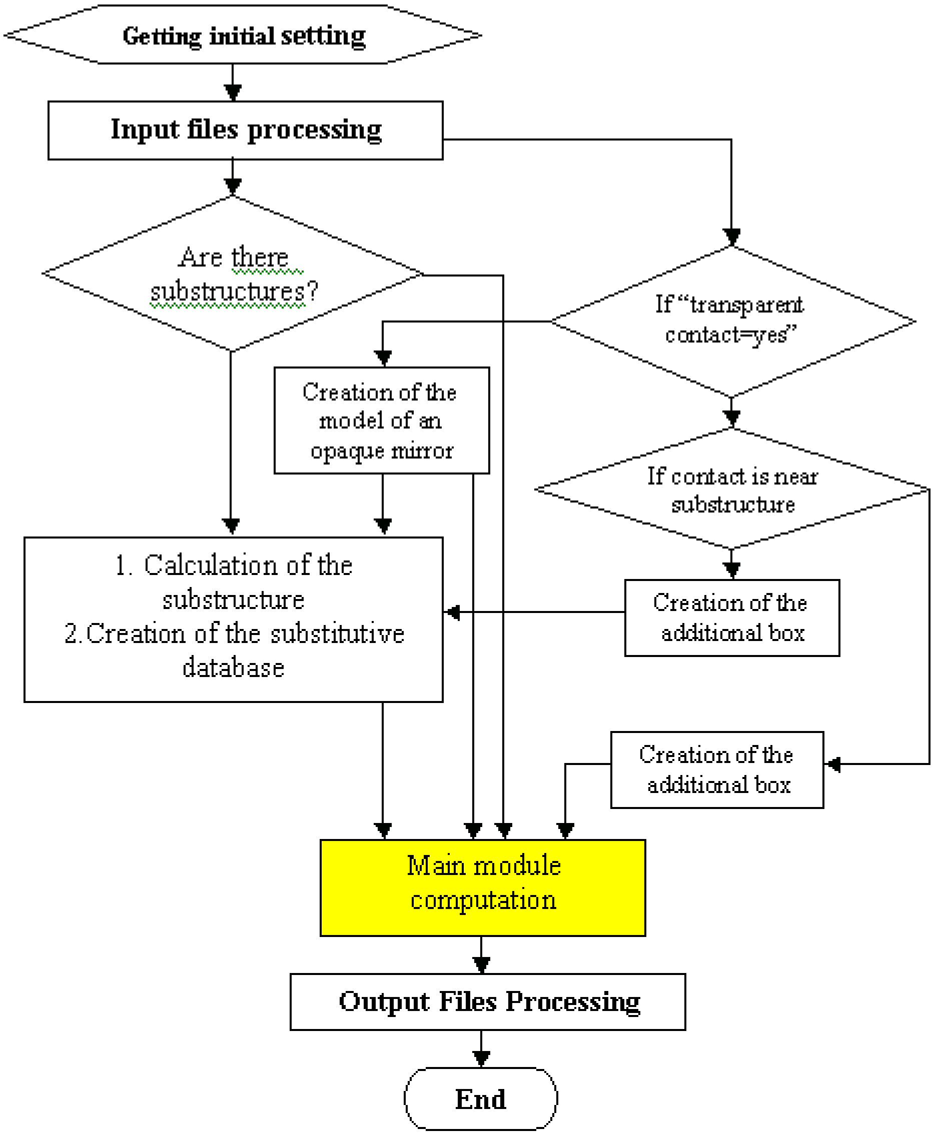


Figure 15.8: Flow-chart shows in general how the RT program works

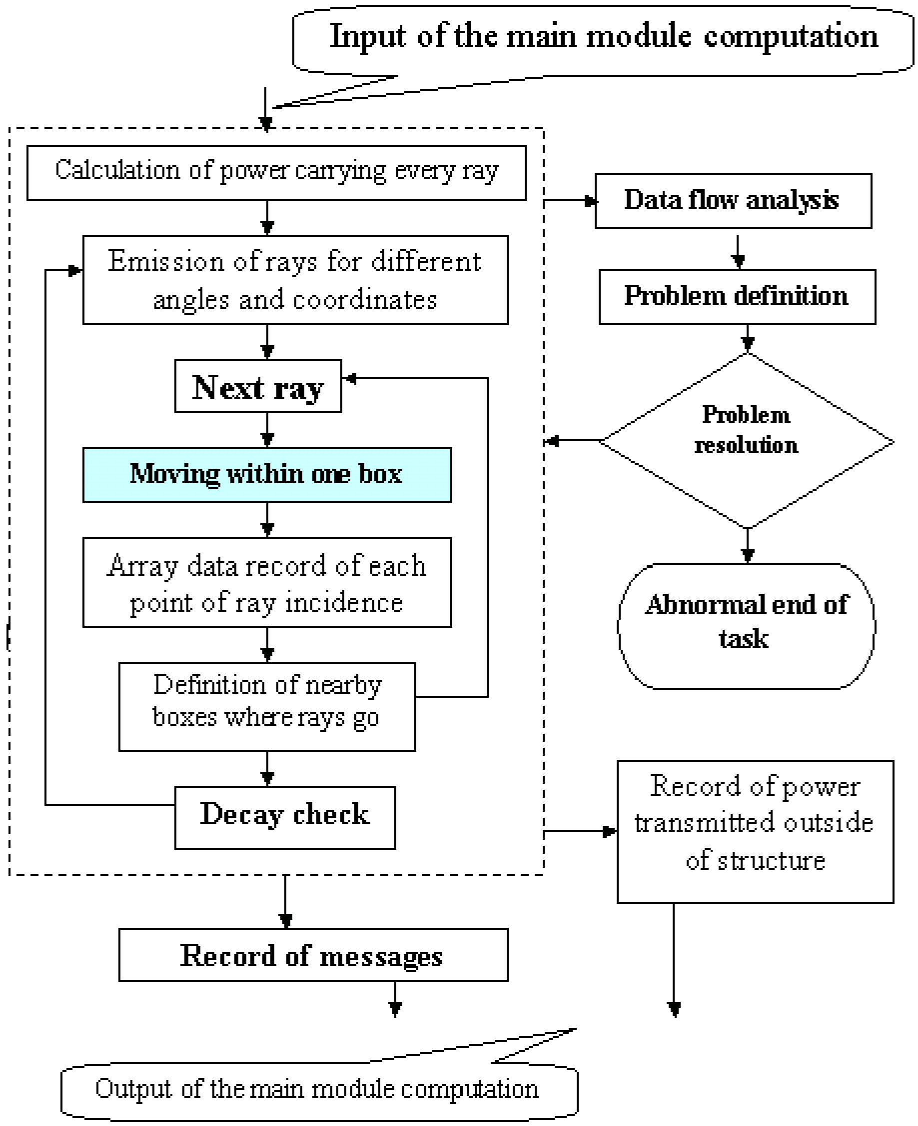


Figure 15.9: Flow-chart of the main RT module.

Substance		Wavelength [micrometer]							
		0.400	0.500	0.600	0.700	0.800	1.000	4.000	10.000
Aluminum	n	0.400	0.667	1.043	1.550	1.990	1.991	6.100	26.000
	k	4.450	5.573	6.568	7.000	7.050	9.300	30.400	67.300
Copper	n	-	0.880	0.186	0.150	0.170	0.197	1.940	10.520
	k	-	2.420	2.980	4.050	4.840	6.272	23.100	59.290
Germanium	n	2.300	3.400	4.500	5.150	5.270	5.100	4.350	4.300
	k	2.800	2.250	1.700	1.300	0.900	0.450	-	-
Gold	n	-	0.840	0.200	0.131	0.149	0.179	2.040	11.500
	k	-	1.840	2.897	3.842	4.654	6.044	27.900	67.500
Indium	n	-	1.020	1.290	1.390	1.500	1.760	7.600	23.800
	k	-	2.080	2.590	5.750	6.690	7.690	26.100	51.700
Lead	n	-	-	-	1.680	1.510	1.410	6.580	21.000
	k	-	-	-	3.670	4.240	5.400	20.800	37.400
Mercury	n	0.730	1.040	1.390	1.760	2.140	-	-	-
	k	3.010	3.700	4.320	4.830	5.330	-	-	-
Silver	n	0.075	0.050	0.060	0.075	0.090	0.250	2.340	10.800
	k	1.930	2.870	3.750	4.620	5.450	6.810	26.900	60.700

Table 15.1: Complex index of refraction of some metals and semiconductors by wavelength

contact, the formulas mentioned previously using complex index of refraction of absorbing media are applicable.

If the thickness of metal film is comparable with light wavelength we should take into account interference between two reflected waves - one reflects from semiconductor-metal interface, another from metal-air interface. There is also interference between transmitted waves. Taking into consideration that multiple layers of metals may be used to improve both the optical transmittance and the electrical conductivity for the contacts, we have implemented a multiple layer metal contact model as part of the transparent contact description.

In our 3D ray tracing module, the reflection, absorption and transmittance of contacts are calculated using the 2x2 matrix method for monochromatic plane waves [72]. Table 15.1 presents a list of refractive indices for some materials that can be used for the semi-transparent contact model.

Let us analyze the three major events of while a ray travels within a device:

1. The emitted photons can be reabsorbed in the semiconductor by creating electron-hole pairs in the active layer.

2. A certain fraction of photons will be reflected back at the semiconductor-air

interface.

3. Some photons impinge upon the surface with angles greater than the critical angle thus suffering total internal reflection.

Let us consider the first event.

It is well known that the internal efficiency of an LED is usually high (close to unity) while the external efficiency is only a small fraction of unity. This is due to the fact that a large fraction of the generated light is never emitted from the device, but trapped by total internal reflection and eventually reabsorbed by the semiconductor. Therefore, we need to consider the re-absorption carefully.

The absorbed emitted photons can produce the electron-hole pairs. As it was mentioned before, this is the photon recycling effect. We refer to the chapter of DRIFT-DIFFUSION MODEL of the General Manual, the governing semiconductor model of all Crosslight's device simulation software packages. The photon recycling effect may be taken into account with addition of a new term G_{phr} in the recombination-generation term in the drift-diffusion as follows.

$$\nabla \cdot J_n - \sum_j R_n^{tj} - R_{sp} - R_{st} - R_{au} + G_{opt}(t) + G_{phr} = \frac{\partial n}{\partial t} + N_D \frac{\partial f_D}{\partial t}, \qquad (15.67)$$

$$\nabla \cdot J_p + \sum_j R_p^{tj} + R_{sp} + R_{st} + R_{au} - G_{opt}(t) - G_{phr} = -\frac{\partial p}{\partial t} + N_A \frac{\partial f_A}{\partial t}. \qquad (15.68)$$

Where G_{phr} is the photon recycling generation rate component.

These equations govern the electrical behavior (*e.g.*, I-V characteristics) of a semiconductor device that in turn influence the internal and external efficiency. G_{phr} depends on the light energy absorbed per volume of semiconductor. That is why the electrical-optical simulation should be self-consistently.

An expression for G_{phr} can be calculated using the formula [**?**]:

$$G_{phr} = \alpha \frac{\Phi_s}{\hbar\omega} \qquad (15.69)$$

Where α is the absorption coefficient, Φ_s is the light intensity or the flux density of the light energy. The formula (15.69) is just the definition of the absorption coefficient: α = (No. of photons absorbed per unit volume per second) / (No. of photons injected per unit area per second).

The equation (15.69) for the electron-hole pairs generation rate is correct if the absorption coefficient is not too large and the variables are not changing with coordinates too sharp. Otherwise it would be better to use the exact formula. The optical generation rate is equal to the number of the injected photons per second, or photon flux ($P_{opt}/\hbar\omega$) per unit volume (V) multiplied by the quantum efficiency η:

$$G_{phr} = \eta \frac{P_{opt}/\hbar\omega}{V} \tag{15.70}$$

Let us consider the second and third events. If n_s is the refractive index of the semiconductor and n_a the index of air, the reflection coefficient for vertical incident light is

$$R = \left(\frac{n_s - n_a}{n_s + n_a}\right)^2 \tag{15.71}$$

For a $GaAs$ LED, if we choose n_s=3.66, n_a=1.0, we get a reflectance of 0.33, i.e., 33% of the photons cannot get through. For non-vertical incident light the situation is even worse. At internal reflection, all power is reflected. To avoid this excessive reflection, the device can be encapsulated in a dielectric dome which has a lower refractive index than the semiconductor.

That is why we have implemented a capability to simulate LEDs encapsulated into epoxy dome. To build a dome around a LED a few simple parameters are used, such as dome width, height, radius of curvature of the emitted side, and the dome refractive index.

For photodetectors the main model of the incident light beam is Gaussian distribution with variable standard deviation of the incident angle. However, this model allows us to create also rigorously parallel beam by using a small enough value of the deviation, or create completely scattered beam using a big enough value of the deviation. Figure 15.10 shows an example of the Gaussian power distribution depending on the incident angle for the beam with angle deviation equals 20 degrees.

15.3.6 Limitations of the ray tracing method

Using the geometrical optics or ray optics is adequate where wave or quantum nature of light can be neglected. Our Ray Tracing Simulator deals with only one wavelength of light, the maximum light spectra of spontaneous emission. The Ray Tracing technique is appropriate where the refractive index varies slowly over the wavelength of light. As a rule for most devices where the Ray Trace method is applicable the error due to such approximations is insignificant.

The most difficult part of the ray tracing simulation is to get accurate results for multi-layer structures in absorbing medium using wave optics approach. Analysis that was carried by [73] showed the difficulties in getting accurate results by solving Maxwell's equation under some conditions. The first restriction is that the incident

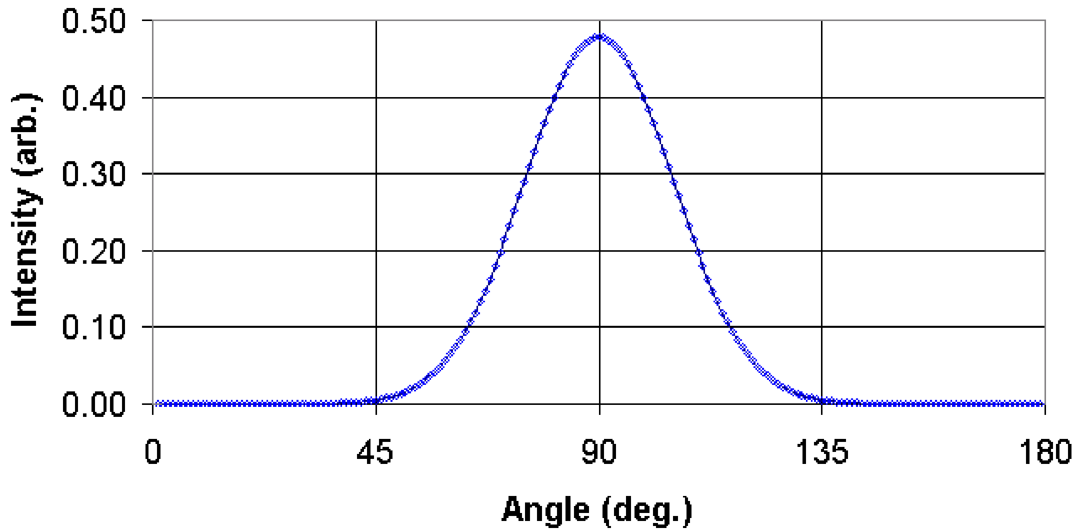


Figure 15.10: An example of the Gaussian power distribution depending on the incident angle for the beam with angle deviation equals 20 degrees

medium must be sufficiently free of absorption for acceptable small errors. Calculation, according to [73], showed that acceptable error (less than 1%) for the ray tracing simulation can be reached at the value of the imaginary part of the optical refractive index less then 0.2.

Two other problems appear when the absorption is too small. At a small absorption, the number of secondary rays due to multi-reflection can reach more than a few million. That puts a huge demand on memory and runtime. Also each reflection adds a small error to the final sum. If we increase given program precision, the number of the secondary ray will also be increased. That is why at a small absorption, the error can be large due to accumulation of the calculation errors.

In our RT program, an acceptable value of the imaginary part of the optical refractive index is in a range $10^{-6}...10^{-1}$ which is sufficient to cover most of most of the multi-layer structures in a semiconductor device.

It should be noted that in order to simulate the metal contact properly one should choose parameters model taking into consideration the existence of an interface layer.

Our model for the domed LED uses the simplification that emitted light does not come back from the dome into semiconductor again. It gives accurate results in most cases when the dome refractive index is less then semiconductor refractive index or/and dimensions of the dome are large compare with device dimensions.

15.3.7 Illustrative results by ray tracing

This subsection is to help better understand the ray tracing technique advantages. Please note the following. The demo diagrams presented in this subsection were made on the base of Ray Tracing Simulator's results with some additional treatment in order to lighten understanding. Simply running the program cannot reproduce some of them directly.

One of the used output data is the angular distribution of the transmitted light power. It is a portion of power in Watts per one degree depending on directional angle in two dimensions simulation. The portion of power emitted into spatial angle measured in Watts per stereo-radian (with scaling factor) depending on angles of spherical coordinate system (θ and ϕ) is used in three dimensions LED simulation.

Other than the absolute value of the transmitted power, one can use the value of the percentage of the transmitted power. It is useful when there is a need to optimize geometry of a certain structure in order to get the most efficient output. Such data can be generated by Ray Tracing program.

Let us consider some simple examples. One is to compare simplified structures with similar sizes and different shapes. For comparative analysis we assign source of light with relative 100% of the power. It was reported before [74], that truncated-inverted-pyramid (TIP) chip geometry provided substantial improvement in light extraction efficiency. We compared cubical and TIP geometry cross-section. Figure 15.11 shows results of such simulations. As you can see, there are significant differences in the angular distribution and transmitted power for these two structures.

Figure 15.12 shows the angular distribution of the transmitted power for the simplest multi-box structure with different complex optical indexes.

Also we simulated LEDs encapsulated into epoxy dome. These simulations allowed us to define how much the difference in optical extraction for domed and non-domed LED due to the Fresnel loss. Figure 15.13 shows an example of the ray tracing simulations for the non-encapsulated and the same encapsulated LED.

Figure 15.14 presents the comparative analysis of the ray tracing simulations for the LEDs with different shape domes.

We simulated resonant cavity LED (RCLED) with different DBR-substructures. The simulations were done also for DBR structures separately and allowed us to estimate edge effects. Figure 15.15 shows the reflectance deviation of some DBR-structure depending on distance between the edge and incident ray summed for all angles.

As well our simulations of different DBR-structures allowed us to reveal a significant optical leakage due to channeling effect [75, 76]. The channeling, or so-called waveguide effect appears near the angle of total internal reflection.

Figure 15.16 explains the origin of the channeling effect. Figure 15.17 shows results

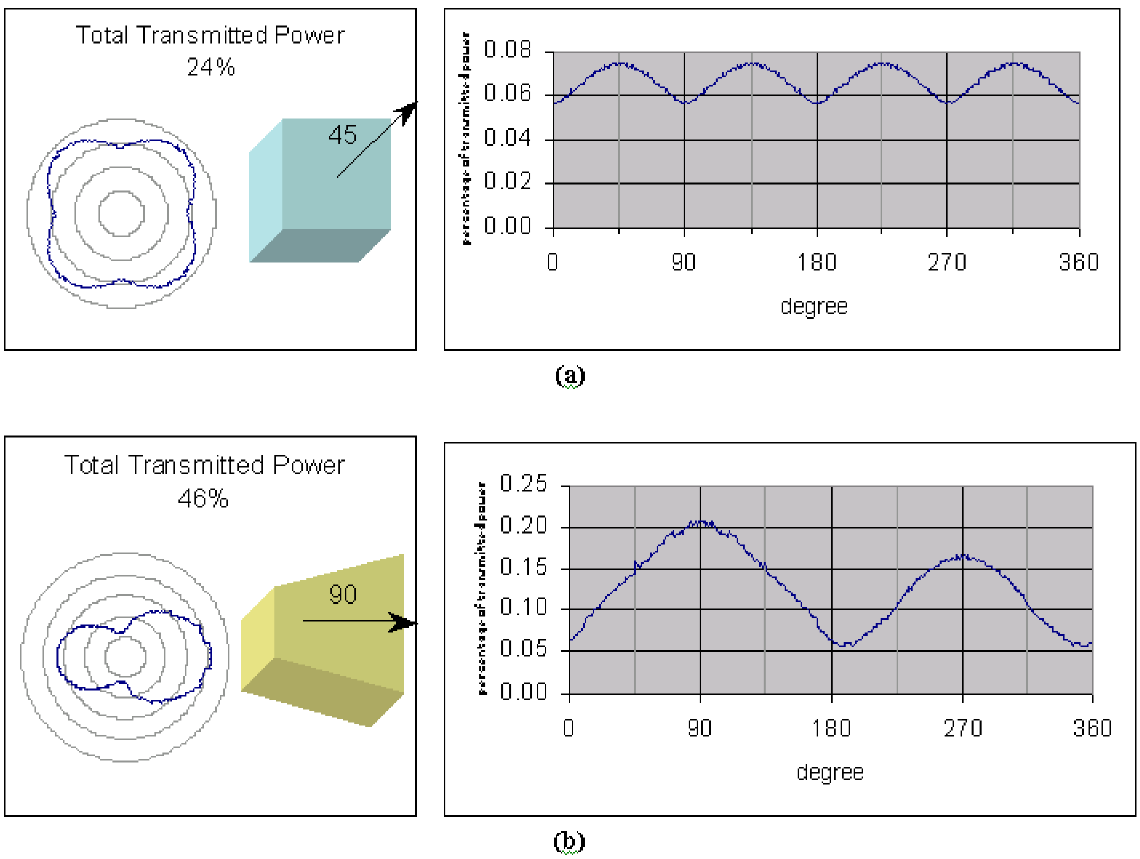


Figure 15.11: Angular distribution of the transmitted power: the cubic (a) and TIP (b) structures

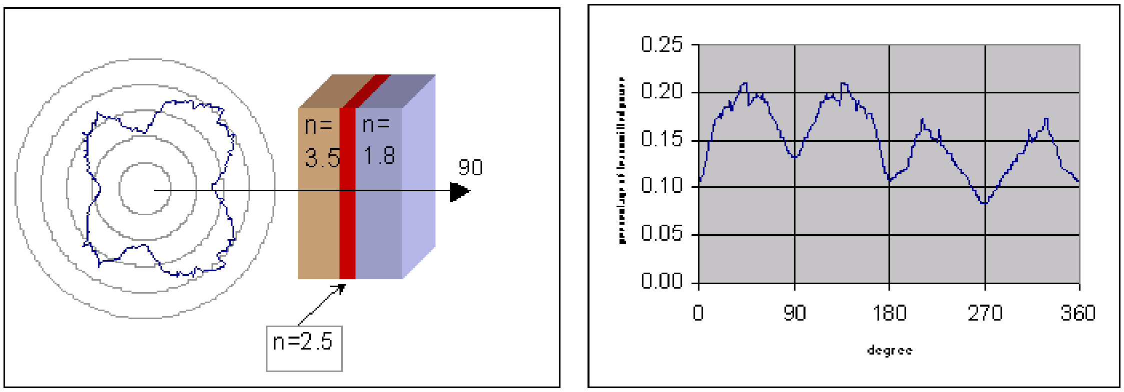


Figure 15.12: A sample of the simplest structure with three different boxes

of our simulations for some DBR structure. One can see the optical leakage due to channeling effect. Using our software can decrease such unwanted effects.

Another useful result one can get is from distribution of the light energy flux density. It can differ distinctly depending on form of the structure, refractive index, absorption, and emission distribution. We analyzed simplified artificial structure to compare the boxes with different refractive indexes. As you can see from figures 15.18, the larger density is in the box with the larger real part of refractive index.

15.3.8 Conclusions

In LEDs the extraction of light from the device is the final step of the ray tracing, where most of the power gets lost as a result of material absorption while undergoing multiple internal reflections. In photodetectors the loss of optical power during ray tracing appears at the transmission of the light into device. The ray tracing technique is an effective method to get detailed light energy distribution for the structures with complicated geometry.

The ray tracing simulation may be used in the following ways:

- To design the LED structures with a certain angular distribution of the emitting power
- To optimize structures to get the most efficient output of the emitting power

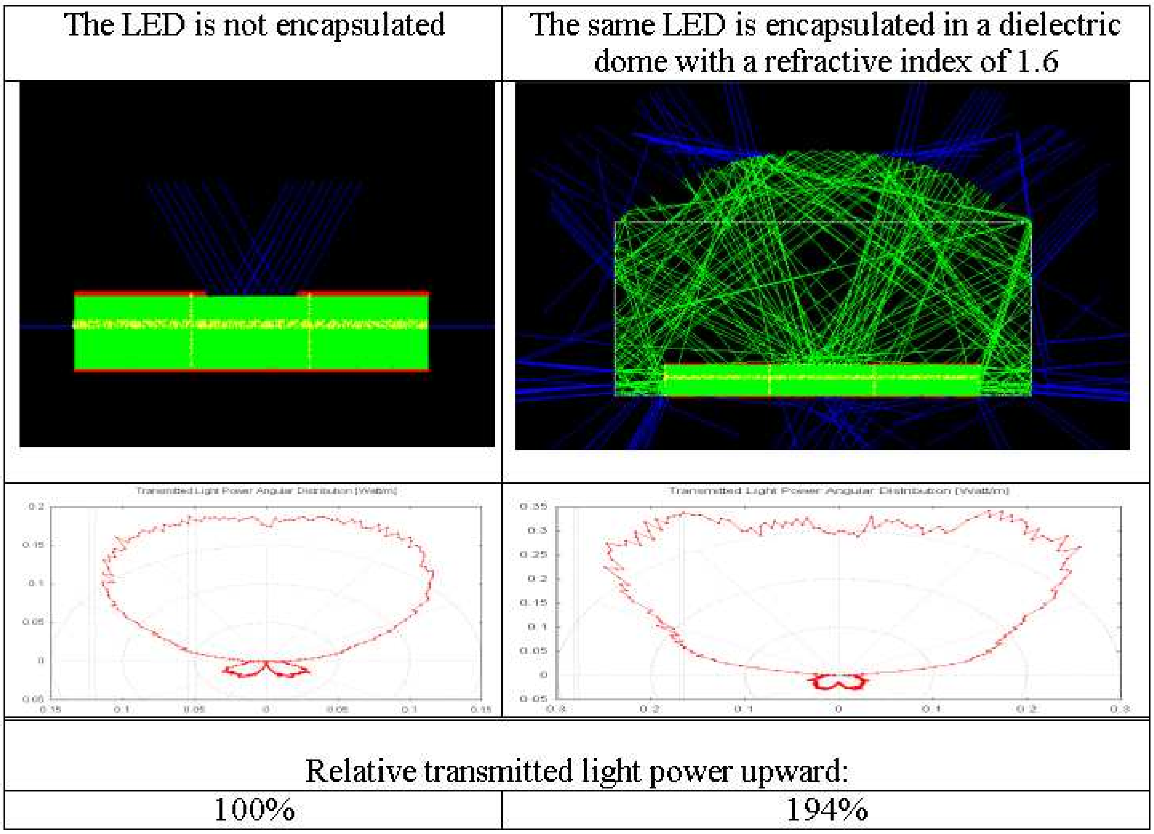


Figure 15.13: An example of the comparative analysis of the ray tracing simulations for the non-encapsulated and encapsulated LED

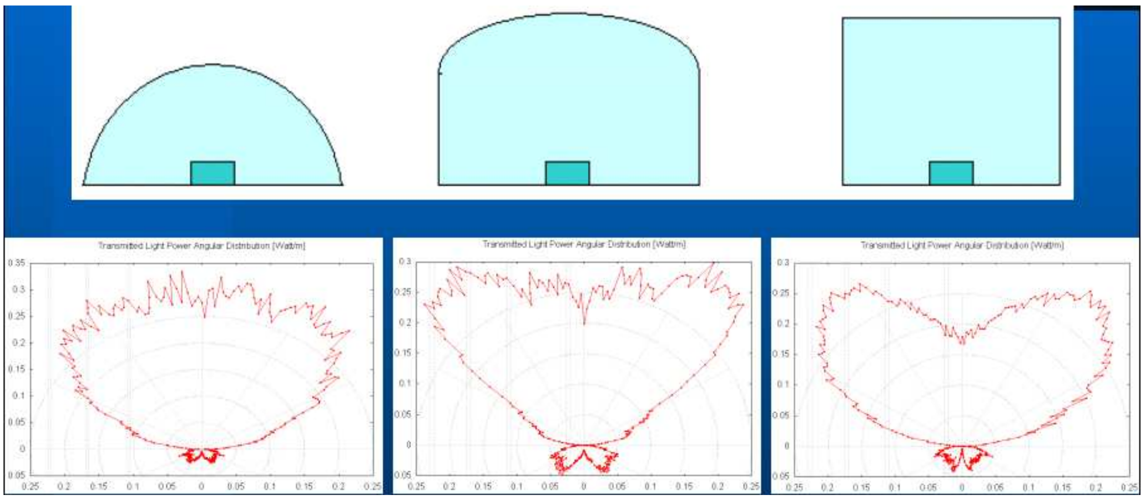


Figure 15.14: An example of the comparative analysis of the ray tracing simulations for the LEDs with different shape domes

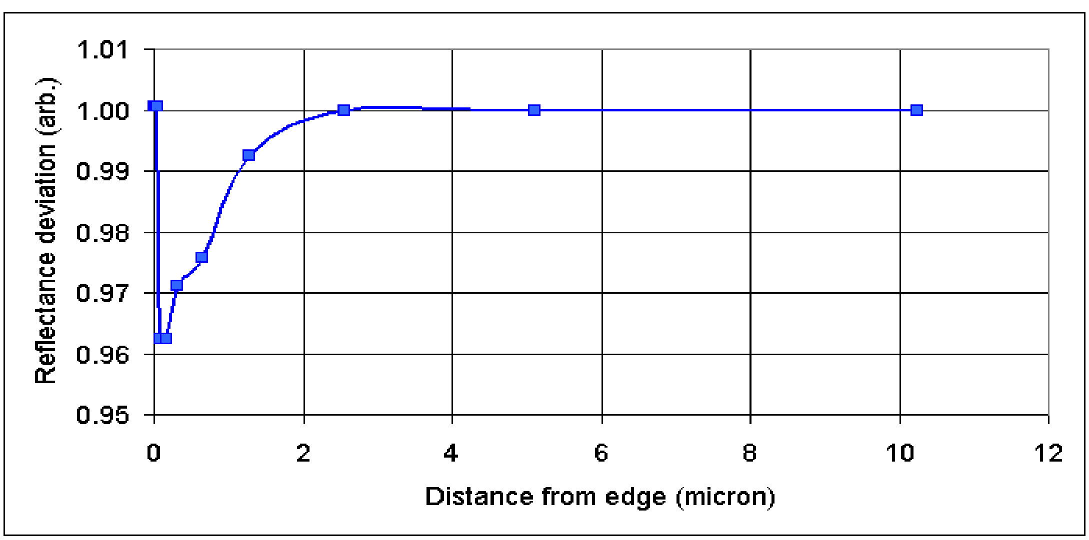


Figure 15.15: Reflectance deviation of some DBR-structure depending on distance between the edge and incident ray summed for all angles

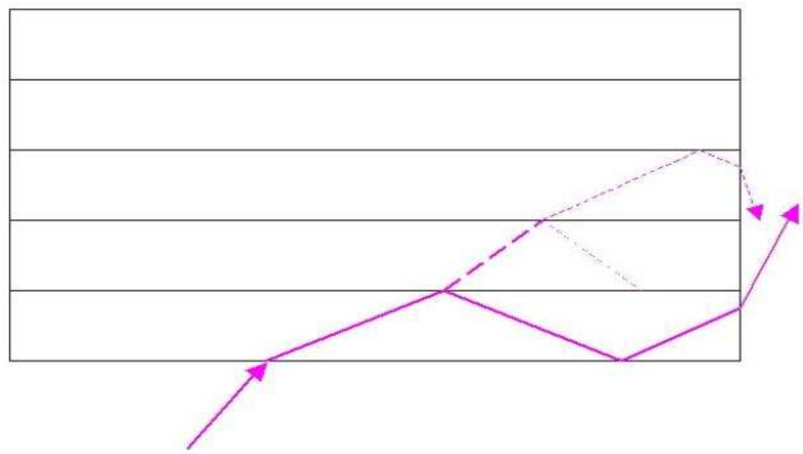


Figure 15.16: Explanation of the channeling effect

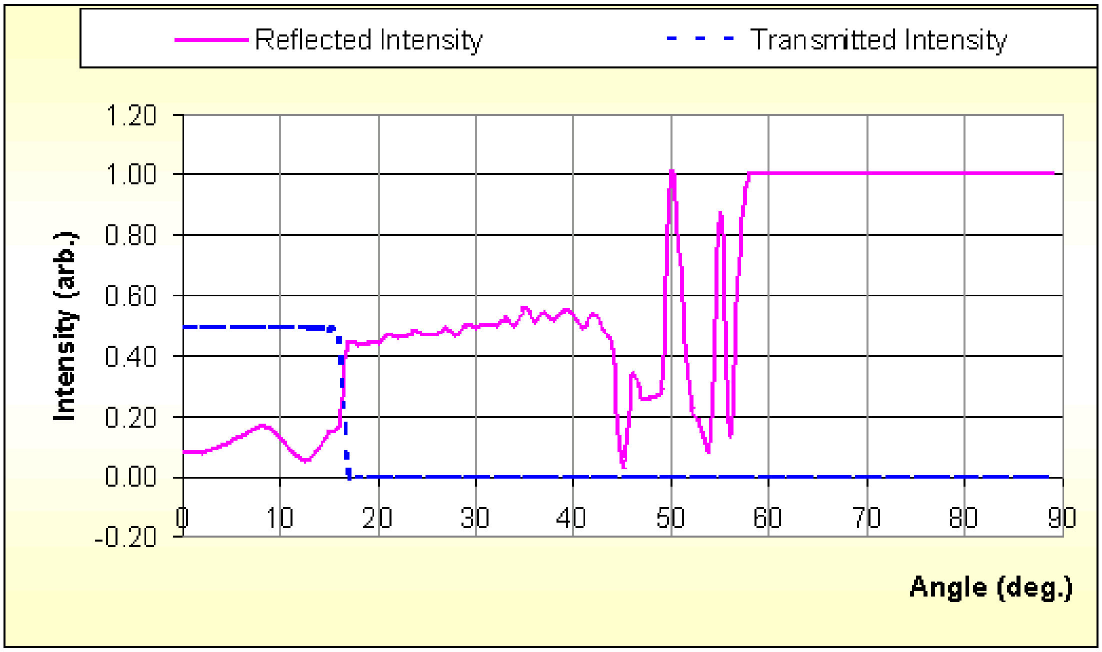
for LEDs

- To get the most sensitive photodetectors
- To obtain accurate results for structures with complicated geometry
- To discover important physical properties of the optoelectronic devices
- To investigate new optoelectronic devices

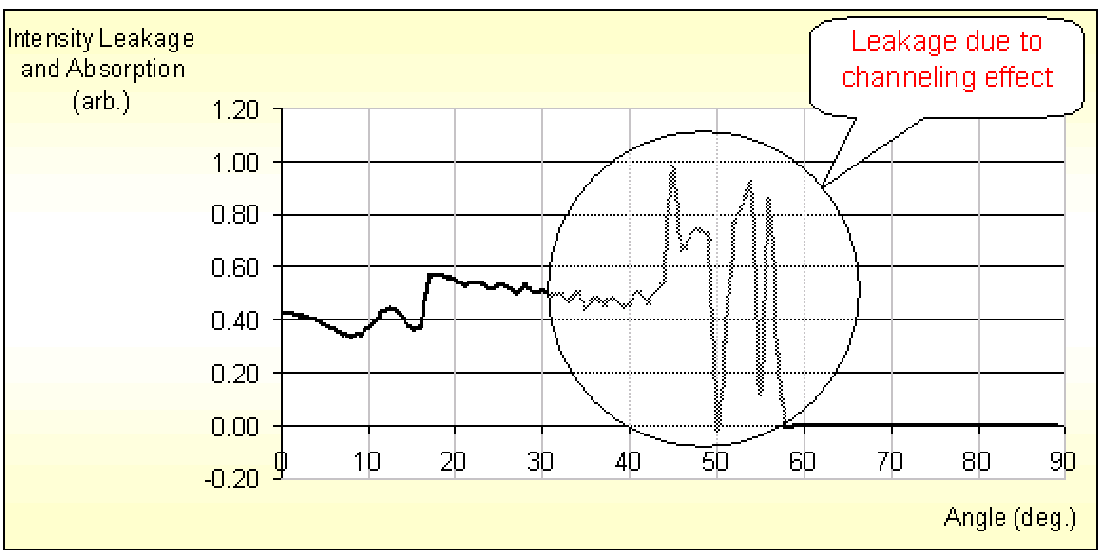
15.4 Strained Si or Si/SiGe Quantum Well Model

15.4.1 Introduction

MOSFET design based on strained Si or Si/SiGe quantum well in the conduction channel has been demonstrated to provide enhanced mobility and thus extend the performance of existing MOS technology[88]. The silicon channel may be strained using a relaxed SiGe material or through stress induced by thin films deposited on the MOS device. The mobility of the strained MOS has been found to depend on the orientation of the strain tensor[89][90]. Thus it is necessary to establish a detailed anisotropic band structure model and associate the mobility enhancement effect with it.



(a)



(b)

Figure 15.17: Reflectance and transmittance (a), absorption and leakage (b) depending on angle of incident ray for some DBR-structure

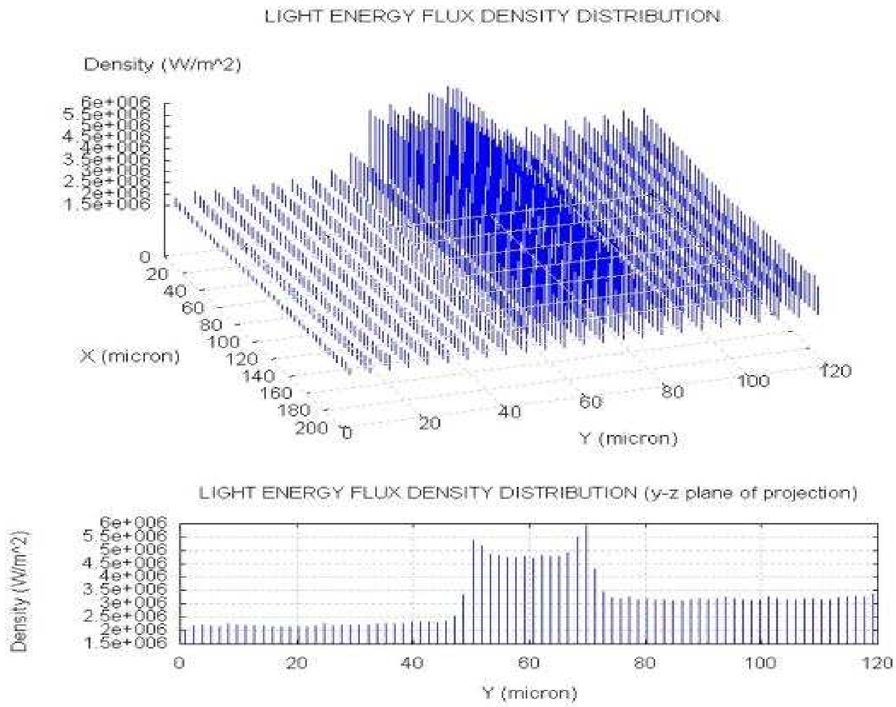


Figure 15.18: Distribution of the light energy flux density inside of device for simplified artificial structure. Real part of refractive index for "y"=0-50 equals 2.5, for 50-70 equals 3.0 (active region), and for 70-120 equals 3.7.

This section is dedicated to the discussion of how Crosslight's simulator takes into account the strain dependence in the band structure and mobility of a quantum MOS device.

15.4.2 Band structure and strain orientation

Let us consider the symmetry of silicon crystal when under stress. A common configuration of strained silicon is to use lattice constant difference between SiGe and Si to apply biaxial tensile stress in the plane of $< 001 >$ (x-y plane in real space). This will cause the lower and upper conduction band valleys to split from the other four valleys. We shall label these two valleys as Δ_2 and other four as Δ_4 with the subscript denoting the degeneracy. Such a situation is represented by Fig. 15.19.

It is also common to apply uniaxial stress in the plane of MOS interface grown in (001) direction. Then, the two valleys along the direction of the strain will have different energy than the others and we shall label these two valleys as Δ_2. The other four valleys lying on a plane perpendicular to the direction of uniaxial stress is labeled Δ_4 (see Fig. 15.20).

In the linear approximation, a biaxial strain induces an uniaxial strain of opposite sign and vice versa[91][92]. For uniaxial in the (001) direction, the relationship is

$$e_b = -D_{001} e_u; D_{001} = 0.771 \tag{15.72}$$

Therefore, if data are available for biaxial strain, it can be converted to those for uniaxial strain.

If the silicon crystal is not gown in (001) direction or if the uniaxial strain is not applied to directions of high symmetry, the six valleys may split into more than two energies. Crosslight's simulator allows a maximum of three different valleys for the conduction band.

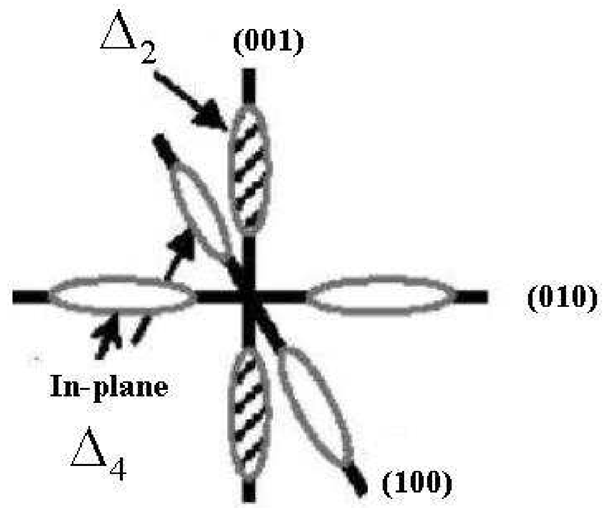


Figure 15.19: Schematics of conduction band valleys of silicon under biaxial in-plane stress.

The strained band structure parameters and mobilities are less well known for the valence band. In general the valence band under strain is highly non-parabolic and this prevents a simple definition of curvature mass[91]. Based on a recent theory, the valence band is also anisotropic[95]. It was found that the higher valence band in (001) direction is light hole while the effective mass appeared to be larger in other directions. There was confusion in the literature whether the higher valence band under tensile biaxial strain should be light hole (LH) or heavy hole (HH). For

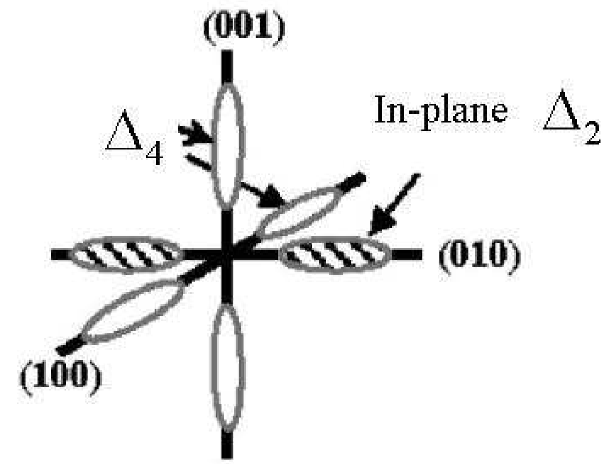


Figure 15.20: Schematics of conduction band valleys of silicon under uniaxial in-plane stress.

example Richard [95] and Fischetti and Laux [91] labeled the upper valence band as LH while other papers did it otherwise (see for example, [96] and [97]).

At this time, we prefer to treat the strain dependence of the valence band mass as a fitting parameter so that the hole mobility behavior can be understand within the frame work of our mobility model as detailed in later subsections.

15.4.3 Band structure parameterization

As compared with its strained counterparts in zincblende and wurtzite crystals, the band structure of strained silicon is more difficult to model within the k.p theory[95]. Therefore, we rely on band theories described in the literature and parameterize the band structure as a function of strain or composition over a large range.

A special type of material macro, called the **general complex strain** macro, or **general_cx_strain** macro has been created to provide a tool to describe the strained silicon bands within the effective mass theory. The strained silicon macro allows the specification of up to three conduction band and valence band valleys with anisotropic effective masses.

For strained silicon grown on unstrained SiGe, we mostly depend on the work of Rieger and Vogl [93] for band edges and electron effective masses. Ref. [93] had already parameterize its results using second order polynomial which can easily be incorporated into our macros. The lower and higher bandgaps based on parameter-

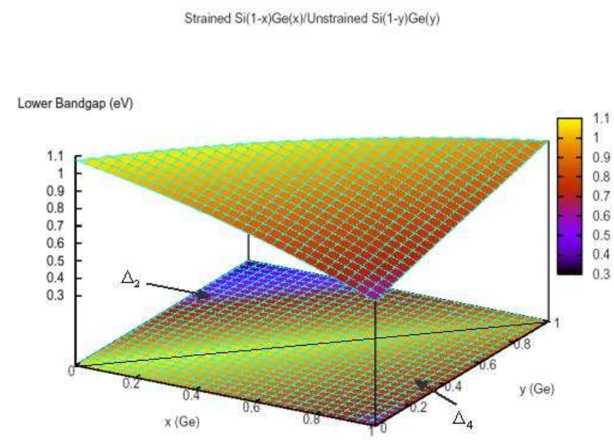


Figure 15.21: Lower bandgap parameterized over a full range of composition for Si(1-x)Ge(x)/relaxed-Si(1-y)Ge(y).

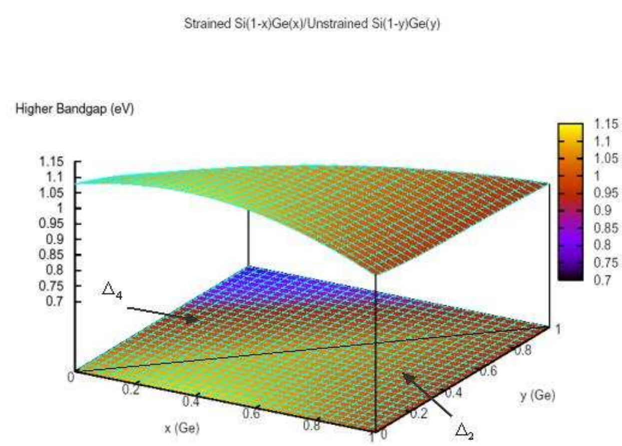


Figure 15.22: Lower bandgap parameterized over a full range of composition for Si(1-x)Ge(x)/relaxed-Si(1-y)Ge(y).

ization by Rieger and Vogl[93] are plotted in Figs. 15.22 and 15.22 over a full range of composition of SiGe/relaxed-SiGe. Thus the current model for Si/relaxed-SiGe is just a special case and can easily be extended to research on strained SiGe/relaxed-SiGe. For silicon under uniaxial strain, we mostly rely on theoretical calculation of Fischetti and Laux [91].

15.4.4 Mobility enhancement

Interest in mobility enhancement by strain has been the driving force behind research in strained silicon. However, the mobility model is also the most complex especially when quantum confinement effect is taken into account. It has been found that for electron mobility in a quantum well (as formed by the potential of the inversion layer in MOS), the mobility can be separated into two parts, one due to intra-valley acoustic phonon scattering and the other due to inter-valley optical phonon scattering[94]. Both parts are complicated functions of subband levels and energy split of the valleys. The complexity of a rigorous theoretical model makes it difficult for experimental data analysis and for CAD purposes.

To reduce the complexity but still maintain the physical intuition needed for understanding mobility enhancement, we propose the following phenomenological model. We start with bulk silicon mobility due to acoustic intra-valley acoustic phonon scattering:

$$\mu_{ac} = \frac{2^{3/2}\sqrt{\pi}\hbar^4 \rho \omega_s^2}{3m^{*5/2} D_{ac}^2 (k_B T)^{3/2}} \tag{15.73}$$

and that due to inter-valley optical phonon scattering:

$$\mu_{op} = \frac{4\sqrt{2\pi} e\hbar^2 \rho sqrt\hbar\omega_0}{3m_d^{*3/2} m^* D_{op}^2} F(x_0) \tag{15.74}$$

where D_{ac} and D_{op} are acoustic and optical phonon deformation potentials, respectively. ρ is crystal density and ω_s and ω_0 are phonon frequencies. F() represents a numerical integral. m_d is the density of state mass and m^* is the conduction mass. For background materials related to the above formulas, please refer to Refs. [98] to [103].

The scattering time constant can be expressed as two terms due to acoustic and optical phonon:

$$1/\tau = 1/\tau_{ac} + 1/\tau_{op} \tag{15.75}$$

and the mobility can be written using the Drude model[104]:

$$\mu = q\tau/m^* \tag{15.76}$$

It is important to notice that the acoustic term and optical term have different dependence on conduction mass. It is commonly believed that mobility enhancement comes from change of conduction mass and suppression of inter-valley optical phonon scattering due to split of valley energies. From energy conservation point of view, carrier transition from one valley to the other via optical phonon scattering will be more difficult as energy difference between valleys are increased. Therefore, we propose the following valley dependent mobility model as follows:

$$1/\mu_k = A_{ac} m_k^{*2.5} + B_{op} m_k^* exp(-\gamma E_{spt}) \tag{15.77}$$

where k is the valley index (denoting one of the six valleys or the HH or LH valleys). A_{ac}, B_{op} and γ are valley independent fitting parameters assume to be independent of the strain. E_{spt} is the valley split between the first and second valleys. The above formula states that mobility enhancement due to strain comes from two terms: an acoustic term depending on conduction effective mass only and another term depending on effective mass and valley splitting. We shall refer to the above phenomenological model as Crosslight's 3-parameter mobility model.

The 3-parameter model in Eq. (15.77) can be conveniently written as

$$1/\mu_k = A_{ac}[m_k^{*2.5} + r_{ba} m_k^* exp(-\gamma E_{spt})] \tag{15.78}$$

where $r_{ba} = B_{op}/A_{ac}$ is a ratio representing the relative importance of acoustic and optical terms in the silicon system.

Using our quantum-MOS model, we solve for the distribution of carriers in each valley k as n_k (or p_k for holes) and the total mobility can be expressed as the following:

$$\mu = \frac{\sum_k n_k \mu_k}{\sum_k n_k} \tag{15.79}$$

To set up the 3-parameter model in a simulation, we need to calibrate the fitting parameters using the following procedure.

- 1) Start with a guess of r_{ba} and γ (both of the order 1).
- 2) Set up an unstrained quantum well subband structure preview (quick simulation requiring no mesh) simulation assuming a typical quantum well carrier concentration (say $1 \times 10^{18} cm^{-3}$).
- 3) Compute the valley occupancy and use formula Eq. (15.78) to determine A_{ac}
- 4) Compared with experimental strain dependence of mobility. Repeat 1) to 4) until good agreement with experiment.

For hole mobility, until a reliable anisotropic effective mass parameterization is achieved, we need to calibrating the mass-strain dependence in addition to the above procedure.

Such a procedure has been demonstrated and documented in the example chapter related to strained silicon. The calculated mobility enhancement as a function strain shows good agreement with experiment without much fitting effort (see Figs. 15.23 and 15.24). Experimental data here are based on Refs. [105] and [88].

Once a reasonable set of parameters are established for strained silicon, the 3-parameter model can easily be extended to include temperature dependence effect since acoustic and optical phonon terms have different temperature behaviors. Also, as more sophisticated mobility models are available, the simple model can be extended to include subband dependence, say between the first two subbands which are more populated than higher subbands.

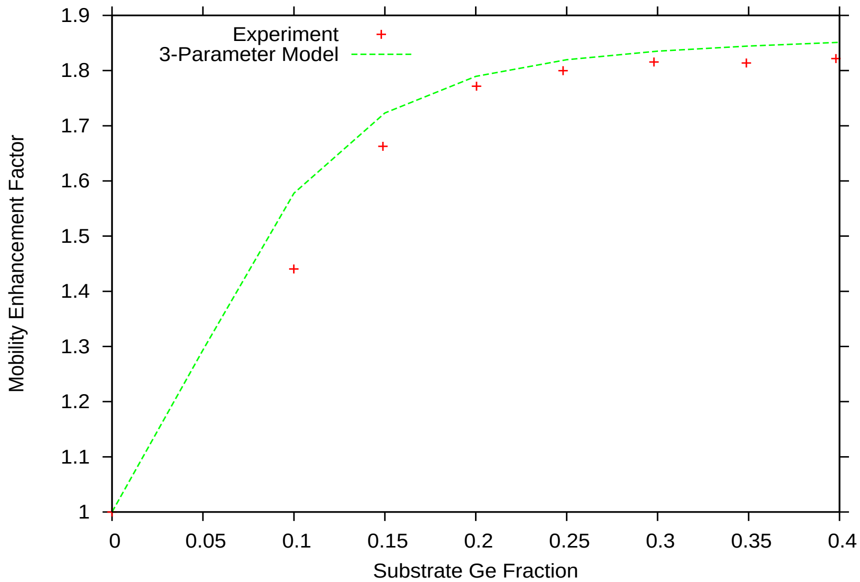


Figure 15.23: Comparison with experiment for Crosslight's 3-parameter mobility model averaged over valley densities in a typical n-MOSFET quantum well.

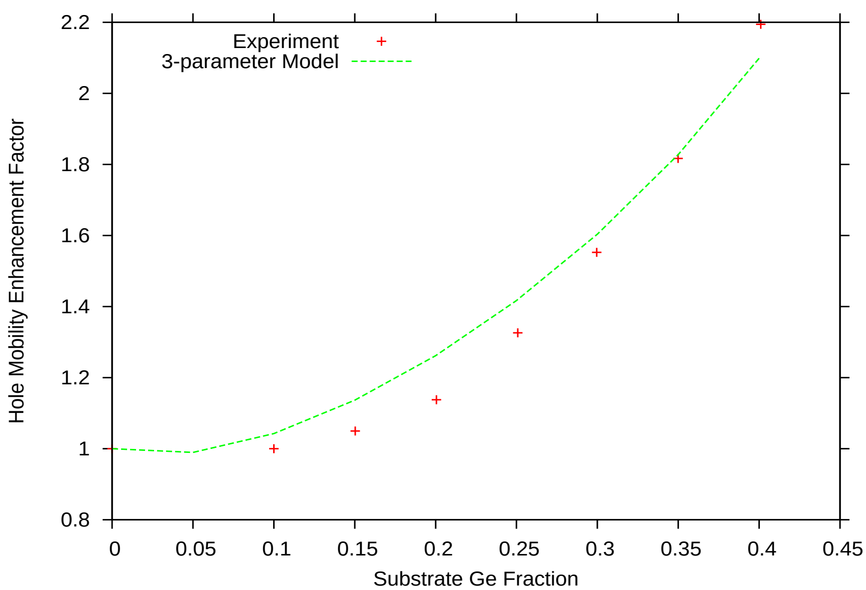


Figure 15.24: Comparison with experiment for Crosslight's 3-parameter mobility model averaged over valley densities in a typical p-MOSFET quantum well.

15.5 Conduction and Recombination in OLED

A detailed description of conduction carriers in organic semiconductor requires the knowledge of molecular orbitals which can be complicated due to the complexity of the atomic configurations of large organic molecules. However, it is possible to use a simplified energy picture to consider only the highest occupied molecular orbitals (HOMO) describing the electron carriers and the lowest unoccupied molecular orbitals (LUMO) describing the hole carriers.

Similarly, we can introduce the concept of density of states for the HOMO and LUMO so that quasi-Fermi levels and Fermi statistics can be used. Having introduced quasi-Fermi levels, the drift-diffusion equation can be established in a similar way as for conventional semiconductors.

Mechanism of carrier conduction is different than conventional semiconductor in that conduction is based on a hopping model. In spite of this difference, we can still approximate the transport using drift diffusion equation if we choose a low value for the mobility. In many cases the carriers will have to go over a high potential barrier and quantum tunneling option must be used.

Since the transport is activated by electrical field in a hopping-like model, we use a Poole-Frenkel-like field dependent mobility based on Ref.[106].

$$\mu = \mu_0 exp(\sqrt{F/F_{pf}}) \tag{15.80}$$

where F_{pf} is an activation field.

The main mechanism of recombination is that electron and holes are attracted to each other via Coulombic forces while the carrier mobility determines the rate at each the recombination occurs. The model is based on the model of Langevin Recombination of carriers [P. Langevin, Ann. Chem. Phys. 28, 289 (1903)].

Carriers are statistically independent, therefore, the e-h recombination is a random process and kinetically bimolecular. The model considers total current attracted by a charge into a certain volume by Coulombic forces. The bimolecular recombination coefficient depends on the mobility as follows.

$$B = \frac{q}{\varepsilon}(\mu_n + \mu_p) \tag{15.81}$$

For more details, we refer to Ref. [106].

15.6 Organic Semiconductor Emission Spectrum Model

15.6.1 Introduction

Organic semiconductor emits light via Frenkel exciton recombination and the mechanism for absorbing and emitting light is different than conventional semiconductor theories based on free-carrier/many-body interband transition. Optical spectrum theory for organic material is generally rather complicated due to the complexity in molecular structure and the advance physical models involved. However from the view point of engineering design, one does not have to understand all the details related to Frenkel exciton to be able to alter and optimize the spectrum. Simplified exciton models with few adjustable parameters correlating to molecular-crystal structure will enable design engineers to achieve such objectives.

We find that the simplified exciton model based on the Holstein model (see Ref. [107, 108, 109, 110, 111, 112]) is suitable for such purposes. The Holstein model assumes a one-dimensional molecular chain with electronic and vibronic interactions between a limited number of nearest neighbors. The advantage of this model is that one may easily alter the optical spectrum by tuning the few physical parameters. For example if we find increasing the interaction parameter between neighours alters the optical spectrum in a desirable manner, we may be motivated to alter the real organic material to increase the number of neighboring molecules.

15.6.2 The Model

Following Refs. [107, 108, 109, 110, 111, 112], we have established optical spectrum model based on exciton-phonon interaction within an organic crystal. A Holstein Hamiltonian is established to include intra-molecular and inter-molecular electronic and vibronic interactions. A phonon cloud of several unit cells is used to represent exciton-phonon interaction. All excited states of the molecular crystal are solved and optical transition dipole moments computed between all states. Please refer details to the above references.

Although mechanisms involved in the Hamiltonian are complex, input parameters are few and fitting to experimental spectrum is easy. Typical input/adjustable parameters: exciton bandgap, molecular vibronic quanta, intra-molecular exciton-phonon interaction constant, and inter-molecular hopping parameter. Doped organic semiconductor may easily be modeled as combination of emission from the host and the dopant separately.

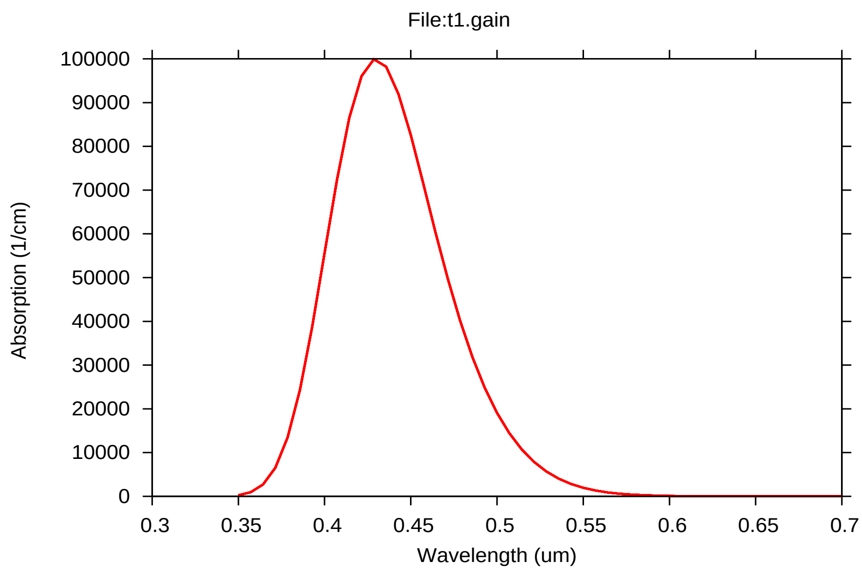


Figure 15.25: Absorption spectrum of Alq3, obtained by fitting exciton parameters with EL spectrum.

15.6.3 Tuning the spectrum

The implementation of the spectrum model is numerically efficient: several minutes per spectrum. We have integrated the spectrum model into APSYS-OLED option so that optical extraction may be calculated based on the spectrum. Bias and current injection dependent spectrum may be simulated to fine tune the color of the OLED.

We use a simple example to provide some insight and a guideline on how to alter the shape of the spectrum. Consider the absorption and EL spectra of Alq3 in Figures 15.25 and 15.26 which were obtained by fitting the EL spectrum to experiments. The smooth and broadened shape corresponds is the results of broadening of tens of spectrum lines as a solution of the Hamiltonian equation. Since the EL spectrum is related to the absorption spectrum, we limit our discussion here to the absorption spectrum which is more simple and it directly correlates to the basic solutions of the Hamiltonian.

To view the actual spectrum lines before broadening, we reduce the Gaussian broadening width (using parameter ox_gaussian_sd1) and obtain the spectrum in Figure 15.27. We see several major lines due to the intra-molecular vibrational excitations. These lines correspond to the major absorption spectrum lines when the unit cell is isolated. We see many lower intensity lines (there are tens of those) arising from electron-electron and electron-phonon interactions within the same unit cell and between the cells.

One way to alter the spectrum is the change the electron-phonon interacton constant (parameter ox_xp_coupling). If we increase the electron-phonon interaction,

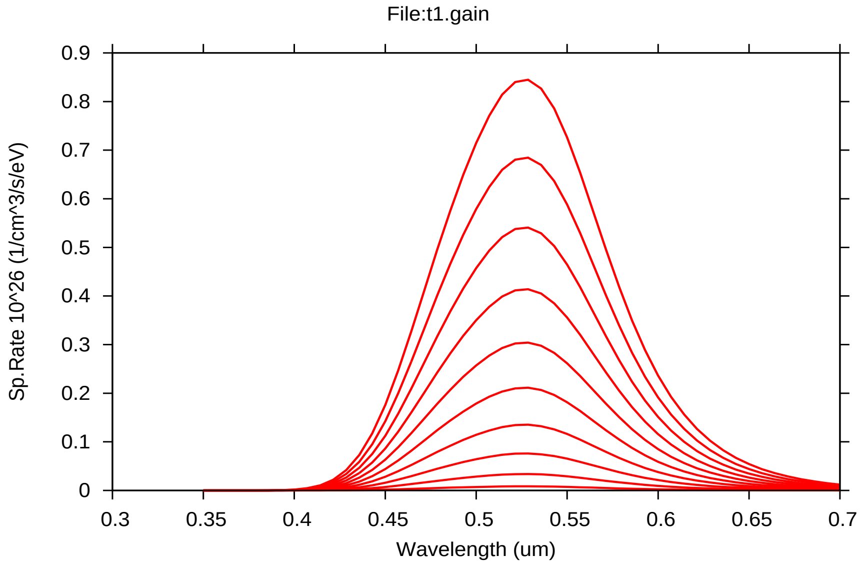


Figure 15.26: EL spectrum of Alq3, obtained by fitting exciton parameters with experimental EL spectrum. Different curves correspond to different carrier injection levels, ranging from 5.E23 to 5.E24 $1/m^3$.

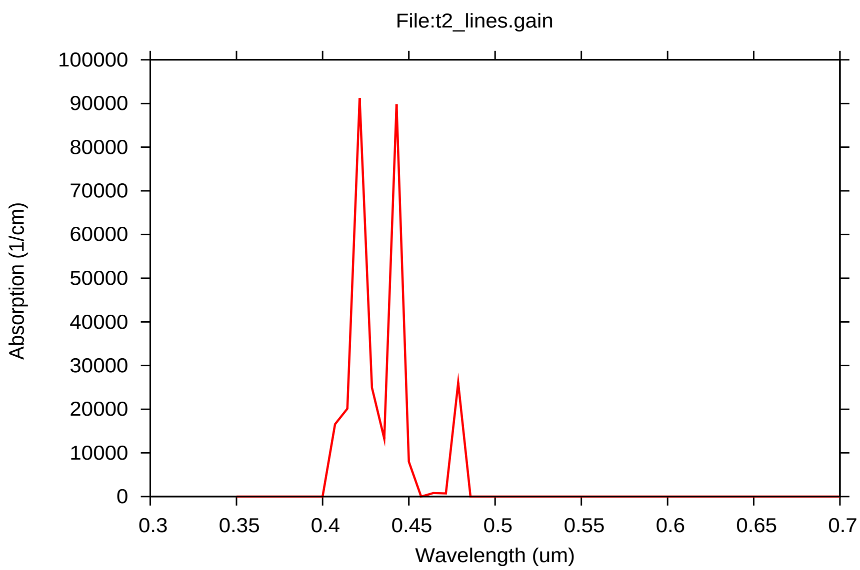


Figure 15.27: Absorption spectrum with reduced Gaussian broadennig.

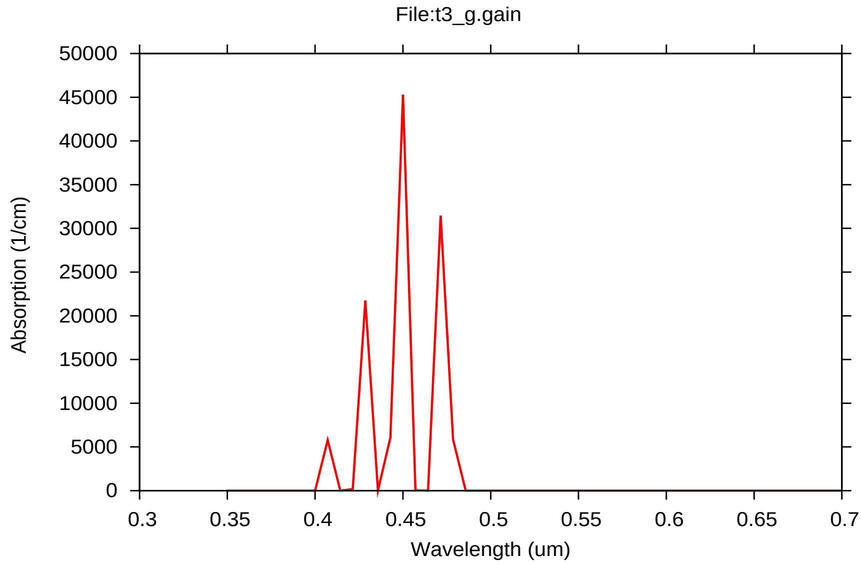


Figure 15.28: Absorption spectrum with increased electron-phonon interaction, showing shift of absorption peak to longer wavelengths.

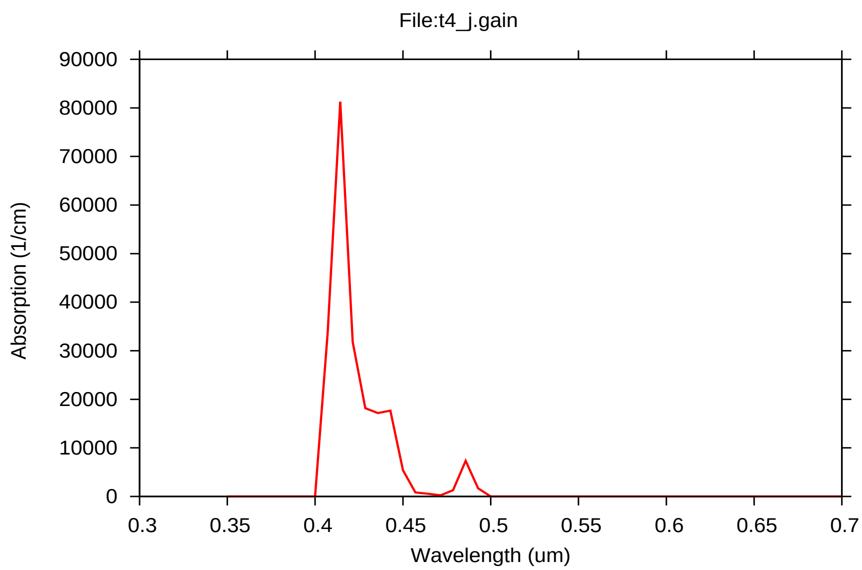


Figure 15.29: Absorption spectrum with increased inter-molecule interaction, showing shift of absorption peak to shorter wavelengths.

the spectrum tends to peak in longer wave lengths (see Figure 15.28.) The interpretation of such a change in shape is that the vibrational energy is much smaller than the electronic energy and a stronger coupling with phonons pulls down the overall energy of the excitations. To achieve such type of tuning, we may be motivated to alter the Franck-Cordon energy of the organic material.

Another way to alter the spectrum is to change the inter-molecule interaction. This can be done by either changing electronic hopping energy (parameter ox_hopping_energy) or by varying the number of interacting nearest neighbors (parameter ox_phonon_cloud). We increase the hopping energy to find that the spectrum tends to peak at higher energies (Figure 15.29). This can be understood as follows: in the limit of zero inter-molecule interaction, the systems prefers to stay at its lowest vibrational state. Higher inter-molecule interaction simply causes a mixture of all the vibrational states and thus tend to equalize the density of states. For such type of tuning, we may be motivated to alter the real organic material to increase the number of neighboring molecules.

15.7 Diffusion of Excitons in Organic Semiconductor

Electron-hole pairs in organic semiconductor form singlet (25 percent) and triplet excitons (75 percent). Both may diffuse over a certain distance before being converted into photons or being quenched via non-radiative means.

We solve the following exciton diffusion equation

$$\frac{dS}{dt} = R_{rad} + D_s \frac{d^2S}{dx^2} - S/\tau_x - S/\tau_q \qquad (15.82)$$

where S is the singlet or triplet exciton density, R_{rad} is the radiative recombination term from the carrier drift-diffusion equation. S/τ_x is photon emission term related to exciton lifetime and S/τ_q is exciton quenching term where τ_q may be a function of S (bi-exciton quenching), electron and hole densities.

The significance of this exciton diffusion model is that OLED does not emit directly from R_{rad} term but through S/τ_x term above. This means deviation in both magnitude and profile from the bi-molecular recombination term of R_{rad} in the drift-diffusion equation.

Using the RCLED model, the $S(x, y, z)/\tau_x$ term acts as a continuous dipole source within the theory of Greens function. The dipole source may be enhanced by optical interference effect to optimize power emission characteristics of an OLED.

15.8 Nomenclature

Symbol	Definition
A_x	Coefficient for band-gap narrowing effects.
a_0, a	Lattice constants of GaAs (a_0) and InGaAs (a).
B	Radiative recombination coefficient.
B_1,B_2,B_z	Complex wave amplitudes of the optical wave propagating from right to left in a laser.
B_k	Constant used in mathematical fittings.
b	Shear deformation potential.
C_n, C_p	Auger recombination coefficients, for electrons (n) and holes (p).
c_{nj}, c_{pj}	Electron (n), hole (p) capture coefficients of the j th deep trap.
c	Velocity of light in vacuum.
c_{11}, c_{12}	Elastic constants.
D_c, D_v	Density of states of the conduction and valence band.
E_D, E_A	Shallow donor (D) and acceptor (A) levels.
E_i^0, E_j^0, E_{ij}^0	Energy minimums of the i th and the j th levels, and the difference between them, . The i refers to the valence band sub-band levels and j refers to the conduction band sub-band levels.
E, E_{ij}	Photon energy and energy difference between the i th and j th levels.
E_{fn}, E_{fp}	Quasi-Fermi energies of electrons (n) and holes (p).
E_g	Bandgap.
E_{g0}	Unstrained bandgap.
ΔE_{PF}	Shift in ionization energy of a dopant due to Poole-Frenkel effect.
δE_{hy}	Hydrostatic strain energy.
δE_{sh}	Energy associated with shear strain.
F	Electric field intensity.
F_l	Lorentzian shape function.
F_{0n},F_{0p}	Threshold electric field used in the electron (n) and hole (p) mobility models.
$F_{1/2}$	Fermi integral of order one-half.
f_D, f_A	Occupancy of the donor (D) and acceptor (A) levels.
f_{tj}	Occupancy of the j th deep trap level.
f_i, f_j	Fermi functions for the i th and the j th level.
f_i', f_j'	Integrated Fermi functions for the i th and the j th levels.
g	Interband local gain.
g_{ij}, g	Local gain due to transition between the i th and the j th levels, and the local gain of the material.
g_d, g_a	Degeneracies of the shallow donor (d) and acceptor (a).
H_{ij}	Hamiltonian matrix element between the i th and the j th states.
$h(x)$	step function of variable x.

h,$\hbar$	Plank's constants.
I_1, I_2	The integrals needed to evaluate the quantum well gain and spontaneous emission rate.
J_n, J_p	Electron (n) and hole (p) current flux densities.
J_{sn}, J_{sp}	Electron (n) and hole (p) current flux densities on the surface.
J_{hn}, J_{hp}	Electron (n) and hole (p) current flux densities across the heterojunction.
J_{source}	Current flux source.
k	Boltzmann constant.
L	Laser cavity length.
$L()$	Lines shape function for gain broadening.
m_0	Electron mass
m_i, m_j, m_{ij}	Relative effective masses of the i th and the j th level and the reduced effective mass between the i th and the j th level. Their relation is defined by $1/m_{ij} = 1/m_i + 1/m_j$. Indices i and j refer here to quantum well sub-bands.
m_n,m_p	Bulk effective masses for electrons (n) and holes (p).
m_e,m_h	The same as m_n and m_p, respectively.
m_{bn},m_{bp}	Bulk effective masses for electrons (n) and holes (p) on the barrier side of the heterojunction.
m_{zl}	Relative effective mass of the L-band used in the quantum level calculation.
m_{vx}, m_{vy}, m_{vz}	Effective hole masses in the x, y and z direction.
M_0, M_{ij}, M_{lh}, M_{hh}	Momentum matrix elements for bulk material, between the i th and the j th states, involving light- and heavy-hole transitions, respectively.
N	Total number of grid points in the simulation space. Also used for the total number of longitudinal layers.
N_b	Number of grid points associated with a boundary of interest.
N_D, N_A	Doping density of shallow donors (D) and shallow acceptors (A).
N_{tj}	Density of the j th deep trap.
n	Electron concentration or density.
n_b	Electron concentration or density on the barrier side of the hetero-junction.
n_{b0}	Electron concentration or density on the barrier side of the hetero-junction when its quasi-Fermi level coincides with that on the other side.
n_s	Electron concentration or density on the surface.
$\overline{n}$	Real part of refractive index.

n_i	Intrinsic carrier density.
n_{2D}	The surface concentration of a quantum well, equal to the bulk density times the layer thickness.
N_{rn}, N_{rp}	Reference doping density in electron (n) and hole (p) mobility equations.
P_{ij}	Probability of a transition from the i th to the j th level.
p	Hole concentration or density.
p_b	Hole concentration or density on the barrier side of the heterojunction.
p_{b0}	Hole concentration or density on the barrier side of the heterojunction when its quasi-Fermi level coincides with that on the other side.
p_s	Hole concentration or density on the surface.
Q^j_{em}	Energy density of the jth emitted longitudinal mode.
Q^j_{cav}	Energy density of the jth longitudinal mode within the laser cavity.
q	Electronic charge.
R^{tj}_n, R^{tj}_p	Electron (n) and hole (p) recombination rates per unit volume through the j th deep trap.
R_{sp}, R^b_{sp}, R^{qw}_{sp}	Spontaneous recombination rate per unit volume, also called the radiative recombination rate, and the same quantity in bulk (b) and in the quantum well (qw).
r^{qw}_{sp}	Frequency dependent spontaneous recombination rate for the quantum well.
R_s	Resistance associated with a boundary condition.
R_{st}	Stimulated recombination rate per unit volume.
R_{au}	Auger recombination rate per unit volume.
r_m,r^{eff}_m	Single facet reflectivity and effective reflectivity of a laser.
T	Absolute temperature.
T_{scat}	The scattering kernel for carrier-carrier scattering.
t	Thickness of the quantum well.
V	Electrical potential.
V_s	Electrical potential on the surface.
$V_{applied}$	Applied electrical potential.
v	Volume.
$\overline{v}_n$, $\overline{v}_p$	Average thermal velocity of electrons (n) and holes (p). The average is taken over all three dimensions.
$\overline{v}^{therm}_n$, $\overline{v}^{therm}_p$	Average thermal velocity of electrons (n) and holes (p). The average is taken only over the dimensional of interest.
$\overline{v}^{therm}_{bn}$, $\overline{v}^{therm}_{bp}$	Average thermal velocity of electrons (n) and holes (p) located on the barrier side of the heterojunction. The average is taken only over the dimensional of interest.

v_{sn},v_{sp}	Saturation velocity for electrons (n) and holes (p).
W	Carrier energy.
W_n	Electron energy.
W_n	Hole energy.
W_j^e, W_i^h	Effective width of the wave functions for electron and hole in the jth and ith subbands, respectively.
x, y	Spatial coordinates used in the model. The x-axis is parallel to the quantum well. x is also used elsewhere as the Al composition in AlGaAs or the In composition in InGaAs.
α α_{qw}	Local loss coefficient due to loss other than interband recombination, and local loss in the quantum well.
α_{int}	Internal loss defined as the weighted average of the local loss α.
α_n, α_p	Constants used in doping dependent mobility functions for electrons (n) and holes (p).
α_{fn}, α_{fp}	Coefficients of the free carrier absorption for electrons (n) and holes (p) in the quantum well.
α_0	Absorption coefficient for regions outside the quantum well.
β, β_1, β_2	Complex eigenvalues of the wave equation and its real (1) and imaginary (2) parts. They are also considered effective indices.
α_{em}	Optical loss coefficient due to emission of light in the lasing mode.
α_{emj}	Optical loss coefficient due to emission of light in the jth longitudinal mode.
β_p	Constant used in the hole mobility equation.
Δ	Spin-orbit splitting of the valence band energy.
Δ_{ij}^a	Transition energy shift term in the Asada broadening model.
δ	Constant. Value is 1 for deep donors and 0 for deep acceptors when used to represent electrical charge. Also used as a δ-function.
ϵ_0	Dielectric constant of vacuum.
ϵ_{dc}	Relative DC or low frequency dielectric constant.
ϵ, ϵ_1, ϵ_2	Complex optical dielectric constant and its real (1) and imaginary (2) parts. ϵ is also used, under a completely context, for non-linear gain coefficient.
ϵ_2^g, ϵ_2^α	Imaginary parts of the dielectric constant corresponding to gain due to interband transition and loss due to other mechanisms, respectively.
ε	Strain.
Γ	Energy level broadening parameter, or the half width of the Lorentzian broadening function.
Γ_0	The maximum energy level broadening parameter, or the half width of the Lorentzian broadening function.

Γ_a	The energy level broadening parameter, or the half width of the Lorentzian broadening function, as used in the Asada's broadening model.
$\gamma_1, \gamma_2, \gamma_3$	Luttinger numbers.
γ	Constant relating the optical wave amplitude to electric field.
γ_n,γ_p	Adjustable constants for the electron (n) and hole (p) thermionic emission model, used to account for tunneling and other effects.
γ_{hn},γ_{hp}	Adjustable constants for the electron (n) and hole (p) thermionic emission model for heterojunctions, used to account for tunneling and other effects.
χ	Affinity.
χ	Affinity of a reference material.
λ	Optical wavelength.
λ_j	Optical wavelength of the jth longitudinal mode.
μ_n, μ_p	Mobility of electrons (n) and holes (p).
μ_{0n}, μ_{0p}	Low field electron (n) and hole (p) mobilities.
μ_{1n}, μ_{1p}	Minimum electron (n) and hole (p) mobilities.
μ_{2n}, μ_{2p}	Maximum electron (n) and hole (p) mobilities.
ν	Function used in the Fermi-Direct statistics.
ρ_i, ρ_j, ρ_{ij}	Density of states for the i th and the j th levels, and the reduced density of states for transition between the j th and the i th levels.
ϕ_n,ϕ_p	Quasi-Fermi potential for electrons (n) and holes (p).
ϕ_b	Schottky barrier height.
ρ_i^0, ρ_j^0, ρ_{ij}^0	Constants for the two-dimensional density of states for the i th and the j th levels, and reduced density of states for transition between the i th and the j th levels. For example $\rho_i^0 = m_i/\pi\hbar^2 t$.
σ_{nj}, σ_{pj}	Electron (n) and hole (p) capture cross sections of the j th deep trap.
τ	Intra-band scattering lifetime.
τ_{nj},τ_{pj}	Electron (n) and hole (p) life time due to the jth trap.
τ_h, τ_v	Hole-hole scattering life time for the Asada gain broadening model. $\tau_h = 2\tau_v$.
ω	Angular frequency of the optical wave.
ω_a,ω_β	SOR parameters used for the wave equation.
ξ	Function used in the Fermi-Dirac statistics.
ζ, ζ_n, ζ_p	Variable related to the effective masses. ζ is also used for half the shear strain energy for a different context.
$<\ >$	Average with $\mid W \mid^2$ as weight.

Chapter 16

PICS3D SPECIFIC MODELS-I: 2D/3D ELECTRIC/OPTICAL MODULE

16.1 Introduction

A basic module of PICS3D is the electrical model described as the drift-diffusion model (DD). An important simulation task is to find the lateral optical mode distribution on the x-y plane of a laser diode or within a vertical cavity surface emitting laser (VCSEL). We will only deal with models or features specific to the PICS3D package in this chapter.

16.2 Simulation of Vertical Cavity Surface Emitting Lasers

A Vertical Cavity Surface Emitting Laser (VCSEL) requires special configuration of the PICS3D simulator and is treated as an option for PICS3D users. We describe some of the theoretical background in this section.

16.2.1 Introduction

There has been increasing interest in VCSEL in recent years. VCSEL offers a number of advantages, such as low divergence output beams desirable for efficient coupling into optical fibers, and inherent single longitudinal mode operation due to large mode spacing because of a short cavity. They can be tested on the wafer and can provide high density 2D self-aligned laser arrays.

The model of VCSEL is different from that of conventional edge emitting lasers and

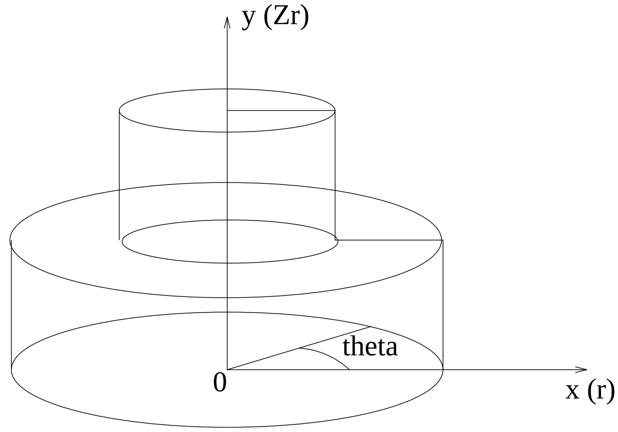


Figure 16.1: Schematic of the coordinate system used for VCSEL model.

requires different theoretical treatments. We provide list some of the difference here.

- 1. A cylindrical coordinate system should be set up to treat the drift-diffusion (DD) problem in a 3D simulation. If the boundary condition is symmetrical, the drift diffusion solver may be reduced to 2D (see Figure 16.1).
- 2. The solution of the optical modes on the reduced 2D plane is different because the direction of wave propagation is also on the reduced 2D plane.
- 3. A multiple longitudinal mode model must be solved on the same reduced 2D plane. The DD solver and longitudinal mode solver are strongly coupled. Efficient treatment of optical properties of the layered structure is required.
- 4. The thermal effects should be included since device heating is the major cause of power limitation for CW operation in VCSEL.

Subsection 16.2.2 describes the cylindrical coordinate system and its use in the DD model. Subsection 16.2.3 discusses the layered model for the longitudinal modes. Subsection 16.2.4 presents a simple model for the optical mode in the lateral direction. Subsection 16.2.5 discusses the integration of various simulation software modules.

16.2.2 Cylindrical Coordinate System and Semiconductor Equations

An accurate model of VCSEL must include the solution of the semiconductor equation or the drift-diffusion (DD) equation:

$$-\nabla \cdot \left(\frac{\epsilon_0 \epsilon_{dc}}{q} \nabla V\right) = -n + p + N_D(1 - f_D) - N_A f_A + \sum_j N_{tj}(\delta_j - f_{tj}),$$

$$\nabla \cdot J_n - \sum_j R_n^{tj} - R_{sp} - R_{st} - R_{au} = \frac{\partial n}{\partial t} + N_D \frac{\partial f_D}{\partial t},$$

$$\nabla \cdot J_p + \sum_j R_p^{tj} + R_{sp} + R_{st} + R_{au} = -\frac{\partial p}{\partial t} + N_A \frac{\partial f_A}{\partial t}.$$

where the potential V, electron concentration n, and hole concentration p are the three variables to be solved. J_n and J_p are the electron and hole current densities. The R terms and the N terms are used to denote the recombination terms and the doping impurities. Other symbols have their usual meanings. These equations govern the electrical behavior and are an important part of the VCSEL model. However, these equations should be re-written and solved in cylindrical coordinate system to fit the boundary condition of a VCSEL.

We show that the following relation can be used to convert the above equations into cylindrical coordinate system:

The gradient is given by

$$\nabla u = \frac{\partial u}{\partial r} \mathbf{e_r} + \frac{1}{r} \frac{\partial u}{\partial \theta} \mathbf{e}_\theta + \frac{\partial u}{\partial z_r} \mathbf{e_z}$$

and the divergence by

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \left[\frac{\partial}{\partial r}(rA_r) + \frac{\partial}{\partial \theta}(A_\theta) + \frac{\partial}{\partial z_r}(rA_z)\right]$$

or

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r}(rA_r) + \frac{1}{r} \frac{\partial}{\partial \theta} A_\theta + \frac{\partial}{\partial z_r} A_z$$

For devices with cylindrical symmetry, $\partial over \partial \theta$ yields zero. The operators gradient and divergence becomes:

$$\nabla u = \frac{\partial u}{\partial r} \mathbf{e_r} + \frac{\partial u}{\partial z_r} \mathbf{e_z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \left[\frac{\partial}{\partial r}(rA_r) + \frac{\partial}{\partial z_r}(rA_z)\right]$$

16.2.3 Longitudinal Modes in VCSEL

In a conventional edge emitting DFB/DBR laser, the number of corrugation gratings is in the order of ≈ 5000 and it is necessary to use the coupled mode theory with

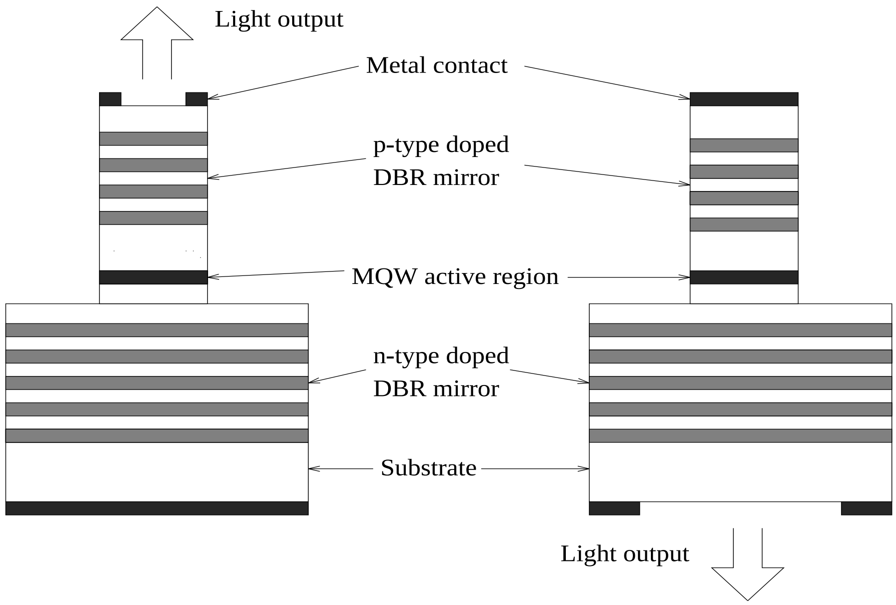


Figure 16.2: Schematics of surface-emitting lasers of front and back emission, respectively.

the coupling coefficient κ to describe the interaction of propagating wave with the gratings.

It is convenient to expand the wave number in Fourier series and keep only the first term:

$$k = \beta_0 + \Delta\tilde{k} + 2\kappa cos[2(\beta_0 + \Delta\beta_{ch})z + j\Omega]$$

where β_0 is a constant reference wave number. As a result of the Fourier expansion, we are able to derive the well known coupled wave equation in 2×2 matrix notation:

$$\frac{\partial}{\partial z}\begin{pmatrix} R(z) \\ L(z) \end{pmatrix} = \begin{pmatrix} -j(\Delta\tilde{k}) & -j\kappa e^{-j\Omega_g} \\ j\kappa e^{j\Omega_g} & j(\Delta\tilde{k}) \end{pmatrix}\begin{pmatrix} R(z) \\ L(z) \end{pmatrix}$$

where R and L are the right-going and the left-going traveling waves.

The above matrix equation can be solved using a numerical approach developed by McCall and Platzman [**?**]. A drawback of the above theory using κL is that the grating must be represented by a sinusoidal function and can be inaccurate for other type of gratings.

In VCSEL (see Figure 16.2), the gratings consist of multiple layers of different materials with large difference in refractive indices. Using a sinusoidal function can be rather inaccurate. We also note that the number of gratings used in VCSEL is relatively small (≈ 100). Therefore it is more convenient to apply the 2×2 matrix method commonly used in multiple layer optics. Following Makino [113], we use the following formulas to solve the coupled wave equation.

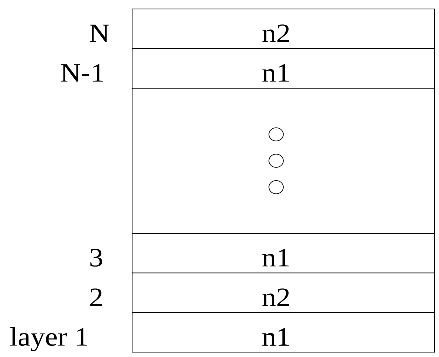


Figure 16.3: Schematic of a simple multi-layer optical model for DFB/DBR gratings.

We use the multi-layer optics/transfer matrix method (TMM) which consists of simple formulas of transmittance reflectance. A basic task of PICS3D is to evaluate the round trip gain of the laser cavity at a particular bias. This may be accomplished with the TMM method which maps the optical field from one point to the other. For example, the following formula can be used for such purpose [113]:

The transfer matrix for layer with index n_1 (see also Figure 16.3):

$$T_1(z|z_k) = \begin{bmatrix} exp[-j\beta_1(z-z_k)] & 0 \\ 0 & exp[j\beta_1(z-z_k)] \end{bmatrix}$$

$$\begin{bmatrix} 1/t_1 & r_1/t_1 \\ r_1/t_1 & 1/t_1 \end{bmatrix}$$

where the transmittance and reflectance are given by

$$r_1 = \frac{n_1 - n_2}{n_1 + n_2}$$

$$t_1 = \sqrt{\frac{n_1'}{n_2'}} \frac{2n_2}{n_1 + n_2}$$

Once the round trip gain is computed, we can use it to search for the roots on the complex frequency plane. The roots correspond to the longitudinal modes. The complex frequencies can be used in Green's function method to compute the optical power spectrum and longitudinal distribution. More details about the Green's function method can be found in the next chapter when discussing longitudinal mode models.

16.2.4 Lateral Modes: Fiber-Like EIM

Exact solution of the optical mode in VCSEL involves vector wave solution in a microcavity of complex boundaries. To see the complexity involved in a vector wave

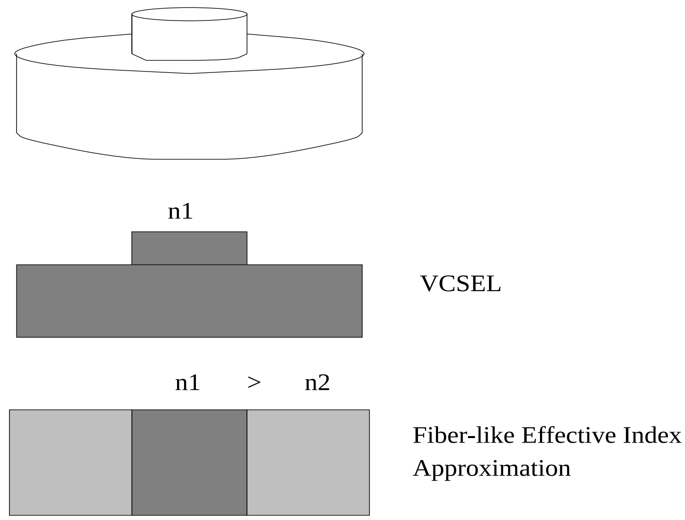


Figure 16.4: Schematic of how the fiber-like effective index approximation is used for a cylindrical VCSEL.

solution even in a simple cylindrical boundary, the reader is referred to Ref. [114]. We have decided to start our VCSEL model from a simple scalar wave solution and build up our capabilities in later versions.

We consider the solution of a scalar wave propagating in the axial direction of a VCSEL structure (see the first and the second structure in Figure 16.4. We assume that the wave distribution in the VCSEL is such that the top cylinder determines the optical confinement in the lateral direction and the lower cylinder only contributes an effective index change in the lateral cladding (outer core). Essentially, we approximate the optical confinement problem to that of an optical fiber. We shall call this approximation a fiber-like effective index method (EIM) in vertical structures [see the third structure in Figure 16.4]. Under the fiber-like EIM, the boundary is reduced to a simple coaxial cylindrical one.

The basic scalar wave equation is given by:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial W}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 W}{\partial \theta^2}+\frac{\partial^2 W}{\partial z_r^2}+k^2 W=0.$$

where z_r is in the axial direction and coincides with the y-axis. It has the following simple solution with cylindrical symmetry.

$$W = W_r(r)exp(-jk_z z_r)$$

where $W_r(r)$ is solved from the Bessel equation of order m:

$$\frac{d^2W_r}{dr^2} + \frac{1}{r}\frac{dW_r}{dr} + \left(k^2 - k_z^2 - \frac{m^2}{r^2}\right) W_r = 0.$$

Using well established Bessel and Hankel functions, the scalar wave equation can be solved efficiently in PICS3D. The fiber-like EIM solution is the default lateral optical model in PICS3D.

16.2.5 Various Simulator Modules in PICS3D

PICS3D should be considered a simulation environment to integrate and coordinate various simulation modules for 3D laser simulation. The following modules are integrated together in PICS3D:

- A two-dimensional (2D) solver for drift-diffusion (DD) equations in cylindrical coordinate system.
- A wave equation solver for the lateral modes.
- A multimode longitudinal mode solver based on results from the 2D module and the lateral mode module.
- A 2D thermal mode solving the heat flow equation in cylindrical coordinate system.

The numerical approach is similar to the 3D simulation procedures for edge emitting lasers. A schematic of the numerical procedure is indicated in the flow diagram in Figure 16.5. The most important part of the coupling between the longitudinal module and 2D DD module is through the evaluation of the modal gain under a certain bias condition. Stimulated recombination and spatial hole burning effects are also important coupling mechanisms.

16.3 Effective Index Method for VCSELs

16.3.1 Introduction

In previous section we describe the fiber-like effective index method solution for the lateral optical mode of a VCSEL. Such a model has the advantage of being easy to interpret and handle. However, when the VCSEL structure becomes more complex, especially when there is oxide confinement layer, more sophisticated lateral model becomes necessary.

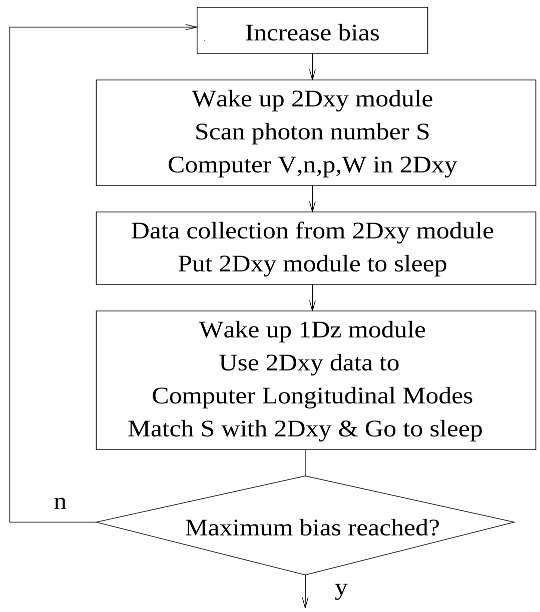


Figure 16.5: Flow diagram of how various modules are integrated.

The Effective index method (EIM) approximation was proposed by Hadley and others (See, for example, G. R. Hadley *et. al.*, "Comprehensive numerical modeling of vertical-cavity surface-emitting lasers," *IEEE J. Quantum Electron.*, Vol. 32, pp. 607-616, 1996.) and has been successfully used to calculate optical modes in VCSEL's. In PICS3D package, we used this EIM calculation method for the calculation of the lateral optical modes in VCSELs. The method is especially useful when a VCSEL has an oxide confinement apecture.

16.3.2 Models

The scalar wave equation in VCSELs is described by the following wave equation:

$$\nabla^2 F(r,z,\phi,t) - \frac{\epsilon(r,z)}{c^2}\frac{\partial^2 F(r,z,\phi,t)}{\partial t^2} = 0, \tag{16.1}$$

where $\epsilon(r,z)$ is the relative permittivity, c is the speed of light in vacuum. The solution of above electric field can be separated into longitudinal ($E_i(z)$) and lateral ($E(r,\phi,t)$) components in VCSELs:

$$F(r,z,\phi,t) \approx E_i(z)E(r,\phi,t)e^{-i\omega t}, \tag{16.2}$$

where i denotes a region in the transverse direction in which the effective index is calculated, and ω_0 is the angular oscillation frequency of the light. Substituting equation (16.2) into (16.1) and writing $\nabla^2 = \partial^2/\partial z^2 + \nabla_\perp^2$, we have

$$\frac{\partial^2 E_i(z)}{\partial z^2} E(r,\phi,t) + E_i(z)\nabla_\perp^2 E(r,\phi,t) + \epsilon_i(r,z)k_0^2 E_i(z)E(r,\phi,t)$$

$$+ 2i\epsilon_i(r,z)k_0 E_i(z)\frac{\partial E(r,\phi,t)}{c\partial t} - \epsilon_i(r,z)E_i(z)\frac{\partial^2 E(r,\phi,t)}{c^2\partial^2 t} = 0, \tag{16.3}$$

where $k_0 = \omega_0/c$. By neglecting the second derivative of $E(r,\phi,t)$ with respect to time (slowly-varying envelope approximation) and write the relative permittivity $\epsilon(r,z)$ as the sum of a structural component $\epsilon_i(z)$ and a nonstructural (depending on the carrier concentration, temperature etc.) component $\epsilon_r(r)$, we can get the following two equations for $E_i(z)$ and $E(r,\phi,t)$ under the effective index (EI) approximation:

$$\frac{\partial^2 E_i(z)}{\partial z^2} + k_0^2(1-\xi_i)\epsilon_i(z)E_i(z) = 0, \tag{16.4}$$

$$\frac{\partial E(r,\phi,t)}{\partial t} = \frac{ic_0}{2k_0\langle\epsilon_i\rangle}[\nabla_\perp^2 + k_0^2\Delta\epsilon_{eff}(r)]E(r,\phi,t), \tag{16.5}$$

where $\langle\epsilon_i\rangle = \int E_i^*(z)\epsilon_i(z)E_i(z)dz / \int E_i^*(z)E_i(z)dz$ and $\Delta\epsilon_{eff}(r) = \xi_i\langle\epsilon_i\rangle + \epsilon_r(r)$. c_0 is the speed of light. The real part of the eigenvalue ξ_i is related to the effective index and the imaginary part is related to the cavity loss (or gain). Using the conventional LP_{mn} notation, the transverse component of the electrical field can be written in the following way:

$$E_{mn}(r,\phi,t) \propto \Psi_{mn}(r,\phi,t)e^{-i\Delta\omega_{mn}t}, \tag{16.6}$$

where Ψ_{mn} is the slowly varying modal field distribution function, and the real part of $\Delta\omega_{mn}$ is the deviation of the LP_{mn} mode from the angular oscillation frequency ω_0. For a system with a cylindrical symmetry with respect to the electrical field, the modal field distribution function Ψ_{mn} must have the following form

$$\Psi_{mn}(r,\phi,t) \equiv \Psi_{mn}(r,t)e^{im\phi}. \tag{16.7}$$

Combining expressions of (16.5)-(16.7) and neglecting the time derivative of the slowly varying modal field distribution function $\Psi_{mn}(r,\phi,t)$, we get the following eigen value equation in cylindrical coordinates:

$$\frac{-c_0}{2k_0\langle\epsilon_i\rangle}\left[\frac{1}{r}\frac{\partial}{\partial r}r\frac{\partial}{\partial r} - \frac{m^2}{r^2} + k_0^2\Delta\epsilon_{eff}\right]\Psi_{mn} = \Delta\omega_{mn}\Psi_{mn}. \tag{16.8}$$

In PICS3D, the equations (16.3) and (16.8) are solved numerically to obtain lateral optical modes in VCSELs.

Chapter 17

PICS3D SPECIFIC MODELS-II: LONGITUDINAL MODES AND 3D SIMULATION

17.1 Update Notes

This chapter describes the longitudinal mode models for a laser diode or VCSEL. Historically, PICS3D started with a decoupled approach as described in the remainder of this chapter. That is to say the longitudinal solver is run only after the main 2/3D electrical solver is completed and all data are collected.

Starting from PICS3D 2007.8, the decoupled solver is being phased out and being replaced by a coupled round trip gain (coupled RTG) solver for better numerical performance. Therefore, this chapter is intended to describe the basic physical models while the numerical methods are yet to be updated. At the printing of this document, the new-RTG method is still being improved while the old decoupled solver is still being supported.

17.2 Introduction

17.3 Basic Equations

Let us start with a wave-guiding structure and define the z-direction as the direction of propagation. The basic equation of concern is, of course, the Maxwell equation. However, we would like to go further and obtain a set of equations directly usable in solving the system. Under the scalar wave assumption, the wave equation can be solved with the technique of variable separation, *i.e.*, we assume that the solution to the wave equation can be written as the product

$$E_\omega(x, y, z) = E_\omega(z)\phi_0(x, y) \tag{17.1}$$

where ω is the optical frequency and z is the direction of the waveguide. Since PICS3D assumes that the laser is operating in a single transverse mode, the exact transverse modal distribution can be left out of the formulation. The modal distribution $\phi_0(x, y)$ can be calculated effectively by other means such as the effective index method or the beam propagation method. It can also be accurately calculated using LASTIP (another program package from Crosslight), which solves the problem self-consistently including all the governing semiconductor equations. The z-dependent part of the electric field $E_\omega(z)$ satisfies the equation

$$\left[\frac{\partial^2}{\partial z^2} + k^2(z)\right] E_\omega(z) = f_\omega(z) \tag{17.2}$$

where $f_\omega(z)$ is the Langevin noise function due to spontaneous emission. We notice that the noise function is extremely important for a laser device since it is a driving force for the solution. For an isolated semiconductor laser, the physical solution for Eq. 17.2 would be zero if the spontaneous noise term was absent. A simple physical explanation is that the spontaneous emission noise generates or excites the photons, which are then amplified by the presence of the optical gain. Lasing oscillation is achieved if the round trip gain approaches unity (as we show later, it may never be unity). The optical power at any bias condition is determined both by the spontaneous emission and the optical gain.

The complex propagation constant $k(z)$ contains information about the solution of the transverse and lateral dimensions, or in other words, it is calculated from the effective index at a specific cross section in the xy-plane. The effective index and the $k(z)$ are dependent on the frequency, material properties and the photon density (as a result of the non-linear gain suppression). Eq. 17.2 is the basic equation which will be used for the optical wave in the PICS3D package.

The second basic equation is the current continuity equation, which is a simple statement of particle conservation. This equation is also solved by the 2D module and is an important part of the Drift-Diffusion model. For the PICS3D simulation, the continuity equation is written as

$$F_{rate}(z, J, S, N) = 0 \tag{17.3}$$

which is a general relation defining the between the injection current, the photon density and carrier densities. This relation is determined by the 2D module of the PICS3D . The physical mechanisms involved are stimulated recombniation and various other recombination terms, drift and diffussion contributions. In many simplified models, this equation many involve the recombnation terms expressed as

$(An + B^2 + Cn^3)$. In PICS3D no assumption is made for the form of the above equation. The equation is sovled numerically based on the 2D module.

Having written down the basic equations we need to solve, we are ready to derive the coupled wave equations and some other important relations used by PICS3D .

17.4 The Green's Function Method

Using Green's function in the analysis of DFB lasers was first proposed by C.H. Henry [116] and later extensively used by Tromborg *et.al.* [117] in deriving analytical formulas for DFB lasers. PICS3D has used the Green's function method because of its accuracy in treating spontaneous emission, and its conceptual simplicity. It is also suitable for numerical implementation.

The Green's function method starts with the wave equation, (17.2). The objective is to obtain a compact expression for the solution to the noise driven wave equation. In the Green's functions method, the solution to Eq. 17.2 can be written as

$$E_\omega(z) = \frac{\int_0^l g(z,z') f_\omega(z') dz'}{W} \tag{17.4}$$

where $f_\omega(z)$ is the local Langevin force, $g(z,z')$ the Green's function, and W the Wronskian of the wave equation. The integration is over the diode cavity length l. The Wronskian W is a *functional* of the *distribution* of the wavenumber $[k(\omega, N(z))]$ in general.

The interpretation of Eq. 17.4 is very simple according to the basic principle of the Green's function method. The Green's function method says that for any linear differential equation with a driving source term, the solution can be found by decomposing the source into many smaller pieces in space. Then the final solution is the sum (or integration) of solutions due to the smaller source terms. In our case, the driving source is the spontaneous noise term, and the electric field solution at location z is a sum (or integration) of solutions due to the spontaneous emission of a single photon at location z'. A more detailed discussion of the Green's function method the the treatment of spontaneous emission is outlined in App. D.

The expressions for both the Green's functions and the Wronskian can be written in terms of the solution for the homogeneous wave equation, *i.e.*, the solution to Eq. 17.2 without the spontaneous noise terms. The Green's function can be written as [116][117]

$$g(z,z') = Z_R(z) Z_L(z') \Theta(z - z') + Z_R(z') Z_L(z) \Theta(z' - z) \ . \tag{17.5}$$

Here $\Theta(z)$ is the Heaviside step function. $Z_L(z)$ is the solution to the homogeneous wave equation [*i.e.* (17.89) with $f_\omega = 0$], which satisfies the boundary condition at the left laser facet and internal interfaces (for multi-section lasers) but may not satisfy the boundary condition at the right laser facet. $Z_R(z)$ is the corresponding solution which satisfies the boundary condition at the right facet and the internal interfaces.

The Wronskian can be written as [116][117]

$$W = Z_L(z)\frac{d}{dz}Z_R(z) - Z_R(z)\frac{d}{dz}Z_L(z) \tag{17.6}$$

Since Z_L and Z_R are solutions to the homogeneous wave equation, it follows from simple algebra that $dW/dz = 0$ in each waveguide section. This means that W is position independent. Therefore, under a particular bias condition, the Wronskian is only a function of the frequency or wavelength. This conclusion is very important and it is the basis for the remaining discussion of modeling techniques.

17.5 Optical Modes and Complex Frequencies

The emission spectrum of a laser is generally continuous. The Green's function method gives a clear definition of a longitudinal mode. According to Eq 17.4, we define a longitudinal mode frequency as one that gives a minimum $\mid W \mid$, or in other words, it is a frequency that results in a maximum in the emission spectrum. Therefore, the task of finding a solution to the basic equations is simplified to one that searches for the minimum of the Wronskian and evaluates the solution using Eq. 17.4 at the minimum of $\mid W \mid$. Note that the ideal of finding the minimum of W in the frequency domain is made possible by the fact that W is independent of position and is only a function of frequency.

It has been shown by Tromborg *et.al.* [117], that with proper choice of normalization,

$$W(\omega) = 1 - R_g \tag{17.7}$$

where R_g is the round trip gain. Since the solution must be finite and the frequency must be real, the Wronskian can never be zero, or in other words, the round trip gain can never reach one. However, we can still force Eq. (17.7) to hold if the frequency is allowed to become complex. In such a case, the complex solution to Eq. (17.7) yields complex ω's with small imaginary parts. The following discussion is based on the above observation.

We concentrate on the stationary multi-mode solution of a laser and neglect the dynamics of the carrier density $N(z)$ at the moment. In Eq. (17.4), the numerator is only weakly dependent on ω, and the field $E_\omega(z)$ peaks at frequencies where $|W|$ is minimum. In a single mode analysis, and when the laser is well above threshold, spontaneous emission can often be neglected (*i.e.* $f_\omega = 0$ in (17.4) and the lasing mode is then characterized by a real frequency ω and carrier density distribution $[N(z)]$ which satisfies the condition $W(\omega, [N(z)]) = 0$). However, the inclusion of spontaneous emission is essential in a multi-mode model. In this case, the Wronskian W is non-zero for all optical frequencies (as the field E has to be finite). Since now $1/W$ is non-singular for *real* optical frequencies, it can be expanded over the poles ω_i of $1/W$ on the *complex frequency* frequency plane

$$\frac{1}{W} \doteq \sum_i \frac{1}{\frac{dW}{d\omega}(\omega - \omega_i)} \tag{17.8}$$

The derivative $dW/d\omega$ in (17.8) is evaluated for ω equals ω_i, which is a solution to the equation

$$W(\omega) = 0 \tag{17.9}$$

In App. E, we show that the optical mode spectrum of a laser can be writtten as

$$I_\omega = \frac{(2\pi)^{-1} R_{sp,i}}{[(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \tag{17.10}$$

where the parameter $R_{sp,i}$ is given by

$$R_{sp,i} c \frac{\left[\int_0^l n(z) n_g |Z_0(z)|^2 dz\right] \left[\int_0^l |Z_0(z')|^2 n(z') g(z') n_{sp} dz'\right]}{\left|\int_0^l n(z) n_g Z_0^2(z) dz\right|^2} \tag{17.11}$$

which is also referred to as the rate of spontaneous emission into the i^{th} mode.

Integrating over z, we get the total number of photons in the cavitty $I_{p,i}$ in the i^{th} mode:

$$I_{p,i} = \frac{R_{sp,i}}{2\omega_{im,i}} \tag{17.12}$$

which agrees with results from other works in the literature (see, e.g., Ref. [117]).

In the above formulas, $Z_0(z) = Z_{0,i}(z)$ is the solution to the homogeneous scalar wave equation at $\omega = \omega_i$, v_g the group velocity, $g(z)$ the modal gain, and n_{sp} the inversion parameter. The expression (17.11) agrees completely with the results in Ref. [117].

The above derivation shows that the longitudinal modes can be represented by a set of poles on the complex frequency plane. The simulator's multi-mode calculation then becomes a task of searching and following such a set of complex poles.

17.6 Complex Frequencies and Wave Distribution

Having found the complex frequencies and the modal spectrum, we must now decide what effect it has on the distribution of longitudinal modes. It is easy to understand that the real part of the frequency is just the frequency observed in the emission spectrum. The issue remains how to interpret the imaginary part of the complex frequency. Our interpretation here is based on the argument of Henry and Kazarinov [119].

Let us use convention of $exp(j\omega t)$. Consider a simple Fabry-Perot laser of cavity length L with modal gain g and intrinsic loss α_i. Under lasing conditions, the round trip gain must be unity, or

$$R_1 R_2 exp[(g-\alpha_i)L - 2j\omega n_{eff} L/c_0] = 1. \tag{17.13}$$

where R_1 and R_2 are the facet reflectivities and c_0 is the speed of light. If ω becomes complex $(\omega_r + j\omega_i)$, the imaginary part effectively modifies the gain by a term

$$\Delta g = 2\omega_i n_{eff}/c_0 \tag{17.14}$$

The same argument can be applied to a DFB or any other type of laser structure because we can always divide an arbitrary structure into small sections (ΔL_k) with uniform material variation. We then use transfer matrix technique to express the round trip gain in terms of the factor $exp[(g-\alpha_i)\Delta L/2 + j\omega n_{eff}\Delta L/c_0]$. By arguing that the round trip gain must remain unity at all times, we reach the same result as in Eq. (17.14).

For a set of longitudinal modes with complex frequencies ($\tilde{\omega}_k$, k=1,2,...), we not only obtain the modal spectrum, but we also get information regarding the spatial distribution of each mode.

17.7 Coupled Wave Equations and Various Laser Structures

17.7.1 Coupled Wave Equations

As was discussed in the previous section, the problem has been reduced to solving for the Green's function and the corresponding Wronskian. These in turn require the knowledge of the solution of the homogeneous wave equation

$$\left[\frac{\partial^2}{\partial z^2} + k^2(z)\right] E_\omega(z) = 0 \tag{17.15}$$

where $k(z)$ is the (complex) wavenumber of the waveguide given by

$$k^2 = n^2 \frac{\omega^2}{c_0^2} \tag{17.16}$$

c_0 is the speed of light in vacuum and n is the refractive index at the optical frequency.

The wave number $k = k(\omega, z, N(z), S(z))$ may depend directly on z due to the grating structure or variation in the material composition and indirectly through the carrier density and photon density variations along the waveguide.

In DFB and DBR lasers, corrugations are made along the wave guides which introduce coupling between forward and backward waves (also referred to as right going and left going waves). The purpose is to perturb the propagation constant $k(z)$ to achieve desirable scattering effects to the propagating waves.

In a laser with a grating of period L_g, its effective refractive index can be written as

$$n = \tilde{n} + 2(\Delta n)cos\left(\frac{2\pi}{L_g(z)}z + \Omega\right) \tag{17.17}$$

where we consider the general case that the grating period may vary as a function of position. $\tilde{n}$ denotes the slow varying part of the complex index and $2(\Delta n)$ is the magnitude of the index variation which is again a complex quantity. We define a reference wave number β_0 (which is usually set to the Bragg wave number in simple grating structures at threshold condition) such that

$$\beta_0 = \frac{\pi}{< L_g >} \tag{17.18}$$

where $<>$ is used to denote the average grating period. β_0 is a constant independent of the frequency and injection conditions.

In general, we can assume that the change in the grating period is smooth and write the following expansion,

$$\frac{\pi}{L_g} \quad = \quad \beta_0 + \Delta\beta_{ch}(z) \tag{17.19}$$

where $\Delta\beta_{ch}(z)$ is caused by some form of chirp grating (variation of grating period).

Based on Eq. (17.17), we re-write the wave number as

$$\begin{aligned} k &= \beta_0 + \Delta\tilde{k} + 2\kappa cos[2(\beta_0 + \Delta\beta_{ch})z + j\Omega] && (17.20) \\ &= \beta_0 + \Delta\tilde{k} + \kappa e^{2j(\beta_0+\Delta\beta_{ch})z+j\Omega} + \kappa e^{-2j(\beta_0+\Delta\beta_{ch})z-j\Omega} && (17.21) \end{aligned}$$

where

$$\Delta\tilde{k} = \tilde{n}\frac{\omega}{c_0} - \beta_0 \tag{17.22}$$

and

$$\kappa = \Delta n\frac{\omega}{c_0} \tag{17.23}$$

Since only the optical frequencies close to the Bragg condition are considered, the second term in Eq. (17.21) is small compared with β_0. Similarly we assume that the coupling coefficient κ is much smaller than β_0. In the following derivation, we will neglect higher order terms of $\Delta\tilde{k}$ and κ.

We propose a trial solution of the form:

$$E_z = R(z)e^{-j\beta_0 z} + L(z)e^{j\beta_0 z} \tag{17.24}$$

where $L(z)$ and $R(z)$ are used to denote the slow varying amplitudes of waves going left and right, respectively. We substitute it into the wave equation to get the following algebra:

$$\begin{aligned}
&-\beta_0^2\left[R(z)e^{-j\beta_0 z} + L(z)e^{j\beta_0 z}\right] \\
&-2j\beta_0\left[\left(\frac{\partial R(z)}{\partial z}\right)e^{-j\beta_0 z} - \left(\frac{\partial L(z)}{\partial z}\right)e^{j\beta_0 z}\right] \\
&+\left(\frac{\partial^2 R(z)}{\partial z^2}\right)e^{-j\beta_0 z} + \left(\frac{\partial^2 L(z)}{\partial z^2}\right)e^{j\beta_0 z} \\
&+k^2\left(R(z)e^{-j\beta_0 z} + L(z)e^{j\beta_0 z}\right) = 0
\end{aligned} \tag{17.25}$$

Treating $\Delta\tilde{k}$ and κ as small quantities, we expand the wave number k^2 as follows.

$$\begin{aligned}
k^2 &= \beta_0^2 + 2\beta_0\left(\Delta\tilde{k} + \kappa e^{2j(\beta_0+\Delta\beta_{ch})z+j\Omega} + \kappa e^{-2j(\beta_0+\Delta\beta_{ch})z-j\Omega}\right) \\
&\quad + \mathit{higher\ order\ terms.}
\end{aligned} \tag{17.26}$$

which is then substituted into Eq. (17.25). To simplify the notation, we introduce a slow varying function

$$\Omega_g(z) = 2\Delta\beta_{ch}(z)z + \Omega \tag{17.27}$$

We further assume that the wave amplitude is a slow varying function of z and neglect the second derivatives involving $\left(\frac{\partial^2 R(z)}{\partial^2 z}\right)$ and $\left(\frac{\partial^2 L(z)}{\partial^2 z}\right)$. After the terms involving β_0^2 are canceled, we are left with the following equation.

$$\begin{aligned}
&-2j\beta_0\left[\left(\frac{\partial R(z)}{\partial z}\right)e^{-j\beta_0 z} - \left(\frac{\partial^2 R(z)}{\partial^2 z}\right)e^{j\beta_0 z}\right] \\
&+2\beta_0\left(\Delta\tilde{k} + \kappa e^{2j\beta_0 z+j\Omega_g} + \kappa e^{-2j\beta_0 z-j\Omega_g}\right)\left(R(z)e^{-j\beta_0 z} + L(z)e^{j\beta_0 z}\right) \\
= &\ 0.
\end{aligned} \tag{17.28}$$

Since $R(z)$ and $L(z)$ are independent functions, this equation makes sense only when terms with the same propagation factor sum up to zero. In other words, the terms with $e^{-j\beta_0 z}$ and $e^{j\beta_0 z}$ must sum up zero, respectively. Collecting these terms, we find

$$-2j\beta_0\left(\frac{\partial R(z)}{\partial z}\right) + 2\beta_0\Delta\tilde{k}R(z) + 2\beta_0\kappa e^{-j\Omega_g}L(z) = 0 \tag{17.29}$$

$$2j\beta_0\left(\frac{\partial L(z)}{\partial z}\right) + 2\beta_0\Delta\tilde{k}L(z) + 2\beta_0\kappa e^{j\Omega_g}R(z) = 0 \tag{17.30}$$

which can be re-written in the familiar form

$$\begin{aligned}\frac{\partial}{\partial z}R(z)+j(\Delta\tilde{k})R(z) &= -j\kappa e^{-j\Omega_g}L(z) && (17.31)\\ \frac{\partial}{\partial z}L(z)-j(\Delta\tilde{k})L(z) &= j\kappa e^{j\Omega_g}R(z) && (17.32)\end{aligned}$$

or in matrix notation:

$$\frac{\partial}{\partial z}\begin{pmatrix}R(z)\\ L(z)\end{pmatrix}=\begin{pmatrix}-j(\Delta\tilde{k}) & -j\kappa e^{-j\Omega_g}\\ j\kappa e^{j\Omega_g} & j(\Delta\tilde{k})\end{pmatrix}\begin{pmatrix}R(z)\\ L(z)\end{pmatrix} \tag{17.33}$$

17.7.2 Transfer Matrix Formulas

In this subsection the technique developed by McCall and Platzman [?] is used to solve the coupled wave equations.

The complex phase factor $e^{j\Omega_g}$ is separated into real and imaginary parts and the wave equation rewritten as:

$$\begin{aligned}\frac{\partial}{\partial z}R(z) &= -j(\Delta\tilde{k})R(z)+[-j\kappa cos(\Omega_g)-\kappa sin(\Omega_g)]L(z) && (17.34)\\ \frac{\partial}{\partial z}L(z) &= [j\kappa cos(\Omega_g)-\kappa sin(\Omega_g)]R(z)+j(\Delta\tilde{k})L(z) && (17.35)\end{aligned}$$

We introduce the Pauli matrices

$$\sigma_1=\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} \tag{17.36}$$

$$\sigma_2=\begin{pmatrix}0 & -j\\ j & 0\end{pmatrix} \tag{17.37}$$

$$\sigma_3=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \tag{17.38}$$

and express the equation in terms of a spinor:

$$\psi=\begin{pmatrix}R(z)\\ L(z)\end{pmatrix} \tag{17.39}$$

We obtain

$$\begin{aligned}\frac{\partial}{\partial z}\psi &= [-\kappa sin(\Omega_g)\sigma_1+\kappa cos(\Omega_g)\sigma_2-j(\Delta\tilde{k})\sigma_3]\psi && (17.40)\\ &= H\psi && (17.41)\\ &= \mu\cdot\sigma\psi && (17.42)\end{aligned}$$

where boldface is used to denote a three component vector. More specifically, in three component notation,

$$\mu = \left[-\kappa sin(\Omega_g); +\kappa cos(\Omega_g); -j(\Delta\tilde{k}) \right] \quad (17.43)$$

and

$$\sigma = \begin{pmatrix} \sigma_1; \\ \sigma_2; \\ \sigma_3 \end{pmatrix} \quad (17.44)$$

An arbitrary function f of the scalar $a + \mathbf{b} \cdot \sigma$, linear in the Pauli matrices, can be reduced to another linear function[120]:

$$f(a + \mathbf{b} \cdot \sigma) = A + \mathbf{B} \cdot \sigma \quad (17.45)$$

where

$$A = \frac{1}{2}[f(a+b) + f(a-b)]; \quad (17.46)$$
$$\mathbf{B} = \frac{\mathbf{b}}{2b}[f(a+b) - f(a-b)]; \quad (17.47)$$

which allows us to expand the solution to Eq. (17.42) which can be written as

$$\psi(z) = cosh[\mu(z - z_0)] + \frac{\mu \cdot \sigma}{\mu} sinh[\mu(z - z_0)] \quad (17.48)$$

where μ is the complex square root of $\mu \cdot \mu$.

$$\mu^2 = \kappa^2 - (\Delta\tilde{k})^2 \quad (17.49)$$

We note that it makes no difference which sign the complex μ takes.

The transfer matrix that maps ψ from one location to another is given by

$$\psi(z) = T\psi(z_0) \quad (17.50)$$

where

$$T_{11} = cosh[\mu(z - z_0)] - j(\Delta\tilde{k}) sinh[\mu(z - z_0)]/\mu \quad (17.51)$$
$$T_{12} = -j\kappa e^{-j\Omega_g} sinh[\mu(z - z_0)]/\mu \quad (17.52)$$
$$T_{21} = j\kappa e^{j\Omega_g} sinh[\mu(z - z_0)]/\mu \quad (17.53)$$
$$T_{22} = cosh[\mu(z - z_0)] + j(\Delta\tilde{k}) sinh[\mu(z - z_0)]/\mu \quad (17.54)$$

where

$$\Omega_g(z) = 2\Delta\beta_{ch}(z)z + \Omega \quad (17.55)$$

17.7.3 Multilayer Optics and Transfer Matrix Formulas

The previous subsection provides an efficient model for edge emitting laser with DFB/DBR gratings. A major approximation is to represent all gratings with sinusoidal functions.

In Vertical Cavity Surface Emitting Lasers (VCSEL), there are fewer grating periods and each grating period consists of layers of different indices. In such cases, it is more accurate to describe the traveling wave using a multilayer optics technique, also involving 2×2 matrices.

PICS3D also implements the layered optics model. The grating structure must be described by periodic layers of two different indices n_1 and n_2 of 1/4 wavelength thickness. The transfer matrix in such a case is given by [113]

$$T_1(z|z_k) = \begin{bmatrix} exp[-j\beta_1(z-z_k)] & 0 \\ 0 & exp[j\beta_1(z-z_k)] \end{bmatrix} \tag{17.56}$$

$$\begin{bmatrix} 1/t_1 & r_1/t_1 \\ r_1/t_1 & 1/t_1 \end{bmatrix} \tag{17.57}$$

where the transmittance and reflectance are given by

$$r_1 = \frac{n_1 - n_2}{n_1 + n_2} \tag{17.58}$$

$$t_1 = \sqrt{\left(\frac{n'_1}{n'_2}\right)}\frac{2n_2}{n_1 + n_2} \tag{17.59}$$

Please note that the field being transferred by the above matrix is normalized such that its amplitude squared equals to the optical power.

In the PICS3D the grating model using kappa is referred to as the kappa grating model while the new layered model described in this section is referred to as the layered grating model. The layered grating model is mostly used for VCSEL's.

17.7.4 Device structures

It is now appropriate to discuss some of the laser structures modeled by PICS3D and how they correspond to the above equations.

Conventional **DFB** lasers are assumed to have AR coating on both facets and a real κ with no z-dependence.

Phase Shifted DFB lasers are assumed to have real κ and zero $\Omega(z)$ but with a phase shifted section (or a section without gratings) somewhere along the waveguide.

Chirp Grating DFB lasers can be represented by a linear change with z in $\Delta\beta_{ch}$. This can be realized with changing the grating period linearly.

Longitudinal Graded Index lasers can be represented by a linear change in $\Delta\tilde{k}$.

DBR lasers can be considered as a special case of DFB lasers with no grating in the active section. The passive mirror section has a regular grating with a low material gain. In such a case, more than one material may need to be defined.

Gain/Loss Coupled DFB lasers can be considered as having complex κ.

Tapered structures can be treated with spatial variation in the optical confinement factor, current pumping rate, cross section of the laser and linear change in index.

PICS3D provides every possibility of defining various variables within the framework of the coupled wave equation. It is interesting to note that a complex κ (*i.e.*, gain coupled structure) has a similar (but not identical) effect on κ as changing the phase $\Omega_g(z)$ (chirp grating). This is consistent with the well known fact that gain coupling, chirp grating and phase shifting all give similar results in the emission characteristics of the laser and help move the main emission peak into the spectrum band-gap.

17.7.5 Bragg wavelength and related reference parameters

In this subsection, we discuss several important parameters used in simulation of waveguiding structures involving DFB/DBR gratings. In a realistic device involving corrugation gratings, the Bragg wavelength in vacuum is generally not well defined. The spacing or the pitch of the grating is fixed by the fabrication procedure but the modal index of the structure varies with bias and is not known before a simulation. We have chosen to fix the parameter of expansion β_0 as a constant at all times for a particular device.

In our simulator, the parameter β_0 is given by the average grating period L_g as

$$\beta_0 = \frac{\pi}{< L_g >} \tag{17.60}$$

Thus in theory, β_0 is well defined once the grating structure is known. However, we often need to plot many parameters in wavelength or frequency domain and we introduce a "reference wavelength" in vacuum

$$\lambda_{ref} = 2n_{ref} < L_g > \tag{17.61}$$

where n_{ref} is a "reference index" which is normally chosen to be close to the modal index at threshold. Thus, when we see a plot of spectrum produced by the simulator, the data is relative to the above "reference wavelength" in vacuum which is generally different from the "Bragg wavelength":

$$\lambda_{Br} = 2n_{eff} < L_g > \tag{17.62}$$

where n_{eff} is the effective index which is a function of bias. From device design point of view, the process is often the other way around: given the Bragg wavelength in vacuum, what is the average pitch (or period)? Thus, it is important that we define n_{ref} clearly (so that we know the meaning of the spectrum data) and compute n_{eff} accurately (so that we can figure out $< L_g >$).

The "reference index" is defined as follows. If the user defines parameter "ref_index" in statement **longitudinal**, this is what we use as reference index. If the user does not define this parameter in the input file, the simulator takes it as the modal index evaluated at an estimated threshold carrier injection density set for the statement **rtg_phase** in the .sol file.

The fundamental assumption in the software for index change is that the real index in the macro library corresponds to the index value at equilibrium (with low carrier density in active regions of many devices). When device is biased, the index change is controlled by the statement **index_model**.

For edge emitting devices, the propagation constant in the coupled mode theory is evaluated by

$$\beta - \beta_0 = (\omega - \omega_{ref})n_{eff}/c + \omega_{ref}[n_{eff} - n_{eff,tab}(N_{init})]/c \tag{17.63}$$

where ω_{ref} is the frequency at λ_{ref} and $n_{eff,tab}(N_{init})$ is the tabulated index change data (as in .gain preview) evaluated at active region density of N_{init} which is set by the parameter "init_conc" in statement **index_wavel**. Thus, if tabulated gain model (or "gain_type3=yes" in older versions) is used, the device should be perfectly tuned at concentration of "init_conc". However, if tabulated gain model is not used, (or "gain_type3=no"), the actual index change at concentration of "init_conc" may deviate from the tabulated index change and we may observe some detuning at "init_conc". Usually this detuning should be small if the hole and electron concentration ratio is estimated correctly for the tabulated gain and index change.

For VCSELs, as we will derive in the following sections, we do not expand the equations in terms of $\beta - \beta_0$ since it can be very large at such a short cavity length. We instead use the transfer matrix method (TMM) where the absolute value of the index n_{eff} as controlled by **index_model** is directly used for each longitudinal optical section. For a longitudinal section containing DBR grating, if the index of each DBR layer is set with a negative sign, only the index difference is used while the average index within the grating is controlled by the **index_model** so that effects such as thermal tuning can be incorporated.

17.8 Self-Consistent Steady State Solution

In previous sections, we have introduced the the rate equation, wave equation, and the transfer matrix method used to solve coupled coupled wave equations. It remains to be determined which variables we are actually solving for steady state.

To account for the spatial hole burning effects, we must divide the longitudinal dimension into N_z subsections. We re-write the Eq. (17.7) to mean round trip being unity:

$$R_g(\omega) - 1 = 0 \tag{17.64}$$

We search for the roots of the above equation and find M complex roots $\tilde{\omega}_i$. Also note that the current continuity equation must be satisfied for each subsection. We are able to write down the following basic equations:

$$\begin{aligned} F_{rate}(z_k, J_k, S_k, N_k) &= 0 & (17.65) \\ Re[R_g(\tilde{\omega}_i, N_k)] - 1 &= 0 & (17.66) \\ Im[R_g(\tilde{\omega}_i, N_k)] &= 0 & (17.67) \end{aligned}$$

where the indices $k = 1, ..., N_z; i = 1, ..., M$. The photon density is computed from the complex frequency with the following total photon number:

$$I_{p,i} = \frac{R_{sp,i}}{2\omega_{im,i}} \tag{17.68}$$

and from the coupled wave equation solver using transfer matrix method:

$$\begin{pmatrix} R(k) \\ L(k) \end{pmatrix} = T_{tot,k} \cdot \begin{pmatrix} R(0) \\ L(0) \end{pmatrix} \tag{17.69}$$

For convenience of discussion, we introduce a relative photon number $S_w(k, i)$ to denote the relative amount of photon at the kth subsection for the ith mode:

$$S_w(k, i) = \frac{(|R_{k,i}|^2 + L_{k,i}^2)\Delta z_k}{\sum_{k=1}^{N_z}(|R_k|^2 + L_k^2)\Delta z_k} \tag{17.70}$$

Then the photon density at the kth subsection can be written as

$$S_k = \sum_{i=1}^{M} S_w(k, i) I_{p,i}/\Delta z_k \tag{17.71}$$

Eqs. (17.65) to (17.67) defines $N_z + 2M$ equations for $N_z + 2M$ real variables. Therefore, we can solve these equations using fully coupled Newton's method. The main task of PICS1D is to solve these equations under various bias conditions.

The above system of equations is highly nonlinear; a good starting guess of the carrier density distribution is necessary for the numerical procedure to converge, and the bias current should be adjusted in small steps. The starting current is chosen such that all the modes are below threshold. We use the following steps to obtain a good guess solution for the fully coupled Newton procedure.

- 1) We ramp up the bias current from below threshold.
- 2) The carrier density distribution is then easily calculated from (17.65) by setting the photon density terms to zero.
- 3) This carrier density distribution and the zero is then used to compute the material gain and the round trip gain at a range of real frequencies.
- 4) If the maximum round trip gain exceeds unity, it means our initial bias current is too large. The simulation is aborted the user is advised to set a smaller initial bias and restart the program.
- 5) If the maximum round trip gain is much less than unity, we go back to 1).
- 6) If the maximum round trip gain is below but close to unity, we continue to the next step.
- 7) We take the frequencies corresponding to the peaks in the round trip gain as the guess for the complex frequencies for the fully coupled Newton procedure. At this point our best guess are the density distribution from 2) and the real frequencies found at this step.

Once the initial guess is obtained, we use the following Newton procedure with automatic bias control.

- 1) Start the full Newton procedure and iterate to solve for the density and complex frequencies.
- 2) If the Newton iteration converges, we increase the bias current and repeat the Newton procedure until the final bias point is reached.
- 3) If the Newton iteration does not converge, we abort the simulation if this is the first bias point. Otherwise we reduce the bias step and go back to 1). If the bias step is reduced to a tiny value set by the user (or a default) because of failure to converge for too many times, the simulation is aborted. The user is advised to improve the mesh of the structure or check for bad frequency ranges.

17.9 Longitudinal Modes and Overlap of Traveling Waves

We have shown that finding longitudinal modes is reduced to finding the complex roots of the Wronskian on the complex omega plane. We have also shown how this is done self-consistently with solving the continuity equations in a full Newton procedure. In this sections, we shall give some details of how to evaluate the longitudinal mode power properly in a practical simulation.

We recall that we must find the complex frequency and determine the ith mode with total photon number:

$$I_{p,i} = \frac{R_{sp,i}}{2\omega_{im,i}} \tag{17.72}$$

Special care must be taken when calculating $R_{sp,i}$. According to App. E the denominator of $R_{sp,i}$ contains the derivative of the Wronskian with respect to the frequency, $\frac{\partial W}{\partial \omega}$. Usually this term is evaluated as follows (see [117]):

$$\frac{\partial W}{\partial \omega} = 2\int_0^L k(z)Z_0^2(z)\frac{\partial k}{\partial \omega}dz \tag{17.73}$$

where Z_0 is the solution to the homogeneous wave equation which can be expressed as the sum of right and left going waves.

$$Z_0 = Z_0^+(z)exp(-j\beta_0 z) + Z_0^-(z)exp(j\beta_0 z) \tag{17.74}$$

where β_0 is a constant reference wave number usually set to be close to the the Bragg wave number. For integration of Z_0^2, the oscillating terms of Z_0^{+2} and Z_0^{-2} are negligible. We are able to use the overlap term to express the Wronskian derivative:

$$\frac{\partial W}{\partial \omega} = 2\int_0^L k(z)2Z_0^+(z)Z_0^-\frac{\partial k}{\partial \omega}dz \tag{17.75}$$

Since this overlap integral appears in the denominators of our theories, it follows that the left and right going waves must overlap significantly for our theory to work.

Fortunately the overlap integral is non-zero for most applications involving semiconductor lasers, because the waves must make many round trips within the laser cavity to get amplification. However, the overlap requirement may cause some problems for solving some of the sidemodes which have very small overlaps between the left and right going waves. In such a case, the formulas for calculating $R_{sp,i}$ breaks down and $R_{sp,i}$ becomes extremely large. As a result, the solver may mistaken the sidemode as the main mode even the complex root shows a very large imaginary part. To overcome this numerical difficulty, we must artificially add a small non-zero term to Eq. (17.75). The issue is how to choose this small term.

We consider two extreme cases for $\frac{\partial W}{\partial \omega}$. The first is an optical amplifier where the overlap integral is zero because the traveling wave never get reflected. According to Henry [116], the Wronskian is given by

$$W = 2jk \tag{17.76}$$

and the derivative is given by

$$\frac{\partial W}{\partial \omega} = 2j/v_g \tag{17.77}$$

The other extreme is a Fabry-Perot laser cavity with reflectivity of r_1:

$$\frac{\partial W}{\partial \omega} = 4kLr_1/v_g \tag{17.78}$$

The derivatives of the two cases can differ by several thousand times.

We expect the $\frac{\partial W}{\partial \omega}$ term for a typical laser to be about 1000-100000 times of that for an optical amplifier. Therefore, it is reasonable to add a non-zero term of 10-100 times of $2j/v_g$ to $\frac{\partial W}{\partial \omega}$ so that stable numerical solution can be found. This explains the parameter **add_overlap** in the **longitudinal** statement from the simulation input file.

17.10 Mode Spectrum

Once the steady state solution for the complex frequencies is found, we can compute the emission spectrum of the laser. An important result from PICS1D is the mode spectrum under various bias conditions and from the two different facets. Here we refer to the "pure spectrum". This is different from the experimental spectrum which is the 'pure spectrum' convoluted with the response function of, say, the spectral analyzer. PICS1D uses rather simple formulas for the computation of the pure modal spectrum.

Here are some simple considerations. We consider the wave equation in its simplest form

$$\nabla^2 E = F_{noise}, \tag{17.79}$$

where E is the electric field of the optical wave and F_{noise} is a general term representing noise of all sources. According to Section 17.5, the wave equation can solved using the Green's function method and the solution can be written in terms of a Wronskian:

$$E = F_{noise}/W \tag{17.80}$$

where W is the Wronskian. In a complete laser model, W includes the effect of boundary conditions, *i.e.*, multiple reflection from the facets. Therefore the solution is expected to give the multi-longitudinal modes produced as a result of the multiple reflection.

Similar to Section 17.5, we take the square of the amplitude of Eq. (17.80) and average over ensemble and spatial variables. The noise term, after appropriate averaging, becomes the spontaneous emission rate into the i^{th} mode, $R_{sp,i}$ [see also Eqs. (17.11) and (17.12)], and the modal spectrum can be written as

$$I_\omega = \frac{R_{sp,i}}{\mid \omega - \omega_i \mid^2}. \tag{17.81}$$

Note that ω_i is complex with a non-vanishing imaginary part. The spectrum peaks at the zero points of the real part of ω_i.

The above formulas for modal spectrum is actually the spectrum for total photon number in the whole cavity. We derive the corresponding expressions for power spectra from the two different facets here.

In our notation, the left facet spectrum should be based on the photon density at subsection number 1 and the right facet at number N_z. The photon density per unity angular frequency is given by

$$S_\omega(1,i) = F_{left,i} \frac{R_{sp,i}}{\mid \omega - \omega_i \mid^2}. \tag{17.82}$$

for the left facet and

$$S_\omega(N_z,i) = F_{right,i} \frac{R_{sp,i}}{\mid \omega - \omega_i \mid^2}. \tag{17.83}$$

for the right facet.

Again, we use the relative photon number to express the facet dependent factor F_{left} and right F_{right} as follows:

where F_{facet} is a factor depending on the facet condition:

$$F_{left,i} = S_w(1,i) v_g (1 - R_{left}) / \Delta z_1 \tag{17.84}$$

and

$$F_{right,i} = S_w(N_z,i) v_g (1 - R_{right}) / \Delta z_{Nz} \tag{17.85}$$

where R_{left} and R_{right} are the facet power reflectivities and the relative photon number at section k is given by

$$S_w(k,i) = \frac{(|R_{k,i}|^2 + L_{k,i}^2)\Delta z_k}{\sum_{k=1}^{N_z} (|R_k|^2 + L_k^2)\Delta z_k} \tag{17.86}$$

17.11 Dynamic Equation for the Mode Amplitude

Having found the steady state solution, we can consider the dynamic properties. We would like to consider how the mode amplitude (or wave envelope) vary with time. In this section, we assume that the system is under small external perturbation so that Taylor expansion can be made for many quantities including carrier and photon densities.

We present the derivation of a general small signal dynamic equation for the mode amplitude of a semiconductor laser. The formulation follows mainly that of [121]. It was originally derived for DFB lasers but should be applicable to other type of lasers.

In the theory of small signal dynamic model outlined in this section, the critical device parameters are the optical gain model and the relationship between injection current and carrier density which is expressed as the ABC formula: $(An+Bn^2+Cn^3)$. In a 2/3D simulation, both the modal gain and the ABC formula can be regarded as a function of the average electron concentration in the active region. The optical gain data is automatically transferred from the 2/3D drift-diffusion solver to the 1Dz module.

In the current version, the the ABC coefficents are determined as follows: We neglect the effect of current overflow and assume all injection current are recombined in the active region. The carrier life time and Auger coefficeint specified in statement **current_conc** are first used to specify the parameters A and C. Then the spontaneous recombination through the quantum levels in the active region is automatically computed and add to the ABC formula. The effect of current spreading can be included through a reduced average carrier density this way. Since the current overflow leakage is neglected, the ABC coefficients are generally underestimated. We hope to improve the current-concentration model in an upgrade in the near future.

We consider a laser waveguide with an arbitrary type of grating corrugation where we may inject an arbitrary current distribution. It is assumed to consist of a finite number of sections within which the laser parameters vary continuously. A longitudinal z-axis is introduced with origin at the left facet. r_1 and r_2 denote the left and right facet reflectivities.

Similar to [116],[122] we assume that the laser oscillates in the fundamental transverse mode $\phi_0(x, y; z)$, which may change from section to section, but is only weakly dependent on z within each section. It is assumed to satisfy the normalization condition

$$\int |\phi_0(x, y)|^2 dxdy = 1\,. \tag{17.87}$$

We can then define a (frequency domain) complex longitudinal electric field $E_\omega(z)$ by

$$E_\omega(z)\phi_0(x, y) = \int_{-\infty}^{\infty} E(\vec{x}, t)e^{-j\omega t} dt \tag{17.88}$$

where $E(\vec{x}, t)$ is the real electric field and $\vec{x} = (x, y, z)$.

The field $E_\omega(z)$ satisfies the inhomogeneous scalar wave equation [116][122]

$$\left[\frac{\partial^2}{\partial z^2} + k^2(z)\right] E_\omega(z) = f_\omega(z) \tag{17.89}$$

within each section. The driving force $f_\omega(z)$ is a Langevin noise function due to spontaneous emission. The complex wave number $k = k(\omega, z, N(z), S(z))$ may depend directly on z, due to the grating structure or variation in the material composition, and indirectly through the carrier density $N(z)$ and photon density $S(z)$. The latter dependence is caused by the nonlinear gain compression. In the following we indicate only the dependence of z and write $k(z)$ for the wave number.

The solution to (17.89) is given by (see *e.g.* [123])

$$W([k(z)])E_\omega(z) = \int_0^l g(z,z')f_\omega(z')dz' \tag{17.90}$$

where $g(z,z')$ is the Green's function:

$$g(z,z') = Z_R(z)Z_L(z')\Theta(z-z') + Z_R(z')Z_L(z)\Theta(z'-z)\ . \tag{17.91}$$

Here $\Theta(z)$ is the Heaviside step function. $Z_L(z)$ is the solution to the homogeneous wave equation [*i.e.* (17.89) with $f_\omega = 0$], which satisfies the boundary condition at the left laser facet and internal interfaces (for multi-section lasers) but may not satisfy the boundary condition at the right laser facet. $Z_R(z)$ is the corresponding solution which satisfies the boundary condition at the right facet and the internal interfaces. For each solution $Z_X(z)$, $X = L, R$, we introduce the right and left traveling waves Z_X^+ and Z_X^- defined by [124]

$$Z_X^\pm(z) = \frac{1}{2}[Z_X(z) \pm \frac{j}{\tilde{k}(z)}\frac{d}{dz}Z_X(z)] \tag{17.92}$$

where $\tilde{k}(z)$ is the slowly varying part of the wave number. In many applications it is sufficient to use a simpler definition of $Z_X^\pm$, where $\tilde{k}(z)$ is replaced by a fixed (real) reference wave number β_0. For DFB structures, β_0 may be chosen as the Bragg wave number. It follows from (17.92) that Z_X is the sum

$$Z_X(z) = Z_X^+(z) + Z_X^-(z) \tag{17.93}$$

of right and left traveling waves.

It is convenient to use the following normalization condition for the 'outgoing' components of Z_L and Z_R at the two ends

$$Z_L^-(0) = 1\ , \qquad\qquad Z_R^+(l) = 1 \tag{17.94}$$

As a consequence the boundary conditions for the reflected components become

$$Z_L^+(0) = r_1\ , \qquad\qquad Z_R^-(l) = r_2 \tag{17.95}$$

The factor W in (17.90) is the Wronskian defined by

$$W = Z_L(z)\frac{d}{dz}Z_R(z) - Z_R(z)\frac{d}{dz}Z_L(z) \tag{17.96}$$

Since Z_L and Z_R are solutions to the homogeneous wave equation, it follows that $dW/dz = 0$ in each waveguide section. This means that W is constant within each section. We shall assume that the waveguide sections are all index guiding, which ensures that W is continuous across the interfaces between the sections and hence constant along the whole waveguide. W may be considered as a mapping of the function $k(z)$ into a complex number $W([k(z)])$, which means that W is a function of $k(z)$.

It follows from (17.90) that (in the limit of vanishing driving force) there can only be a non-zero solution when $W = 0$. For the stationary case with DC current injection, the modes of the system are therefore the states for which

$$W([k(z)]) = 0 \tag{17.97}$$

is fulfilled together with the rate equation for the stationary carrier density. The actual solutions (including spontaneous emission) are treated as perturbations of the solutions to (17.97).

The oscillation condition (17.97) is equivalent to round trip gain being unity. A mathematical proof of this can be found in [117]

In general there can be multiple solutions to (17.97). However, here we only concentrate on the single mode case, and the corresponding wave number is denoted by $k_0(z)$. Since $W([k_0(z)]) = 0$, the two solutions $Z_L(z)$ and $Z_R(z)$ are proportional, and in this case the expression for $g(z, z')$ reduces to

$$g(z, z') = Z_L(z)Z_R(z') \tag{17.98}$$

If the laser is oscillating in the mode $n = 0$, we have $W \simeq 0$ so that we can use the above Green's function. To first order in the (small) driving force the field of the lasing mode is given by

$$W([k(z)])E_\omega(z) = Z_L(z)\int_0^l Z_R(z')f_\omega(z')dz' \tag{17.99}$$

$E_\omega(z)$ is therefore proportional to $Z_L(z)$, and hence also to $Z_R(z)$. Following Henry [116] we define a mode amplitude A_ω by

$$E_\omega(z) = A_\omega Z_L(z) \tag{17.100}$$

and the corresponding envelope function $A(t)$ by

$$A(t) = \frac{1}{2\pi}\int_0^\infty A_\omega e^{j(\omega-\omega_0)t}d\omega\ . \tag{17.101}$$

Due to the normalization $Z_L^-(0) = 1$ the envelope $A(t)$ represents the outgoing field at the inner left facet.

In order to derive a rate equation for the envelope function $A(t)$ we expand the function $W([k(z)])$ around the stationary solution $k_0(z)$ [116]. As discussed previously, the wave number $k(z)$ depends on frequency ω, carrier density $N(z)$ and photon density $S(z)$. Hence

$$\Delta k = \frac{\partial k}{\partial \omega}(\omega - \omega_0) + \frac{\partial k}{\partial N}\Delta N + \frac{\partial k}{\partial S}\Delta S\ , \tag{17.102}$$

where $\Delta k = k - k_0$, $\Delta N = N - N_0$ and $\Delta S = S - S_0$. The first order expansion of the Wronskian can be written as

$$W([k(z)]) = \int_0^l \frac{\delta W}{\delta k(z')} \Delta k(z') dz' \tag{17.103}$$

where $\delta W/\delta k(z)$ is the functional derivative. It is shown [117] to be given by the simple expression

$$\frac{\delta W}{\delta k(z)} = 2k_0(z) Z_L(z) Z_R(z) \tag{17.104}$$

By (17.100) and (17.99) the mode amplitude satisfies the equation

$$W([k(z)]) A_\omega = \int_0^l Z_R(z') f_\omega(z') dz' \tag{17.105}$$

in the frequency domain. We have the following derivation:

$$\begin{aligned}
W([k(z)]) &= (\omega - \omega_0) \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial \omega} dz' \\
&+ \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial N} \Delta N dz' \\
&+ \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial S} \Delta S dz' \\
&= (\omega - \omega_0) \frac{\partial W}{\partial \omega} \\
&+ \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial N} \Delta N dz' \\
&+ \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial S} \Delta S \, dz'
\end{aligned} \tag{17.106}$$

Our basic equation from the Green's function becomes:

$$-W([k(z)]) A_\omega + \int_0^l Z_R(z') f_\omega(z') dz' = 0 \tag{17.107}$$

or

$$\begin{aligned}
(\omega - \omega_0) A_\omega \frac{\partial W}{\partial \omega} &= -A_\omega \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial N} \Delta N dz' \\
&- A_\omega \int_0^l \frac{\delta W}{\delta k(z')} \frac{\partial k}{\partial S} \Delta S \, dz' \\
&+ \int_0^l Z_R(z') f_\omega(z') dz' = 0
\end{aligned} \tag{17.108}$$

We make an inverse Fourier transform (noting $(\omega - \omega_0) = \frac{\partial}{\partial t}$), we find the following rate equation for $A(t)$

$$\frac{d}{dt}A(t) = A(t)\int_0^l [C_N(z)\Delta N(z) + C_S(z)\Delta S(z)]dz + F_A(t) \tag{17.109}$$

where the important weight functions $C_N(z)$ and $C_S(z)$ are defined by

$$C_X(z) = -j\frac{\delta W}{\delta k(z)}\frac{\partial k}{\partial X}(z)\Bigg/\frac{\partial W}{\partial \omega} \tag{17.110}$$

for $X = N, S$. The term

$$F_A(t) = j\int_0^l Z_R(z)f(z,t)dz\Bigg/\frac{\partial W}{\partial \omega} \tag{17.111}$$

is a Langevin noise function representing the integrated effect of spontaneous emission along the cavity, where

$$f(z,t) = \frac{1}{2\pi}\int_0^\infty f_\omega(z)e^{j(\omega-\omega_0)t}d\omega\ . \tag{17.112}$$

is the corresponding *local* Langevin force.

The field equation (17.109) for the mode amplitude $A(t)$ is a very important result and is used extensively by PICS1D. As demonstrated in [125] , it is a generalization of the usual field equation (see *e.g.* [126]), which allows us to obtain analytical expressions for the linewidth and noise spectra in the case of spatial hole-burning and nonlinear gain compression. The influence of these effects is governed by the longitudinal variation of the weight functions $C_X(z)$.

It follows from (17.103) and (17.104) that

$$\frac{\partial W}{\partial \omega} = 2\int_0^l k_0(z)Z_L(z)Z_R(z)\frac{\partial k}{\partial \omega}(z)dz \tag{17.113}$$

which by (17.110) leads to the expression

$$C_X(z) = -j\frac{k_0(z)Z_0^2(z)\frac{\partial k}{\partial X}(z)}{\int_0^l k_0(z')Z_0^2(z')\frac{\partial k}{\partial \omega}(z')dz'} \tag{17.114}$$

for $X = N, S$. We have used the fact that the constants of proportionality Z_L and Z_R cancel in (17.110), so that we can replace Z_L and Z_R by any solution $Z_0(z)$ proportional to Z_L and Z_R. Note that the $C_X(z)$ factor is rapidly oscillating due to the standing-wave part of $Z_0^2(z)$. For the time being we can ignore the rapidly oscillating terms in Z_0^2 and replace it by $2Z_0^+Z_0^-$. This finally gives the expression

$$C_X(z) = -j\frac{Z_0^+(z)Z_0^-(z)\frac{\partial k}{\partial X}(z)}{\int_0^l Z_0^+(z')Z_0^-(z')\frac{\partial k}{\partial \omega}(z')dz'} \tag{17.115}$$

The functions $C_N(z)$ and $C_S(z)$ are proportional to the product of right and left traveling fields and the local derivatives of the complex wave number. The derivative of k with respect to the variable X ($X = N, S$) is given by

$$\frac{\partial k}{\partial X} = \tfrac{1}{2} j(1 + j\alpha_X)\frac{\partial g}{\partial X} \tag{17.116}$$

where $-\alpha_X$ gives the ratio between changes in the real and imaginary part of k due to a change in X. α_X is a *material* parameter, which may depend on both ω, N and S. The modal gain g is usually represented as

$$g(\omega, N, S) = \frac{g_0(\omega, N)}{1 + \varepsilon S} \tag{17.117}$$

where $g_0(\omega, N)$ is the linear gain, and ε is the nonlinear gain coefficient. Nonlinearity of the refractive index is often ignored, which corresponds to setting $\alpha_S = 0$. For g given by (17.117) this implies – due to the reduction of the differential gain – that $\alpha_N = \alpha_0(1 + \varepsilon S)$, where α_0 is the "usual" linewidth enhancement factor without nonlinear gain.

In summary, we have derived a small signal dynamic equation based on the expansion of the Green's function solution in terms of frequency, carrier and photon densities. The frequency expansion has been used to make an inverse Fourier transform so that time domain small signal dependence can be obtained. We will use this dynamic equation to derive our formulas for AM, FM modulation responses and noise behavior later in this chapter.

17.12 Modulation Responses

Based on the dynamic equations described previously, the AM and FM modulation responses can be derived with a small signal analysis. Here we use a shorthand of dot product to denote the integral over the z, *i.e.*, $\cdot$ stands for $\int dz$.

Introducing important weight functions C_N and C_S, we re-write the dynamic equations as follows:

$$\frac{1}{2P_0}\frac{d\Delta P}{dt} = \mathbf{C}_{Nr} \cdot \Delta\mathbf{N} + \mathbf{C}_{Sr} \cdot \Delta\mathbf{S} \tag{17.118}$$

$$\frac{d\Delta\phi}{dt} = C_{Ni} \cdot \Delta N + C_{Si} \cdot \Delta S \tag{17.119}$$

where we write the envelope field $A(t)$ as $\sqrt{P_0 + \Delta P(t)} exp[j\phi(t)]$. $S(z)$ is the distribution of photon density. P_0 is the output power in steady state and $\Delta P(t)$ is the small signal response. Other symbols have been defined previously when introducing the dynamic equations.

Since the source of modulation comes from the current modulation of the carrier density, we must consider the continuity equation:

$$\frac{d\Delta N}{dt} = \Delta J - \Delta N/\tau_R(N) - v_g g \Delta S \tag{17.120}$$

where $\tau_R(N)$ is the carrier lifetime representing all recombination mechanisms except stimulated emission.

We assume that the output power P (a scalar) and $S(z)$ (a vector) are related by:

$$\Delta S(z)/S_0(z) = \Delta P/P_0 \tag{17.121}$$

or in other words, the relative response of the photon density to external modulation is the same everywhere. Such an assumption greatly simplifies the mathematics and allow us to eliminate the $\Delta S(z)$ immediately. The continuity equation becomes

$$\frac{d\Delta N}{dt} = \Delta J - \Delta N/\tau_R - v_g g S_0 \Delta P/P_0 \tag{17.122}$$

and the real part of the dynamic equation becomes

$$\frac{1}{2P_0}\frac{d\Delta P}{dt} = C_{Nr} \cdot \Delta N + C_{Sr} \cdot \Delta S_0 \Delta P/P_0 \tag{17.123}$$

Assume the modulation current has the form

$$\Delta J = \Delta J_0(z) exp(j\Omega t) \tag{17.124}$$

The following algebra results.

$$\begin{aligned} j\Omega\Delta N &= \Delta J - \Delta N/\Delta\tau_R - v_g g S_0 \Delta P/P_0 && (17.125)\\ (j\Omega + 1/\tau_R)\Delta N &= \Delta J - v_g g S_0 \Delta P/P_0 && (17.126)\\ \Delta N &= \frac{\Delta J - v_g g S_0 \Delta P/P_0]}{j\Omega + 1/\tau_R} && (17.127) \end{aligned}$$

Eliminating the concentration variation from the following amplitude rate equation,

$$j\Omega\Delta P = 2P_0 \left(C_{Nr} \cdot \Delta N + C_{Sr} \cdot S_0 \Delta P/P_0\right) \tag{17.128}$$

we obtain the AM response as follows. The algebra is shown in detail.

$$j\Omega\Delta P/P_0 = \frac{2\mathbf{C}_{Nr}\cdot[\Delta J]}{j\Omega+1/\tau_R} - \frac{2\mathbf{C}_{Nr}\cdot v_g g\mathbf{S}_0\Delta P/P_0}{j\Omega+1/\tau_R} + 2\mathbf{C}_{Sr}\cdot\mathbf{S}_0\Delta P/P_0 \tag{17.129}$$

$$j\Omega\Delta P/P_0 + \frac{2\mathbf{C}_{Nr}\cdot v_g g\mathbf{S}_0\Delta P/P_0}{j\Omega+1/\tau_R} - 2\mathbf{C}_{Sr}\cdot\mathbf{S}_0\Delta P/P_0 = \frac{2\mathbf{C}_{Nr}\cdot\Delta\mathbf{J}}{j\Omega+1/\tau_R} \tag{17.130}$$

$$[j\Omega + 2\mathbf{C}_{Nr}\cdot v_g g\mathbf{S}_0/(j\Omega+1/\tau_R) - 2\mathbf{C}_{Sr}\cdot\mathbf{S}_0]\,\Delta P/P_0 = 2\mathbf{C}_{Nr}\cdot\Delta\mathbf{J}/(j\Omega+1/\tau_R) \tag{17.131}$$

We have finally arrived at the useful expression for AM response:

$$\Delta P/P_0 = \frac{2\mathbf{C}_{Nr}\cdot\Delta\mathbf{J}/(j\Omega+1/\tau_R)}{j\Omega + 2\mathbf{C}_{Nr}\cdot v_g g\mathbf{S}_0/(j\Omega+1/\tau_R) - 2\mathbf{C}_{Sr}\cdot\mathbf{S}_0} \tag{17.132}$$

Let us look at the frequency variation and write the phase variation as

$$\phi(t) = \phi_0 + \Delta\phi(t) \tag{17.133}$$

or

$$\frac{d\phi(t)}{dt} = \frac{d\Delta\phi(t)}{dt} \tag{17.134}$$

Thus the frequency variation can be written as

$$\Delta\omega = \frac{d\Delta\phi(t)}{dt} \tag{17.135}$$

Eliminating $\Delta N(z)$ and $\Delta S(z)$ from the second dynamic equation we obtain

$$\begin{aligned}
\Delta\omega &= C_{Ni}\cdot\frac{\Delta J - v_g g S_0\Delta P/P_0}{j\Omega+1/\tau_R} + C_{Si}\cdot\mathbf{S_0}\Delta P/P_0 && (17.136)\\
&= C_{Ni}\cdot\Delta J/(j\Omega+1/\tau_R) - C_{Ni}\cdot v_g g\mathbf{S}_0\frac{\Delta P}{P_0}/(j\Omega+1/\tau_R) + C_{Si}\cdot\mathbf{S}_0\frac{\Delta P}{P_0} && (17.137)\\
&= C_{Ni}\cdot\Delta J/(j\Omega+1/\tau_R)\,[-C_{Ni}\cdot v_g g\mathbf{S}_0/(j\Omega+1/\tau_R) + C_{Si}\cdot\mathbf{S}_0]\,\frac{\Delta P}{P_0} && (17.138)
\end{aligned}$$

The FM response is then

$$\Delta\omega = \frac{C_{Ni}\cdot\Delta J}{j\Omega+1/\tau_R} + \left[C_{Si}\cdot\mathbf{S}_0 - \frac{C_{Ni}\cdot v_g g\mathbf{S}_0}{j\Omega+1/\tau_R}\right]\left(\frac{\Delta P}{P_0}\right) \tag{17.139}$$

where $\left(\frac{\Delta P}{P_0}\right)$ is the AM response and has been given in Eq. (17.132). We have shown that the FM is related to the AM response.

17.13 Noise Spectra

In this section, we derive the FM and RIN noise spectra from a small signal analysis of the basic dynamic equations. The approach is similar to that in Section 17.12, except that the driving force here is a set of noise terms instead of external applied modulation. Since the linewidth is a special case of the FM noise spectrum, the analysis here also provides an elegant alternative method of calculating the linewidth and the effective alpha factor.

We first introduce the concept of power spectrum used to describe the noise process. Secondly, we summarize several useful correlation relations for the Langevin noise functions. Finally, we derive the formulas for FM and RIN noise spectrum. As in other sections, an attempt is made to give all the details of our theoretical derivation so that our readers can check the derivation.

17.13.1 Power spectrum

In the derivation in this section, we are concerned with two frequencies, ω and Ω. The former is the laser emission frequency and the latter is the frequency of the noise or the external modulation (if any). For simplicity we ignore a trivial factor of $(2\pi)^{-1}$ when we perform the Fourier transforms.

We define the power spectrum $V_p(\Omega)$ of an arbitrary variable V(t) as the Fourier transform of its correlation function $V_c(t)$, where

$$V_c(t) = < V(0)V(t) > \tag{17.140}$$

Using a Fourier transform, we express the variables in the frequency domain.

$$V(t) = \int V_{\Omega_1} exp(j\Omega_1 t) d\Omega \tag{17.141}$$

$$V(0) = \int V_{\Omega_2} d\Omega_2 \tag{17.142}$$

$$V_c(t) = \int\int < V_{\Omega_1} V_{\Omega_2} > exp(j\Omega_1 t) d\Omega_1 d\Omega_2 \tag{17.143}$$

Taking the Fourier transform of the above equation yields

$$V_p(\Omega) = \int exp(-j\Omega t) dt \int\int < V_{\Omega_1} V_{\Omega_2} > exp(j\Omega_1 t) d\Omega_1 d\Omega_2 \tag{17.144}$$

$$V_p(\Omega) = \int\int < V_{\Omega_1} V_{\Omega_2} > \int dt exp(j(\Omega_1 - \Omega)t) d\Omega_1 d\Omega_2 \tag{17.145}$$

$$V_p(\Omega) = \int\int < V_{\Omega_1} V_{\Omega_2} > \delta(\Omega_1 - \Omega) d\Omega_1 d\Omega_2 \tag{17.146}$$

$$V_p(\Omega) = \int < V_{\Omega} V_{\Omega_2} > d\Omega_2 \tag{17.147}$$

The RIN is defined as the power spectrum of the relative power fluctuation

$$RIN = power\ spectrum\ of\ \Delta P(t)/P0 \tag{17.148}$$

or

$$RIN = power\ spectrum\ of\ \frac{2\Delta a(t)}{a0} \tag{17.149}$$

where $a(t)$ is the real amplitude of the wave envelope function. The FM noise spectrum is defined as the power spectrum of the lasing frequency deviation

$$FM - noise = power\ spectrum\ of\ \Delta\omega(t) \tag{17.150}$$

or

$$FM - noise = power - spectrum\ of\ \frac{d(\Delta\phi(t))}{dt} \tag{17.151}$$

Therefore, we need to find the Fourier transform of the two variables $\frac{2\Delta a(\Omega)}{a0}$ and $\frac{d(\Delta\phi(t))}{dt}$ for the RIN and FM noise spectra.

17.13.2 Correlation relations

We will need the following correlation relations for the Langevin noise functions based on the work of Tromborg *et.al.* [117]:

$$< F_r(t')^* F_r(t) > \;=\; 2D_{SS}\delta(t-t') \tag{17.152}$$
$$< F_i(t')^* F_i(t) > \;=\; 2D_{SS}\delta(t-t') \tag{17.153}$$
$$< F_N(t')^* F_N(t) > \;=\; 2D_{NN}\delta(t-t') \tag{17.154}$$
$$< F_N(t')^* F_r(t) > \;=\; 2D_{NS}\delta(t-t') \tag{17.155}$$
$$< F_N(t')^* F_i(t) > \;=\; 0 \tag{17.156}$$

where

$$2D_{SS} = \frac{R_{sp}}{2I_0} \tag{17.157}$$

$$2D_{NN}(z) = 2\left\{v_g g_{mat}(z) n_{sp} S_0(z) + R(N(z))\right\} \tag{17.158}$$

$$2D_{NS}(z) = -v_g g_{mat}(z) n_{sp} F_{ng} \frac{S_0(z)}{I_0} \tag{17.159}$$

$$F_{ng} = \frac{(Z_0^* \cdot n n_g Z_0)^2}{\mid Z_0 \cdot n n_g Z_0 \mid^2} \tag{17.160}$$

Here g_{mat} is the material gain in the active region. Other symbols are consistent with previous sections.

We will need to express the correlation relations in frequency domain, for example,

$$
\begin{aligned}
& < F_r^*(\Omega_1) F_r(\Omega) > && (17.161)\\
= & < \int dt_1 F_r^*(t_1) exp(-j\Omega_1 t_1) \int dt F_r(t) exp(j\Omega t) > && (17.162)\\
= & \int dt_1 \int dt < F_r^*(t_1) F_r(t) > exp(-j\Omega_1 t_1) exp(j\Omega t) && (17.163)\\
= & \int dt_1 \int dt 2D_{SS}\delta(t-t_1) exp(-j\Omega_1 t_1) exp(j\Omega t) && (17.164)\\
= & \int dt 2D_{SS} exp(-j\Omega_1 t) exp(j\Omega t) && (17.165)\\
= & 2D_{SS}\delta(\Omega_1 - \Omega) && (17.166)
\end{aligned}
$$

The other relations in the frequency domain may be expressed in a similar manner.

17.13.3 FM noise, linewidth and effective alpha factor

As in the previous section where we dealt with external modulations, we start with our basic equations in small signal expansion. This time, however, we have all the noise terms.

$$\frac{1}{a_0}\frac{d\Delta a}{dt} = \mathbf{C}_{Nr} \cdot \Delta \mathbf{N} + \mathbf{C}_{Sr} \cdot \Delta \mathbf{S} + \frac{R_{sp}}{2I_0} + F_r(t) \tag{17.167}$$

As in Ref. [117], we neglect the term $\frac{R_{sp}}{2I_0}$ which contributes a small background and is only significant very close to the threshold:

$$\frac{1}{a_0}\frac{d\Delta a}{dt} = \mathbf{C}_{Nr} \cdot \Delta \mathbf{N} + \mathbf{C}_{Sr} \cdot \Delta \mathbf{S} + F_r(t) \tag{17.168}$$

$$\Delta\omega(t) = C_{Ni} \cdot \Delta N + C_{Si} \cdot \Delta S + F_i(t) \tag{17.169}$$

where F_r and F_i are noise terms associated with the amplitude and phase, respectively. Note that these terms average to zero. The continuity equation (with zero external modulation) is written as:

$$\frac{d\Delta N}{dt} = -\Delta N/\tau_R - v_g g(z)\Delta S + F_N(t) \tag{17.170}$$

The relation between the amplitude a and S is assumed to be proportional.

$$\Delta S = 2S_0(z)\Delta a/a_0 \tag{17.171}$$

The equations become:

$$\frac{d\Delta N}{dt} = -\Delta N/\tau_R - v_g g 2S_0 \Delta a/a_0 + F_N(t) \tag{17.172}$$

$$\frac{1}{a_0}\frac{d\Delta a}{dt} = C_{Nr}\cdot\Delta N + C_{Sr}\cdot\Delta 2S_0\Delta a/a_0 + F_r(t) \tag{17.173}$$

Now we are ready to take the Fourier transform of the above equations.

$$j\Omega\Delta a/a_0 = C_{Nr}\cdot\Delta N + C_{Sr}2\cdot S_0\Delta a/a_0 + F_r(\Omega) \tag{17.174}$$

$$j\Omega\Delta\phi = C_{Ni}\cdot\Delta N + C_{Si}\cdot[2S_0\Delta a/a_0] + F_i(\Omega) \tag{17.175}$$

Similarly for the continuity equation:

$$j\Omega\Delta N = -\Delta N/\tau_R - v_g g 2S_0\Delta a/a_0 + F_N(\Omega) \tag{17.176}$$

or

$$(j\Omega + 1/\tau_R)\Delta N = -v_g g 2S_0\Delta a/a_0 + F_N(\Omega) \tag{17.177}$$

or

$$\Delta N = \frac{-v_g g(z)2S_0(z)\Delta a/a_0 + F_N(\Omega)}{j\Omega + 1/\tau_R} \tag{17.178}$$

Define

$$Q_t = (j\Omega + 1/\tau_R)^{-1} \tag{17.179}$$

$$\Delta N = -Q_t v_g g 2S_0\Delta a/a_0 + Q_t F_N(\Omega) \tag{17.180}$$

Eliminating the concentration variation from the amplitude rate equation, we obtain

$$\begin{aligned} j\Omega(\Delta a/a_0) \quad = \quad & -2[Q_t(z)v_g g(z)C_{Nr}\cdot S_0](\Delta a/a_0) + [Q_t C_{Nr}\cdot F_N(\Omega)] \\ & +2(C_{Sr}\cdot S_0)(\Delta a/a_0) + F_r(\Omega) \end{aligned} \tag{17.181}$$

$$\begin{aligned} & j\Omega(\Delta a/a_0) + Q_t 2v_g g(C_{Nr}\cdot S_0)(\Delta a/a_0) - 2(C_{Sr}\cdot S_0)(\Delta a/a_0) \\ = \quad & Q_t(C_{Nr}\cdot F_N(\Omega)) + F_r(\Omega) \end{aligned} \tag{17.182}$$

$$(\Delta a/a_0) = G_f^{-1}[Q_t C_{Nr}\cdot F_N(\Omega)] + G_f^{-1}F_r(\Omega) \tag{17.183}$$

where

$$G_f(\Omega) = j\Omega + [Q_t 2 v_g g C_{Nr} \cdot S_0] - 2(C_{Sr} \cdot S_0) \tag{17.184}$$

We perform the same derivation for the phase equation:

$$\Delta\omega(t) = C_{Ni} \cdot \Delta N + C_{Si} \cdot \Delta S + F_i(\Omega) \tag{17.185}$$

$$j\Omega\Delta\phi = C_{Ni} \cdot \Delta N + C_{Si} \cdot [2S_0 \Delta a/a_0] + F_i(\Omega) \tag{17.186}$$

$$j\Omega\Delta\phi = [-v_g g 2 C_{Ni} \cdot S_0 \Delta a/a_0 + C_{Ni} \cdot F_N(\Omega)]Q_t + C_{Si} \cdot [2S_0 \Delta a/a_0] + F_i(\Omega) \tag{17.187}$$

Define

$$U_f(\Omega) = 2(C_{Si} \cdot S_0) - 2 v_g g (C_{Ni} \cdot S_0) Q_t \tag{17.188}$$

$$j\Omega\Delta\phi = Q_t C_{Ni} \cdot F_N(\Omega) + U_f(\Delta a/a_0) + F_i(\Omega) \tag{17.189}$$

$$j\Omega\Delta\phi = Q_t C_{Ni} \cdot F_N(\Omega) + U_f[G_f^{-1} Q_t (C_{Nr} \cdot F_N(\Omega)) + G_f^{-1} F_r(\Omega)] + F_i(\Omega) \tag{17.190}$$

$$j\Omega\Delta\phi = Q_t C_{Ni} \cdot F_N(\Omega) + U_f G_f^{-1} Q_t (C_{Nr} \cdot F_N(\Omega)) + U_f G_f^{-1} F_r(\Omega) + F_i(\Omega) \tag{17.191}$$

Define

$$V_f(z) = Q_t C_{Ni} + U_f G_f^{-1} Q_t C_{Nr} \tag{17.192}$$

$$j\Omega\Delta\phi = (V_f \cdot F_N(\Omega)) + U_f G_f^{-1} F_r(\Omega) + F_i(\Omega) \tag{17.193}$$

The FM noise is given by

$$\begin{aligned} FM &= \int < \Omega_1 \Delta\phi^*(\Omega_1) \Omega \Delta\phi(\Omega) > d\Omega_1 & (17.194) \\ &= FM_{NN} + FM_{rr} + FM_{ii} + FM_{Nr} & (17.195) \end{aligned}$$

Consider the term

$$\begin{aligned}
FM_{NN} &= \int d\Omega_1 < \int dz_1 V_f^*(z_1) F_N^*(\Omega_1, z_1) \int dz V_f(z) F_N(\Omega, z) > & (17.196)\\
&= \int d\Omega_1 \int dz_1 V_f^*(z_1) \int dz V_f(z) < F_N^*(\Omega_1, z_1) F_N(\Omega, z) > & (17.197)\\
&= \int d\Omega_1 \int dz_1 V_f^*(z_1) \int dz V_f(z) 2D_{NN}(z)\delta(z - z_1)\delta(\Omega_1 - \Omega) & (17.198)\\
&= \int dz \mid V_f(z) \mid^2 2D_{NN}(z) & (17.199)
\end{aligned}$$

Similarly

$$FM_{rr} = \mid U_f \mid^2 \mid G_f \mid^{-2} 2D_{SS} \quad (17.200)$$

$$FM_{ii} = 2D_{SS} \quad (17.201)$$

The cross term is given by:

$$\begin{aligned}
FM_{rN} &= \int d\Omega_1 < [(V_f^* \cdot F_N^*(\Omega_1)) U_f G_f^{-1} F_r(\Omega) \\
&\quad + U_f^* G_f^{-1*} F_r^*(\Omega_1)(V_f \cdot F_N(\Omega))] > & (17.202)\\
&= \int d\Omega_1 < [U_f G_f^{-1} F_r(\Omega) \int dz V_f^*(z) F_N^*(\Omega_1, z) \\
&\quad + U_f^* G_f^{-1*} F_r^*(\Omega_1) \int dz V_f(z) F_N(\Omega, z)] > & (17.203)\\
&= \int d\Omega_1 [U_f G_f^{-1} \int dz V_f^*(z) < F_N^*(\Omega_1, z) F_r(\Omega) > \\
&\quad + U_f^* G_f^{-1*} \int dz V_f(z) < F_N(\Omega, z) F_r^*(\Omega_1) > & (17.204)\\
&= \int d\Omega_1 [U_f G_f^{-1} \int dz V_f^*(z) 2D_{NS}(z) \delta(\Omega_1 - \Omega) \\
&\quad + U_f^* G_f^{-1*} \int dz V_f(z) 2D_{NS}(z) \delta(\Omega_1 - \Omega)] & (17.205)\\
&= 2Re[U_f^* G_f^{-1*} \int dz V_f(z) 2D_{NS}(z)] & (17.206)
\end{aligned}$$

The final expression for the FM noise is then:

$$\begin{aligned}
FM &= \int dz \mid V_f(z) \mid^2 2D_{NN}(z) + \mid U_f \mid^2 \mid G_f \mid^{-2} 2D_{SS} \\
&\quad + 2D_{SS} + 2Re[U_f^* G_f^{-1*} \int dz V_f(z) 2D_{NS}(z)] & (17.207)
\end{aligned}$$

The linewidth is a special case of FM noise; it is the FM noise at zero modulation frequency. Or in other words, the linewidth is the frequency fluctuation (or deviation) at zero external modulation frequency.

$$\Delta\omega = FM(0) \quad (17.208)$$

The effective linewidth enhancement factor (α_{eff}) is defined as the enhancement factor to the linewidth term $2D_{SS} = R_{sp}/(2I_0)$ due to photon fluctuation or the Langevin force in the wave equation.

$$\Delta\omega_{SS} = 2D_{SS}+ \mid U_f \mid^2 \mid G_f \mid^{-2} 2D_{SS} \tag{17.209}$$

$$= 2D_{SS}(1 + \alpha_{eff}^2) \tag{17.210}$$

where

$$\alpha_{eff} = \mid U_f/G_f \mid \tag{17.211}$$

So far, we have considered the noise and linewidth under one oscillation frequency. The formulas should be good when the side mode power is much less than the dominant mode. If the side mode is very strong, we must consider derive the formulas for more than one frequencies and consider the effects of cross correlation between different lasing modes.

In the theory up to the present version of PICS1D, the main mode linewidth is calculated under the assumption of single mode or weak side mode. We do not consider the interaction between different longitudinal modes. In the weak side mode assumption, the side mode contribution to linewidth scales with the third power of the photon number of the side mode. This trend can not continue indefinitely as the side mode intensity gets larger. In a multi-mode model, the side mode intensity may become comparable to that of the main mode. In such a case, both the main and side modes are quite stable, which results in a reduction of the linewidth, and our previous formulas for side mode contribution to the main mode linewidth are no longer applicable.

We hope to provide formulas for a rigorous multimode treatment in future versions of PICS1D. Fortunately, the linewidth in a multimode environment is not very interesting, since if the side mode suppression is bad, the linewidth is of secondary importance.

17.13.4 RIN formulas

The formula for the RIN noise is defined as the spectrum of the variable $(2\Delta a/a_0)$:

$$(2\Delta a/a_0) = 2G_f^{-1}[Q_t C_{Nr} \cdot F_N(\Omega)] + 2G_f^{-1} F_r(\Omega) \tag{17.212}$$

$$RIN = \int d\Omega_1 < (2\Delta a^*(\Omega_1)/a_0^*)(2\Delta a(\Omega)/a_0) > \tag{17.213}$$

$$= RIN_{NN} + RIN_{rr} + RIN_{rN} \tag{17.214}$$

$$RIN_{NN} = 4 \mid G_f \mid^{-2} \int dz \mid Q_t(z) C_{Nr}(z) \mid^2 2D_{NN}(z) \tag{17.215}$$

$$RIN_{rr} = 4 \mid G_f \mid^{-2} 2D_{SS} \tag{17.216}$$

$$RIN_{rN} = 4 \mid G_f \mid^{-2} 2Re[\int dz Q_t(z) C_{Nr}(z) 2D_{NS}(z)] \tag{17.217}$$

The final result for the RIN is as follows:

$$\begin{aligned} RIN &= 4 \mid G_f \mid^{-2} \int dz \mid Q_t(z) C_{Nr}(z) \mid^2 2D_{NN}(z) \\ &\quad +4 \mid G_f \mid^{-2} 2D_{SS} + 4 \mid G_f \mid^{-2} 2Re[\int dz Q_t(z) C_{Nr}(z) 2D_{NS}(z)] \end{aligned} \tag{17.218}$$

17.14 Second Harmonic Distortion of a DFB Laser

The rate equations for the photon number $I_p(t)$ and the carrier density $N(z,t)$ read

$$\frac{dI_p(t)}{dt} = I_p(t) \left\{ \int 2C_{Nr}(z) \Delta N(z,t) dz + \int 2C_{Sr}(z) \Delta S(z,t) dz \right\} \tag{17.219}$$

$$\frac{dN(z,t)}{dt} = J(z,t) - R(N) - G(N,S) S(z,t) \tag{17.220}$$

where notations are the same as in previous sections. The injection current density is modulated at an angular frequency ω

$$J(z,t) = J_s(z) + \Delta J(z) e^{j\omega t} \tag{17.221}$$

To evaluate the second harmonic distortion (SHD), the carrier density and the photon number are expanded to second order

$$N(z,t) = N_s(z) + \Delta N_1(z) e^{j\omega t} + \Delta N_2(z) e^{j2\omega t} \tag{17.222}$$

$$I_p(t) = I_{ps} + \Delta I_{p1} e^{j\omega t} + \Delta I_{p2} e^{j2\omega t} \tag{17.223}$$

In addition, to simplify the calculation, we make the assumption that

$$S(z,t) = \frac{S_s(z)}{I_{ps}} I_p(t) \tag{17.224}$$

Equations (17.219)–(17.224) are solved for ΔI_{p1} and ΔI_{p2}, following a straight forward small signal analysis. We define the SHD as

$$SHD = \frac{\Delta I_{p2}}{\Delta I_{p1}} \tag{17.225}$$

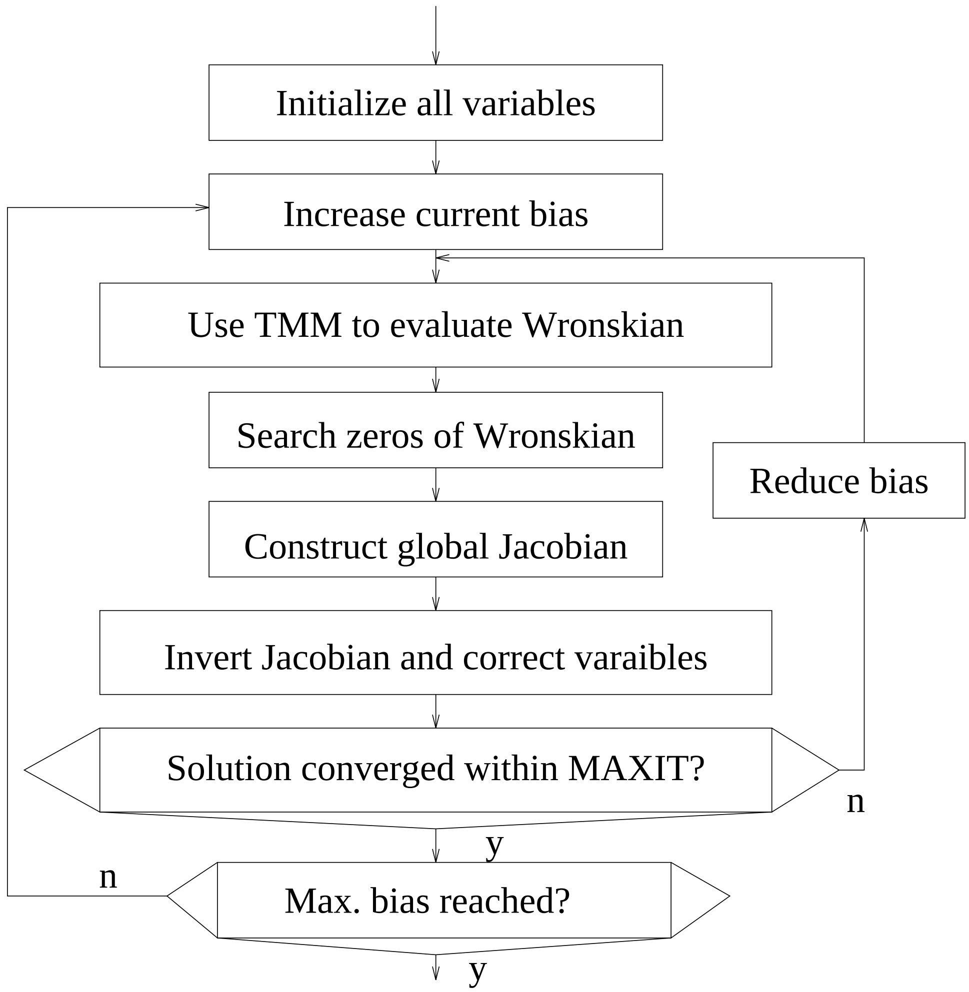


Figure 17.1: A flow diagram for the longitudinal module of PICS3D

17.15 Longitudinal Modes of PICS3D

The solution for longitudinal modes can be regarded as a relatively independent solution module in PICS3D. The self-consistent solution of the longitudinal module is described as follows (also see the flow diagram in Fig. 17.1).

- 1. Initialize all variables and parameters for the longitudinal mode solution. This includes the optical confinement, optical gain and refractive index model for each cross section on the xy-planes. The step is mostly done by the 2D module.
- 2. Increase the bias current.
- 3. Use the transfer matrix method to evaluate the Wronskian.
- 4. Search the zeros of the Wronskian.
- 5. Evaluate the continuity equation.
- 6. Construct the global Jacobian matrix.
- 7. Invert the Jacobian and correct variables such as the linear photon density and electron concentration in the active region.

17.16 Slope Efficiency Issues: 3D vs. 2D Simulation

17.16.1 Introduction

For laser diodes with strong longitudinal variation such as DBF and DBR lasers, 3D simulation is necessary. For Fabry-Perot (FP) lasers, it would be interesting to compare 2D and 3D model of power emission characteristics. To be more specific, we are interested in comparing the slope efficiency of 2D (as in LASTIP simulation) and 3D (as in PICS3D). The purpose of this section is to provide insight and guidance to users who use both LASTIP and PICS3D would like to know exactly what kind of difference to expect when comparing the results of 2D and 3D simulations.

In the following discussion, we assume the laser has been well driven into lasing with stimulated emission dominating.

Let us recall power emission models in LASTIP (2D model) for a Fabry-Perot laser. We use the mirror loss α_m:

$$\alpha_m = 1/(2L)ln[1/(R_1R_2)] \tag{17.226}$$

where L is the cavity length; R_1 and R_2 are the facet power reflectivities. Following standard practice, our 2D simulator LASTIP computes the power emitted (P_e) from the laser diode as the photon number in the cavity (S) times the mirror loss:

$$P_e/(\hbar\omega) = v_g \alpha_m S \tag{17.227}$$

where v_g is the group velocity of light within the laser.

In a 2D simulation, we usually assume the modal optical gain is uniform in the z-dimension. Thus the injected current/power converts into stimulated recombination:

$$P_{inj}(z)/(\hbar\omega) = v_g g_m(z) P_{tot}(z) \tag{17.228}$$

The lasing condition requires that:

$$g_{m0} = \alpha_i + \alpha_m \tag{17.229}$$

where α_i is the internal material loss.

The efficiency can thus be written as,

$$\eta_{2D} = P_e/P_{inj} = 1/(1 + \alpha_i/\alpha_m) = 1/(1 + F_r) \tag{17.230}$$

where, for the convenience of discussions later, we defined a fraction F_r:

$$F_{r2D} = \alpha_i/\alpha_m = \frac{2L\alpha_i}{ln[1/(R_1R_2)]} \tag{17.231}$$

17.16.2 3D simulation: general consideration

Now let use examine the situation in 3D simulation with consideration of spatial hole burning. Here we define spatial hole burning (SHB) as gain (and index variation) in the z-direction caused by inhomogeneous stimulated recombination In the case of FP laser with uniform injection (as is the case for default setting in PICS3D) along the cavity of an edge laser, any variation in optical gain must be accompanied by variation in wave intensity when the laser is well above threshold [see Eq. (17.228)].

In the following derivation, we neglect the factor of $\frac{1}{\hbar\omega}$ so that the power should be regarded as having units of $\hbar\omega$.

We separate the linear power density in an arbitrary laser into right-going and left-going waves:

$$P_{tot}(z) = P_r(z) + P_l(z) \tag{17.232}$$

By definition of gain and loss, we have the following relation if we neglect any coupling between the left and right-going waves.

$$\begin{aligned} dP_r/dz &= (g_m(z) - \alpha_i)P_r. \\ dP_l/dz &= -(g_m(z) - \alpha_i)P_l. \end{aligned} \tag{17.233}$$

The injected power through stimulated recombination for the right going power is as follows,

$$\begin{aligned}\int v_g P_r(z) g_m(z) dz &= \int v_g \alpha_i P_r(z) dz + \int v_g P_r(z)(g_m(z) - \alpha_i) dz \\ &= v_g \int \alpha_i P_r(z) dz + v_g \int d[P_r(z)] \\ &= v_g \int \alpha_i P_r(z) dz + v_g [P_r(z2) - P_r(z1)] \end{aligned} \qquad (17.234)$$

Please note that we have merely used the definition in Eq. (17.233) and no assumptions of uniformity in any quantities are made.

Similarly, integrated injected power for the left going wave is as follows:

$$v_g \int P_l \alpha_i dz + v_g (P_l(z1) - P_l(z2)) \qquad (17.235)$$

By simple algebraic manipulations, we can rewrite the injection power as follows:

$$P_{inj} = v_g \int P_{tot}(z) \alpha_i dz + v_g P_l(z_1)(1 - R_1) + v_g P_r(z_2)(1 - R_2) \qquad (17.236)$$

whose last two terms we identify as the power emission from both laser facets:

$$\begin{aligned} P_e &= v_g P_l(z_1)(1 - R_1) + v_g P_r(z_2)(1 - R_2) \\ &= v_g P_{tot}(z_1)(1 - R_1)/(1 + R_1) + v_g P_{tot}(z_2)(1 - R_2)/(1 + R_2) \end{aligned} \qquad (17.237)$$

Therefore we obtain the following expression of efficiency for a general laser diode:

$$\eta_{3D} = 1/(1 + F_{r3D}) \qquad (17.238)$$

where, using definition of F_r,

$$F_{r3D} = \frac{\int \alpha_i P_{tot}(z) dz}{P_{tot}(z_1)(1 - R_1)/(1 + R_1) + P_{tot}(z_2)(1 - R_2)/(1 + R_2)} \qquad (17.239)$$

Please note that the above derivation neglects any grating/coupling effects. However, it can be shown that if the coupling coefficient is real, the formula in Eq. (17.239)applies to any type of lasers including FP, DFB, DBR and VCSEL with variations in all quantities.

This expression leads to some useful conclusions. Suppose we fix the photon density at the facets $P_{tot}(z_1)$ and $P_{tot}(z_2)$ and allow the intensity to vary inside the cavity. If the intensity inside is low (such as in FP laser), the integral in the numerator will be small and thus lead to a higher efficiency.

On the other hand, if the power is high in the mid-section and small near the facets, such as in 1/4 wavelength shifted DFB laser and VCSEL , the laser will be a less efficient laser.

17.16.3 Fabry-Perot laser with uniform gain

We shall show that for FP laser with uniform gain, Eq. (17.239) reduces to our standard 2D expression in Eq. (17.231).

For simplicity, we assume that $R_1 = R_2$. If the optical gain is uniform in the longitudinal direction, we can write the wave intensity as follows.

$$P_{tot}(z) = P_0 \{exp[gz] + exp[g(L-z)]\} \tag{17.240}$$

where g is the net gain: $g = g_m - \alpha_i$. The integral in the numerator of Eq. (17.239) becomes:

$$\int \alpha_i P_{tot}(z)dz = 2\alpha_i P_0[exp(gL) - 1]/g \tag{17.241}$$

Using lasing condition $gL = ln(1/R)$, we obtain

$$\begin{aligned}\int \alpha_i P_{tot}(z)dz &= \frac{2\alpha_i P_0[(1/R)-1]}{(1/L)ln(1/R)} \\ &= 2\alpha_i P_0[(1/R)-1]/\alpha_m \end{aligned} \tag{17.242}$$

We notice that in the denominator of Eq. (17.239),

$$P_{tot}(z_1) = P_0[1 + exp(gL)] = P_0[1 + (1/R)] \tag{17.243}$$

Therefore, we obtain the following results for FP laser under uniform gain:

$$\begin{aligned}F_{r3D} &= \frac{\int \alpha_i P_{tot}(z)dz}{2P_{tot}(z_1)(1-R)/(1+R)} \\ &= \frac{2\alpha_i P_0[(1/R)-1]/\alpha_m}{2P_0[1+(1/R)](1-R)/(1+R)} \\ &= \alpha_i/\alpha_m \end{aligned} \tag{17.244}$$

which is exactly the model we use in 2D model.

17.16.4 Fabry-Perot laser with non-uniform gain

In this subsection, we allow the optical gain in FP laser to vary. In general, there is no analytical expression if optical gain is allowed to vary. But we can assume a reasonable form for the gain variation in order to study its effect on laser emission characteristics. We assume the following parabolic variation of the net optical gain (see Fig. 17.2):

$$g(z) = g_0 + \Delta g - \Delta g \left(\frac{2}{L}\right)\left(z - \frac{L}{2}\right)^2 \tag{17.245}$$

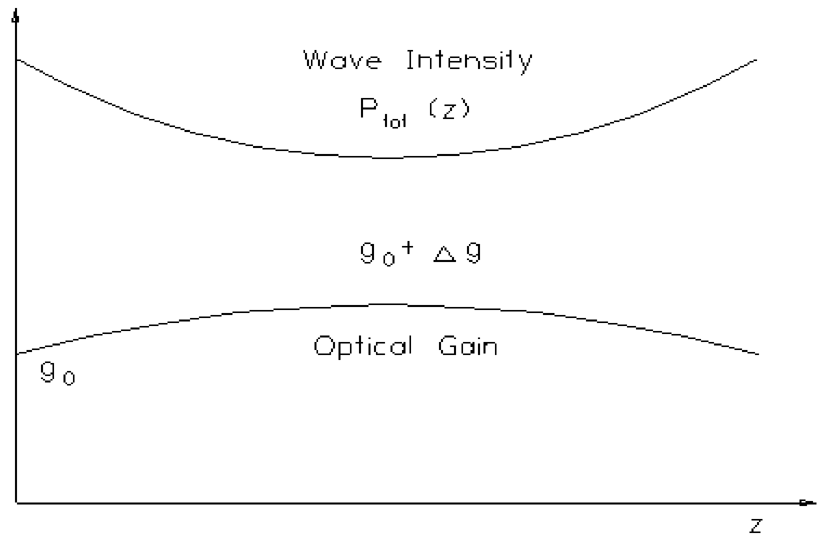


Figure 17.2: Schematics of variation of power and gain within a Fabry-Perot laser.

The optical power of the right-going wave can be expressed as follows from the definition of the net optical gain:

$$ln\left(\frac{P_r(z)}{P_r(0)}\right) = (g_0 + \Delta g)z - \frac{\Delta g}{3}\left(\frac{2}{L}\right)^2\left[\left(z - \frac{L}{2}\right)^3 + \left(\frac{L}{2}\right)^3\right] \tag{17.246}$$

By symmetry consideration, we can write down similar expression for the optical power of the left-going wave.

The lasing condition becomes:

$$\int_0^L g(z)dz = ln(1/R) \tag{17.247}$$

or

$$g_0 + 2\Delta g/3 = (1/L)ln(1/R) \tag{17.248}$$

which is to be compared with the case of uniform gain: $g_0 = (1/L)ln(1/R)$.

Since the stimulated recombination is dominant above threshold, we require that the product of gain and total optical power is uniform because of uniform injection assumption. This is in general not possible for parabolic gain variation. However, we believe it will be reasonable to require that the product of gain-power be equal at facet and at the middle point of the cavity as follows:

$$g(0)[P_r(0) + P_l(0)] = g(L/2)[P_r(L/2) + P_l(L/2)] \tag{17.249}$$

Noting that

$$P_r(0) = P_0[1 + exp[(g_0L + 2\Delta gL/3)] \tag{17.250}$$

and

$$P_r(L/2) = P_l(L/2) = P_0 exp[(g_0 + 2\Delta g/3)(L/2)] \tag{17.251}$$

Eq. (17.249) becomes

$$g_0[1 + exp(g_0 L + 2\Delta g L/3)] = (g_0 + \Delta g) 2 exp[(g_0 + 2\Delta g/3)(L/2)] \tag{17.252}$$

Using the lasing condition in Eq. (17.248), the above becomes:

$$g_0(1 + 1/R) = 2(g_0 + \Delta g)\sqrt{1/R} \tag{17.253}$$

Combing the lasing condition in Eq. (17.248) with with Eq. (17.253), we obtain the following solution of spatial hole burning under the assumption of parabolic gain variation:

$$\Delta g = \left(\frac{1}{L} ln \frac{1}{R}\right) \times \left[\frac{2}{3} + \frac{2\sqrt{1/R}}{1 + (1/R) - 2\sqrt{1/R}}\right]^{-1} \tag{17.254}$$

$$g_0 = \frac{1}{L} ln \frac{1}{R} - \frac{2}{3}\Delta g \tag{17.255}$$

Having successfully solved the power distribution within the FP laser with spatial hole burning, we may now proceed to compute the quantum efficiency using Eq (17.239).

$$\begin{aligned} F_{r3D} &= \frac{\alpha_i \int P_{tot}(z) dz}{2P_{tot}(z_1)(1 - R)/(1 + R)} \\ &= \frac{\alpha_i R}{1 - R} \int_0^L exp\left\{(g_0 + \Delta g)z - \frac{\Delta g}{3}\left(\frac{2}{L}\right)^2 \left[\left(z - \frac{L}{2}\right)^3 + \left(\frac{L}{2}\right)^3\right] dz\right\} \end{aligned} \tag{17.256}$$

We find it convenient to write the above integral in the following form:

$$F_{r3D} = \frac{\alpha_i R}{1 - R} \int_0^L exp[(g_0 + 2\Delta g/3)z + h(z)] dz \tag{17.257}$$

$$h(z) = \frac{\Delta g z}{3} - \frac{\Delta g}{3}\left(\frac{2}{L}\right)^2 \left[\left(z - \frac{L}{2}\right)^3 + \left(\frac{L}{2}\right)^3\right] \tag{17.258}$$

Since $h(z)$ is a small quantity (linear in Δg) equal to zero at facet and at the mid-point of the cavity, we may use the following Taylor expansion:

$$\begin{aligned} F_{r3D} &= \frac{\alpha_i R}{1 - R} \int_0^L exp[(g_0 + 2\Delta g/3)z] dz \\ &\quad + \frac{\alpha_i R}{1 - R} \int_0^L exp[(g_0 + 2\Delta g/3)z] h(z) dz \end{aligned} \tag{17.259}$$

$$= term1 + term2 \tag{17.260}$$

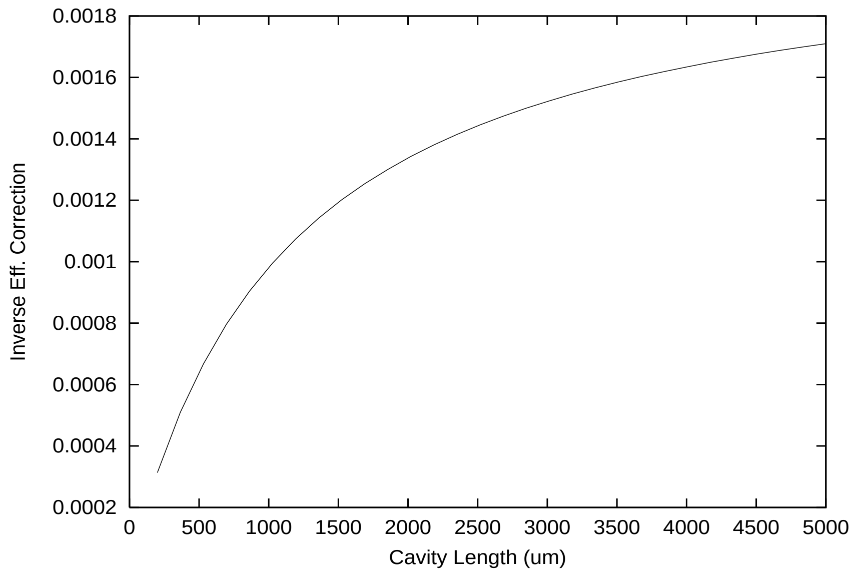


Figure 17.3: Correction to the inverse efficiency of a Fabry Perot laser due to spatial hole burning. The facet power reflectivity is taken to be 0.32. The internal loss is set to be 1000 1/m

We find the first term above reduces to that of the usual mirror loss:

$$term1 = \frac{\alpha_i R}{1-R}\int_0^L exp[(g_0 + 2\Delta g/3)z]dz \quad (17.261)$$

$$= \frac{\alpha_i R}{(1-R)(g_0 + 2\Delta g/3)}\{exp[(g_0 + 2\Delta g/3)L] - 1\} \quad (17.262)$$

Using the lasing condition again,

$$term1 = \frac{\alpha_i R}{(1-R)(1/L)ln(1/R)}[(1/R) - 1]$$

$$= \frac{\alpha_i}{(1/L)ln(1/R)} \quad (17.263)$$

The second term $term2$ may be regarded as a correction term to the standard mirror loss model for the computation of quantum efficiency. It may be evaluated analytically. Since the final expression is rather lengthy, we shall only plot the numerical results of the correction term for a typical case of $R = 0.32$ and $\alpha_i = 1000$ 1/m in Fig. 17.3. Please note that the correction term, $term2$, is proportional to the internal loss. The correction term becomes larger as the cavity length increases, as we may expect from increased SHB. However, the magnitude of the correction term to the conventional mirror loss in FP laser is typically around 0.1%.

17.16.5 Conclusions

In conclusion, we have derived a convenient formula for computation of slope efficiency of lasing characteristics from a given power distribution. It is convenient to be used to analyze the emission efficiency of an arbitrary laser under condition of spatial hole burning (SHB). For a Fabry-Perot laser, use of mirror loss (1/L)ln(1/R) is equivalent to a model of power distribution with uniform gain. The effect of SHB on the efficiency of the standard mirror is in general rather small (about 0.1%) for a Fabry-Perot laser. Thus, as far as FP lasers are concerned, 2D simulation (as in LASTIP) and 3D simulation (as in PICS3D) should yield the same result for practical purposes.

In the case of a phase shifted DFB the SHB influence can be very significant. In that case both the gain distribution and index distribution will greatly affect the mirror loss. Thus it is important to perform longitudinal modelling.

The author of this section (Simon Li) wishes to acknowledge helpful discussions on the topic of spatial hole burning with Richard Schatz and Sebastian Mogg, Royal Institute of Technology, Stockholm."

17.17 Time-Dependent Solution of Longitudinal Modes

In the previous sections, we derived the coupled wave equation for steady state (CW) operation of a wave guiding system. We now consider a time-dependent (or large signal model) situation where the mode amplitude varies with time as a result of external time dependent bias. We assume that the external bias source has a frequency much smaller than the optical frequency ω. This assumption is reasonable since the typical optical frequency is 3×10^{14} Hz, while a typical external modulation biasing source is no more than 10 GHz (1×10^{10} Hz).

For any theory of large signal, our requirement is that it must be consistent with our model in CW. That is, at $t = \infty$, the equation must be reduced to our CW theory. Therefore, it is useful for us to go back to our CW theory again so that modification can be made for the large signal model.

17.17.1 Alternative CW Solution of Longitudinal Modes

Let us consider a laser from 0 to L. We shall choose a reference point where the photon densities for the left and right going waves are non-zero. Let $L(z)$ and $R(z)$ denote left-going and right-going wave functions, respectively. We have the following equations:

$$L(z) = r_R R(z) + F_L \tag{17.264}$$

r_R is reflectivity looking to the right. Similarly,

$$R(z) = r_L L(z) + F_R \tag{17.265}$$

where F_L and F_R are noise cources terms. We have

$$R(z) = r_L r_R R(z) + r_L F_L + F_R \tag{17.266}$$

or

$$(1 - r_L r_R) R(z) = F_\omega \tag{17.267}$$

Without loss of generality, we choose z=L so that $r_R = r_2$ is a constant.

$$(1 - r_2 r_L(\omega)) R = F_\omega \tag{17.268}$$

r_L is a function of propagation constant which depends on ω and optical gain/loss. The above results are consistent with the well known results from Green's function method:

$$R = F/(1 - r_L r_2) \tag{17.269}$$

and we can use complex omega to evaluate the above equation and obtain the power spectrum. Under CW, laser oscillates at the complex solution ω_s:

$$1 - r_2 r_L(\omega_s) = 0 \tag{17.270}$$

17.17.2 Longitudinal Mode Transient Model

As we mentioned earlier, the time scale involved can be separated into two different types. The optical frequency causes the modes to vary with $exp(j\omega t)$ while the slow external modulation causes a much slower change $exp(j\Omega t)$ in the wave amplitude. We assume that above time scale are so different that they can be treated as decoupled (ie., treat one time scale as if the other does not exist).

If we apply external modulation of $exp(j\Omega t)$, the round trip gain as calculated by the optical frequency only may not be unity since photon density can change with time when the optical round trip gain is not unity. However the derivation in the previous section is still valid if we assume the reflectivity is not the reflectivity at optical frequency but the instantaneous reflectivity at the transient state, ie, we can still use $r_L(\omega + \Omega)$ which includes the transient effect.

In this case, we shall go back to our basic equation and regard the frequency as $\omega + \Omega$ where $\omega >> \Omega$. We can expand r_L as follows:

$$r_L(\omega + \Omega) = r_L(\omega) \left[1 + \frac{\partial ln(r_L)}{\partial \omega} \Omega \right] \tag{17.271}$$

Our basic equation becomes:

$$(r_2 r_L(\omega + \Omega) - 1)R(\omega + \Omega) = -F_{\omega+\Omega} \tag{17.272}$$

$$\left[r_2 r_L(\omega) + r_2 r_L(\omega)\frac{\partial ln(r_L)}{\partial \omega}\Omega - 1\right] R(\omega + \Omega) = -F_{\omega+\Omega} \tag{17.273}$$

Since $r_2 r_L(\omega) \approx 1$, the above can be written as

$$\left[r_2 r_L(\omega) + \frac{\partial ln(r_L)}{\partial \omega}\Omega - 1\right] R(\omega + \Omega) = -F_{\omega+\Omega} \tag{17.274}$$

We can derive the coefficient $\frac{\partial ln(r_L)}{\partial \omega}$ from a FP laser as follows:

$$r_L = r_1 r_2 exp(-j2Lk_r + g_m L) \tag{17.275}$$

Neglecting the weak dependence of g_m on frequency we have:

$$\frac{\partial ln(r_L)}{\partial \omega} = -j2L/v_g \tag{17.276}$$

where the group velocity can be written as $1/v_g = \frac{\partial k_r}{\partial \omega}$ Then our basic equation becomes:

$$\left[r_2 r_L(\omega) - j2L/v_g\Omega - 1\right] R(\omega + \Omega) = -F_{\omega+\Omega} \tag{17.277}$$

Taking the inverse Fourier transform to get the slow time dependence and noting $j\Omega = \frac{\partial}{\partial t}$, we obtain

$$\left[r_2 r_L(\omega, t) - 1 - (2L/v_g)\frac{1}{R(\omega, t)}\frac{\partial R(\omega, t)}{\partial t}\right] R(\omega, t) = -F_{\omega,t} \tag{17.278}$$

We can use the photon number $S(t)$ to express $R = \sqrt{S(t)}exp(j\phi(t))$ and to rewrite the above as

$$\frac{2L}{v_g}\frac{1}{R(t)}\frac{\partial R(t)}{\partial t} = \frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} + j\frac{2L}{v_g}\frac{\partial \phi(t)}{\partial t} \tag{17.279}$$

Our key result is as follows:

$$\left[r_2 r_L(t) - 1 - \frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} - j\frac{2L}{v_g}\frac{\partial \phi(t)}{\partial t}\right] R(t) = F_t \tag{17.280}$$

In our simulation, we are not directly concerned with the phase variation but we are more interested in the emission frequency. The phase factor can be written as $exp[j\omega t + \phi(t)]$. We consider a short time interval $\Delta t = t - t_0$.

The phase at time t_0 is given by $j\omega_0 t_0 + \phi(t_0)$. Assuming the fast oscillation frequency does not change substantially, the new phase is

$$j\omega_0 t + \phi(t_0) + \frac{\partial \phi(t)}{\partial t}\Delta t \tag{17.281}$$

Therefore the new frequency should be

$$\omega(t) = \omega_0 + \frac{\partial \phi(t)}{\partial t} \tag{17.282}$$

Our basic equation in terms of $S(t)$ and $\omega(t)$ can be written in a form easy for numerical implementation:

$$\left[r_2 r_L(t) - 1 - \frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} - j\frac{2L}{v_g}(\omega(t) - \omega(t_0)) \right] R(t) = F_t \tag{17.283}$$

The above equation leads to the new oscillation condition:

$$r_2 r_L(t) - 1 - \frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} - j\frac{2L}{v_g}(\omega(t) - \omega(t_0)) = 0 \tag{17.284}$$

We have implemented Eq. (17.284) in the longitudinal model solver of PICS3D simulator.

17.17.3 Comparison with Conventional Photon Rate Equation

We are going to make sure the above is consistent with the commonly used rate equation for a FP laser:

$$g_m - \left(\frac{1}{L}\right) ln\left(\frac{1}{r_2 r_1}\right) = \frac{1}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} \tag{17.285}$$

Please note that we use amplitude reflectivity which may be expressed in terms of power reflectivity as $r = \sqrt{R}$. The derivation may proceed as follows.

$$g_m L - ln\left(\frac{1}{r_2 r_1}\right) = \frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} \tag{17.286}$$

$$r_1 r_2 exp(g_m L) = exp\left[\frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t}\right] \tag{17.287}$$

$$1 - r_1 r_2 exp(g_m L) = 1 - exp\left[\frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t}\right] \tag{17.288}$$

Expanding the quantity on the right hand side we obtain:

$$1 - r_1 r_2 exp(g_m L) = 1 - exp\left[\frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t}\right] \tag{17.289}$$

$$1 - Re[r_L r_2] = -\frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} \tag{17.290}$$

or

$$1 - Re[r_L r_2] + \frac{L}{v_g}\frac{1}{S(t)}\frac{\partial S(t)}{\partial t} = 0 \tag{17.291}$$

which is identical to the real part of the oscillation condition we show in Eq. (17.284).

17.18 Self-Consistent Solution for 3D Simulator

17.18.1 Introduction

The semiconductor laser is a unique three dimensional (3D) device, in that both the lateral and longitudinal dimensions are crucial to the operation of the device as a light emitter. The lateral dimension provides the important mechanisms of optical gain, spontaneous and stimulated recombinations, while the longitudinal dimension provides the amplification of the spontaneous emission and produces the emission characteristics of the device.

Missing either the lateral or longitudinal dimension is a handicap for any laser simulator. The 2D module provides detailed modeling capability for the material and layer geometry in the lateral dimension (x and y directions) while the longitudinal dimension (z-direction) is a simplified rate equation assuming uniform distribution in z direction and single longitudinal mode. The longitudinal spatial hole burning effect is thereby ignored, resulting in improperly treated cases where longitudinal behavior is crucial. Common examples of this type are DFB and DBR lasers where the distributed effect in the z-direction plays an important role.

17.18.2 Numerical Approach

A straight-forward approach to 3D simulation would be to put the 1D longitudinal mode solver (1Dz) and 2D simulator on the x-y plane (2Dxy) together by cutting up the device into many x-y planes and letting longitudinal solver be the driver to control the 2Dxy simulation on these x-y planes. For each x-y plane, the 2Dxy module solver is applied. The 1Dz and the 2Dxy module would have a master-slave relationship. However, such an approach is not realistic. A single 2Dxy module already takes up to an hour to run, and 1Dz calls for hundreds of such runs in a self-consistent multiple longitudinal mode model.

We have adopted a more realistic approach of treating the two packages in a sleep-awake relationship. In other words, the two packages are equally important and are called about the same number of times in a 3D simulation. In one time interval, the 1Dz solver is active while the 2Dxy module is asleep. 1Dz solver determines what the longitudinal mode distribution is in the many x-y planes based on information generated by a previous 2Dxy run. In the next time interval, the 2Dxy module is active while the 1Dz is asleep. The 2Dxy module is used to scan through the different x-y planes to gather all types of information needed for the next 1Dz run. Based on results generated from the previous 1Dz run, the 2Dxy module knows exactly what type of information to gather when it is active. To be more specific, the previous 1Dz run determines the photon and current flow condition for each x-y plane so that the 2Dxy module knows what to do with each x-y plane. A flow diagram illustrate the idea of sleep-wake-up approach is given in Fig. 17.4.

17.18.3 Time Requirements

As was described in the previous subsection, PICS3D turns on and off the 2Dxy and 1Dz modules self-consistently and information is exchanged between the two packages dynamically. Therefore the computation time should be the sum of the two packages plus some administrative overheads. For a mid-level workstation the following timing is typical:

- 1. For a PICS3D simulation treating only the y-z dimensions (the x-direction is uniform as in broad area lasers), the longitudinal dimension is cut up into 20 or so sections. The CW simulation takes approximately one hour.
- 2. Consider a full 3D simulation of a strained multiple quantum well DFB laser. The z-dimension is cut up into 20 or so sections. The CW simulation takes about 5 hours.

17.19 Coupled Wave Equations For Second Order Grating

17.19.1 Introduction

When the grating period of the structure is 1/2 of the wavelength, we have a first order grating structure. This is commonly used for DFB and DBR lasers. In a first order grating structure, the forward wave will only couples with the backward wave in 180 degree from forward wave.

A grating structure with a period that equals the wavelength is a second order grating. The forward wave will experience a first order diffraction in 90 degrees which causes radiation loss and a second order diffraction in 180 degrees (or the

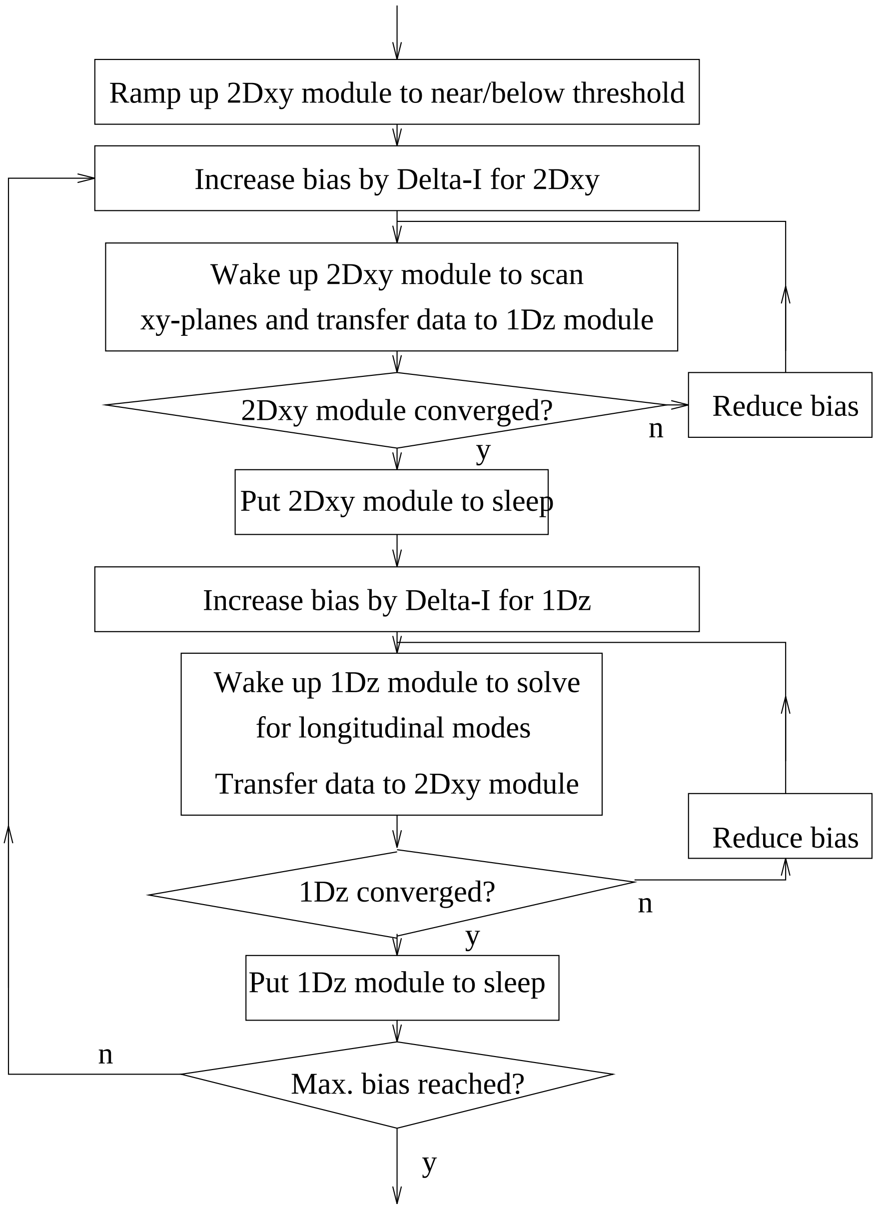


Figure 17.4: A flow diagram to illustrate the idea of sleep/awake approach for PICS3D.

backward wave). Therefore, we must modify the coupled wave equation to have a term to account for the radiation loss. In subsection 17.19.2, we derive the coupled wave equations here based on Ref. [127]. Subsection 17.19.3 is used to formulate the radiating waves. Subsection 17.19.4 derives the formulas for a commonly used grating structure and compare the results with those in Ref. [127].

17.19.2 Coupling coefficients

To be consistent with our convention for propagating waves, we assume a time dependence of $exp(j\omega t)$ so that the forward wave has a factor $exp(-jk_z z)$.

We choose a reference wave number β_0 such that the grating property can be expanded in term of it. The dielectric function at the optical frequency can be written as

$$\varepsilon(y,z) = \varepsilon_0(y) + \Delta\varepsilon(y,z) \tag{17.292}$$

where $\Delta\varepsilon(y,z)$ is non zero only in the grating layers.

$$\begin{aligned}\Delta\varepsilon(y,z) &= \Delta\varepsilon[\zeta_{+1}(y)exp(-j\beta_0 z) + \zeta_{-1}(y)exp(+j\beta_0 z) \\ &+ \zeta_{+2}(y)exp(-j2\beta_0 z) + \zeta_{-2}(y)exp(+j2\beta_0 z)]\end{aligned} \tag{17.293}$$

Our starting point is the following wave equation in three dimensions:

$$\left[\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \frac{\omega^2}{c_0^2}\varepsilon(y,z)\right]E(y,z) = 0. \tag{17.294}$$

where c_0 is the speed of light.

We consider a propagating field of the following form:

$$E(y,z) = [R(z)exp(-j\beta_0 z) + L(z)exp(j\beta_0 z)]\phi(y) + \Delta E(y,z) \tag{17.295}$$

where $\Delta E(y,z)$ is the radiating wave. The wave equation becomes

$$\begin{aligned}&\left[\frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + \frac{\omega^2}{c_0^2}\varepsilon_0(y)\right. \\ &+\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{+1}(y)exp(-j\beta_0 z) + \frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-1}(y)exp(+j\beta_0 z) \\ &\left.+\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{+2}(y)exp(-j2\beta_0 z) + \frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-2}(y)exp(+j2\beta_0 z)\right]E(y,z) = 0\end{aligned} \tag{17.296}$$

Let us substitute the expression for E(y,z) and work out all the terms as follows:

$$\left[\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+\frac{\omega^2}{c_0^2}\varepsilon_0(y)\right.$$
$$+\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{+1}(y)exp(-j\beta_0 z)+\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-1}(y)exp(+j\beta_0 z)$$
$$\left.+\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{+2}(y)exp(-j2\beta_0 z)+\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-2}(y)exp(+j2\beta_0 z)\right]$$
$$[R(z)exp(-j\beta_0 z)\phi(y)+L(z)exp(j\beta_0 z)\phi(y)+\Delta E(y,z)]=0. \quad (17.297)$$

If we collect terms of different exponential factors, we obtain the following equations. For terms with no exponential wave factor:

$$\left[\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+\frac{\omega^2}{c_0^2}\varepsilon_0(y)\right]\Delta E(y,z)=-\frac{\omega^2}{c_0^2}\Delta\varepsilon[\zeta_{+1}(y)L(z)+\zeta_{-1}(y)R(z)]\phi(y) \quad (17.298)$$

For terms with $exp(-j\beta_0 z)$:

$$\left[\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+\frac{\omega^2}{c_0^2}\varepsilon_0(y)\right]R(z)exp(-j\beta_0 z)\phi(y)$$
$$= -\frac{\omega^2}{c_0^2}\Delta\varepsilon\left[\zeta_{+1}(y)\Delta E(y,z)+\zeta_{+2}(y)L(z)\phi(y)\right]exp(-j\beta_0 z) \quad (17.299)$$

We drop the terms with $\frac{\partial^2 R}{\partial z^2}$ and $\frac{\partial^2 L}{\partial z^2}$:

$$\left[\frac{\partial^2}{\partial y^2}-2j\beta_0\frac{\partial}{\partial z}-\beta_0^2+\frac{\omega^2}{c_0^2}\varepsilon_0(y)\right]R(z)\phi(y)$$
$$= -\frac{\omega^2}{c_0^2}\Delta\varepsilon[\zeta_{+1}(y)\Delta E(y,z)+\zeta_{+2}(y)L(z)\phi(y)] \quad (17.300)$$

Similarly for terms with $exp(j\beta_0 z)$:

$$\left[\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+\frac{\omega^2}{c_0^2}\varepsilon_0(y)\right]L(z)exp(j\beta_0 z)\phi(y)$$
$$= -\frac{\omega^2}{c_0^2}\Delta\varepsilon[\zeta_{-1}(y)\Delta E(y,z)+\zeta_{-2}(y)R(z)\phi(y)]exp(j\beta_0 z) \quad (17.301)$$

or

$$\left[\frac{\partial^2}{\partial y^2}+2j\beta_0\frac{\partial}{\partial z}-\beta_0^2+\frac{\omega^2}{c_0^2}\varepsilon_0(y)\right]L(z)\phi(y)$$
$$= -\frac{\omega^2}{c_0^2}\Delta\varepsilon[\zeta_{-1}(y)\Delta E(y,z)+\zeta_{-2}(y)R(z)\phi(y)] \quad (17.302)$$

The left hand side of equation Eq. (17.298) is very familiar to us. Assuming a fast varying factor $exp(\pm j\beta_0 z)$ for $\Delta E(y,z)$, we solve the following Green's function for it y-dependence.

$$[\frac{\partial^2}{\partial y^2} - \beta_0^2 + \frac{\omega^2}{c_0^2}\varepsilon_0(y)]G(y,y_1) = \delta(y-y_1) \tag{17.303}$$

$$\Delta E(y,z) = -\frac{\omega^2}{c_0^2}\Delta\varepsilon\int G(y,y_1)[\zeta_{+1}(y_1)L(z)+\zeta_{-1}(y_1)R(z)]\phi(y_1)dy_1 \tag{17.304}$$

$$\begin{aligned}\Delta E(y,z) &= -\frac{\omega^2}{c_0^2}\Delta\varepsilon\int G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)]dy_1 L(z)\\ &- \frac{\omega^2}{c_0^2}\Delta\varepsilon\int G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dy_1 R(z)\end{aligned} \tag{17.305}$$

We note that $\phi(y)$ satisfies the following equation for a single transverse mode:

$$\left[\frac{\partial^2}{\partial y^2} - \beta_0^2 + \frac{\omega_0^2}{c_0^2}Re[\varepsilon_0(y)]\right]\phi(y) = 0 \tag{17.306}$$

This allows us to re-write Eq. (17.300) as follows:

$$\begin{aligned}&\left[\frac{\omega^2}{c_0^2}\varepsilon_0(y) - \frac{\omega_0^2}{c_0^2}\varepsilon_0(y) + \frac{\omega^2}{c_0^2}Im[\varepsilon_0(y)] - 2j\beta_0\frac{\partial}{\partial z}\right]R(z)\phi(y)\\ =& -\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{+1}(y)\left[\frac{\omega^2}{c_0^2}\Delta\varepsilon\int G(y,y_1)\zeta_{+1}(y)\phi(y_1)]dy_1 L(z)\right.\\ -& \left.\frac{\omega^2}{c_0^2}\Delta\varepsilon\int G(y,y_1)\zeta_{-1}(y)\phi(y_1)dy_1 R(z)\right]\\ -& \frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{+2}(y)L(z)\phi(y)\end{aligned} \tag{17.307}$$

We multiply the above with $\phi(y)$ and integrate over y:

$$\begin{aligned}&\left\{\beta^2 - \beta_0^2 + j\frac{\omega_0^2}{c_0^2}\int\phi(y)Im[\varepsilon_0(y)]\phi(y)dy - 2j\beta_0\frac{\partial}{\partial z}\right\}R(z)\\ =& -\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2\int\int\phi(y)\zeta_{+1}(y)G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)dydy_1L(z)\\ -& \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2\int\int\phi(y)\zeta_{+1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1R(z)\\ -& \frac{\omega^2}{c_0^2}\Delta\varepsilon\int\zeta_{+2}(y)\phi(y)^2dyL(z)\end{aligned} \tag{17.308}$$

where

$$\begin{aligned}\beta^2 - \beta_0^2 &= 2\beta_0 \frac{\partial \beta}{\partial \omega} \Delta\omega \\ &= 2\beta_0 \frac{\Delta\omega}{v_g}\end{aligned} \tag{17.309}$$

We also note that we can use modal gain and internal loss to express the following integral:

$$\int \phi(y) Im[\varepsilon_0(y)] \phi(y) dy = \frac{n' c_0}{\omega_0}(g - \alpha_i) \tag{17.310}$$

Then

$$\begin{aligned}&\left\{2\beta_0 \frac{\Delta\omega}{v_g} + j\beta_0(g-\alpha_i) - 2j\beta_0 \frac{\partial}{\partial z}\right. \\ + &\left.\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int\int \phi(y)\zeta_{+1}(y) G(y,y_1)\zeta_{-1}(y_1)\phi(y_1) dy dy_1\right\} R(z) \\ = &-\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int\int \phi(y)\zeta_{+1}(y) G(y,y_1)\zeta_{+1}(y_1)\phi(y_1) dy dy_1 L(z) \\ - &\frac{\omega^2}{c_0^2}\Delta\varepsilon \int \zeta_{+2}(y)\phi(y)^2 dy L(z)\end{aligned} \tag{17.311}$$

Similar derivation can be made for Eq. (17.302):

$$\begin{aligned}&\left[\frac{\omega^2}{c_0^2}\varepsilon_0(y) - \frac{\omega_0^2}{c_0^2}\varepsilon_0(y) + 2j\beta_0\frac{\partial}{\partial z} + j\frac{\omega^2}{c_0^2} Im[\varepsilon_0(y)]\right] L(z)\phi(y) \\ = &-\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-1}(y)\Delta E(y,z) \\ - &\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-2}(y) R(z)\phi(y)\end{aligned} \tag{17.312}$$

$$\begin{aligned}&\left[\frac{\omega^2}{c_0^2}\varepsilon_0(y) - \frac{\omega_0^2}{c_0^2}\varepsilon_0(y) + 2j\beta_0\frac{\partial}{\partial z} + j\frac{\omega^2}{c_0^2} Im[\varepsilon_0(y)]\right] L(z)\phi(y) \\ = &-\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int \zeta_{-1}(y) G(y,y_1)\zeta_{+1}(y_1)\phi(y_1) dy_1 L(z) \\ - &\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int \zeta_{-1}(y) G(y,y_1)\zeta_{-1}(y_1)\phi(y_1) dy_1 R(z) \\ - &\frac{\omega^2}{c_0^2}\Delta\varepsilon\zeta_{-2} R(z)\phi(y)\end{aligned} \tag{17.313}$$

We multiply $\phi(y)$ and integrate over y:

$$\begin{aligned}
& \left[\beta^2 - \beta_0^2 + 2j\beta_0\frac{\partial}{\partial z} + j\frac{\omega^2}{c_0^2}\int \phi(y)Im[\varepsilon_0(y)]\phi(y)dy\right]L(z) \\
= & -\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)dydy_1L(z) \\
- & \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1R(z) \\
- & \frac{\omega^2}{c_0^2}\Delta\varepsilon\int \zeta_{-2}(y)\phi(y)^2dyR(z)
\end{aligned} \tag{17.314}$$

$$\begin{aligned}
& \left\{2\beta_0\frac{\Delta\omega}{v_g} + 2j\beta_0\frac{\partial}{\partial z} + j\frac{\omega}{c_0}n'(g-\alpha_i)\right. \\
+ & \left.[\frac{\omega^2}{c_0^2}\Delta\varepsilon]^2 \int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)dydy_1\right\}L(z) \\
= & -\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1R(z) \\
- & \frac{\omega^2}{c_0^2}\Delta\varepsilon\int \zeta_{-2}(y)\phi(y)^2dyR(z)
\end{aligned} \tag{17.315}$$

Our key results are of the following form:

$$\begin{aligned}
& \left\{\frac{\Delta\omega}{v_g} + j\frac{g-\alpha_i}{2} - j\frac{\partial}{\partial z}\right. \\
+ & \left.\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{+1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1\right\}R(z) \\
+ & \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{+1}(y)G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)dydy_1L(z) \\
+ & \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2\beta_0}\int \zeta_{+2}\phi(y)^2dyL(z) \\
= & \ 0
\end{aligned} \tag{17.316}$$

$$\left\{\frac{\Delta\omega}{v_g} + j\frac{\partial}{\partial z} + j\frac{g-\alpha_i}{2}\right.$$

$$
\begin{aligned}
&+ \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)dydy_1\Big\}L(z) \\
&+ \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1 R(z) \\
&+ \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2\beta_0}\int \zeta_{-2}(y)\phi(y)^2 dy R(z) \\
&= 0
\end{aligned}
\tag{17.317}
$$

These are rewritten as

$$\left\{\frac{\partial}{\partial z} + j\frac{\Delta\omega}{v_g} - \frac{g-\alpha_i}{2} + h_{1a}\right\}R(z) + h_{1b}L(z) + jh_{2b}L(z) = 0 \tag{17.318}$$

$$\left\{\frac{\partial}{\partial z} - j\frac{\Delta\omega}{v_g} + \frac{g-\alpha_i}{2} - h_{1a}\right\}L(z) - h_{1c}R(z) - jh_{2a}R(z) = 0 \tag{17.319}$$

$$h_{1a} = -j\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{+1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1 \tag{17.320}$$

$$h_{1b} = -j\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{+1}(y)G(y,y_1)\zeta_{+1}(y_1)\phi(y_1)dydy_1 \tag{17.321}$$

$$h_{1c} = -j\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{2\beta_0}\int\int \phi(y)\zeta_{-1}(y)G(y,y_1)\zeta_{-1}(y_1)\phi(y_1)dydy_1 \tag{17.322}$$

$$h_{2a} = \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2\beta_0}\int \zeta_{-2}(y)\phi(y)^2 dy \tag{17.323}$$

$$h_{2b} = \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2\beta_0}\int \zeta_{+2}(y)\phi(y)^2 dy \tag{17.324}$$

The Green's function can be written in terms of the solutions of the defining equation with the right hand side equal to zero. These solutions are written as $\phi(y)_+$ and $\phi(y)_-$, which are the radiating waves to the positive and negative direction of y, respectively:

$$G(y,y_1) = \frac{1}{W}\begin{cases} \phi(y)_+\phi(y_1)_- & y > y_1 \\ \phi(y)_-\phi(y_1)_+ & y < y_1 \end{cases} \tag{17.325}$$

where the Wronskian is given by

$$W = \frac{d\phi(y)_+}{dy}\phi(y)_- - \frac{d\phi(y)_-}{dy}\phi(y)_+ \tag{17.326}$$

For simplicity, we assume the radiating waves are plane waves in y direction with dependence $exp(\pm jk_y y)$. Then

$$W = -2jk_y \tag{17.327}$$

$$G(y, y_1) = \frac{1}{W}\begin{cases} exp[-jk_y(y - y_1)] & y > y_1 \\ exp[-jk_y(y_1 - y)] & y < y_1 \end{cases} \tag{17.328}$$

or

$$G(y, y_1) = -\frac{1}{2jk_y}\begin{cases} cos[-k_y(y - y_1)] + jsin[-k_y(y - y_1)] & y > y_1 \\ cos[-k_y(y_1 - y)] + jsin[-k_y(y_1 - y)] & y < y_1 \end{cases} \tag{17.329}$$

For simplicity, we only consider the case of symmetric grating with $\zeta_{+m} = \zeta_{-m}$. Then $h_{1a} = h_{1b} = h_{1c} = h_1$ and $h_{2a} = h_{2b} = h_2$.

We consider the following integral:

$$\begin{aligned}
& h_1 4\beta_0 k_y \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^{-2} \\
= & \int_{y>y_1}\int \phi(y)\zeta_1(y)\{cos[-k_y(y - y_1)] + jsin[-k_y(y - y_1)]\}\zeta_1(y_1)\phi(y_1)dydy_1 \\
+ & \int_{y<y_1}\int \phi(y)\zeta_1(y)\{cos[-k_y(y_1 - y)] + jsin[-k_y(y_1 - y)]\}\zeta_1(y_1)\phi(y_1)dydy_1 \\
= & \int_{y>y_1}\int \phi(y)\zeta_1(y)\{cos[-k_y(y - y_1)] + jsin[-k_y(y - y_1)]\}\zeta_1(y_1)\phi(y_1)dydy_1 \\
+ & \int_{y<y_1}\int \phi(y)\zeta_1(y)\{cos[-k_y(y_1 - y)] + jsin[-k_y(y - y_1)]\}\zeta_1(y_1)\phi(y_1)dydy_1 \\
+ & \int_{y<y_1}\int \phi(y)\zeta_1(y)\{-jsin[-k_y(y - y_1)] + jsin[-k_y(y_1 - y)]\}\zeta_1(y_1)\phi(y_1)dydy_1 \\
= & \int\int \phi(y)\zeta_1(y)exp[-jk_y(y - y_1)]\zeta_1(y_1)\phi(y_1)dydy_1 \\
- & 2j\int_{y_1>y}\int \phi(y)\zeta_1(y)sin[k_y(y_1 - y)]\zeta_1(y_1)\phi(y_1)dydy_1 \\
= & \left|\int \phi(y)\zeta_1(y)exp[-jk_y y]\right|^2 dy \\
- & 2j\int_{y_1>y}\int \phi(y)\zeta_1(y)sin[k_y(y_1 - y)]\zeta_1(y_1)\phi(y_1)dydy_1
\end{aligned} \tag{17.330}$$

In many cases, the grating layer thickness is much smaller than the wavelength. As a result, the $\zeta(y)$ function is much like a delta-function in y. Therefore, the 2nd term in the above equation is negligible and we obtain the following.

$$h_1 = \left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{4\beta_0 k_y} \mid \int \phi(y)\zeta_1(y)exp(-jk_y y)dy \mid^2 \tag{17.331}$$

$$h_2 = \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2\beta_0}\int \zeta_2(y)\phi(y)^2 dy \tag{17.332}$$

The above equations can be further simplified assuming $\beta_0 = k_y = \frac{n\omega}{c} = \frac{2\pi n}{\lambda}$ $\Delta\varepsilon = 2n\Delta n$

The factor for h_1 becomes:

$$\left(\frac{\omega^2}{c_0^2}\Delta\varepsilon\right)^2 \frac{1}{4\beta_0 k_y} = \frac{\omega^2}{c_0^2}(\Delta\varepsilon)^2\frac{1}{4n^2} = \left(\frac{2\pi}{\lambda}\right)^2 (\Delta n)^2 \tag{17.333}$$

The factor for h_2 becomes

$$\frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2\beta_0} = \frac{\omega}{c_0}2n\Delta n\frac{1}{2n} = \frac{2\pi}{\lambda}\Delta n \tag{17.334}$$

Our final results are as follows:

$$h_1 = \left(\frac{2\pi}{\lambda}\right)^2 (\Delta n)^2 \mid \int \phi(y)\zeta_1(y)exp(-jk_y y)dy \mid^2 \tag{17.335}$$

$$h_2 = \frac{2\pi}{\lambda}\Delta n \int \zeta_2(y)\phi(y)^2 dy \tag{17.336}$$

For a symmetric grating structure, the coupled wave equations are as follows:

$$\left\{\frac{\partial}{\partial z} + j\frac{\Delta\omega}{v_g} - \frac{g-\alpha_i}{2} + h_1\right\} R(z) + h_1 L(z) + jh_2 L(z) = 0 \tag{17.337}$$

$$\left\{\frac{\partial}{\partial z} - j\frac{\Delta\omega}{v_g} + \frac{g-\alpha_i}{2} - h_1\right\} L(z) - h_1 R(z) - jh_2 R(z) = 0 \tag{17.338}$$

For a real index 2nd order grating, h_1 is real and positive. The effect of the first order diffraction (at 90 degrees) is to introduce an additional loss term of $2h_1$ to the internal loss and an imaginary coupling term of $-h_1$ to the conventional coupling coefficient h_2 or (κ). Therefore the effect is not only to make the laser lossy but also to introduce a lossy coupling term which may affect the longitudinal modes significantly.

For a complex index 2nd order grating, the imaginary part of h_1 adds a term $Im(h_1)$ to the frequency deviation $\frac{\Delta\omega}{v_g}$. It also adds a term $Im(h_1)$ to the real part of κ. Therefore a loss or gain coupled 2nd order grating can also enhance the real kappa and shift the emission frequency.

17.19.3 Radiating Waves

In many applications, it is desirable to calculate the radiating wave. We recall from previous subsection that the radiating field can be written as follows,

$$\Delta E(y,z) = -\frac{\omega^2}{c_0^2}\Delta\varepsilon \int G(y,y_1)[\zeta_{+1}(y_1)L(z)+\zeta_{-1}(y_1)R(z)]\phi(y_1)dy_1 \tag{17.339}$$

where

$$G(y,y_1) = -\frac{1}{2jk_y}\begin{cases} exp[-jk_y(y-y_1)] & y>y_1 \\ exp[-jk_y(y_1-y)] & y<y_1 \end{cases} \tag{17.340}$$

We obtain:

$$\begin{aligned}
\Delta E(y,z) &= -\frac{\omega^2}{c_0^2}\Delta\varepsilon \int G(y,y_1)\zeta_{+1}(y_1)[L(z)+R(z)]\phi(y_1)dy_1 \\
&= -\frac{\omega^2}{c_0^2}\Delta\varepsilon \int_{y>y_1} G(y,y_1)\zeta_{+1}(y_1)[L(z)+R(z)]\phi(y_1)dy_1 \\
&- \frac{\omega^2}{c_0^2}\Delta\varepsilon \int_{y<y_1} G(y,y_1)\zeta_{+1}(y_1)[L(z)+R(z)]\phi(y_1)dy_1 \\
&= \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2jk_y} \int_{y>y_1} exp[-jk_y(y-y_1)]\zeta_{+1}(y_1)[L(z)+R(z)]\phi(y_1)dy_1 \\
&+ \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2jk_y} \int_{y<y_1} exp[-jk_y(y_1-y)]\zeta_{+1}(y_1)[L(z)+R(z)]\phi(y_1)dy_1
\end{aligned} \tag{17.341}$$

We assume that the 90 degree diffraction downwards gets absorbed and we only consider the upward waves:

$$\Delta E(y,z) = \frac{\omega^2}{c_0^2}\Delta\varepsilon\frac{1}{2jk_y} exp[-jk_y(y-y_0)]\zeta_{+1}(y_0)[L(z)+R(z)]\phi(y_0)d_g \tag{17.342}$$

where we assume a thin grating layer of d_g thickness located at y_0.

If the grating is on the surface (case A) of the device, $k_y = \frac{\omega}{c_0}$ and we have:

$$\Delta E(y,z) = \frac{\pi d_g}{j\lambda}\Delta\varepsilon\zeta_{+1}(y_0)\phi(y_0)[L(z)+R(z)]exp[-jk_y(y-y_0)] \tag{17.343}$$

where Γ_g is the confinement factor for the grating layer. If the grating is buried within the device (case B), $k_y = \frac{n\omega}{c_0}$ and we have:

$$\Delta E(y,z) = \frac{\pi d_g}{jn\lambda}\Delta\varepsilon\zeta_{+1}(y_0)\phi(y_0)[L(z)+R(z)]exp[-jk_y(y-y_0)] \tag{17.344}$$

To evaluate the emitted power of the radiating field, we consider the expression for the edge emitting power first:

$$E(y,z) = [R(z)exp(-j\beta_0 z) + L(z)exp(j\beta_0 z)]\phi(y) \tag{17.345}$$

The bulk photon density is written as

$$s(y,z) = \epsilon_0 n^2 |E|^2 = \epsilon_0 n^2 |R(z)|^2 + |L(z)|^2]|\phi(y)|^2 \tag{17.346}$$

where we have neglected the interference term. The linear photon density is as follows:

$$s_l(z) = \epsilon_0 n^2 \int |E|^2 dxdy = \epsilon_0 n^2 |R(z)|^2 + |L(z)|^2]| \tag{17.347}$$

We assume that

$$\int \phi(y)dydx = 1 \tag{17.348}$$

In our simulator, many of power quantities are expressed in terms of the total photon number of the mode, S, which can be directly calculated from the Green's function.

$$S = \int s_l(z)dz = \epsilon_0 n^2 \int |R(z)|^2 + |L(z)|^2]|dz \tag{17.349}$$

from which we can calculate the amplitude of R and L given the photon number in the cavity.

The emitted power from the surface for the case A is as follows:

$$\begin{aligned}
P_{top} &= \hbar\omega c_0 \epsilon_0 \int |\Delta E(y,z)|^2 dxdz \\
&= \hbar\omega c_0 \epsilon_0 \frac{\pi^2 d_g^2}{\lambda^2} (\Delta\varepsilon)^2 \zeta_1^2 \phi(y_0)^2 d_{wid} \int [L(z) + R(z)]^2 dz \\
&= \hbar\omega c_0 \epsilon_0 \frac{\pi^2 d_g}{\lambda^2} (\Delta\varepsilon)^2 \zeta_1^2 \Gamma_g \int |L(z) + R(z)|^2 dz \\
&= \hbar\omega c_0 \epsilon_0 \frac{\pi^2 d_g}{\lambda^2} (\Delta\varepsilon)^2 \zeta_1^2 \Gamma_g S \left\{ \frac{\int |L(z) + R(z)|^2 dz}{\epsilon_0 n^2 \int |R(z)|^2 + |L(z)|^2]|dz} \right\}
\end{aligned} \tag{17.350}$$

or

$$P_{top} = \hbar\omega c_0 \frac{\pi^2 d_g}{n^2\lambda^2} (\Delta\varepsilon\zeta_1)^2 \Gamma_g S \left\{ \frac{\int |L(z) + R(z)|^2 dz}{\int |R(z)|^2 + |L(z)|^2]|dz} \right\} \tag{17.351}$$

The upwards power within the device for the case B is as follows:

$$\begin{aligned}
P_{up} &= \hbar\omega v_g \epsilon_0 n^2 \int |\Delta E(y,z)|^2 dxdz \\
&= \hbar\omega v_g \epsilon_0 n^2 \frac{\pi^2 d_g^2}{n^2\lambda^2} (\Delta\varepsilon)^2 \zeta_1^2 \phi(y_0)^2 d_{wid} \int [L(z) + R(z)]^2 dz \\
&= \hbar\omega v_g \epsilon_0 \frac{\pi^2 d_g}{\lambda^2} (\Delta\varepsilon)^2 \zeta_1^2 \Gamma_g \int |L(z) + R(z)|^2 dz
\end{aligned}$$

$$
\begin{aligned}
&= \hbar\omega v_g \epsilon_0 \frac{\pi^2 d_g}{\lambda^2}(\Delta\varepsilon)^2 \zeta_1^2 \Gamma_g S \left\{ \frac{\int |L(z)+R(z)|^2 dz}{\epsilon_0 n^2 \int |R(z)|^2 + |L(z)|^2]|dz} \right\} \\
&= \hbar\omega v_g \frac{\pi^2 d_g}{n^2\lambda^2}(\Delta\varepsilon\zeta_1)^2 \Gamma_g S \left\{ \frac{\int |L(z)+R(z)|^2 dz}{\int |R(z)|^2 + |L(z)|^2]|dz} \right\} \qquad (17.352)
\end{aligned}
$$

The surface emitting power (case B) is as follows:

$$
P_{top} = (1 - r_{top})\hbar\omega v_g \frac{\pi^2 d_g}{n^2\lambda^2}(\Delta\varepsilon\zeta_1)^2 \Gamma_g S \left\{ \frac{\int |L(z)+R(z)|^2 dz}{\int |R(z)|^2 + |L(z)|^2]|dz} \right\} \qquad (17.353)
$$

where r_{top} is the power reflectivity of the top facet.

We define a unitless coefficient (surface power emisison coefficient) to measure the surface emission intensity:

$$
C_{surf} = \frac{\pi^2 d_g}{n^2\lambda}(\Delta\varepsilon\zeta_1)^2 \Gamma_g \qquad (17.354)
$$

so that the emisison power for case A is given by

$$
P_{top} = \frac{\hbar\omega c_0}{\lambda} C_{surf} S \left\{ \frac{\int |L(z)+R(z)|^2 dz}{\int |R(z)|^2 + |L(z)|^2]|dz} \right\} \qquad (17.355)
$$

and for case B:

$$
P_{top} = (1 - r_{top})\frac{\hbar\omega v_g}{\lambda} C_{surf} S \left\{ \frac{\int |L(z)+R(z)|^2 dz}{\int |R(z)|^2 + |L(z)|^2]|dz} \right\} \qquad (17.356)
$$

17.19.4 A common grating structure

We are going to solve for a commonly used grating structure with the following index distribution (see Fig. 17.5).

$$
\begin{aligned}
\Delta n(z) &= \Delta n/2; \qquad 0 < z < a/2 \\
\Delta n(z) &= \Delta n/2 - \frac{2\Delta n}{c}(z - a/2); \qquad a/2 < z < a/2 + c/2 \\
\Delta n(z) &= -\Delta n/2; \qquad a/2 + c/2 < z < a/2 + c/2 + b/2 \qquad (17.357)
\end{aligned}
$$

$$
\Delta n(z) = \Delta n[2\zeta_1 cos(\beta_0 z) + 2\zeta_2 cos(2\beta_0 z)] \qquad (17.358)
$$

where $\beta_0 = \frac{2\pi}{a+c+b}$

Integrate the above equation by $\int_0^{a/2+c/2+b/2} cos(\beta_0 z)dz$ to obtain:

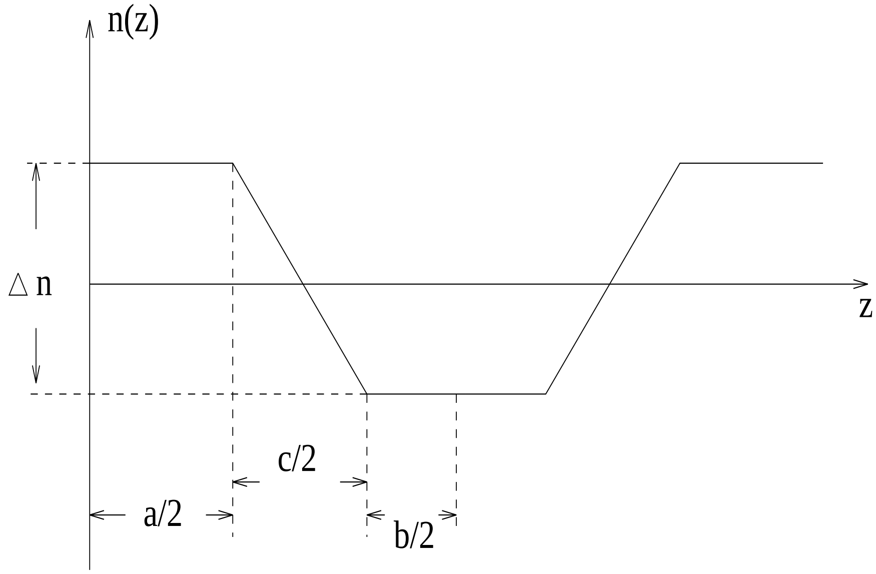


Figure 17.5: Schematics of a common grating structure.

$$
\begin{aligned}
& \int_0^{a/2+c/2+b/2} \Delta n(z) cos(\beta_0 z) dz \\
= & \int_0^{a/2} \Delta n/2 cos(\beta_0 z) dz \\
+ & \int_{a/2}^{a/2+c/2} \left[\Delta n/2 - \frac{2\Delta n}{c}(z-a/2)\right] cos(\beta_0 z) dz \\
+ & \int_{a/2+c/2}^{a/2+c/2+b/2} (-\Delta n/2) cos(\beta_0 z) dz \\
= & \Delta n/2 \frac{1}{\beta_0} sin(\beta_0 z)\Big|_0^{a/2} \\
+ & \left[\Delta n/2 + \frac{\Delta n}{c} a\right] \frac{1}{\beta_0} sin(\beta_0 z)\Big|_{a/2}^{a/2+c/2} \\
- & \frac{2\Delta n}{c}\frac{1}{\beta_0^2}[(\beta_0 z) sin(\beta_0 z) + cos(\beta_0 z)]\Big|_{a/2}^{a/2+c/2} \\
- & \Delta n/2 \frac{1}{\beta_0} sin(\beta_0 z)\Big|_{a/2+c/2}^{a/2+c/2+b/2}
\end{aligned}
\tag{17.359}
$$

Our result is as follows:

$$
\int_0^{a/2+c/2+b/2} \Delta n(z) cos(\beta_0 z) dz
$$

$$
\begin{aligned}
&= \Delta n/2\frac{1}{\beta_0}sin(\beta_0 a/2) \\
&+ \left[\Delta n/2 + \frac{\Delta n}{c}a\right]\frac{1}{\beta_0}[sin(\beta_0(a/2+c/2)) - sin(\beta_0 a/2)] \\
&- \frac{2\Delta n}{c}\frac{1}{\beta_0^2}[\beta_0(a/2+c/2)sin(\beta_0(a/2+c/2)) - (\beta_0 a/2)sin(\beta_0 a/2) \\
&+ cos(\beta_0(a/2+c/2)) - cos(\beta_0 a/2)] \\
&- \Delta n/2\frac{1}{\beta_0}[sin(\beta_0(a/2+c/2+b/2)) - sin(\beta_0(a/2+c/2))] \qquad (17.360)
\end{aligned}
$$

We derive the expression for ζ_1 here.

$$\int \Delta n(z)cos(\beta_0 z)dz = \Delta n 2\zeta_1 \int cos^2(\beta_0 z)dz \qquad (17.361)$$

$$\int_0^{a/2+c/2+b/2} cos^2(\beta_0 z)dz = (a/2+c/2+b/2)/2 \qquad (17.362)$$

$$\int \Delta n(z)cos(\beta_0 z)dz = \Delta n\zeta_1(a/2+c/2+b/2) \qquad (17.363)$$

$$\zeta_1 = \frac{1}{\Delta n(a/2+c/2+b/2)}\int \Delta n(z)cos(\beta_0 z)dz \qquad (17.364)$$

Similarly,

$$
\begin{aligned}
&\int_0^{a/2+c/2+b/2} \Delta n(z)cos(2\beta_0 z)dz \\
&= \Delta n/2\frac{1}{2\beta_0}sin(2\beta_0 a/2) \\
&+ \left[\Delta n/2 + \frac{\Delta n}{c}a\right]\frac{1}{2\beta_0}[sin(2\beta_0(a/2+c/2)) - sin(2\beta_0 a/2)] \\
&- \frac{2\Delta n}{c}\frac{1}{2\beta_0^2}[2\beta_0(a/2+c/2)sin(2\beta_0(a/2+c/2)) - (2\beta_0 a/2)sin(2\beta_0 a/2) \\
&+ cos(2\beta_0(a/2+c/2)) - cos(2\beta_0 a/2)] \\
&- \Delta n/2\frac{1}{2\beta_0}[sin(2\beta_0(a/2+c/2+b/2)) - sin(2\beta_0(a/2+c/2))] \qquad (17.365)
\end{aligned}
$$

$$\zeta_2 = \frac{1}{\Delta n(a/2+c/2+b/2)}\int \Delta n(z)cos(2\beta_0 z)dz \qquad (17.366)$$

In the special case of $c = 0$:

$$\begin{aligned}
&\int_0^{a/2+b/2} \Delta n(z) cos(\beta_0 z) dz \\
= \;& \frac{\Delta n}{\beta_0} sin(\pi a/(a+b))
\end{aligned} \tag{17.367}$$

$$\zeta_1 = \frac{1}{(a/2+c/2+b/2)} \frac{a+b+c}{2\pi} sin(\pi a/(a+b)) \tag{17.368}$$

$$\zeta_1 = \frac{1}{\pi} sin(\pi a/(a+b)) \tag{17.369}$$

$$\zeta_2 = \frac{1}{\Delta n(a/2+c/2+b/2)} \int \Delta n(z) cos(2\beta_0 z) dz \tag{17.370}$$

$$\begin{aligned}
&\int_0^{a/2+c/2+b/2} \Delta n(z) cos(2\beta_0 z) dz \\
= \;& \Delta n \frac{1}{2\beta_0} sin(2\beta_0 a/2)
\end{aligned} \tag{17.371}$$

$$\zeta_2 = \frac{1}{2\pi} sin(2\pi a/(a+b)) \tag{17.372}$$

In the case of thin grating layer of d_g with a confining factor for the grating layer of Γ_g, $exp(jk_y y) = 1$ and we have the following results:

$$\begin{aligned}
h_1 &= \left(\frac{2\pi}{\lambda}\right)^2 (\Delta n)^2 \zeta_1^2 \Gamma_g d_g \\
&= \left(\frac{2\pi}{\lambda}\right)^2 (\Delta n)^2 \frac{1}{\pi^2} sin^2(\pi a/(a+b)) \Gamma_g d_g \\
&= \left(\frac{2}{\lambda}\right)^2 (\Delta n)^2 sin^2(\pi a/(a+b)) \Gamma_g d_g
\end{aligned} \tag{17.373}$$

$$h_2 = \frac{2\pi}{\lambda} \Delta n \int \zeta_2(y) \phi(y)^2 dy = \frac{1}{\lambda} \Delta n sin(2\pi a/(a+b)) \Gamma_g \tag{17.374}$$

$$h_1/h_2 = \lambda \left(\frac{2}{\lambda}\right)^2 \Delta n sin^2(\pi a/(a+b)) d_g / sin(2\pi a/(a+b)) \tag{17.375}$$

$$
\begin{aligned}
h_1/h_2 = &= \frac{2}{\lambda}\Delta n sin^2(\pi a/(a+b))d_g/[sin(\pi a/(a+b))cos(\pi a/(a+b))] \\
&= \frac{2\Delta n d_g}{\lambda} tan(\pi a/(a+b))
\end{aligned}
\tag{17.376}
$$

The ratio above is identical to that derived in Ref. [127].

17.19.5 Case of thicker grating layer

In case of thicker grating layer (comparable or greater than wavelength) we need to evaluate the coupling coefficient and surfurce emitting power more accurately. We need to modify Eqs. 17.354 and 17.373. In both of these equations, the factor $\Gamma_g d_g$ should be replaced by the following integral:

$$
\Gamma_g d_g = |\int \phi(x,y)exp[-jk_y(y-y_c)]dxdy|^2/d_{wid} \tag{17.377}
$$

where d_{wid} is the width of the grating layer. The intergral is performed over the x-y plane on the grating region. The y_c is the y-position within the grating chosen such that the 2nd term in Eq. 17.330 is zero. It is usually reasonable to choose y_c as the center of the grating. The above integral can be performed numerically.

17.20 3D Optical Amplifier Model

PICS3D is ideal for the study of semicondcutor optical amplifier. The simulation approach is similiar to that for lasers. The lack of optical feedback results in simplification of our numerical models. We can skip the longitudinal model search and only deal with the spatial hole burning effects.

To use PICS3D for optical amplifier, we assume that the input light is from the right facet with a single wavelength. We ramp up the bias current as in a laser simulation. As the optical gain increases we expect the output power from the left facet to increase. The statement to activate the amplifier model is **3d_amplifier_model**.

17.21 Interplanar Current Flow - Full 3D Simulation

When simulating an edge type of laser diode, the default setting is to prohibit the current flow between 2D planes. The quasi-3D treatment speeds up the simulation substantially without sacrifying accuracy for most cases.

However, one may need to consider interplanar current flow under certain situations. For example, if longitidinal spatial hole burning is strong and/or facet heating at high power operation is severe, current flow in the longitudinal direction becomes important. Here, current flow means both electrical current and heat current.

PICS3D has an option (Interplanar 3D flow option) to treat the simulation as a full blown 3D electrical and optical simulation. It is no longer possible to treat each plane as independent. Since the interplanar current flow makes each 2D plane completely different from the other, the procedure of data collection for longitudinal mode model is more complicated and we recommend that smaller bias steps be used to guarantee convergence and self-consistency.

Part IV

REFERENCE

Appendix A

Wave Equation for Perfectly Matched Layer

We shall derive the wave equation for the PML starting from the Maxwell equations:

$$\begin{aligned}\nabla \cdot \varepsilon[\Lambda]\overrightarrow{E} &= 0\\ \nabla \cdot \mu[\Lambda]\overrightarrow{H} &= 0\\ \nabla \times \overrightarrow{E} &= -j\omega\mu[\Lambda]\overrightarrow{H}\\ \nabla \times \overrightarrow{H} &= j\omega\varepsilon[\Lambda]\overrightarrow{E}\end{aligned} \tag{A.1}$$

Based on the 3rd and 4th equations above, we obtain the wave equation in the following form.

$$[\Lambda]^{-1}\nabla \times [\Lambda]^{-1}\nabla \times \overrightarrow{E} - k^2\overrightarrow{E} = 0 \tag{A.2}$$

Using a, b, c to define the $[\Lambda]$ tensor, we can make the following expansion and derivation.

$$\begin{aligned}[\Lambda]^{-1}\nabla \times [\Lambda]^{-1}\nabla \times \overrightarrow{E} &= \frac{1}{a}\left[\frac{1}{c}\left(\frac{\partial^2 E_y}{\partial x \partial y} - \frac{\partial^2 E_x}{\partial y^2}\right) - \frac{1}{b}\left(\frac{\partial^2 E_x}{\partial z^2} - \frac{\partial^2 E_z}{\partial x \partial z}\right)\right]\overrightarrow{i}\\ &+ \frac{1}{b}\left[\frac{1}{a}\left(\frac{\partial^2 E_z}{\partial y \partial z} - \frac{\partial^2 E_y}{\partial z^2}\right) - \frac{1}{c}\left(\frac{\partial^2 E_y}{\partial x^2} - \frac{\partial^2 E_x}{\partial x \partial y}\right)\right]\overrightarrow{j}\\ &+ \frac{1}{c}\left[\frac{1}{b}\left(\frac{\partial^2 E_x}{\partial x \partial z} - \frac{\partial^2 E_z}{\partial x^2}\right) - \frac{1}{a}\left(\frac{\partial^2 E_z}{\partial y^2} - \frac{\partial^2 E_y}{\partial y \partial z}\right)\right]\overrightarrow{k}\end{aligned} \tag{A.3}$$

For anisotropic PML with interface normal to x-axis, the coefficients should be related by

$$c = b = 1/a \tag{A.4}$$

We may simplify the above equation as follows.

$$\begin{aligned}[\Lambda]^{-1}\nabla\times[\Lambda]^{-1}\nabla\times\vec{E} &= \left[\left(\frac{\partial^2 E_y}{\partial x\partial y}-\frac{\partial^2 E_x}{\partial y^2}\right)-\left(\frac{\partial^2 E_x}{\partial z^2}-\frac{\partial^2 E_z}{\partial x\partial z}\right)\right]\vec{i}\\ &+ \left[\left(\frac{\partial^2 E_z}{\partial y\partial z}-\frac{\partial^2 E_y}{\partial z^2}\right)-\frac{1}{c^2}\left(\frac{\partial^2 E_y}{\partial x^2}-\frac{\partial^2 E_x}{\partial x\partial y}\right)\right]\vec{j}\\ &+ \left[\frac{1}{c^2}\left(\frac{\partial^2 E_x}{\partial x\partial z}-\frac{\partial^2 E_z}{\partial x^2}\right)-\left(\frac{\partial^2 E_z}{\partial y^2}-\frac{\partial^2 E_y}{\partial y\partial z}\right)\right]\vec{k}\end{aligned}\tag{A.5}$$

It is useful to consider the following expression.

$$\begin{aligned}\frac{1}{c}\nabla(\nabla\cdot[\Lambda]\vec{E}) &= \nabla\left(\frac{1}{c^2}\frac{\partial E_x}{\partial x}+\frac{\partial E_y}{\partial y}+\frac{\partial E_x}{\partial z}\right)\\ &= \vec{i}\,\frac{\partial}{\partial x}\left(\frac{1}{c^2}\frac{\partial E_x}{\partial x}+\frac{\partial E_y}{\partial y}+\frac{\partial E_x}{\partial z}\right)\\ &+ \vec{j}\,\frac{\partial}{\partial y}\left(\frac{1}{c^2}\frac{\partial E_x}{\partial x}+\frac{\partial E_y}{\partial y}+\frac{\partial E_x}{\partial z}\right)\\ &+ \vec{k}\,\frac{\partial}{\partial z}\left(\frac{1}{c^2}\frac{\partial E_x}{\partial x}+\frac{\partial E_y}{\partial y}+\frac{\partial E_x}{\partial z}\right)\end{aligned}\tag{A.6}$$

Based on the first Maxwell equation, the above equation should be zero. Thus we may subtract it from Eq. A.5 so that the cross terms may be cancelled out. We obtain the following.

$$\begin{aligned}[\Lambda]^{-1}\nabla\times[\Lambda]^{-1}\nabla\times\vec{E} &= -\left(\frac{1}{c^2}\frac{\partial^2 E_x}{\partial x^2}+\frac{\partial^2 E_x}{\partial y^2}+\frac{\partial^2 E_x}{\partial z^2}\right)\vec{i}\\ &- \left(\frac{1}{c^2}\frac{\partial^2 E_y}{\partial x^2}+\frac{\partial^2 E_y}{\partial y^2}+\frac{\partial^2 E_y}{\partial z^2}\right)\vec{j}\\ &- \left(\frac{1}{c^2}\frac{\partial^2 E_z}{\partial x^2}+\frac{\partial^2 E_z}{\partial y^2}+\frac{\partial^2 E_z}{\partial z^2}\right)\vec{k}\end{aligned}\tag{A.7}$$

Or simply:

$$[\Lambda]^{-1}\nabla\times[\Lambda]^{-1}\nabla\times\vec{E} = -\left(\frac{1}{c^2}\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\vec{E}\tag{A.8}$$

The wave equation in the PML can be thus written in the following form:

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\vec{E}+k^2\vec{E}=0\tag{A.9}$$

Appendix B

Material Parameters

Most material parameters implemented in the simulation software are well established data taken from Refs. such as [1], [35] and [36]. They are specified outside the program in the form of input statements. Since there are a large number of semiconductor parameters for a device, these material statements are collected together as macros. To ensure a rapid update of the most recent material parameters, the material macro library is directly included here without modification. Should the user have better knowledge of the material parameters than those listed in the macro library, he/she is encouraged to revise the library or override the parameters in the macro by re-issuing an input material statement of proper keyword after the call of the material macro.

In this appendix, we only provide a full listing of several typical material macro parameters for the purpose of illustration. All others are listed briefly so that users may know their existence and can also have a brief description. For a full listing of all macros, users may find it in files "crosslight.mac" and "more.mac" which comes with software installation.

```
$ IMPORTANT: Please do not use any invisible characters, such as tabs in macro
$****************MATERIAL MACRO********************************
$
$      Copyright (c) 1995-present Crosslight Software Inc..
$                All rights reserved.
$
$     Part II for Wurtzite Structure semiconductors, organic material (OLED),
$     strained-Si, strained-SiGe and Metals.
$
$  -------------SOME INFORMATION ABOUT THIS MACRO FILE--------
$
$
$                  Macro Styles and Function Formats
```

```
$
$  An advanced feature of Crosslight Software macro is its ability
$  to define mathematical functions external to the simulation
$  program.  Such functions are evaluated dynamically at run-time.
$
$  Currently, there are two kinds of macros with different formats of
$  functions:  a) the older "fix-arguement-style" macros and
$  b) the more recent "free-arguement-style" macros.  The older macros
$  are included here for compatibility reason and the newer formats are
$  highly recommended.  The two formats are explained as follows.
$
$  a) Fix-Arguement-Style Macros
$
$  The rules for function variables/arguements are rather strict.
$  If a function is used for a physical quantity (presented as
$  a statement), there should be exactly the same number of
$  of function variables as that used in the "load_macro" or
$  "get_active_layer" statement.  For example,  if you use
$  (x,temper) as the variables in "load_macro", all functions
$  within the macro must have these two variables even if
$  all of them do not use the two variables.  When calling such a macro,
$  the following can be used:
$
$  load_macro name=algaas.temp var1=0.3 var2=300
$
$  Since all functions of such a style have the same number of
$  arguements, the program knows which value supplied belongs to which
$  arguement in a function.  Please know that when calling these kinds
$  of macros,  the user must supply all values of the arguements even
$  if they are reserved symbols such as "temper", "doping_n", etc.
$
$  b) Free-Arguement-Style Macros
$
$  The use of free-style format is more based on the need of variables.
$  A function passes and uses as many function arguements as it needs.
$  For example, the bandgap can pass and use both the composition
$  and temperature as (x,temper).  But for the effective mass, only
$  the composition is neede and passed as (x).  The evaluation of such
$  a macro is based on symbol matching, i.e., the user must specify
$  both the symbol and the value of the symbol when calling the macro,
$  such as in the following call.
$
$  load_macro name=algaas.temp var1=0.3 var2=300 &&
```

```
$    var_symbol1=x var_symbol2=temper
$
$  The user may skip the specification of reserved internal arguements
$  such as "temper", "doping_n", etc., because the program already
$  knows their values.  For example, the following may be used instead of
$  the above.
$
$  load_macro name=algaas.temp var1=0.3 var_symbol1=x
$
$
$                Searching For a Macros
$
$  The macro files can contain both the older and newer style macros
$  with the same macro name.  The search of the correct version depends
$  on how the macro is called.  If "var_symbol" is not specified in a
$  macro call, older style macro is assumed.   The user is encouraged
$  to convert to the newer style because Crosslight does not intend to
$  maintain/support the older style in the future.
$
$
$==============================================================
```

B.1 A Full Listing of a Zinc-Blende Active Layer Macro

This section lists a typical zinc-blende active layer macro which is constructed from binary compound parameters and bowing parameters.

Please note that both parameters for the well and barrier must be correctly supplied for the macro call.

```
$ **************************************************************
$
$  Active layer macro of strained well/barrier of
$  In(1-xw-yw)Ga(xw)Al(yw)P/In(1-xb-yb)Ga(xb)Al(yb)P
$  Grown on material lattice matched to GaAs
$  [free-style]
$
$  Most of the formulas here were derived from Vegard's law.
$  The compound comes from two ternarires
$  (In(1-x1)Ga(x1)P)(z)+(In(1-y1)Al(y1)P)(1-z)
$  where x1*z=x and y1*(1-z)=y
$
```

```
$ Please note the minus sign for Bowing coefficient.
$
$ Ref: A.A. Mbaye and C. Verie, "Electronic structure of
$      trimetallic III-V alloys:  The Al(1-x-z)Ga(x)In(z)P system",
$      Phys. Rev. B, vol 29, No. 6, pp. 3756-3758, 1984.
$
$   Modified/contributed  May 1996   Stefan Mohrdiek
$ Typical use:
$   get_active_layer name=InGaAlP/GaAs mater=#m &&
$     var1=#xw var2=#yw var3=#xb var4=#yb &&
$     var_symbol1=xw var_symbol2=yw var_symbol3=xb var_symbol4=yb
$ ************************************************************
begin_active_layer InGaAlP/GaAs
layer_type type=strained_well valley_gamma=1 valley_l=4 &&
  valley_hh=1 valley_lh=1
$
$ direct gap is used here.
$
eg0_well variation=function
function(xw,yw,temper)
shift=-4.1d-4*temper**2/(temper+136.) ;
shift0=-4.1d-4*300.**2/(300.+136.) ;
E_InP=1.347;
E_GaP=2.780;
E_AlP=3.62;
K_InGaP=0.69;
K_InAlP=0.4;
K_GaAlP=0.01;
Linear=E_InP*(1-xw-yw)+xw*E_GaP+yw*E_AlP;
Bowing=(1-xw-yw)*xw*K_InGaP+(1-xw-yw)*yw*K_InAlP+xw*yw*K_GaAlP;
Linear-Bowing+shift-shift0
end_function
$
eg0_bar variation=function
function(xb,yb,temper)
shift=-4.1d-4*temper**2/(temper+136.) ;
shift0=-4.1d-4*300.**2/(300.+136.) ;
E_InP=1.347;
E_GaP=2.780;
E_AlP=3.62;
K_InGaP=0.69;
K_InAlP=0.4;
K_GaAlP=0.01;
```

```
Linear=E_InP*(1-xb-yb)+xb*E_GaP+yb*E_AlP;
Bowing=(1-xb-yb)*xb*K_InGaP+(1-xb-yb)*yb*K_InAlP+xb*yb*K_GaAlP;
Linear-Bowing+shift-shift0
end_function
$
lband_well value=0.28
$
lband_bar value=0.28
$
band_offset value=0.5
$
$  The compound comes from two ternarires
$  (In(1-x1)Ga(x1)P)(z)+(In(1-y1)Al(y1)P)(1-z)
$  where x1*z=x and y1*(1-z)=y
$
$  Compressive strain is negative:
$
strain_well variation=function
function(xw,yw)
a0_InP=5.8688 ;
a0_GaP=5.45052 ;
a0_AlP=5.46354 ;
a0_GaAs=5.65325 ;
axy=a0_InP*(1-xw-yw)+xw*a0_GaP+yw*a0_AlP;
(a0_GaAs-axy)/axy
end_function
$
strain_bar variation=function
function(xb,yb)
a0_InP=5.8688 ;
a0_GaP=5.45052 ;
a0_AlP=5.46354 ;
a0_GaAs=5.65325 ;
axy=a0_InP*(1-xb-yb)+xb*a0_GaP+yb*a0_AlP;
(a0_GaAs-axy)/axy
end_function
$
$
delta_so_well variation=function
function(xw,yw)
del_InP=0.10 ;
del_GaP=0.08 ;
del_AlP=0.01 ;
```

```
del_InP*(1-xw-yw)+xw*del_GaP+yw*del_AlP
end_function
$
delta_so_bar  variation=function
function(xb,yb)
del_InP=0.10 ;
del_GaP=0.08 ;
del_AlP=0.01 ;
del_InP*(1-xb-yb)+xb*del_GaP+yb*del_AlP
end_function
$
mass_gamma_well variation=function
function(xw,yw)
emn_InP=0.078;
emn_GaP=0.17;
emn_AlP=0.2;
emn_InP*(1-xw-yw)+xw*emn_GaP+yw*emn_AlP
end_function
$
mass_l_well value=0.56
$
mass_gamma_bar variation=function
function(xb,yb)
emn_InP=0.078;
emn_GaP=0.17;
emn_AlP=0.2;
emn_InP*(1-xb-yb)+xb*emn_GaP+yb*emn_AlP
end_function
$
mass_l_bar value=0.56
$
gamma1_well  variation=function
function(xw,yw)
g1_GaP=4.20;
g1_InP=5.0 ;
g1_AlP=3.47;
g1_InP*(1-xw-yw)+xw*g1_GaP+yw*g1_AlP
end_function
$
gamma1_bar variation=function
function(xb,yb)
g1_GaP=4.20;
g1_InP=5.0 ;
```

```
g1_AlP=3.47;
g1_InP*(1-xb-yb)+xb*g1_GaP+yb*g1_AlP
end_function
$
gamma2_well  variation=function
function(xw,yw)
g2_GaP=0.98;
g2_InP=1.5 ;
g2_AlP=0.06;
g2_InP*(1-xw-yw)+xw*g2_GaP+yw*g2_AlP
end_function
$
gamma2_bar variation=function
function(xb,yb)
g2_GaP=0.98;
g2_InP=1.5 ;
g2_AlP=0.06;
g2_InP*(1-xb-yb)+xb*g2_GaP+yb*g2_AlP
end_function
$
gamma3_well  variation=function
function(xw,yw)
g3_InP=2.76;
g3_GaP=1.66;
g3_AlP=1.15;
g3_InP*(1-xw-yw)+xw*g3_GaP+yw*g3_AlP
end_function
$
gamma3_bar variation=function
function(xb,yb)
g3_InP=2.76;
g3_GaP=1.66;
g3_AlP=1.15;
g3_InP*(1-xb-yb)+xb*g3_GaP+yb*g3_AlP
end_function
$
a_well  variation=function
function(xw,yw)
a_InP=-6.35 ;
a_GaP=-9.3 ;
a_AlP=-8;
a_InP*(1-xw-yw)+xw*a_GaP+yw*a_AlP
end_function
```

```
$
a_bar variation=function
function(xb,yb)
a_InP=-6.35 ;
a_GaP=-9.3 ;
a_AlP=-8;
a_InP*(1-xb-yb)+xb*a_GaP+yb*a_AlP
end_function
$
b_well  variation=function
function(xw,yw)
b_InP=-2.0 ;
b_GaP=-1.8 ;
b_AlP=-2 ;
b_InP*(1-xw-yw)+xw*b_GaP+yw*b_AlP
end_function
$
b_bar variation=function
function(xb,yb)
b_InP=-2.0 ;
b_GaP=-1.8 ;
b_AlP=-2 ;
b_InP*(1-xb-yb)+xb*b_GaP+yb*b_AlP
end_function
$
c11_well  variation=function
function(xw,yw)
c11_InP=10.11 ;
c11_GaP=14.11 ;
c11_AlP=13.2 ;
c11_InP*(1-xw-yw)+xw*c11_GaP+yw*c11_AlP
end_function
$
c11_bar variation=function
function(xb,yb)
c11_InP=10.11 ;
c11_GaP=14.11 ;
c11_AlP=13.2 ;
c11_InP*(1-xb-yb)+xb*c11_GaP+yb*c11_AlP
end_function
$
c12_well  variation=function
function(xw,yw)
```

```
c12_InP=5.61 ;
c12_GaP=6.398 ;
c12_AlP=6.30 ;
c12_InP*(1-xw-yw)+xw*c12_GaP+yw*c12_AlP
end_function
$
c12_bar variation=function
function(xb,yb)
c12_InP=5.61 ;
c12_GaP=6.398 ;
c12_AlP=6.30 ;
c12_InP*(1-xb-yb)+xb*c12_GaP+yb*c12_AlP
end_function
lattice_constant value=5.65325
end_active_layer InGaAlP/GaAs
```

B.2 A Full Listing of a Zinc-Blende Bulk Macro

This section gives a full listing of a typical zinc-blende bulk macro of materal Al(x)Ga(1-x)As as a function of aluminum commposition.

In bulk macros it is also possible to use previously defined external functions. These must be named "bulk_xfunck" (k=1,2,...5). In the following listing, "bulk_xfunc1" is used to define the bandgap which may be used in the definition of the wavelength dependent version of the absorption coefficient. "bulk_xfunc2" is used to define the background absorption which will be later used in the defintion of the wavelength dependence of the absorption function.

```
$***********************************
$ macro algaas
$ Bulk Al(x)Ga(1-x)As at 300K
$ [free-style]
$ Typical use:
$   load_macro name=algaas var1=#x var2=#xlam mater=#m &&
$     var_symbol1=x var_symbol2=xlam
$ where xlam is wavelength in microns.
$***********************************
begin_macro algaas
material type=semicond band_valleys=(1 1) &&
  el_vel_model=n.gaas hole_vel_model=beta
dielectric_constant variation=function
function(x)
```

```
13.1 - 3 * x
end_function
electron_mass variation=function
function(x)
for   0.<x<0.45
0.067 + 0.083 * x
for  0.45<x<1.
0.85 - 0.14 * x
end_function
hole_mass variation=function
function(x)
   (    ( 0.087 + 0.063 * x ) ** (3 / 2)
 + ( 0.62 + 0.14 * x  ) ** (3 / 2)  ) ** (2 / 3)
end_function
band_gap variation=function
function(x)
for 0.<x<0.45
1.424 + 1.247 * x
for 0.45<x<1.
1.9 + 0.125 * x + 0.143 *x * x
end_function
$
$ band gap as external function to be used by absorption
bulk_xfunc1 variation=function
function(x)
for 0.<x<0.45
1.424 + 1.247 * x
for 0.45<x<1.
1.9 + 0.125 * x + 0.143 *x * x
end_function
affinity variation=function
function(x)
for 0<x<0.45
4.07 - 0.748 * x
for 0.45<x<1.
3.7964 - 0.14 * x
end_function


$
$electron_mobility variation=function
$function(x,doping_n,doping_p,trap_1)
$for 0<x<0.45
```

```
$mu_max=0.85 * exp(-18.516 * x ** 2 );
$mu_min=0.;
$ref_dens=1.69d23;
$alpha=0.436;
$total_doping=doping_n+doping_p+trap_1;
$mu_min+(mu_max-mu_min)/(1+(total_doping/ref_dens)**alpha)
$for 0.45<x<1.
$mu_max=0.02;
$mu_min=0.;
$ref_dens=1.69d23;
$alpha=0.436;
$total_doping=doping_n+doping_p+trap_1;
$mu_min+(mu_max-mu_min)/(1+(total_doping/ref_dens)**alpha)
$end_function

$
$hole_mobility variation=function
$function(x,doping_n,doping_p,trap_1)
$mu_max=0.04 - 0.048 * x + 0.02 * x * x;
$mu_min=0.;
$ref_dens=2.75d23;
$alpha=0.395;
$total_doping=doping_n+doping_p+trap_1;
$mu_min+(mu_max-mu_min)/(1+(total_doping/ref_dens)**alpha)
$end_function
$
$ Equivalent in theory to the following:

max_electron_mob variation=function
function(x)
for 0<x<0.45
0.85 * exp(-18.516 * x ** 2 )
for 0.45<x<1.
0.02
end_function
min_electron_mob value=0.
electron_ref_dens value=1.69d23
alpha_n value=0.436

max_hole_mob variation=function
function(x)
0.04 - 0.048 * x + 0.02 * x * x
end_function
```

```
min_hole_mob value=0.
hole_ref_dens value=2.75d23
alpha_p value=0.395


beta_n value=2.
electron_sat_vel variation=function
function(x)
for 0<x<0.45
0.77e5 * (1 - 0.44 * x )
for 0.45<x<1.
8.e4
end_function
beta_p value=1.
hole_sat_vel value=1.d5
norm_field value=4.e5
tau_energy value=1.e-13
radiative_recomb value=1.d-16
auger_n value=1.5d-42
auger_p value=1.5d-42
lifetime_n value=1.e-7
lifetime_p value=1.e-7

$real_index variation=function
$function(x)
$3.65-0.73*x
$end_function
$
$ Wavelength dep. real index.  You may use the following simple
$ linear interpolation.  But you may also use the more complicated
$ fit to experiment.
$
$real_index variation=function
$function(x,xlam)
$xev=1.24/xlam;
$n_GaAs=4.177137-3.512283*xev+5.163323*xev**2
$  -3.242022*xev**3+0.7935377*xev**4;
$n_AlAs=2.835238+0.1040565*xev-8.6519323e-2*xev**2
$  +8.2142524e-2*xev**3 -1.4580269e-2*xev**4;
$n_GaAs*(1-x)+n_AlAs*x
$end_function
$
$ Fit to experiment.
```

```
$
real_index variation=function
function(x,xlam)
for 0<x<0.1
xev=1.24/xlam;
n1=4.177137-3.512283*xev+5.163323*xev**2
  -3.242022*xev**3+0.7935377*xev**4;
n2=3.1715+0.8963*xev-1.4964*xev**2+1.0665*xev**3-0.2310*xev**4;
n1+(n2-n1)*(x-0.)/0.1
for 0.1<x<0.2
xev=1.24/xlam;
n1=3.1715+0.8963*xev-1.4964*xev**2+1.0665*xev**3-0.2310*xev**4;
n2=2.3080+3.5046*xev-4.4066*xev**2+2.4046*xev**3-0.4481*xev**4;
n1+(n2-n1)*(x-0.1)/0.1
for 0.2<x<0.3
xev=1.24/xlam;
n1=2.3080+3.5046*xev-4.4066*xev**2+2.4046*xev**3-0.4481*xev**4;
n2=2.4397+2.7106*xev-3.2388*xev**2+1.6831*xev**3-0.2919*xev**4;
n1+(n2-n1)*(x-0.2)/0.1
for 0.3<x<0.4
xev=1.24/xlam;
n1=2.4397+2.7106*xev-3.2388*xev**2+1.6831*xev**3-0.2919*xev**4;
n2=3.3968-0.8732*xev+1.3142*xev**2-0.7823*xev**3+0.1881*xev**4;
n1+(n2-n1)*(x-0.3)/0.1
for 0.4<x<0.5
xev=1.24/xlam;
n1=3.3968-0.8732*xev+1.3142*xev**2-0.7823*xev**3+0.1881*xev**4;
n2=4.1143 -3.5076*xev +4.5832*xev**2 -2.4916*xev**3 +0.5027*xev**4;
n1+(n2-n1)*(x-0.4)/0.1
for 0.5<x<0.6
xev=1.24/xlam;
n1=4.1143 -3.5076*xev +4.5832*xev**2 -2.4916*xev**3 +0.5027*xev**4;
n2=3.2169 -0.5468*xev +0.8554*xev**2 -0.4952*xev**3 +0.1178*xev**4;
n1+(n2-n1)*(x-0.5)/0.1
for 0.6<x<0.7
xev=1.24/xlam;
n1=3.2169 -0.5468*xev +0.8554*xev**2 -0.4952*xev**3 +0.1178*xev**4;
n2=3.3236 -1.1576*xev +1.6762*xev**2 -0.9628*xev**3 +0.2096*xev**4;
n1+(n2-n1)*(x-0.6)/0.1
for 0.7<x<0.8
xev=1.24/xlam;
n1=3.3236 -1.1576*xev +1.6762*xev**2 -0.9628*xev**3 +0.2096*xev**4;
n2=3.1106 -0.4848*xev +0.6741*xev**2 -0.3351*xev**3 +0.0700*xev**4;
```

```
n1+(n2-n1)*(x-0.7)/0.1
for 0.8<x<0.9
xev=1.24/xlam;
n1=3.1106 -0.4848*xev +0.6741*xev**2 -0.3351*xev**3 +0.0700*xev**4;
n2=3.0202 -0.3050*xev +0.4331*xev**2 -0.2045*xev**3 +0.0447*xev**4;
n1+(n2-n1)*(x-0.8)/0.1
for 0.9<x<1.
xev=1.24/xlam;
n1=3.0202 -0.3050*xev +0.4331*xev**2 -0.2045*xev**3 +0.0447*xev**4;
n2=2.835238+0.1040565*xev-8.6519323e-2*xev**2
  +8.2142524e-2*xev**3 -1.4580269e-2*xev**4;
n1+(n2-n1)*(x-0.9)/0.1
end_function

$absorption value=0.
$ wavelength dependent absorption
$ define backgroud aborption as bulk_xfunc2 in units 1/m
bulk_xfunc2 variation=function
function(x,xlam)
500.
end_function

absorption variation=function
function(x,xlam)
for 1.24/xlam<bulk_xfunc1
bulk_xfunc2
for 0<x<0.1
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=-59938.21*(xev-eg)+561976.8*(xev-eg)**2
  -855408.5*(xev-eg)**3+430717.0*(xev-eg)**4;
a2=-18043.81*(xev-eg)+470037.7*(xev-eg)**2
  -762741.0*(xev-eg)**3+409836.0*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.)/0.1;
bulk_xfunc2+alpha*100.
for 0.1<x<0.2
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=-18043.81*(xev-eg)+470037.7*(xev-eg)**2
  -762741.0*(xev-eg)**3+409836.0*(xev-eg)**4;
a2=91721.91*(xev-eg)+107193.1*(xev-eg)**2
  -399274.5*(xev-eg)**3+323630.5*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.1)/0.1;
```

```
bulk_xfunc2+alpha*100.
for 0.2<x<0.3
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=91721.91*(xev-eg)+107193.1*(xev-eg)**2
  -399274.5*(xev-eg)**3+323630.5*(xev-eg)**4;
a2=103024.9*(xev-eg)+285712.4*(xev-eg)**2
  -821371.1*(xev-eg)**3+592386.0*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.2)/0.1;
bulk_xfunc2+alpha*100.
for 0.3<x<0.4
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=103024.9*(xev-eg)+285712.4*(xev-eg)**2
  -821371.1*(xev-eg)**3+592386.0*(xev-eg)**4;
a2=95788.52*(xev-eg)+13612.16*(xev-eg)**2
  -176194.3*(xev-eg)**3+275328.6*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.3)/0.1;
bulk_xfunc2+alpha*100.
for 0.4<x<0.5
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=95788.52*(xev-eg)+13612.16*(xev-eg)**2
  -176194.3*(xev-eg)**3+275328.6*(xev-eg)**4;
a2=90497.19*(xev-eg)-118715.9*(xev-eg)**2
  +136604.8*(xev-eg)**3+105422.4*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.4)/0.1;
bulk_xfunc2+alpha*100.
for 0.5<x<0.6
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=90497.19*(xev-eg)-118715.9*(xev-eg)**2
  +136604.8*(xev-eg)**3+105422.4*(xev-eg)**4;
a2=76853.11*(xev-eg)+161217.3*(xev-eg)**2
  -360155.9*(xev-eg)**3+318543.7*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.5)/0.1;
bulk_xfunc2+alpha*100.
for 0.6<x<0.7
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=76853.11*(xev-eg)+161217.3*(xev-eg)**2
  -360155.9*(xev-eg)**3+318543.7*(xev-eg)**4;
a2=32454.24*(xev-eg)+50188.12*(xev-eg)**2
```

```
  -16573.62*(xev-eg)**3+106340.0*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.6)/0.1;
bulk_xfunc2+alpha*100.
for 0.7<x<0.8
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=32454.24*(xev-eg)+50188.12*(xev-eg)**2
  -16573.62*(xev-eg)**3+106340.0*(xev-eg)**4;
a2=17639.67*(xev-eg)-31695.61*(xev-eg)**2
  +116720.0*(xev-eg)**3+33719.42*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.7)/0.1;
bulk_xfunc2+alpha*100.
for 0.8<x<0.9
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=17639.67*(xev-eg)-31695.61*(xev-eg)**2
  +116720.0*(xev-eg)**3+33719.42*(xev-eg)**4;
a2=6957.562*(xev-eg)+964.3477*(xev-eg)**2
  -74366.28*(xev-eg)**3+101554.3*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.8)/0.1;
bulk_xfunc2+alpha*100.
for else
xev=1.24/xlam;
eg=bulk_xfunc1;
a1=6957.562*(xev-eg)+964.3477*(xev-eg)**2
  -74366.28*(xev-eg)**3+101554.3*(xev-eg)**4;
a2=6957.562*(xev-eg)+964.3477*(xev-eg)**2
  -74366.28*(xev-eg)**3+101554.3*(xev-eg)**4;
alpha=a1+(a2-a1)*(x-0.9)/0.1;
bulk_xfunc2+alpha*100.
end_function
thermal_kappa value=46.
end_macro algaas
```

B.3 A Full Listing of a Wurtzite Active Layer Macro

This section lists a typical wurtzite active layer InGaN/InGaN, i.e., InGaN well grown on InGaN barrier. The parameters for wurtzite semiconductor is substantially different from those of zinc-blende structure due to the complex coupling of the three valence bands. The numerous parameters are mostly used to define the valence band Hamiltonian appearing in Ref. [5] listed in the macro below.

Since the whole structure is grown on saphite, it is unclear exactly what the base lattice size should be. Thus the user is required to investigate this issue based on information of the growth on saphite. Unless experimental data indicates otherwise, we suggest setting base lattice to that of the thickest layer on saphite since mechanically, this lattice force from this massive layer wins over all other layers and sets the base lattice.

```
$ **********************************************
$ active layer macro for
$ quantum well system of In(xw)Ga(1-xw)N/In(xb)Ga(1-xb)N
$  [free-style]
$ xw=In composition in well, xb=In composition in barrier,
$ temper=well temperature
$
$ revised by J. Piprek 9/9/2001
$
$ Strained quantum well grown on strained quantum barrier.
$ The substrate is assumed to be lattice matched to GaN
$ Typical use:
$   get_active_layer name=InGaN/InGaN  mater=#m &&
$     var1=#xw var2=#yw var_symbol1=xw var_symbol2=yw
$ *********************************************
$
begin_active_layer InGaN/InGaN
$
layer_type type=wurtzite_well valley_gamma=1 &&
  valley_hh=1 valley_lh=1 valley_ch=1
$
$ Lattice constant of InN is 3.548
$ Lattice constant of GaN is 3.189
$
$ We need to compute the strain based on lattice constant
$   of the substrate
lattice_constant value=3.189
$
lattice_well variation=function
function(xw)
3.189*(1-xw)+3.548*xw
end_function
$
lattice_bar variation=function
function(xb)
3.189*(1-xb)+3.548*xb
```

```
end_function
$ --- energy parameters
eg0_well variation=function
function(xw,temper)
eg0GaN=3.42;
bowing=3.8;
eg300=(1.0-xw)*eg0GaN+1.89*xw-bowing*xw*(1-xw);
eg300-6.e-4*(temper-300.)
end_function
$
eg0_bar variation=function
function(xb,temper)
eg0GaN=3.42;
bowing=3.8;
eg300=(1.0-xb)*eg0GaN+1.89*xb-bowing*xb*(1-xb);
eg300-6.e-4*(temper-300.)
end_function
$
band_offset value=0.7
$
delta_so_well variation=function
function(xw)
gan=0.013;
inn=0.001;
gan+(inn-gan)*xw
end_function
$
delta_so_bar variation=function
function(xb)
gan=0.013;
inn=0.001;
gan+(inn-gan)*xb
end_function
$
delta1_well variation=function
function(xw)
gan=0.042;
inn=0.041;
gan+(inn-gan)*xw
end_function
$
delta1_bar variation=function
function(xb)
```

```
gan=0.042;
inn=0.041;
gan+(inn-gan)*xb
end_function
$
delta2_well variation=function
function(xw)
gan=0.0043;
inn=0.0003;
gan+(inn-gan)*xw
end_function
$
delta2_bar variation=function
function(xb)
gan=0.0043;
inn=0.0003;
gan+(inn-gan)*xb
end_function
$
delta3_well variation=function
function(xw)
gan=0.0043;
inn=0.0003;
gan+(inn-gan)*xw
end_function
$
delta3_bar variation=function
function(xb)
gan=0.0043;
inn=0.0003;
gan+(inn-gan)*xb
end_function
$
$ ----conduction band masses
mass_gamma_well variation=function
function(xw)
gan=.20;
inn=.11;
gan+(inn-gan)*xw
end_function
$
mass_gamma_bar variation=function
function(xb)
```

```
gan=.20;
inn=.11;
gan+(inn-gan)*xb
end_function
$
tmass_gamma_well variation=function
function(xw)
gan=.18;
inn=.11;
gan+(inn-gan)*xw
end_function
$
tmass_gamma_bar variation=function
function(xb)
gan=.18;
inn=.11;
gan+(inn-gan)*xb
end_function
$
a1_well variation=function
function(xw)
inn=-9.28;
gan=-7.24;
gan+(inn-gan)*xw
end_function
$
a2_well variation=function
function(xw)
inn=-0.60;
gan=-0.51;
gan+(inn-gan)*xw
end_function
$
a3_well variation=function
function(xw)
inn=8.68;
gan=6.73;
gan+(inn-gan)*xw
end_function
$
a4_well variation=function
function(xw)
inn=-4.34;
```

```
gan=-3.36;
gan+(inn-gan)*xw
end_function
$
a5_well variation=function
function(xw)
inn=-4.32;
gan=-3.35;
gan+(inn-gan)*xw
end_function
$
a6_well variation=function
function(xw)
inn=-6.08;
gan=-4.72;
gan+(inn-gan)*xw
end_function
$
$
a1_bar variation=function
function(xb)
inn=-9.28;
gan=-7.24;
gan+(inn-gan)*xb
end_function
$
a2_bar variation=function
function(xb)
inn=-0.60;
gan=-0.51;
gan+(inn-gan)*xb
end_function
$
a3_bar variation=function
function(xb)
inn=8.68;
gan=6.73;
gan+(inn-gan)*xb
end_function
$
a4_bar variation=function
function(xw,xb,temper)
inn=-4.34;
```

```
gan=-3.36;
gan+(inn-gan)*xb
end_function
$
a5_bar variation=function
function(xb)
inn=-4.32;
gan=-3.35;
gan+(inn-gan)*xb
end_function
$
a6_bar variation=function
function(xb)
inn=-6.08;
gan=-4.72;
gan+(inn-gan)*xb
end_function
$
c33_well variation=function
function(xw)
gan=39.2;
inn=20.0;
gan+(inn-gan)*xw
end_function
$
c33_bar variation=function
function(xb)
gan=39.2;
inn=20.0;
gan+(inn-gan)*xb
end_function
$
c13_well variation=function
function(xw)
gan=10.;
inn=9.4;
gan+(inn-gan)*xw
end_function
$
c13_bar variation=function
function(xb)
gan=10.;
inn=9.4;
```

```
gan+(inn-gan)*xb
end_function
$
a_well value=-8.16
ac_well value=-4.08
a_bar value=-8.16
ac_bar value=-4.08
d1_well value=-0.89
d1_bar value=-0.89
d2_well value=4.27
d2_bar value=4.27
d3_well value=5.18
d3_bar value=5.18
d4_well value=-2.59
d4_bar value=-2.59
$
end_active_layer InGaN/InGaN
```

B.4 A Full Listing of a Wurtzite Bulk Macro

This section lists a typical wurtzite bulk macro for InGaN.

Please note that the bulk macro of wurtzite structure material parameters are similar to those of active layer macros except that the parameters for the barrier are not needed. Also note that the lattice base setting is critical since it affects the strain.

```
$*********************************************
$ macro ingan
$ Bulk In(x)Ga(1-x)N parameters
$  [free-style]
$ Contributors: Peter Mensz, Khalid Shahzad (and maybe) others.
$ Changed by Kehl Sink 6/10/99
$ changed by Joachim Piprek 12/2/99 + 1/7/00
$ Typical use:
$   load_macro name=ingan var1=#x mater=#m var_symbol1=x
$ ********************************************
begin_macro ingan
material type=wurtzite band_valleys=(1 1) &&
  el_vel_model=beta  hole_vel_model=beta

$---wurtzite parameters-----------
$---These are the same as in corresponding Active Layer Macro
```

```
$---Please see references therein

lattice_base value=3.189
lattice_bulk variation=function
function(x)
3.189*(1-x)+3.548*x
end_function
$
eg0_bulk variation=function
function(x)
eg0GaN=3.42;
bowing=3.8;
(1.0-x)*eg0GaN+1.89*x-bowing*x*(1-x)
end_function
$
delta1_bulk variation=function
function(x)
gan=0.042;
inn=0.041;
gan+(inn-gan)*x
end_function
$
delta2_bulk variation=function
function(x)
gan=0.0043;
inn=0.0003;
gan+(inn-gan)*x
end_function
$
delta3_bulk variation=function
function(x)
gan=0.0043;
inn=0.0003;
gan+(inn-gan)*x
end_function
$
mass_gamma_bulk variation=function
function(x)
gan=.20;
inn=.11;
gan+(inn-gan)*x
end_function
$
```

```
tmass_gamma_bulk variation=function
function(x)
gan=.18;
inn=.11;
gan+(inn-gan)*x
end_function
$
a1_bulk variation=function
function(x)
inn=-9.28;
gan=-7.24;
gan+(inn-gan)*x
end_function
$
a2_bulk variation=function
function(x)
inn=-0.60;
gan=-0.51;
gan+(inn-gan)*x
end_function
$
a3_bulk variation=function
function(x)
inn=8.68;
gan=6.73;
gan+(inn-gan)*x
end_function
$
a4_bulk variation=function
function(x)
inn=-4.34;
gan=-3.36;
gan+(inn-gan)*x
end_function
$
a5_bulk variation=function
function(x)
inn=-4.32;
gan=-3.35;
gan+(inn-gan)*x
end_function
$
a6_bulk variation=function
```

```
function(x)
inn=-6.08;
gan=-4.72;
gan+(inn-gan)*x
end_function

$
c33_bulk variation=function
function(x)
gan=39.2;
inn=20.0;
gan+(inn-gan)*x
end_function
$
c13_bulk variation=function
function(x)
gan=10.;
inn=9.4;
gan+(inn-gan)*x
end_function

$ GaN values for def. pot.
d1_bulk value=-0.89
d2_bulk value=4.27
d3_bulk value=5.18
d4_bulk value=-2.59
a_bulk value=-8.16
ac_bulk value=-4.08
$---end of wurtzite parameters-----------
dielectric_constant variation=function
function(x)
gan=9.5;
inn=15;
gan+(inn-gan)*x
end_function
$
$ It appears that the actual strained HH bandgap of InGaN decreases
$ much faster as a function of x than the unstrained bandgap.
$ Thus, we should reduce the affinity increase here by a factor.
$ Revised Jun03 by Crosslight
$
affinity variation=function
function(x)
```

```
eg0GaN=3.42;
bowing=3.8;
egx=(1.0-x)*eg0GaN+1.89*x-bowing*x*(1-x);
aff_gan=4.0;
eg_gan=3.42;
cbpercent=0.70;
factor=0.8;
aff_gan+(eg_gan-egx)*cbpercent*factor
end_function

$
electron_mobility variation=function
function(x,temper,doping_n,doping_p,trap_1)
mu_max=684.d-4*(300/temper)**1.5;
mu_min=386.d-4*(300/temper)**1.5;
ref_dens=1.d23;
alpha=1.37;
total_doping=doping_n+doping_p+trap_1;
mu_min+(mu_max-mu_min)/(1+(total_doping/ref_dens)**alpha)
end_function

beta_n value=1.
electron_sat_vel value=1.d5
norm_field value=2.1e7

$ from [7]
hole_mobility variation=function
function(doping_n,doping_p,trap_1)
mu_max=2.0d-4;
mu_min=2.0d-4;
ref_dens=2.75d23;
alpha=0.395;
total_doping=doping_n+doping_p+trap_1;
mu_min+(mu_max-mu_min)/(1+(total_doping/ref_dens)**alpha)
end_function

beta_p value=1.
hole_sat_vel value=1.d5
$
tau_energy value=1.d-13
$
lifetime_n value=1.e-9
lifetime_p value=1.e-9
```

```
$ adjusted 1/7/00
radiative_recomb  value=0.2d-16
auger_n  value=1.d-46
auger_p  value=1.d-46
$
real_index value=2.7
absorption value=0.
$ rough estimate for thin layers:
thermal_kappa value=20.
$
$
$
$ References:
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$     "Electron overflow to the AlGaN p-cladding layer in
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$     photoreflection spectroscopy," Applied Physics Letters, v. 73,
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$
end_macro ingan
```

Appendix C

Modified Scharfetter-Gummel Formula for Hot Electrons

C.1 Current Flow in Hydrodynamic Model

The expression for hot electron current flow has been derived by Azoff [5][7]. Here we use a very initiative and less rigorous approach to derive the expression for hot electron current flow to gain insight into how it relates to the corresponding model in the conventional drift-diffusion model. We give all the details so that our user can check all the steps.

Our purpose is to convert our expression of current flow so that the spatial variation in the electron temperature (or electron energy) can be taken into account. We start by defining a ratio to indicate the effect of Fermi-Dirac statistics.

$$\gamma_F/kT = ln\left\{\frac{F_{1/2}[(E_{fn}-E_c)/kT]}{exp(E_{fn}-E_c)/kT]}\right\} \tag{C.1}$$

We recall that the electron concentration is given by

$$n = N_c F_{1/2}[(E_{fn}-E_c)/kT] \tag{C.2}$$

or in terms of the γ_F constant we just introduced,

$$n = exp[ln(N_c)]exp(\gamma_F/kT)exp[(E_{fn}-E_c)/kT] \tag{C.3}$$

or

$$kTln(n) = kTln(N_c) + \gamma_F + (E_{fn}-E_c) \tag{C.4}$$

We choose the vacuum as the zero energy and express the conduction band in terms of the potention and the electron affinity.

$$E_c = -\psi - \chi \tag{C.5}$$

Then we have

$$kTln(n) = kTln(N_c) + \gamma_F + E_{fn} + \psi + \chi \tag{C.6}$$

or

$$E_{fn} = kTln(n) - kTln(N_c) - \gamma_F - \psi - \chi \tag{C.7}$$

A well known formula in drift and diffusion theory relates the current flow to the quasi-Fermi level:

$$J_n q/\mu_n = n\nabla E_{fn} \tag{C.8}$$

where

$$E_{fn} = kTln(n) - kTln(N_c) - \gamma_F - \psi - \chi \tag{C.9}$$

We split this into drift and diffusion parts:

$$J_n q/\mu_n[drift] = n\nabla[-kTln(N_c) - \gamma_F - \psi - \chi] \tag{C.10}$$

$$J_n q/\mu_n[diff] = n\nabla[kTln(n)] \tag{C.11}$$

or

$$J_n q/\mu_n[diff] = \nabla[kTn] \tag{C.12}$$

We recall that the density of states is given by

$$N_c = M_x 2.5 \times 10^{25}\left(\frac{m_c kT}{km_0 300.}\right)^{3/2} \tag{C.13}$$

where M_x is the number of band valleys (or degeneracy) in k-space. The drift part under constant electron temperature is rewritten as

$$\begin{aligned} J_n q/\mu_n[drift] &= n\nabla[-kTln(N_c) - \gamma_F - \psi - \chi] && \text{(C.14)} \\ &= -nkT(3/2)\nabla ln(m_c) - n\nabla[\gamma_F + \psi + \chi] && \text{(C.15)} \end{aligned}$$

Here we have neglected the effect of temperature gradient to the drift part. We also assume that the temperature gradient contributes an extra diffusion term $n\nabla(kT)$. Therefore the diffusion current is proportional to the gradient of (kTn) instead of n alone:

$$J_n q/\mu_n[diff] = \nabla[kTn] \tag{C.16}$$

To avoid confusion with lattice temperature, we use T_e to denote electron carrier temperature. Our final results are

$$J_n q/\mu_n = -n\nabla[\gamma_F + \psi + \chi] - nkT_e(3/2)\nabla ln(m_c) + \nabla[kT_e n] \tag{C.17}$$

which is the same as that derived by Azoff [5][7] from Boltsman equation.

We define the electron energy as $w = (3/2)kT_e$ and we have the following final result for current flow:

$$J_n q/\mu_n = -n\nabla[\gamma_F + \psi + \chi] - nw\nabla ln(m_c) + (2/3)\nabla[wn] \tag{C.18}$$

C.2 Discretization for Hot Electron Current

In the conventional drift-diffusion model, the discretization of the current flow on a grid is best handled by the formula derived by Scharfetter and Gummel (SG-formula) in 1969 [34]. For hot electron model, the discretization formula must be modified to relfect the spatial variation of electron temperature. Following Azoff [7], we derive the new discretization as follwos.

We consider two grid points along the x-direction and express the current flow as follows:

$$(3/2)J_n q/\mu_n = -(3/2)n\frac{\partial}{\partial x}[\gamma_F + \psi + \chi] - (3/2)nw\frac{\partial}{\partial x}ln(m_c) + \frac{\partial}{\partial x}[wn] \tag{C.19}$$

We assume that the electric field between mesh points is constant. The electron temperature may be different between the two points, but we assume that the temparture variation is smooth enough so that $\frac{\partial w}{\partial x}$ and $\frac{\partial ln(w)}{\partial x}$ may be regarded as constants. We re-write the current:

$$(3/2)J_n q/\mu_n = \frac{\partial w}{\partial x}n\{-(3/2)\frac{\partial}{\partial x}[\gamma_F + \psi + \chi] - (3/2)w\frac{\partial}{\partial x}ln(m_c)\}\left(\frac{\partial w}{\partial x}\right)^{-1} + \frac{\partial}{\partial x}[wn] \tag{C.20}$$

or

$$(3/2)J_n q/\mu_n = \frac{\partial w}{\partial x}n\alpha + \frac{\partial}{\partial x}[wn], \tag{C.21}$$

where α is a position independent parameter based on our previous assumptions,

$$\begin{aligned} \alpha &= \{(3/2)\frac{\partial}{\partial x}[-\gamma_F - \psi - \chi] - (3/2)w\frac{\partial}{\partial x}ln(m_c)\}\left(\frac{\partial w}{\partial x}\right)^{-1} && \text{(C.22)} \\ &= \{(3/2)\frac{\partial}{\partial x}(-\gamma_F - \psi - \chi)\left(\frac{\partial w}{\partial x}\right)^{-1} \\ &\quad -(3/2)\frac{\partial}{\partial x}ln(m_c)\left(\frac{\partial}{\partial x}ln(w)\right)^{-1} && \text{(C.23)} \end{aligned}$$

We apply the basic assumption of the original SG-formula that the electron current is constant between two grid piints. Using w^α as an integration factor on Eq. (C.21), we obtain

$$\begin{aligned} (3/2)J_n q/\mu_n &= w^{-\alpha}\left[\frac{\partial w}{\partial x}n\alpha w^\alpha + \frac{\partial}{\partial x}(wn)w^\alpha\right] && \text{(C.24)} \\ (3/2)J_n q/\mu_n &= w^{-\alpha}\left[\frac{\partial w}{\partial x}n\alpha w^\alpha + \frac{\partial w}{\partial x}nw^\alpha + \frac{\partial n}{\partial x}w^{\alpha+1}\right] && \text{(C.25)} \end{aligned}$$

$$
\begin{aligned}
(3/2)J_n q/\mu_n &= w^{-\alpha}\left[\frac{\partial w}{\partial x} n(\alpha+1)w^{\alpha} + \frac{\partial n}{\partial x} w^{\alpha+1}\right] && \text{(C.26)}\\
(3/2)J_n q/\mu_n &= w^{-\alpha}\left[\frac{\partial}{\partial x} w^{\alpha+1} n + \frac{\partial}{\partial x} n w^{\alpha+1}\right] && \text{(C.27)}\\
(3/2)J_n q/\mu_n &= w^{-\alpha}\frac{\partial}{\partial x}(w^{\alpha+1} n) && \text{(C.28)}
\end{aligned}
$$

We assume w is a linear function between the two grid points so that

$$w = w_a x + w_b \tag{C.29}$$

We re-write the equation as

$$(3/2)J_n q/\mu_n w^{\alpha} dx = d(w^{\alpha+1} n) \tag{C.30}$$

We integrate from point 1 to point 2 and get:

$$(3/2)J_n q/\mu_n \left[\frac{w^{\alpha+1}}{(1+\alpha)w_a}\right]_1^2 = [w^{\alpha+1} n]_1^2 \tag{C.31}$$

$$
\begin{aligned}
J_n q/\mu_n &= \frac{(2/3)[w^{\alpha+1} n]_1^2}{\left[\frac{w^{\alpha+1}}{(1+\alpha)w_a}\right]_1^2} && \text{(C.32)}\\
&= (2/3)(x_2-x_1)^{-1}\frac{(w_2-w_1)[w^{\alpha+1} n]_1^2}{\left[\frac{w^{\alpha+1}}{(1+\alpha)}\right]_1^2} && \text{(C.33)}\\
&= (2/3)(x_2-x_1)^{-1}\frac{(w_2-w_1)(w_2^{\alpha+1} n_2 - w_1^{\alpha+1} n_1)(1+\alpha)}{w_2^{\alpha+1} - w_1^{\alpha+1}} && \text{(C.34)}
\end{aligned}
$$

$$
\begin{aligned}
J_n q/\mu_n &= (2/3)(x_2-x_1)^{-1}(w_2-w_1)(1+\alpha)\frac{w_2^{\alpha+1} n_2 - w_1^{\alpha+1} n_1}{w_2^{\alpha+1} - w_1^{\alpha+1}} && \text{(C.35)}\\
&= (2/3)(x_2-x_1)^{-1}(w_2-w_1)(1+\alpha)\\
&\quad \left[\frac{w_2^{\alpha+1} n_2}{w_2^{\alpha+1} - w_1^{\alpha+1}} + \frac{w_1^{\alpha+1} n_1}{w_1^{\alpha+1} - w_2^{\alpha+1}}\right] && \text{(C.36)}
\end{aligned}
$$

We obtain our result:

$$J_n q/\mu_n = (2/3)(x_2-x_1)^{-1}(w_2-w_1)(1+\alpha)\left[\frac{n_1}{1-(w_2/w_1)^{\alpha+1}} + \frac{n_2}{1-(w_1/w_2)^{\alpha+1}}\right] \tag{C.37}$$

Furthermore, we wish to use the lattice temperature kT_L as the unit for $W = w/kT_L$ so that the equilibrium electron temperature is $W_0 = 3/2$. We obtain:

$$J_n = \mu_n \left(\frac{kT_L}{q}\right)(2/3)(x_2-x_1)^{-1}(W_2-W_1)(1+\alpha)$$

$$\left[\frac{n_1}{1-(W_2/W_1)^{\alpha+1}}+\frac{n_2}{1-(W_1/W_2)^{\alpha+1}}\right] \tag{C.38}$$

$$= \mu_n v_t (2/3)(x_2-x_1)^{-1}(W_2-W_1)(1+\alpha)$$
$$\left[\frac{n_1}{1-(W_2/W_1)^{\alpha+1}}+\frac{n_2}{1-(W_1/W_2)^{\alpha+1}}\right] \tag{C.39}$$

where $v_t = kT_L/q$ is the lattice thermal voltage.

C.3 Hot Electron Current with Equal Energy

When the electron energy at the two points are equal or nearly equal, some approximation can be made. This is important in numerical simulation to avoid floating point problem when the carrier temperature of the adjacent points are nearly equal. We start from our basic equation before any discretization:

$$J_n q/\mu_n = -n\nabla(\gamma_F+\psi+\chi) - nw\nabla ln(m_c) + (2/3)\nabla(wn) \tag{C.40}$$

When w can be considered constant (but not necessarily equal to the lattice temperature), we have

$$J_n q/\mu_n = -n\nabla[\gamma_F+\psi+\chi] - n(3/2)kT_e\nabla ln(m_c) + kT_e\nabla n \tag{C.41}$$

$$J_n q/(kT_e\mu_n) = -n\nabla[\gamma_F+\psi+\chi+(3/2)kT_e\nabla ln(m_c)]/kT_e + \nabla n \tag{C.42}$$

Again, we apply the basic assumption of the SG-formula that the current is constant between two points. Then we obtain a differential equation in the x-direction of the form:

$$a = -bn + \frac{dn}{dx} \tag{C.43}$$

where

$$J_n q/(kT_e\mu_n) = -n\nabla[\gamma_F+\psi+\chi+(3/2)kT_e\nabla ln(m_c)]/kT_e + \nabla n \tag{C.44}$$

$$a = J_n q/(kT_e\mu_n) \tag{C.45}$$

$$b = \nabla[\gamma_F+\psi+\chi+(3/2)kT_e\nabla ln(m_c)]/kT_e \tag{C.46}$$

Solving the simple differential equation from x_1 to x_2, we have

$$a = 1/(x_2-x_1)b(x_2-x_1)[n_2 - n_1 exp[b(x_2-x_1)]]/[exp[b(x_2-x_1)]-1] \tag{C.47}$$

$$\begin{aligned} a(x_2 - x_1) &= b(x_2 - x_1)n_2/[exp[b(x_2 - x_1)] - 1] \\ &\quad -b(x_2 - x_1)n_1 exp[b(x_2 - x_1)]/[exp[b(x_2 - x_1)] - 1] \\ &= b(x_2 - x_1)n_2/[exp[b(x_2 - x_1)] - 1] \\ &\quad -b(x_2 - x_1)n_1/[1 - exp[b(x_1 - x_2)]] \\ &= b(x_2 - x_1)n_2/[exp[b(x_2 - x_1)] - 1] \\ &\quad -b(x_1 - x_2)n_1/[exp[b(x_1 - x_2)] - 1] \end{aligned} \tag{C.48}$$

$$a = \frac{1}{x_2 - x_1}[n_2 B[b(x_2 - x_1)] - n_1 B[b(x_1 - X_2)]] \tag{C.49}$$

where B() is the Bernouli function. Substituting a and b from Eqs. (C.45) and (C.46)

$$J_n q/(kT_e \mu_n) = \frac{1}{x_2 - x_1}[n_2 B(\eta_2^h - \eta_1^h) - n_1 B(\eta_1^h - \eta_2^h)] \tag{C.50}$$

$$J_n = V_{te}\frac{\mu_n}{x_2 - x_1}[n_2 B(\eta_2^h - \eta_1^h) - n_1 B(\eta_1^h - \eta_2^h)] \tag{C.51}$$

This is to be compared with the cool electron expression:

$$J_n = V_t \frac{\mu_n}{x_2 - x_1}\left[(n_2 B(\eta_2 - \eta_1) - n_1 B(\eta_1 - \eta_2)\right] \tag{C.52}$$

It is most convenient to relate the η^h to η

$$\begin{aligned} \eta^h &= [\gamma_F + \psi + \chi + kT_L ln(N_c)]/kT_e \\ &\quad +(3/2)\nabla ln(m_c) - ln(N_c)T_L/T_e && \text{(C.53)} \\ &= [\gamma_F + \psi + \chi + kT_L ln(N_c)]/kT_L(T_L/T_e) \\ &\quad +(3/2)\nabla ln(m_c) - ln(N_c)T_L/T_e && \text{(C.54)} \\ &= \eta(T_L/T_e) + (3/2)ln(m_c) - ln(N_c)T_L/T_e && \text{(C.55)} \end{aligned}$$

Using the unitless W:

$$\begin{aligned} \eta^h &= \eta 1.5(kT_L/1.5kT_e) \\ &\quad +(3/2)ln(m_c) - 1.5ln(N_c)(kT_L/1.5kT_e) && \text{(C.56)} \\ &= \eta 1.5/W + (3/2)ln(m_c) - 1.5ln(N_c)/W && \text{(C.57)} \end{aligned}$$

Similarly

$$V_{te} = kT_e/q \tag{C.58}$$

$$V_{te} = 1.5kT_e kT_L/(1.5kT_L q) \tag{C.59}$$

$$V_{te} = (1.5kT_e/kT_L)/1.5(kT_L/q) \tag{C.60}$$

$$V_{te} = W/1.5V_t \tag{C.61}$$

Appendix D

GREEN'S FUNCTION AND SPONTANEOUS EMISSION

This chapter starts from the Maxwell equation to establish some notations. Then we recall some basic theories of partial differential equations. Finally, we derive the theories for spontaneous recombination noise power which is critical for our understanding of linewidth and mode spectrum.

D.1 Maxwell Equations

The four fundamental equations of Maxwell are as follows:

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \rho, && \text{(D.1)} \\ \nabla \cdot \mathbf{B} &= 0, && \text{(D.2)} \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, && \text{(D.3)} \\ \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} &= \mathbf{J}. && \text{(D.4)} \end{aligned}$$

The notations are defined as follows,

- $\mathbf{D}$ is the electrical displacement vector in $coulombs/meter^2$.
- $\mathbf{B}$ is the magnetic induction in $webers/meter^2$.
- $\mathbf{E}$ is the electric field intensity in $volts/meter$.
- $\mathbf{H}$ is the magnetic field in $ampere/meter$.

- ρ is the free charge density in $coulombs/meter^3$.
- $\mathbf{J}$ is the current density in $coulombs/meter^2$.

With the introduction of the polarization vector $\mathbf{P}$ and the magnetization vector $\mathbf{M}$, it is possible to eliminate the vectors $\mathbf{D}$ and $\mathbf{H}$ from the fundamental equations by substituting:

$$\begin{aligned} \mathbf{D} &= \epsilon_0 \mathbf{E} + \mathbf{P}, && \text{(D.5)} \\ \mathbf{H} &= \frac{\mathbf{B}}{\mu_0} - \mathbf{M}. && \text{(D.6)} \end{aligned}$$

Then we re-write the Maxwell equations in the following form:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{\epsilon}(\rho - \nabla \cdot \mathbf{P}) && \text{(D.7)} \\ \nabla \cdot \mathbf{B} &= 0, && \text{(D.8)} \\ \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} &= 0, && \text{(D.9)} \\ \nabla \times \mathbf{B} - \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t} &= \mu_0 \left(\mathbf{J} + \frac{\partial \mathbf{P}}{\partial t} + \nabla \times \mathbf{M} \right). && \text{(D.10)} \end{aligned}$$

In our applications, no magnetization is involved. This allows us to set the $\mathbf{M}$ to be zero in the above equations. The Maxwell equations can be simplified further if we concentrate on the Fourier component at the optical frequency, also referred to as $\mathbf{E}_\omega$ elsewhere in the book. Using a time dependent factor of $exp(-j\omega t)$ for all the variables, we obtained the following expressions for the last two equations of the Maxwell equation set:

$$\begin{aligned} \nabla \times \mathbf{E} + j\omega \mathbf{B} &= 0, && \text{(D.11)} \\ \nabla \times \mathbf{B} - j\omega \epsilon_0 \mu_0 \mathbf{E} &= \mu_0 \left(\mathbf{J} + j\omega \mathbf{P} \right). && \text{(D.12)} \end{aligned}$$

In a waveguiding structure, we can consider the free current $\mathbf{J}$ to be zero at the optical frequencies. The polarization vector can be decomposed into two components as follows:

$$\mathbf{P} = \epsilon_0(\epsilon_r - 1)\mathbf{E} + P_{sp}. \tag{D.13}$$

where the first term is the polarization induced by the electric field which includes the usual effects from the dielectric constants and also effects like loss and gain. The second term is polarization induced by spontaneous emission of photons associated with radiative recombination of electron-hole pairs. Such a spontaneous emission process is random and not directly related to the external electric field intensity. As a results, we expect the second term to have a zero mean in both the spatial and time domain.

Elimination of the **B** vector from the Maxwell equations results in the following

$$-\nabla \times \nabla \times \mathbf{E} + \omega^2 \epsilon_0 \epsilon_r \mu_0 \mathbf{E} = -\mu_0 \omega^2 \mathbf{P_{sp}}. \tag{D.14}$$

With the assumption $\nabla \cdot \mathbf{E} = 0$, we finally arrived at the vector wave equation we use most often in our applications,

$$\nabla^2 \mathbf{E} + \omega^2 \epsilon_0 \epsilon_r \mu_0 \mathbf{E} = -\mu_0 \omega^2 \mathbf{P_{sp}}. \tag{D.15}$$

Using the speed of light $c_0 = 1/\sqrt{\epsilon_0 \mu_0}$ the wave equation can be written in a more familiar form:

$$\nabla^2 \mathbf{E} + \frac{\omega^2}{c_0^2} \epsilon_r \mathbf{E} = \mathbf{F}_\omega. \tag{D.16}$$

The right hand side term is also called the Langevin force F_ω:

$$F_\omega = -\mu_0 \omega^2 \mathbf{P_{sp}}. \tag{D.17}$$

D.2 Conventions

In this work we shall assume the oscillating field takes the form $exp(-j\omega t)$. As a result, the forward (or right going) wave propagates with a factor $exp(+jkz)$ and backward (or left going) wave with factor $exp(-jkz)$. Under such a convention, a loss material has positive imaginary wave number k'' where $k = k' + jk''$. Any time dependence quantities can be expanded in Fourier component with $exp(-j\omega t)$ and we assume that the real field can be expressed as

$$E(t) = \int_0^\infty E_\omega exp(-j\omega t) d\omega + c.c. \tag{D.18}$$

where c.c. is used to denote the complex conjugate. Using this notation, the field with single frequency can be written as

$$E(t) = E_\omega exp(-j\omega t) + E_\omega^* exp(j\omega t) \tag{D.19}$$

In some cases, we adopt the bracket notation from quantum mechanics such that

$$\begin{aligned} \phi(\mathbf{r}) &= |\phi >= |n > && \text{(D.20)} \\ \phi^*(\mathbf{r}) &= <\phi| = |n > && \text{(D.21)} \\ <\psi|\phi> &= <n|n>= \int \psi^* \phi d\mathbf{r} && \text{(D.22)} \end{aligned}$$

D.3 Basic Wave Equation

For simplicity we are only concerned with the scalar field in the optical frequency domain ω. The scalar wave equation is written as:

$$\nabla^2 E_\omega + \frac{\omega^2}{c_0^2}\epsilon_r E_\omega = F_\omega. \tag{D.23}$$

In a wave guide environment, it common to solve the above wave equation by separating variables in the field (we drop the subscript ω and it is understood that we deal with single frequency):

$$E(x, y, z) = Z(z)\phi(x, y) \tag{D.24}$$

or simply

$$E(x, y, z) = E(z)\phi(x, y) \tag{D.25}$$

We assume that there is a orthogonal and complete set of transverse modes $\phi_n(x, y)$ satisfying

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\omega^2}{c_0^2}\epsilon_r(\mathbf{r})\right]\phi_n(x, y) = \frac{\omega^2}{c_0^2}n_{eff}^2(z)\phi_n(x, y). \tag{D.26}$$

where the z-dependent effective index $n_{eff}(z)^2$ is also used to define the z-dependent wave number (or propagation constant):

$$k(z) = \frac{\omega}{c_0}n_{eff}(z). \tag{D.27}$$

We notice that the lateral solution $|\phi_n>$ may still be z-dependent. However, if the z-dependence is weak compared with the x-y dependence, we can treat it as a parameter and evade the differential operator $\frac{\partial}{\partial z}$. For this reason we temporarily hide its z-dependence here.

It is well known that solution to Eq. (D.26) has nice properties such as orthogonality and completeness. The orthogonality is

$$\int \phi_n\phi_m =< \phi_n\phi_m >= \delta_{nm} < \phi_n\phi_m > . \tag{D.28}$$

And the completeness is

$$\sum_n \frac{\phi_n(x, y)\phi_n(x_s, y_s)}{< \phi_n\phi_n >} = \delta(x - x_s)\delta(y - y_s). \tag{D.29}$$

For an arbitrary solution for $\phi(x, y)$, we expand it in terms of the complete set of ϕ_n

$$|\phi >= \sum_n A_n|\phi_n > \tag{D.30}$$

where

$$A_n = \frac{< \phi_n|\phi >}{< \phi_n|\phi_n >} \tag{D.31}$$

Using this expansion, we are able to rewrite the wave equation as follows:

$$E(z)\left(\frac{\partial^2}{x^2}+\frac{\partial^2}{y^2}\right)|\phi > +|\phi > \frac{\partial^2}{z^2}E(z)+\frac{\omega^2}{c_0^2}\epsilon_r E(z)|\phi > \quad = \quad F \tag{D.32}$$

$$E(z)\left(\frac{\partial^2}{x^2}+\frac{\partial^2}{y^2}\right)(\sum_n A_n|\phi_n >) + |\phi > \frac{\partial^2}{z^2}E(z)+\frac{\omega^2}{c_0^2}\epsilon_r E(z)(\sum_n A_n|\phi_n >) \quad = \quad F \tag{D.33}$$

$$|\phi > \frac{\partial^2}{z^2}E(z) + E_z\left[\left(\frac{\partial^2}{x^2}+\frac{\partial^2}{y^2}\right)(\sum_n A_n|\phi_n >)+\frac{\omega^2}{c_0^2}\epsilon_r(\sum_n A_n|\phi_n >)\right] \quad = \quad F \tag{D.34}$$

$$|\phi > \frac{\partial^2}{z^2}E(z) + E(z)\left[\left(\frac{\partial^2}{x^2}+\frac{\partial^2}{y^2}\right)(\sum_n A_n|\phi_n >)+\frac{\omega^2}{c_0^2}\epsilon_r(\sum_n A_n|\phi_n >)\right] \quad = \quad F \tag{D.35}$$

$$|\phi > \frac{\partial^2}{z^2}E(z) + (\sum_n A_n k^2 \phi_n)E(z) \quad = \quad F \tag{D.36}$$

Let us multiply the above equation by $< \phi_n|$ and obtain

$$< \phi_n|\phi > \frac{\partial^2}{z^2}E_z + A_n k^2 < \phi_n|\phi_n > E_z \quad = \quad < \phi_n|F > . \tag{D.37}$$

$$< \phi_n|\phi > \frac{\partial^2}{z^2}E_z + A_n k^2 < \phi_n|\phi_n > E_z \quad = \quad < \phi_n|F > . \tag{D.38}$$

$$< \phi_n|\phi > \frac{\partial^2}{z^2}E_z + k^2 < \phi_n|\phi > E_z \quad = \quad < \phi_n|F > . \tag{D.39}$$

$$\frac{\partial^2}{z^2}E_z + k^2 E_z \quad = \quad \frac{< \phi_n|F(x,y,z) >}{< \phi_n|\phi >} \tag{D.40}$$

$$\frac{\partial^2}{z^2}E_z + k^2 E_z \quad = \quad f(z) \tag{D.41}$$

where

$$f(z) = \frac{< \phi_n|F(x,y,z) >}{< \phi_n|\phi >} \tag{D.42}$$

We have finally reduced the 3-variable equation to a single variable differential equation. If we are lucky enough to find the solution for E_z, our final solution for the three dimensional equation can be written as

$$E_z(z)[\sum_n A_n(z)\phi_n(x, y; z)] \tag{D.43}$$

Here we have explicitly show the dependence of the lateral solution to z. One good example of z-dependent solution is beam propagation in a tapered wave guiding structure.

D.4 The Green's Function

A well know technique for solving partial differential equation is the Green's function method. The basic idea is that for a differential equation of the form

$$\mathcal{L}(Z) = f(z) \tag{D.44}$$

where $\mathcal{L}$ is a linear differential operator. The solution can be written as

$$Z(z) = \int f(z_s)g(z, z_s)dz_s \tag{D.45}$$

where $g(z, z_s)$ is the Green's function satisfying

$$\mathcal{L}(Z) = \delta(z - z_s) \tag{D.46}$$

The interpretation of the Green function technique is simple enough: If we can find a solution due to a single point solution located at z_s, the solution due to a continuous distribution of sources $f(z)$ is just the superposition of all the solutions from the points sources contained in $f(z)$.

The Green's function technique is ideal for the analysis of our waveguiding system because the the spontaneous noise can be considered as a random ensemble of point sources. The Green's function allows us to give a compact formal solution in terms of the noise. Note that the formal solution is all we need here because it is impossible to write down an analytical expression for random noise.

D.5 Longitudinal Green's Function

Our final goal is to find the solution to the Green's function in Eq. (D.41). Several derivation steps should be taken before we arrive at the solution of the Green's function. We consider the following ordinary differential equation:

$$\mathcal{L}(Z) = \left(\frac{d^2}{dz^2} + k^2(z)\right) Z(z) = f(z). \tag{D.47}$$

The longitudinal Green's function is just a special case of the above equation with $f(z) = \delta(z - z_s)$.

D.5.1 The Wronskian

It is convenient to introduce the Wronskian from functions $Z_1(z)$ and $Z_2(z)$:

$$\begin{aligned} W &= \begin{vmatrix} Z_1(z) & Z_1'(z) \\ Z_2(z) & Z_2'(z) \end{vmatrix} && \text{(D.48)} \\ &= Z_1(z)Z_2'(z) - Z_2(z)Z_1'(z) && \text{(D.49)} \end{aligned}$$

where we have used $'$ to denote the derivative with respect to z.

We notice a couple of interesting properties of the Wronskian. First the Wronskian can be used to test the dependency of two functions. If the two functions Z_1 and Z_2 are dependent on each other, i.e., if they are proportional to each other, then the Wronskian is zero everywhere. If the Wronskian differs from zero for any range of of values of z, then Z_1 is a completely different function from Z_2 and the two functions are said to be independent.

The second properties of the Wronskian is that, if Z_1 and Z_2 are two independent solutions of the homogeneous equation $\mathcal{L}(\mathcal{Z}) = 0$, then the Wronskian should be independent of z, or in another word,

$$\begin{aligned} \frac{dW}{dz} &= Z_1 Z_2'' - Z_2 Z_1'', && \text{(D.50)} \\ &= Z_1[-k^2(z)Z_2] - Z_2[-k^2(z)Z_1], && \text{(D.51)} \\ &= 0. && \text{(D.52)} \end{aligned}$$

If W is not zero but a constant independent of z, and if we already find one solution Z_1 to the homogeneous equation $\mathcal{L}(Z) = 0$, then we can use the Wronskian to find another independent solution to the homogeneous equation as follows,

$$\begin{aligned} W &= Z_1(z)Z_2'(z) - Z_2(z)Z_1'(z), && \text{(D.53)} \\ &= Z_1^2 \frac{d}{dz}\left(\frac{Z_2}{Z_1}\right). && \text{(D.54)} \\ Z_2(z) &= W Z_1(z) \int_{z_0}^{z} \frac{du}{Z_1^2(u)}. && \text{(D.55)} \end{aligned}$$

D.5.2 Solution of Inhomogeneous Equation

Let us try to find a solution to the inhomogeneous equation Eq. (D.47) by setting $Z(z) = u(z)v(z)$ in $\mathcal{L}(Z) = f(z)$, obtaining

$$v\mathcal{L}(u) + uv'' + 2u'v' = f(z). \qquad \text{(D.56)}$$

If now we set u equal to one of the solutions Z_1 of the homogeneous equation, $\mathcal{L}(Z) = 0$, we obtain

$$v'' + 2(Z_1'/Z_1)v' = f(z)/Z_1 \qquad \text{(D.57)}$$

On the other hand, we already know that the second solution from is related to first by the Wronskian by the following [see Eq. (D.55)]

$$\frac{d}{dz}\left(\frac{Z_2(z)}{Z_1(z)}\right) = \frac{W}{Z_1^2(z)}. \tag{D.58}$$

Taking another derivative, we have

$$\begin{aligned}\frac{d}{dz^2}\left(\frac{Z_2(z)}{Z_1(z)}\right) &= \frac{W}{Z_1^2(z)} &\text{(D.59)}\\ &= \left(\frac{W}{Z_z^2}\right)' &\text{(D.60)}\\ &= -2\frac{WZ_1'}{Z_1^3} &\text{(D.61)}\end{aligned}$$

Using Eq. (D.58) so that

$$(Z_2/Z_1)'' + 2(Z_1/Z_1)(Z_2/Z_1)' = 0 \tag{D.62}$$

Multiplying this equation by v' and Eq. (D.57) by $(Z_2/Z_1)'$ and subtracting so that the second term on both equations cancel out, we obtain

$$\begin{aligned}\left(\frac{Z_2}{Z_1}\right)' v'' - v'\left(\frac{Z_2}{Z_1}\right)'' &= \left[\left(\frac{Z_2}{Z_1}\right)'\right]^2 \frac{d}{dz}\left[\frac{v'}{(Z_2/Z1)'}\right] &\text{(D.63)}\\ &= \left(\frac{f(z)}{Z_1}\right)\left(\frac{Z_2}{Z_1}\right)' &\text{(D.64)}\\ &= \left(\frac{f(z)}{Z_1}\right)\left(\frac{W}{Z_1^2}\right) &\text{(D.65)}\end{aligned}$$

By now, we have reduced our original second order inhomogeneous equation $\mathcal{L}(Z) = f(z)$ to a simple first order inhomogeneous equation

$$\frac{d}{dz}\left[\frac{v'}{(Z_2/Z_1)'}\right] = \frac{fZ_1}{W} \tag{D.66}$$

Integrating this first order equation, we obtain

$$v' = \left[\frac{d}{dz}\left(\frac{Z_2}{Z_1}\right)\right]\int \frac{fZ_1}{W}dz \tag{D.67}$$

or by adding another term on both side of the above equation

$$v' + \frac{fZ_2}{W} = \frac{d}{dz}\left[\left(\frac{Z_2}{Z_1}\right)\int \frac{fZ_1}{W}dz\right] \tag{D.68}$$

Integration of the above equation gives our final solution:

$$Z = vZ_1 \tag{D.69}$$
$$= Z_1 \int_z^{z_2} \frac{fZ_2 dz}{W} + Z_2 \int_{z_1}^{z} \frac{fZ_1 dz}{W} \tag{D.70}$$

where z_1 and z_2 are constants used to fit boundary conditions. It is interesting to note that z_1 and z_2 can be regarded as the starting and ending points along the longitudinal direction and Z_1 and Z_2 may be regarded as solutions to the homogeneous equation satisfying the boundary condition at z_1 and z_2. To be more specific, we examine Eq. (D.70) in more details. If we take the limit of $z->z_1$, the second term goes to zero and the first one is proportional to Z_1. Similarly if $z->z_2$, then the first term is zero and the second term is proportional to Z_2. Therefore, if Z_1 and Z_2 satisfy the boundary conditions at $z = z_1$ and $z = z_2$, respectively, then we have found a solution to the inhomogeneous equation which satisfies all boundary conditions.

In the case of Green's function, we set $f(z) = \delta(z - z_s)$ and obtain:

$$g(z, z_s) = Z_1 \int_z^{z_2} \frac{\delta(z - z_s) Z_2 dz}{W} + Z_2 \int_{z_1}^{z} \frac{\delta(z - z_s) Z_1 dz}{W} \tag{D.71}$$
$$g(z, z_s) = [Z_1(z) Z_2(z_s) \theta(z_s - z) + Z_2(z) Z_1(z_s) \theta(z - z_s)]/W \tag{D.72}$$

D.6 Green's Function for an AR Coated Waveguide

We consider a simple example of an unending waveguide (or a traveling wave amplifier with perfect anti-reflection AR coatings). This example has special significance because we need to use the result to derive a more general relation between spontaneous emission power and the diffusion coefficient.

We need to choose Z_1 and Z_2 to construct the Green's function and the Wronskian. For such a waveguide with perfect AR coating, the boundary condition at the two facets are waves traveling out of the device.

$$Z_1 = exp(-jkz) \tag{D.73}$$
$$Z_2 = exp(+jkz) \tag{D.74}$$

The Wronskian is

$$W_n = 2jk \tag{D.75}$$

and the Green's function is

$$g(z, z_s) W_n = Z_1(z) Z_2(z_s) \theta(z_s - z) + Z_2(z) Z_1(z_s) \theta(z - z_s) \tag{D.76}$$

$$
\begin{aligned}
&= exp[-jk(z-z_s)]\theta(z_s-z)+exp[-jk(z-z_s)]\theta(z-z_s) \qquad \text{(D.77)}\\
&= exp[-jk(z-z_s)]\theta(z_s-z)+exp[jk(z-z_s)]\theta(z-z_s) \qquad \text{(D.78)}\\
&= exp(jk|z-z_s|) \qquad \text{(D.79)}
\end{aligned}
$$

Let us try to find the solution of the wave equation in this waveguide if we assume the driving noise term is uncorrelated and has a δ function like correlation:

$$< f(z)f^*(z') >= f_0\delta(z-z') \qquad \text{(D.80)}$$

where $<>$ is used denote ensemble average and

$$f_0(z) = \left|\frac{< \phi_n|F(x,y,z) >}{< \phi_n|\phi >}\right|^2 \qquad \text{(D.81)}$$

The field E_z can be expressed as a function of the Green's function:

$$
\begin{aligned}
E(z) &= \int f(z_s)g(z,z_s)dz_s \qquad \text{(D.82)}\\
&= \int \frac{f(z_s)exp(jk|z-z_s|)}{W_n}dz_s \qquad \text{(D.83)}\\
&= \int \frac{f(z_s)exp(jk|z-z_s|)}{2jk}dz_s \qquad \text{(D.84)}
\end{aligned}
$$

Since $f(z)$ is used to represent noise, it is not possible to express E_z analytically. However, we can consider the average field amplitude due to the noise:

$$
\begin{aligned}
&< E_z(z)E_z^*(z) >\\
&= \int_0^L\int_0^L < f(z_s)f^*(z_s')g(z,z_s)g^*(z,z_s')dz_s dz_s' \qquad \text{(D.85)}\\
&= \int_0^L\int_0^L \frac{< f(z_s)f^*(z_s') > exp(jk|z-z_s|-jk^*|z-z_s'|)}{4|k|^2}dz_s dz_s' \qquad \text{(D.86)}\\
&= \int_0^L \frac{f_0\delta(z_s-z_s')exp(jk|z-z_s|-jk^*|z-z_s'|)}{4|k|^2}dz_s dz_s' \qquad \text{(D.87)}\\
&= f_0\int_0^L \frac{exp(-2k''|z-z_s|)}{4|k|}dz_s \qquad \text{(D.88)}\\
&= f_0\int_0^z \frac{exp[-2k''(z-z_s)]}{4|k|}dz_s + f_0\int_z^L \frac{exp[+2k''(z-z_s)]}{4|k|}dz_s \qquad \text{(D.89)}\\
&= \frac{f_0}{4|k|^2}\left\{\int_0^z exp[-2k''(z-z_s)]dz_s + \int_z^L exp[+2k''(z-z_s)]dz_s\right\}
\end{aligned}
$$

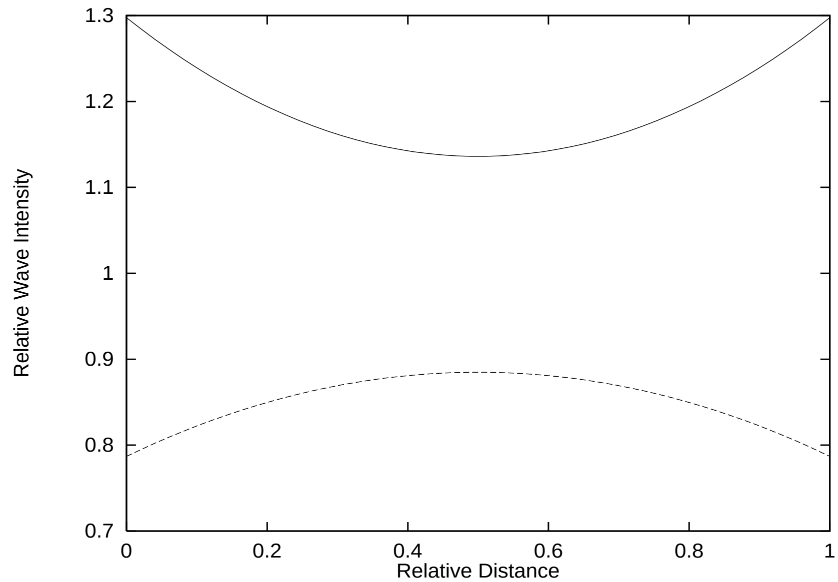

Figure D.1: The relative wave intensity of wave in a AR coated waveguide with gain (solid) and loss (dash). The relative gain (g_r)and loss (α_r) are set at 0.5 each.

$$\tag{D.90}$$

$$= \frac{f_0}{4|k|^2}\left\{\frac{1}{2k''}\left\{1-exp[-2k''z]\right\}+\frac{1}{2k''}\left\{1-exp[-2k''(L-z)]\right\}\right\} \tag{D.91}$$

$$= \frac{f_0}{8|k|^2k''}\left\{\left\{1-exp[-2k''z]\right\}+\left\{1-exp[-2k''(L-z)]\right\}\right\} \tag{D.92}$$

If we plot the above quantity as a function of the relative distance $z_r = z/L$, we will have the following interesting observations. In a gain medium, we define $g_r = -2k''L$ and

$$< E_z(z)E_z^*(z) > \frac{4|k|^2}{f_0} = \frac{1}{g_r}\left\{exp[g_r z_r]-1\right\}+\frac{1}{g_r}\left\{exp[g_r(1-z_r)]-1\right\} \tag{D.93}$$

Similarly in the case of a loss material, we set $\alpha_r = \alpha L = 2kL$ and obtain

$$< E_z(z)E_z^*(z) > \frac{4|k|^2}{f_0} = \frac{1}{\alpha_r}\left\{1-exp[-\alpha_r z_r]\right\}+\frac{1}{\alpha_r}\left\{1-exp[-\alpha_r(1-z_r)]\right\} \tag{D.94}$$

We plot the amplitude of the field due to the noise in Fig D.1. It shows that for a gain medium, the noise field amplitude (or the noise power) gets amplified and shows a maximum value at both side of the device, while a lossy material attenuates the noise field and has minimum amplitudes at both ends.

In the following sections, we need to estimate the pure effect of spontaneous emission. This requires the elimination of boundary condition and any amplification effect. A long and lossy waveguide is regarded to have this pure effect at one end. Consider z=L and L being large, we have

$$< E_z(z)E_z^*(z) >= \frac{f_0}{8|k|^2k''} \tag{D.95}$$

or

$$< E_z(z)E_z^*(z) >= \frac{2D_{FF}(z)}{8|k|^2k''} \tag{D.96}$$

D.7 Diffusion Coefficient of the Langevin Forces

The driving noise term F appearing on the right hand side of wave equation is random in nature and uncorrelated over time or distance. Our starting point is that the ensemble average

$$\begin{aligned} < F_\omega(\mathbf{r})F_{\omega'}(\mathbf{r}') > &= 0 \quad (D.97) \\ < F_\omega^*(\mathbf{r})F_{\omega'}^*(\mathbf{r}') > &= 0 \quad (D.98) \end{aligned}$$

and the correlation function is of δ function like:

$$< F_\omega(\mathbf{r})F_{\omega'}^*(\mathbf{r}') >= 2D_{FF^*}(\mathbf{r})\delta(\mathbf{r}-\mathbf{r}')\delta(\omega-\omega') \tag{D.99}$$

where the coefficient $2D_{FF^*}(\mathbf{r})$ was introduced by C. H. Henry [116] and was called "diffusion coefficient". For convenience we drop the delta function in ω and always use the same frequency for all the field quantities. The driving noise term can be expressed as

$$\begin{aligned} f(z) &= \frac{< \phi_n|F(x,y,z) >}{< \phi_n|\phi >} \quad (D.100) \\ f(z')^* &= \frac{< F(x',y',z')|\phi_n >,}{< \phi|\phi_n >} \quad (D.101) \end{aligned}$$

$$\begin{aligned} < f(z)f(z')^* > &= \left\langle \frac{< \phi_n(\mathbf{r}_{xy})|F(\mathbf{r}) >}{< \phi_n|\phi >} \frac{< F(\mathbf{r}')|\phi_n(\mathbf{r}'_{xy}) >}{< \phi|\phi_n >} \right\rangle \quad (D.102) \\ &= \left\langle \frac{< \phi_n(\mathbf{r}_{xy})| < F(\mathbf{r}')|F(\mathbf{r}) > |\phi_n(\mathbf{r}'_{xy}) >}{|< \phi_n|\phi >|^2} \right\rangle \quad (D.103) \\ &= \frac{< \phi_n(\mathbf{r}_{xy})|2D_{FF^*}(\mathbf{r})\delta(\mathbf{r}'-\mathbf{r})|\phi_n(\mathbf{r}'_{xy}) >}{|< \phi_n|\phi >|^2} \quad (D.104) \end{aligned}$$

$$= \frac{<\phi_n(\mathbf{r}_{xy})|2D_{FF^*}|\phi_n(\mathbf{r}_{xy})>}{|<\phi_n|\phi>|^2}\delta(z-z') \tag{D.105}$$

$$= f_0(z)\delta(z-z') \tag{D.106}$$

In order to determine the diffusion coefficient, we wish to evaluate the emitted optical power within $\Delta\omega$ from the bulk photon energy density as follows:

$$S_\omega\hbar\omega = 2\varepsilon_0 n n_g < |E(z)|^2 > |\phi(x,y)|^2 \tag{D.107}$$

The linear energy density is as follows

$$s_\omega\hbar\omega = 2\varepsilon_0 n n_g < |E(z)|^2 > \int dxdy|\phi(x,y)|^2 \tag{D.108}$$

or

$$s_\omega\hbar\omega = 2\varepsilon_0 n n_g < |E(z)|^2 > \tag{D.109}$$

From the result of the previous section

$$s_\omega\hbar\omega = 2\varepsilon_0 n n_g \frac{2D_{FF}(z)}{8|k|^2k''} \tag{D.110}$$

The power emitted is just the above times v_g:

$$P_n = v_g 2\varepsilon_0 n n_g \frac{2D_{FF}(z)}{8|k|^2k''} \tag{D.111}$$

$$P_n = -v_g 2\varepsilon_0 n n_g \frac{2D_{FF}(z)}{4|k|^2 g} \tag{D.112}$$

$$P_n = -c_0\varepsilon_0 n \frac{2D_{FF}(z)}{2|k|^2 g} \tag{D.113}$$

Please note the minus sign comes in because $-g = 2k''$.

Now let us derive the optical power from a completely different view point. When the photons are in equilibrium with the environment, the average noise power within $\Delta\omega$ can be expressed in terms of the mode occupation number $< n_\omega >$:

$$P_n\Delta\omega = \sum_n v_{gn}\hbar\omega < n_\omega > \Delta\omega D_1(E) \tag{D.114}$$

where $D_1(E)$ is the 1D mode density (modes per unit energy per unit length) which is derived as follows.

The mode spacing is

$$\Delta k_z = \frac{2\pi}{L_z} \tag{D.115}$$

Therefore the mode number per unit length per unit ω is $D_1(E) = \frac{1}{2\pi}\frac{dk}{d\omega}$. To a good approximation, the group velocity may be used:

$$\frac{dk}{d\omega} = 1/v_g \tag{D.116}$$

This gives a power

$$P_n = \hbar\omega < n_\omega > /(2\pi) \tag{D.117}$$

Comparison of the two different expressions of optical power, we finally obtain the following relation for the diffusion coefficient:

$$2D_{FF}(z) = -|k|^2\hbar\omega g < n_\omega > /(\pi c_0 \varepsilon_0 n) \tag{D.118}$$

or, if we neglect the imaginary part of the refractive index in $|k|$,

$$2D_{FF}(z) = -\frac{\hbar\omega^3 n'(z) g(z) < n_\omega >}{\pi c_0^3 \varepsilon_0} \tag{D.119}$$

Please note that a difference of factor $(2\pi)^2$ than that from Ref. [116] is due to the use of different unit system.

According to Henry [116], the mode occupation number is given by

$$< n_\omega >= -n_{sp} = \frac{1}{exp\left(\frac{\hbar\omega - eV_F}{kT}\right) - 1} \tag{D.120}$$

Using the well-known relation between optical gain and spontaneous emission, we can also use an alternative (or more convenient) form for the diffusion coefficient in terms of bulk spontaneous recombination rate.

$$g = \frac{r_{sp}(E)}{v_g D(E)} \left\{1 - exp[(\hbar\omega - eV_F)/kT]\right\}. \tag{D.121}$$

where

$$D(E) = \frac{\pi k^2 \Delta k}{\pi^3 \Delta E}. \tag{D.122}$$

$$= \left(\frac{n}{\hbar c}\right)^3 \frac{E^2}{\pi^2} \tag{D.123}$$

is the photon mode density. Thus, the product gn_{sp} be re-written as

$$gn_{sp} = \frac{r_{sp}(E)}{v_g D(E)} \tag{D.124}$$

Using the above for the n-th lateral mode, we have

$$2D_{FF}(z) = |k|^2 E < \phi_n | r_{sp} | \phi_n > /(D(E) v_g \pi c_0 \varepsilon_0 n) \tag{D.125}$$

$$2D_{FF}(z) = n^2\omega^2 E < n|r_{sp}|n > \pi^2\hbar^3 c_0^3/(n^3 E^2 v_g \pi c_0^3 \varepsilon_0 n) \tag{D.126}$$

$$2D_{FF}(z) = (\hbar\omega) < n|r_{sp}(E)|n > \pi\hbar/(n^2 v_g \varepsilon_0) \tag{D.127}$$

Please note that $r_{sp}(E)$ must be in units of rate per volume per energy interval.

Appendix E

THEORY OF POWER SPECTRUM

It has been shown in Ref. [117] that near lasing the Wronskian is independent of the position z and is only a function of frequency. Thus we can expand it over a complex omega plane as follows:

$$\frac{1}{W} \approx \sum_i \frac{1}{\frac{dW(\omega_i)}{d\omega}(\omega - \omega_i)} \tag{E.1}$$

Let us derive the photon density in terms of the electric field. According to Landau and Lifschitz [118], the energy density, U, of a dispersive medium can be written as

$$U = \varepsilon_0 \frac{\partial}{\partial\omega}(\varepsilon'\omega)E^2 + \varepsilon_0(\varepsilon')E^2 \tag{E.2}$$

where ε' is the real part of the dielectric constant at the optical frequencies. We go on and derive the following:

$$U = 2\varepsilon_0 n'\omega\frac{\partial n'}{\partial\omega}E^2 + \varepsilon_0 n'^2 E^2 + \varepsilon_0\varepsilon' E^2 \tag{E.3}$$

which can be expressed in terms of the group index as follows:

$$\begin{aligned} U &= 2\varepsilon_0\omega\frac{\partial n'}{\partial\omega}E^2 + 2\varepsilon_0 n' E^2 && \text{(E.4)} \\ &= 2\varepsilon_0 n' n_g E^2 && \text{(E.5)} \end{aligned}$$

where

$$n_g = n' + \omega\frac{\partial n'}{\partial\omega} \tag{E.6}$$

Using the Fourier transform of the electric field, we write down our expression of photon density as

$$S(\omega) = \frac{U}{\hbar\omega} = \frac{2\varepsilon_0 n' n_g}{\hbar\omega}\left|E_\omega(z)\phi(x,y)\right|^2 \tag{E.7}$$

Integration over x-y leads to the linear photon density:

$$s_l(\omega, z) = \frac{2\varepsilon_0 n' n_g}{\hbar\omega} |E_\omega(z)|^2 \tag{E.8}$$

We recall that the spectrum of the electric field can be repressed in terms of the Green's function as follows:

$$E_\omega(z) = \frac{\int_0^l g(z, z') f_\omega(z') dz'}{W(\omega)} \tag{E.9}$$

We perform the spatial integral

$$\begin{aligned} s_l(\omega, z) &= \frac{2\varepsilon_0 n'(z) n_g}{\hbar\omega} |E_\omega(z)|^2 \\ &= \frac{2\varepsilon_0 n'(z) n_g}{\hbar\omega |W(\omega)|^2} \int_0^l g(z, z')^* f_\omega^*(z') dz' \int_0^l g(z, z'') f_\omega(z'') dz'' \\ &= \frac{2\varepsilon_0 n'(z) n_g}{\hbar\omega |W(\omega)|^2} \int_0^l \int_0^l g(z, z')^* g(z, z'') f_\omega^*(z') f_\omega(z'') dz' dz'' \qquad \text{(E.10)} \end{aligned} \tag{E.11}$$

We take the ensemble average of the above to obtain:

$$\begin{aligned} s_l(\omega, z) &= \frac{2\varepsilon_0 n'(z) n_g}{\hbar\omega |W(\omega)|^2} \int_0^l \int_0^l g(z, z')^* g(z, z'') < f_\omega^*(z') f_\omega(z'') > dz' dz'' \\ &= \frac{2\varepsilon_0 n'(z) n_g}{\hbar\omega |W(\omega)|^2} \int_0^l \int_0^l g(z, z')^* g(z, z'') 2D_\omega(z') \delta(z' - z'') dz' dz'' \\ &= \frac{2\varepsilon_0 n'(z) n_g}{\hbar\omega |W(\omega)|^2} \int_0^l |g(z, z')|^2 2D_\omega(z') dz' \end{aligned} \tag{E.12}$$

Let us transform the Wronskin as follows:

$$\begin{aligned} \frac{1}{|W(\omega)|^2} &= \sum_i \sum_j \frac{1}{\frac{dW^*(\omega_i)}{d\omega}(\omega - \omega_i^*)} \frac{1}{\frac{dW(\omega_j)}{d\omega}(\omega - \omega_j)} \\ &= \sum_i \frac{1}{|\ \frac{dW(\omega_i)}{d\omega}\ |^2\ (\omega - \omega_i^*)(\omega - \omega_i)} \\ &= \sum_i \frac{1}{|\ \frac{dW(\omega_i)}{d\omega}\ |^2\ [(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \end{aligned} \tag{E.13}$$

Please note that we have dropped the cross terms of $\frac{1}{(\omega-\omega_i)(\omega-\omega_j)^*}$ because these are normally much smaller than the sqared terms.

Using and n' and n interchangeably and combining Eq. (E.12) and Eq. (E.13), we obtain

$$s_l(\omega, z) = \frac{2\epsilon_0 n(z) n_g}{\hbar\omega} \left[\sum_i \frac{1}{|\frac{dW(\omega_i)}{d\omega}|^2 [(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \right] \int_0^l |g(z, z')|^2 2D_\omega(z') dz' \tag{E.14}$$

where $\omega_{re,i}$ and $\omega_{im,i}$ are the real and imaginary parts of a pole respectively. ϵ_0, n and n_g are the vacuum permitivity, refractive index and group index, respectively. $D_\omega(z)$ is the diffusion coefficient of the Langevin force $f_\omega(z)$ [117]. The above expression shows that the spectrum of the photon density is a sum of Lorentzian lines centered at $\omega_{re,i}$, with peak intensity proportional to $1/\omega_{im,i}^2$.

We integrate over z and obtain the modal spectrum:

$$\begin{aligned} I_\omega &= \int_0^l s_l(\omega, z) dz \\ &= \frac{4\pi\epsilon_0}{\hbar\omega} \sum_i \left[\left| \frac{dW(\omega_i)}{d\omega} \right|^{-2} \int_0^l \int_0^l n(z) n_g |g(z, z')|^2 2D_\omega(z') dz' dz \right] \\ &\times \frac{(2\pi)^{-1}}{[(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \end{aligned} \tag{E.15}$$

The term related to the Wronskian has been worked out in Ref. [117] as

$$\frac{dW(\omega_i)}{d\omega} = 2 \int_0^l k(z) Z_L(z) Z_R(z) \frac{\partial k(\omega_i, z)}{\partial \omega} dz \tag{E.16}$$

Near the material gains peak, to a good approximation we obtain

$$\frac{\partial k}{\partial \omega} = \frac{1}{v_g} \tag{E.17}$$

Assuming the system is near threshold, both $Z_L(z)$ and $Z_R(z)$ reduce to a common solution $Z_{0,i}(z)$ where the label i indicates wave function at frequency ω_i:

$$\frac{dW(\omega_i)}{d\omega} = \frac{2}{v_g} \int_0^l k(z) Z_{0,i}^2(z) dz \tag{E.18}$$

The Green's function is reduced to a simpler form:

$$g(z, z') = Z_0(z) Z_0(z') \tag{E.19}$$

where $Z_0(z)$ is understood as $Z_{0,i}$.

We have arrived at the following spectrum

$$\begin{aligned} I_\omega &= \frac{4\pi\epsilon_0}{\hbar\omega} \sum_i \frac{\int_0^l \int_0^l n(z) n_g |Z_0(z)|^2 |Z_0(z')|^2 2D_\omega(z') dz' dz}{\left| \frac{2\omega}{c^2} \int_0^l n(z) n_g Z_0^2(z) dz \right|^2} \\ &\times \frac{(2\pi)^{-1}}{[(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \end{aligned} \tag{E.20}$$

From Appendix D, we recall that the diffusion constant is related to the optical gain by

$$2D_{FF}(z) = \frac{\hbar\omega^3 n(z) g(z) n_{sp}}{\pi c^3 \varepsilon_0} \tag{E.21}$$

We thus obtain:

$$\begin{aligned} I_\omega &= \frac{4\pi\epsilon_0}{\hbar\omega} \frac{\hbar\omega^3}{\pi c^3 \varepsilon_0} \sum_i \frac{\int_0^l \int_0^l n(z) n_g |Z_0(z)|^2 |Z_0(z')|^2 n(z') g(z') n_{sp} dz' dz}{\left| \frac{2\omega}{c^2} \int_0^l n(z) n_g Z_0^2(z) dz \right|^2} \\ &\times \frac{(2\pi)^{-1}}{[(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \end{aligned} \tag{E.22}$$

which may be written in our final form of the spectrum:

$$\begin{aligned} I_\omega &= c \sum_i \frac{\left[\int_0^l n(z) n_g |Z_0(z)|^2 dz\right] \left[\int_0^l |Z_0(z')|^2 n(z') g(z') n_{sp} dz'\right]}{\left| \int_0^l n(z) n_g Z_0^2(z) dz \right|^2} \\ &\times \frac{(2\pi)^{-1}}{[(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \end{aligned} \tag{E.23}$$

Introducing the parameter $R_{sp,i}$:

$$R_{sp,i} = c \frac{\left[\int_0^l n(z) n_g |Z_0(z)|^2 dz\right] \left[\int_0^l |Z_0(z')|^2 n(z') g(z') n_{sp} dz'\right]}{\left| \int_0^l n(z) n_g Z_0^2(z) dz \right|^2} \tag{E.24}$$

which is also referred to as the rate of spontaneous emission into the i^{th} mode. The spectrum may be written as

$$I_\omega = \frac{(2\pi)^{-1} R_{sp,i}}{[(\omega - \omega_{re,i})^2 + \omega_{im,i}^2]} \tag{E.25}$$

Integrating over all frequencies, we get the total number of photons in the cavity $I_{p,i}$ in the i^{th} mode:

$$I_{p,i} = \frac{R_{sp,i}}{2\omega_{im,i}} \tag{E.26}$$

which agrees with results from other works in the literature (see, e.g., Ref. [117]).

In the above formulas, $Z_0(z) = Z_{0,i}(z)$ is the solution to the homogeneous scalar wave equation at $\omega = \omega_i$, v_g the group velocity, $g(z)$ the modal gain, and $n_{sp}(z)$ the inversion parameter. The expression (E.24) agrees completely with results in Ref. [117].

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